



```
In [ ]: from google.colab import drive
drive.mount('/content/drive')
```

Mounted at /content/drive

Thesis Project by Group CLS

THESIS 1-3

Thesis Project

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Section 1. Introduction

1.1 Problem Overview

Large-scale global optimization (LSGO) presents a critical challenge in computational intelligence, where the "curse of dimensionality" leads to an explosive increase in local optima and a high risk of premature convergence. While foundational algorithms like Differential Evolution (DE) are robust, their dependence on crossover and random mutation may restrict guided exploration — reducing particle diversity over time and leading to premature convergence.

This notebook implements and evaluates the proposed **Diversity-Preserving Level-Based Learning Differential Evolution (DP-LLDE)** algorithm, tested against the **CEC 2017 Large-Scale Global Optimization Benchmark** across 29 benchmark functions (F1-F30, excluding deprecated F2) at **10 dimensions**.

1.2 Research Objectives

1. Design and implement DP-LLDE, integrating sequential relative particle mutation and dynamic level-based learning into the Differential Evolution framework.

2. Evaluate DP-LLDE on the CEC 2017 benchmark suite (F1–F30) at 10 dimensions.
3. Examine convergence behavior, solution quality, and the effect of diversity-preserving strategies across unimodal, multimodal, hybrid, and composition functions.

1.3 Algorithm Overview - DP-LLDE

DP-LLDE extends LBLDE (Qiao et al., 2022) with three core contributions:

1. **Sequential Level Assignment** — each individual is assigned a learning source level drawn randomly from $\{2, \dots, NL\}$ on the first generation, decaying toward the elite level (level 1) in subsequent generations.
2. **Diversity-Aware Exemplar Selection** — if two exemplars e_1, e_2 fall below a diversity threshold, one is replaced with the most distant candidate in that level, preventing exemplar collapse.
3. **Normalized Mutation Pressure (Eq. 10):**

$$V_i = e_1 + F_i \times (e_2 - X_i) \times \frac{NL - k}{NL}$$

The factor $(NL-k)/NL$ decays to zero as $k \rightarrow NL$, naturally reducing mutation pressure as learning quality improves.

Parameter	Value
Population Size (NP)	100
Number of Levels (NL)	4
Initial μ_{CR}	0.35
Max FES	$10,000 \times D$
Dimensions (D)	10
Seed	42

Section 2. Benchmark Function Description

2.1 Overview

The **CEC 2017 Large-Scale Global Optimization Benchmark** (Awad et al., 2017) is a standardized suite of 30 test functions designed to evaluate optimization algorithms on complex, high-dimensional search spaces. The benchmark tests an algorithm's ability to avoid false local optima, convergence speed, and solution accuracy.

All functions are defined over the search space $[-100, 100]^D$ with their respective global optima offset by a function-specific bias value.

Library used: `cec2017` by Duncan Tilley — provides the official shift vectors, rotation matrices, and shuffle permutations from the CEC 2017 specification.

2.2 Unimodal Function Overview

Unimodal functions contain a **single global optimum and no local optima**. They are used to test the convergence speed and accuracy of an algorithm in a straightforward landscape.

No.	Function Name	Global Optimum
F1	Shifted and Rotated Bent Cigar	100
F2	Shifted and Rotated Sum of Different Power (<i>Deprecated</i>)	200
F3	Shifted and Rotated Zakharov	300

Note: F2 is officially deprecated from the CEC 2017 benchmark but is included here for completeness as it remains implemented in the `cec2017` library.

2.3 Multimodal Function Overview

Multimodal functions contain **multiple local optima**, making them significantly harder to solve. They are used to evaluate an algorithm's capacity to avoid premature convergence and escape false local optima.

No.	Function Name	Global Optimum
F4	Shifted and Rotated Rosenbrock's	400
F5	Shifted and Rotated Rastrigin's	500

No.	Function Name	Global Optimum
F6	Shifted and Rotated Expanded Schaffer's F6	600
F7	Shifted and Rotated Lunacek Bi-Rastrigin	700
F8	Shifted and Rotated Non-Continuous Rastrigin's	800
F9	Shifted and Rotated Levy	900
F10	Shifted and Rotated Schwefel's	1000

2.4 Hybrid Function Overview

Hybrid functions divide the search space into several regions, each governed by a **different basic function**. They assess an algorithm's adaptability to heterogeneous landscapes and mixed search behaviors.

No.	Component Functions	Global Optimum
F11	Zakharov; Rosenbrock; Rastrigin	1100
F12	Elliptic; Schwefel; Bent Cigar	1200
F13	Bent Cigar; Rosenbrock; Bi-Rastrigin	1300
F14	Elliptic; Ackley; Schaffer F7; Rastrigin	1400
F15	Bent Cigar; HGBat; Rastrigin; Rosenbrock	1500
F16	Schaffer F6; HGBat; Rosenbrock; Schwefel	1600
F17	Katsuura; Ackley; Griewank+Rosenbrock; Schwefel; Rastrigin	1700
F18	Elliptic; Ackley; Rastrigin; HGBat; Discus	1800
F19	Bent Cigar; Rastrigin; Griewank+Rosenbrock; Weierstrass; Schaffer F6	1900
F20	HappyCat; Katsuura; Ackley; Rastrigin; Schwefel; Schaffer F7	2000

2.5 Composite Function Overview

Composition functions **combine multiple shifted and rotated basic functions** into a single search space, creating highly complex and deceptive optimization landscapes. They are the most challenging category in the CEC 2017 suite and significantly test an algorithm's overall robustness and global search capability.

No.	Component Functions	Global Optimum
F21	Rosenbrock; Elliptic; Rastrigin	2100
F22	Rastrigin; Griewank; Schwefel	2200
F23	Rosenbrock; Ackley; Schwefel; Rastrigin	2300
F24	Ackley; Elliptic; Griewank; Rastrigin	2400
F25	Rastrigin; HappyCat; Ackley; Discus; Rosenbrock	2500
F26	Schaffer F6; Schwefel; Griewank; Rosenbrock; Rastrigin	2600
F27	HGBat; Rastrigin; Schwefel; Bent Cigar; Elliptic; Schaffer F6	2700
F28	Ackley; Griewank; Discus; Rosenbrock; HappyCat; Schaffer F6	2800
F29	F15; F16; F17	2900
F30	F15; F18; F19	3000

Section 3. List of Required Libraries

3.1 List of Libraries and Versions

The following libraries and modules were used for optimization, benchmarking, and visualization in this project:

Library / Module	Purpose	Version
<code>python</code>	Core programming language	3.10+
<code>numpy</code>	Numerical computing and array operations	2.0.2
<code>matplotlib</code>	Plotting and saving convergence graphs	3.10.0
<code>cec2017</code>	CEC 2017 benchmark functions (simple, hybrid, composition)	Latest (pip)
<code>cec2017.simple</code>	Simple benchmark functions F1-F10	—
<code>cec2017.hybrid</code>	Hybrid benchmark functions F11-F20	—
<code>cec2017.composition</code>	Composition benchmark functions F21-F30	—
<code>dplldc</code>	DP-LLDE optimization algorithm (uploaded locally)	Custom
<code>google.colab.files</code>	File upload utility for loading <code>dplldc.py</code> into the Colab environment	Built-in
<code>sys</code>	Clears cached module before re-importing <code>dplldc.py</code>	Built-in

Section 4. Environment Setup and Validation

4.1 Installation

Install `cec2017`

```
In [ ]: !pip install cec2017 --quiet
```

ERROR: Could not find a version that satisfies the requirement cec2017 (from versions: none)
ERROR: No matching distribution found for cec2017

4.2 File Validation

Import `simple.py`, `hybrid.py`, and `composition.py` python files for benchmark functions

```
In [ ]: from cec2017 import simple, hybrid, composition
print('cec2017 loaded!')
```

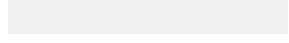
cec2017 loaded!

Import `sys` control over warning messages for verification that `lblde.py` exists

```
In [ ]: from google.colab import files
import sys

uploaded = files.upload()

if 'lblde_mutation_rule.py' in uploaded:
    if 'lblde_mutation_rule' in sys.modules:
        del sys.modules['lblde']
    import lblde_mutation_rule
    from lblde_mutation_rule import LBLDE_MutationRule
    print('lblde_mutation_rule.py uploaded and imported successfully!')
else:
    print('ERROR: Please upload a file named lblde.py')
```

 Upload widget is only available when the cell has been executed in the current browser session. Please rerun this cell to enable.
Saving `lblde_mutation_rule.py` to `lblde_mutation_rule.py`
`lblde_mutation_rule.py` uploaded and imported successfully!

Section 5. Configuration and Helper Functions

5.1 Library Imports

Import libraries `numpy`, `matplotlib`, and `cec2017` for numerical computation, visualization, and CEC 2017 benchmark functions.

```
In [ ]: import numpy as np
import matplotlib.pyplot as plt
from cec2017 import simple, hybrid, composition
```

5.2 Global Parameters

Define global parameters

```
In [ ]: D = 100 # Change this when testing different dimensions
```

```
In [ ]: bounds = np.array([[-100.0, 100.0]] * D)
MAX_FES = 10000 * D
NP, NL, MU_CR = 100, 4, 0.35
SEED = 42
```

Store all results for summary table under `all_results` dict

```
In [ ]: all_results = {}
```

```
In [ ]: import pickle
import os
from google.colab import drive

# Save functions
def save_results_checkpoint(results_dict, filename='all_results_checkpoint.pkl'):
    """Save all results to Google Drive."""
    drive_path = '/content/drive/My Drive/thesis_results_2'
    os.makedirs(drive_path, exist_ok=True)
    filepath = os.path.join(drive_path, filename)

    with open(filepath, 'wb') as f:
        pickle.dump(results_dict, f)
    print(f'✓ Checkpoint saved to: {filepath}')

def save_plots_to_drive(plot_filename):
    """Copy a single plot to Google Drive."""
```

```

drive_path = '/content/drive/My Drive/thesis_results_2/plots'
os.makedirs(drive_path, exist_ok=True)

if os.path.exists(plot_filename):
    import shutil
    dest = os.path.join(drive_path, plot_filename)
    shutil.copy(plot_filename, dest)
    print(f'✓ Plot saved to: {dest}')

def load_results_checkpoint(filename='all_results_checkpoint.pkl'):
    """Load previous results from Google Drive if they exist."""
    drive_path = '/content/drive/My Drive/thesis_results_2'
    filepath = os.path.join(drive_path, filename)

    if os.path.exists(filepath):
        with open(filepath, 'rb') as f:
            return pickle.load(f)
    return {}

```

5.3 Function Definitions

Define `make_wrapper` to adapt CEC 2017 functions to accept a 1D input array from DP-LLDE, returning a large penalty value if the output is invalid (nan/inf).

```

In [ ]: def make_wrapper(fn):
    """Wrap a cec2017 function (expects (1,D) input) to accept (D,) input."""
    def wrapper(x):
        val = fn(x.reshape(1, -1))
        if hasattr(val, '__len__'):
            val = float(val[0])
        else:
            val = float(val)
        return val if np.isfinite(val) else 1e18
    return wrapper

```

Define `run_and_plot` to execute a single DP-LLDE optimization run on a given benchmark function, print the results, store them for the summary table, and save the convergence graph.

```

In [ ]: def run_and_plot(fn_name, fn_eval, f_global, fn_label):
    """Run DP-LLDE_m on a function and plot convergence + diversity graphs."""
    print('=' * 80)
    print(f'DP-LLDE_m on CEC 2017 {fn_name} – Single Run')
    print('=' * 80)
    print(f'Function      : {fn_label} ({D}D)')
    print(f'Bounds        : [-100, 100]')
    print(f'Global optimum : {f_global}')

    opt = LBLDE_MutationRule(
        objective_func=fn_eval,

```



```

        bounds=bounds,
        NP=NP, NL=NL, mu_CR_ini=MU_CR,
        max_fes=MAX_FES,
        seed=SEED
    )
    # ← Unpack 4 values now
    best_sol, best_fit, history, diversity_history = opt.optimize(verbose=True)

    error = abs(best_fit - f_global)
    print(f'\n{"="*80}')
    print('Results:')
    print(f'    Best Fitness           : {best_fit:.6e}')
    print(f'    Global Optimum          : {f_global:.6e}')
    print(f'    Error (|best-optimum|) : {error:.6e}')
    print(f'    Best Solution (5 dim)   : {best_sol[:5]}')
    print(f'    D^(0) (init. diversity): {opt.D_0:.4e}')
    print(f'{"="*80}')
```

```

    # Store for summary
    all_results[fn_name] = {
        'label': fn_label,
        'best_fit': best_fit,
        'f_global': f_global,
        'error': error
    }

    save_results_checkpoint(all_results)

    fes_axis = np.linspace(0, MAX_FES, len(history))
    gen_axis = np.arange(len(diversity_history)) # Generations (0, 1, 2, ...)

    # — Two-panel figure —
    fig, (ax1, ax2) = plt.subplots(2, 1, figsize=(10, 10))

    # Convergence curve (top) – FEs on x-axis
    ax1.plot(fes_axis, history, color='steelblue', linewidth=1.5)
    ax1.set_ylabel('Best Fitness', fontsize=12)
    ax1.set_xlabel('Function Evaluations (FEs)', fontsize=12)
    ax1.set_title(
        f'DP-LLDE_m – CEC 2017 {fn_name}: {fn_label} ({D}D)',
        fontsize=12
    )
    ax1.grid(True, which='both', linestyle='--', alpha=0.6)

    # Diversity curve (bottom) – Generations on x-axis
    ax2.plot(gen_axis, diversity_history, color='darkorange', linewidth=1.5)
    ax2.set_xlabel('Generation', fontsize=12)
    ax2.set_ylabel('Population Diversity\n(Mean Std Dev)', fontsize=12)
    ax2.set_title(f'Diversity Across Generations – {fn_name}', fontsize=12)
    ax2.grid(True, which='both', linestyle='--', alpha=0.6)

    plt.tight_layout()
    fname = f'convergence_{fn_name}.png'

```

```
plt.savefig(fname, dpi=150)
plt.show()
print(f'Graph saved as {fname}\n')

save_plots_to_drive(fname)
```

```
In [ ]: def load_results_checkpoint(filename='all_results_checkpoint.pkl'):
        """Load previous results from local Colab storage."""
        filepath = f'/content/{filename}'

        if os.path.exists(filepath):
            with open(filepath, 'rb') as f:
                return pickle.load(f)
        return {}
```

```
In [ ]: all_results = load_results_checkpoint()
        if all_results:
            print(f'✓ Loaded {len(all_results)} previous results')
```

✓ Loaded 12 previous results

```
In [ ]: all_results = load_results_checkpoint()

        if all_results:
            print(f'✓ Loaded {len(all_results)} previous results')
            print('Already completed:')
            for fn_name in sorted(all_results.keys()):
                print(f'✓ {fn_name}')
            completed = set(all_results.keys())
        else:
            print('No checkpoint found')
            completed = set()
```

✓ Loaded 12 previous results

Already completed:

- ✓ F1
- ✓ F10
- ✓ F11
- ✓ F12
- ✓ F2
- ✓ F3
- ✓ F4
- ✓ F5
- ✓ F6
- ✓ F7
- ✓ F8
- ✓ F9

Section 6. CEC2017 Benchmark Functions

6.1 Unimodal Functions (F1-F3)

```
In [ ]: # =====  
# F1 – Shifted and Rotated Bent Cigar  
# =====  
run_and_plot('F1', make_wrapper(simple.f1), 100.0, 'Shifted and Rotated Bent C
```

```

=====
=
DP-LLDE_m on CEC 2017 F1 – Single Run
=====
=
Function      : Shifted and Rotated Bent Cigar (100D)
Bounds       : [-100, 100]
Global optimum : 100.0
Initial D^(0) = 5.7220e+01 | Diversity threshold = 5.7220e+00
Generation 0: Best Fitness = 6.191709e+12 | Diversity = 5.7220e+01
Generation 10: Best = 2.025431e+12 FES = 1100 Diversity = 2.0516e+01
Generation 20: Best = 1.425275e+12 FES = 2100 Diversity = 8.6047e+00
Generation 30: Best = 1.267832e+12 FES = 3100 Diversity = 3.9056e+00
Generation 40: Best = 1.179406e+12 FES = 4100 Diversity = 2.2337e+00
Generation 50: Best = 1.116157e+12 FES = 5100 Diversity = 1.5602e+00
Generation 60: Best = 1.075613e+12 FES = 6100 Diversity = 1.0523e+00
Generation 70: Best = 1.049188e+12 FES = 7100 Diversity = 8.2219e-01
Generation 80: Best = 1.013439e+12 FES = 8100 Diversity = 9.7248e-01
Generation 90: Best = 9.825452e+11 FES = 9100 Diversity = 8.6516e-01
Generation 100: Best = 9.545606e+11 FES = 10100 Diversity = 8.3506e-01
Generation 110: Best = 9.231353e+11 FES = 11100 Diversity = 8.7559e-01
Generation 120: Best = 9.009362e+11 FES = 12100 Diversity = 6.8016e-01
Generation 130: Best = 8.810622e+11 FES = 13100 Diversity = 5.6814e-01
Generation 140: Best = 8.558641e+11 FES = 14100 Diversity = 6.9263e-01
Generation 150: Best = 8.398966e+11 FES = 15100 Diversity = 5.7680e-01
Generation 160: Best = 8.205322e+11 FES = 16100 Diversity = 5.9823e-01
Generation 170: Best = 8.049693e+11 FES = 17100 Diversity = 4.9728e-01
Generation 180: Best = 7.892133e+11 FES = 18100 Diversity = 4.7119e-01
Generation 190: Best = 7.748646e+11 FES = 19100 Diversity = 4.8321e-01
Generation 200: Best = 7.627164e+11 FES = 20100 Diversity = 4.8952e-01
Generation 210: Best = 7.473136e+11 FES = 21100 Diversity = 5.2872e-01
Generation 220: Best = 7.329606e+11 FES = 22100 Diversity = 4.7499e-01
Generation 230: Best = 7.179016e+11 FES = 23100 Diversity = 5.1258e-01
Generation 240: Best = 7.042879e+11 FES = 24100 Diversity = 4.8635e-01
Generation 250: Best = 6.905528e+11 FES = 25100 Diversity = 3.9220e-01
Generation 260: Best = 6.773683e+11 FES = 26100 Diversity = 4.3536e-01
Generation 270: Best = 6.643233e+11 FES = 27100 Diversity = 3.7765e-01
Generation 280: Best = 6.557536e+11 FES = 28100 Diversity = 3.8312e-01
Generation 290: Best = 6.459645e+11 FES = 29100 Diversity = 3.4432e-01
Generation 300: Best = 6.379861e+11 FES = 30100 Diversity = 3.0809e-01
Generation 310: Best = 6.299517e+11 FES = 31100 Diversity = 3.0505e-01
Generation 320: Best = 6.241754e+11 FES = 32100 Diversity = 2.7786e-01
Generation 330: Best = 6.180698e+11 FES = 33100 Diversity = 2.9018e-01
Generation 340: Best = 6.135685e+11 FES = 34100 Diversity = 2.7335e-01
Generation 350: Best = 6.079938e+11 FES = 35100 Diversity = 2.8226e-01
Generation 360: Best = 6.030251e+11 FES = 36100 Diversity = 2.4075e-01
Generation 370: Best = 5.996834e+11 FES = 37100 Diversity = 1.9155e-01
Generation 380: Best = 5.962214e+11 FES = 38100 Diversity = 1.5827e-01
Generation 390: Best = 5.935080e+11 FES = 39100 Diversity = 1.9126e-01
Generation 400: Best = 5.900055e+11 FES = 40100 Diversity = 1.8896e-01
Generation 410: Best = 5.871194e+11 FES = 41100 Diversity = 1.5910e-01
Generation 420: Best = 5.848781e+11 FES = 42100 Diversity = 1.3934e-01
Generation 430: Best = 5.827437e+11 FES = 43100 Diversity = 1.2081e-01
Generation 440: Best = 5.810704e+11 FES = 44100 Diversity = 1.1715e-01

```

Generation	450:	Best = 5.796076e+11	FES = 45100	Diversity = 8.1030e-02
Generation	460:	Best = 5.789598e+11	FES = 46100	Diversity = 7.0018e-02
Generation	470:	Best = 5.785740e+11	FES = 47100	Diversity = 4.0624e-02
Generation	480:	Best = 5.784153e+11	FES = 48100	Diversity = 2.6825e-02
Generation	490:	Best = 5.783022e+11	FES = 49100	Diversity = 1.7052e-02
Generation	500:	Best = 5.782526e+11	FES = 50100	Diversity = 9.6590e-03
Generation	510:	Best = 5.782134e+11	FES = 51100	Diversity = 7.4297e-03
Generation	520:	Best = 5.781845e+11	FES = 52100	Diversity = 5.2904e-03
Generation	530:	Best = 5.781618e+11	FES = 53100	Diversity = 6.3002e-03
Generation	540:	Best = 5.781377e+11	FES = 54100	Diversity = 3.8221e-03
Generation	550:	Best = 5.781244e+11	FES = 55100	Diversity = 2.8768e-03
Generation	560:	Best = 5.781168e+11	FES = 56100	Diversity = 2.2222e-03
Generation	570:	Best = 5.781121e+11	FES = 57100	Diversity = 1.5421e-03
Generation	580:	Best = 5.781082e+11	FES = 58100	Diversity = 1.6941e-03
Generation	590:	Best = 5.781054e+11	FES = 59100	Diversity = 7.6509e-04
Generation	600:	Best = 5.781027e+11	FES = 60100	Diversity = 1.3271e-03
Generation	610:	Best = 5.781012e+11	FES = 61100	Diversity = 8.9932e-04
Generation	620:	Best = 5.780999e+11	FES = 62100	Diversity = 8.1520e-04
Generation	630:	Best = 5.780993e+11	FES = 63100	Diversity = 5.6022e-04
Generation	640:	Best = 5.780987e+11	FES = 64100	Diversity = 4.9685e-04
Generation	650:	Best = 5.780983e+11	FES = 65100	Diversity = 3.6989e-04
Generation	660:	Best = 5.780979e+11	FES = 66100	Diversity = 3.7629e-04
Generation	670:	Best = 5.780976e+11	FES = 67100	Diversity = 3.5213e-04
Generation	680:	Best = 5.780973e+11	FES = 68100	Diversity = 2.9456e-04
Generation	690:	Best = 5.780971e+11	FES = 69100	Diversity = 2.9802e-04
Generation	700:	Best = 5.780970e+11	FES = 70100	Diversity = 2.1781e-04
Generation	710:	Best = 5.780969e+11	FES = 71100	Diversity = 1.6089e-04
Generation	720:	Best = 5.780969e+11	FES = 72100	Diversity = 1.0363e-04
Generation	730:	Best = 5.780968e+11	FES = 73100	Diversity = 1.1380e-04
Generation	740:	Best = 5.780968e+11	FES = 74100	Diversity = 6.5080e-05
Generation	750:	Best = 5.780968e+11	FES = 75100	Diversity = 7.7357e-05
Generation	760:	Best = 5.780968e+11	FES = 76100	Diversity = 3.6897e-05
Generation	770:	Best = 5.780968e+11	FES = 77100	Diversity = 1.7364e-05
Generation	780:	Best = 5.780968e+11	FES = 78100	Diversity = 2.2367e-05
Generation	790:	Best = 5.780968e+11	FES = 79100	Diversity = 1.5553e-05
Generation	800:	Best = 5.780968e+11	FES = 80100	Diversity = 1.0129e-05
Generation	810:	Best = 5.780968e+11	FES = 81100	Diversity = 8.5622e-06
Generation	820:	Best = 5.780968e+11	FES = 82100	Diversity = 7.1730e-06
Generation	830:	Best = 5.780968e+11	FES = 83100	Diversity = 4.5059e-06
Generation	840:	Best = 5.780968e+11	FES = 84100	Diversity = 3.2544e-06
Generation	850:	Best = 5.780968e+11	FES = 85100	Diversity = 3.7597e-06
Generation	860:	Best = 5.780968e+11	FES = 86100	Diversity = 3.1440e-06
Generation	870:	Best = 5.780968e+11	FES = 87100	Diversity = 1.7129e-06
Generation	880:	Best = 5.780968e+11	FES = 88100	Diversity = 1.6910e-06
Generation	890:	Best = 5.780968e+11	FES = 89100	Diversity = 7.5794e-07
Generation	900:	Best = 5.780968e+11	FES = 90100	Diversity = 5.6420e-07
Generation	910:	Best = 5.780968e+11	FES = 91100	Diversity = 2.3272e-07
Generation	920:	Best = 5.780968e+11	FES = 92100	Diversity = 2.1573e-07
Generation	930:	Best = 5.780968e+11	FES = 93100	Diversity = 1.5705e-07
Generation	940:	Best = 5.780968e+11	FES = 94100	Diversity = 9.5790e-08
Generation	950:	Best = 5.780968e+11	FES = 95100	Diversity = 5.0537e-08
Generation	960:	Best = 5.780968e+11	FES = 96100	Diversity = 4.4754e-08
Generation	970:	Best = 5.780968e+11	FES = 97100	Diversity = 4.5424e-08
Generation	980:	Best = 5.780968e+11	FES = 98100	Diversity = 5.4428e-08

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[illegible]

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```

Generation 9630: Best = 5.780968e+11 FES = 963100 Diversity = 3.1202e-14
Generation 9640: Best = 5.780968e+11 FES = 964100 Diversity = 3.1202e-14
Generation 9650: Best = 5.780968e+11 FES = 965100 Diversity = 3.1202e-14
Generation 9660: Best = 5.780968e+11 FES = 966100 Diversity = 3.1202e-14
Generation 9670: Best = 5.780968e+11 FES = 967100 Diversity = 3.1202e-14
Generation 9680: Best = 5.780968e+11 FES = 968100 Diversity = 3.1202e-14
Generation 9690: Best = 5.780968e+11 FES = 969100 Diversity = 3.1202e-14
Generation 9700: Best = 5.780968e+11 FES = 970100 Diversity = 3.1202e-14
Generation 9710: Best = 5.780968e+11 FES = 971100 Diversity = 3.1202e-14
Generation 9720: Best = 5.780968e+11 FES = 972100 Diversity = 3.1202e-14
Generation 9730: Best = 5.780968e+11 FES = 973100 Diversity = 3.1202e-14
Generation 9740: Best = 5.780968e+11 FES = 974100 Diversity = 3.1202e-14
Generation 9750: Best = 5.780968e+11 FES = 975100 Diversity = 3.1202e-14
Generation 9760: Best = 5.780968e+11 FES = 976100 Diversity = 3.1202e-14
Generation 9770: Best = 5.780968e+11 FES = 977100 Diversity = 3.1202e-14
Generation 9780: Best = 5.780968e+11 FES = 978100 Diversity = 3.1202e-14
Generation 9790: Best = 5.780968e+11 FES = 979100 Diversity = 3.1202e-14
Generation 9800: Best = 5.780968e+11 FES = 980100 Diversity = 3.1202e-14
Generation 9810: Best = 5.780968e+11 FES = 981100 Diversity = 3.1202e-14
Generation 9820: Best = 5.780968e+11 FES = 982100 Diversity = 3.1202e-14
Generation 9830: Best = 5.780968e+11 FES = 983100 Diversity = 3.1202e-14
Generation 9840: Best = 5.780968e+11 FES = 984100 Diversity = 3.1202e-14
Generation 9850: Best = 5.780968e+11 FES = 985100 Diversity = 3.1202e-14
Generation 9860: Best = 5.780968e+11 FES = 986100 Diversity = 3.1202e-14
Generation 9870: Best = 5.780968e+11 FES = 987100 Diversity = 3.1202e-14
Generation 9880: Best = 5.780968e+11 FES = 988100 Diversity = 3.1202e-14
Generation 9890: Best = 5.780968e+11 FES = 989100 Diversity = 3.1202e-14
Generation 9900: Best = 5.780968e+11 FES = 990100 Diversity = 3.1202e-14
Generation 9910: Best = 5.780968e+11 FES = 991100 Diversity = 3.1202e-14
Generation 9920: Best = 5.780968e+11 FES = 992100 Diversity = 3.1202e-14
Generation 9930: Best = 5.780968e+11 FES = 993100 Diversity = 3.1202e-14
Generation 9940: Best = 5.780968e+11 FES = 994100 Diversity = 3.1202e-14
Generation 9950: Best = 5.780968e+11 FES = 995100 Diversity = 3.1202e-14
Generation 9960: Best = 5.780968e+11 FES = 996100 Diversity = 3.1202e-14
Generation 9970: Best = 5.780968e+11 FES = 997100 Diversity = 3.1202e-14
Generation 9980: Best = 5.780968e+11 FES = 998100 Diversity = 3.1202e-14
Generation 9990: Best = 5.780968e+11 FES = 999100 Diversity = 3.1202e-14

```

```

Optimization completed!
Final Best Fitness = 5.780968e+11
Total Function Evaluations = 1000000

```

```

=====
=
Results:
  Best Fitness          : 5.780968e+11
  Global Optimum        : 1.000000e+02
  Error (|best-optimum|) : 5.780968e+11
  Best Solution (5 dim) : [-21.8105716 -50.8574945 8.70791665 -17.29683838
15.10922178]
  D^(0) (init. diversity): 5.7220e+01
=====
=

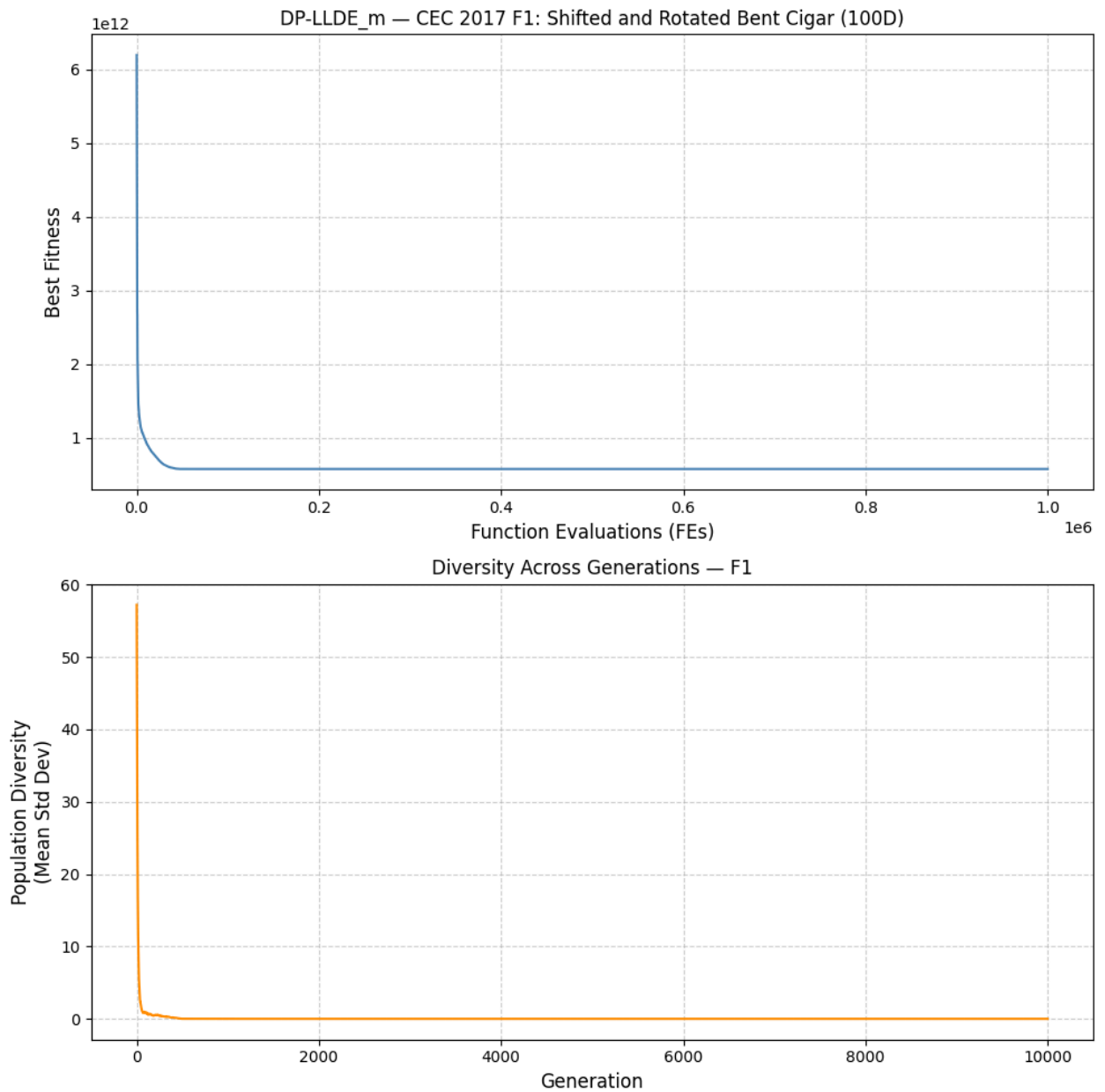
```

```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results_2/all_results_che

```

ckpoint.pkl



Graph saved as convergence_F1.png

✓ Plot saved to: /content/drive/My Drive/thesis_results_2/plots/convergence_F1.png

```
In [ ]: # =====  
# F2 — Shifted and Rotated Sum of Different Power  
# =====  
run_and_plot('F2', make_wrapper(simple.f2), 200.0, 'Shifted and Rotated Sum of
```

=====

=

DP-LLDE_m on CEC 2017 F2 – Single Run

=====

=

Function : Shifted and Rotated Sum of Different Power (Deprecated) (100D)
Bounds : [-100, 100]
Global optimum : 200.0

WARNING: f2 has been deprecated from the CEC 2017 benchmark suite

Initial $D^*(0) = 5.7220e+01$ | Diversity threshold = 5.7220e+00

Generation 0: Best Fitness = 1.048956e+189 | Diversity = 5.7220e+01

Generation	10:	Best = 4.162568e+159	FES = 1100	Diversity = 2.9319e+01
Generation	20:	Best = 2.748188e+144	FES = 2100	Diversity = 1.5019e+01
Generation	30:	Best = 2.315204e+139	FES = 3100	Diversity = 8.3374e+00
Generation	40:	Best = 5.722929e+135	FES = 4100	Diversity = 5.4108e+00
Generation	50:	Best = 7.758144e+132	FES = 5100	Diversity = 3.3518e+00
Generation	60:	Best = 4.100384e+131	FES = 6100	Diversity = 2.6626e+00
Generation	70:	Best = 4.344871e+130	FES = 7100	Diversity = 2.0960e+00
Generation	80:	Best = 8.753456e+129	FES = 8100	Diversity = 1.4691e+00
Generation	90:	Best = 3.051455e+129	FES = 9100	Diversity = 1.2189e+00
Generation	100:	Best = 1.229981e+129	FES = 10100	Diversity = 9.2230e-01
Generation	110:	Best = 5.921953e+128	FES = 11100	Diversity = 8.4761e-01
Generation	120:	Best = 3.070052e+128	FES = 12100	Diversity = 7.0894e-01
Generation	130:	Best = 1.772464e+128	FES = 13100	Diversity = 5.5764e-01
Generation	140:	Best = 1.334325e+128	FES = 14100	Diversity = 5.5676e-01
Generation	150:	Best = 1.114793e+128	FES = 15100	Diversity = 4.1937e-01
Generation	160:	Best = 9.693719e+127	FES = 16100	Diversity = 4.2020e-01
Generation	170:	Best = 8.970599e+127	FES = 17100	Diversity = 3.7530e-01
Generation	180:	Best = 8.432413e+127	FES = 18100	Diversity = 3.9970e-01
Generation	190:	Best = 7.936459e+127	FES = 19100	Diversity = 4.2514e-01
Generation	200:	Best = 7.588771e+127	FES = 20100	Diversity = 4.6969e-01
Generation	210:	Best = 7.281581e+127	FES = 21100	Diversity = 4.6067e-01
Generation	220:	Best = 7.055724e+127	FES = 22100	Diversity = 5.0966e-01
Generation	230:	Best = 6.890120e+127	FES = 23100	Diversity = 5.1730e-01
Generation	240:	Best = 6.752214e+127	FES = 24100	Diversity = 5.6403e-01
Generation	250:	Best = 6.647448e+127	FES = 25100	Diversity = 5.2466e-01
Generation	260:	Best = 6.568390e+127	FES = 26100	Diversity = 4.5115e-01
Generation	270:	Best = 6.466984e+127	FES = 27100	Diversity = 4.8824e-01
Generation	280:	Best = 6.412428e+127	FES = 28100	Diversity = 5.0597e-01
Generation	290:	Best = 6.368411e+127	FES = 29100	Diversity = 3.7799e-01
Generation	300:	Best = 6.317585e+127	FES = 30100	Diversity = 3.9810e-01
Generation	310:	Best = 6.279229e+127	FES = 31100	Diversity = 4.3545e-01
Generation	320:	Best = 6.252425e+127	FES = 32100	Diversity = 3.6771e-01
Generation	330:	Best = 6.219202e+127	FES = 33100	Diversity = 4.1404e-01
Generation	340:	Best = 6.189839e+127	FES = 34100	Diversity = 5.4474e-01
Generation	350:	Best = 6.163822e+127	FES = 35100	Diversity = 4.2945e-01
Generation	360:	Best = 6.144168e+127	FES = 36100	Diversity = 3.7004e-01
Generation	370:	Best = 6.131362e+127	FES = 37100	Diversity = 3.5867e-01
Generation	380:	Best = 6.118595e+127	FES = 38100	Diversity = 4.0973e-01
Generation	390:	Best = 6.108853e+127	FES = 39100	Diversity = 3.5396e-01
Generation	400:	Best = 6.101807e+127	FES = 40100	Diversity = 3.4167e-01
Generation	410:	Best = 6.097475e+127	FES = 41100	Diversity = 3.5728e-01
Generation	420:	Best = 6.092170e+127	FES = 42100	Diversity = 3.4103e-01
Generation	430:	Best = 6.087572e+127	FES = 43100	Diversity = 3.8968e-01

Generation	440:	Best = 6.083662e+127	FES = 44100	Diversity = 4.1619e-01
Generation	450:	Best = 6.080249e+127	FES = 45100	Diversity = 4.1637e-01
Generation	460:	Best = 6.077399e+127	FES = 46100	Diversity = 3.0326e-01
Generation	470:	Best = 6.074507e+127	FES = 47100	Diversity = 4.4137e-01
Generation	480:	Best = 6.073000e+127	FES = 48100	Diversity = 4.6719e-01
Generation	490:	Best = 6.072087e+127	FES = 49100	Diversity = 4.4686e-01
Generation	500:	Best = 6.070968e+127	FES = 50100	Diversity = 5.6140e-01
Generation	510:	Best = 6.070054e+127	FES = 51100	Diversity = 6.4521e-01
Generation	520:	Best = 6.069602e+127	FES = 52100	Diversity = 4.5828e-01
Generation	530:	Best = 6.069195e+127	FES = 53100	Diversity = 4.1902e-01
Generation	540:	Best = 6.068750e+127	FES = 54100	Diversity = 5.3562e-01
Generation	550:	Best = 6.068449e+127	FES = 55100	Diversity = 3.2782e-01
Generation	560:	Best = 6.068193e+127	FES = 56100	Diversity = 5.1644e-01
Generation	570:	Best = 6.068081e+127	FES = 57100	Diversity = 2.8974e-01
Generation	580:	Best = 6.067958e+127	FES = 58100	Diversity = 4.3991e-01
Generation	590:	Best = 6.067829e+127	FES = 59100	Diversity = 4.4789e-01
Generation	600:	Best = 6.067678e+127	FES = 60100	Diversity = 4.0393e-01
Generation	610:	Best = 6.067573e+127	FES = 61100	Diversity = 2.6654e-01
Generation	620:	Best = 6.067472e+127	FES = 62100	Diversity = 2.7916e-01
Generation	630:	Best = 6.067391e+127	FES = 63100	Diversity = 3.8612e-01
Generation	640:	Best = 6.067321e+127	FES = 64100	Diversity = 3.1785e-01
Generation	650:	Best = 6.067262e+127	FES = 65100	Diversity = 1.2295e-01
Generation	660:	Best = 6.067212e+127	FES = 66100	Diversity = 1.1307e-01
Generation	670:	Best = 6.067164e+127	FES = 67100	Diversity = 8.3297e-02
Generation	680:	Best = 6.067126e+127	FES = 68100	Diversity = 7.9552e-02
Generation	690:	Best = 6.067108e+127	FES = 69100	Diversity = 7.6287e-02
Generation	700:	Best = 6.067102e+127	FES = 70100	Diversity = 6.4436e-02
Generation	710:	Best = 6.067098e+127	FES = 71100	Diversity = 4.3309e-02
Generation	720:	Best = 6.067095e+127	FES = 72100	Diversity = 5.9225e-02
Generation	730:	Best = 6.067093e+127	FES = 73100	Diversity = 6.0560e-02
Generation	740:	Best = 6.067091e+127	FES = 74100	Diversity = 7.0908e-02
Generation	750:	Best = 6.067090e+127	FES = 75100	Diversity = 7.9204e-02
Generation	760:	Best = 6.067089e+127	FES = 76100	Diversity = 6.2927e-02
Generation	770:	Best = 6.067088e+127	FES = 77100	Diversity = 1.0719e-01
Generation	780:	Best = 6.067087e+127	FES = 78100	Diversity = 1.7417e-01
Generation	790:	Best = 6.067086e+127	FES = 79100	Diversity = 7.7232e-02
Generation	800:	Best = 6.067085e+127	FES = 80100	Diversity = 8.3400e-02
Generation	810:	Best = 6.067085e+127	FES = 81100	Diversity = 1.1385e-01
Generation	820:	Best = 6.067084e+127	FES = 82100	Diversity = 1.0648e-01
Generation	830:	Best = 6.067084e+127	FES = 83100	Diversity = 6.1432e-02
Generation	840:	Best = 6.067084e+127	FES = 84100	Diversity = 7.0034e-02
Generation	850:	Best = 6.067084e+127	FES = 85100	Diversity = 7.4408e-02
Generation	860:	Best = 6.067084e+127	FES = 86100	Diversity = 8.7797e-02
Generation	870:	Best = 6.067084e+127	FES = 87100	Diversity = 6.7922e-02
Generation	880:	Best = 6.067084e+127	FES = 88100	Diversity = 1.1296e-01
Generation	890:	Best = 6.067084e+127	FES = 89100	Diversity = 9.8141e-02
Generation	900:	Best = 6.067084e+127	FES = 90100	Diversity = 1.0178e-01
Generation	910:	Best = 6.067084e+127	FES = 91100	Diversity = 2.6645e-02
Generation	920:	Best = 6.067083e+127	FES = 92100	Diversity = 3.1659e-03
Generation	930:	Best = 6.067083e+127	FES = 93100	Diversity = 6.8913e-04
Generation	940:	Best = 6.067083e+127	FES = 94100	Diversity = 7.2704e-03
Generation	950:	Best = 6.067083e+127	FES = 95100	Diversity = 1.3320e-02
Generation	960:	Best = 6.067083e+127	FES = 96100	Diversity = 7.4619e-03
Generation	970:	Best = 6.067083e+127	FES = 97100	Diversity = 4.7767e-03

Generation 980:	Best = 6.067083e+127	FES = 98100	Diversity = 9.9663e-03
Generation 990:	Best = 6.067083e+127	FES = 99100	Diversity = 9.0569e-03
Generation 1000:	Best = 6.067083e+127	FES = 100100	Diversity = 6.4144e-03
Generation 1010:	Best = 6.067083e+127	FES = 101100	Diversity = 1.0439e-02
Generation 1020:	Best = 6.067083e+127	FES = 102100	Diversity = 1.0282e-02
Generation 1030:	Best = 6.067083e+127	FES = 103100	Diversity = 2.5283e-02
Generation 1040:	Best = 6.067083e+127	FES = 104100	Diversity = 3.1822e-02
Generation 1050:	Best = 6.067083e+127	FES = 105100	Diversity = 4.3634e-02
Generation 1060:	Best = 6.067083e+127	FES = 106100	Diversity = 3.7459e-02
Generation 1070:	Best = 6.067083e+127	FES = 107100	Diversity = 2.6235e-02
Generation 1080:	Best = 6.067083e+127	FES = 108100	Diversity = 1.6215e-02
Generation 1090:	Best = 6.067083e+127	FES = 109100	Diversity = 1.1918e-02
Generation 1100:	Best = 6.067083e+127	FES = 110100	Diversity = 9.5302e-03
Generation 1110:	Best = 6.067083e+127	FES = 111100	Diversity = 5.8419e-03
Generation 1120:	Best = 6.067083e+127	FES = 112100	Diversity = 3.9451e-03
Generation 1130:	Best = 6.067083e+127	FES = 113100	Diversity = 1.5256e-03
Generation 1140:	Best = 6.067083e+127	FES = 114100	Diversity = 9.2045e-04
Generation 1150:	Best = 6.067083e+127	FES = 115100	Diversity = 7.7025e-04
Generation 1160:	Best = 6.067083e+127	FES = 116100	Diversity = 3.8323e-04
Generation 1170:	Best = 6.067083e+127	FES = 117100	Diversity = 1.5088e-04
Generation 1180:	Best = 6.067083e+127	FES = 118100	Diversity = 9.0659e-05
Generation 1190:	Best = 6.067083e+127	FES = 119100	Diversity = 5.6254e-05
Generation 1200:	Best = 6.067083e+127	FES = 120100	Diversity = 5.6134e-05
Generation 1210:	Best = 6.067083e+127	FES = 121100	Diversity = 2.1110e-05
Generation 1220:	Best = 6.067083e+127	FES = 122100	Diversity = 3.8655e-06
Generation 1230:	Best = 6.067083e+127	FES = 123100	Diversity = 2.2748e-06
Generation 1240:	Best = 6.067083e+127	FES = 124100	Diversity = 3.5689e-13
Generation 1250:	Best = 6.067083e+127	FES = 125100	Diversity = 2.6028e-14
Generation 1260:	Best = 6.067083e+127	FES = 126100	Diversity = 2.5408e-14
Generation 1270:	Best = 6.067083e+127	FES = 127100	Diversity = 2.6010e-14
Generation 1280:	Best = 6.067083e+127	FES = 128100	Diversity = 2.5995e-14
Generation 1290:	Best = 6.067083e+127	FES = 129100	Diversity = 2.5995e-14
Generation 1300:	Best = 6.067083e+127	FES = 130100	Diversity = 2.5995e-14
Generation 1310:	Best = 6.067083e+127	FES = 131100	Diversity = 2.5995e-14
Generation 1320:	Best = 6.067083e+127	FES = 132100	Diversity = 2.5995e-14
Generation 1330:	Best = 6.067083e+127	FES = 133100	Diversity = 2.5995e-14
Generation 1340:	Best = 6.067083e+127	FES = 134100	Diversity = 2.5995e-14
Generation 1350:	Best = 6.067083e+127	FES = 135100	Diversity = 2.5995e-14
Generation 1360:	Best = 6.067083e+127	FES = 136100	Diversity = 2.5995e-14
Generation 1370:	Best = 6.067083e+127	FES = 137100	Diversity = 2.5995e-14
Generation 1380:	Best = 6.067083e+127	FES = 138100	Diversity = 2.5995e-14
Generation 1390:	Best = 6.067083e+127	FES = 139100	Diversity = 2.5995e-14
Generation 1400:	Best = 6.067083e+127	FES = 140100	Diversity = 2.5995e-14
Generation 1410:	Best = 6.067083e+127	FES = 141100	Diversity = 2.5995e-14
Generation 1420:	Best = 6.067083e+127	FES = 142100	Diversity = 2.5995e-14
Generation 1430:	Best = 6.067083e+127	FES = 143100	Diversity = 2.5995e-14
Generation 1440:	Best = 6.067083e+127	FES = 144100	Diversity = 2.5995e-14
Generation 1450:	Best = 6.067083e+127	FES = 145100	Diversity = 2.5995e-14
Generation 1460:	Best = 6.067083e+127	FES = 146100	Diversity = 2.5995e-14
Generation 1470:	Best = 6.067083e+127	FES = 147100	Diversity = 2.5995e-14
Generation 1480:	Best = 6.067083e+127	FES = 148100	Diversity = 2.5995e-14
Generation 1490:	Best = 6.067083e+127	FES = 149100	Diversity = 2.5995e-14
Generation 1500:	Best = 6.067083e+127	FES = 150100	Diversity = 2.5995e-14
Generation 1510:	Best = 6.067083e+127	FES = 151100	Diversity = 2.5995e-14

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Generation 9620:	Best = 6.067083e+127	FES = 962100	Diversity = 2.5995e-14
Generation 9630:	Best = 6.067083e+127	FES = 963100	Diversity = 2.5995e-14
Generation 9640:	Best = 6.067083e+127	FES = 964100	Diversity = 2.5995e-14
Generation 9650:	Best = 6.067083e+127	FES = 965100	Diversity = 2.5995e-14
Generation 9660:	Best = 6.067083e+127	FES = 966100	Diversity = 2.5995e-14
Generation 9670:	Best = 6.067083e+127	FES = 967100	Diversity = 2.5995e-14
Generation 9680:	Best = 6.067083e+127	FES = 968100	Diversity = 2.5995e-14
Generation 9690:	Best = 6.067083e+127	FES = 969100	Diversity = 2.5995e-14
Generation 9700:	Best = 6.067083e+127	FES = 970100	Diversity = 2.5995e-14
Generation 9710:	Best = 6.067083e+127	FES = 971100	Diversity = 2.5995e-14
Generation 9720:	Best = 6.067083e+127	FES = 972100	Diversity = 2.5995e-14
Generation 9730:	Best = 6.067083e+127	FES = 973100	Diversity = 2.5995e-14
Generation 9740:	Best = 6.067083e+127	FES = 974100	Diversity = 2.5995e-14
Generation 9750:	Best = 6.067083e+127	FES = 975100	Diversity = 2.5995e-14
Generation 9760:	Best = 6.067083e+127	FES = 976100	Diversity = 2.5995e-14
Generation 9770:	Best = 6.067083e+127	FES = 977100	Diversity = 2.5995e-14
Generation 9780:	Best = 6.067083e+127	FES = 978100	Diversity = 2.5995e-14
Generation 9790:	Best = 6.067083e+127	FES = 979100	Diversity = 2.5995e-14
Generation 9800:	Best = 6.067083e+127	FES = 980100	Diversity = 2.5995e-14
Generation 9810:	Best = 6.067083e+127	FES = 981100	Diversity = 2.5995e-14
Generation 9820:	Best = 6.067083e+127	FES = 982100	Diversity = 2.5995e-14
Generation 9830:	Best = 6.067083e+127	FES = 983100	Diversity = 2.5995e-14
Generation 9840:	Best = 6.067083e+127	FES = 984100	Diversity = 2.5995e-14
Generation 9850:	Best = 6.067083e+127	FES = 985100	Diversity = 2.5995e-14
Generation 9860:	Best = 6.067083e+127	FES = 986100	Diversity = 2.5995e-14
Generation 9870:	Best = 6.067083e+127	FES = 987100	Diversity = 2.5995e-14
Generation 9880:	Best = 6.067083e+127	FES = 988100	Diversity = 2.5995e-14
Generation 9890:	Best = 6.067083e+127	FES = 989100	Diversity = 2.5995e-14
Generation 9900:	Best = 6.067083e+127	FES = 990100	Diversity = 2.5995e-14
Generation 9910:	Best = 6.067083e+127	FES = 991100	Diversity = 2.5995e-14
Generation 9920:	Best = 6.067083e+127	FES = 992100	Diversity = 2.5995e-14
Generation 9930:	Best = 6.067083e+127	FES = 993100	Diversity = 2.5995e-14
Generation 9940:	Best = 6.067083e+127	FES = 994100	Diversity = 2.5995e-14
Generation 9950:	Best = 6.067083e+127	FES = 995100	Diversity = 2.5995e-14
Generation 9960:	Best = 6.067083e+127	FES = 996100	Diversity = 2.5995e-14
Generation 9970:	Best = 6.067083e+127	FES = 997100	Diversity = 2.5995e-14
Generation 9980:	Best = 6.067083e+127	FES = 998100	Diversity = 2.5995e-14
Generation 9990:	Best = 6.067083e+127	FES = 999100	Diversity = 2.5995e-14

Optimization completed!

Final Best Fitness = 6.067083e+127

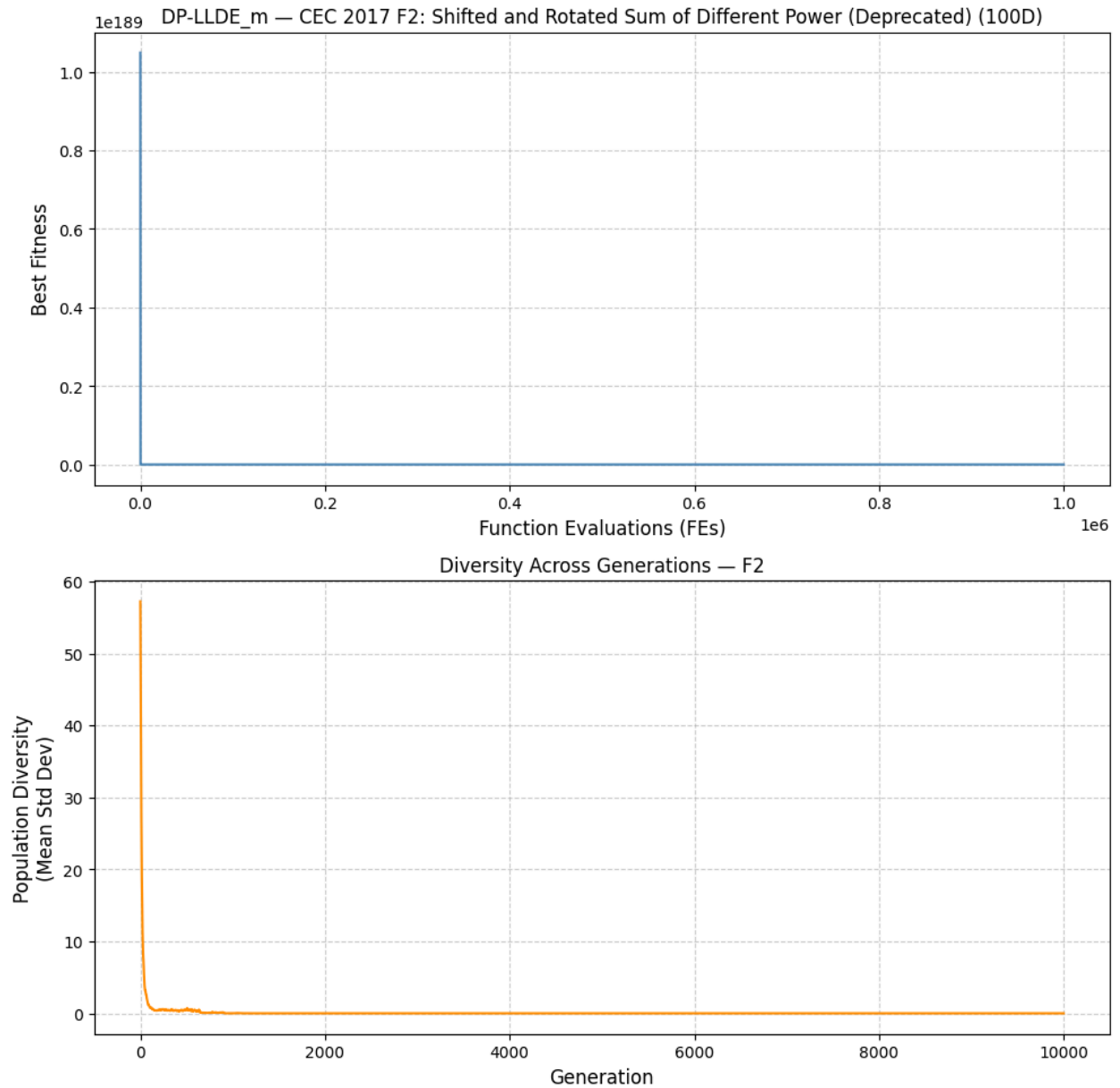
Total Function Evaluations = 1000000

```

=====
=
Results:
  Best Fitness           : 6.067083e+127
  Global Optimum         : 2.000000e+02
  Error (|best-optimum|) : 6.067083e+127
  Best Solution (5 dim)  : [-16.15694626 -27.53136374  7.89147825 -1.06395907
-15.77190888]
  D^(0) (init. diversity): 5.7220e+01
=====
=

```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results_2/all_results_checkpoint.pkl



Graph saved as convergence_F2.png

✓ Plot saved to: /content/drive/My Drive/thesis_results_2/plots/convergence_F2.png

```
In [ ]: # =====  
# F3 — Shifted and Rotated Zakharov  
# =====  
run_and_plot('F3', make_wrapper(simple.f3), 300.0, 'Shifted and Rotated Zakharov')
```

```

=====
=
DP-LLDE_m on CEC 2017 F3 – Single Run
=====
=
Function      : Shifted and Rotated Zakharov (100D)
Bounds       : [-100, 100]
Global optimum : 300.0
Initial D^(0) = 5.7220e+01 | Diversity threshold = 5.7220e+00
Generation 0: Best Fitness = 7.602487e+06 | Diversity = 5.7220e+01
Generation 10: Best = 5.247016e+05 FES = 1100 Diversity = 4.5868e+01
Generation 20: Best = 5.222593e+05 FES = 2100 Diversity = 4.0272e+01
Generation 30: Best = 4.012172e+05 FES = 3100 Diversity = 3.1584e+01
Generation 40: Best = 3.390407e+05 FES = 4100 Diversity = 2.1803e+01
Generation 50: Best = 3.267406e+05 FES = 5100 Diversity = 1.4728e+01
Generation 60: Best = 3.135844e+05 FES = 6100 Diversity = 1.0065e+01
Generation 70: Best = 3.053230e+05 FES = 7100 Diversity = 7.6364e+00
Generation 80: Best = 3.023392e+05 FES = 8100 Diversity = 5.4913e+00
Generation 90: Best = 2.993134e+05 FES = 9100 Diversity = 4.0212e+00
Generation 100: Best = 2.963837e+05 FES = 10100 Diversity = 2.7564e+00
Generation 110: Best = 2.946000e+05 FES = 11100 Diversity = 2.2962e+00
Generation 120: Best = 2.945261e+05 FES = 12100 Diversity = 2.2282e+00
Generation 130: Best = 2.932592e+05 FES = 13100 Diversity = 2.0838e+00
Generation 140: Best = 2.915426e+05 FES = 14100 Diversity = 1.9720e+00
Generation 150: Best = 2.890451e+05 FES = 15100 Diversity = 1.8540e+00
Generation 160: Best = 2.886212e+05 FES = 16100 Diversity = 1.9695e+00
Generation 170: Best = 2.874616e+05 FES = 17100 Diversity = 1.8623e+00
Generation 180: Best = 2.837866e+05 FES = 18100 Diversity = 1.8408e+00
Generation 190: Best = 2.790974e+05 FES = 19100 Diversity = 1.9019e+00
Generation 200: Best = 2.785345e+05 FES = 20100 Diversity = 2.1793e+00
Generation 210: Best = 2.762286e+05 FES = 21100 Diversity = 2.2291e+00
Generation 220: Best = 2.761022e+05 FES = 22100 Diversity = 2.0494e+00
Generation 230: Best = 2.756583e+05 FES = 23100 Diversity = 1.8361e+00
Generation 240: Best = 2.739754e+05 FES = 24100 Diversity = 1.6260e+00
Generation 250: Best = 2.724187e+05 FES = 25100 Diversity = 1.5237e+00
Generation 260: Best = 2.714347e+05 FES = 26100 Diversity = 1.5185e+00
Generation 270: Best = 2.703202e+05 FES = 27100 Diversity = 1.5191e+00
Generation 280: Best = 2.698978e+05 FES = 28100 Diversity = 1.4684e+00
Generation 290: Best = 2.697210e+05 FES = 29100 Diversity = 1.2023e+00
Generation 300: Best = 2.678663e+05 FES = 30100 Diversity = 1.4004e+00
Generation 310: Best = 2.675816e+05 FES = 31100 Diversity = 1.3029e+00
Generation 320: Best = 2.670510e+05 FES = 32100 Diversity = 1.2463e+00
Generation 330: Best = 2.649851e+05 FES = 33100 Diversity = 1.4435e+00
Generation 340: Best = 2.626105e+05 FES = 34100 Diversity = 1.5866e+00
Generation 350: Best = 2.614523e+05 FES = 35100 Diversity = 1.6233e+00
Generation 360: Best = 2.610912e+05 FES = 36100 Diversity = 1.6691e+00
Generation 370: Best = 2.593396e+05 FES = 37100 Diversity = 1.5935e+00
Generation 380: Best = 2.589565e+05 FES = 38100 Diversity = 1.6514e+00
Generation 390: Best = 2.577569e+05 FES = 39100 Diversity = 1.4700e+00
Generation 400: Best = 2.566890e+05 FES = 40100 Diversity = 1.5807e+00
Generation 410: Best = 2.558110e+05 FES = 41100 Diversity = 1.6106e+00
Generation 420: Best = 2.542419e+05 FES = 42100 Diversity = 1.5047e+00
Generation 430: Best = 2.515018e+05 FES = 43100 Diversity = 1.4353e+00
Generation 440: Best = 2.486860e+05 FES = 44100 Diversity = 1.6708e+00

```

Generation	450:	Best = 2.480821e+05	FES = 45100	Diversity = 1.9647e+00
Generation	460:	Best = 2.462648e+05	FES = 46100	Diversity = 1.8006e+00
Generation	470:	Best = 2.445996e+05	FES = 47100	Diversity = 1.8005e+00
Generation	480:	Best = 2.445996e+05	FES = 48100	Diversity = 1.8749e+00
Generation	490:	Best = 2.438880e+05	FES = 49100	Diversity = 1.7954e+00
Generation	500:	Best = 2.424970e+05	FES = 50100	Diversity = 1.6851e+00
Generation	510:	Best = 2.422744e+05	FES = 51100	Diversity = 1.4588e+00
Generation	520:	Best = 2.410633e+05	FES = 52100	Diversity = 1.4096e+00
Generation	530:	Best = 2.394252e+05	FES = 53100	Diversity = 1.3271e+00
Generation	540:	Best = 2.392600e+05	FES = 54100	Diversity = 1.5216e+00
Generation	550:	Best = 2.391472e+05	FES = 55100	Diversity = 1.3792e+00
Generation	560:	Best = 2.388760e+05	FES = 56100	Diversity = 1.1492e+00
Generation	570:	Best = 2.374966e+05	FES = 57100	Diversity = 1.1369e+00
Generation	580:	Best = 2.369469e+05	FES = 58100	Diversity = 1.0624e+00
Generation	590:	Best = 2.365219e+05	FES = 59100	Diversity = 1.0931e+00
Generation	600:	Best = 2.363449e+05	FES = 60100	Diversity = 1.0899e+00
Generation	610:	Best = 2.348031e+05	FES = 61100	Diversity = 1.1831e+00
Generation	620:	Best = 2.346583e+05	FES = 62100	Diversity = 1.3960e+00
Generation	630:	Best = 2.339749e+05	FES = 63100	Diversity = 1.3850e+00
Generation	640:	Best = 2.332336e+05	FES = 64100	Diversity = 1.3012e+00
Generation	650:	Best = 2.327996e+05	FES = 65100	Diversity = 1.3056e+00
Generation	660:	Best = 2.320169e+05	FES = 66100	Diversity = 1.3063e+00
Generation	670:	Best = 2.309959e+05	FES = 67100	Diversity = 1.2945e+00
Generation	680:	Best = 2.295856e+05	FES = 68100	Diversity = 1.4296e+00
Generation	690:	Best = 2.292150e+05	FES = 69100	Diversity = 1.4803e+00
Generation	700:	Best = 2.290275e+05	FES = 70100	Diversity = 1.3029e+00
Generation	710:	Best = 2.284569e+05	FES = 71100	Diversity = 1.2026e+00
Generation	720:	Best = 2.272310e+05	FES = 72100	Diversity = 1.1177e+00
Generation	730:	Best = 2.255582e+05	FES = 73100	Diversity = 1.2649e+00
Generation	740:	Best = 2.255582e+05	FES = 74100	Diversity = 1.2746e+00
Generation	750:	Best = 2.246051e+05	FES = 75100	Diversity = 1.2015e+00
Generation	760:	Best = 2.231401e+05	FES = 76100	Diversity = 1.1412e+00
Generation	770:	Best = 2.221424e+05	FES = 77100	Diversity = 1.3008e+00
Generation	780:	Best = 2.205690e+05	FES = 78100	Diversity = 1.5177e+00
Generation	790:	Best = 2.203583e+05	FES = 79100	Diversity = 1.6359e+00
Generation	800:	Best = 2.195368e+05	FES = 80100	Diversity = 1.7672e+00
Generation	810:	Best = 2.190592e+05	FES = 81100	Diversity = 1.5769e+00
Generation	820:	Best = 2.169535e+05	FES = 82100	Diversity = 1.6621e+00
Generation	830:	Best = 2.158036e+05	FES = 83100	Diversity = 1.6521e+00
Generation	840:	Best = 2.138487e+05	FES = 84100	Diversity = 1.6239e+00
Generation	850:	Best = 2.130322e+05	FES = 85100	Diversity = 1.5161e+00
Generation	860:	Best = 2.127061e+05	FES = 86100	Diversity = 1.4404e+00
Generation	870:	Best = 2.119452e+05	FES = 87100	Diversity = 1.3529e+00
Generation	880:	Best = 2.096188e+05	FES = 88100	Diversity = 1.5844e+00
Generation	890:	Best = 2.082815e+05	FES = 89100	Diversity = 1.6099e+00
Generation	900:	Best = 2.072389e+05	FES = 90100	Diversity = 1.8102e+00
Generation	910:	Best = 2.067548e+05	FES = 91100	Diversity = 1.7007e+00
Generation	920:	Best = 2.062488e+05	FES = 92100	Diversity = 1.7017e+00
Generation	930:	Best = 2.050266e+05	FES = 93100	Diversity = 1.5793e+00
Generation	940:	Best = 2.045138e+05	FES = 94100	Diversity = 1.3075e+00
Generation	950:	Best = 2.040345e+05	FES = 95100	Diversity = 1.4058e+00
Generation	960:	Best = 2.023488e+05	FES = 96100	Diversity = 1.3590e+00
Generation	970:	Best = 2.003157e+05	FES = 97100	Diversity = 1.7489e+00
Generation	980:	Best = 1.993913e+05	FES = 98100	Diversity = 1.8536e+00

Generation 990:	Best = 1.988401e+05	FES = 99100	Diversity = 1.8159e+00
Generation 1000:	Best = 1.978913e+05	FES = 100100	Diversity = 1.7038e+00
Generation 1010:	Best = 1.972224e+05	FES = 101100	Diversity = 1.5898e+00
Generation 1020:	Best = 1.964986e+05	FES = 102100	Diversity = 1.3730e+00
Generation 1030:	Best = 1.959459e+05	FES = 103100	Diversity = 1.4045e+00
Generation 1040:	Best = 1.955983e+05	FES = 104100	Diversity = 1.3876e+00
Generation 1050:	Best = 1.943017e+05	FES = 105100	Diversity = 1.3350e+00
Generation 1060:	Best = 1.937962e+05	FES = 106100	Diversity = 1.3592e+00
Generation 1070:	Best = 1.922219e+05	FES = 107100	Diversity = 1.4665e+00
Generation 1080:	Best = 1.915148e+05	FES = 108100	Diversity = 1.5024e+00
Generation 1090:	Best = 1.896469e+05	FES = 109100	Diversity = 1.8289e+00
Generation 1100:	Best = 1.886766e+05	FES = 110100	Diversity = 1.7232e+00
Generation 1110:	Best = 1.872591e+05	FES = 111100	Diversity = 1.7267e+00
Generation 1120:	Best = 1.864056e+05	FES = 112100	Diversity = 1.8633e+00
Generation 1130:	Best = 1.845585e+05	FES = 113100	Diversity = 1.7092e+00
Generation 1140:	Best = 1.845585e+05	FES = 114100	Diversity = 1.6748e+00
Generation 1150:	Best = 1.843736e+05	FES = 115100	Diversity = 1.6707e+00
Generation 1160:	Best = 1.834863e+05	FES = 116100	Diversity = 1.5531e+00
Generation 1170:	Best = 1.832120e+05	FES = 117100	Diversity = 1.3735e+00
Generation 1180:	Best = 1.824988e+05	FES = 118100	Diversity = 1.1831e+00
Generation 1190:	Best = 1.816389e+05	FES = 119100	Diversity = 1.1558e+00
Generation 1200:	Best = 1.811027e+05	FES = 120100	Diversity = 1.0686e+00
Generation 1210:	Best = 1.805617e+05	FES = 121100	Diversity = 1.2860e+00
Generation 1220:	Best = 1.801180e+05	FES = 122100	Diversity = 1.3247e+00
Generation 1230:	Best = 1.794129e+05	FES = 123100	Diversity = 1.2372e+00
Generation 1240:	Best = 1.793390e+05	FES = 124100	Diversity = 1.1484e+00
Generation 1250:	Best = 1.781479e+05	FES = 125100	Diversity = 1.0795e+00
Generation 1260:	Best = 1.771388e+05	FES = 126100	Diversity = 1.1324e+00
Generation 1270:	Best = 1.757916e+05	FES = 127100	Diversity = 1.4681e+00
Generation 1280:	Best = 1.752947e+05	FES = 128100	Diversity = 1.4719e+00
Generation 1290:	Best = 1.741376e+05	FES = 129100	Diversity = 1.6054e+00
Generation 1300:	Best = 1.730975e+05	FES = 130100	Diversity = 1.6513e+00
Generation 1310:	Best = 1.730975e+05	FES = 131100	Diversity = 1.5491e+00
Generation 1320:	Best = 1.725367e+05	FES = 132100	Diversity = 1.3712e+00
Generation 1330:	Best = 1.714831e+05	FES = 133100	Diversity = 1.2987e+00
Generation 1340:	Best = 1.710600e+05	FES = 134100	Diversity = 1.4002e+00
Generation 1350:	Best = 1.697062e+05	FES = 135100	Diversity = 1.3465e+00
Generation 1360:	Best = 1.691036e+05	FES = 136100	Diversity = 1.4185e+00
Generation 1370:	Best = 1.685971e+05	FES = 137100	Diversity = 1.4818e+00
Generation 1380:	Best = 1.681465e+05	FES = 138100	Diversity = 1.4062e+00
Generation 1390:	Best = 1.671104e+05	FES = 139100	Diversity = 1.3700e+00
Generation 1400:	Best = 1.646249e+05	FES = 140100	Diversity = 1.3477e+00
Generation 1410:	Best = 1.641202e+05	FES = 141100	Diversity = 1.8250e+00
Generation 1420:	Best = 1.610975e+05	FES = 142100	Diversity = 1.7003e+00
Generation 1430:	Best = 1.610975e+05	FES = 143100	Diversity = 1.7729e+00
Generation 1440:	Best = 1.598300e+05	FES = 144100	Diversity = 1.6102e+00
Generation 1450:	Best = 1.591456e+05	FES = 145100	Diversity = 1.6258e+00
Generation 1460:	Best = 1.591078e+05	FES = 146100	Diversity = 1.4385e+00
Generation 1470:	Best = 1.590220e+05	FES = 147100	Diversity = 1.2824e+00
Generation 1480:	Best = 1.574946e+05	FES = 148100	Diversity = 1.2549e+00
Generation 1490:	Best = 1.572695e+05	FES = 149100	Diversity = 1.3102e+00
Generation 1500:	Best = 1.565152e+05	FES = 150100	Diversity = 1.2539e+00
Generation 1510:	Best = 1.546490e+05	FES = 151100	Diversity = 1.3923e+00
Generation 1520:	Best = 1.544087e+05	FES = 152100	Diversity = 1.4940e+00

Generation 1530:	Best = 1.540988e+05	FES = 153100	Diversity = 1.4014e+00
Generation 1540:	Best = 1.538079e+05	FES = 154100	Diversity = 1.1741e+00
Generation 1550:	Best = 1.527900e+05	FES = 155100	Diversity = 1.1520e+00
Generation 1560:	Best = 1.517547e+05	FES = 156100	Diversity = 1.3723e+00
Generation 1570:	Best = 1.512615e+05	FES = 157100	Diversity = 1.3906e+00
Generation 1580:	Best = 1.504835e+05	FES = 158100	Diversity = 1.2925e+00
Generation 1590:	Best = 1.502373e+05	FES = 159100	Diversity = 1.2499e+00
Generation 1600:	Best = 1.492338e+05	FES = 160100	Diversity = 1.1722e+00
Generation 1610:	Best = 1.492338e+05	FES = 161100	Diversity = 1.1442e+00
Generation 1620:	Best = 1.480753e+05	FES = 162100	Diversity = 1.0262e+00
Generation 1630:	Best = 1.474196e+05	FES = 163100	Diversity = 1.2149e+00
Generation 1640:	Best = 1.467819e+05	FES = 164100	Diversity = 1.3551e+00
Generation 1650:	Best = 1.459072e+05	FES = 165100	Diversity = 1.3492e+00
Generation 1660:	Best = 1.451961e+05	FES = 166100	Diversity = 1.5627e+00
Generation 1670:	Best = 1.447945e+05	FES = 167100	Diversity = 1.3615e+00
Generation 1680:	Best = 1.442767e+05	FES = 168100	Diversity = 1.2343e+00
Generation 1690:	Best = 1.439199e+05	FES = 169100	Diversity = 1.1573e+00
Generation 1700:	Best = 1.426427e+05	FES = 170100	Diversity = 1.1373e+00
Generation 1710:	Best = 1.426427e+05	FES = 171100	Diversity = 1.1708e+00
Generation 1720:	Best = 1.424541e+05	FES = 172100	Diversity = 1.0213e+00
Generation 1730:	Best = 1.421474e+05	FES = 173100	Diversity = 9.7245e-01
Generation 1740:	Best = 1.418175e+05	FES = 174100	Diversity = 8.8809e-01
Generation 1750:	Best = 1.408738e+05	FES = 175100	Diversity = 9.0302e-01
Generation 1760:	Best = 1.407837e+05	FES = 176100	Diversity = 9.1594e-01
Generation 1770:	Best = 1.401717e+05	FES = 177100	Diversity = 9.2638e-01
Generation 1780:	Best = 1.400972e+05	FES = 178100	Diversity = 8.6939e-01
Generation 1790:	Best = 1.398344e+05	FES = 179100	Diversity = 8.9687e-01
Generation 1800:	Best = 1.395736e+05	FES = 180100	Diversity = 8.7946e-01
Generation 1810:	Best = 1.385705e+05	FES = 181100	Diversity = 8.8600e-01
Generation 1820:	Best = 1.368368e+05	FES = 182100	Diversity = 1.0353e+00
Generation 1830:	Best = 1.368368e+05	FES = 183100	Diversity = 1.2222e+00
Generation 1840:	Best = 1.368082e+05	FES = 184100	Diversity = 1.1408e+00
Generation 1850:	Best = 1.356801e+05	FES = 185100	Diversity = 1.0772e+00
Generation 1860:	Best = 1.356801e+05	FES = 186100	Diversity = 1.0785e+00
Generation 1870:	Best = 1.351382e+05	FES = 187100	Diversity = 1.1085e+00
Generation 1880:	Best = 1.340408e+05	FES = 188100	Diversity = 1.1770e+00
Generation 1890:	Best = 1.337027e+05	FES = 189100	Diversity = 1.2500e+00
Generation 1900:	Best = 1.326953e+05	FES = 190100	Diversity = 1.1694e+00
Generation 1910:	Best = 1.314694e+05	FES = 191100	Diversity = 1.2737e+00
Generation 1920:	Best = 1.308533e+05	FES = 192100	Diversity = 1.3659e+00
Generation 1930:	Best = 1.303219e+05	FES = 193100	Diversity = 1.2727e+00
Generation 1940:	Best = 1.292077e+05	FES = 194100	Diversity = 1.1883e+00
Generation 1950:	Best = 1.286744e+05	FES = 195100	Diversity = 1.1273e+00
Generation 1960:	Best = 1.276004e+05	FES = 196100	Diversity = 1.2979e+00
Generation 1970:	Best = 1.271289e+05	FES = 197100	Diversity = 1.3660e+00
Generation 1980:	Best = 1.265702e+05	FES = 198100	Diversity = 1.4238e+00
Generation 1990:	Best = 1.255346e+05	FES = 199100	Diversity = 1.3291e+00
Generation 2000:	Best = 1.254720e+05	FES = 200100	Diversity = 1.4145e+00
Generation 2010:	Best = 1.251150e+05	FES = 201100	Diversity = 1.1881e+00
Generation 2020:	Best = 1.242291e+05	FES = 202100	Diversity = 1.1603e+00
Generation 2030:	Best = 1.235556e+05	FES = 203100	Diversity = 1.0876e+00
Generation 2040:	Best = 1.231880e+05	FES = 204100	Diversity = 1.0097e+00
Generation 2050:	Best = 1.230052e+05	FES = 205100	Diversity = 1.1008e+00
Generation 2060:	Best = 1.228701e+05	FES = 206100	Diversity = 1.0213e+00

Generation 2070:	Best = 1.221961e+05	FES = 207100	Diversity = 9.2373e-01
Generation 2080:	Best = 1.216832e+05	FES = 208100	Diversity = 9.2720e-01
Generation 2090:	Best = 1.212363e+05	FES = 209100	Diversity = 9.3139e-01
Generation 2100:	Best = 1.211352e+05	FES = 210100	Diversity = 1.0329e+00
Generation 2110:	Best = 1.207744e+05	FES = 211100	Diversity = 1.0378e+00
Generation 2120:	Best = 1.202834e+05	FES = 212100	Diversity = 9.3426e-01
Generation 2130:	Best = 1.201139e+05	FES = 213100	Diversity = 9.4132e-01
Generation 2140:	Best = 1.200237e+05	FES = 214100	Diversity = 8.6817e-01
Generation 2150:	Best = 1.192892e+05	FES = 215100	Diversity = 8.6771e-01
Generation 2160:	Best = 1.188936e+05	FES = 216100	Diversity = 9.5130e-01
Generation 2170:	Best = 1.185225e+05	FES = 217100	Diversity = 1.0069e+00
Generation 2180:	Best = 1.178643e+05	FES = 218100	Diversity = 9.7007e-01
Generation 2190:	Best = 1.177030e+05	FES = 219100	Diversity = 1.0079e+00
Generation 2200:	Best = 1.170525e+05	FES = 220100	Diversity = 1.0189e+00
Generation 2210:	Best = 1.170415e+05	FES = 221100	Diversity = 9.7553e-01
Generation 2220:	Best = 1.165581e+05	FES = 222100	Diversity = 8.1442e-01
Generation 2230:	Best = 1.163880e+05	FES = 223100	Diversity = 9.0821e-01
Generation 2240:	Best = 1.158649e+05	FES = 224100	Diversity = 8.5316e-01
Generation 2250:	Best = 1.155343e+05	FES = 225100	Diversity = 8.3714e-01
Generation 2260:	Best = 1.150887e+05	FES = 226100	Diversity = 8.8764e-01
Generation 2270:	Best = 1.147345e+05	FES = 227100	Diversity = 9.9791e-01
Generation 2280:	Best = 1.143117e+05	FES = 228100	Diversity = 1.0048e+00
Generation 2290:	Best = 1.121451e+05	FES = 229100	Diversity = 1.2932e+00
Generation 2300:	Best = 1.115114e+05	FES = 230100	Diversity = 1.2841e+00
Generation 2310:	Best = 1.113583e+05	FES = 231100	Diversity = 1.3127e+00
Generation 2320:	Best = 1.106117e+05	FES = 232100	Diversity = 1.2404e+00
Generation 2330:	Best = 1.098838e+05	FES = 233100	Diversity = 1.0429e+00
Generation 2340:	Best = 1.097491e+05	FES = 234100	Diversity = 1.1731e+00
Generation 2350:	Best = 1.094124e+05	FES = 235100	Diversity = 1.0888e+00
Generation 2360:	Best = 1.090090e+05	FES = 236100	Diversity = 8.6967e-01
Generation 2370:	Best = 1.088172e+05	FES = 237100	Diversity = 8.2191e-01
Generation 2380:	Best = 1.086081e+05	FES = 238100	Diversity = 8.5673e-01
Generation 2390:	Best = 1.082910e+05	FES = 239100	Diversity = 9.1434e-01
Generation 2400:	Best = 1.080886e+05	FES = 240100	Diversity = 9.4946e-01
Generation 2410:	Best = 1.075191e+05	FES = 241100	Diversity = 8.7147e-01
Generation 2420:	Best = 1.074946e+05	FES = 242100	Diversity = 7.8016e-01
Generation 2430:	Best = 1.069520e+05	FES = 243100	Diversity = 8.0522e-01
Generation 2440:	Best = 1.069040e+05	FES = 244100	Diversity = 7.9788e-01
Generation 2450:	Best = 1.067582e+05	FES = 245100	Diversity = 7.3334e-01
Generation 2460:	Best = 1.063508e+05	FES = 246100	Diversity = 6.7828e-01
Generation 2470:	Best = 1.061003e+05	FES = 247100	Diversity = 7.7298e-01
Generation 2480:	Best = 1.059876e+05	FES = 248100	Diversity = 8.0262e-01
Generation 2490:	Best = 1.057645e+05	FES = 249100	Diversity = 7.7131e-01
Generation 2500:	Best = 1.053519e+05	FES = 250100	Diversity = 7.9200e-01
Generation 2510:	Best = 1.051913e+05	FES = 251100	Diversity = 8.6009e-01
Generation 2520:	Best = 1.047981e+05	FES = 252100	Diversity = 9.3146e-01
Generation 2530:	Best = 1.044347e+05	FES = 253100	Diversity = 9.8094e-01
Generation 2540:	Best = 1.037520e+05	FES = 254100	Diversity = 1.0432e+00
Generation 2550:	Best = 1.029669e+05	FES = 255100	Diversity = 1.1692e+00
Generation 2560:	Best = 1.023990e+05	FES = 256100	Diversity = 1.3717e+00
Generation 2570:	Best = 1.019932e+05	FES = 257100	Diversity = 1.3237e+00
Generation 2580:	Best = 1.014176e+05	FES = 258100	Diversity = 1.2082e+00
Generation 2590:	Best = 1.004886e+05	FES = 259100	Diversity = 1.1052e+00
Generation 2600:	Best = 1.004886e+05	FES = 260100	Diversity = 1.0804e+00

Generation 2610:	Best = 1.001612e+05	FES = 261100	Diversity = 9.9105e-01
Generation 2620:	Best = 9.970093e+04	FES = 262100	Diversity = 1.0120e+00
Generation 2630:	Best = 9.960623e+04	FES = 263100	Diversity = 9.6534e-01
Generation 2640:	Best = 9.893135e+04	FES = 264100	Diversity = 9.2312e-01
Generation 2650:	Best = 9.869533e+04	FES = 265100	Diversity = 9.7455e-01
Generation 2660:	Best = 9.834342e+04	FES = 266100	Diversity = 1.0406e+00
Generation 2670:	Best = 9.781073e+04	FES = 267100	Diversity = 9.0578e-01
Generation 2680:	Best = 9.737811e+04	FES = 268100	Diversity = 9.7635e-01
Generation 2690:	Best = 9.707481e+04	FES = 269100	Diversity = 1.0012e+00
Generation 2700:	Best = 9.690650e+04	FES = 270100	Diversity = 9.3756e-01
Generation 2710:	Best = 9.671675e+04	FES = 271100	Diversity = 8.8824e-01
Generation 2720:	Best = 9.658661e+04	FES = 272100	Diversity = 7.4793e-01
Generation 2730:	Best = 9.634058e+04	FES = 273100	Diversity = 6.5036e-01
Generation 2740:	Best = 9.614261e+04	FES = 274100	Diversity = 6.7962e-01
Generation 2750:	Best = 9.587537e+04	FES = 275100	Diversity = 6.7672e-01
Generation 2760:	Best = 9.568838e+04	FES = 276100	Diversity = 7.3070e-01
Generation 2770:	Best = 9.546010e+04	FES = 277100	Diversity = 6.6730e-01
Generation 2780:	Best = 9.531906e+04	FES = 278100	Diversity = 7.1788e-01
Generation 2790:	Best = 9.513421e+04	FES = 279100	Diversity = 7.4215e-01
Generation 2800:	Best = 9.500313e+04	FES = 280100	Diversity = 6.7013e-01
Generation 2810:	Best = 9.455481e+04	FES = 281100	Diversity = 6.3087e-01
Generation 2820:	Best = 9.455481e+04	FES = 282100	Diversity = 6.0153e-01
Generation 2830:	Best = 9.416486e+04	FES = 283100	Diversity = 6.7452e-01
Generation 2840:	Best = 9.390086e+04	FES = 284100	Diversity = 7.4125e-01
Generation 2850:	Best = 9.380605e+04	FES = 285100	Diversity = 7.2202e-01
Generation 2860:	Best = 9.348985e+04	FES = 286100	Diversity = 7.4300e-01
Generation 2870:	Best = 9.304706e+04	FES = 287100	Diversity = 7.1227e-01
Generation 2880:	Best = 9.300903e+04	FES = 288100	Diversity = 7.8843e-01
Generation 2890:	Best = 9.275376e+04	FES = 289100	Diversity = 8.1800e-01
Generation 2900:	Best = 9.262568e+04	FES = 290100	Diversity = 7.5889e-01
Generation 2910:	Best = 9.227170e+04	FES = 291100	Diversity = 7.6498e-01
Generation 2920:	Best = 9.209997e+04	FES = 292100	Diversity = 8.7368e-01
Generation 2930:	Best = 9.179859e+04	FES = 293100	Diversity = 8.3932e-01
Generation 2940:	Best = 9.156486e+04	FES = 294100	Diversity = 8.5362e-01
Generation 2950:	Best = 9.132206e+04	FES = 295100	Diversity = 7.7335e-01
Generation 2960:	Best = 9.057136e+04	FES = 296100	Diversity = 9.2263e-01
Generation 2970:	Best = 9.031141e+04	FES = 297100	Diversity = 1.0565e+00
Generation 2980:	Best = 9.000620e+04	FES = 298100	Diversity = 9.1847e-01
Generation 2990:	Best = 8.983216e+04	FES = 299100	Diversity = 9.6226e-01
Generation 3000:	Best = 8.927172e+04	FES = 300100	Diversity = 8.5627e-01
Generation 3010:	Best = 8.880643e+04	FES = 301100	Diversity = 9.2660e-01
Generation 3020:	Best = 8.856133e+04	FES = 302100	Diversity = 9.2190e-01
Generation 3030:	Best = 8.856109e+04	FES = 303100	Diversity = 8.8488e-01
Generation 3040:	Best = 8.846960e+04	FES = 304100	Diversity = 8.5310e-01
Generation 3050:	Best = 8.830238e+04	FES = 305100	Diversity = 7.1349e-01
Generation 3060:	Best = 8.825442e+04	FES = 306100	Diversity = 6.4549e-01
Generation 3070:	Best = 8.804799e+04	FES = 307100	Diversity = 6.3885e-01
Generation 3080:	Best = 8.787337e+04	FES = 308100	Diversity = 6.2617e-01
Generation 3090:	Best = 8.742722e+04	FES = 309100	Diversity = 8.1996e-01
Generation 3100:	Best = 8.718409e+04	FES = 310100	Diversity = 9.4279e-01
Generation 3110:	Best = 8.676604e+04	FES = 311100	Diversity = 9.1973e-01
Generation 3120:	Best = 8.648102e+04	FES = 312100	Diversity = 9.5083e-01
Generation 3130:	Best = 8.618915e+04	FES = 313100	Diversity = 8.9235e-01
Generation 3140:	Best = 8.595230e+04	FES = 314100	Diversity = 7.8701e-01

Generation 3150:	Best = 8.549757e+04	FES = 315100	Diversity = 8.2240e-01
Generation 3160:	Best = 8.539060e+04	FES = 316100	Diversity = 9.4342e-01
Generation 3170:	Best = 8.520688e+04	FES = 317100	Diversity = 8.9768e-01
Generation 3180:	Best = 8.482076e+04	FES = 318100	Diversity = 7.7628e-01
Generation 3190:	Best = 8.472439e+04	FES = 319100	Diversity = 8.4120e-01
Generation 3200:	Best = 8.405100e+04	FES = 320100	Diversity = 9.5556e-01
Generation 3210:	Best = 8.333917e+04	FES = 321100	Diversity = 9.6649e-01
Generation 3220:	Best = 8.328392e+04	FES = 322100	Diversity = 9.9006e-01
Generation 3230:	Best = 8.282123e+04	FES = 323100	Diversity = 9.2537e-01
Generation 3240:	Best = 8.250907e+04	FES = 324100	Diversity = 9.4383e-01
Generation 3250:	Best = 8.188330e+04	FES = 325100	Diversity = 9.1197e-01
Generation 3260:	Best = 8.162456e+04	FES = 326100	Diversity = 9.8226e-01
Generation 3270:	Best = 8.124257e+04	FES = 327100	Diversity = 9.6919e-01
Generation 3280:	Best = 8.070781e+04	FES = 328100	Diversity = 1.0131e+00
Generation 3290:	Best = 8.051211e+04	FES = 329100	Diversity = 9.9502e-01
Generation 3300:	Best = 7.997066e+04	FES = 330100	Diversity = 1.0355e+00
Generation 3310:	Best = 7.981845e+04	FES = 331100	Diversity = 1.0949e+00
Generation 3320:	Best = 7.975288e+04	FES = 332100	Diversity = 9.3823e-01
Generation 3330:	Best = 7.939100e+04	FES = 333100	Diversity = 8.5238e-01
Generation 3340:	Best = 7.924957e+04	FES = 334100	Diversity = 7.8967e-01
Generation 3350:	Best = 7.916419e+04	FES = 335100	Diversity = 8.1784e-01
Generation 3360:	Best = 7.895399e+04	FES = 336100	Diversity = 7.4067e-01
Generation 3370:	Best = 7.857577e+04	FES = 337100	Diversity = 7.5798e-01
Generation 3380:	Best = 7.849400e+04	FES = 338100	Diversity = 7.6632e-01
Generation 3390:	Best = 7.830875e+04	FES = 339100	Diversity = 7.9103e-01
Generation 3400:	Best = 7.803999e+04	FES = 340100	Diversity = 7.3985e-01
Generation 3410:	Best = 7.775645e+04	FES = 341100	Diversity = 7.3000e-01
Generation 3420:	Best = 7.768591e+04	FES = 342100	Diversity = 7.9155e-01
Generation 3430:	Best = 7.721057e+04	FES = 343100	Diversity = 7.2047e-01
Generation 3440:	Best = 7.710120e+04	FES = 344100	Diversity = 7.8752e-01
Generation 3450:	Best = 7.703132e+04	FES = 345100	Diversity = 6.6973e-01
Generation 3460:	Best = 7.669918e+04	FES = 346100	Diversity = 6.3407e-01
Generation 3470:	Best = 7.657413e+04	FES = 347100	Diversity = 7.6432e-01
Generation 3480:	Best = 7.606647e+04	FES = 348100	Diversity = 7.6443e-01
Generation 3490:	Best = 7.594191e+04	FES = 349100	Diversity = 7.5111e-01
Generation 3500:	Best = 7.565482e+04	FES = 350100	Diversity = 7.6196e-01
Generation 3510:	Best = 7.534721e+04	FES = 351100	Diversity = 8.1824e-01
Generation 3520:	Best = 7.505740e+04	FES = 352100	Diversity = 8.2922e-01
Generation 3530:	Best = 7.471550e+04	FES = 353100	Diversity = 9.0838e-01
Generation 3540:	Best = 7.412836e+04	FES = 354100	Diversity = 8.9898e-01
Generation 3550:	Best = 7.391858e+04	FES = 355100	Diversity = 9.3121e-01
Generation 3560:	Best = 7.384924e+04	FES = 356100	Diversity = 8.8418e-01
Generation 3570:	Best = 7.378510e+04	FES = 357100	Diversity = 7.5987e-01
Generation 3580:	Best = 7.346175e+04	FES = 358100	Diversity = 7.3619e-01
Generation 3590:	Best = 7.313544e+04	FES = 359100	Diversity = 8.4710e-01
Generation 3600:	Best = 7.298675e+04	FES = 360100	Diversity = 8.3801e-01
Generation 3610:	Best = 7.280222e+04	FES = 361100	Diversity = 7.8241e-01
Generation 3620:	Best = 7.252360e+04	FES = 362100	Diversity = 7.7812e-01
Generation 3630:	Best = 7.244206e+04	FES = 363100	Diversity = 7.6243e-01
Generation 3640:	Best = 7.220351e+04	FES = 364100	Diversity = 6.6905e-01
Generation 3650:	Best = 7.197305e+04	FES = 365100	Diversity = 7.8093e-01
Generation 3660:	Best = 7.181761e+04	FES = 366100	Diversity = 8.6396e-01
Generation 3670:	Best = 7.151821e+04	FES = 367100	Diversity = 8.8928e-01
Generation 3680:	Best = 7.143033e+04	FES = 368100	Diversity = 7.9164e-01

Generation 3690:	Best = 7.133164e+04	FES = 369100	Diversity = 7.5369e-01
Generation 3700:	Best = 7.112393e+04	FES = 370100	Diversity = 7.3652e-01
Generation 3710:	Best = 7.081721e+04	FES = 371100	Diversity = 7.9166e-01
Generation 3720:	Best = 7.063906e+04	FES = 372100	Diversity = 7.5688e-01
Generation 3730:	Best = 7.043431e+04	FES = 373100	Diversity = 7.5515e-01
Generation 3740:	Best = 7.011788e+04	FES = 374100	Diversity = 6.7334e-01
Generation 3750:	Best = 6.999261e+04	FES = 375100	Diversity = 6.8790e-01
Generation 3760:	Best = 6.987010e+04	FES = 376100	Diversity = 7.2086e-01
Generation 3770:	Best = 6.967675e+04	FES = 377100	Diversity = 7.3332e-01
Generation 3780:	Best = 6.937881e+04	FES = 378100	Diversity = 7.6830e-01
Generation 3790:	Best = 6.924004e+04	FES = 379100	Diversity = 8.0411e-01
Generation 3800:	Best = 6.905242e+04	FES = 380100	Diversity = 7.9973e-01
Generation 3810:	Best = 6.876939e+04	FES = 381100	Diversity = 7.9031e-01
Generation 3820:	Best = 6.846559e+04	FES = 382100	Diversity = 7.5451e-01
Generation 3830:	Best = 6.830449e+04	FES = 383100	Diversity = 8.3189e-01
Generation 3840:	Best = 6.805039e+04	FES = 384100	Diversity = 7.4654e-01
Generation 3850:	Best = 6.798158e+04	FES = 385100	Diversity = 7.8332e-01
Generation 3860:	Best = 6.764151e+04	FES = 386100	Diversity = 7.2110e-01
Generation 3870:	Best = 6.713955e+04	FES = 387100	Diversity = 7.2625e-01
Generation 3880:	Best = 6.671234e+04	FES = 388100	Diversity = 8.0838e-01
Generation 3890:	Best = 6.656699e+04	FES = 389100	Diversity = 8.9109e-01
Generation 3900:	Best = 6.639477e+04	FES = 390100	Diversity = 8.5958e-01
Generation 3910:	Best = 6.628154e+04	FES = 391100	Diversity = 7.3035e-01
Generation 3920:	Best = 6.606813e+04	FES = 392100	Diversity = 7.0746e-01
Generation 3930:	Best = 6.574341e+04	FES = 393100	Diversity = 7.0522e-01
Generation 3940:	Best = 6.569985e+04	FES = 394100	Diversity = 6.5384e-01
Generation 3950:	Best = 6.554632e+04	FES = 395100	Diversity = 6.1014e-01
Generation 3960:	Best = 6.550919e+04	FES = 396100	Diversity = 5.4390e-01
Generation 3970:	Best = 6.493708e+04	FES = 397100	Diversity = 5.4795e-01
Generation 3980:	Best = 6.492627e+04	FES = 398100	Diversity = 7.5711e-01
Generation 3990:	Best = 6.467260e+04	FES = 399100	Diversity = 6.2917e-01
Generation 4000:	Best = 6.459158e+04	FES = 400100	Diversity = 6.2537e-01
Generation 4010:	Best = 6.431311e+04	FES = 401100	Diversity = 7.1116e-01
Generation 4020:	Best = 6.420174e+04	FES = 402100	Diversity = 7.6713e-01
Generation 4030:	Best = 6.386694e+04	FES = 403100	Diversity = 7.6509e-01
Generation 4040:	Best = 6.320122e+04	FES = 404100	Diversity = 8.6191e-01
Generation 4050:	Best = 6.320122e+04	FES = 405100	Diversity = 8.7319e-01
Generation 4060:	Best = 6.300719e+04	FES = 406100	Diversity = 8.4278e-01
Generation 4070:	Best = 6.283451e+04	FES = 407100	Diversity = 7.6344e-01
Generation 4080:	Best = 6.253861e+04	FES = 408100	Diversity = 6.7389e-01
Generation 4090:	Best = 6.237490e+04	FES = 409100	Diversity = 7.5491e-01
Generation 4100:	Best = 6.216220e+04	FES = 410100	Diversity = 7.7711e-01
Generation 4110:	Best = 6.206803e+04	FES = 411100	Diversity = 7.0349e-01
Generation 4120:	Best = 6.184143e+04	FES = 412100	Diversity = 6.3446e-01
Generation 4130:	Best = 6.154993e+04	FES = 413100	Diversity = 6.4826e-01
Generation 4140:	Best = 6.132955e+04	FES = 414100	Diversity = 7.1657e-01
Generation 4150:	Best = 6.114581e+04	FES = 415100	Diversity = 7.3167e-01
Generation 4160:	Best = 6.098879e+04	FES = 416100	Diversity = 7.5948e-01
Generation 4170:	Best = 6.089571e+04	FES = 417100	Diversity = 6.8988e-01
Generation 4180:	Best = 6.067710e+04	FES = 418100	Diversity = 6.7830e-01
Generation 4190:	Best = 6.040014e+04	FES = 419100	Diversity = 6.8792e-01
Generation 4200:	Best = 6.025000e+04	FES = 420100	Diversity = 7.1012e-01
Generation 4210:	Best = 6.007006e+04	FES = 421100	Diversity = 6.7337e-01
Generation 4220:	Best = 5.992019e+04	FES = 422100	Diversity = 6.5539e-01

Generation 4230:	Best = 5.968320e+04	FES = 423100	Diversity = 6.4908e-01
Generation 4240:	Best = 5.960686e+04	FES = 424100	Diversity = 6.3666e-01
Generation 4250:	Best = 5.954652e+04	FES = 425100	Diversity = 5.7049e-01
Generation 4260:	Best = 5.935077e+04	FES = 426100	Diversity = 5.4832e-01
Generation 4270:	Best = 5.924137e+04	FES = 427100	Diversity = 6.1214e-01
Generation 4280:	Best = 5.894142e+04	FES = 428100	Diversity = 5.5481e-01
Generation 4290:	Best = 5.864250e+04	FES = 429100	Diversity = 6.1362e-01
Generation 4300:	Best = 5.843646e+04	FES = 430100	Diversity = 7.1066e-01
Generation 4310:	Best = 5.815316e+04	FES = 431100	Diversity = 7.3050e-01
Generation 4320:	Best = 5.799564e+04	FES = 432100	Diversity = 7.7504e-01
Generation 4330:	Best = 5.768333e+04	FES = 433100	Diversity = 7.6611e-01
Generation 4340:	Best = 5.749384e+04	FES = 434100	Diversity = 7.6020e-01
Generation 4350:	Best = 5.727967e+04	FES = 435100	Diversity = 7.1886e-01
Generation 4360:	Best = 5.717121e+04	FES = 436100	Diversity = 6.7521e-01
Generation 4370:	Best = 5.699736e+04	FES = 437100	Diversity = 6.5423e-01
Generation 4380:	Best = 5.679832e+04	FES = 438100	Diversity = 6.4755e-01
Generation 4390:	Best = 5.661732e+04	FES = 439100	Diversity = 6.2187e-01
Generation 4400:	Best = 5.648783e+04	FES = 440100	Diversity = 6.5391e-01
Generation 4410:	Best = 5.617183e+04	FES = 441100	Diversity = 5.7726e-01
Generation 4420:	Best = 5.607334e+04	FES = 442100	Diversity = 6.5176e-01
Generation 4430:	Best = 5.598259e+04	FES = 443100	Diversity = 6.7400e-01
Generation 4440:	Best = 5.539178e+04	FES = 444100	Diversity = 6.2072e-01
Generation 4450:	Best = 5.534331e+04	FES = 445100	Diversity = 6.3093e-01
Generation 4460:	Best = 5.526599e+04	FES = 446100	Diversity = 6.1624e-01
Generation 4470:	Best = 5.498710e+04	FES = 447100	Diversity = 5.9694e-01
Generation 4480:	Best = 5.480960e+04	FES = 448100	Diversity = 6.8821e-01
Generation 4490:	Best = 5.454167e+04	FES = 449100	Diversity = 7.0218e-01
Generation 4500:	Best = 5.423664e+04	FES = 450100	Diversity = 7.3758e-01
Generation 4510:	Best = 5.395296e+04	FES = 451100	Diversity = 7.7848e-01
Generation 4520:	Best = 5.357493e+04	FES = 452100	Diversity = 7.3849e-01
Generation 4530:	Best = 5.343386e+04	FES = 453100	Diversity = 7.1229e-01
Generation 4540:	Best = 5.300316e+04	FES = 454100	Diversity = 6.4637e-01
Generation 4550:	Best = 5.290739e+04	FES = 455100	Diversity = 6.9788e-01
Generation 4560:	Best = 5.282281e+04	FES = 456100	Diversity = 6.6912e-01
Generation 4570:	Best = 5.270371e+04	FES = 457100	Diversity = 6.1447e-01
Generation 4580:	Best = 5.257686e+04	FES = 458100	Diversity = 5.9461e-01
Generation 4590:	Best = 5.185874e+04	FES = 459100	Diversity = 7.6126e-01
Generation 4600:	Best = 5.158835e+04	FES = 460100	Diversity = 9.1500e-01
Generation 4610:	Best = 5.135262e+04	FES = 461100	Diversity = 8.6985e-01
Generation 4620:	Best = 5.098756e+04	FES = 462100	Diversity = 8.2926e-01
Generation 4630:	Best = 5.086745e+04	FES = 463100	Diversity = 7.6066e-01
Generation 4640:	Best = 5.050975e+04	FES = 464100	Diversity = 8.1497e-01
Generation 4650:	Best = 5.041511e+04	FES = 465100	Diversity = 8.9029e-01
Generation 4660:	Best = 5.019555e+04	FES = 466100	Diversity = 7.4111e-01
Generation 4670:	Best = 4.982270e+04	FES = 467100	Diversity = 6.6661e-01
Generation 4680:	Best = 4.977166e+04	FES = 468100	Diversity = 7.1170e-01
Generation 4690:	Best = 4.961297e+04	FES = 469100	Diversity = 7.2354e-01
Generation 4700:	Best = 4.930106e+04	FES = 470100	Diversity = 6.3715e-01
Generation 4710:	Best = 4.923163e+04	FES = 471100	Diversity = 6.7278e-01
Generation 4720:	Best = 4.903390e+04	FES = 472100	Diversity = 5.7573e-01
Generation 4730:	Best = 4.891771e+04	FES = 473100	Diversity = 5.9225e-01
Generation 4740:	Best = 4.879161e+04	FES = 474100	Diversity = 6.0839e-01
Generation 4750:	Best = 4.871321e+04	FES = 475100	Diversity = 5.2600e-01
Generation 4760:	Best = 4.855620e+04	FES = 476100	Diversity = 5.3340e-01

Generation 4770:	Best = 4.842705e+04	FES = 477100	Diversity = 5.7078e-01
Generation 4780:	Best = 4.809104e+04	FES = 478100	Diversity = 5.3768e-01
Generation 4790:	Best = 4.804323e+04	FES = 479100	Diversity = 6.1151e-01
Generation 4800:	Best = 4.784867e+04	FES = 480100	Diversity = 5.6680e-01
Generation 4810:	Best = 4.782138e+04	FES = 481100	Diversity = 5.5540e-01
Generation 4820:	Best = 4.764712e+04	FES = 482100	Diversity = 4.9226e-01
Generation 4830:	Best = 4.757149e+04	FES = 483100	Diversity = 4.8917e-01
Generation 4840:	Best = 4.751835e+04	FES = 484100	Diversity = 4.8270e-01
Generation 4850:	Best = 4.741669e+04	FES = 485100	Diversity = 5.2912e-01
Generation 4860:	Best = 4.718759e+04	FES = 486100	Diversity = 4.7304e-01
Generation 4870:	Best = 4.710965e+04	FES = 487100	Diversity = 5.0517e-01
Generation 4880:	Best = 4.663782e+04	FES = 488100	Diversity = 4.8772e-01
Generation 4890:	Best = 4.641671e+04	FES = 489100	Diversity = 6.5762e-01
Generation 4900:	Best = 4.627501e+04	FES = 490100	Diversity = 7.6657e-01
Generation 4910:	Best = 4.599043e+04	FES = 491100	Diversity = 7.9487e-01
Generation 4920:	Best = 4.586915e+04	FES = 492100	Diversity = 6.8784e-01
Generation 4930:	Best = 4.561398e+04	FES = 493100	Diversity = 6.7877e-01
Generation 4940:	Best = 4.557217e+04	FES = 494100	Diversity = 6.1819e-01
Generation 4950:	Best = 4.528408e+04	FES = 495100	Diversity = 5.4935e-01
Generation 4960:	Best = 4.513441e+04	FES = 496100	Diversity = 6.0119e-01
Generation 4970:	Best = 4.501604e+04	FES = 497100	Diversity = 5.9376e-01
Generation 4980:	Best = 4.492632e+04	FES = 498100	Diversity = 6.2575e-01
Generation 4990:	Best = 4.472824e+04	FES = 499100	Diversity = 5.6591e-01
Generation 5000:	Best = 4.466500e+04	FES = 500100	Diversity = 5.9147e-01
Generation 5010:	Best = 4.455091e+04	FES = 501100	Diversity = 5.7568e-01
Generation 5020:	Best = 4.434110e+04	FES = 502100	Diversity = 5.4743e-01
Generation 5030:	Best = 4.431431e+04	FES = 503100	Diversity = 5.0704e-01
Generation 5040:	Best = 4.414969e+04	FES = 504100	Diversity = 4.3190e-01
Generation 5050:	Best = 4.387045e+04	FES = 505100	Diversity = 4.6317e-01
Generation 5060:	Best = 4.378128e+04	FES = 506100	Diversity = 5.5037e-01
Generation 5070:	Best = 4.352245e+04	FES = 507100	Diversity = 6.8503e-01
Generation 5080:	Best = 4.339362e+04	FES = 508100	Diversity = 6.8185e-01
Generation 5090:	Best = 4.325440e+04	FES = 509100	Diversity = 6.7746e-01
Generation 5100:	Best = 4.308536e+04	FES = 510100	Diversity = 5.9880e-01
Generation 5110:	Best = 4.297350e+04	FES = 511100	Diversity = 6.1870e-01
Generation 5120:	Best = 4.279665e+04	FES = 512100	Diversity = 5.3688e-01
Generation 5130:	Best = 4.266153e+04	FES = 513100	Diversity = 5.4378e-01
Generation 5140:	Best = 4.255326e+04	FES = 514100	Diversity = 5.7584e-01
Generation 5150:	Best = 4.246006e+04	FES = 515100	Diversity = 5.9591e-01
Generation 5160:	Best = 4.231134e+04	FES = 516100	Diversity = 5.7456e-01
Generation 5170:	Best = 4.222430e+04	FES = 517100	Diversity = 5.8492e-01
Generation 5180:	Best = 4.198066e+04	FES = 518100	Diversity = 5.7692e-01
Generation 5190:	Best = 4.198066e+04	FES = 519100	Diversity = 5.4154e-01
Generation 5200:	Best = 4.171957e+04	FES = 520100	Diversity = 5.4506e-01
Generation 5210:	Best = 4.166643e+04	FES = 521100	Diversity = 5.4805e-01
Generation 5220:	Best = 4.153208e+04	FES = 522100	Diversity = 5.1310e-01
Generation 5230:	Best = 4.138055e+04	FES = 523100	Diversity = 5.1133e-01
Generation 5240:	Best = 4.135159e+04	FES = 524100	Diversity = 5.2477e-01
Generation 5250:	Best = 4.128407e+04	FES = 525100	Diversity = 4.7779e-01
Generation 5260:	Best = 4.125025e+04	FES = 526100	Diversity = 4.4465e-01
Generation 5270:	Best = 4.102300e+04	FES = 527100	Diversity = 4.3633e-01
Generation 5280:	Best = 4.102300e+04	FES = 528100	Diversity = 4.3768e-01
Generation 5290:	Best = 4.102300e+04	FES = 529100	Diversity = 4.6836e-01
Generation 5300:	Best = 4.085944e+04	FES = 530100	Diversity = 4.7487e-01

Generation 5310:	Best = 4.073491e+04	FES = 531100	Diversity = 5.2143e-01
Generation 5320:	Best = 4.055349e+04	FES = 532100	Diversity = 5.5151e-01
Generation 5330:	Best = 4.051234e+04	FES = 533100	Diversity = 5.8533e-01
Generation 5340:	Best = 4.008554e+04	FES = 534100	Diversity = 6.2306e-01
Generation 5350:	Best = 3.990879e+04	FES = 535100	Diversity = 7.6925e-01
Generation 5360:	Best = 3.986003e+04	FES = 536100	Diversity = 7.3051e-01
Generation 5370:	Best = 3.968129e+04	FES = 537100	Diversity = 6.6618e-01
Generation 5380:	Best = 3.964642e+04	FES = 538100	Diversity = 6.0231e-01
Generation 5390:	Best = 3.958438e+04	FES = 539100	Diversity = 5.3543e-01
Generation 5400:	Best = 3.932647e+04	FES = 540100	Diversity = 4.7635e-01
Generation 5410:	Best = 3.932647e+04	FES = 541100	Diversity = 4.8321e-01
Generation 5420:	Best = 3.927081e+04	FES = 542100	Diversity = 4.7856e-01
Generation 5430:	Best = 3.917911e+04	FES = 543100	Diversity = 4.9651e-01
Generation 5440:	Best = 3.909565e+04	FES = 544100	Diversity = 4.6002e-01
Generation 5450:	Best = 3.877633e+04	FES = 545100	Diversity = 4.9769e-01
Generation 5460:	Best = 3.868995e+04	FES = 546100	Diversity = 5.8060e-01
Generation 5470:	Best = 3.852140e+04	FES = 547100	Diversity = 5.7045e-01
Generation 5480:	Best = 3.839761e+04	FES = 548100	Diversity = 5.9398e-01
Generation 5490:	Best = 3.832824e+04	FES = 549100	Diversity = 6.1483e-01
Generation 5500:	Best = 3.824614e+04	FES = 550100	Diversity = 5.3631e-01
Generation 5510:	Best = 3.808381e+04	FES = 551100	Diversity = 5.1685e-01
Generation 5520:	Best = 3.796308e+04	FES = 552100	Diversity = 5.0995e-01
Generation 5530:	Best = 3.779071e+04	FES = 553100	Diversity = 5.3063e-01
Generation 5540:	Best = 3.771150e+04	FES = 554100	Diversity = 5.8710e-01
Generation 5550:	Best = 3.756112e+04	FES = 555100	Diversity = 5.7207e-01
Generation 5560:	Best = 3.751653e+04	FES = 556100	Diversity = 5.8512e-01
Generation 5570:	Best = 3.741804e+04	FES = 557100	Diversity = 5.5319e-01
Generation 5580:	Best = 3.724679e+04	FES = 558100	Diversity = 5.7403e-01
Generation 5590:	Best = 3.711340e+04	FES = 559100	Diversity = 5.3119e-01
Generation 5600:	Best = 3.709459e+04	FES = 560100	Diversity = 5.8380e-01
Generation 5610:	Best = 3.695033e+04	FES = 561100	Diversity = 5.8756e-01
Generation 5620:	Best = 3.682030e+04	FES = 562100	Diversity = 5.6203e-01
Generation 5630:	Best = 3.678705e+04	FES = 563100	Diversity = 5.3625e-01
Generation 5640:	Best = 3.663449e+04	FES = 564100	Diversity = 5.4673e-01
Generation 5650:	Best = 3.654641e+04	FES = 565100	Diversity = 5.5439e-01
Generation 5660:	Best = 3.642043e+04	FES = 566100	Diversity = 5.2069e-01
Generation 5670:	Best = 3.622380e+04	FES = 567100	Diversity = 5.8106e-01
Generation 5680:	Best = 3.604650e+04	FES = 568100	Diversity = 7.0763e-01
Generation 5690:	Best = 3.594518e+04	FES = 569100	Diversity = 7.2916e-01
Generation 5700:	Best = 3.579834e+04	FES = 570100	Diversity = 6.5734e-01
Generation 5710:	Best = 3.573083e+04	FES = 571100	Diversity = 5.9653e-01
Generation 5720:	Best = 3.555847e+04	FES = 572100	Diversity = 5.6269e-01
Generation 5730:	Best = 3.548447e+04	FES = 573100	Diversity = 6.2775e-01
Generation 5740:	Best = 3.526852e+04	FES = 574100	Diversity = 6.0957e-01
Generation 5750:	Best = 3.523466e+04	FES = 575100	Diversity = 6.1609e-01
Generation 5760:	Best = 3.506692e+04	FES = 576100	Diversity = 5.6815e-01
Generation 5770:	Best = 3.491617e+04	FES = 577100	Diversity = 5.8938e-01
Generation 5780:	Best = 3.485834e+04	FES = 578100	Diversity = 5.8760e-01
Generation 5790:	Best = 3.480591e+04	FES = 579100	Diversity = 5.2934e-01
Generation 5800:	Best = 3.472682e+04	FES = 580100	Diversity = 4.7167e-01
Generation 5810:	Best = 3.470774e+04	FES = 581100	Diversity = 4.9455e-01
Generation 5820:	Best = 3.457854e+04	FES = 582100	Diversity = 4.6283e-01
Generation 5830:	Best = 3.450491e+04	FES = 583100	Diversity = 4.7464e-01
Generation 5840:	Best = 3.447174e+04	FES = 584100	Diversity = 5.1123e-01

Generation 5850:	Best = 3.421170e+04	FES = 585100	Diversity = 4.8749e-01
Generation 5860:	Best = 3.415368e+04	FES = 586100	Diversity = 5.2350e-01
Generation 5870:	Best = 3.406400e+04	FES = 587100	Diversity = 5.1442e-01
Generation 5880:	Best = 3.397305e+04	FES = 588100	Diversity = 5.1735e-01
Generation 5890:	Best = 3.393376e+04	FES = 589100	Diversity = 4.6101e-01
Generation 5900:	Best = 3.387552e+04	FES = 590100	Diversity = 4.2384e-01
Generation 5910:	Best = 3.378102e+04	FES = 591100	Diversity = 4.2106e-01
Generation 5920:	Best = 3.377381e+04	FES = 592100	Diversity = 4.1269e-01
Generation 5930:	Best = 3.370584e+04	FES = 593100	Diversity = 4.2051e-01
Generation 5940:	Best = 3.362814e+04	FES = 594100	Diversity = 4.0075e-01
Generation 5950:	Best = 3.346257e+04	FES = 595100	Diversity = 4.3592e-01
Generation 5960:	Best = 3.346257e+04	FES = 596100	Diversity = 4.2884e-01
Generation 5970:	Best = 3.338638e+04	FES = 597100	Diversity = 4.5481e-01
Generation 5980:	Best = 3.328143e+04	FES = 598100	Diversity = 4.5570e-01
Generation 5990:	Best = 3.325594e+04	FES = 599100	Diversity = 4.8144e-01
Generation 6000:	Best = 3.317330e+04	FES = 600100	Diversity = 4.4641e-01
Generation 6010:	Best = 3.308273e+04	FES = 601100	Diversity = 4.7663e-01
Generation 6020:	Best = 3.303633e+04	FES = 602100	Diversity = 5.1978e-01
Generation 6030:	Best = 3.291600e+04	FES = 603100	Diversity = 5.0134e-01
Generation 6040:	Best = 3.275097e+04	FES = 604100	Diversity = 5.4949e-01
Generation 6050:	Best = 3.262558e+04	FES = 605100	Diversity = 5.4738e-01
Generation 6060:	Best = 3.241003e+04	FES = 606100	Diversity = 6.0353e-01
Generation 6070:	Best = 3.225747e+04	FES = 607100	Diversity = 5.9949e-01
Generation 6080:	Best = 3.221129e+04	FES = 608100	Diversity = 6.4736e-01
Generation 6090:	Best = 3.209177e+04	FES = 609100	Diversity = 7.0172e-01
Generation 6100:	Best = 3.196657e+04	FES = 610100	Diversity = 6.3758e-01
Generation 6110:	Best = 3.184841e+04	FES = 611100	Diversity = 6.0423e-01
Generation 6120:	Best = 3.162583e+04	FES = 612100	Diversity = 5.8325e-01
Generation 6130:	Best = 3.152106e+04	FES = 613100	Diversity = 6.3498e-01
Generation 6140:	Best = 3.147940e+04	FES = 614100	Diversity = 6.2152e-01
Generation 6150:	Best = 3.145460e+04	FES = 615100	Diversity = 5.7232e-01
Generation 6160:	Best = 3.131993e+04	FES = 616100	Diversity = 5.4899e-01
Generation 6170:	Best = 3.126737e+04	FES = 617100	Diversity = 5.4309e-01
Generation 6180:	Best = 3.111277e+04	FES = 618100	Diversity = 5.3283e-01
Generation 6190:	Best = 3.096783e+04	FES = 619100	Diversity = 5.3801e-01
Generation 6200:	Best = 3.085374e+04	FES = 620100	Diversity = 5.8336e-01
Generation 6210:	Best = 3.069528e+04	FES = 621100	Diversity = 4.9639e-01
Generation 6220:	Best = 3.068010e+04	FES = 622100	Diversity = 5.0542e-01
Generation 6230:	Best = 3.044955e+04	FES = 623100	Diversity = 5.0424e-01
Generation 6240:	Best = 3.022757e+04	FES = 624100	Diversity = 5.7152e-01
Generation 6250:	Best = 3.022757e+04	FES = 625100	Diversity = 5.2150e-01
Generation 6260:	Best = 3.012485e+04	FES = 626100	Diversity = 6.0009e-01
Generation 6270:	Best = 3.010054e+04	FES = 627100	Diversity = 5.4869e-01
Generation 6280:	Best = 2.991459e+04	FES = 628100	Diversity = 5.5394e-01
Generation 6290:	Best = 2.982414e+04	FES = 629100	Diversity = 5.6258e-01
Generation 6300:	Best = 2.976396e+04	FES = 630100	Diversity = 5.6322e-01
Generation 6310:	Best = 2.969009e+04	FES = 631100	Diversity = 5.2814e-01
Generation 6320:	Best = 2.964041e+04	FES = 632100	Diversity = 5.3720e-01
Generation 6330:	Best = 2.957189e+04	FES = 633100	Diversity = 4.7435e-01
Generation 6340:	Best = 2.954023e+04	FES = 634100	Diversity = 4.7004e-01
Generation 6350:	Best = 2.945840e+04	FES = 635100	Diversity = 4.2684e-01
Generation 6360:	Best = 2.943326e+04	FES = 636100	Diversity = 4.3763e-01
Generation 6370:	Best = 2.936915e+04	FES = 637100	Diversity = 4.3316e-01
Generation 6380:	Best = 2.930508e+04	FES = 638100	Diversity = 4.5396e-01

Generation 6390:	Best = 2.928910e+04	FES = 639100	Diversity = 4.1732e-01
Generation 6400:	Best = 2.924925e+04	FES = 640100	Diversity = 3.8775e-01
Generation 6410:	Best = 2.918018e+04	FES = 641100	Diversity = 3.5840e-01
Generation 6420:	Best = 2.917009e+04	FES = 642100	Diversity = 3.3932e-01
Generation 6430:	Best = 2.913208e+04	FES = 643100	Diversity = 3.4492e-01
Generation 6440:	Best = 2.910003e+04	FES = 644100	Diversity = 3.4593e-01
Generation 6450:	Best = 2.906171e+04	FES = 645100	Diversity = 3.5538e-01
Generation 6460:	Best = 2.903330e+04	FES = 646100	Diversity = 3.8811e-01
Generation 6470:	Best = 2.900050e+04	FES = 647100	Diversity = 3.8248e-01
Generation 6480:	Best = 2.884707e+04	FES = 648100	Diversity = 4.0284e-01
Generation 6490:	Best = 2.883960e+04	FES = 649100	Diversity = 4.3508e-01
Generation 6500:	Best = 2.873861e+04	FES = 650100	Diversity = 4.4649e-01
Generation 6510:	Best = 2.865170e+04	FES = 651100	Diversity = 4.3649e-01
Generation 6520:	Best = 2.859948e+04	FES = 652100	Diversity = 4.2690e-01
Generation 6530:	Best = 2.852908e+04	FES = 653100	Diversity = 4.2461e-01
Generation 6540:	Best = 2.847379e+04	FES = 654100	Diversity = 4.5415e-01
Generation 6550:	Best = 2.838517e+04	FES = 655100	Diversity = 4.3666e-01
Generation 6560:	Best = 2.835068e+04	FES = 656100	Diversity = 4.2820e-01
Generation 6570:	Best = 2.829639e+04	FES = 657100	Diversity = 4.3628e-01
Generation 6580:	Best = 2.823372e+04	FES = 658100	Diversity = 4.4546e-01
Generation 6590:	Best = 2.810325e+04	FES = 659100	Diversity = 4.4467e-01
Generation 6600:	Best = 2.810141e+04	FES = 660100	Diversity = 4.6233e-01
Generation 6610:	Best = 2.803586e+04	FES = 661100	Diversity = 4.4289e-01
Generation 6620:	Best = 2.796215e+04	FES = 662100	Diversity = 4.2653e-01
Generation 6630:	Best = 2.783274e+04	FES = 663100	Diversity = 4.3652e-01
Generation 6640:	Best = 2.778123e+04	FES = 664100	Diversity = 5.0212e-01
Generation 6650:	Best = 2.766269e+04	FES = 665100	Diversity = 4.7031e-01
Generation 6660:	Best = 2.765339e+04	FES = 666100	Diversity = 4.8899e-01
Generation 6670:	Best = 2.756754e+04	FES = 667100	Diversity = 4.9893e-01
Generation 6680:	Best = 2.752104e+04	FES = 668100	Diversity = 4.8698e-01
Generation 6690:	Best = 2.745381e+04	FES = 669100	Diversity = 4.9439e-01
Generation 6700:	Best = 2.734028e+04	FES = 670100	Diversity = 4.6704e-01
Generation 6710:	Best = 2.733679e+04	FES = 671100	Diversity = 4.3540e-01
Generation 6720:	Best = 2.727448e+04	FES = 672100	Diversity = 4.2790e-01
Generation 6730:	Best = 2.715580e+04	FES = 673100	Diversity = 4.3369e-01
Generation 6740:	Best = 2.715580e+04	FES = 674100	Diversity = 4.4337e-01
Generation 6750:	Best = 2.706404e+04	FES = 675100	Diversity = 4.1434e-01
Generation 6760:	Best = 2.700417e+04	FES = 676100	Diversity = 4.2501e-01
Generation 6770:	Best = 2.699875e+04	FES = 677100	Diversity = 3.9804e-01
Generation 6780:	Best = 2.688195e+04	FES = 678100	Diversity = 4.0792e-01
Generation 6790:	Best = 2.686686e+04	FES = 679100	Diversity = 4.1114e-01
Generation 6800:	Best = 2.681371e+04	FES = 680100	Diversity = 4.1041e-01
Generation 6810:	Best = 2.681371e+04	FES = 681100	Diversity = 3.7246e-01
Generation 6820:	Best = 2.676955e+04	FES = 682100	Diversity = 4.0178e-01
Generation 6830:	Best = 2.668090e+04	FES = 683100	Diversity = 3.8777e-01
Generation 6840:	Best = 2.657590e+04	FES = 684100	Diversity = 4.2245e-01
Generation 6850:	Best = 2.657590e+04	FES = 685100	Diversity = 4.0407e-01
Generation 6860:	Best = 2.655835e+04	FES = 686100	Diversity = 3.9537e-01
Generation 6870:	Best = 2.645511e+04	FES = 687100	Diversity = 3.6494e-01
Generation 6880:	Best = 2.643948e+04	FES = 688100	Diversity = 3.7455e-01
Generation 6890:	Best = 2.635529e+04	FES = 689100	Diversity = 3.7242e-01
Generation 6900:	Best = 2.635529e+04	FES = 690100	Diversity = 3.7890e-01
Generation 6910:	Best = 2.632828e+04	FES = 691100	Diversity = 3.2697e-01
Generation 6920:	Best = 2.632308e+04	FES = 692100	Diversity = 3.3880e-01

Generation 6930:	Best = 2.627025e+04	FES = 693100	Diversity = 3.3092e-01
Generation 6940:	Best = 2.620880e+04	FES = 694100	Diversity = 3.7030e-01
Generation 6950:	Best = 2.615029e+04	FES = 695100	Diversity = 3.6523e-01
Generation 6960:	Best = 2.612508e+04	FES = 696100	Diversity = 4.1218e-01
Generation 6970:	Best = 2.606419e+04	FES = 697100	Diversity = 4.6608e-01
Generation 6980:	Best = 2.600321e+04	FES = 698100	Diversity = 4.3933e-01
Generation 6990:	Best = 2.595705e+04	FES = 699100	Diversity = 4.4701e-01
Generation 7000:	Best = 2.585899e+04	FES = 700100	Diversity = 4.3345e-01
Generation 7010:	Best = 2.584585e+04	FES = 701100	Diversity = 4.6651e-01
Generation 7020:	Best = 2.561508e+04	FES = 702100	Diversity = 4.1352e-01
Generation 7030:	Best = 2.554312e+04	FES = 703100	Diversity = 4.6051e-01
Generation 7040:	Best = 2.550133e+04	FES = 704100	Diversity = 5.2431e-01
Generation 7050:	Best = 2.543268e+04	FES = 705100	Diversity = 4.7467e-01
Generation 7060:	Best = 2.535826e+04	FES = 706100	Diversity = 5.0718e-01
Generation 7070:	Best = 2.533917e+04	FES = 707100	Diversity = 4.7492e-01
Generation 7080:	Best = 2.523565e+04	FES = 708100	Diversity = 4.5761e-01
Generation 7090:	Best = 2.514681e+04	FES = 709100	Diversity = 4.7584e-01
Generation 7100:	Best = 2.510951e+04	FES = 710100	Diversity = 4.5482e-01
Generation 7110:	Best = 2.503361e+04	FES = 711100	Diversity = 4.3709e-01
Generation 7120:	Best = 2.500153e+04	FES = 712100	Diversity = 4.2260e-01
Generation 7130:	Best = 2.495707e+04	FES = 713100	Diversity = 4.0235e-01
Generation 7140:	Best = 2.488267e+04	FES = 714100	Diversity = 3.9414e-01
Generation 7150:	Best = 2.486590e+04	FES = 715100	Diversity = 3.8145e-01
Generation 7160:	Best = 2.477599e+04	FES = 716100	Diversity = 3.8872e-01
Generation 7170:	Best = 2.469363e+04	FES = 717100	Diversity = 4.4671e-01
Generation 7180:	Best = 2.460050e+04	FES = 718100	Diversity = 4.4035e-01
Generation 7190:	Best = 2.456144e+04	FES = 719100	Diversity = 4.6034e-01
Generation 7200:	Best = 2.448564e+04	FES = 720100	Diversity = 4.5278e-01
Generation 7210:	Best = 2.441432e+04	FES = 721100	Diversity = 4.6758e-01
Generation 7220:	Best = 2.433926e+04	FES = 722100	Diversity = 5.1496e-01
Generation 7230:	Best = 2.429208e+04	FES = 723100	Diversity = 4.5635e-01
Generation 7240:	Best = 2.414682e+04	FES = 724100	Diversity = 4.2307e-01
Generation 7250:	Best = 2.414682e+04	FES = 725100	Diversity = 4.3898e-01
Generation 7260:	Best = 2.409974e+04	FES = 726100	Diversity = 3.8929e-01
Generation 7270:	Best = 2.400783e+04	FES = 727100	Diversity = 3.4700e-01
Generation 7280:	Best = 2.396714e+04	FES = 728100	Diversity = 4.2204e-01
Generation 7290:	Best = 2.390179e+04	FES = 729100	Diversity = 4.0892e-01
Generation 7300:	Best = 2.380775e+04	FES = 730100	Diversity = 4.0580e-01
Generation 7310:	Best = 2.369656e+04	FES = 731100	Diversity = 4.6077e-01
Generation 7320:	Best = 2.367908e+04	FES = 732100	Diversity = 4.5345e-01
Generation 7330:	Best = 2.360101e+04	FES = 733100	Diversity = 4.4820e-01
Generation 7340:	Best = 2.357194e+04	FES = 734100	Diversity = 4.5520e-01
Generation 7350:	Best = 2.345373e+04	FES = 735100	Diversity = 4.2889e-01
Generation 7360:	Best = 2.344847e+04	FES = 736100	Diversity = 4.3545e-01
Generation 7370:	Best = 2.336665e+04	FES = 737100	Diversity = 3.9941e-01
Generation 7380:	Best = 2.327548e+04	FES = 738100	Diversity = 4.1152e-01
Generation 7390:	Best = 2.325295e+04	FES = 739100	Diversity = 4.6127e-01
Generation 7400:	Best = 2.320198e+04	FES = 740100	Diversity = 4.3996e-01
Generation 7410:	Best = 2.317354e+04	FES = 741100	Diversity = 4.1485e-01
Generation 7420:	Best = 2.301272e+04	FES = 742100	Diversity = 4.2190e-01
Generation 7430:	Best = 2.299069e+04	FES = 743100	Diversity = 4.3316e-01
Generation 7440:	Best = 2.298355e+04	FES = 744100	Diversity = 4.6396e-01
Generation 7450:	Best = 2.290635e+04	FES = 745100	Diversity = 4.0348e-01
Generation 7460:	Best = 2.287862e+04	FES = 746100	Diversity = 3.7011e-01

Generation 7470:	Best = 2.286550e+04	FES = 747100	Diversity = 3.3860e-01
Generation 7480:	Best = 2.275951e+04	FES = 748100	Diversity = 3.3312e-01
Generation 7490:	Best = 2.275951e+04	FES = 749100	Diversity = 3.3611e-01
Generation 7500:	Best = 2.275951e+04	FES = 750100	Diversity = 3.3316e-01
Generation 7510:	Best = 2.269248e+04	FES = 751100	Diversity = 3.2065e-01
Generation 7520:	Best = 2.260487e+04	FES = 752100	Diversity = 3.3489e-01
Generation 7530:	Best = 2.256721e+04	FES = 753100	Diversity = 3.6219e-01
Generation 7540:	Best = 2.253596e+04	FES = 754100	Diversity = 3.8567e-01
Generation 7550:	Best = 2.247129e+04	FES = 755100	Diversity = 3.6369e-01
Generation 7560:	Best = 2.246344e+04	FES = 756100	Diversity = 3.5953e-01
Generation 7570:	Best = 2.236994e+04	FES = 757100	Diversity = 4.1595e-01
Generation 7580:	Best = 2.231167e+04	FES = 758100	Diversity = 4.3937e-01
Generation 7590:	Best = 2.223489e+04	FES = 759100	Diversity = 4.3216e-01
Generation 7600:	Best = 2.220781e+04	FES = 760100	Diversity = 4.1332e-01
Generation 7610:	Best = 2.210526e+04	FES = 761100	Diversity = 4.0962e-01
Generation 7620:	Best = 2.210526e+04	FES = 762100	Diversity = 4.1094e-01
Generation 7630:	Best = 2.186455e+04	FES = 763100	Diversity = 4.7289e-01
Generation 7640:	Best = 2.179626e+04	FES = 764100	Diversity = 4.6933e-01
Generation 7650:	Best = 2.167725e+04	FES = 765100	Diversity = 4.9443e-01
Generation 7660:	Best = 2.161734e+04	FES = 766100	Diversity = 4.8290e-01
Generation 7670:	Best = 2.161426e+04	FES = 767100	Diversity = 4.2931e-01
Generation 7680:	Best = 2.155910e+04	FES = 768100	Diversity = 3.7593e-01
Generation 7690:	Best = 2.149921e+04	FES = 769100	Diversity = 3.9804e-01
Generation 7700:	Best = 2.141193e+04	FES = 770100	Diversity = 4.5597e-01
Generation 7710:	Best = 2.135851e+04	FES = 771100	Diversity = 4.6553e-01
Generation 7720:	Best = 2.131386e+04	FES = 772100	Diversity = 3.9880e-01
Generation 7730:	Best = 2.130573e+04	FES = 773100	Diversity = 3.8581e-01
Generation 7740:	Best = 2.124981e+04	FES = 774100	Diversity = 3.5584e-01
Generation 7750:	Best = 2.124981e+04	FES = 775100	Diversity = 3.5635e-01
Generation 7760:	Best = 2.121303e+04	FES = 776100	Diversity = 3.4356e-01
Generation 7770:	Best = 2.117812e+04	FES = 777100	Diversity = 3.4898e-01
Generation 7780:	Best = 2.114598e+04	FES = 778100	Diversity = 3.5988e-01
Generation 7790:	Best = 2.112950e+04	FES = 779100	Diversity = 3.4572e-01
Generation 7800:	Best = 2.108657e+04	FES = 780100	Diversity = 3.3394e-01
Generation 7810:	Best = 2.108304e+04	FES = 781100	Diversity = 3.1515e-01
Generation 7820:	Best = 2.104082e+04	FES = 782100	Diversity = 3.0258e-01
Generation 7830:	Best = 2.096698e+04	FES = 783100	Diversity = 3.1684e-01
Generation 7840:	Best = 2.096698e+04	FES = 784100	Diversity = 3.3716e-01
Generation 7850:	Best = 2.094034e+04	FES = 785100	Diversity = 3.2964e-01
Generation 7860:	Best = 2.089670e+04	FES = 786100	Diversity = 3.3415e-01
Generation 7870:	Best = 2.084596e+04	FES = 787100	Diversity = 3.6676e-01
Generation 7880:	Best = 2.078602e+04	FES = 788100	Diversity = 3.7338e-01
Generation 7890:	Best = 2.077779e+04	FES = 789100	Diversity = 3.6942e-01
Generation 7900:	Best = 2.070370e+04	FES = 790100	Diversity = 3.6188e-01
Generation 7910:	Best = 2.065592e+04	FES = 791100	Diversity = 3.6455e-01
Generation 7920:	Best = 2.061655e+04	FES = 792100	Diversity = 3.7316e-01
Generation 7930:	Best = 2.056852e+04	FES = 793100	Diversity = 3.8511e-01
Generation 7940:	Best = 2.052707e+04	FES = 794100	Diversity = 3.7348e-01
Generation 7950:	Best = 2.052111e+04	FES = 795100	Diversity = 3.6567e-01
Generation 7960:	Best = 2.048937e+04	FES = 796100	Diversity = 3.4846e-01
Generation 7970:	Best = 2.044612e+04	FES = 797100	Diversity = 3.4621e-01
Generation 7980:	Best = 2.038388e+04	FES = 798100	Diversity = 3.4006e-01
Generation 7990:	Best = 2.038388e+04	FES = 799100	Diversity = 3.3994e-01
Generation 8000:	Best = 2.035704e+04	FES = 800100	Diversity = 3.3888e-01

Generation 8010:	Best = 2.034896e+04	FES = 801100	Diversity = 3.5305e-01
Generation 8020:	Best = 2.025898e+04	FES = 802100	Diversity = 4.1873e-01
Generation 8030:	Best = 2.022369e+04	FES = 803100	Diversity = 4.2253e-01
Generation 8040:	Best = 2.021093e+04	FES = 804100	Diversity = 4.0470e-01
Generation 8050:	Best = 2.019838e+04	FES = 805100	Diversity = 3.5057e-01
Generation 8060:	Best = 2.014528e+04	FES = 806100	Diversity = 3.2776e-01
Generation 8070:	Best = 2.012592e+04	FES = 807100	Diversity = 3.3509e-01
Generation 8080:	Best = 2.004869e+04	FES = 808100	Diversity = 3.4003e-01
Generation 8090:	Best = 2.004722e+04	FES = 809100	Diversity = 3.4961e-01
Generation 8100:	Best = 1.998406e+04	FES = 810100	Diversity = 3.3520e-01
Generation 8110:	Best = 1.995221e+04	FES = 811100	Diversity = 3.5318e-01
Generation 8120:	Best = 1.994339e+04	FES = 812100	Diversity = 3.7351e-01
Generation 8130:	Best = 1.986029e+04	FES = 813100	Diversity = 3.5132e-01
Generation 8140:	Best = 1.981169e+04	FES = 814100	Diversity = 3.7000e-01
Generation 8150:	Best = 1.981169e+04	FES = 815100	Diversity = 3.4893e-01
Generation 8160:	Best = 1.974351e+04	FES = 816100	Diversity = 3.3486e-01
Generation 8170:	Best = 1.971771e+04	FES = 817100	Diversity = 3.4460e-01
Generation 8180:	Best = 1.961242e+04	FES = 818100	Diversity = 3.4510e-01
Generation 8190:	Best = 1.956250e+04	FES = 819100	Diversity = 3.8353e-01
Generation 8200:	Best = 1.956250e+04	FES = 820100	Diversity = 3.6843e-01
Generation 8210:	Best = 1.954906e+04	FES = 821100	Diversity = 3.5916e-01
Generation 8220:	Best = 1.950681e+04	FES = 822100	Diversity = 3.1528e-01
Generation 8230:	Best = 1.950603e+04	FES = 823100	Diversity = 3.0736e-01
Generation 8240:	Best = 1.940778e+04	FES = 824100	Diversity = 3.1345e-01
Generation 8250:	Best = 1.937408e+04	FES = 825100	Diversity = 3.3363e-01
Generation 8260:	Best = 1.937408e+04	FES = 826100	Diversity = 3.5025e-01
Generation 8270:	Best = 1.934416e+04	FES = 827100	Diversity = 3.3945e-01
Generation 8280:	Best = 1.927380e+04	FES = 828100	Diversity = 3.4536e-01
Generation 8290:	Best = 1.925176e+04	FES = 829100	Diversity = 3.3478e-01
Generation 8300:	Best = 1.920915e+04	FES = 830100	Diversity = 3.3633e-01
Generation 8310:	Best = 1.918564e+04	FES = 831100	Diversity = 3.3663e-01
Generation 8320:	Best = 1.910014e+04	FES = 832100	Diversity = 3.4827e-01
Generation 8330:	Best = 1.907055e+04	FES = 833100	Diversity = 3.4982e-01
Generation 8340:	Best = 1.905974e+04	FES = 834100	Diversity = 3.5910e-01
Generation 8350:	Best = 1.901742e+04	FES = 835100	Diversity = 3.3034e-01
Generation 8360:	Best = 1.900522e+04	FES = 836100	Diversity = 3.1219e-01
Generation 8370:	Best = 1.896072e+04	FES = 837100	Diversity = 3.1460e-01
Generation 8380:	Best = 1.894577e+04	FES = 838100	Diversity = 3.2530e-01
Generation 8390:	Best = 1.888914e+04	FES = 839100	Diversity = 3.3630e-01
Generation 8400:	Best = 1.883548e+04	FES = 840100	Diversity = 3.2846e-01
Generation 8410:	Best = 1.876712e+04	FES = 841100	Diversity = 3.5323e-01
Generation 8420:	Best = 1.867208e+04	FES = 842100	Diversity = 4.0912e-01
Generation 8430:	Best = 1.867208e+04	FES = 843100	Diversity = 3.9339e-01
Generation 8440:	Best = 1.858048e+04	FES = 844100	Diversity = 3.8145e-01
Generation 8450:	Best = 1.858048e+04	FES = 845100	Diversity = 3.6272e-01
Generation 8460:	Best = 1.856272e+04	FES = 846100	Diversity = 3.4357e-01
Generation 8470:	Best = 1.854889e+04	FES = 847100	Diversity = 3.2376e-01
Generation 8480:	Best = 1.853460e+04	FES = 848100	Diversity = 3.0825e-01
Generation 8490:	Best = 1.849020e+04	FES = 849100	Diversity = 2.8567e-01
Generation 8500:	Best = 1.849020e+04	FES = 850100	Diversity = 2.7408e-01
Generation 8510:	Best = 1.848626e+04	FES = 851100	Diversity = 2.4691e-01
Generation 8520:	Best = 1.846061e+04	FES = 852100	Diversity = 2.5283e-01
Generation 8530:	Best = 1.843826e+04	FES = 853100	Diversity = 2.4977e-01
Generation 8540:	Best = 1.841095e+04	FES = 854100	Diversity = 2.6771e-01

Generation 8550:	Best = 1.839514e+04	FES = 855100	Diversity = 2.5731e-01
Generation 8560:	Best = 1.835190e+04	FES = 856100	Diversity = 2.6086e-01
Generation 8570:	Best = 1.832795e+04	FES = 857100	Diversity = 2.9328e-01
Generation 8580:	Best = 1.823811e+04	FES = 858100	Diversity = 2.9197e-01
Generation 8590:	Best = 1.823811e+04	FES = 859100	Diversity = 2.8777e-01
Generation 8600:	Best = 1.822563e+04	FES = 860100	Diversity = 2.9362e-01
Generation 8610:	Best = 1.814373e+04	FES = 861100	Diversity = 2.8187e-01
Generation 8620:	Best = 1.811028e+04	FES = 862100	Diversity = 3.1505e-01
Generation 8630:	Best = 1.807197e+04	FES = 863100	Diversity = 3.1956e-01
Generation 8640:	Best = 1.803096e+04	FES = 864100	Diversity = 3.1986e-01
Generation 8650:	Best = 1.797705e+04	FES = 865100	Diversity = 3.0225e-01
Generation 8660:	Best = 1.795296e+04	FES = 866100	Diversity = 2.8775e-01
Generation 8670:	Best = 1.793160e+04	FES = 867100	Diversity = 3.0615e-01
Generation 8680:	Best = 1.793160e+04	FES = 868100	Diversity = 3.0153e-01
Generation 8690:	Best = 1.792347e+04	FES = 869100	Diversity = 2.7587e-01
Generation 8700:	Best = 1.791507e+04	FES = 870100	Diversity = 2.5368e-01
Generation 8710:	Best = 1.788539e+04	FES = 871100	Diversity = 2.4686e-01
Generation 8720:	Best = 1.786902e+04	FES = 872100	Diversity = 2.5051e-01
Generation 8730:	Best = 1.786438e+04	FES = 873100	Diversity = 2.4232e-01
Generation 8740:	Best = 1.784873e+04	FES = 874100	Diversity = 2.3674e-01
Generation 8750:	Best = 1.784873e+04	FES = 875100	Diversity = 2.2093e-01
Generation 8760:	Best = 1.780936e+04	FES = 876100	Diversity = 2.1305e-01
Generation 8770:	Best = 1.780936e+04	FES = 877100	Diversity = 2.2516e-01
Generation 8780:	Best = 1.780342e+04	FES = 878100	Diversity = 2.2349e-01
Generation 8790:	Best = 1.779792e+04	FES = 879100	Diversity = 2.0584e-01
Generation 8800:	Best = 1.779231e+04	FES = 880100	Diversity = 1.9598e-01
Generation 8810:	Best = 1.777542e+04	FES = 881100	Diversity = 1.9442e-01
Generation 8820:	Best = 1.777217e+04	FES = 882100	Diversity = 1.8545e-01
Generation 8830:	Best = 1.774216e+04	FES = 883100	Diversity = 2.0112e-01
Generation 8840:	Best = 1.772709e+04	FES = 884100	Diversity = 2.1626e-01
Generation 8850:	Best = 1.772405e+04	FES = 885100	Diversity = 2.3224e-01
Generation 8860:	Best = 1.771696e+04	FES = 886100	Diversity = 2.1984e-01
Generation 8870:	Best = 1.770609e+04	FES = 887100	Diversity = 2.0450e-01
Generation 8880:	Best = 1.770476e+04	FES = 888100	Diversity = 1.8112e-01
Generation 8890:	Best = 1.770026e+04	FES = 889100	Diversity = 1.6255e-01
Generation 8900:	Best = 1.768739e+04	FES = 890100	Diversity = 1.5661e-01
Generation 8910:	Best = 1.767538e+04	FES = 891100	Diversity = 1.5119e-01
Generation 8920:	Best = 1.767483e+04	FES = 892100	Diversity = 1.5912e-01
Generation 8930:	Best = 1.766083e+04	FES = 893100	Diversity = 1.5962e-01
Generation 8940:	Best = 1.766027e+04	FES = 894100	Diversity = 1.4868e-01
Generation 8950:	Best = 1.766008e+04	FES = 895100	Diversity = 1.3044e-01
Generation 8960:	Best = 1.765028e+04	FES = 896100	Diversity = 1.3521e-01
Generation 8970:	Best = 1.763752e+04	FES = 897100	Diversity = 1.3521e-01
Generation 8980:	Best = 1.763752e+04	FES = 898100	Diversity = 1.3722e-01
Generation 8990:	Best = 1.763752e+04	FES = 899100	Diversity = 1.2657e-01
Generation 9000:	Best = 1.763351e+04	FES = 900100	Diversity = 1.2054e-01
Generation 9010:	Best = 1.762885e+04	FES = 901100	Diversity = 1.1837e-01
Generation 9020:	Best = 1.762210e+04	FES = 902100	Diversity = 1.1760e-01
Generation 9030:	Best = 1.762140e+04	FES = 903100	Diversity = 1.1859e-01
Generation 9040:	Best = 1.761896e+04	FES = 904100	Diversity = 1.1976e-01
Generation 9050:	Best = 1.761190e+04	FES = 905100	Diversity = 1.2275e-01
Generation 9060:	Best = 1.761129e+04	FES = 906100	Diversity = 1.2641e-01
Generation 9070:	Best = 1.760959e+04	FES = 907100	Diversity = 1.1959e-01
Generation 9080:	Best = 1.760459e+04	FES = 908100	Diversity = 1.0697e-01

Generation 9090:	Best = 1.760015e+04	FES = 909100	Diversity = 1.0066e-01
Generation 9100:	Best = 1.759406e+04	FES = 910100	Diversity = 1.0463e-01
Generation 9110:	Best = 1.758356e+04	FES = 911100	Diversity = 1.0864e-01
Generation 9120:	Best = 1.758356e+04	FES = 912100	Diversity = 1.1228e-01
Generation 9130:	Best = 1.757852e+04	FES = 913100	Diversity = 1.0887e-01
Generation 9140:	Best = 1.757852e+04	FES = 914100	Diversity = 1.0604e-01
Generation 9150:	Best = 1.757425e+04	FES = 915100	Diversity = 1.0705e-01
Generation 9160:	Best = 1.756168e+04	FES = 916100	Diversity = 1.0818e-01
Generation 9170:	Best = 1.755845e+04	FES = 917100	Diversity = 1.0892e-01
Generation 9180:	Best = 1.755816e+04	FES = 918100	Diversity = 1.1220e-01
Generation 9190:	Best = 1.753817e+04	FES = 919100	Diversity = 1.1318e-01
Generation 9200:	Best = 1.753817e+04	FES = 920100	Diversity = 1.1853e-01
Generation 9210:	Best = 1.753817e+04	FES = 921100	Diversity = 1.2852e-01
Generation 9220:	Best = 1.753292e+04	FES = 922100	Diversity = 1.1950e-01
Generation 9230:	Best = 1.752739e+04	FES = 923100	Diversity = 1.0939e-01
Generation 9240:	Best = 1.752617e+04	FES = 924100	Diversity = 1.0612e-01
Generation 9250:	Best = 1.752227e+04	FES = 925100	Diversity = 1.0219e-01
Generation 9260:	Best = 1.750947e+04	FES = 926100	Diversity = 1.0063e-01
Generation 9270:	Best = 1.750835e+04	FES = 927100	Diversity = 1.0029e-01
Generation 9280:	Best = 1.750758e+04	FES = 928100	Diversity = 1.0596e-01
Generation 9290:	Best = 1.750195e+04	FES = 929100	Diversity = 1.0829e-01
Generation 9300:	Best = 1.749533e+04	FES = 930100	Diversity = 1.1178e-01
Generation 9310:	Best = 1.748766e+04	FES = 931100	Diversity = 1.1523e-01
Generation 9320:	Best = 1.748488e+04	FES = 932100	Diversity = 1.2465e-01
Generation 9330:	Best = 1.748488e+04	FES = 933100	Diversity = 1.2194e-01
Generation 9340:	Best = 1.747553e+04	FES = 934100	Diversity = 1.1013e-01
Generation 9350:	Best = 1.747270e+04	FES = 935100	Diversity = 1.1312e-01
Generation 9360:	Best = 1.746722e+04	FES = 936100	Diversity = 1.1423e-01
Generation 9370:	Best = 1.746722e+04	FES = 937100	Diversity = 1.0341e-01
Generation 9380:	Best = 1.746525e+04	FES = 938100	Diversity = 9.9622e-02
Generation 9390:	Best = 1.746070e+04	FES = 939100	Diversity = 9.4196e-02
Generation 9400:	Best = 1.745313e+04	FES = 940100	Diversity = 9.5067e-02
Generation 9410:	Best = 1.745313e+04	FES = 941100	Diversity = 9.1679e-02
Generation 9420:	Best = 1.744816e+04	FES = 942100	Diversity = 9.3482e-02
Generation 9430:	Best = 1.744664e+04	FES = 943100	Diversity = 8.4467e-02
Generation 9440:	Best = 1.744595e+04	FES = 944100	Diversity = 7.5772e-02
Generation 9450:	Best = 1.744366e+04	FES = 945100	Diversity = 7.1953e-02
Generation 9460:	Best = 1.743951e+04	FES = 946100	Diversity = 7.3055e-02
Generation 9470:	Best = 1.743736e+04	FES = 947100	Diversity = 6.8516e-02
Generation 9480:	Best = 1.743283e+04	FES = 948100	Diversity = 6.5844e-02
Generation 9490:	Best = 1.743221e+04	FES = 949100	Diversity = 5.9724e-02
Generation 9500:	Best = 1.742510e+04	FES = 950100	Diversity = 6.2633e-02
Generation 9510:	Best = 1.742510e+04	FES = 951100	Diversity = 6.0722e-02
Generation 9520:	Best = 1.742215e+04	FES = 952100	Diversity = 5.9123e-02
Generation 9530:	Best = 1.741805e+04	FES = 953100	Diversity = 6.5566e-02
Generation 9540:	Best = 1.741649e+04	FES = 954100	Diversity = 6.2729e-02
Generation 9550:	Best = 1.741179e+04	FES = 955100	Diversity = 6.7691e-02
Generation 9560:	Best = 1.740287e+04	FES = 956100	Diversity = 7.3446e-02
Generation 9570:	Best = 1.740206e+04	FES = 957100	Diversity = 7.8779e-02
Generation 9580:	Best = 1.738721e+04	FES = 958100	Diversity = 8.6508e-02
Generation 9590:	Best = 1.738374e+04	FES = 959100	Diversity = 8.9108e-02
Generation 9600:	Best = 1.738294e+04	FES = 960100	Diversity = 8.7232e-02
Generation 9610:	Best = 1.738261e+04	FES = 961100	Diversity = 7.9128e-02
Generation 9620:	Best = 1.737876e+04	FES = 962100	Diversity = 8.7660e-02

Generation 9630:	Best = 1.737780e+04	FES = 963100	Diversity = 8.0908e-02
Generation 9640:	Best = 1.736172e+04	FES = 964100	Diversity = 7.7415e-02
Generation 9650:	Best = 1.735684e+04	FES = 965100	Diversity = 8.2821e-02
Generation 9660:	Best = 1.735092e+04	FES = 966100	Diversity = 1.0196e-01
Generation 9670:	Best = 1.734448e+04	FES = 967100	Diversity = 1.0733e-01
Generation 9680:	Best = 1.733110e+04	FES = 968100	Diversity = 1.1887e-01
Generation 9690:	Best = 1.733102e+04	FES = 969100	Diversity = 1.1714e-01
Generation 9700:	Best = 1.733102e+04	FES = 970100	Diversity = 1.0892e-01
Generation 9710:	Best = 1.732487e+04	FES = 971100	Diversity = 1.0095e-01
Generation 9720:	Best = 1.731936e+04	FES = 972100	Diversity = 1.0134e-01
Generation 9730:	Best = 1.730928e+04	FES = 973100	Diversity = 1.0639e-01
Generation 9740:	Best = 1.730455e+04	FES = 974100	Diversity = 1.1087e-01
Generation 9750:	Best = 1.730384e+04	FES = 975100	Diversity = 1.0801e-01
Generation 9760:	Best = 1.729821e+04	FES = 976100	Diversity = 1.0266e-01
Generation 9770:	Best = 1.729606e+04	FES = 977100	Diversity = 9.8608e-02
Generation 9780:	Best = 1.729606e+04	FES = 978100	Diversity = 9.4253e-02
Generation 9790:	Best = 1.729273e+04	FES = 979100	Diversity = 8.8075e-02
Generation 9800:	Best = 1.728897e+04	FES = 980100	Diversity = 7.9767e-02
Generation 9810:	Best = 1.728083e+04	FES = 981100	Diversity = 8.1884e-02
Generation 9820:	Best = 1.727715e+04	FES = 982100	Diversity = 7.4871e-02
Generation 9830:	Best = 1.727715e+04	FES = 983100	Diversity = 7.1920e-02
Generation 9840:	Best = 1.727715e+04	FES = 984100	Diversity = 6.7235e-02
Generation 9850:	Best = 1.727665e+04	FES = 985100	Diversity = 6.3452e-02
Generation 9860:	Best = 1.727226e+04	FES = 986100	Diversity = 6.1137e-02
Generation 9870:	Best = 1.726688e+04	FES = 987100	Diversity = 6.1225e-02
Generation 9880:	Best = 1.726688e+04	FES = 988100	Diversity = 5.9723e-02
Generation 9890:	Best = 1.726688e+04	FES = 989100	Diversity = 5.9669e-02
Generation 9900:	Best = 1.726442e+04	FES = 990100	Diversity = 5.5672e-02
Generation 9910:	Best = 1.726262e+04	FES = 991100	Diversity = 5.5074e-02
Generation 9920:	Best = 1.726036e+04	FES = 992100	Diversity = 4.8075e-02
Generation 9930:	Best = 1.725837e+04	FES = 993100	Diversity = 4.9286e-02
Generation 9940:	Best = 1.725764e+04	FES = 994100	Diversity = 4.8195e-02
Generation 9950:	Best = 1.725397e+04	FES = 995100	Diversity = 4.5274e-02
Generation 9960:	Best = 1.725371e+04	FES = 996100	Diversity = 4.4273e-02
Generation 9970:	Best = 1.725163e+04	FES = 997100	Diversity = 4.7596e-02
Generation 9980:	Best = 1.724510e+04	FES = 998100	Diversity = 4.7653e-02
Generation 9990:	Best = 1.724380e+04	FES = 999100	Diversity = 4.7747e-02

Optimization completed!

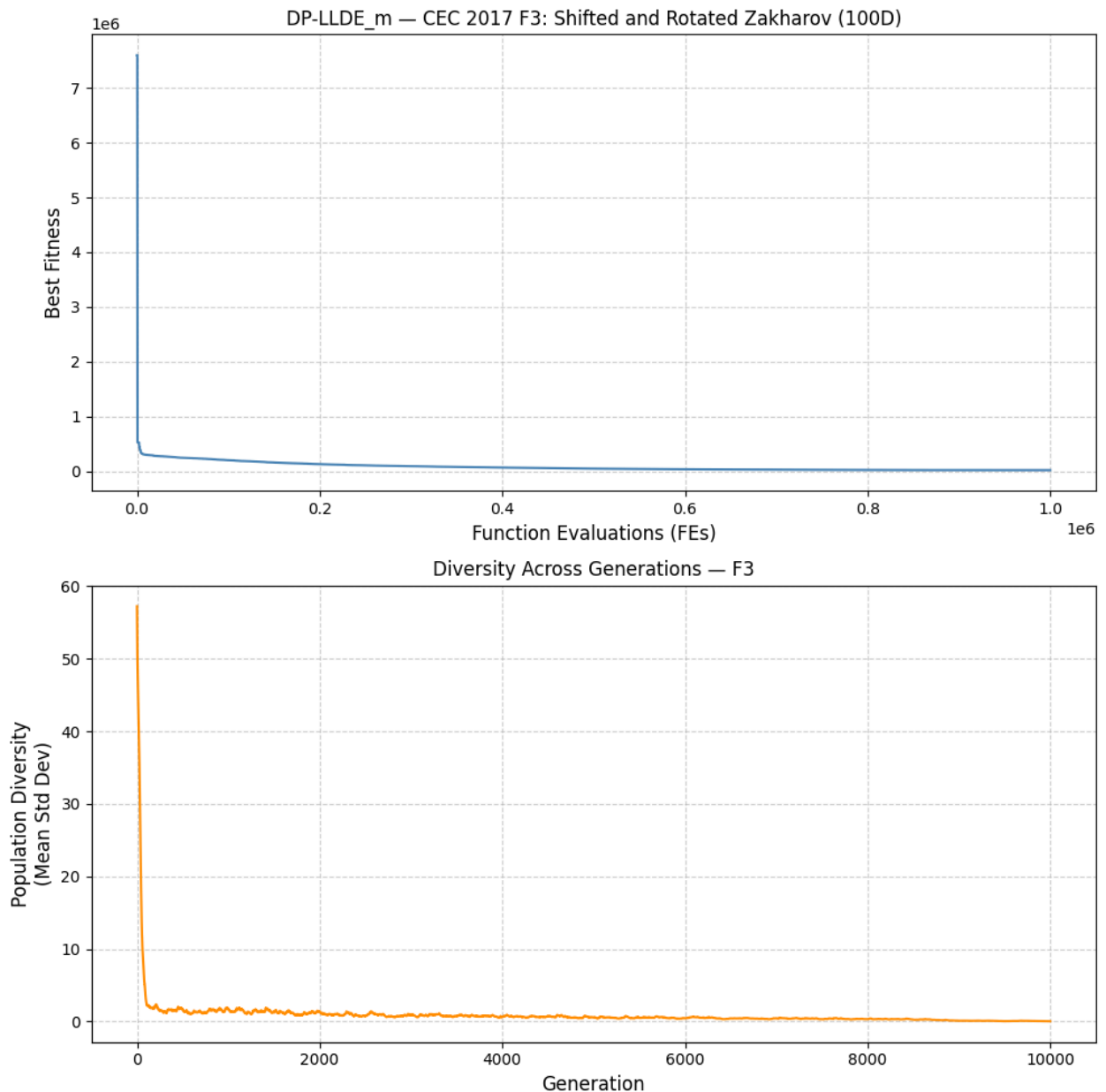
Final Best Fitness = 1.724335e+04

Total Function Evaluations = 1000000

```
=====
=
Results:
  Best Fitness           : 1.724335e+04
  Global Optimum         : 3.000000e+02
  Error (|best-optimum|) : 1.694335e+04
  Best Solution (5 dim)  : [-57.27233684   5.95281723  35.36386606   1.31133564
-31.74239653]
  D^(0) (init. diversity): 5.7220e+01
=====
```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results_2/all_results_che

ckpoint.pkl



Graph saved as convergence_F3.png

✓ Plot saved to: /content/drive/My Drive/thesis_results_2/plots/convergence_F3.png

6.2 Multimodal Functions (F4-F10)

```
In [ ]: # =====  
# F4 — Shifted and Rotated Rosenbrock  
# =====  
run_and_plot('F4', make_wrapper(simple.f4), 400.0, "Shifted and Rotated Rosenb
```

=====

=

DP-LLDE_m on CEC 2017 F4 – Single Run

=====

=

Function : Shifted and Rotated Rosenbrock's (100D)

Bounds : [-100, 100]

Global optimum : 400.0

Initial $D^{(0)} = 5.7220e+01$ | Diversity threshold = 5.7220e+00

Generation 0: Best Fitness = 2.335375e+05 | Diversity = 5.7220e+01

Generation 10: Best = 4.824759e+04 FES = 1100 Diversity = 2.3473e+01

Generation 20: Best = 3.201676e+04 FES = 2100 Diversity = 9.6503e+00

Generation 30: Best = 2.547574e+04 FES = 3100 Diversity = 4.6399e+00

Generation 40: Best = 2.165888e+04 FES = 4100 Diversity = 2.8884e+00

Generation 50: Best = 1.970775e+04 FES = 5100 Diversity = 1.9360e+00

Generation 60: Best = 1.833925e+04 FES = 6100 Diversity = 1.6712e+00

Generation 70: Best = 1.685983e+04 FES = 7100 Diversity = 1.3814e+00

Generation 80: Best = 1.546046e+04 FES = 8100 Diversity = 1.2639e+00

Generation 90: Best = 1.447298e+04 FES = 9100 Diversity = 1.2004e+00

Generation 100: Best = 1.369610e+04 FES = 10100 Diversity = 1.0165e+00

Generation 110: Best = 1.304784e+04 FES = 11100 Diversity = 9.8497e-01

Generation 120: Best = 1.237194e+04 FES = 12100 Diversity = 9.3319e-01

Generation 130: Best = 1.162826e+04 FES = 13100 Diversity = 9.1236e-01

Generation 140: Best = 1.103210e+04 FES = 14100 Diversity = 1.0220e+00

Generation 150: Best = 1.049174e+04 FES = 15100 Diversity = 8.3670e-01

Generation 160: Best = 1.006789e+04 FES = 16100 Diversity = 7.4870e-01

Generation 170: Best = 9.736619e+03 FES = 17100 Diversity = 6.6227e-01

Generation 180: Best = 9.462596e+03 FES = 18100 Diversity = 5.7227e-01

Generation 190: Best = 9.261643e+03 FES = 19100 Diversity = 4.9863e-01

Generation 200: Best = 9.054194e+03 FES = 20100 Diversity = 4.9206e-01

Generation 210: Best = 8.841696e+03 FES = 21100 Diversity = 5.0036e-01

Generation 220: Best = 8.631276e+03 FES = 22100 Diversity = 4.2573e-01

Generation 230: Best = 8.464415e+03 FES = 23100 Diversity = 4.6548e-01

Generation 240: Best = 8.298012e+03 FES = 24100 Diversity = 4.0765e-01

Generation 250: Best = 8.166573e+03 FES = 25100 Diversity = 3.6935e-01

Generation 260: Best = 8.069439e+03 FES = 26100 Diversity = 3.3201e-01

Generation 270: Best = 7.976823e+03 FES = 27100 Diversity = 3.3216e-01

Generation 280: Best = 7.893933e+03 FES = 28100 Diversity = 3.1833e-01

Generation 290: Best = 7.803671e+03 FES = 29100 Diversity = 3.4549e-01

Generation 300: Best = 7.687819e+03 FES = 30100 Diversity = 3.3929e-01

Generation 310: Best = 7.583165e+03 FES = 31100 Diversity = 3.2722e-01

Generation 320: Best = 7.474304e+03 FES = 32100 Diversity = 2.9651e-01

Generation 330: Best = 7.395636e+03 FES = 33100 Diversity = 2.7878e-01

Generation 340: Best = 7.323961e+03 FES = 34100 Diversity = 2.7388e-01

Generation 350: Best = 7.281809e+03 FES = 35100 Diversity = 1.8999e-01

Generation 360: Best = 7.255634e+03 FES = 36100 Diversity = 1.4054e-01

Generation 370: Best = 7.234036e+03 FES = 37100 Diversity = 1.2597e-01

Generation 380: Best = 7.220423e+03 FES = 38100 Diversity = 8.1536e-02

Generation 390: Best = 7.211915e+03 FES = 39100 Diversity = 5.8267e-02

Generation 400: Best = 7.202974e+03 FES = 40100 Diversity = 6.0176e-02

Generation 410: Best = 7.193354e+03 FES = 41100 Diversity = 5.0358e-02

Generation 420: Best = 7.188412e+03 FES = 42100 Diversity = 4.3060e-02

Generation 430: Best = 7.185170e+03 FES = 43100 Diversity = 3.4347e-02

Generation 440: Best = 7.182632e+03 FES = 44100 Diversity = 2.1959e-02

Generation	450:	Best = 7.180822e+03	FES = 45100	Diversity = 1.6746e-02
Generation	460:	Best = 7.179569e+03	FES = 46100	Diversity = 1.3676e-02
Generation	470:	Best = 7.178609e+03	FES = 47100	Diversity = 1.2170e-02
Generation	480:	Best = 7.177907e+03	FES = 48100	Diversity = 1.4559e-02
Generation	490:	Best = 7.177157e+03	FES = 49100	Diversity = 1.2358e-02
Generation	500:	Best = 7.176594e+03	FES = 50100	Diversity = 9.0441e-03
Generation	510:	Best = 7.176138e+03	FES = 51100	Diversity = 7.7897e-03
Generation	520:	Best = 7.175845e+03	FES = 52100	Diversity = 5.6980e-03
Generation	530:	Best = 7.175676e+03	FES = 53100	Diversity = 3.9816e-03
Generation	540:	Best = 7.175566e+03	FES = 54100	Diversity = 2.4864e-03
Generation	550:	Best = 7.175503e+03	FES = 55100	Diversity = 2.1847e-03
Generation	560:	Best = 7.175473e+03	FES = 56100	Diversity = 1.3908e-03
Generation	570:	Best = 7.175448e+03	FES = 57100	Diversity = 1.1799e-03
Generation	580:	Best = 7.175436e+03	FES = 58100	Diversity = 7.8320e-04
Generation	590:	Best = 7.175425e+03	FES = 59100	Diversity = 7.9782e-04
Generation	600:	Best = 7.175415e+03	FES = 60100	Diversity = 8.0304e-04
Generation	610:	Best = 7.175407e+03	FES = 61100	Diversity = 6.4701e-04
Generation	620:	Best = 7.175403e+03	FES = 62100	Diversity = 4.6066e-04
Generation	630:	Best = 7.175400e+03	FES = 63100	Diversity = 4.1051e-04
Generation	640:	Best = 7.175397e+03	FES = 64100	Diversity = 3.4836e-04
Generation	650:	Best = 7.175394e+03	FES = 65100	Diversity = 3.7860e-04
Generation	660:	Best = 7.175393e+03	FES = 66100	Diversity = 2.6982e-04
Generation	670:	Best = 7.175392e+03	FES = 67100	Diversity = 2.1309e-04
Generation	680:	Best = 7.175391e+03	FES = 68100	Diversity = 1.9354e-04
Generation	690:	Best = 7.175390e+03	FES = 69100	Diversity = 1.9367e-04
Generation	700:	Best = 7.175390e+03	FES = 70100	Diversity = 1.6403e-04
Generation	710:	Best = 7.175389e+03	FES = 71100	Diversity = 9.3174e-05
Generation	720:	Best = 7.175389e+03	FES = 72100	Diversity = 9.0617e-05
Generation	730:	Best = 7.175389e+03	FES = 73100	Diversity = 6.6404e-05
Generation	740:	Best = 7.175389e+03	FES = 74100	Diversity = 6.3687e-05
Generation	750:	Best = 7.175389e+03	FES = 75100	Diversity = 4.2237e-05
Generation	760:	Best = 7.175389e+03	FES = 76100	Diversity = 3.6322e-05
Generation	770:	Best = 7.175389e+03	FES = 77100	Diversity = 3.2918e-05
Generation	780:	Best = 7.175389e+03	FES = 78100	Diversity = 2.0099e-05
Generation	790:	Best = 7.175389e+03	FES = 79100	Diversity = 1.3869e-05
Generation	800:	Best = 7.175389e+03	FES = 80100	Diversity = 9.7373e-06
Generation	810:	Best = 7.175389e+03	FES = 81100	Diversity = 6.7745e-06
Generation	820:	Best = 7.175389e+03	FES = 82100	Diversity = 4.0685e-06
Generation	830:	Best = 7.175389e+03	FES = 83100	Diversity = 2.1893e-06
Generation	840:	Best = 7.175389e+03	FES = 84100	Diversity = 2.3287e-06
Generation	850:	Best = 7.175389e+03	FES = 85100	Diversity = 1.4635e-06
Generation	860:	Best = 7.175389e+03	FES = 86100	Diversity = 1.1860e-06
Generation	870:	Best = 7.175389e+03	FES = 87100	Diversity = 7.0552e-07
Generation	880:	Best = 7.175389e+03	FES = 88100	Diversity = 5.0121e-07
Generation	890:	Best = 7.175389e+03	FES = 89100	Diversity = 4.2745e-07
Generation	900:	Best = 7.175389e+03	FES = 90100	Diversity = 2.9144e-07
Generation	910:	Best = 7.175389e+03	FES = 91100	Diversity = 2.2038e-07
Generation	920:	Best = 7.175389e+03	FES = 92100	Diversity = 2.0689e-07
Generation	930:	Best = 7.175389e+03	FES = 93100	Diversity = 1.4209e-07
Generation	940:	Best = 7.175389e+03	FES = 94100	Diversity = 9.7176e-08
Generation	950:	Best = 7.175389e+03	FES = 95100	Diversity = 4.2851e-08
Generation	960:	Best = 7.175389e+03	FES = 96100	Diversity = 2.9105e-08
Generation	970:	Best = 7.175389e+03	FES = 97100	Diversity = 2.5484e-08
Generation	980:	Best = 7.175389e+03	FES = 98100	Diversity = 2.6358e-08


```

Generation 9630: Best = 7.175389e+03 FES = 963100 Diversity = 3.2287e-14
Generation 9640: Best = 7.175389e+03 FES = 964100 Diversity = 3.2287e-14
Generation 9650: Best = 7.175389e+03 FES = 965100 Diversity = 3.2287e-14
Generation 9660: Best = 7.175389e+03 FES = 966100 Diversity = 3.2287e-14
Generation 9670: Best = 7.175389e+03 FES = 967100 Diversity = 3.2287e-14
Generation 9680: Best = 7.175389e+03 FES = 968100 Diversity = 3.2287e-14
Generation 9690: Best = 7.175389e+03 FES = 969100 Diversity = 3.2287e-14
Generation 9700: Best = 7.175389e+03 FES = 970100 Diversity = 3.2287e-14
Generation 9710: Best = 7.175389e+03 FES = 971100 Diversity = 3.2287e-14
Generation 9720: Best = 7.175389e+03 FES = 972100 Diversity = 3.2287e-14
Generation 9730: Best = 7.175389e+03 FES = 973100 Diversity = 3.2287e-14
Generation 9740: Best = 7.175389e+03 FES = 974100 Diversity = 3.2287e-14
Generation 9750: Best = 7.175389e+03 FES = 975100 Diversity = 3.2287e-14
Generation 9760: Best = 7.175389e+03 FES = 976100 Diversity = 3.2287e-14
Generation 9770: Best = 7.175389e+03 FES = 977100 Diversity = 3.2287e-14
Generation 9780: Best = 7.175389e+03 FES = 978100 Diversity = 3.2287e-14
Generation 9790: Best = 7.175389e+03 FES = 979100 Diversity = 3.2287e-14
Generation 9800: Best = 7.175389e+03 FES = 980100 Diversity = 3.2287e-14
Generation 9810: Best = 7.175389e+03 FES = 981100 Diversity = 3.2287e-14
Generation 9820: Best = 7.175389e+03 FES = 982100 Diversity = 3.2287e-14
Generation 9830: Best = 7.175389e+03 FES = 983100 Diversity = 3.2287e-14
Generation 9840: Best = 7.175389e+03 FES = 984100 Diversity = 3.2287e-14
Generation 9850: Best = 7.175389e+03 FES = 985100 Diversity = 3.2287e-14
Generation 9860: Best = 7.175389e+03 FES = 986100 Diversity = 3.2287e-14
Generation 9870: Best = 7.175389e+03 FES = 987100 Diversity = 3.2287e-14
Generation 9880: Best = 7.175389e+03 FES = 988100 Diversity = 3.2287e-14
Generation 9890: Best = 7.175389e+03 FES = 989100 Diversity = 3.2287e-14
Generation 9900: Best = 7.175389e+03 FES = 990100 Diversity = 3.2287e-14
Generation 9910: Best = 7.175389e+03 FES = 991100 Diversity = 3.2287e-14
Generation 9920: Best = 7.175389e+03 FES = 992100 Diversity = 3.2287e-14
Generation 9930: Best = 7.175389e+03 FES = 993100 Diversity = 3.2287e-14
Generation 9940: Best = 7.175389e+03 FES = 994100 Diversity = 3.2287e-14
Generation 9950: Best = 7.175389e+03 FES = 995100 Diversity = 3.2287e-14
Generation 9960: Best = 7.175389e+03 FES = 996100 Diversity = 3.2287e-14
Generation 9970: Best = 7.175389e+03 FES = 997100 Diversity = 3.2287e-14
Generation 9980: Best = 7.175389e+03 FES = 998100 Diversity = 3.2287e-14
Generation 9990: Best = 7.175389e+03 FES = 999100 Diversity = 3.2287e-14

```

Optimization completed!
 Final Best Fitness = 7.175389e+03
 Total Function Evaluations = 1000000

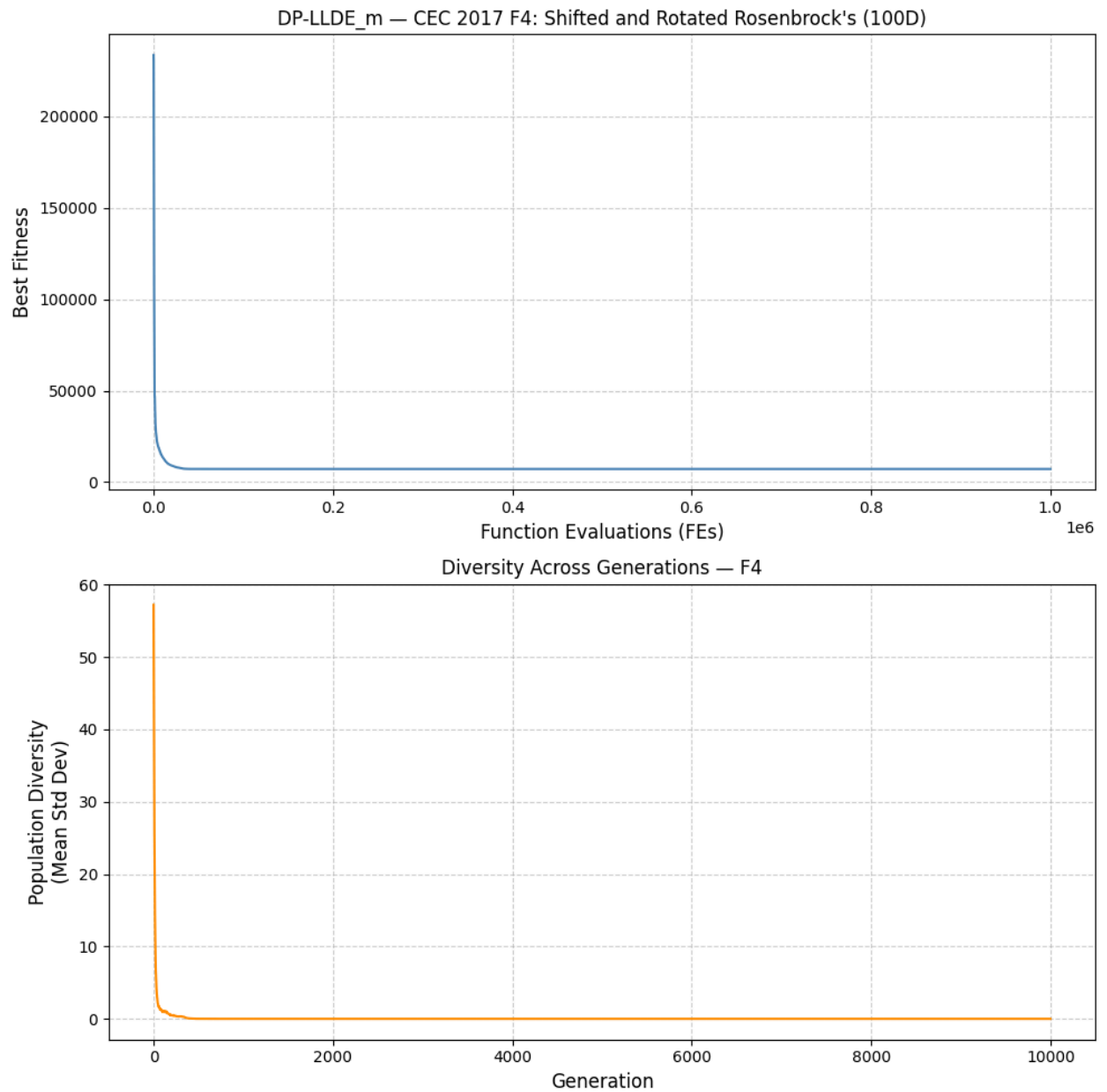
```

=====
=
Results:
  Best Fitness           : 7.175389e+03
  Global Optimum         : 4.000000e+02
  Error (|best-optimum|) : 6.775389e+03
  Best Solution (5 dim)  : [ 44.0737308  39.06982203 -48.92459878  38.89580742
1.99944662]
  D^(0) (init. diversity): 5.7220e+01
=====
=

```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results_2/all_results_che

ckpoint.pkl



Graph saved as convergence_F4.png

✓ Plot saved to: /content/drive/My Drive/thesis_results_2/plots/convergence_F4.png

```
In [ ]: # =====  
# F5 — Shifted and Rotated Rastrigin  
# =====  
run_and_plot('F5', make_wrapper(simple.f5), 500.0, "Shifted and Rotated Rastrigin")
```

```

=====
=
DP-LLDE_m on CEC 2017 F5 – Single Run
=====
=
Function      : Shifted and Rotated Rastrigin's (100D)
Bounds       : [-100, 100]
Global optimum : 500.0
Initial D^(0) = 5.7220e+01 | Diversity threshold = 5.7220e+00
Generation 0: Best Fitness = 3.063799e+03 | Diversity = 5.7220e+01
Generation 10: Best = 2.127790e+03 FES = 1100 Diversity = 2.3494e+01
Generation 20: Best = 1.944859e+03 FES = 2100 Diversity = 1.6879e+01
Generation 30: Best = 1.826706e+03 FES = 3100 Diversity = 1.3861e+01
Generation 40: Best = 1.783905e+03 FES = 4100 Diversity = 1.1979e+01
Generation 50: Best = 1.745340e+03 FES = 5100 Diversity = 1.0848e+01
Generation 60: Best = 1.724716e+03 FES = 6100 Diversity = 1.0036e+01
Generation 70: Best = 1.717908e+03 FES = 7100 Diversity = 9.8171e+00
Generation 80: Best = 1.676624e+03 FES = 8100 Diversity = 9.2579e+00
Generation 90: Best = 1.643466e+03 FES = 9100 Diversity = 8.8864e+00
Generation 100: Best = 1.643466e+03 FES = 10100 Diversity = 8.7502e+00
Generation 110: Best = 1.616055e+03 FES = 11100 Diversity = 8.5668e+00
Generation 120: Best = 1.602188e+03 FES = 12100 Diversity = 8.4018e+00
Generation 130: Best = 1.577333e+03 FES = 13100 Diversity = 8.2934e+00
Generation 140: Best = 1.577333e+03 FES = 14100 Diversity = 8.0543e+00
Generation 150: Best = 1.539228e+03 FES = 15100 Diversity = 8.0505e+00
Generation 160: Best = 1.537115e+03 FES = 16100 Diversity = 7.8190e+00
Generation 170: Best = 1.532024e+03 FES = 17100 Diversity = 7.7501e+00
Generation 180: Best = 1.516435e+03 FES = 18100 Diversity = 7.7348e+00
Generation 190: Best = 1.516435e+03 FES = 19100 Diversity = 7.6469e+00
Generation 200: Best = 1.516435e+03 FES = 20100 Diversity = 7.5729e+00
Generation 210: Best = 1.516435e+03 FES = 21100 Diversity = 7.4184e+00
Generation 220: Best = 1.492404e+03 FES = 22100 Diversity = 7.3970e+00
Generation 230: Best = 1.492404e+03 FES = 23100 Diversity = 7.3832e+00
Generation 240: Best = 1.467747e+03 FES = 24100 Diversity = 7.3775e+00
Generation 250: Best = 1.466362e+03 FES = 25100 Diversity = 7.2776e+00
Generation 260: Best = 1.466362e+03 FES = 26100 Diversity = 7.2247e+00
Generation 270: Best = 1.466362e+03 FES = 27100 Diversity = 7.1577e+00
Generation 280: Best = 1.460773e+03 FES = 28100 Diversity = 7.1474e+00
Generation 290: Best = 1.454543e+03 FES = 29100 Diversity = 7.1394e+00
Generation 300: Best = 1.432595e+03 FES = 30100 Diversity = 7.0798e+00
Generation 310: Best = 1.432595e+03 FES = 31100 Diversity = 7.0603e+00
Generation 320: Best = 1.432595e+03 FES = 32100 Diversity = 7.0164e+00
Generation 330: Best = 1.420983e+03 FES = 33100 Diversity = 7.0210e+00
Generation 340: Best = 1.420983e+03 FES = 34100 Diversity = 6.9898e+00
Generation 350: Best = 1.420983e+03 FES = 35100 Diversity = 6.9581e+00
Generation 360: Best = 1.416405e+03 FES = 36100 Diversity = 6.9379e+00
Generation 370: Best = 1.416405e+03 FES = 37100 Diversity = 6.8632e+00
Generation 380: Best = 1.409531e+03 FES = 38100 Diversity = 6.8718e+00
Generation 390: Best = 1.384025e+03 FES = 39100 Diversity = 6.8430e+00
Generation 400: Best = 1.382343e+03 FES = 40100 Diversity = 6.7854e+00
Generation 410: Best = 1.382343e+03 FES = 41100 Diversity = 6.7793e+00
Generation 420: Best = 1.373728e+03 FES = 42100 Diversity = 6.7702e+00
Generation 430: Best = 1.373396e+03 FES = 43100 Diversity = 6.7664e+00
Generation 440: Best = 1.373396e+03 FES = 44100 Diversity = 6.6331e+00

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Generation	450:	Best = 1.371947e+03	FES = 45100	Diversity = 6.6027e+00
Generation	460:	Best = 1.371947e+03	FES = 46100	Diversity = 6.6155e+00
Generation	470:	Best = 1.371947e+03	FES = 47100	Diversity = 6.5579e+00
Generation	480:	Best = 1.371781e+03	FES = 48100	Diversity = 6.5708e+00
Generation	490:	Best = 1.371781e+03	FES = 49100	Diversity = 6.5548e+00
Generation	500:	Best = 1.371781e+03	FES = 50100	Diversity = 6.5453e+00
Generation	510:	Best = 1.370170e+03	FES = 51100	Diversity = 6.5307e+00
Generation	520:	Best = 1.365523e+03	FES = 52100	Diversity = 6.5551e+00
Generation	530:	Best = 1.365523e+03	FES = 53100	Diversity = 6.5180e+00
Generation	540:	Best = 1.365523e+03	FES = 54100	Diversity = 6.4886e+00
Generation	550:	Best = 1.365523e+03	FES = 55100	Diversity = 6.4902e+00
Generation	560:	Best = 1.365523e+03	FES = 56100	Diversity = 6.4480e+00
Generation	570:	Best = 1.365523e+03	FES = 57100	Diversity = 6.4522e+00
Generation	580:	Best = 1.365523e+03	FES = 58100	Diversity = 6.4415e+00
Generation	590:	Best = 1.362946e+03	FES = 59100	Diversity = 6.4245e+00
Generation	600:	Best = 1.362946e+03	FES = 60100	Diversity = 6.4037e+00
Generation	610:	Best = 1.353592e+03	FES = 61100	Diversity = 6.3888e+00
Generation	620:	Best = 1.353592e+03	FES = 62100	Diversity = 6.3804e+00
Generation	630:	Best = 1.353592e+03	FES = 63100	Diversity = 6.3598e+00
Generation	640:	Best = 1.353592e+03	FES = 64100	Diversity = 6.3469e+00
Generation	650:	Best = 1.350378e+03	FES = 65100	Diversity = 6.3053e+00
Generation	660:	Best = 1.350378e+03	FES = 66100	Diversity = 6.3031e+00
Generation	670:	Best = 1.350378e+03	FES = 67100	Diversity = 6.2805e+00
Generation	680:	Best = 1.348670e+03	FES = 68100	Diversity = 6.2724e+00
Generation	690:	Best = 1.345013e+03	FES = 69100	Diversity = 6.2711e+00
Generation	700:	Best = 1.329009e+03	FES = 70100	Diversity = 6.2606e+00
Generation	710:	Best = 1.326907e+03	FES = 71100	Diversity = 6.2558e+00
Generation	720:	Best = 1.320632e+03	FES = 72100	Diversity = 6.2493e+00
Generation	730:	Best = 1.310861e+03	FES = 73100	Diversity = 6.2060e+00
Generation	740:	Best = 1.304085e+03	FES = 74100	Diversity = 6.1455e+00
Generation	750:	Best = 1.303874e+03	FES = 75100	Diversity = 6.1290e+00
Generation	760:	Best = 1.303874e+03	FES = 76100	Diversity = 6.1068e+00
Generation	770:	Best = 1.300271e+03	FES = 77100	Diversity = 6.0986e+00
Generation	780:	Best = 1.274194e+03	FES = 78100	Diversity = 6.0959e+00
Generation	790:	Best = 1.269634e+03	FES = 79100	Diversity = 6.1013e+00
Generation	800:	Best = 1.269400e+03	FES = 80100	Diversity = 6.0600e+00
Generation	810:	Best = 1.267792e+03	FES = 81100	Diversity = 6.0484e+00
Generation	820:	Best = 1.267792e+03	FES = 82100	Diversity = 6.0457e+00
Generation	830:	Best = 1.262129e+03	FES = 83100	Diversity = 5.9915e+00
Generation	840:	Best = 1.251293e+03	FES = 84100	Diversity = 5.9062e+00
Generation	850:	Best = 1.251182e+03	FES = 85100	Diversity = 5.8883e+00
Generation	860:	Best = 1.249060e+03	FES = 86100	Diversity = 5.7739e+00
Generation	870:	Best = 1.247127e+03	FES = 87100	Diversity = 5.7644e+00
Generation	880:	Best = 1.230199e+03	FES = 88100	Diversity = 5.7624e+00
Generation	890:	Best = 1.228099e+03	FES = 89100	Diversity = 5.7526e+00
Generation	900:	Best = 1.223412e+03	FES = 90100	Diversity = 5.7459e+00
Generation	910:	Best = 1.222829e+03	FES = 91100	Diversity = 5.6719e+00
Generation	920:	Best = 1.215966e+03	FES = 92100	Diversity = 5.6553e+00
Generation	930:	Best = 1.215743e+03	FES = 93100	Diversity = 5.6177e+00
Generation	940:	Best = 1.211004e+03	FES = 94100	Diversity = 5.5977e+00
Generation	950:	Best = 1.208731e+03	FES = 95100	Diversity = 5.4918e+00
Generation	960:	Best = 1.208731e+03	FES = 96100	Diversity = 5.5056e+00
Generation	970:	Best = 1.207643e+03	FES = 97100	Diversity = 5.4900e+00
Generation	980:	Best = 1.207643e+03	FES = 98100	Diversity = 5.3907e+00

Generation 990:	Best = 1.207255e+03	FES = 99100	Diversity = 5.3797e+00
Generation 1000:	Best = 1.200204e+03	FES = 100100	Diversity = 5.3960e+00
Generation 1010:	Best = 1.187907e+03	FES = 101100	Diversity = 5.3534e+00
Generation 1020:	Best = 1.186558e+03	FES = 102100	Diversity = 5.2767e+00
Generation 1030:	Best = 1.181960e+03	FES = 103100	Diversity = 5.1349e+00
Generation 1040:	Best = 1.180434e+03	FES = 104100	Diversity = 4.8471e+00
Generation 1050:	Best = 1.176695e+03	FES = 105100	Diversity = 4.6491e+00
Generation 1060:	Best = 1.174288e+03	FES = 106100	Diversity = 3.7736e+00
Generation 1070:	Best = 1.153108e+03	FES = 107100	Diversity = 1.5890e+00
Generation 1080:	Best = 1.143275e+03	FES = 108100	Diversity = 4.9717e-01
Generation 1090:	Best = 1.110906e+03	FES = 109100	Diversity = 3.9818e-01
Generation 1100:	Best = 1.090637e+03	FES = 110100	Diversity = 3.5186e-01
Generation 1110:	Best = 1.078362e+03	FES = 111100	Diversity = 2.6473e-01
Generation 1120:	Best = 1.067551e+03	FES = 112100	Diversity = 2.3085e-01
Generation 1130:	Best = 1.062399e+03	FES = 113100	Diversity = 1.7307e-01
Generation 1140:	Best = 1.057521e+03	FES = 114100	Diversity = 1.4041e-01
Generation 1150:	Best = 1.053463e+03	FES = 115100	Diversity = 1.0721e-01
Generation 1160:	Best = 1.050931e+03	FES = 116100	Diversity = 8.6218e-02
Generation 1170:	Best = 1.048593e+03	FES = 117100	Diversity = 6.8888e-02
Generation 1180:	Best = 1.046277e+03	FES = 118100	Diversity = 6.1202e-02
Generation 1190:	Best = 1.043921e+03	FES = 119100	Diversity = 5.3939e-02
Generation 1200:	Best = 1.042265e+03	FES = 120100	Diversity = 4.7389e-02
Generation 1210:	Best = 1.041093e+03	FES = 121100	Diversity = 3.9606e-02
Generation 1220:	Best = 1.039610e+03	FES = 122100	Diversity = 3.8960e-02
Generation 1230:	Best = 1.038289e+03	FES = 123100	Diversity = 3.4818e-02
Generation 1240:	Best = 1.037423e+03	FES = 124100	Diversity = 2.9919e-02
Generation 1250:	Best = 1.036459e+03	FES = 125100	Diversity = 2.7365e-02
Generation 1260:	Best = 1.035923e+03	FES = 126100	Diversity = 2.1971e-02
Generation 1270:	Best = 1.034984e+03	FES = 127100	Diversity = 2.3973e-02
Generation 1280:	Best = 1.034029e+03	FES = 128100	Diversity = 2.4366e-02
Generation 1290:	Best = 1.033048e+03	FES = 129100	Diversity = 2.3512e-02
Generation 1300:	Best = 1.032169e+03	FES = 130100	Diversity = 2.4868e-02
Generation 1310:	Best = 1.031592e+03	FES = 131100	Diversity = 1.8035e-02
Generation 1320:	Best = 1.031225e+03	FES = 132100	Diversity = 1.3445e-02
Generation 1330:	Best = 1.030900e+03	FES = 133100	Diversity = 1.1597e-02
Generation 1340:	Best = 1.030557e+03	FES = 134100	Diversity = 1.0934e-02
Generation 1350:	Best = 1.030259e+03	FES = 135100	Diversity = 9.4819e-03
Generation 1360:	Best = 1.029988e+03	FES = 136100	Diversity = 9.7858e-03
Generation 1370:	Best = 1.029704e+03	FES = 137100	Diversity = 8.7304e-03
Generation 1380:	Best = 1.029525e+03	FES = 138100	Diversity = 7.2688e-03
Generation 1390:	Best = 1.029374e+03	FES = 139100	Diversity = 6.9546e-03
Generation 1400:	Best = 1.029223e+03	FES = 140100	Diversity = 6.4017e-03
Generation 1410:	Best = 1.029095e+03	FES = 141100	Diversity = 5.8085e-03
Generation 1420:	Best = 1.028974e+03	FES = 142100	Diversity = 5.7844e-03
Generation 1430:	Best = 1.028865e+03	FES = 143100	Diversity = 5.5833e-03
Generation 1440:	Best = 1.028764e+03	FES = 144100	Diversity = 5.3037e-03
Generation 1450:	Best = 1.028669e+03	FES = 145100	Diversity = 4.9004e-03
Generation 1460:	Best = 1.028564e+03	FES = 146100	Diversity = 5.0098e-03
Generation 1470:	Best = 1.028473e+03	FES = 147100	Diversity = 4.7498e-03
Generation 1480:	Best = 1.028397e+03	FES = 148100	Diversity = 4.4347e-03
Generation 1490:	Best = 1.028331e+03	FES = 149100	Diversity = 3.9245e-03
Generation 1500:	Best = 1.028303e+03	FES = 150100	Diversity = 2.6100e-03
Generation 1510:	Best = 1.028264e+03	FES = 151100	Diversity = 2.3835e-03
Generation 1520:	Best = 1.028239e+03	FES = 152100	Diversity = 1.8078e-03

Generation 1530:	Best = 1.028212e+03	FES = 153100	Diversity = 1.6583e-03
Generation 1540:	Best = 1.028195e+03	FES = 154100	Diversity = 1.5476e-03
Generation 1550:	Best = 1.028166e+03	FES = 155100	Diversity = 1.8978e-03
Generation 1560:	Best = 1.028143e+03	FES = 156100	Diversity = 1.5006e-03
Generation 1570:	Best = 1.028121e+03	FES = 157100	Diversity = 1.4016e-03
Generation 1580:	Best = 1.028100e+03	FES = 158100	Diversity = 1.3403e-03
Generation 1590:	Best = 1.028083e+03	FES = 159100	Diversity = 1.2934e-03
Generation 1600:	Best = 1.028063e+03	FES = 160100	Diversity = 1.4366e-03
Generation 1610:	Best = 1.028050e+03	FES = 161100	Diversity = 1.2762e-03
Generation 1620:	Best = 1.028036e+03	FES = 162100	Diversity = 1.5080e-03
Generation 1630:	Best = 1.028027e+03	FES = 163100	Diversity = 9.7080e-04
Generation 1640:	Best = 1.028022e+03	FES = 164100	Diversity = 7.4502e-04
Generation 1650:	Best = 1.028017e+03	FES = 165100	Diversity = 6.7841e-04
Generation 1660:	Best = 1.028012e+03	FES = 166100	Diversity = 5.8203e-04
Generation 1670:	Best = 1.028007e+03	FES = 167100	Diversity = 6.9737e-04
Generation 1680:	Best = 1.028002e+03	FES = 168100	Diversity = 5.8975e-04
Generation 1690:	Best = 1.027998e+03	FES = 169100	Diversity = 4.7353e-04
Generation 1700:	Best = 1.027995e+03	FES = 170100	Diversity = 3.8096e-04
Generation 1710:	Best = 1.027993e+03	FES = 171100	Diversity = 3.1756e-04
Generation 1720:	Best = 1.027991e+03	FES = 172100	Diversity = 3.0264e-04
Generation 1730:	Best = 1.027989e+03	FES = 173100	Diversity = 2.3669e-04
Generation 1740:	Best = 1.027988e+03	FES = 174100	Diversity = 2.4170e-04
Generation 1750:	Best = 1.027987e+03	FES = 175100	Diversity = 1.6878e-04
Generation 1760:	Best = 1.027986e+03	FES = 176100	Diversity = 1.4923e-04
Generation 1770:	Best = 1.027986e+03	FES = 177100	Diversity = 1.1571e-04
Generation 1780:	Best = 1.027985e+03	FES = 178100	Diversity = 9.5336e-05
Generation 1790:	Best = 1.027985e+03	FES = 179100	Diversity = 8.3734e-05
Generation 1800:	Best = 1.027985e+03	FES = 180100	Diversity = 7.5338e-05
Generation 1810:	Best = 1.027985e+03	FES = 181100	Diversity = 5.8639e-05
Generation 1820:	Best = 1.027985e+03	FES = 182100	Diversity = 4.5387e-05
Generation 1830:	Best = 1.027985e+03	FES = 183100	Diversity = 4.4026e-05
Generation 1840:	Best = 1.027984e+03	FES = 184100	Diversity = 4.1552e-05
Generation 1850:	Best = 1.027984e+03	FES = 185100	Diversity = 3.1923e-05
Generation 1860:	Best = 1.027984e+03	FES = 186100	Diversity = 2.8284e-05
Generation 1870:	Best = 1.027984e+03	FES = 187100	Diversity = 2.2370e-05
Generation 1880:	Best = 1.027984e+03	FES = 188100	Diversity = 1.5399e-05
Generation 1890:	Best = 1.027984e+03	FES = 189100	Diversity = 1.1334e-05
Generation 1900:	Best = 1.027984e+03	FES = 190100	Diversity = 1.2114e-05
Generation 1910:	Best = 1.027984e+03	FES = 191100	Diversity = 1.0919e-05
Generation 1920:	Best = 1.027984e+03	FES = 192100	Diversity = 8.4206e-06
Generation 1930:	Best = 1.027984e+03	FES = 193100	Diversity = 7.6796e-06
Generation 1940:	Best = 1.027984e+03	FES = 194100	Diversity = 6.6486e-06
Generation 1950:	Best = 1.027984e+03	FES = 195100	Diversity = 6.9271e-06
Generation 1960:	Best = 1.027984e+03	FES = 196100	Diversity = 5.2955e-06
Generation 1970:	Best = 1.027984e+03	FES = 197100	Diversity = 3.0473e-06
Generation 1980:	Best = 1.027984e+03	FES = 198100	Diversity = 3.0476e-06
Generation 1990:	Best = 1.027984e+03	FES = 199100	Diversity = 2.5059e-06
Generation 2000:	Best = 1.027984e+03	FES = 200100	Diversity = 1.9990e-06
Generation 2010:	Best = 1.027984e+03	FES = 201100	Diversity = 1.4237e-06
Generation 2020:	Best = 1.027984e+03	FES = 202100	Diversity = 1.2358e-06
Generation 2030:	Best = 1.027984e+03	FES = 203100	Diversity = 8.8886e-07
Generation 2040:	Best = 1.027984e+03	FES = 204100	Diversity = 5.7929e-07
Generation 2050:	Best = 1.027984e+03	FES = 205100	Diversity = 6.1931e-07
Generation 2060:	Best = 1.027984e+03	FES = 206100	Diversity = 5.8377e-07

Generation 2070:	Best = 1.027984e+03	FES = 207100	Diversity = 3.8469e-07
Generation 2080:	Best = 1.027984e+03	FES = 208100	Diversity = 2.8034e-07
Generation 2090:	Best = 1.027984e+03	FES = 209100	Diversity = 2.8554e-07
Generation 2100:	Best = 1.027984e+03	FES = 210100	Diversity = 2.2062e-07
Generation 2110:	Best = 1.027984e+03	FES = 211100	Diversity = 1.7005e-07
Generation 2120:	Best = 1.027984e+03	FES = 212100	Diversity = 8.4446e-08
Generation 2130:	Best = 1.027984e+03	FES = 213100	Diversity = 8.9836e-08
Generation 2140:	Best = 1.027984e+03	FES = 214100	Diversity = 5.9708e-08
Generation 2150:	Best = 1.027984e+03	FES = 215100	Diversity = 3.4324e-08
Generation 2160:	Best = 1.027984e+03	FES = 216100	Diversity = 4.7299e-08
Generation 2170:	Best = 1.027984e+03	FES = 217100	Diversity = 2.2867e-08
Generation 2180:	Best = 1.027984e+03	FES = 218100	Diversity = 1.6824e-08
Generation 2190:	Best = 1.027984e+03	FES = 219100	Diversity = 1.4985e-08
Generation 2200:	Best = 1.027984e+03	FES = 220100	Diversity = 1.3065e-08
Generation 2210:	Best = 1.027984e+03	FES = 221100	Diversity = 9.3901e-09
Generation 2220:	Best = 1.027984e+03	FES = 222100	Diversity = 5.4839e-09
Generation 2230:	Best = 1.027984e+03	FES = 223100	Diversity = 5.7890e-09
Generation 2240:	Best = 1.027984e+03	FES = 224100	Diversity = 4.4708e-09
Generation 2250:	Best = 1.027984e+03	FES = 225100	Diversity = 2.8013e-09
Generation 2260:	Best = 1.027984e+03	FES = 226100	Diversity = 1.8693e-09
Generation 2270:	Best = 1.027984e+03	FES = 227100	Diversity = 1.8368e-09
Generation 2280:	Best = 1.027984e+03	FES = 228100	Diversity = 1.0365e-09
Generation 2290:	Best = 1.027984e+03	FES = 229100	Diversity = 7.3912e-10
Generation 2300:	Best = 1.027984e+03	FES = 230100	Diversity = 3.5615e-10
Generation 2310:	Best = 1.027984e+03	FES = 231100	Diversity = 8.0856e-14
Generation 2320:	Best = 1.027984e+03	FES = 232100	Diversity = 2.2015e-14
Generation 2330:	Best = 1.027984e+03	FES = 233100	Diversity = 2.2095e-14
Generation 2340:	Best = 1.027984e+03	FES = 234100	Diversity = 2.2103e-14
Generation 2350:	Best = 1.027984e+03	FES = 235100	Diversity = 2.2103e-14
Generation 2360:	Best = 1.027984e+03	FES = 236100	Diversity = 2.2103e-14
Generation 2370:	Best = 1.027984e+03	FES = 237100	Diversity = 2.2103e-14
Generation 2380:	Best = 1.027984e+03	FES = 238100	Diversity = 2.2103e-14
Generation 2390:	Best = 1.027984e+03	FES = 239100	Diversity = 2.2103e-14
Generation 2400:	Best = 1.027984e+03	FES = 240100	Diversity = 2.2103e-14
Generation 2410:	Best = 1.027984e+03	FES = 241100	Diversity = 2.2103e-14
Generation 2420:	Best = 1.027984e+03	FES = 242100	Diversity = 2.2103e-14
Generation 2430:	Best = 1.027984e+03	FES = 243100	Diversity = 2.2103e-14
Generation 2440:	Best = 1.027984e+03	FES = 244100	Diversity = 2.2103e-14
Generation 2450:	Best = 1.027984e+03	FES = 245100	Diversity = 2.2103e-14
Generation 2460:	Best = 1.027984e+03	FES = 246100	Diversity = 2.2103e-14
Generation 2470:	Best = 1.027984e+03	FES = 247100	Diversity = 2.2103e-14
Generation 2480:	Best = 1.027984e+03	FES = 248100	Diversity = 2.2103e-14
Generation 2490:	Best = 1.027984e+03	FES = 249100	Diversity = 2.2103e-14
Generation 2500:	Best = 1.027984e+03	FES = 250100	Diversity = 2.2103e-14
Generation 2510:	Best = 1.027984e+03	FES = 251100	Diversity = 2.2103e-14
Generation 2520:	Best = 1.027984e+03	FES = 252100	Diversity = 2.2103e-14
Generation 2530:	Best = 1.027984e+03	FES = 253100	Diversity = 2.2103e-14
Generation 2540:	Best = 1.027984e+03	FES = 254100	Diversity = 2.2103e-14
Generation 2550:	Best = 1.027984e+03	FES = 255100	Diversity = 2.2103e-14
Generation 2560:	Best = 1.027984e+03	FES = 256100	Diversity = 2.2103e-14
Generation 2570:	Best = 1.027984e+03	FES = 257100	Diversity = 2.2103e-14
Generation 2580:	Best = 1.027984e+03	FES = 258100	Diversity = 2.2103e-14
Generation 2590:	Best = 1.027984e+03	FES = 259100	Diversity = 2.2103e-14
Generation 2600:	Best = 1.027984e+03	FES = 260100	Diversity = 2.2103e-14

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Generation 9630: Best = 1.027984e+03 FES = 963100 Diversity = 2.2103e-14
Generation 9640: Best = 1.027984e+03 FES = 964100 Diversity = 2.2103e-14
Generation 9650: Best = 1.027984e+03 FES = 965100 Diversity = 2.2103e-14
Generation 9660: Best = 1.027984e+03 FES = 966100 Diversity = 2.2103e-14
Generation 9670: Best = 1.027984e+03 FES = 967100 Diversity = 2.2103e-14
Generation 9680: Best = 1.027984e+03 FES = 968100 Diversity = 2.2103e-14
Generation 9690: Best = 1.027984e+03 FES = 969100 Diversity = 2.2103e-14
Generation 9700: Best = 1.027984e+03 FES = 970100 Diversity = 2.2103e-14
Generation 9710: Best = 1.027984e+03 FES = 971100 Diversity = 2.2103e-14
Generation 9720: Best = 1.027984e+03 FES = 972100 Diversity = 2.2103e-14
Generation 9730: Best = 1.027984e+03 FES = 973100 Diversity = 2.2103e-14
Generation 9740: Best = 1.027984e+03 FES = 974100 Diversity = 2.2103e-14
Generation 9750: Best = 1.027984e+03 FES = 975100 Diversity = 2.2103e-14
Generation 9760: Best = 1.027984e+03 FES = 976100 Diversity = 2.2103e-14
Generation 9770: Best = 1.027984e+03 FES = 977100 Diversity = 2.2103e-14
Generation 9780: Best = 1.027984e+03 FES = 978100 Diversity = 2.2103e-14
Generation 9790: Best = 1.027984e+03 FES = 979100 Diversity = 2.2103e-14
Generation 9800: Best = 1.027984e+03 FES = 980100 Diversity = 2.2103e-14
Generation 9810: Best = 1.027984e+03 FES = 981100 Diversity = 2.2103e-14
Generation 9820: Best = 1.027984e+03 FES = 982100 Diversity = 2.2103e-14
Generation 9830: Best = 1.027984e+03 FES = 983100 Diversity = 2.2103e-14
Generation 9840: Best = 1.027984e+03 FES = 984100 Diversity = 2.2103e-14
Generation 9850: Best = 1.027984e+03 FES = 985100 Diversity = 2.2103e-14
Generation 9860: Best = 1.027984e+03 FES = 986100 Diversity = 2.2103e-14
Generation 9870: Best = 1.027984e+03 FES = 987100 Diversity = 2.2103e-14
Generation 9880: Best = 1.027984e+03 FES = 988100 Diversity = 2.2103e-14
Generation 9890: Best = 1.027984e+03 FES = 989100 Diversity = 2.2103e-14
Generation 9900: Best = 1.027984e+03 FES = 990100 Diversity = 2.2103e-14
Generation 9910: Best = 1.027984e+03 FES = 991100 Diversity = 2.2103e-14
Generation 9920: Best = 1.027984e+03 FES = 992100 Diversity = 2.2103e-14
Generation 9930: Best = 1.027984e+03 FES = 993100 Diversity = 2.2103e-14
Generation 9940: Best = 1.027984e+03 FES = 994100 Diversity = 2.2103e-14
Generation 9950: Best = 1.027984e+03 FES = 995100 Diversity = 2.2103e-14
Generation 9960: Best = 1.027984e+03 FES = 996100 Diversity = 2.2103e-14
Generation 9970: Best = 1.027984e+03 FES = 997100 Diversity = 2.2103e-14
Generation 9980: Best = 1.027984e+03 FES = 998100 Diversity = 2.2103e-14
Generation 9990: Best = 1.027984e+03 FES = 999100 Diversity = 2.2103e-14

```

```

Optimization completed!
Final Best Fitness = 1.027984e+03
Total Function Evaluations = 1000000

```

```

=====
=
Results:
  Best Fitness          : 1.027984e+03
  Global Optimum        : 5.000000e+02
  Error (|best-optimum|) : 5.279842e+02
  Best Solution (5 dim) : [ -0.13079281  19.97358757   7.04217585  12.35092616
-26.43931773]
  D^(0) (init. diversity): 5.7220e+01
=====
=

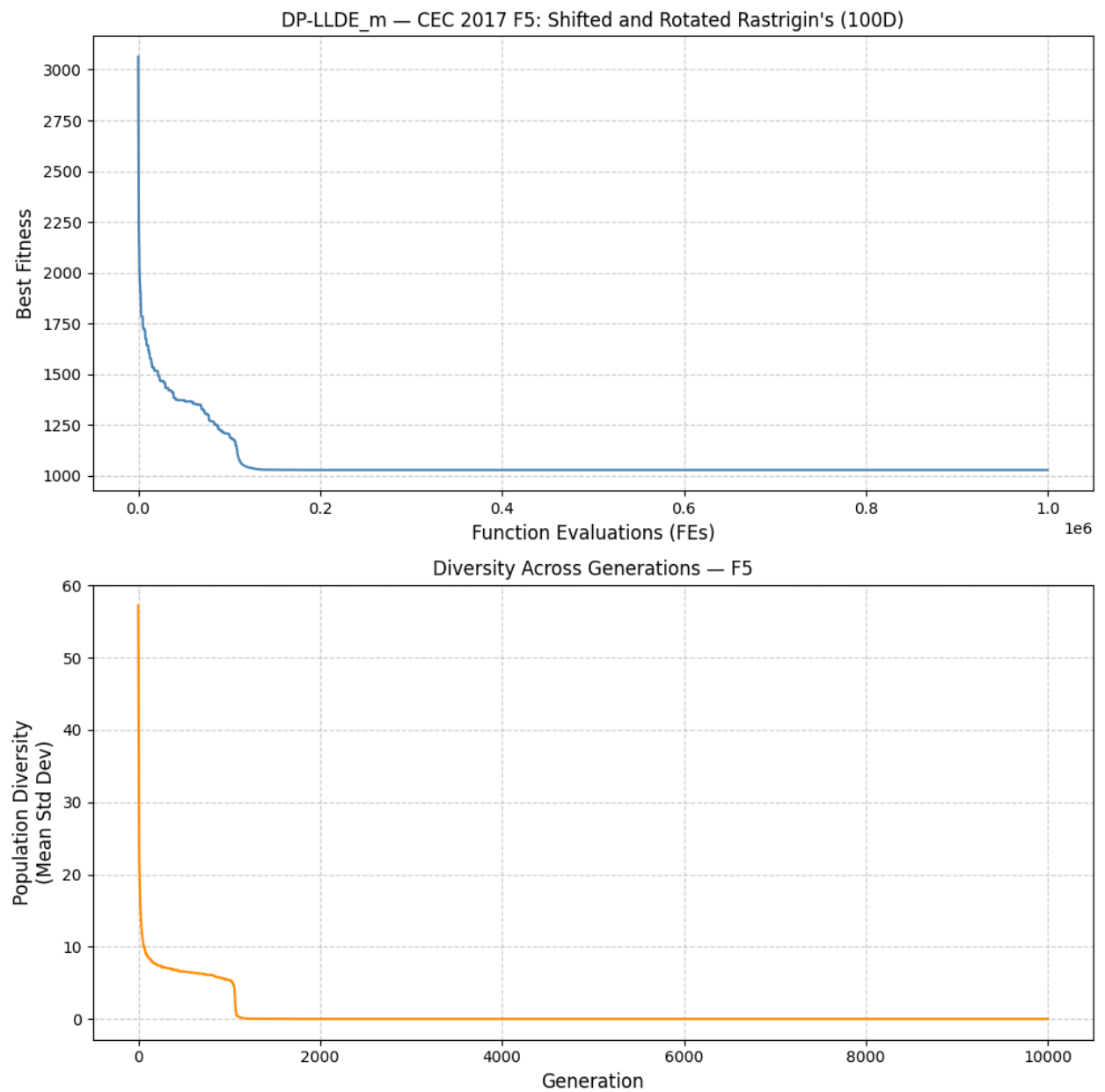
```

```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results_2/all_results_che

```

ckpoint.pkl



Graph saved as convergence_F5.png

✓ Plot saved to: /content/drive/My Drive/thesis_results_2/plots/convergence_F5.png

```
In [ ]: # =====  
# F6 — Shifted and Rotated Schaffer's F7  
# =====  
run_and_plot('F6', make_wrapper(simple.f6), 600.0, "Shifted and Rotated Schaff
```

```

=====
=
DP-LLDE_m on CEC 2017 F6 – Single Run
=====
=
Function      : Shifted and Rotated Schaffer's F7 (100D)
Bounds       : [-100, 100]
Global optimum : 600.0
Initial D^(0) = 5.7220e+01 | Diversity threshold = 5.7220e+00
Generation 0: Best Fitness = 7.936410e+02 | Diversity = 5.7220e+01
Generation 10: Best = 7.230730e+02 FES = 1100 Diversity = 2.9172e+01
Generation 20: Best = 6.981165e+02 FES = 2100 Diversity = 1.7603e+01
Generation 30: Best = 6.930351e+02 FES = 3100 Diversity = 1.3445e+01
Generation 40: Best = 6.880962e+02 FES = 4100 Diversity = 1.1238e+01
Generation 50: Best = 6.800144e+02 FES = 5100 Diversity = 1.0100e+01
Generation 60: Best = 6.785985e+02 FES = 6100 Diversity = 8.7942e+00
Generation 70: Best = 6.671024e+02 FES = 7100 Diversity = 7.9824e+00
Generation 80: Best = 6.671024e+02 FES = 8100 Diversity = 7.2824e+00
Generation 90: Best = 6.671024e+02 FES = 9100 Diversity = 6.5381e+00
Generation 100: Best = 6.640510e+02 FES = 10100 Diversity = 6.0435e+00
Generation 110: Best = 6.624097e+02 FES = 11100 Diversity = 5.5903e+00
Generation 120: Best = 6.618265e+02 FES = 12100 Diversity = 5.4693e+00
Generation 130: Best = 6.604523e+02 FES = 13100 Diversity = 5.2116e+00
Generation 140: Best = 6.587447e+02 FES = 14100 Diversity = 5.0408e+00
Generation 150: Best = 6.587447e+02 FES = 15100 Diversity = 4.7626e+00
Generation 160: Best = 6.565122e+02 FES = 16100 Diversity = 4.6068e+00
Generation 170: Best = 6.559519e+02 FES = 17100 Diversity = 4.5035e+00
Generation 180: Best = 6.548354e+02 FES = 18100 Diversity = 4.4492e+00
Generation 190: Best = 6.532247e+02 FES = 19100 Diversity = 4.3605e+00
Generation 200: Best = 6.531692e+02 FES = 20100 Diversity = 4.3270e+00
Generation 210: Best = 6.531692e+02 FES = 21100 Diversity = 4.3189e+00
Generation 220: Best = 6.531692e+02 FES = 22100 Diversity = 4.2566e+00
Generation 230: Best = 6.523558e+02 FES = 23100 Diversity = 4.2266e+00
Generation 240: Best = 6.518891e+02 FES = 24100 Diversity = 4.1991e+00
Generation 250: Best = 6.516894e+02 FES = 25100 Diversity = 4.1141e+00
Generation 260: Best = 6.512991e+02 FES = 26100 Diversity = 4.1055e+00
Generation 270: Best = 6.505466e+02 FES = 27100 Diversity = 4.0482e+00
Generation 280: Best = 6.501348e+02 FES = 28100 Diversity = 3.9712e+00
Generation 290: Best = 6.501348e+02 FES = 29100 Diversity = 3.9570e+00
Generation 300: Best = 6.487522e+02 FES = 30100 Diversity = 3.9649e+00
Generation 310: Best = 6.487522e+02 FES = 31100 Diversity = 3.9658e+00
Generation 320: Best = 6.487522e+02 FES = 32100 Diversity = 3.9222e+00
Generation 330: Best = 6.487522e+02 FES = 33100 Diversity = 3.8641e+00
Generation 340: Best = 6.480602e+02 FES = 34100 Diversity = 3.8347e+00
Generation 350: Best = 6.480602e+02 FES = 35100 Diversity = 3.7660e+00
Generation 360: Best = 6.480602e+02 FES = 36100 Diversity = 3.7550e+00
Generation 370: Best = 6.480602e+02 FES = 37100 Diversity = 3.7261e+00
Generation 380: Best = 6.479770e+02 FES = 38100 Diversity = 3.7201e+00
Generation 390: Best = 6.477434e+02 FES = 39100 Diversity = 3.7308e+00
Generation 400: Best = 6.468166e+02 FES = 40100 Diversity = 3.7189e+00
Generation 410: Best = 6.465004e+02 FES = 41100 Diversity = 3.7011e+00
Generation 420: Best = 6.461814e+02 FES = 42100 Diversity = 3.7024e+00
Generation 430: Best = 6.461814e+02 FES = 43100 Diversity = 3.6987e+00
Generation 440: Best = 6.461814e+02 FES = 44100 Diversity = 3.6954e+00

```


Generation	450:	Best = 6.461814e+02	FES = 45100	Diversity = 3.7386e+00
Generation	460:	Best = 6.461814e+02	FES = 46100	Diversity = 3.7380e+00
Generation	470:	Best = 6.461814e+02	FES = 47100	Diversity = 3.7249e+00
Generation	480:	Best = 6.458067e+02	FES = 48100	Diversity = 3.7114e+00
Generation	490:	Best = 6.458067e+02	FES = 49100	Diversity = 3.7314e+00
Generation	500:	Best = 6.457209e+02	FES = 50100	Diversity = 3.7204e+00
Generation	510:	Best = 6.455113e+02	FES = 51100	Diversity = 3.7051e+00
Generation	520:	Best = 6.455113e+02	FES = 52100	Diversity = 3.7027e+00
Generation	530:	Best = 6.455113e+02	FES = 53100	Diversity = 3.6684e+00
Generation	540:	Best = 6.453638e+02	FES = 54100	Diversity = 3.6957e+00
Generation	550:	Best = 6.453638e+02	FES = 55100	Diversity = 3.7042e+00
Generation	560:	Best = 6.453638e+02	FES = 56100	Diversity = 3.7035e+00
Generation	570:	Best = 6.453638e+02	FES = 57100	Diversity = 3.7017e+00
Generation	580:	Best = 6.453638e+02	FES = 58100	Diversity = 3.6814e+00
Generation	590:	Best = 6.453638e+02	FES = 59100	Diversity = 3.6753e+00
Generation	600:	Best = 6.453638e+02	FES = 60100	Diversity = 3.6734e+00
Generation	610:	Best = 6.445101e+02	FES = 61100	Diversity = 3.6716e+00
Generation	620:	Best = 6.445101e+02	FES = 62100	Diversity = 3.6656e+00
Generation	630:	Best = 6.445101e+02	FES = 63100	Diversity = 3.6700e+00
Generation	640:	Best = 6.445101e+02	FES = 64100	Diversity = 3.6717e+00
Generation	650:	Best = 6.442167e+02	FES = 65100	Diversity = 3.6692e+00
Generation	660:	Best = 6.442167e+02	FES = 66100	Diversity = 3.6685e+00
Generation	670:	Best = 6.442124e+02	FES = 67100	Diversity = 3.6756e+00
Generation	680:	Best = 6.442124e+02	FES = 68100	Diversity = 3.6715e+00
Generation	690:	Best = 6.442124e+02	FES = 69100	Diversity = 3.6639e+00
Generation	700:	Best = 6.442124e+02	FES = 70100	Diversity = 3.6622e+00
Generation	710:	Best = 6.442124e+02	FES = 71100	Diversity = 3.6552e+00
Generation	720:	Best = 6.442124e+02	FES = 72100	Diversity = 3.6525e+00
Generation	730:	Best = 6.442124e+02	FES = 73100	Diversity = 3.6496e+00
Generation	740:	Best = 6.442124e+02	FES = 74100	Diversity = 3.6470e+00
Generation	750:	Best = 6.442124e+02	FES = 75100	Diversity = 3.6223e+00
Generation	760:	Best = 6.442124e+02	FES = 76100	Diversity = 3.6179e+00
Generation	770:	Best = 6.442124e+02	FES = 77100	Diversity = 3.6057e+00
Generation	780:	Best = 6.442124e+02	FES = 78100	Diversity = 3.6071e+00
Generation	790:	Best = 6.442124e+02	FES = 79100	Diversity = 3.6149e+00
Generation	800:	Best = 6.442124e+02	FES = 80100	Diversity = 3.5987e+00
Generation	810:	Best = 6.442124e+02	FES = 81100	Diversity = 3.5752e+00
Generation	820:	Best = 6.442114e+02	FES = 82100	Diversity = 3.5651e+00
Generation	830:	Best = 6.442114e+02	FES = 83100	Diversity = 3.5365e+00
Generation	840:	Best = 6.442114e+02	FES = 84100	Diversity = 3.5315e+00
Generation	850:	Best = 6.442114e+02	FES = 85100	Diversity = 3.5298e+00
Generation	860:	Best = 6.442114e+02	FES = 86100	Diversity = 3.4876e+00
Generation	870:	Best = 6.440728e+02	FES = 87100	Diversity = 3.4705e+00
Generation	880:	Best = 6.440728e+02	FES = 88100	Diversity = 3.4691e+00
Generation	890:	Best = 6.440728e+02	FES = 89100	Diversity = 3.4685e+00
Generation	900:	Best = 6.440728e+02	FES = 90100	Diversity = 3.4548e+00
Generation	910:	Best = 6.440728e+02	FES = 91100	Diversity = 3.4500e+00
Generation	920:	Best = 6.440728e+02	FES = 92100	Diversity = 3.4500e+00
Generation	930:	Best = 6.440728e+02	FES = 93100	Diversity = 3.4342e+00
Generation	940:	Best = 6.440728e+02	FES = 94100	Diversity = 3.4328e+00
Generation	950:	Best = 6.440728e+02	FES = 95100	Diversity = 3.4041e+00
Generation	960:	Best = 6.440728e+02	FES = 96100	Diversity = 3.4007e+00
Generation	970:	Best = 6.440728e+02	FES = 97100	Diversity = 3.3975e+00
Generation	980:	Best = 6.440728e+02	FES = 98100	Diversity = 3.3964e+00

Generation 990:	Best = 6.440728e+02	FES = 99100	Diversity = 3.3964e+00
Generation 1000:	Best = 6.440728e+02	FES = 100100	Diversity = 3.3946e+00
Generation 1010:	Best = 6.440062e+02	FES = 101100	Diversity = 3.3947e+00
Generation 1020:	Best = 6.438939e+02	FES = 102100	Diversity = 3.3924e+00
Generation 1030:	Best = 6.438939e+02	FES = 103100	Diversity = 3.3895e+00
Generation 1040:	Best = 6.438939e+02	FES = 104100	Diversity = 3.3819e+00
Generation 1050:	Best = 6.438939e+02	FES = 105100	Diversity = 3.3975e+00
Generation 1060:	Best = 6.438939e+02	FES = 106100	Diversity = 3.3888e+00
Generation 1070:	Best = 6.438939e+02	FES = 107100	Diversity = 3.3767e+00
Generation 1080:	Best = 6.438939e+02	FES = 108100	Diversity = 3.3854e+00
Generation 1090:	Best = 6.438939e+02	FES = 109100	Diversity = 3.3935e+00
Generation 1100:	Best = 6.438939e+02	FES = 110100	Diversity = 3.3908e+00
Generation 1110:	Best = 6.438939e+02	FES = 111100	Diversity = 3.3776e+00
Generation 1120:	Best = 6.438939e+02	FES = 112100	Diversity = 3.3741e+00
Generation 1130:	Best = 6.438939e+02	FES = 113100	Diversity = 3.3736e+00
Generation 1140:	Best = 6.437543e+02	FES = 114100	Diversity = 3.3885e+00
Generation 1150:	Best = 6.437543e+02	FES = 115100	Diversity = 3.4034e+00
Generation 1160:	Best = 6.437543e+02	FES = 116100	Diversity = 3.4004e+00
Generation 1170:	Best = 6.437543e+02	FES = 117100	Diversity = 3.3865e+00
Generation 1180:	Best = 6.437543e+02	FES = 118100	Diversity = 3.3860e+00
Generation 1190:	Best = 6.437543e+02	FES = 119100	Diversity = 3.3856e+00
Generation 1200:	Best = 6.437543e+02	FES = 120100	Diversity = 3.3769e+00
Generation 1210:	Best = 6.437543e+02	FES = 121100	Diversity = 3.3746e+00
Generation 1220:	Best = 6.431311e+02	FES = 122100	Diversity = 3.3726e+00
Generation 1230:	Best = 6.431311e+02	FES = 123100	Diversity = 3.3714e+00
Generation 1240:	Best = 6.427312e+02	FES = 124100	Diversity = 3.3725e+00
Generation 1250:	Best = 6.427312e+02	FES = 125100	Diversity = 3.3721e+00
Generation 1260:	Best = 6.427312e+02	FES = 126100	Diversity = 3.3714e+00
Generation 1270:	Best = 6.427312e+02	FES = 127100	Diversity = 3.3710e+00
Generation 1280:	Best = 6.427312e+02	FES = 128100	Diversity = 3.3703e+00
Generation 1290:	Best = 6.427312e+02	FES = 129100	Diversity = 3.3543e+00
Generation 1300:	Best = 6.427312e+02	FES = 130100	Diversity = 3.3530e+00
Generation 1310:	Best = 6.427312e+02	FES = 131100	Diversity = 3.3529e+00
Generation 1320:	Best = 6.427312e+02	FES = 132100	Diversity = 3.3526e+00
Generation 1330:	Best = 6.427312e+02	FES = 133100	Diversity = 3.3523e+00
Generation 1340:	Best = 6.427312e+02	FES = 134100	Diversity = 3.3519e+00
Generation 1350:	Best = 6.427312e+02	FES = 135100	Diversity = 3.3504e+00
Generation 1360:	Best = 6.427312e+02	FES = 136100	Diversity = 3.3502e+00
Generation 1370:	Best = 6.427312e+02	FES = 137100	Diversity = 3.3488e+00
Generation 1380:	Best = 6.425727e+02	FES = 138100	Diversity = 3.3381e+00
Generation 1390:	Best = 6.425727e+02	FES = 139100	Diversity = 3.3354e+00
Generation 1400:	Best = 6.423636e+02	FES = 140100	Diversity = 3.3262e+00
Generation 1410:	Best = 6.423636e+02	FES = 141100	Diversity = 3.3262e+00
Generation 1420:	Best = 6.423636e+02	FES = 142100	Diversity = 3.3190e+00
Generation 1430:	Best = 6.423636e+02	FES = 143100	Diversity = 3.3180e+00
Generation 1440:	Best = 6.423636e+02	FES = 144100	Diversity = 3.3180e+00
Generation 1450:	Best = 6.423636e+02	FES = 145100	Diversity = 3.3185e+00
Generation 1460:	Best = 6.423636e+02	FES = 146100	Diversity = 3.3335e+00
Generation 1470:	Best = 6.423636e+02	FES = 147100	Diversity = 3.3303e+00
Generation 1480:	Best = 6.423636e+02	FES = 148100	Diversity = 3.3304e+00
Generation 1490:	Best = 6.423636e+02	FES = 149100	Diversity = 3.3295e+00
Generation 1500:	Best = 6.423636e+02	FES = 150100	Diversity = 3.3268e+00
Generation 1510:	Best = 6.423636e+02	FES = 151100	Diversity = 3.3259e+00
Generation 1520:	Best = 6.423636e+02	FES = 152100	Diversity = 3.3255e+00

Generation 1530:	Best = 6.420538e+02	FES = 153100	Diversity = 3.3240e+00
Generation 1540:	Best = 6.420538e+02	FES = 154100	Diversity = 3.3243e+00
Generation 1550:	Best = 6.417540e+02	FES = 155100	Diversity = 3.3233e+00
Generation 1560:	Best = 6.417540e+02	FES = 156100	Diversity = 3.3240e+00
Generation 1570:	Best = 6.417540e+02	FES = 157100	Diversity = 3.3236e+00
Generation 1580:	Best = 6.417540e+02	FES = 158100	Diversity = 3.3240e+00
Generation 1590:	Best = 6.417540e+02	FES = 159100	Diversity = 3.3231e+00
Generation 1600:	Best = 6.417540e+02	FES = 160100	Diversity = 3.3225e+00
Generation 1610:	Best = 6.417540e+02	FES = 161100	Diversity = 3.3215e+00
Generation 1620:	Best = 6.417540e+02	FES = 162100	Diversity = 3.3203e+00
Generation 1630:	Best = 6.417540e+02	FES = 163100	Diversity = 3.3186e+00
Generation 1640:	Best = 6.417540e+02	FES = 164100	Diversity = 3.3210e+00
Generation 1650:	Best = 6.417540e+02	FES = 165100	Diversity = 3.3176e+00
Generation 1660:	Best = 6.417540e+02	FES = 166100	Diversity = 3.3149e+00
Generation 1670:	Best = 6.417540e+02	FES = 167100	Diversity = 3.3111e+00
Generation 1680:	Best = 6.417540e+02	FES = 168100	Diversity = 3.3110e+00
Generation 1690:	Best = 6.417540e+02	FES = 169100	Diversity = 3.3110e+00
Generation 1700:	Best = 6.417540e+02	FES = 170100	Diversity = 3.3105e+00
Generation 1710:	Best = 6.417540e+02	FES = 171100	Diversity = 3.3098e+00
Generation 1720:	Best = 6.417540e+02	FES = 172100	Diversity = 3.3093e+00
Generation 1730:	Best = 6.417540e+02	FES = 173100	Diversity = 3.3087e+00
Generation 1740:	Best = 6.417465e+02	FES = 174100	Diversity = 3.3078e+00
Generation 1750:	Best = 6.417465e+02	FES = 175100	Diversity = 3.3055e+00
Generation 1760:	Best = 6.417465e+02	FES = 176100	Diversity = 3.3074e+00
Generation 1770:	Best = 6.417465e+02	FES = 177100	Diversity = 3.3076e+00
Generation 1780:	Best = 6.416782e+02	FES = 178100	Diversity = 3.3045e+00
Generation 1790:	Best = 6.416782e+02	FES = 179100	Diversity = 3.3109e+00
Generation 1800:	Best = 6.416415e+02	FES = 180100	Diversity = 3.3108e+00
Generation 1810:	Best = 6.416415e+02	FES = 181100	Diversity = 3.3108e+00
Generation 1820:	Best = 6.416415e+02	FES = 182100	Diversity = 3.3107e+00
Generation 1830:	Best = 6.413307e+02	FES = 183100	Diversity = 3.3101e+00
Generation 1840:	Best = 6.413307e+02	FES = 184100	Diversity = 3.3094e+00
Generation 1850:	Best = 6.413307e+02	FES = 185100	Diversity = 3.3095e+00
Generation 1860:	Best = 6.413307e+02	FES = 186100	Diversity = 3.3048e+00
Generation 1870:	Best = 6.413188e+02	FES = 187100	Diversity = 3.3037e+00
Generation 1880:	Best = 6.413188e+02	FES = 188100	Diversity = 3.3036e+00
Generation 1890:	Best = 6.413188e+02	FES = 189100	Diversity = 3.3034e+00
Generation 1900:	Best = 6.413188e+02	FES = 190100	Diversity = 3.3029e+00
Generation 1910:	Best = 6.413188e+02	FES = 191100	Diversity = 3.3011e+00
Generation 1920:	Best = 6.413188e+02	FES = 192100	Diversity = 3.2999e+00
Generation 1930:	Best = 6.413188e+02	FES = 193100	Diversity = 3.2997e+00
Generation 1940:	Best = 6.405767e+02	FES = 194100	Diversity = 3.2988e+00
Generation 1950:	Best = 6.405621e+02	FES = 195100	Diversity = 3.2988e+00
Generation 1960:	Best = 6.405621e+02	FES = 196100	Diversity = 3.3097e+00
Generation 1970:	Best = 6.405621e+02	FES = 197100	Diversity = 3.3093e+00
Generation 1980:	Best = 6.405621e+02	FES = 198100	Diversity = 3.3097e+00
Generation 1990:	Best = 6.405621e+02	FES = 199100	Diversity = 3.3084e+00
Generation 2000:	Best = 6.405621e+02	FES = 200100	Diversity = 3.3082e+00
Generation 2010:	Best = 6.405621e+02	FES = 201100	Diversity = 3.3081e+00
Generation 2020:	Best = 6.405621e+02	FES = 202100	Diversity = 3.3079e+00
Generation 2030:	Best = 6.405621e+02	FES = 203100	Diversity = 3.3071e+00
Generation 2040:	Best = 6.405621e+02	FES = 204100	Diversity = 3.3059e+00
Generation 2050:	Best = 6.405621e+02	FES = 205100	Diversity = 3.3051e+00
Generation 2060:	Best = 6.405621e+02	FES = 206100	Diversity = 3.3025e+00

Generation 2070:	Best = 6.405621e+02	FES = 207100	Diversity = 3.3020e+00
Generation 2080:	Best = 6.405621e+02	FES = 208100	Diversity = 3.3023e+00
Generation 2090:	Best = 6.405621e+02	FES = 209100	Diversity = 3.3019e+00
Generation 2100:	Best = 6.405621e+02	FES = 210100	Diversity = 3.3006e+00
Generation 2110:	Best = 6.405621e+02	FES = 211100	Diversity = 3.3004e+00
Generation 2120:	Best = 6.405621e+02	FES = 212100	Diversity = 3.2973e+00
Generation 2130:	Best = 6.405621e+02	FES = 213100	Diversity = 3.2962e+00
Generation 2140:	Best = 6.404902e+02	FES = 214100	Diversity = 3.2958e+00
Generation 2150:	Best = 6.404902e+02	FES = 215100	Diversity = 3.2952e+00
Generation 2160:	Best = 6.404902e+02	FES = 216100	Diversity = 3.2949e+00
Generation 2170:	Best = 6.404902e+02	FES = 217100	Diversity = 3.2943e+00
Generation 2180:	Best = 6.404902e+02	FES = 218100	Diversity = 3.2907e+00
Generation 2190:	Best = 6.404902e+02	FES = 219100	Diversity = 3.3003e+00
Generation 2200:	Best = 6.404902e+02	FES = 220100	Diversity = 3.2747e+00
Generation 2210:	Best = 6.404902e+02	FES = 221100	Diversity = 3.2748e+00
Generation 2220:	Best = 6.404902e+02	FES = 222100	Diversity = 3.2745e+00
Generation 2230:	Best = 6.404845e+02	FES = 223100	Diversity = 3.2735e+00
Generation 2240:	Best = 6.404845e+02	FES = 224100	Diversity = 3.2723e+00
Generation 2250:	Best = 6.403007e+02	FES = 225100	Diversity = 3.2733e+00
Generation 2260:	Best = 6.403007e+02	FES = 226100	Diversity = 3.2757e+00
Generation 2270:	Best = 6.403007e+02	FES = 227100	Diversity = 3.2750e+00
Generation 2280:	Best = 6.403007e+02	FES = 228100	Diversity = 3.2742e+00
Generation 2290:	Best = 6.403007e+02	FES = 229100	Diversity = 3.2710e+00
Generation 2300:	Best = 6.403007e+02	FES = 230100	Diversity = 3.2695e+00
Generation 2310:	Best = 6.403007e+02	FES = 231100	Diversity = 3.2694e+00
Generation 2320:	Best = 6.403007e+02	FES = 232100	Diversity = 3.2695e+00
Generation 2330:	Best = 6.403007e+02	FES = 233100	Diversity = 3.2690e+00
Generation 2340:	Best = 6.400707e+02	FES = 234100	Diversity = 3.2689e+00
Generation 2350:	Best = 6.400707e+02	FES = 235100	Diversity = 3.2681e+00
Generation 2360:	Best = 6.400707e+02	FES = 236100	Diversity = 3.2680e+00
Generation 2370:	Best = 6.400707e+02	FES = 237100	Diversity = 3.2673e+00
Generation 2380:	Best = 6.400707e+02	FES = 238100	Diversity = 3.2671e+00
Generation 2390:	Best = 6.400707e+02	FES = 239100	Diversity = 3.2672e+00
Generation 2400:	Best = 6.400707e+02	FES = 240100	Diversity = 3.2671e+00
Generation 2410:	Best = 6.400707e+02	FES = 241100	Diversity = 3.2669e+00
Generation 2420:	Best = 6.400707e+02	FES = 242100	Diversity = 3.2670e+00
Generation 2430:	Best = 6.400707e+02	FES = 243100	Diversity = 3.2672e+00
Generation 2440:	Best = 6.400707e+02	FES = 244100	Diversity = 3.2662e+00
Generation 2450:	Best = 6.400707e+02	FES = 245100	Diversity = 3.2638e+00
Generation 2460:	Best = 6.400707e+02	FES = 246100	Diversity = 3.2638e+00
Generation 2470:	Best = 6.400707e+02	FES = 247100	Diversity = 3.2600e+00
Generation 2480:	Best = 6.400707e+02	FES = 248100	Diversity = 3.2599e+00
Generation 2490:	Best = 6.400707e+02	FES = 249100	Diversity = 3.2597e+00
Generation 2500:	Best = 6.400707e+02	FES = 250100	Diversity = 3.2590e+00
Generation 2510:	Best = 6.400707e+02	FES = 251100	Diversity = 3.2588e+00
Generation 2520:	Best = 6.400707e+02	FES = 252100	Diversity = 3.2583e+00
Generation 2530:	Best = 6.400707e+02	FES = 253100	Diversity = 3.2579e+00
Generation 2540:	Best = 6.400707e+02	FES = 254100	Diversity = 3.2573e+00
Generation 2550:	Best = 6.400707e+02	FES = 255100	Diversity = 3.2569e+00
Generation 2560:	Best = 6.400707e+02	FES = 256100	Diversity = 3.2554e+00
Generation 2570:	Best = 6.400707e+02	FES = 257100	Diversity = 3.2572e+00
Generation 2580:	Best = 6.400707e+02	FES = 258100	Diversity = 3.2528e+00
Generation 2590:	Best = 6.400638e+02	FES = 259100	Diversity = 3.2526e+00
Generation 2600:	Best = 6.400638e+02	FES = 260100	Diversity = 3.2525e+00

Generation 2610:	Best = 6.400638e+02	FES = 261100	Diversity = 3.2521e+00
Generation 2620:	Best = 6.400638e+02	FES = 262100	Diversity = 3.2516e+00
Generation 2630:	Best = 6.400638e+02	FES = 263100	Diversity = 3.2510e+00
Generation 2640:	Best = 6.400638e+02	FES = 264100	Diversity = 3.2495e+00
Generation 2650:	Best = 6.400638e+02	FES = 265100	Diversity = 3.2496e+00
Generation 2660:	Best = 6.400638e+02	FES = 266100	Diversity = 3.2493e+00
Generation 2670:	Best = 6.400638e+02	FES = 267100	Diversity = 3.2467e+00
Generation 2680:	Best = 6.400638e+02	FES = 268100	Diversity = 3.2467e+00
Generation 2690:	Best = 6.400638e+02	FES = 269100	Diversity = 3.2461e+00
Generation 2700:	Best = 6.400638e+02	FES = 270100	Diversity = 3.2462e+00
Generation 2710:	Best = 6.400638e+02	FES = 271100	Diversity = 3.2463e+00
Generation 2720:	Best = 6.400638e+02	FES = 272100	Diversity = 3.2463e+00
Generation 2730:	Best = 6.400638e+02	FES = 273100	Diversity = 3.2463e+00
Generation 2740:	Best = 6.400638e+02	FES = 274100	Diversity = 3.2463e+00
Generation 2750:	Best = 6.400638e+02	FES = 275100	Diversity = 3.2453e+00
Generation 2760:	Best = 6.400638e+02	FES = 276100	Diversity = 3.2445e+00
Generation 2770:	Best = 6.400638e+02	FES = 277100	Diversity = 3.2446e+00
Generation 2780:	Best = 6.400638e+02	FES = 278100	Diversity = 3.2446e+00
Generation 2790:	Best = 6.400638e+02	FES = 279100	Diversity = 3.2445e+00
Generation 2800:	Best = 6.400638e+02	FES = 280100	Diversity = 3.2445e+00
Generation 2810:	Best = 6.400638e+02	FES = 281100	Diversity = 3.2447e+00
Generation 2820:	Best = 6.400638e+02	FES = 282100	Diversity = 3.2456e+00
Generation 2830:	Best = 6.400638e+02	FES = 283100	Diversity = 3.2456e+00
Generation 2840:	Best = 6.399781e+02	FES = 284100	Diversity = 3.2454e+00
Generation 2850:	Best = 6.399781e+02	FES = 285100	Diversity = 3.2454e+00
Generation 2860:	Best = 6.399781e+02	FES = 286100	Diversity = 3.2453e+00
Generation 2870:	Best = 6.399781e+02	FES = 287100	Diversity = 3.2369e+00
Generation 2880:	Best = 6.399508e+02	FES = 288100	Diversity = 3.2369e+00
Generation 2890:	Best = 6.399508e+02	FES = 289100	Diversity = 3.2365e+00
Generation 2900:	Best = 6.399508e+02	FES = 290100	Diversity = 3.2366e+00
Generation 2910:	Best = 6.399508e+02	FES = 291100	Diversity = 3.2364e+00
Generation 2920:	Best = 6.399508e+02	FES = 292100	Diversity = 3.2362e+00
Generation 2930:	Best = 6.399508e+02	FES = 293100	Diversity = 3.2338e+00
Generation 2940:	Best = 6.399508e+02	FES = 294100	Diversity = 3.2334e+00
Generation 2950:	Best = 6.399508e+02	FES = 295100	Diversity = 3.2328e+00
Generation 2960:	Best = 6.399508e+02	FES = 296100	Diversity = 3.2330e+00
Generation 2970:	Best = 6.399508e+02	FES = 297100	Diversity = 3.2331e+00
Generation 2980:	Best = 6.399508e+02	FES = 298100	Diversity = 3.2322e+00
Generation 2990:	Best = 6.399508e+02	FES = 299100	Diversity = 3.2320e+00
Generation 3000:	Best = 6.399508e+02	FES = 300100	Diversity = 3.2319e+00
Generation 3010:	Best = 6.399508e+02	FES = 301100	Diversity = 3.2318e+00
Generation 3020:	Best = 6.399508e+02	FES = 302100	Diversity = 3.2318e+00
Generation 3030:	Best = 6.399508e+02	FES = 303100	Diversity = 3.2317e+00
Generation 3040:	Best = 6.399508e+02	FES = 304100	Diversity = 3.2318e+00
Generation 3050:	Best = 6.399508e+02	FES = 305100	Diversity = 3.2317e+00
Generation 3060:	Best = 6.399508e+02	FES = 306100	Diversity = 3.2318e+00
Generation 3070:	Best = 6.399508e+02	FES = 307100	Diversity = 3.2317e+00
Generation 3080:	Best = 6.399508e+02	FES = 308100	Diversity = 3.2318e+00
Generation 3090:	Best = 6.399508e+02	FES = 309100	Diversity = 3.2318e+00
Generation 3100:	Best = 6.399508e+02	FES = 310100	Diversity = 3.2316e+00
Generation 3110:	Best = 6.399508e+02	FES = 311100	Diversity = 3.2316e+00
Generation 3120:	Best = 6.399508e+02	FES = 312100	Diversity = 3.2314e+00
Generation 3130:	Best = 6.399508e+02	FES = 313100	Diversity = 3.2312e+00
Generation 3140:	Best = 6.397278e+02	FES = 314100	Diversity = 3.2135e+00

Generation 3150:	Best = 6.397278e+02	FES = 315100	Diversity = 3.2136e+00
Generation 3160:	Best = 6.397278e+02	FES = 316100	Diversity = 3.2134e+00
Generation 3170:	Best = 6.397278e+02	FES = 317100	Diversity = 3.2136e+00
Generation 3180:	Best = 6.397278e+02	FES = 318100	Diversity = 3.2134e+00
Generation 3190:	Best = 6.397278e+02	FES = 319100	Diversity = 3.2140e+00
Generation 3200:	Best = 6.397278e+02	FES = 320100	Diversity = 3.2137e+00
Generation 3210:	Best = 6.397278e+02	FES = 321100	Diversity = 3.2137e+00
Generation 3220:	Best = 6.397278e+02	FES = 322100	Diversity = 3.2134e+00
Generation 3230:	Best = 6.397278e+02	FES = 323100	Diversity = 3.2129e+00
Generation 3240:	Best = 6.397278e+02	FES = 324100	Diversity = 3.2123e+00
Generation 3250:	Best = 6.397278e+02	FES = 325100	Diversity = 3.2106e+00
Generation 3260:	Best = 6.397278e+02	FES = 326100	Diversity = 3.2075e+00
Generation 3270:	Best = 6.397278e+02	FES = 327100	Diversity = 3.2054e+00
Generation 3280:	Best = 6.397278e+02	FES = 328100	Diversity = 3.2051e+00
Generation 3290:	Best = 6.397278e+02	FES = 329100	Diversity = 3.2030e+00
Generation 3300:	Best = 6.397278e+02	FES = 330100	Diversity = 3.2026e+00
Generation 3310:	Best = 6.397278e+02	FES = 331100	Diversity = 3.2027e+00
Generation 3320:	Best = 6.397278e+02	FES = 332100	Diversity = 3.2028e+00
Generation 3330:	Best = 6.394417e+02	FES = 333100	Diversity = 3.2028e+00
Generation 3340:	Best = 6.394417e+02	FES = 334100	Diversity = 3.2029e+00
Generation 3350:	Best = 6.394417e+02	FES = 335100	Diversity = 3.2014e+00
Generation 3360:	Best = 6.394417e+02	FES = 336100	Diversity = 3.2014e+00
Generation 3370:	Best = 6.394417e+02	FES = 337100	Diversity = 3.2001e+00
Generation 3380:	Best = 6.394417e+02	FES = 338100	Diversity = 3.2051e+00
Generation 3390:	Best = 6.394417e+02	FES = 339100	Diversity = 3.2049e+00
Generation 3400:	Best = 6.394417e+02	FES = 340100	Diversity = 3.2050e+00
Generation 3410:	Best = 6.394417e+02	FES = 341100	Diversity = 3.2048e+00
Generation 3420:	Best = 6.394417e+02	FES = 342100	Diversity = 3.2018e+00
Generation 3430:	Best = 6.394417e+02	FES = 343100	Diversity = 3.2013e+00
Generation 3440:	Best = 6.393957e+02	FES = 344100	Diversity = 3.1962e+00
Generation 3450:	Best = 6.393957e+02	FES = 345100	Diversity = 3.1954e+00
Generation 3460:	Best = 6.393957e+02	FES = 346100	Diversity = 3.1947e+00
Generation 3470:	Best = 6.393957e+02	FES = 347100	Diversity = 3.1948e+00
Generation 3480:	Best = 6.393957e+02	FES = 348100	Diversity = 3.1949e+00
Generation 3490:	Best = 6.393699e+02	FES = 349100	Diversity = 3.1949e+00
Generation 3500:	Best = 6.393699e+02	FES = 350100	Diversity = 3.2023e+00
Generation 3510:	Best = 6.393699e+02	FES = 351100	Diversity = 3.2022e+00
Generation 3520:	Best = 6.393699e+02	FES = 352100	Diversity = 3.2022e+00
Generation 3530:	Best = 6.393699e+02	FES = 353100	Diversity = 3.2015e+00
Generation 3540:	Best = 6.393699e+02	FES = 354100	Diversity = 3.2014e+00
Generation 3550:	Best = 6.393699e+02	FES = 355100	Diversity = 3.2016e+00
Generation 3560:	Best = 6.393699e+02	FES = 356100	Diversity = 3.2016e+00
Generation 3570:	Best = 6.393699e+02	FES = 357100	Diversity = 3.2016e+00
Generation 3580:	Best = 6.393699e+02	FES = 358100	Diversity = 3.2010e+00
Generation 3590:	Best = 6.393699e+02	FES = 359100	Diversity = 3.1998e+00
Generation 3600:	Best = 6.393699e+02	FES = 360100	Diversity = 3.1998e+00
Generation 3610:	Best = 6.393699e+02	FES = 361100	Diversity = 3.1996e+00
Generation 3620:	Best = 6.393699e+02	FES = 362100	Diversity = 3.1994e+00
Generation 3630:	Best = 6.393699e+02	FES = 363100	Diversity = 3.1994e+00
Generation 3640:	Best = 6.393699e+02	FES = 364100	Diversity = 3.1990e+00
Generation 3650:	Best = 6.393699e+02	FES = 365100	Diversity = 3.1973e+00
Generation 3660:	Best = 6.393699e+02	FES = 366100	Diversity = 3.1972e+00
Generation 3670:	Best = 6.393699e+02	FES = 367100	Diversity = 3.1977e+00
Generation 3680:	Best = 6.393699e+02	FES = 368100	Diversity = 3.1975e+00

Generation 3690:	Best = 6.393699e+02	FES = 369100	Diversity = 3.1964e+00
Generation 3700:	Best = 6.393699e+02	FES = 370100	Diversity = 3.1964e+00
Generation 3710:	Best = 6.393699e+02	FES = 371100	Diversity = 3.1964e+00
Generation 3720:	Best = 6.393699e+02	FES = 372100	Diversity = 3.1966e+00
Generation 3730:	Best = 6.393377e+02	FES = 373100	Diversity = 3.1967e+00
Generation 3740:	Best = 6.393377e+02	FES = 374100	Diversity = 3.1957e+00
Generation 3750:	Best = 6.393342e+02	FES = 375100	Diversity = 3.1952e+00
Generation 3760:	Best = 6.393342e+02	FES = 376100	Diversity = 3.1951e+00
Generation 3770:	Best = 6.393342e+02	FES = 377100	Diversity = 3.1951e+00
Generation 3780:	Best = 6.393294e+02	FES = 378100	Diversity = 3.1884e+00
Generation 3790:	Best = 6.393294e+02	FES = 379100	Diversity = 3.1980e+00
Generation 3800:	Best = 6.393294e+02	FES = 380100	Diversity = 3.1970e+00
Generation 3810:	Best = 6.393294e+02	FES = 381100	Diversity = 3.1965e+00
Generation 3820:	Best = 6.393294e+02	FES = 382100	Diversity = 3.1965e+00
Generation 3830:	Best = 6.393294e+02	FES = 383100	Diversity = 3.1965e+00
Generation 3840:	Best = 6.393294e+02	FES = 384100	Diversity = 3.1964e+00
Generation 3850:	Best = 6.393294e+02	FES = 385100	Diversity = 3.1962e+00
Generation 3860:	Best = 6.393294e+02	FES = 386100	Diversity = 3.1963e+00
Generation 3870:	Best = 6.393294e+02	FES = 387100	Diversity = 3.1963e+00
Generation 3880:	Best = 6.393294e+02	FES = 388100	Diversity = 3.1963e+00
Generation 3890:	Best = 6.393294e+02	FES = 389100	Diversity = 3.1963e+00
Generation 3900:	Best = 6.393294e+02	FES = 390100	Diversity = 3.1963e+00
Generation 3910:	Best = 6.393294e+02	FES = 391100	Diversity = 3.1963e+00
Generation 3920:	Best = 6.393294e+02	FES = 392100	Diversity = 3.1959e+00
Generation 3930:	Best = 6.393294e+02	FES = 393100	Diversity = 3.1960e+00
Generation 3940:	Best = 6.393294e+02	FES = 394100	Diversity = 3.1960e+00
Generation 3950:	Best = 6.393294e+02	FES = 395100	Diversity = 3.1959e+00
Generation 3960:	Best = 6.393294e+02	FES = 396100	Diversity = 3.1937e+00
Generation 3970:	Best = 6.393294e+02	FES = 397100	Diversity = 3.1932e+00
Generation 3980:	Best = 6.392730e+02	FES = 398100	Diversity = 3.1928e+00
Generation 3990:	Best = 6.392730e+02	FES = 399100	Diversity = 3.1924e+00
Generation 4000:	Best = 6.392730e+02	FES = 400100	Diversity = 3.1925e+00
Generation 4010:	Best = 6.392730e+02	FES = 401100	Diversity = 3.1907e+00
Generation 4020:	Best = 6.392730e+02	FES = 402100	Diversity = 3.1905e+00
Generation 4030:	Best = 6.392730e+02	FES = 403100	Diversity = 3.1907e+00
Generation 4040:	Best = 6.392730e+02	FES = 404100	Diversity = 3.1916e+00
Generation 4050:	Best = 6.392730e+02	FES = 405100	Diversity = 3.1916e+00
Generation 4060:	Best = 6.392730e+02	FES = 406100	Diversity = 3.1916e+00
Generation 4070:	Best = 6.390708e+02	FES = 407100	Diversity = 3.1916e+00
Generation 4080:	Best = 6.390708e+02	FES = 408100	Diversity = 3.1915e+00
Generation 4090:	Best = 6.390708e+02	FES = 409100	Diversity = 3.1915e+00
Generation 4100:	Best = 6.390708e+02	FES = 410100	Diversity = 3.1913e+00
Generation 4110:	Best = 6.390708e+02	FES = 411100	Diversity = 3.1883e+00
Generation 4120:	Best = 6.390708e+02	FES = 412100	Diversity = 3.1883e+00
Generation 4130:	Best = 6.390708e+02	FES = 413100	Diversity = 3.1878e+00
Generation 4140:	Best = 6.390708e+02	FES = 414100	Diversity = 3.1873e+00
Generation 4150:	Best = 6.390708e+02	FES = 415100	Diversity = 3.1873e+00
Generation 4160:	Best = 6.390708e+02	FES = 416100	Diversity = 3.1872e+00
Generation 4170:	Best = 6.390708e+02	FES = 417100	Diversity = 3.1873e+00
Generation 4180:	Best = 6.390708e+02	FES = 418100	Diversity = 3.1843e+00
Generation 4190:	Best = 6.390708e+02	FES = 419100	Diversity = 3.1844e+00
Generation 4200:	Best = 6.390708e+02	FES = 420100	Diversity = 3.1843e+00
Generation 4210:	Best = 6.390708e+02	FES = 421100	Diversity = 3.1842e+00
Generation 4220:	Best = 6.390708e+02	FES = 422100	Diversity = 3.1841e+00

Generation 4230:	Best = 6.390708e+02	FES = 423100	Diversity = 3.1841e+00
Generation 4240:	Best = 6.390708e+02	FES = 424100	Diversity = 3.1840e+00
Generation 4250:	Best = 6.390708e+02	FES = 425100	Diversity = 3.1928e+00
Generation 4260:	Best = 6.390708e+02	FES = 426100	Diversity = 3.1933e+00
Generation 4270:	Best = 6.390708e+02	FES = 427100	Diversity = 3.1933e+00
Generation 4280:	Best = 6.390708e+02	FES = 428100	Diversity = 3.1933e+00
Generation 4290:	Best = 6.390697e+02	FES = 429100	Diversity = 3.1933e+00
Generation 4300:	Best = 6.390697e+02	FES = 430100	Diversity = 3.1933e+00
Generation 4310:	Best = 6.390697e+02	FES = 431100	Diversity = 3.1933e+00
Generation 4320:	Best = 6.390697e+02	FES = 432100	Diversity = 3.1932e+00
Generation 4330:	Best = 6.390697e+02	FES = 433100	Diversity = 3.1932e+00
Generation 4340:	Best = 6.390697e+02	FES = 434100	Diversity = 3.1929e+00
Generation 4350:	Best = 6.390697e+02	FES = 435100	Diversity = 3.1928e+00
Generation 4360:	Best = 6.390697e+02	FES = 436100	Diversity = 3.1928e+00
Generation 4370:	Best = 6.390697e+02	FES = 437100	Diversity = 3.1928e+00
Generation 4380:	Best = 6.390697e+02	FES = 438100	Diversity = 3.1927e+00
Generation 4390:	Best = 6.390697e+02	FES = 439100	Diversity = 3.1927e+00
Generation 4400:	Best = 6.390697e+02	FES = 440100	Diversity = 3.1927e+00
Generation 4410:	Best = 6.390697e+02	FES = 441100	Diversity = 3.1927e+00
Generation 4420:	Best = 6.390697e+02	FES = 442100	Diversity = 3.1931e+00
Generation 4430:	Best = 6.390697e+02	FES = 443100	Diversity = 3.1932e+00
Generation 4440:	Best = 6.390697e+02	FES = 444100	Diversity = 3.1932e+00
Generation 4450:	Best = 6.390697e+02	FES = 445100	Diversity = 3.1930e+00
Generation 4460:	Best = 6.390697e+02	FES = 446100	Diversity = 3.1931e+00
Generation 4470:	Best = 6.390490e+02	FES = 447100	Diversity = 3.1930e+00
Generation 4480:	Best = 6.390490e+02	FES = 448100	Diversity = 3.1930e+00
Generation 4490:	Best = 6.390490e+02	FES = 449100	Diversity = 3.1804e+00
Generation 4500:	Best = 6.390490e+02	FES = 450100	Diversity = 3.1803e+00
Generation 4510:	Best = 6.390490e+02	FES = 451100	Diversity = 3.1804e+00
Generation 4520:	Best = 6.390490e+02	FES = 452100	Diversity = 3.1804e+00
Generation 4530:	Best = 6.390490e+02	FES = 453100	Diversity = 3.1798e+00
Generation 4540:	Best = 6.390490e+02	FES = 454100	Diversity = 3.1794e+00
Generation 4550:	Best = 6.390490e+02	FES = 455100	Diversity = 3.1860e+00
Generation 4560:	Best = 6.390490e+02	FES = 456100	Diversity = 3.1859e+00
Generation 4570:	Best = 6.390490e+02	FES = 457100	Diversity = 3.1861e+00
Generation 4580:	Best = 6.390490e+02	FES = 458100	Diversity = 3.1858e+00
Generation 4590:	Best = 6.390490e+02	FES = 459100	Diversity = 3.1857e+00
Generation 4600:	Best = 6.390490e+02	FES = 460100	Diversity = 3.1852e+00
Generation 4610:	Best = 6.390490e+02	FES = 461100	Diversity = 3.1851e+00
Generation 4620:	Best = 6.389473e+02	FES = 462100	Diversity = 3.1851e+00
Generation 4630:	Best = 6.389473e+02	FES = 463100	Diversity = 3.1851e+00
Generation 4640:	Best = 6.389473e+02	FES = 464100	Diversity = 3.1841e+00
Generation 4650:	Best = 6.389473e+02	FES = 465100	Diversity = 3.1835e+00
Generation 4660:	Best = 6.389473e+02	FES = 466100	Diversity = 3.1836e+00
Generation 4670:	Best = 6.389473e+02	FES = 467100	Diversity = 3.1834e+00
Generation 4680:	Best = 6.389473e+02	FES = 468100	Diversity = 3.1831e+00
Generation 4690:	Best = 6.389473e+02	FES = 469100	Diversity = 3.1830e+00
Generation 4700:	Best = 6.389473e+02	FES = 470100	Diversity = 3.1829e+00
Generation 4710:	Best = 6.389473e+02	FES = 471100	Diversity = 3.1828e+00
Generation 4720:	Best = 6.389473e+02	FES = 472100	Diversity = 3.1827e+00
Generation 4730:	Best = 6.389473e+02	FES = 473100	Diversity = 3.1815e+00
Generation 4740:	Best = 6.389473e+02	FES = 474100	Diversity = 3.1810e+00
Generation 4750:	Best = 6.389473e+02	FES = 475100	Diversity = 3.1811e+00
Generation 4760:	Best = 6.389473e+02	FES = 476100	Diversity = 3.1810e+00

Generation 4770:	Best = 6.389473e+02	FES = 477100	Diversity = 3.1809e+00
Generation 4780:	Best = 6.389473e+02	FES = 478100	Diversity = 3.1808e+00
Generation 4790:	Best = 6.389473e+02	FES = 479100	Diversity = 3.1807e+00
Generation 4800:	Best = 6.389473e+02	FES = 480100	Diversity = 3.1805e+00
Generation 4810:	Best = 6.389473e+02	FES = 481100	Diversity = 3.1800e+00
Generation 4820:	Best = 6.389473e+02	FES = 482100	Diversity = 3.1800e+00
Generation 4830:	Best = 6.389473e+02	FES = 483100	Diversity = 3.1790e+00
Generation 4840:	Best = 6.388695e+02	FES = 484100	Diversity = 3.1785e+00
Generation 4850:	Best = 6.388695e+02	FES = 485100	Diversity = 3.1780e+00
Generation 4860:	Best = 6.388695e+02	FES = 486100	Diversity = 3.1778e+00
Generation 4870:	Best = 6.388695e+02	FES = 487100	Diversity = 3.1773e+00
Generation 4880:	Best = 6.388154e+02	FES = 488100	Diversity = 3.1769e+00
Generation 4890:	Best = 6.388154e+02	FES = 489100	Diversity = 3.1757e+00
Generation 4900:	Best = 6.387039e+02	FES = 490100	Diversity = 3.1755e+00
Generation 4910:	Best = 6.387039e+02	FES = 491100	Diversity = 3.1755e+00
Generation 4920:	Best = 6.387039e+02	FES = 492100	Diversity = 3.1754e+00
Generation 4930:	Best = 6.387039e+02	FES = 493100	Diversity = 3.1751e+00
Generation 4940:	Best = 6.387039e+02	FES = 494100	Diversity = 3.1752e+00
Generation 4950:	Best = 6.387039e+02	FES = 495100	Diversity = 3.1752e+00
Generation 4960:	Best = 6.387039e+02	FES = 496100	Diversity = 3.1745e+00
Generation 4970:	Best = 6.387039e+02	FES = 497100	Diversity = 3.1751e+00
Generation 4980:	Best = 6.387039e+02	FES = 498100	Diversity = 3.1751e+00
Generation 4990:	Best = 6.387039e+02	FES = 499100	Diversity = 3.1808e+00
Generation 5000:	Best = 6.387039e+02	FES = 500100	Diversity = 3.1797e+00
Generation 5010:	Best = 6.387039e+02	FES = 501100	Diversity = 3.1797e+00
Generation 5020:	Best = 6.386747e+02	FES = 502100	Diversity = 3.1796e+00
Generation 5030:	Best = 6.386747e+02	FES = 503100	Diversity = 3.1789e+00
Generation 5040:	Best = 6.386747e+02	FES = 504100	Diversity = 3.1783e+00
Generation 5050:	Best = 6.386747e+02	FES = 505100	Diversity = 3.1782e+00
Generation 5060:	Best = 6.386747e+02	FES = 506100	Diversity = 3.1781e+00
Generation 5070:	Best = 6.386747e+02	FES = 507100	Diversity = 3.1774e+00
Generation 5080:	Best = 6.386747e+02	FES = 508100	Diversity = 3.1773e+00
Generation 5090:	Best = 6.386747e+02	FES = 509100	Diversity = 3.1773e+00
Generation 5100:	Best = 6.386747e+02	FES = 510100	Diversity = 3.1773e+00
Generation 5110:	Best = 6.386747e+02	FES = 511100	Diversity = 3.1773e+00
Generation 5120:	Best = 6.386747e+02	FES = 512100	Diversity = 3.1773e+00
Generation 5130:	Best = 6.386747e+02	FES = 513100	Diversity = 3.1772e+00
Generation 5140:	Best = 6.386747e+02	FES = 514100	Diversity = 3.1772e+00
Generation 5150:	Best = 6.386747e+02	FES = 515100	Diversity = 3.1770e+00
Generation 5160:	Best = 6.386747e+02	FES = 516100	Diversity = 3.1770e+00
Generation 5170:	Best = 6.386747e+02	FES = 517100	Diversity = 3.1762e+00
Generation 5180:	Best = 6.386747e+02	FES = 518100	Diversity = 3.1762e+00
Generation 5190:	Best = 6.386747e+02	FES = 519100	Diversity = 3.1931e+00
Generation 5200:	Best = 6.386747e+02	FES = 520100	Diversity = 3.1931e+00
Generation 5210:	Best = 6.386747e+02	FES = 521100	Diversity = 3.1929e+00
Generation 5220:	Best = 6.386747e+02	FES = 522100	Diversity = 3.1933e+00
Generation 5230:	Best = 6.386747e+02	FES = 523100	Diversity = 3.1933e+00
Generation 5240:	Best = 6.386747e+02	FES = 524100	Diversity = 3.1933e+00
Generation 5250:	Best = 6.386747e+02	FES = 525100	Diversity = 3.1894e+00
Generation 5260:	Best = 6.386747e+02	FES = 526100	Diversity = 3.1895e+00
Generation 5270:	Best = 6.386747e+02	FES = 527100	Diversity = 3.1897e+00
Generation 5280:	Best = 6.386747e+02	FES = 528100	Diversity = 3.1897e+00
Generation 5290:	Best = 6.386747e+02	FES = 529100	Diversity = 3.1896e+00
Generation 5300:	Best = 6.386747e+02	FES = 530100	Diversity = 3.1897e+00

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Generation 5850:	Best = 6.386747e+02	FES = 585100	Diversity = 3.1699e+00
Generation 5860:	Best = 6.386747e+02	FES = 586100	Diversity = 3.1699e+00
Generation 5870:	Best = 6.386747e+02	FES = 587100	Diversity = 3.1702e+00
Generation 5880:	Best = 6.386747e+02	FES = 588100	Diversity = 3.1695e+00
Generation 5890:	Best = 6.386747e+02	FES = 589100	Diversity = 3.1691e+00
Generation 5900:	Best = 6.386747e+02	FES = 590100	Diversity = 3.1692e+00
Generation 5910:	Best = 6.386747e+02	FES = 591100	Diversity = 3.1692e+00
Generation 5920:	Best = 6.386747e+02	FES = 592100	Diversity = 3.1692e+00
Generation 5930:	Best = 6.386747e+02	FES = 593100	Diversity = 3.1690e+00
Generation 5940:	Best = 6.386747e+02	FES = 594100	Diversity = 3.1680e+00
Generation 5950:	Best = 6.386747e+02	FES = 595100	Diversity = 3.1666e+00
Generation 5960:	Best = 6.386747e+02	FES = 596100	Diversity = 3.1655e+00
Generation 5970:	Best = 6.386747e+02	FES = 597100	Diversity = 3.1656e+00
Generation 5980:	Best = 6.386747e+02	FES = 598100	Diversity = 3.1654e+00
Generation 5990:	Best = 6.386747e+02	FES = 599100	Diversity = 3.1656e+00
Generation 6000:	Best = 6.386747e+02	FES = 600100	Diversity = 3.1656e+00
Generation 6010:	Best = 6.386747e+02	FES = 601100	Diversity = 3.1657e+00
Generation 6020:	Best = 6.386747e+02	FES = 602100	Diversity = 3.1704e+00
Generation 6030:	Best = 6.386747e+02	FES = 603100	Diversity = 3.1704e+00
Generation 6040:	Best = 6.386747e+02	FES = 604100	Diversity = 3.1701e+00
Generation 6050:	Best = 6.386747e+02	FES = 605100	Diversity = 3.1700e+00
Generation 6060:	Best = 6.386747e+02	FES = 606100	Diversity = 3.1700e+00
Generation 6070:	Best = 6.386747e+02	FES = 607100	Diversity = 3.1700e+00
Generation 6080:	Best = 6.386726e+02	FES = 608100	Diversity = 3.1698e+00
Generation 6090:	Best = 6.386726e+02	FES = 609100	Diversity = 3.1698e+00
Generation 6100:	Best = 6.386726e+02	FES = 610100	Diversity = 3.1697e+00
Generation 6110:	Best = 6.386726e+02	FES = 611100	Diversity = 3.1696e+00
Generation 6120:	Best = 6.386726e+02	FES = 612100	Diversity = 3.1695e+00
Generation 6130:	Best = 6.386726e+02	FES = 613100	Diversity = 3.1694e+00
Generation 6140:	Best = 6.386726e+02	FES = 614100	Diversity = 3.1695e+00
Generation 6150:	Best = 6.386726e+02	FES = 615100	Diversity = 3.1695e+00
Generation 6160:	Best = 6.386726e+02	FES = 616100	Diversity = 3.1695e+00
Generation 6170:	Best = 6.386726e+02	FES = 617100	Diversity = 3.1698e+00
Generation 6180:	Best = 6.386726e+02	FES = 618100	Diversity = 3.1698e+00
Generation 6190:	Best = 6.386726e+02	FES = 619100	Diversity = 3.1697e+00
Generation 6200:	Best = 6.386726e+02	FES = 620100	Diversity = 3.1695e+00
Generation 6210:	Best = 6.386606e+02	FES = 621100	Diversity = 3.1694e+00
Generation 6220:	Best = 6.386606e+02	FES = 622100	Diversity = 3.1693e+00
Generation 6230:	Best = 6.386606e+02	FES = 623100	Diversity = 3.1693e+00
Generation 6240:	Best = 6.386606e+02	FES = 624100	Diversity = 3.1693e+00
Generation 6250:	Best = 6.384110e+02	FES = 625100	Diversity = 3.1693e+00
Generation 6260:	Best = 6.384110e+02	FES = 626100	Diversity = 3.1692e+00
Generation 6270:	Best = 6.384110e+02	FES = 627100	Diversity = 3.1691e+00
Generation 6280:	Best = 6.384110e+02	FES = 628100	Diversity = 3.1689e+00
Generation 6290:	Best = 6.383984e+02	FES = 629100	Diversity = 3.1682e+00
Generation 6300:	Best = 6.383984e+02	FES = 630100	Diversity = 3.1682e+00
Generation 6310:	Best = 6.383984e+02	FES = 631100	Diversity = 3.1681e+00
Generation 6320:	Best = 6.383984e+02	FES = 632100	Diversity = 3.1673e+00
Generation 6330:	Best = 6.383984e+02	FES = 633100	Diversity = 3.1671e+00
Generation 6340:	Best = 6.383984e+02	FES = 634100	Diversity = 3.1670e+00
Generation 6350:	Best = 6.383984e+02	FES = 635100	Diversity = 3.1670e+00
Generation 6360:	Best = 6.383984e+02	FES = 636100	Diversity = 3.1669e+00
Generation 6370:	Best = 6.383984e+02	FES = 637100	Diversity = 3.1668e+00
Generation 6380:	Best = 6.383984e+02	FES = 638100	Diversity = 3.1668e+00

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

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Generation 9630: Best = 6.381420e+02 FES = 963100 Diversity = 3.1419e+00
Generation 9640: Best = 6.381420e+02 FES = 964100 Diversity = 3.1419e+00
Generation 9650: Best = 6.381420e+02 FES = 965100 Diversity = 3.1419e+00
Generation 9660: Best = 6.381420e+02 FES = 966100 Diversity = 3.1419e+00
Generation 9670: Best = 6.381420e+02 FES = 967100 Diversity = 3.1419e+00
Generation 9680: Best = 6.381420e+02 FES = 968100 Diversity = 3.1419e+00
Generation 9690: Best = 6.381420e+02 FES = 969100 Diversity = 3.1412e+00
Generation 9700: Best = 6.381420e+02 FES = 970100 Diversity = 3.1412e+00
Generation 9710: Best = 6.381420e+02 FES = 971100 Diversity = 3.1412e+00
Generation 9720: Best = 6.381420e+02 FES = 972100 Diversity = 3.1412e+00
Generation 9730: Best = 6.381420e+02 FES = 973100 Diversity = 3.1412e+00
Generation 9740: Best = 6.381420e+02 FES = 974100 Diversity = 3.1412e+00
Generation 9750: Best = 6.381420e+02 FES = 975100 Diversity = 3.1412e+00
Generation 9760: Best = 6.381420e+02 FES = 976100 Diversity = 3.1412e+00
Generation 9770: Best = 6.381420e+02 FES = 977100 Diversity = 3.1412e+00
Generation 9780: Best = 6.381420e+02 FES = 978100 Diversity = 3.1412e+00
Generation 9790: Best = 6.381420e+02 FES = 979100 Diversity = 3.1412e+00
Generation 9800: Best = 6.381420e+02 FES = 980100 Diversity = 3.1412e+00
Generation 9810: Best = 6.381420e+02 FES = 981100 Diversity = 3.1410e+00
Generation 9820: Best = 6.381420e+02 FES = 982100 Diversity = 3.1410e+00
Generation 9830: Best = 6.381420e+02 FES = 983100 Diversity = 3.1409e+00
Generation 9840: Best = 6.381420e+02 FES = 984100 Diversity = 3.1409e+00
Generation 9850: Best = 6.381420e+02 FES = 985100 Diversity = 3.1409e+00
Generation 9860: Best = 6.381420e+02 FES = 986100 Diversity = 3.1409e+00
Generation 9870: Best = 6.381420e+02 FES = 987100 Diversity = 3.1409e+00
Generation 9880: Best = 6.381420e+02 FES = 988100 Diversity = 3.1409e+00
Generation 9890: Best = 6.381420e+02 FES = 989100 Diversity = 3.1408e+00
Generation 9900: Best = 6.381420e+02 FES = 990100 Diversity = 3.1408e+00
Generation 9910: Best = 6.381420e+02 FES = 991100 Diversity = 3.1408e+00
Generation 9920: Best = 6.381420e+02 FES = 992100 Diversity = 3.1408e+00
Generation 9930: Best = 6.381420e+02 FES = 993100 Diversity = 3.1408e+00
Generation 9940: Best = 6.381420e+02 FES = 994100 Diversity = 3.1408e+00
Generation 9950: Best = 6.381420e+02 FES = 995100 Diversity = 3.1408e+00
Generation 9960: Best = 6.381420e+02 FES = 996100 Diversity = 3.1407e+00
Generation 9970: Best = 6.381420e+02 FES = 997100 Diversity = 3.1407e+00
Generation 9980: Best = 6.381420e+02 FES = 998100 Diversity = 3.1408e+00
Generation 9990: Best = 6.381420e+02 FES = 999100 Diversity = 3.1408e+00

```

Optimization completed!

Final Best Fitness = 6.381420e+02

Total Function Evaluations = 1000000

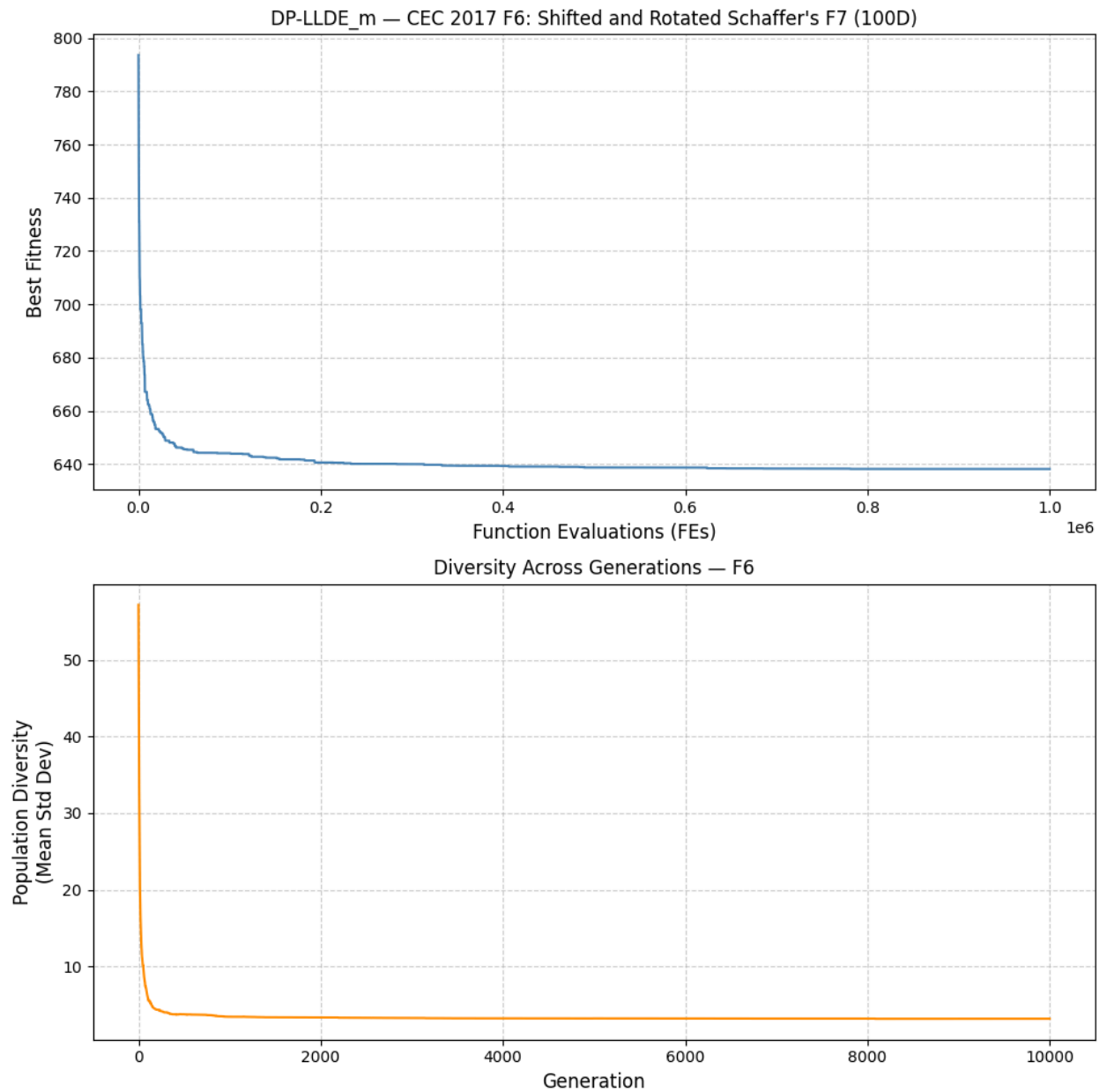
```

=====
=
Results:
  Best Fitness           : 6.381420e+02
  Global Optimum         : 6.000000e+02
  Error (|best-optimum|) : 3.814201e+01
  Best Solution (5 dim)  : [ 33.41394703 -4.00612651  4.84749052 -30.46829069
55.67753003]
  D^(0) (init. diversity): 5.7220e+01
=====
=

```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results_2/all_results_che

ckpoint.pkl



Graph saved as convergence_F6.png

✓ Plot saved to: /content/drive/My Drive/thesis_results_2/plots/convergence_F6.png

```
In [ ]: # =====  
# F7 — Shifted and Rotated Lunacek Bi-Rastrigin  
# =====  
run_and_plot('F7', make_wrapper(simple.f7), 700.0, "Shifted and Rotated Lunacek Bi-Rastrigin")
```

```

=====
=
DP-LLDE_m on CEC 2017 F7 – Single Run
=====
=
Function      : Shifted and Rotated Lunacek Bi-Rastrigin's (100D)
Bounds       : [-100, 100]
Global optimum : 700.0
Initial D^(0) = 5.7220e+01 | Diversity threshold = 5.7220e+00
Generation 0: Best Fitness = 1.400178e+04 | Diversity = 5.7220e+01
Generation 10: Best = 4.491958e+03 FES = 1100 Diversity = 1.9566e+01
Generation 20: Best = 3.333676e+03 FES = 2100 Diversity = 8.0242e+00
Generation 30: Best = 3.048735e+03 FES = 3100 Diversity = 5.4340e+00
Generation 40: Best = 2.962713e+03 FES = 4100 Diversity = 4.1419e+00
Generation 50: Best = 2.791334e+03 FES = 5100 Diversity = 3.5843e+00
Generation 60: Best = 2.711379e+03 FES = 6100 Diversity = 3.2349e+00
Generation 70: Best = 2.711379e+03 FES = 7100 Diversity = 3.1303e+00
Generation 80: Best = 2.666847e+03 FES = 8100 Diversity = 3.1569e+00
Generation 90: Best = 2.626009e+03 FES = 9100 Diversity = 3.3015e+00
Generation 100: Best = 2.572081e+03 FES = 10100 Diversity = 3.3432e+00
Generation 110: Best = 2.543696e+03 FES = 11100 Diversity = 3.4410e+00
Generation 120: Best = 2.505401e+03 FES = 12100 Diversity = 3.3924e+00
Generation 130: Best = 2.420922e+03 FES = 13100 Diversity = 3.4450e+00
Generation 140: Best = 2.392600e+03 FES = 14100 Diversity = 3.4108e+00
Generation 150: Best = 2.368740e+03 FES = 15100 Diversity = 3.3783e+00
Generation 160: Best = 2.348105e+03 FES = 16100 Diversity = 3.3094e+00
Generation 170: Best = 2.317851e+03 FES = 17100 Diversity = 3.2993e+00
Generation 180: Best = 2.271093e+03 FES = 18100 Diversity = 3.2508e+00
Generation 190: Best = 2.266961e+03 FES = 19100 Diversity = 3.2380e+00
Generation 200: Best = 2.240429e+03 FES = 20100 Diversity = 3.2162e+00
Generation 210: Best = 2.240429e+03 FES = 21100 Diversity = 3.1640e+00
Generation 220: Best = 2.204374e+03 FES = 22100 Diversity = 3.1082e+00
Generation 230: Best = 2.171575e+03 FES = 23100 Diversity = 3.0985e+00
Generation 240: Best = 2.155565e+03 FES = 24100 Diversity = 3.1594e+00
Generation 250: Best = 2.144572e+03 FES = 25100 Diversity = 3.2384e+00
Generation 260: Best = 2.134819e+03 FES = 26100 Diversity = 3.1673e+00
Generation 270: Best = 2.120971e+03 FES = 27100 Diversity = 3.1859e+00
Generation 280: Best = 2.114769e+03 FES = 28100 Diversity = 3.1623e+00
Generation 290: Best = 2.114769e+03 FES = 29100 Diversity = 3.1603e+00
Generation 300: Best = 2.102646e+03 FES = 30100 Diversity = 3.1723e+00
Generation 310: Best = 2.100284e+03 FES = 31100 Diversity = 3.1612e+00
Generation 320: Best = 2.094665e+03 FES = 32100 Diversity = 3.1829e+00
Generation 330: Best = 2.094665e+03 FES = 33100 Diversity = 3.2132e+00
Generation 340: Best = 2.078142e+03 FES = 34100 Diversity = 3.1873e+00
Generation 350: Best = 2.078142e+03 FES = 35100 Diversity = 3.1671e+00
Generation 360: Best = 2.078142e+03 FES = 36100 Diversity = 3.1653e+00
Generation 370: Best = 2.074170e+03 FES = 37100 Diversity = 3.2085e+00
Generation 380: Best = 2.073056e+03 FES = 38100 Diversity = 3.1709e+00
Generation 390: Best = 2.073056e+03 FES = 39100 Diversity = 3.1621e+00
Generation 400: Best = 2.073056e+03 FES = 40100 Diversity = 3.1555e+00
Generation 410: Best = 2.073056e+03 FES = 41100 Diversity = 3.1386e+00
Generation 420: Best = 2.056802e+03 FES = 42100 Diversity = 3.1492e+00
Generation 430: Best = 2.056802e+03 FES = 43100 Diversity = 3.1434e+00
Generation 440: Best = 2.051028e+03 FES = 44100 Diversity = 3.1268e+00

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Generation	450:	Best = 2.051028e+03	FES = 45100	Diversity = 3.1253e+00
Generation	460:	Best = 2.042036e+03	FES = 46100	Diversity = 3.1452e+00
Generation	470:	Best = 2.028373e+03	FES = 47100	Diversity = 3.1674e+00
Generation	480:	Best = 2.028373e+03	FES = 48100	Diversity = 3.1414e+00
Generation	490:	Best = 2.028373e+03	FES = 49100	Diversity = 3.1212e+00
Generation	500:	Best = 2.028373e+03	FES = 50100	Diversity = 3.1215e+00
Generation	510:	Best = 2.028373e+03	FES = 51100	Diversity = 3.1178e+00
Generation	520:	Best = 2.028373e+03	FES = 52100	Diversity = 3.1115e+00
Generation	530:	Best = 2.025656e+03	FES = 53100	Diversity = 3.1220e+00
Generation	540:	Best = 2.017016e+03	FES = 54100	Diversity = 3.1171e+00
Generation	550:	Best = 2.017016e+03	FES = 55100	Diversity = 3.0946e+00
Generation	560:	Best = 1.992578e+03	FES = 56100	Diversity = 3.0842e+00
Generation	570:	Best = 1.991812e+03	FES = 57100	Diversity = 3.1178e+00
Generation	580:	Best = 1.989502e+03	FES = 58100	Diversity = 3.1303e+00
Generation	590:	Best = 1.989294e+03	FES = 59100	Diversity = 3.1139e+00
Generation	600:	Best = 1.989294e+03	FES = 60100	Diversity = 3.1094e+00
Generation	610:	Best = 1.988525e+03	FES = 61100	Diversity = 3.0931e+00
Generation	620:	Best = 1.940146e+03	FES = 62100	Diversity = 3.0893e+00
Generation	630:	Best = 1.940146e+03	FES = 63100	Diversity = 3.0758e+00
Generation	640:	Best = 1.940146e+03	FES = 64100	Diversity = 3.0619e+00
Generation	650:	Best = 1.934965e+03	FES = 65100	Diversity = 3.0475e+00
Generation	660:	Best = 1.934965e+03	FES = 66100	Diversity = 3.0374e+00
Generation	670:	Best = 1.934965e+03	FES = 67100	Diversity = 3.0472e+00
Generation	680:	Best = 1.931245e+03	FES = 68100	Diversity = 3.0631e+00
Generation	690:	Best = 1.915846e+03	FES = 69100	Diversity = 3.0492e+00
Generation	700:	Best = 1.912254e+03	FES = 70100	Diversity = 3.0435e+00
Generation	710:	Best = 1.912244e+03	FES = 71100	Diversity = 3.0232e+00
Generation	720:	Best = 1.909387e+03	FES = 72100	Diversity = 3.0127e+00
Generation	730:	Best = 1.898516e+03	FES = 73100	Diversity = 3.0103e+00
Generation	740:	Best = 1.888241e+03	FES = 74100	Diversity = 3.0055e+00
Generation	750:	Best = 1.888241e+03	FES = 75100	Diversity = 2.9998e+00
Generation	760:	Best = 1.887952e+03	FES = 76100	Diversity = 3.0083e+00
Generation	770:	Best = 1.883252e+03	FES = 77100	Diversity = 3.0086e+00
Generation	780:	Best = 1.883119e+03	FES = 78100	Diversity = 2.9963e+00
Generation	790:	Best = 1.881819e+03	FES = 79100	Diversity = 2.9946e+00
Generation	800:	Best = 1.879613e+03	FES = 80100	Diversity = 2.9891e+00
Generation	810:	Best = 1.879613e+03	FES = 81100	Diversity = 2.9871e+00
Generation	820:	Best = 1.879463e+03	FES = 82100	Diversity = 2.9787e+00
Generation	830:	Best = 1.879270e+03	FES = 83100	Diversity = 2.9759e+00
Generation	840:	Best = 1.879270e+03	FES = 84100	Diversity = 2.9903e+00
Generation	850:	Best = 1.879270e+03	FES = 85100	Diversity = 2.9842e+00
Generation	860:	Best = 1.879270e+03	FES = 86100	Diversity = 2.9784e+00
Generation	870:	Best = 1.878815e+03	FES = 87100	Diversity = 2.9759e+00
Generation	880:	Best = 1.877341e+03	FES = 88100	Diversity = 2.9774e+00
Generation	890:	Best = 1.870650e+03	FES = 89100	Diversity = 2.9719e+00
Generation	900:	Best = 1.870650e+03	FES = 90100	Diversity = 2.9851e+00
Generation	910:	Best = 1.870650e+03	FES = 91100	Diversity = 2.9889e+00
Generation	920:	Best = 1.867696e+03	FES = 92100	Diversity = 2.9691e+00
Generation	930:	Best = 1.867587e+03	FES = 93100	Diversity = 2.9666e+00
Generation	940:	Best = 1.867258e+03	FES = 94100	Diversity = 2.9637e+00
Generation	950:	Best = 1.867258e+03	FES = 95100	Diversity = 2.9634e+00
Generation	960:	Best = 1.867258e+03	FES = 96100	Diversity = 2.9639e+00
Generation	970:	Best = 1.867056e+03	FES = 97100	Diversity = 2.9521e+00
Generation	980:	Best = 1.866832e+03	FES = 98100	Diversity = 2.9335e+00

Generation 990:	Best = 1.866509e+03	FES = 99100	Diversity = 2.9318e+00
Generation 1000:	Best = 1.861605e+03	FES = 100100	Diversity = 2.9400e+00
Generation 1010:	Best = 1.861605e+03	FES = 101100	Diversity = 2.9384e+00
Generation 1020:	Best = 1.861605e+03	FES = 102100	Diversity = 2.9271e+00
Generation 1030:	Best = 1.853355e+03	FES = 103100	Diversity = 2.9253e+00
Generation 1040:	Best = 1.847483e+03	FES = 104100	Diversity = 2.9237e+00
Generation 1050:	Best = 1.846766e+03	FES = 105100	Diversity = 2.9098e+00
Generation 1060:	Best = 1.842059e+03	FES = 106100	Diversity = 2.9079e+00
Generation 1070:	Best = 1.841307e+03	FES = 107100	Diversity = 2.9065e+00
Generation 1080:	Best = 1.840784e+03	FES = 108100	Diversity = 2.9077e+00
Generation 1090:	Best = 1.840257e+03	FES = 109100	Diversity = 2.9067e+00
Generation 1100:	Best = 1.840257e+03	FES = 110100	Diversity = 2.9065e+00
Generation 1110:	Best = 1.839839e+03	FES = 111100	Diversity = 2.9067e+00
Generation 1120:	Best = 1.839839e+03	FES = 112100	Diversity = 2.9047e+00
Generation 1130:	Best = 1.819080e+03	FES = 113100	Diversity = 2.9018e+00
Generation 1140:	Best = 1.819080e+03	FES = 114100	Diversity = 2.8935e+00
Generation 1150:	Best = 1.818991e+03	FES = 115100	Diversity = 2.8966e+00
Generation 1160:	Best = 1.818991e+03	FES = 116100	Diversity = 2.8947e+00
Generation 1170:	Best = 1.818991e+03	FES = 117100	Diversity = 2.9124e+00
Generation 1180:	Best = 1.812740e+03	FES = 118100	Diversity = 2.9104e+00
Generation 1190:	Best = 1.812721e+03	FES = 119100	Diversity = 2.9098e+00
Generation 1200:	Best = 1.802561e+03	FES = 120100	Diversity = 2.9084e+00
Generation 1210:	Best = 1.801003e+03	FES = 121100	Diversity = 2.9079e+00
Generation 1220:	Best = 1.801003e+03	FES = 122100	Diversity = 2.9079e+00
Generation 1230:	Best = 1.800963e+03	FES = 123100	Diversity = 2.9075e+00
Generation 1240:	Best = 1.799022e+03	FES = 124100	Diversity = 2.9069e+00
Generation 1250:	Best = 1.799022e+03	FES = 125100	Diversity = 2.9066e+00
Generation 1260:	Best = 1.799022e+03	FES = 126100	Diversity = 2.9062e+00
Generation 1270:	Best = 1.799022e+03	FES = 127100	Diversity = 2.9012e+00
Generation 1280:	Best = 1.798219e+03	FES = 128100	Diversity = 2.9012e+00
Generation 1290:	Best = 1.791139e+03	FES = 129100	Diversity = 2.9000e+00
Generation 1300:	Best = 1.790614e+03	FES = 130100	Diversity = 2.8969e+00
Generation 1310:	Best = 1.790303e+03	FES = 131100	Diversity = 2.8952e+00
Generation 1320:	Best = 1.790303e+03	FES = 132100	Diversity = 2.8941e+00
Generation 1330:	Best = 1.789747e+03	FES = 133100	Diversity = 2.8934e+00
Generation 1340:	Best = 1.789747e+03	FES = 134100	Diversity = 2.8921e+00
Generation 1350:	Best = 1.785078e+03	FES = 135100	Diversity = 2.8990e+00
Generation 1360:	Best = 1.783348e+03	FES = 136100	Diversity = 2.8974e+00
Generation 1370:	Best = 1.782462e+03	FES = 137100	Diversity = 2.8900e+00
Generation 1380:	Best = 1.782451e+03	FES = 138100	Diversity = 2.8827e+00
Generation 1390:	Best = 1.782451e+03	FES = 139100	Diversity = 2.8928e+00
Generation 1400:	Best = 1.782451e+03	FES = 140100	Diversity = 2.8899e+00
Generation 1410:	Best = 1.782376e+03	FES = 141100	Diversity = 2.8900e+00
Generation 1420:	Best = 1.781358e+03	FES = 142100	Diversity = 2.8890e+00
Generation 1430:	Best = 1.781225e+03	FES = 143100	Diversity = 2.8636e+00
Generation 1440:	Best = 1.781225e+03	FES = 144100	Diversity = 2.8633e+00
Generation 1450:	Best = 1.780952e+03	FES = 145100	Diversity = 2.8674e+00
Generation 1460:	Best = 1.779890e+03	FES = 146100	Diversity = 2.8521e+00
Generation 1470:	Best = 1.779778e+03	FES = 147100	Diversity = 2.8520e+00
Generation 1480:	Best = 1.779770e+03	FES = 148100	Diversity = 2.8462e+00
Generation 1490:	Best = 1.779770e+03	FES = 149100	Diversity = 2.8445e+00
Generation 1500:	Best = 1.779770e+03	FES = 150100	Diversity = 2.8442e+00
Generation 1510:	Best = 1.779770e+03	FES = 151100	Diversity = 2.8437e+00
Generation 1520:	Best = 1.777451e+03	FES = 152100	Diversity = 2.8401e+00

Generation 1530:	Best = 1.776541e+03	FES = 153100	Diversity = 2.8348e+00
Generation 1540:	Best = 1.776508e+03	FES = 154100	Diversity = 2.8339e+00
Generation 1550:	Best = 1.776508e+03	FES = 155100	Diversity = 2.8340e+00
Generation 1560:	Best = 1.776168e+03	FES = 156100	Diversity = 2.8486e+00
Generation 1570:	Best = 1.774926e+03	FES = 157100	Diversity = 2.8479e+00
Generation 1580:	Best = 1.774747e+03	FES = 158100	Diversity = 2.8473e+00
Generation 1590:	Best = 1.774747e+03	FES = 159100	Diversity = 2.8475e+00
Generation 1600:	Best = 1.774747e+03	FES = 160100	Diversity = 2.8376e+00
Generation 1610:	Best = 1.773814e+03	FES = 161100	Diversity = 2.8366e+00
Generation 1620:	Best = 1.771095e+03	FES = 162100	Diversity = 2.8355e+00
Generation 1630:	Best = 1.771095e+03	FES = 163100	Diversity = 2.8336e+00
Generation 1640:	Best = 1.771095e+03	FES = 164100	Diversity = 2.8332e+00
Generation 1650:	Best = 1.771095e+03	FES = 165100	Diversity = 2.8304e+00
Generation 1660:	Best = 1.771095e+03	FES = 166100	Diversity = 2.8300e+00
Generation 1670:	Best = 1.770961e+03	FES = 167100	Diversity = 2.8302e+00
Generation 1680:	Best = 1.768687e+03	FES = 168100	Diversity = 2.8287e+00
Generation 1690:	Best = 1.766178e+03	FES = 169100	Diversity = 2.8275e+00
Generation 1700:	Best = 1.765754e+03	FES = 170100	Diversity = 2.8267e+00
Generation 1710:	Best = 1.765754e+03	FES = 171100	Diversity = 2.8267e+00
Generation 1720:	Best = 1.765754e+03	FES = 172100	Diversity = 2.8263e+00
Generation 1730:	Best = 1.765754e+03	FES = 173100	Diversity = 2.8262e+00
Generation 1740:	Best = 1.765742e+03	FES = 174100	Diversity = 2.8258e+00
Generation 1750:	Best = 1.762013e+03	FES = 175100	Diversity = 2.8253e+00
Generation 1760:	Best = 1.762013e+03	FES = 176100	Diversity = 2.8214e+00
Generation 1770:	Best = 1.761998e+03	FES = 177100	Diversity = 2.8211e+00
Generation 1780:	Best = 1.750477e+03	FES = 178100	Diversity = 2.8137e+00
Generation 1790:	Best = 1.750477e+03	FES = 179100	Diversity = 2.8134e+00
Generation 1800:	Best = 1.750038e+03	FES = 180100	Diversity = 2.8131e+00
Generation 1810:	Best = 1.750038e+03	FES = 181100	Diversity = 2.8124e+00
Generation 1820:	Best = 1.750038e+03	FES = 182100	Diversity = 2.8031e+00
Generation 1830:	Best = 1.750038e+03	FES = 183100	Diversity = 2.8025e+00
Generation 1840:	Best = 1.750038e+03	FES = 184100	Diversity = 2.8016e+00
Generation 1850:	Best = 1.750038e+03	FES = 185100	Diversity = 2.7997e+00
Generation 1860:	Best = 1.750038e+03	FES = 186100	Diversity = 2.8071e+00
Generation 1870:	Best = 1.742621e+03	FES = 187100	Diversity = 2.8012e+00
Generation 1880:	Best = 1.742217e+03	FES = 188100	Diversity = 2.7986e+00
Generation 1890:	Best = 1.742217e+03	FES = 189100	Diversity = 2.7893e+00
Generation 1900:	Best = 1.742217e+03	FES = 190100	Diversity = 2.7895e+00
Generation 1910:	Best = 1.742217e+03	FES = 191100	Diversity = 2.7893e+00
Generation 1920:	Best = 1.742217e+03	FES = 192100	Diversity = 2.7888e+00
Generation 1930:	Best = 1.742217e+03	FES = 193100	Diversity = 2.7884e+00
Generation 1940:	Best = 1.742217e+03	FES = 194100	Diversity = 2.7881e+00
Generation 1950:	Best = 1.742217e+03	FES = 195100	Diversity = 2.7874e+00
Generation 1960:	Best = 1.742217e+03	FES = 196100	Diversity = 2.7863e+00
Generation 1970:	Best = 1.742217e+03	FES = 197100	Diversity = 2.7853e+00
Generation 1980:	Best = 1.742217e+03	FES = 198100	Diversity = 2.7853e+00
Generation 1990:	Best = 1.742217e+03	FES = 199100	Diversity = 2.7797e+00
Generation 2000:	Best = 1.742217e+03	FES = 200100	Diversity = 2.7515e+00
Generation 2010:	Best = 1.742217e+03	FES = 201100	Diversity = 2.7517e+00
Generation 2020:	Best = 1.741723e+03	FES = 202100	Diversity = 2.7518e+00
Generation 2030:	Best = 1.741723e+03	FES = 203100	Diversity = 2.7511e+00
Generation 2040:	Best = 1.741689e+03	FES = 204100	Diversity = 2.7512e+00
Generation 2050:	Best = 1.741689e+03	FES = 205100	Diversity = 2.7508e+00
Generation 2060:	Best = 1.741689e+03	FES = 206100	Diversity = 2.7422e+00

Generation 2070:	Best = 1.740010e+03	FES = 207100	Diversity = 2.7306e+00
Generation 2080:	Best = 1.739606e+03	FES = 208100	Diversity = 2.7305e+00
Generation 2090:	Best = 1.739574e+03	FES = 209100	Diversity = 2.7409e+00
Generation 2100:	Best = 1.739395e+03	FES = 210100	Diversity = 2.7404e+00
Generation 2110:	Best = 1.739395e+03	FES = 211100	Diversity = 2.7394e+00
Generation 2120:	Best = 1.739395e+03	FES = 212100	Diversity = 2.7542e+00
Generation 2130:	Best = 1.739395e+03	FES = 213100	Diversity = 2.7544e+00
Generation 2140:	Best = 1.739395e+03	FES = 214100	Diversity = 2.7536e+00
Generation 2150:	Best = 1.739395e+03	FES = 215100	Diversity = 2.7323e+00
Generation 2160:	Best = 1.739395e+03	FES = 216100	Diversity = 2.7308e+00
Generation 2170:	Best = 1.739395e+03	FES = 217100	Diversity = 2.7302e+00
Generation 2180:	Best = 1.739389e+03	FES = 218100	Diversity = 2.7300e+00
Generation 2190:	Best = 1.739389e+03	FES = 219100	Diversity = 2.7299e+00
Generation 2200:	Best = 1.739341e+03	FES = 220100	Diversity = 2.7288e+00
Generation 2210:	Best = 1.739341e+03	FES = 221100	Diversity = 2.7288e+00
Generation 2220:	Best = 1.739223e+03	FES = 222100	Diversity = 2.7097e+00
Generation 2230:	Best = 1.739223e+03	FES = 223100	Diversity = 2.7097e+00
Generation 2240:	Best = 1.739223e+03	FES = 224100	Diversity = 2.7117e+00
Generation 2250:	Best = 1.738790e+03	FES = 225100	Diversity = 2.7166e+00
Generation 2260:	Best = 1.738790e+03	FES = 226100	Diversity = 2.7161e+00
Generation 2270:	Best = 1.738790e+03	FES = 227100	Diversity = 2.7268e+00
Generation 2280:	Best = 1.738384e+03	FES = 228100	Diversity = 2.7257e+00
Generation 2290:	Best = 1.738381e+03	FES = 229100	Diversity = 2.6977e+00
Generation 2300:	Best = 1.738381e+03	FES = 230100	Diversity = 2.6966e+00
Generation 2310:	Best = 1.738322e+03	FES = 231100	Diversity = 2.6984e+00
Generation 2320:	Best = 1.736122e+03	FES = 232100	Diversity = 2.6846e+00
Generation 2330:	Best = 1.736028e+03	FES = 233100	Diversity = 2.6841e+00
Generation 2340:	Best = 1.735812e+03	FES = 234100	Diversity = 2.6836e+00
Generation 2350:	Best = 1.735812e+03	FES = 235100	Diversity = 2.6829e+00
Generation 2360:	Best = 1.734289e+03	FES = 236100	Diversity = 2.6824e+00
Generation 2370:	Best = 1.734289e+03	FES = 237100	Diversity = 2.6825e+00
Generation 2380:	Best = 1.734289e+03	FES = 238100	Diversity = 2.6808e+00
Generation 2390:	Best = 1.734289e+03	FES = 239100	Diversity = 2.6803e+00
Generation 2400:	Best = 1.734289e+03	FES = 240100	Diversity = 2.6793e+00
Generation 2410:	Best = 1.734289e+03	FES = 241100	Diversity = 2.6768e+00
Generation 2420:	Best = 1.733850e+03	FES = 242100	Diversity = 2.6769e+00
Generation 2430:	Best = 1.733850e+03	FES = 243100	Diversity = 2.6755e+00
Generation 2440:	Best = 1.733850e+03	FES = 244100	Diversity = 2.6739e+00
Generation 2450:	Best = 1.733823e+03	FES = 245100	Diversity = 2.6746e+00
Generation 2460:	Best = 1.733823e+03	FES = 246100	Diversity = 2.6744e+00
Generation 2470:	Best = 1.733823e+03	FES = 247100	Diversity = 2.6753e+00
Generation 2480:	Best = 1.733823e+03	FES = 248100	Diversity = 2.6745e+00
Generation 2490:	Best = 1.733823e+03	FES = 249100	Diversity = 2.6743e+00
Generation 2500:	Best = 1.733823e+03	FES = 250100	Diversity = 2.6615e+00
Generation 2510:	Best = 1.733823e+03	FES = 251100	Diversity = 2.6608e+00
Generation 2520:	Best = 1.733823e+03	FES = 252100	Diversity = 2.6607e+00
Generation 2530:	Best = 1.733768e+03	FES = 253100	Diversity = 2.6609e+00
Generation 2540:	Best = 1.733768e+03	FES = 254100	Diversity = 2.6607e+00
Generation 2550:	Best = 1.733768e+03	FES = 255100	Diversity = 2.6609e+00
Generation 2560:	Best = 1.733768e+03	FES = 256100	Diversity = 2.6607e+00
Generation 2570:	Best = 1.733768e+03	FES = 257100	Diversity = 2.6606e+00
Generation 2580:	Best = 1.733768e+03	FES = 258100	Diversity = 2.6600e+00
Generation 2590:	Best = 1.733768e+03	FES = 259100	Diversity = 2.6597e+00
Generation 2600:	Best = 1.733768e+03	FES = 260100	Diversity = 2.6595e+00

Generation 2610:	Best = 1.733768e+03	FES = 261100	Diversity = 2.6596e+00
Generation 2620:	Best = 1.733768e+03	FES = 262100	Diversity = 2.6599e+00
Generation 2630:	Best = 1.731017e+03	FES = 263100	Diversity = 2.6576e+00
Generation 2640:	Best = 1.730793e+03	FES = 264100	Diversity = 2.6573e+00
Generation 2650:	Best = 1.730733e+03	FES = 265100	Diversity = 2.6569e+00
Generation 2660:	Best = 1.727358e+03	FES = 266100	Diversity = 2.6565e+00
Generation 2670:	Best = 1.727358e+03	FES = 267100	Diversity = 2.6563e+00
Generation 2680:	Best = 1.726980e+03	FES = 268100	Diversity = 2.6558e+00
Generation 2690:	Best = 1.726980e+03	FES = 269100	Diversity = 2.6442e+00
Generation 2700:	Best = 1.726980e+03	FES = 270100	Diversity = 2.6441e+00
Generation 2710:	Best = 1.726913e+03	FES = 271100	Diversity = 2.6436e+00
Generation 2720:	Best = 1.726913e+03	FES = 272100	Diversity = 2.6432e+00
Generation 2730:	Best = 1.726548e+03	FES = 273100	Diversity = 2.6430e+00
Generation 2740:	Best = 1.726548e+03	FES = 274100	Diversity = 2.6450e+00
Generation 2750:	Best = 1.726338e+03	FES = 275100	Diversity = 2.6557e+00
Generation 2760:	Best = 1.722021e+03	FES = 276100	Diversity = 2.6557e+00
Generation 2770:	Best = 1.721937e+03	FES = 277100	Diversity = 2.6555e+00
Generation 2780:	Best = 1.721937e+03	FES = 278100	Diversity = 2.6556e+00
Generation 2790:	Best = 1.721937e+03	FES = 279100	Diversity = 2.6647e+00
Generation 2800:	Best = 1.721815e+03	FES = 280100	Diversity = 2.6517e+00
Generation 2810:	Best = 1.721668e+03	FES = 281100	Diversity = 2.6520e+00
Generation 2820:	Best = 1.721668e+03	FES = 282100	Diversity = 2.6512e+00
Generation 2830:	Best = 1.721668e+03	FES = 283100	Diversity = 2.6508e+00
Generation 2840:	Best = 1.721668e+03	FES = 284100	Diversity = 2.6507e+00
Generation 2850:	Best = 1.721665e+03	FES = 285100	Diversity = 2.6507e+00
Generation 2860:	Best = 1.721665e+03	FES = 286100	Diversity = 2.6506e+00
Generation 2870:	Best = 1.721302e+03	FES = 287100	Diversity = 2.6506e+00
Generation 2880:	Best = 1.720788e+03	FES = 288100	Diversity = 2.6500e+00
Generation 2890:	Best = 1.720788e+03	FES = 289100	Diversity = 2.6487e+00
Generation 2900:	Best = 1.720788e+03	FES = 290100	Diversity = 2.6482e+00
Generation 2910:	Best = 1.718566e+03	FES = 291100	Diversity = 2.6435e+00
Generation 2920:	Best = 1.718564e+03	FES = 292100	Diversity = 2.6097e+00
Generation 2930:	Best = 1.718564e+03	FES = 293100	Diversity = 2.6094e+00
Generation 2940:	Best = 1.718405e+03	FES = 294100	Diversity = 2.6105e+00
Generation 2950:	Best = 1.718405e+03	FES = 295100	Diversity = 2.6108e+00
Generation 2960:	Best = 1.718405e+03	FES = 296100	Diversity = 2.6109e+00
Generation 2970:	Best = 1.718405e+03	FES = 297100	Diversity = 2.6080e+00
Generation 2980:	Best = 1.718405e+03	FES = 298100	Diversity = 2.6048e+00
Generation 2990:	Best = 1.718403e+03	FES = 299100	Diversity = 2.6205e+00
Generation 3000:	Best = 1.718403e+03	FES = 300100	Diversity = 2.6206e+00
Generation 3010:	Best = 1.715687e+03	FES = 301100	Diversity = 2.6214e+00
Generation 3020:	Best = 1.715687e+03	FES = 302100	Diversity = 2.6215e+00
Generation 3030:	Best = 1.715687e+03	FES = 303100	Diversity = 2.6251e+00
Generation 3040:	Best = 1.715687e+03	FES = 304100	Diversity = 2.6257e+00
Generation 3050:	Best = 1.715687e+03	FES = 305100	Diversity = 2.6257e+00
Generation 3060:	Best = 1.715687e+03	FES = 306100	Diversity = 2.6257e+00
Generation 3070:	Best = 1.715687e+03	FES = 307100	Diversity = 2.6274e+00
Generation 3080:	Best = 1.715687e+03	FES = 308100	Diversity = 2.6279e+00
Generation 3090:	Best = 1.715622e+03	FES = 309100	Diversity = 2.6278e+00
Generation 3100:	Best = 1.715622e+03	FES = 310100	Diversity = 2.6269e+00
Generation 3110:	Best = 1.715622e+03	FES = 311100	Diversity = 2.6267e+00
Generation 3120:	Best = 1.715592e+03	FES = 312100	Diversity = 2.6300e+00
Generation 3130:	Best = 1.715592e+03	FES = 313100	Diversity = 2.6213e+00
Generation 3140:	Best = 1.715591e+03	FES = 314100	Diversity = 2.6217e+00

Generation 3150:	Best = 1.715539e+03	FES = 315100	Diversity = 2.6206e+00
Generation 3160:	Best = 1.715539e+03	FES = 316100	Diversity = 2.6211e+00
Generation 3170:	Best = 1.715539e+03	FES = 317100	Diversity = 2.6212e+00
Generation 3180:	Best = 1.715539e+03	FES = 318100	Diversity = 2.6213e+00
Generation 3190:	Best = 1.715539e+03	FES = 319100	Diversity = 2.6266e+00
Generation 3200:	Best = 1.715491e+03	FES = 320100	Diversity = 2.6265e+00
Generation 3210:	Best = 1.715491e+03	FES = 321100	Diversity = 2.6368e+00
Generation 3220:	Best = 1.715491e+03	FES = 322100	Diversity = 2.6367e+00
Generation 3230:	Best = 1.714933e+03	FES = 323100	Diversity = 2.6339e+00
Generation 3240:	Best = 1.714812e+03	FES = 324100	Diversity = 2.6312e+00
Generation 3250:	Best = 1.714812e+03	FES = 325100	Diversity = 2.6213e+00
Generation 3260:	Best = 1.714812e+03	FES = 326100	Diversity = 2.6213e+00
Generation 3270:	Best = 1.714812e+03	FES = 327100	Diversity = 2.6215e+00
Generation 3280:	Best = 1.714812e+03	FES = 328100	Diversity = 2.6211e+00
Generation 3290:	Best = 1.714812e+03	FES = 329100	Diversity = 2.6208e+00
Generation 3300:	Best = 1.714791e+03	FES = 330100	Diversity = 2.6210e+00
Generation 3310:	Best = 1.712028e+03	FES = 331100	Diversity = 2.6210e+00
Generation 3320:	Best = 1.712028e+03	FES = 332100	Diversity = 2.6200e+00
Generation 3330:	Best = 1.712028e+03	FES = 333100	Diversity = 2.6184e+00
Generation 3340:	Best = 1.712028e+03	FES = 334100	Diversity = 2.6178e+00
Generation 3350:	Best = 1.712028e+03	FES = 335100	Diversity = 2.6173e+00
Generation 3360:	Best = 1.711978e+03	FES = 336100	Diversity = 2.6171e+00
Generation 3370:	Best = 1.711978e+03	FES = 337100	Diversity = 2.6159e+00
Generation 3380:	Best = 1.711978e+03	FES = 338100	Diversity = 2.6153e+00
Generation 3390:	Best = 1.710828e+03	FES = 339100	Diversity = 2.6136e+00
Generation 3400:	Best = 1.710828e+03	FES = 340100	Diversity = 2.6130e+00
Generation 3410:	Best = 1.710828e+03	FES = 341100	Diversity = 2.6131e+00
Generation 3420:	Best = 1.710827e+03	FES = 342100	Diversity = 2.6125e+00
Generation 3430:	Best = 1.710827e+03	FES = 343100	Diversity = 2.6130e+00
Generation 3440:	Best = 1.710736e+03	FES = 344100	Diversity = 2.6126e+00
Generation 3450:	Best = 1.710736e+03	FES = 345100	Diversity = 2.6108e+00
Generation 3460:	Best = 1.710699e+03	FES = 346100	Diversity = 2.6097e+00
Generation 3470:	Best = 1.710699e+03	FES = 347100	Diversity = 2.6094e+00
Generation 3480:	Best = 1.710699e+03	FES = 348100	Diversity = 2.6091e+00
Generation 3490:	Best = 1.710699e+03	FES = 349100	Diversity = 2.6078e+00
Generation 3500:	Best = 1.708909e+03	FES = 350100	Diversity = 2.6072e+00
Generation 3510:	Best = 1.708410e+03	FES = 351100	Diversity = 2.6059e+00
Generation 3520:	Best = 1.708410e+03	FES = 352100	Diversity = 2.6046e+00
Generation 3530:	Best = 1.708409e+03	FES = 353100	Diversity = 2.5913e+00
Generation 3540:	Best = 1.708409e+03	FES = 354100	Diversity = 2.5909e+00
Generation 3550:	Best = 1.708409e+03	FES = 355100	Diversity = 2.5908e+00
Generation 3560:	Best = 1.708024e+03	FES = 356100	Diversity = 2.5906e+00
Generation 3570:	Best = 1.708024e+03	FES = 357100	Diversity = 2.5903e+00
Generation 3580:	Best = 1.706894e+03	FES = 358100	Diversity = 2.5902e+00
Generation 3590:	Best = 1.697287e+03	FES = 359100	Diversity = 2.5889e+00
Generation 3600:	Best = 1.697278e+03	FES = 360100	Diversity = 2.5890e+00
Generation 3610:	Best = 1.697278e+03	FES = 361100	Diversity = 2.5909e+00
Generation 3620:	Best = 1.697278e+03	FES = 362100	Diversity = 2.5909e+00
Generation 3630:	Best = 1.697278e+03	FES = 363100	Diversity = 2.5910e+00
Generation 3640:	Best = 1.697277e+03	FES = 364100	Diversity = 2.5912e+00
Generation 3650:	Best = 1.697277e+03	FES = 365100	Diversity = 2.5910e+00
Generation 3660:	Best = 1.697277e+03	FES = 366100	Diversity = 2.5913e+00
Generation 3670:	Best = 1.697277e+03	FES = 367100	Diversity = 2.5914e+00
Generation 3680:	Best = 1.697277e+03	FES = 368100	Diversity = 2.6012e+00

Generation 3690:	Best = 1.697277e+03	FES = 369100	Diversity = 2.6008e+00
Generation 3700:	Best = 1.695171e+03	FES = 370100	Diversity = 2.6004e+00
Generation 3710:	Best = 1.693571e+03	FES = 371100	Diversity = 2.6001e+00
Generation 3720:	Best = 1.693252e+03	FES = 372100	Diversity = 2.5972e+00
Generation 3730:	Best = 1.689908e+03	FES = 373100	Diversity = 2.5960e+00
Generation 3740:	Best = 1.688753e+03	FES = 374100	Diversity = 2.5956e+00
Generation 3750:	Best = 1.688753e+03	FES = 375100	Diversity = 2.6044e+00
Generation 3760:	Best = 1.688753e+03	FES = 376100	Diversity = 2.6046e+00
Generation 3770:	Best = 1.688753e+03	FES = 377100	Diversity = 2.6046e+00
Generation 3780:	Best = 1.688753e+03	FES = 378100	Diversity = 2.6150e+00
Generation 3790:	Best = 1.688753e+03	FES = 379100	Diversity = 2.6247e+00
Generation 3800:	Best = 1.687526e+03	FES = 380100	Diversity = 2.6244e+00
Generation 3810:	Best = 1.687526e+03	FES = 381100	Diversity = 2.6243e+00
Generation 3820:	Best = 1.687526e+03	FES = 382100	Diversity = 2.6240e+00
Generation 3830:	Best = 1.686293e+03	FES = 383100	Diversity = 2.6240e+00
Generation 3840:	Best = 1.686293e+03	FES = 384100	Diversity = 2.6240e+00
Generation 3850:	Best = 1.686293e+03	FES = 385100	Diversity = 2.6239e+00
Generation 3860:	Best = 1.686293e+03	FES = 386100	Diversity = 2.6238e+00
Generation 3870:	Best = 1.686293e+03	FES = 387100	Diversity = 2.6238e+00
Generation 3880:	Best = 1.686293e+03	FES = 388100	Diversity = 2.6217e+00
Generation 3890:	Best = 1.686293e+03	FES = 389100	Diversity = 2.6214e+00
Generation 3900:	Best = 1.682812e+03	FES = 390100	Diversity = 2.6212e+00
Generation 3910:	Best = 1.682811e+03	FES = 391100	Diversity = 2.6214e+00
Generation 3920:	Best = 1.682808e+03	FES = 392100	Diversity = 2.6216e+00
Generation 3930:	Best = 1.682808e+03	FES = 393100	Diversity = 2.6220e+00
Generation 3940:	Best = 1.682808e+03	FES = 394100	Diversity = 2.6214e+00
Generation 3950:	Best = 1.682768e+03	FES = 395100	Diversity = 2.6212e+00
Generation 3960:	Best = 1.682768e+03	FES = 396100	Diversity = 2.6211e+00
Generation 3970:	Best = 1.682768e+03	FES = 397100	Diversity = 2.6211e+00
Generation 3980:	Best = 1.682768e+03	FES = 398100	Diversity = 2.6212e+00
Generation 3990:	Best = 1.682187e+03	FES = 399100	Diversity = 2.6210e+00
Generation 4000:	Best = 1.682187e+03	FES = 400100	Diversity = 2.6221e+00
Generation 4010:	Best = 1.682173e+03	FES = 401100	Diversity = 2.6210e+00
Generation 4020:	Best = 1.682173e+03	FES = 402100	Diversity = 2.6207e+00
Generation 4030:	Best = 1.682173e+03	FES = 403100	Diversity = 2.6207e+00
Generation 4040:	Best = 1.682173e+03	FES = 404100	Diversity = 2.6205e+00
Generation 4050:	Best = 1.682171e+03	FES = 405100	Diversity = 2.6207e+00
Generation 4060:	Best = 1.682171e+03	FES = 406100	Diversity = 2.6210e+00
Generation 4070:	Best = 1.682171e+03	FES = 407100	Diversity = 2.6205e+00
Generation 4080:	Best = 1.682171e+03	FES = 408100	Diversity = 2.6215e+00
Generation 4090:	Best = 1.681987e+03	FES = 409100	Diversity = 2.6213e+00
Generation 4100:	Best = 1.681987e+03	FES = 410100	Diversity = 2.6215e+00
Generation 4110:	Best = 1.681987e+03	FES = 411100	Diversity = 2.6214e+00
Generation 4120:	Best = 1.681952e+03	FES = 412100	Diversity = 2.6215e+00
Generation 4130:	Best = 1.681952e+03	FES = 413100	Diversity = 2.6204e+00
Generation 4140:	Best = 1.681952e+03	FES = 414100	Diversity = 2.6189e+00
Generation 4150:	Best = 1.681948e+03	FES = 415100	Diversity = 2.6188e+00
Generation 4160:	Best = 1.681865e+03	FES = 416100	Diversity = 2.6189e+00
Generation 4170:	Best = 1.676833e+03	FES = 417100	Diversity = 2.6196e+00
Generation 4180:	Best = 1.676833e+03	FES = 418100	Diversity = 2.6181e+00
Generation 4190:	Best = 1.676827e+03	FES = 419100	Diversity = 2.6185e+00
Generation 4200:	Best = 1.675425e+03	FES = 420100	Diversity = 2.6184e+00
Generation 4210:	Best = 1.675425e+03	FES = 421100	Diversity = 2.6180e+00
Generation 4220:	Best = 1.675425e+03	FES = 422100	Diversity = 2.6178e+00

Generation 4230:	Best = 1.675385e+03	FES = 423100	Diversity = 2.6173e+00
Generation 4240:	Best = 1.675273e+03	FES = 424100	Diversity = 2.6176e+00
Generation 4250:	Best = 1.675036e+03	FES = 425100	Diversity = 2.6172e+00
Generation 4260:	Best = 1.675036e+03	FES = 426100	Diversity = 2.6171e+00
Generation 4270:	Best = 1.675036e+03	FES = 427100	Diversity = 2.6153e+00
Generation 4280:	Best = 1.673181e+03	FES = 428100	Diversity = 2.6144e+00
Generation 4290:	Best = 1.673124e+03	FES = 429100	Diversity = 2.6140e+00
Generation 4300:	Best = 1.672540e+03	FES = 430100	Diversity = 2.6133e+00
Generation 4310:	Best = 1.672540e+03	FES = 431100	Diversity = 2.6123e+00
Generation 4320:	Best = 1.672540e+03	FES = 432100	Diversity = 2.6118e+00
Generation 4330:	Best = 1.672540e+03	FES = 433100	Diversity = 2.6101e+00
Generation 4340:	Best = 1.672169e+03	FES = 434100	Diversity = 2.6097e+00
Generation 4350:	Best = 1.672169e+03	FES = 435100	Diversity = 2.6097e+00
Generation 4360:	Best = 1.671890e+03	FES = 436100	Diversity = 2.6091e+00
Generation 4370:	Best = 1.669554e+03	FES = 437100	Diversity = 2.6095e+00
Generation 4380:	Best = 1.669554e+03	FES = 438100	Diversity = 2.6094e+00
Generation 4390:	Best = 1.667965e+03	FES = 439100	Diversity = 2.6083e+00
Generation 4400:	Best = 1.667713e+03	FES = 440100	Diversity = 2.6085e+00
Generation 4410:	Best = 1.667713e+03	FES = 441100	Diversity = 2.6083e+00
Generation 4420:	Best = 1.667713e+03	FES = 442100	Diversity = 2.6083e+00
Generation 4430:	Best = 1.667713e+03	FES = 443100	Diversity = 2.6075e+00
Generation 4440:	Best = 1.667713e+03	FES = 444100	Diversity = 2.6075e+00
Generation 4450:	Best = 1.666919e+03	FES = 445100	Diversity = 2.6075e+00
Generation 4460:	Best = 1.666919e+03	FES = 446100	Diversity = 2.6067e+00
Generation 4470:	Best = 1.666919e+03	FES = 447100	Diversity = 2.6064e+00
Generation 4480:	Best = 1.666919e+03	FES = 448100	Diversity = 2.6064e+00
Generation 4490:	Best = 1.666919e+03	FES = 449100	Diversity = 2.6060e+00
Generation 4500:	Best = 1.666919e+03	FES = 450100	Diversity = 2.6060e+00
Generation 4510:	Best = 1.665932e+03	FES = 451100	Diversity = 2.6061e+00
Generation 4520:	Best = 1.665917e+03	FES = 452100	Diversity = 2.6058e+00
Generation 4530:	Best = 1.665500e+03	FES = 453100	Diversity = 2.6048e+00
Generation 4540:	Best = 1.665366e+03	FES = 454100	Diversity = 2.6047e+00
Generation 4550:	Best = 1.665366e+03	FES = 455100	Diversity = 2.6042e+00
Generation 4560:	Best = 1.665163e+03	FES = 456100	Diversity = 2.6037e+00
Generation 4570:	Best = 1.665163e+03	FES = 457100	Diversity = 2.5999e+00
Generation 4580:	Best = 1.665163e+03	FES = 458100	Diversity = 2.5953e+00
Generation 4590:	Best = 1.665163e+03	FES = 459100	Diversity = 2.5954e+00
Generation 4600:	Best = 1.665049e+03	FES = 460100	Diversity = 2.5951e+00
Generation 4610:	Best = 1.665049e+03	FES = 461100	Diversity = 2.5950e+00
Generation 4620:	Best = 1.665022e+03	FES = 462100	Diversity = 2.5958e+00
Generation 4630:	Best = 1.665022e+03	FES = 463100	Diversity = 2.5958e+00
Generation 4640:	Best = 1.665022e+03	FES = 464100	Diversity = 2.5957e+00
Generation 4650:	Best = 1.665022e+03	FES = 465100	Diversity = 2.5957e+00
Generation 4660:	Best = 1.664886e+03	FES = 466100	Diversity = 2.5953e+00
Generation 4670:	Best = 1.664886e+03	FES = 467100	Diversity = 2.5953e+00
Generation 4680:	Best = 1.664886e+03	FES = 468100	Diversity = 2.5954e+00
Generation 4690:	Best = 1.662273e+03	FES = 469100	Diversity = 2.5952e+00
Generation 4700:	Best = 1.662273e+03	FES = 470100	Diversity = 2.6010e+00
Generation 4710:	Best = 1.662007e+03	FES = 471100	Diversity = 2.6010e+00
Generation 4720:	Best = 1.662007e+03	FES = 472100	Diversity = 2.6008e+00
Generation 4730:	Best = 1.662007e+03	FES = 473100	Diversity = 2.6007e+00
Generation 4740:	Best = 1.661999e+03	FES = 474100	Diversity = 2.6007e+00
Generation 4750:	Best = 1.661999e+03	FES = 475100	Diversity = 2.6004e+00
Generation 4760:	Best = 1.661999e+03	FES = 476100	Diversity = 2.5996e+00

Generation 4770:	Best = 1.661999e+03	FES = 477100	Diversity = 2.6002e+00
Generation 4780:	Best = 1.661999e+03	FES = 478100	Diversity = 2.6000e+00
Generation 4790:	Best = 1.661999e+03	FES = 479100	Diversity = 2.5995e+00
Generation 4800:	Best = 1.661999e+03	FES = 480100	Diversity = 2.5995e+00
Generation 4810:	Best = 1.661968e+03	FES = 481100	Diversity = 2.5953e+00
Generation 4820:	Best = 1.661968e+03	FES = 482100	Diversity = 2.5948e+00
Generation 4830:	Best = 1.661968e+03	FES = 483100	Diversity = 2.5949e+00
Generation 4840:	Best = 1.661968e+03	FES = 484100	Diversity = 2.5948e+00
Generation 4850:	Best = 1.661938e+03	FES = 485100	Diversity = 2.5948e+00
Generation 4860:	Best = 1.661938e+03	FES = 486100	Diversity = 2.5948e+00
Generation 4870:	Best = 1.661928e+03	FES = 487100	Diversity = 2.5948e+00
Generation 4880:	Best = 1.661928e+03	FES = 488100	Diversity = 2.5947e+00
Generation 4890:	Best = 1.661928e+03	FES = 489100	Diversity = 2.5948e+00
Generation 4900:	Best = 1.661928e+03	FES = 490100	Diversity = 2.5855e+00
Generation 4910:	Best = 1.661259e+03	FES = 491100	Diversity = 2.5854e+00
Generation 4920:	Best = 1.661182e+03	FES = 492100	Diversity = 2.5852e+00
Generation 4930:	Best = 1.661181e+03	FES = 493100	Diversity = 2.5914e+00
Generation 4940:	Best = 1.661178e+03	FES = 494100	Diversity = 2.5914e+00
Generation 4950:	Best = 1.661178e+03	FES = 495100	Diversity = 2.5876e+00
Generation 4960:	Best = 1.661178e+03	FES = 496100	Diversity = 2.5851e+00
Generation 4970:	Best = 1.661027e+03	FES = 497100	Diversity = 2.5844e+00
Generation 4980:	Best = 1.661015e+03	FES = 498100	Diversity = 2.5846e+00
Generation 4990:	Best = 1.661012e+03	FES = 499100	Diversity = 2.5846e+00
Generation 5000:	Best = 1.660960e+03	FES = 500100	Diversity = 2.5845e+00
Generation 5010:	Best = 1.660954e+03	FES = 501100	Diversity = 2.5845e+00
Generation 5020:	Best = 1.660954e+03	FES = 502100	Diversity = 2.5845e+00
Generation 5030:	Best = 1.660848e+03	FES = 503100	Diversity = 2.5843e+00
Generation 5040:	Best = 1.660502e+03	FES = 504100	Diversity = 2.5841e+00
Generation 5050:	Best = 1.660502e+03	FES = 505100	Diversity = 2.5854e+00
Generation 5060:	Best = 1.660502e+03	FES = 506100	Diversity = 2.6011e+00
Generation 5070:	Best = 1.660502e+03	FES = 507100	Diversity = 2.6009e+00
Generation 5080:	Best = 1.660387e+03	FES = 508100	Diversity = 2.6009e+00
Generation 5090:	Best = 1.660387e+03	FES = 509100	Diversity = 2.6009e+00
Generation 5100:	Best = 1.660387e+03	FES = 510100	Diversity = 2.6006e+00
Generation 5110:	Best = 1.660387e+03	FES = 511100	Diversity = 2.6006e+00
Generation 5120:	Best = 1.660108e+03	FES = 512100	Diversity = 2.6005e+00
Generation 5130:	Best = 1.660085e+03	FES = 513100	Diversity = 2.6085e+00
Generation 5140:	Best = 1.660085e+03	FES = 514100	Diversity = 2.6084e+00
Generation 5150:	Best = 1.660066e+03	FES = 515100	Diversity = 2.6084e+00
Generation 5160:	Best = 1.657807e+03	FES = 516100	Diversity = 2.6085e+00
Generation 5170:	Best = 1.657807e+03	FES = 517100	Diversity = 2.6085e+00
Generation 5180:	Best = 1.657807e+03	FES = 518100	Diversity = 2.6087e+00
Generation 5190:	Best = 1.657793e+03	FES = 519100	Diversity = 2.6070e+00
Generation 5200:	Best = 1.657792e+03	FES = 520100	Diversity = 2.6073e+00
Generation 5210:	Best = 1.657792e+03	FES = 521100	Diversity = 2.6081e+00
Generation 5220:	Best = 1.657792e+03	FES = 522100	Diversity = 2.5973e+00
Generation 5230:	Best = 1.657792e+03	FES = 523100	Diversity = 2.5973e+00
Generation 5240:	Best = 1.657792e+03	FES = 524100	Diversity = 2.5967e+00
Generation 5250:	Best = 1.657792e+03	FES = 525100	Diversity = 2.5967e+00
Generation 5260:	Best = 1.657787e+03	FES = 526100	Diversity = 2.5966e+00
Generation 5270:	Best = 1.657787e+03	FES = 527100	Diversity = 2.5966e+00
Generation 5280:	Best = 1.657787e+03	FES = 528100	Diversity = 2.5957e+00
Generation 5290:	Best = 1.657787e+03	FES = 529100	Diversity = 2.5957e+00
Generation 5300:	Best = 1.657787e+03	FES = 530100	Diversity = 2.5956e+00

Generation 5310:	Best = 1.657787e+03	FES = 531100	Diversity = 2.5946e+00
Generation 5320:	Best = 1.657784e+03	FES = 532100	Diversity = 2.5941e+00
Generation 5330:	Best = 1.657784e+03	FES = 533100	Diversity = 2.5940e+00
Generation 5340:	Best = 1.657784e+03	FES = 534100	Diversity = 2.5930e+00
Generation 5350:	Best = 1.656928e+03	FES = 535100	Diversity = 2.5929e+00
Generation 5360:	Best = 1.656928e+03	FES = 536100	Diversity = 2.5927e+00
Generation 5370:	Best = 1.656928e+03	FES = 537100	Diversity = 2.5928e+00
Generation 5380:	Best = 1.656928e+03	FES = 538100	Diversity = 2.5928e+00
Generation 5390:	Best = 1.656927e+03	FES = 539100	Diversity = 2.5927e+00
Generation 5400:	Best = 1.656927e+03	FES = 540100	Diversity = 2.5928e+00
Generation 5410:	Best = 1.656927e+03	FES = 541100	Diversity = 2.5929e+00
Generation 5420:	Best = 1.656927e+03	FES = 542100	Diversity = 2.5930e+00
Generation 5430:	Best = 1.656838e+03	FES = 543100	Diversity = 2.5930e+00
Generation 5440:	Best = 1.656774e+03	FES = 544100	Diversity = 2.5925e+00
Generation 5450:	Best = 1.656774e+03	FES = 545100	Diversity = 2.5922e+00
Generation 5460:	Best = 1.656774e+03	FES = 546100	Diversity = 2.5914e+00
Generation 5470:	Best = 1.655587e+03	FES = 547100	Diversity = 2.5916e+00
Generation 5480:	Best = 1.655587e+03	FES = 548100	Diversity = 2.5955e+00
Generation 5490:	Best = 1.655223e+03	FES = 549100	Diversity = 2.5955e+00
Generation 5500:	Best = 1.655223e+03	FES = 550100	Diversity = 2.5941e+00
Generation 5510:	Best = 1.655223e+03	FES = 551100	Diversity = 2.5941e+00
Generation 5520:	Best = 1.655223e+03	FES = 552100	Diversity = 2.5940e+00
Generation 5530:	Best = 1.655223e+03	FES = 553100	Diversity = 2.5941e+00
Generation 5540:	Best = 1.654965e+03	FES = 554100	Diversity = 2.5941e+00
Generation 5550:	Best = 1.654965e+03	FES = 555100	Diversity = 2.5941e+00
Generation 5560:	Best = 1.654944e+03	FES = 556100	Diversity = 2.5937e+00
Generation 5570:	Best = 1.652938e+03	FES = 557100	Diversity = 2.5935e+00
Generation 5580:	Best = 1.652938e+03	FES = 558100	Diversity = 2.5936e+00
Generation 5590:	Best = 1.652938e+03	FES = 559100	Diversity = 2.5936e+00
Generation 5600:	Best = 1.652938e+03	FES = 560100	Diversity = 2.5936e+00
Generation 5610:	Best = 1.649756e+03	FES = 561100	Diversity = 2.5938e+00
Generation 5620:	Best = 1.649736e+03	FES = 562100	Diversity = 2.5937e+00
Generation 5630:	Best = 1.649736e+03	FES = 563100	Diversity = 2.5937e+00
Generation 5640:	Best = 1.649736e+03	FES = 564100	Diversity = 2.5936e+00
Generation 5650:	Best = 1.648396e+03	FES = 565100	Diversity = 2.5936e+00
Generation 5660:	Best = 1.646308e+03	FES = 566100	Diversity = 2.5933e+00
Generation 5670:	Best = 1.646308e+03	FES = 567100	Diversity = 2.5932e+00
Generation 5680:	Best = 1.646308e+03	FES = 568100	Diversity = 2.5931e+00
Generation 5690:	Best = 1.646308e+03	FES = 569100	Diversity = 2.5927e+00
Generation 5700:	Best = 1.646308e+03	FES = 570100	Diversity = 2.5925e+00
Generation 5710:	Best = 1.646308e+03	FES = 571100	Diversity = 2.5885e+00
Generation 5720:	Best = 1.646308e+03	FES = 572100	Diversity = 2.5885e+00
Generation 5730:	Best = 1.646308e+03	FES = 573100	Diversity = 2.5886e+00
Generation 5740:	Best = 1.646308e+03	FES = 574100	Diversity = 2.5808e+00
Generation 5750:	Best = 1.646308e+03	FES = 575100	Diversity = 2.5874e+00
Generation 5760:	Best = 1.646308e+03	FES = 576100	Diversity = 2.5866e+00
Generation 5770:	Best = 1.646308e+03	FES = 577100	Diversity = 2.5867e+00
Generation 5780:	Best = 1.646308e+03	FES = 578100	Diversity = 2.5873e+00
Generation 5790:	Best = 1.646258e+03	FES = 579100	Diversity = 2.5873e+00
Generation 5800:	Best = 1.646258e+03	FES = 580100	Diversity = 2.5873e+00
Generation 5810:	Best = 1.646258e+03	FES = 581100	Diversity = 2.5873e+00
Generation 5820:	Best = 1.646258e+03	FES = 582100	Diversity = 2.5874e+00
Generation 5830:	Best = 1.645508e+03	FES = 583100	Diversity = 2.5878e+00
Generation 5840:	Best = 1.645508e+03	FES = 584100	Diversity = 2.5880e+00

Generation 5850:	Best = 1.645508e+03	FES = 585100	Diversity = 2.5880e+00
Generation 5860:	Best = 1.641426e+03	FES = 586100	Diversity = 2.5880e+00
Generation 5870:	Best = 1.641426e+03	FES = 587100	Diversity = 2.5880e+00
Generation 5880:	Best = 1.641426e+03	FES = 588100	Diversity = 2.5880e+00
Generation 5890:	Best = 1.641426e+03	FES = 589100	Diversity = 2.5880e+00
Generation 5900:	Best = 1.641426e+03	FES = 590100	Diversity = 2.5880e+00
Generation 5910:	Best = 1.641426e+03	FES = 591100	Diversity = 2.5880e+00
Generation 5920:	Best = 1.641426e+03	FES = 592100	Diversity = 2.5879e+00
Generation 5930:	Best = 1.641426e+03	FES = 593100	Diversity = 2.5878e+00
Generation 5940:	Best = 1.641426e+03	FES = 594100	Diversity = 2.5877e+00
Generation 5950:	Best = 1.641426e+03	FES = 595100	Diversity = 2.5877e+00
Generation 5960:	Best = 1.641426e+03	FES = 596100	Diversity = 2.5874e+00
Generation 5970:	Best = 1.641277e+03	FES = 597100	Diversity = 2.5873e+00
Generation 5980:	Best = 1.640728e+03	FES = 598100	Diversity = 2.5875e+00
Generation 5990:	Best = 1.640728e+03	FES = 599100	Diversity = 2.5864e+00
Generation 6000:	Best = 1.640728e+03	FES = 600100	Diversity = 2.5865e+00
Generation 6010:	Best = 1.637556e+03	FES = 601100	Diversity = 2.5861e+00
Generation 6020:	Best = 1.637556e+03	FES = 602100	Diversity = 2.5853e+00
Generation 6030:	Best = 1.637556e+03	FES = 603100	Diversity = 2.5853e+00
Generation 6040:	Best = 1.637556e+03	FES = 604100	Diversity = 2.5856e+00
Generation 6050:	Best = 1.637556e+03	FES = 605100	Diversity = 2.5856e+00
Generation 6060:	Best = 1.636639e+03	FES = 606100	Diversity = 2.5856e+00
Generation 6070:	Best = 1.631386e+03	FES = 607100	Diversity = 2.5817e+00
Generation 6080:	Best = 1.628929e+03	FES = 608100	Diversity = 2.5818e+00
Generation 6090:	Best = 1.628929e+03	FES = 609100	Diversity = 2.5815e+00
Generation 6100:	Best = 1.628929e+03	FES = 610100	Diversity = 2.5816e+00
Generation 6110:	Best = 1.628929e+03	FES = 611100	Diversity = 2.5815e+00
Generation 6120:	Best = 1.628929e+03	FES = 612100	Diversity = 2.5815e+00
Generation 6130:	Best = 1.628929e+03	FES = 613100	Diversity = 2.5814e+00
Generation 6140:	Best = 1.628860e+03	FES = 614100	Diversity = 2.5812e+00
Generation 6150:	Best = 1.628860e+03	FES = 615100	Diversity = 2.5812e+00
Generation 6160:	Best = 1.618380e+03	FES = 616100	Diversity = 2.5820e+00
Generation 6170:	Best = 1.618380e+03	FES = 617100	Diversity = 2.5857e+00
Generation 6180:	Best = 1.618071e+03	FES = 618100	Diversity = 2.5857e+00
Generation 6190:	Best = 1.618071e+03	FES = 619100	Diversity = 2.5857e+00
Generation 6200:	Best = 1.618071e+03	FES = 620100	Diversity = 2.5857e+00
Generation 6210:	Best = 1.618071e+03	FES = 621100	Diversity = 2.5857e+00
Generation 6220:	Best = 1.618071e+03	FES = 622100	Diversity = 2.5857e+00
Generation 6230:	Best = 1.618071e+03	FES = 623100	Diversity = 2.5924e+00
Generation 6240:	Best = 1.618071e+03	FES = 624100	Diversity = 2.5925e+00
Generation 6250:	Best = 1.618071e+03	FES = 625100	Diversity = 2.5727e+00
Generation 6260:	Best = 1.618071e+03	FES = 626100	Diversity = 2.5726e+00
Generation 6270:	Best = 1.618071e+03	FES = 627100	Diversity = 2.5726e+00
Generation 6280:	Best = 1.616639e+03	FES = 628100	Diversity = 2.5725e+00
Generation 6290:	Best = 1.616639e+03	FES = 629100	Diversity = 2.5726e+00
Generation 6300:	Best = 1.616639e+03	FES = 630100	Diversity = 2.5714e+00
Generation 6310:	Best = 1.616639e+03	FES = 631100	Diversity = 2.5715e+00
Generation 6320:	Best = 1.616639e+03	FES = 632100	Diversity = 2.5714e+00
Generation 6330:	Best = 1.614984e+03	FES = 633100	Diversity = 2.5715e+00
Generation 6340:	Best = 1.614835e+03	FES = 634100	Diversity = 2.5714e+00
Generation 6350:	Best = 1.614835e+03	FES = 635100	Diversity = 2.5694e+00
Generation 6360:	Best = 1.614835e+03	FES = 636100	Diversity = 2.5757e+00
Generation 6370:	Best = 1.614575e+03	FES = 637100	Diversity = 2.5750e+00
Generation 6380:	Best = 1.614372e+03	FES = 638100	Diversity = 2.5727e+00

Generation 6390:	Best = 1.614372e+03	FES = 639100	Diversity = 2.5710e+00
Generation 6400:	Best = 1.614372e+03	FES = 640100	Diversity = 2.5710e+00
Generation 6410:	Best = 1.614365e+03	FES = 641100	Diversity = 2.5710e+00
Generation 6420:	Best = 1.614365e+03	FES = 642100	Diversity = 2.5701e+00
Generation 6430:	Best = 1.614365e+03	FES = 643100	Diversity = 2.5672e+00
Generation 6440:	Best = 1.614299e+03	FES = 644100	Diversity = 2.5672e+00
Generation 6450:	Best = 1.613635e+03	FES = 645100	Diversity = 2.5667e+00
Generation 6460:	Best = 1.613635e+03	FES = 646100	Diversity = 2.5662e+00
Generation 6470:	Best = 1.613635e+03	FES = 647100	Diversity = 2.5661e+00
Generation 6480:	Best = 1.613361e+03	FES = 648100	Diversity = 2.5662e+00
Generation 6490:	Best = 1.613361e+03	FES = 649100	Diversity = 2.5663e+00
Generation 6500:	Best = 1.612258e+03	FES = 650100	Diversity = 2.5663e+00
Generation 6510:	Best = 1.612258e+03	FES = 651100	Diversity = 2.5664e+00
Generation 6520:	Best = 1.612258e+03	FES = 652100	Diversity = 2.5661e+00
Generation 6530:	Best = 1.611368e+03	FES = 653100	Diversity = 2.5654e+00
Generation 6540:	Best = 1.610947e+03	FES = 654100	Diversity = 2.5655e+00
Generation 6550:	Best = 1.610947e+03	FES = 655100	Diversity = 2.5652e+00
Generation 6560:	Best = 1.610947e+03	FES = 656100	Diversity = 2.5652e+00
Generation 6570:	Best = 1.610947e+03	FES = 657100	Diversity = 2.5653e+00
Generation 6580:	Best = 1.610947e+03	FES = 658100	Diversity = 2.5653e+00
Generation 6590:	Best = 1.610947e+03	FES = 659100	Diversity = 2.5641e+00
Generation 6600:	Best = 1.610947e+03	FES = 660100	Diversity = 2.5640e+00
Generation 6610:	Best = 1.610183e+03	FES = 661100	Diversity = 2.5641e+00
Generation 6620:	Best = 1.610053e+03	FES = 662100	Diversity = 2.5634e+00
Generation 6630:	Best = 1.610053e+03	FES = 663100	Diversity = 2.5633e+00
Generation 6640:	Best = 1.608598e+03	FES = 664100	Diversity = 2.5634e+00
Generation 6650:	Best = 1.608598e+03	FES = 665100	Diversity = 2.5634e+00
Generation 6660:	Best = 1.608598e+03	FES = 666100	Diversity = 2.5634e+00
Generation 6670:	Best = 1.608598e+03	FES = 667100	Diversity = 2.5634e+00
Generation 6680:	Best = 1.608515e+03	FES = 668100	Diversity = 2.5633e+00
Generation 6690:	Best = 1.607777e+03	FES = 669100	Diversity = 2.5634e+00
Generation 6700:	Best = 1.607777e+03	FES = 670100	Diversity = 2.5634e+00
Generation 6710:	Best = 1.607337e+03	FES = 671100	Diversity = 2.5615e+00
Generation 6720:	Best = 1.607337e+03	FES = 672100	Diversity = 2.5615e+00
Generation 6730:	Best = 1.607337e+03	FES = 673100	Diversity = 2.5612e+00
Generation 6740:	Best = 1.607337e+03	FES = 674100	Diversity = 2.5612e+00
Generation 6750:	Best = 1.607337e+03	FES = 675100	Diversity = 2.5610e+00
Generation 6760:	Best = 1.607337e+03	FES = 676100	Diversity = 2.5610e+00
Generation 6770:	Best = 1.607337e+03	FES = 677100	Diversity = 2.5610e+00
Generation 6780:	Best = 1.606003e+03	FES = 678100	Diversity = 2.5607e+00
Generation 6790:	Best = 1.605803e+03	FES = 679100	Diversity = 2.5605e+00
Generation 6800:	Best = 1.605803e+03	FES = 680100	Diversity = 2.5526e+00
Generation 6810:	Best = 1.603396e+03	FES = 681100	Diversity = 2.5526e+00
Generation 6820:	Best = 1.603396e+03	FES = 682100	Diversity = 2.5523e+00
Generation 6830:	Best = 1.603396e+03	FES = 683100	Diversity = 2.5516e+00
Generation 6840:	Best = 1.602849e+03	FES = 684100	Diversity = 2.5518e+00
Generation 6850:	Best = 1.602849e+03	FES = 685100	Diversity = 2.5575e+00
Generation 6860:	Best = 1.602849e+03	FES = 686100	Diversity = 2.5575e+00
Generation 6870:	Best = 1.602849e+03	FES = 687100	Diversity = 2.5576e+00
Generation 6880:	Best = 1.602848e+03	FES = 688100	Diversity = 2.5574e+00
Generation 6890:	Best = 1.602848e+03	FES = 689100	Diversity = 2.5570e+00
Generation 6900:	Best = 1.602848e+03	FES = 690100	Diversity = 2.5570e+00
Generation 6910:	Best = 1.599016e+03	FES = 691100	Diversity = 2.5570e+00
Generation 6920:	Best = 1.599016e+03	FES = 692100	Diversity = 2.5569e+00

Generation 6930:	Best = 1.598906e+03	FES = 693100	Diversity = 2.5569e+00
Generation 6940:	Best = 1.598906e+03	FES = 694100	Diversity = 2.5562e+00
Generation 6950:	Best = 1.598154e+03	FES = 695100	Diversity = 2.5561e+00
Generation 6960:	Best = 1.598154e+03	FES = 696100	Diversity = 2.5561e+00
Generation 6970:	Best = 1.598154e+03	FES = 697100	Diversity = 2.5506e+00
Generation 6980:	Best = 1.598154e+03	FES = 698100	Diversity = 2.5572e+00
Generation 6990:	Best = 1.597374e+03	FES = 699100	Diversity = 2.5575e+00
Generation 7000:	Best = 1.597374e+03	FES = 700100	Diversity = 2.5575e+00
Generation 7010:	Best = 1.597374e+03	FES = 701100	Diversity = 2.5576e+00
Generation 7020:	Best = 1.597220e+03	FES = 702100	Diversity = 2.5631e+00
Generation 7030:	Best = 1.593597e+03	FES = 703100	Diversity = 2.5630e+00
Generation 7040:	Best = 1.593509e+03	FES = 704100	Diversity = 2.5630e+00
Generation 7050:	Best = 1.593509e+03	FES = 705100	Diversity = 2.5489e+00
Generation 7060:	Best = 1.593509e+03	FES = 706100	Diversity = 2.5488e+00
Generation 7070:	Best = 1.593509e+03	FES = 707100	Diversity = 2.5491e+00
Generation 7080:	Best = 1.593509e+03	FES = 708100	Diversity = 2.5485e+00
Generation 7090:	Best = 1.592583e+03	FES = 709100	Diversity = 2.5485e+00
Generation 7100:	Best = 1.592583e+03	FES = 710100	Diversity = 2.5486e+00
Generation 7110:	Best = 1.592583e+03	FES = 711100	Diversity = 2.5486e+00
Generation 7120:	Best = 1.592583e+03	FES = 712100	Diversity = 2.5503e+00
Generation 7130:	Best = 1.592583e+03	FES = 713100	Diversity = 2.5502e+00
Generation 7140:	Best = 1.592583e+03	FES = 714100	Diversity = 2.5509e+00
Generation 7150:	Best = 1.592583e+03	FES = 715100	Diversity = 2.5509e+00
Generation 7160:	Best = 1.589767e+03	FES = 716100	Diversity = 2.5509e+00
Generation 7170:	Best = 1.589767e+03	FES = 717100	Diversity = 2.5509e+00
Generation 7180:	Best = 1.589767e+03	FES = 718100	Diversity = 2.5512e+00
Generation 7190:	Best = 1.589767e+03	FES = 719100	Diversity = 2.5460e+00
Generation 7200:	Best = 1.589767e+03	FES = 720100	Diversity = 2.5459e+00
Generation 7210:	Best = 1.589767e+03	FES = 721100	Diversity = 2.5462e+00
Generation 7220:	Best = 1.589767e+03	FES = 722100	Diversity = 2.5463e+00
Generation 7230:	Best = 1.589767e+03	FES = 723100	Diversity = 2.5476e+00
Generation 7240:	Best = 1.589767e+03	FES = 724100	Diversity = 2.5477e+00
Generation 7250:	Best = 1.589767e+03	FES = 725100	Diversity = 2.5477e+00
Generation 7260:	Best = 1.589767e+03	FES = 726100	Diversity = 2.5477e+00
Generation 7270:	Best = 1.589767e+03	FES = 727100	Diversity = 2.5476e+00
Generation 7280:	Best = 1.589767e+03	FES = 728100	Diversity = 2.5477e+00
Generation 7290:	Best = 1.589767e+03	FES = 729100	Diversity = 2.5476e+00
Generation 7300:	Best = 1.589767e+03	FES = 730100	Diversity = 2.5547e+00
Generation 7310:	Best = 1.589767e+03	FES = 731100	Diversity = 2.5597e+00
Generation 7320:	Best = 1.589767e+03	FES = 732100	Diversity = 2.5593e+00
Generation 7330:	Best = 1.589767e+03	FES = 733100	Diversity = 2.5562e+00
Generation 7340:	Best = 1.589767e+03	FES = 734100	Diversity = 2.5563e+00
Generation 7350:	Best = 1.589421e+03	FES = 735100	Diversity = 2.5595e+00
Generation 7360:	Best = 1.589421e+03	FES = 736100	Diversity = 2.5581e+00
Generation 7370:	Best = 1.589421e+03	FES = 737100	Diversity = 2.5581e+00
Generation 7380:	Best = 1.587825e+03	FES = 738100	Diversity = 2.5579e+00
Generation 7390:	Best = 1.587825e+03	FES = 739100	Diversity = 2.5579e+00
Generation 7400:	Best = 1.587825e+03	FES = 740100	Diversity = 2.5580e+00
Generation 7410:	Best = 1.587825e+03	FES = 741100	Diversity = 2.5579e+00
Generation 7420:	Best = 1.587825e+03	FES = 742100	Diversity = 2.5659e+00
Generation 7430:	Best = 1.587825e+03	FES = 743100	Diversity = 2.5660e+00
Generation 7440:	Best = 1.587825e+03	FES = 744100	Diversity = 2.5661e+00
Generation 7450:	Best = 1.587825e+03	FES = 745100	Diversity = 2.5661e+00
Generation 7460:	Best = 1.587825e+03	FES = 746100	Diversity = 2.5662e+00

Generation 7470:	Best = 1.586631e+03	FES = 747100	Diversity = 2.5571e+00
Generation 7480:	Best = 1.586631e+03	FES = 748100	Diversity = 2.5576e+00
Generation 7490:	Best = 1.586631e+03	FES = 749100	Diversity = 2.5575e+00
Generation 7500:	Best = 1.586631e+03	FES = 750100	Diversity = 2.5574e+00
Generation 7510:	Best = 1.586612e+03	FES = 751100	Diversity = 2.5574e+00
Generation 7520:	Best = 1.586605e+03	FES = 752100	Diversity = 2.5528e+00
Generation 7530:	Best = 1.586590e+03	FES = 753100	Diversity = 2.5527e+00
Generation 7540:	Best = 1.586590e+03	FES = 754100	Diversity = 2.5525e+00
Generation 7550:	Best = 1.580527e+03	FES = 755100	Diversity = 2.5520e+00
Generation 7560:	Best = 1.580511e+03	FES = 756100	Diversity = 2.5522e+00
Generation 7570:	Best = 1.580466e+03	FES = 757100	Diversity = 2.5522e+00
Generation 7580:	Best = 1.580465e+03	FES = 758100	Diversity = 2.5521e+00
Generation 7590:	Best = 1.580465e+03	FES = 759100	Diversity = 2.5522e+00
Generation 7600:	Best = 1.580465e+03	FES = 760100	Diversity = 2.5523e+00
Generation 7610:	Best = 1.580465e+03	FES = 761100	Diversity = 2.5521e+00
Generation 7620:	Best = 1.580465e+03	FES = 762100	Diversity = 2.5523e+00
Generation 7630:	Best = 1.580465e+03	FES = 763100	Diversity = 2.5524e+00
Generation 7640:	Best = 1.579586e+03	FES = 764100	Diversity = 2.5523e+00
Generation 7650:	Best = 1.579542e+03	FES = 765100	Diversity = 2.5475e+00
Generation 7660:	Best = 1.576895e+03	FES = 766100	Diversity = 2.5475e+00
Generation 7670:	Best = 1.576895e+03	FES = 767100	Diversity = 2.5472e+00
Generation 7680:	Best = 1.576895e+03	FES = 768100	Diversity = 2.5456e+00
Generation 7690:	Best = 1.576895e+03	FES = 769100	Diversity = 2.5457e+00
Generation 7700:	Best = 1.576895e+03	FES = 770100	Diversity = 2.5460e+00
Generation 7710:	Best = 1.576840e+03	FES = 771100	Diversity = 2.5461e+00
Generation 7720:	Best = 1.576840e+03	FES = 772100	Diversity = 2.5462e+00
Generation 7730:	Best = 1.576840e+03	FES = 773100	Diversity = 2.5390e+00
Generation 7740:	Best = 1.576840e+03	FES = 774100	Diversity = 2.5374e+00
Generation 7750:	Best = 1.576840e+03	FES = 775100	Diversity = 2.5486e+00
Generation 7760:	Best = 1.576840e+03	FES = 776100	Diversity = 2.5488e+00
Generation 7770:	Best = 1.576840e+03	FES = 777100	Diversity = 2.5622e+00
Generation 7780:	Best = 1.576840e+03	FES = 778100	Diversity = 2.5622e+00
Generation 7790:	Best = 1.575618e+03	FES = 779100	Diversity = 2.5612e+00
Generation 7800:	Best = 1.575618e+03	FES = 780100	Diversity = 2.5609e+00
Generation 7810:	Best = 1.575618e+03	FES = 781100	Diversity = 2.5609e+00
Generation 7820:	Best = 1.575618e+03	FES = 782100	Diversity = 2.5607e+00
Generation 7830:	Best = 1.575618e+03	FES = 783100	Diversity = 2.5607e+00
Generation 7840:	Best = 1.575618e+03	FES = 784100	Diversity = 2.5576e+00
Generation 7850:	Best = 1.575618e+03	FES = 785100	Diversity = 2.5575e+00
Generation 7860:	Best = 1.574361e+03	FES = 786100	Diversity = 2.5575e+00
Generation 7870:	Best = 1.574361e+03	FES = 787100	Diversity = 2.5575e+00
Generation 7880:	Best = 1.574361e+03	FES = 788100	Diversity = 2.5579e+00
Generation 7890:	Best = 1.574194e+03	FES = 789100	Diversity = 2.5578e+00
Generation 7900:	Best = 1.574194e+03	FES = 790100	Diversity = 2.5564e+00
Generation 7910:	Best = 1.574194e+03	FES = 791100	Diversity = 2.5582e+00
Generation 7920:	Best = 1.573800e+03	FES = 792100	Diversity = 2.5586e+00
Generation 7930:	Best = 1.573800e+03	FES = 793100	Diversity = 2.5587e+00
Generation 7940:	Best = 1.573800e+03	FES = 794100	Diversity = 2.5588e+00
Generation 7950:	Best = 1.573800e+03	FES = 795100	Diversity = 2.5586e+00
Generation 7960:	Best = 1.573800e+03	FES = 796100	Diversity = 2.5584e+00
Generation 7970:	Best = 1.573245e+03	FES = 797100	Diversity = 2.5589e+00
Generation 7980:	Best = 1.573245e+03	FES = 798100	Diversity = 2.5589e+00
Generation 7990:	Best = 1.573245e+03	FES = 799100	Diversity = 2.5591e+00
Generation 8000:	Best = 1.573245e+03	FES = 800100	Diversity = 2.5621e+00

Generation 8010:	Best = 1.573245e+03	FES = 801100	Diversity = 2.5622e+00
Generation 8020:	Best = 1.573245e+03	FES = 802100	Diversity = 2.5621e+00
Generation 8030:	Best = 1.573229e+03	FES = 803100	Diversity = 2.5571e+00
Generation 8040:	Best = 1.572463e+03	FES = 804100	Diversity = 2.5572e+00
Generation 8050:	Best = 1.572436e+03	FES = 805100	Diversity = 2.5572e+00
Generation 8060:	Best = 1.572436e+03	FES = 806100	Diversity = 2.5572e+00
Generation 8070:	Best = 1.572436e+03	FES = 807100	Diversity = 2.5573e+00
Generation 8080:	Best = 1.572436e+03	FES = 808100	Diversity = 2.5575e+00
Generation 8090:	Best = 1.572436e+03	FES = 809100	Diversity = 2.5572e+00
Generation 8100:	Best = 1.572428e+03	FES = 810100	Diversity = 2.5571e+00
Generation 8110:	Best = 1.572428e+03	FES = 811100	Diversity = 2.5607e+00
Generation 8120:	Best = 1.572368e+03	FES = 812100	Diversity = 2.5608e+00
Generation 8130:	Best = 1.572368e+03	FES = 813100	Diversity = 2.5612e+00
Generation 8140:	Best = 1.572368e+03	FES = 814100	Diversity = 2.5581e+00
Generation 8150:	Best = 1.572357e+03	FES = 815100	Diversity = 2.5581e+00
Generation 8160:	Best = 1.572357e+03	FES = 816100	Diversity = 2.5584e+00
Generation 8170:	Best = 1.572357e+03	FES = 817100	Diversity = 2.5583e+00
Generation 8180:	Best = 1.572357e+03	FES = 818100	Diversity = 2.5583e+00
Generation 8190:	Best = 1.572357e+03	FES = 819100	Diversity = 2.5582e+00
Generation 8200:	Best = 1.572348e+03	FES = 820100	Diversity = 2.5678e+00
Generation 8210:	Best = 1.572348e+03	FES = 821100	Diversity = 2.5498e+00
Generation 8220:	Best = 1.569720e+03	FES = 822100	Diversity = 2.5495e+00
Generation 8230:	Best = 1.569720e+03	FES = 823100	Diversity = 2.5312e+00
Generation 8240:	Best = 1.569687e+03	FES = 824100	Diversity = 2.5303e+00
Generation 8250:	Best = 1.568086e+03	FES = 825100	Diversity = 2.5309e+00
Generation 8260:	Best = 1.568053e+03	FES = 826100	Diversity = 2.5312e+00
Generation 8270:	Best = 1.566977e+03	FES = 827100	Diversity = 2.5230e+00
Generation 8280:	Best = 1.566977e+03	FES = 828100	Diversity = 2.5245e+00
Generation 8290:	Best = 1.566977e+03	FES = 829100	Diversity = 2.5245e+00
Generation 8300:	Best = 1.566736e+03	FES = 830100	Diversity = 2.5245e+00
Generation 8310:	Best = 1.563047e+03	FES = 831100	Diversity = 2.5245e+00
Generation 8320:	Best = 1.563047e+03	FES = 832100	Diversity = 2.5245e+00
Generation 8330:	Best = 1.563047e+03	FES = 833100	Diversity = 2.5243e+00
Generation 8340:	Best = 1.563047e+03	FES = 834100	Diversity = 2.5243e+00
Generation 8350:	Best = 1.563047e+03	FES = 835100	Diversity = 2.5114e+00
Generation 8360:	Best = 1.563047e+03	FES = 836100	Diversity = 2.5038e+00
Generation 8370:	Best = 1.563047e+03	FES = 837100	Diversity = 2.5004e+00
Generation 8380:	Best = 1.563047e+03	FES = 838100	Diversity = 2.5001e+00
Generation 8390:	Best = 1.563047e+03	FES = 839100	Diversity = 2.5007e+00
Generation 8400:	Best = 1.563047e+03	FES = 840100	Diversity = 2.4955e+00
Generation 8410:	Best = 1.563047e+03	FES = 841100	Diversity = 2.4956e+00
Generation 8420:	Best = 1.563047e+03	FES = 842100	Diversity = 2.4956e+00
Generation 8430:	Best = 1.563047e+03	FES = 843100	Diversity = 2.4957e+00
Generation 8440:	Best = 1.563047e+03	FES = 844100	Diversity = 2.4957e+00
Generation 8450:	Best = 1.563047e+03	FES = 845100	Diversity = 2.4828e+00
Generation 8460:	Best = 1.563047e+03	FES = 846100	Diversity = 2.4657e+00
Generation 8470:	Best = 1.563047e+03	FES = 847100	Diversity = 2.4648e+00
Generation 8480:	Best = 1.563047e+03	FES = 848100	Diversity = 2.4495e+00
Generation 8490:	Best = 1.563047e+03	FES = 849100	Diversity = 2.4391e+00
Generation 8500:	Best = 1.563047e+03	FES = 850100	Diversity = 2.4395e+00
Generation 8510:	Best = 1.563047e+03	FES = 851100	Diversity = 2.4395e+00
Generation 8520:	Best = 1.563047e+03	FES = 852100	Diversity = 2.4395e+00
Generation 8530:	Best = 1.563047e+03	FES = 853100	Diversity = 2.4395e+00
Generation 8540:	Best = 1.563047e+03	FES = 854100	Diversity = 2.4394e+00

Generation 8550:	Best = 1.563047e+03	FES = 855100	Diversity = 2.4394e+00
Generation 8560:	Best = 1.563047e+03	FES = 856100	Diversity = 2.4394e+00
Generation 8570:	Best = 1.563047e+03	FES = 857100	Diversity = 2.4397e+00
Generation 8580:	Best = 1.563047e+03	FES = 858100	Diversity = 2.4397e+00
Generation 8590:	Best = 1.563047e+03	FES = 859100	Diversity = 2.4397e+00
Generation 8600:	Best = 1.563047e+03	FES = 860100	Diversity = 2.4357e+00
Generation 8610:	Best = 1.563047e+03	FES = 861100	Diversity = 2.4355e+00
Generation 8620:	Best = 1.563047e+03	FES = 862100	Diversity = 2.4355e+00
Generation 8630:	Best = 1.563047e+03	FES = 863100	Diversity = 2.4355e+00
Generation 8640:	Best = 1.563047e+03	FES = 864100	Diversity = 2.4374e+00
Generation 8650:	Best = 1.563047e+03	FES = 865100	Diversity = 2.4374e+00
Generation 8660:	Best = 1.563047e+03	FES = 866100	Diversity = 2.4374e+00
Generation 8670:	Best = 1.563047e+03	FES = 867100	Diversity = 2.4374e+00
Generation 8680:	Best = 1.563047e+03	FES = 868100	Diversity = 2.4434e+00
Generation 8690:	Best = 1.563047e+03	FES = 869100	Diversity = 2.4437e+00
Generation 8700:	Best = 1.563047e+03	FES = 870100	Diversity = 2.4436e+00
Generation 8710:	Best = 1.563047e+03	FES = 871100	Diversity = 2.4436e+00
Generation 8720:	Best = 1.563047e+03	FES = 872100	Diversity = 2.4430e+00
Generation 8730:	Best = 1.563047e+03	FES = 873100	Diversity = 2.4424e+00
Generation 8740:	Best = 1.563047e+03	FES = 874100	Diversity = 2.4426e+00
Generation 8750:	Best = 1.563047e+03	FES = 875100	Diversity = 2.4426e+00
Generation 8760:	Best = 1.563047e+03	FES = 876100	Diversity = 2.4462e+00
Generation 8770:	Best = 1.563047e+03	FES = 877100	Diversity = 2.4461e+00
Generation 8780:	Best = 1.563047e+03	FES = 878100	Diversity = 2.4462e+00
Generation 8790:	Best = 1.563047e+03	FES = 879100	Diversity = 2.4462e+00
Generation 8800:	Best = 1.563047e+03	FES = 880100	Diversity = 2.4462e+00
Generation 8810:	Best = 1.563047e+03	FES = 881100	Diversity = 2.4462e+00
Generation 8820:	Best = 1.563047e+03	FES = 882100	Diversity = 2.4462e+00
Generation 8830:	Best = 1.563047e+03	FES = 883100	Diversity = 2.4401e+00
Generation 8840:	Best = 1.563047e+03	FES = 884100	Diversity = 2.4398e+00
Generation 8850:	Best = 1.563047e+03	FES = 885100	Diversity = 2.4397e+00
Generation 8860:	Best = 1.563047e+03	FES = 886100	Diversity = 2.4397e+00
Generation 8870:	Best = 1.563047e+03	FES = 887100	Diversity = 2.4439e+00
Generation 8880:	Best = 1.562301e+03	FES = 888100	Diversity = 2.4436e+00
Generation 8890:	Best = 1.562301e+03	FES = 889100	Diversity = 2.4421e+00
Generation 8900:	Best = 1.556371e+03	FES = 890100	Diversity = 2.4327e+00
Generation 8910:	Best = 1.545576e+03	FES = 891100	Diversity = 2.4160e+00
Generation 8920:	Best = 1.545161e+03	FES = 892100	Diversity = 2.4101e+00
Generation 8930:	Best = 1.540684e+03	FES = 893100	Diversity = 2.4099e+00
Generation 8940:	Best = 1.539862e+03	FES = 894100	Diversity = 2.4009e+00
Generation 8950:	Best = 1.538851e+03	FES = 895100	Diversity = 2.4000e+00
Generation 8960:	Best = 1.536757e+03	FES = 896100	Diversity = 2.3999e+00
Generation 8970:	Best = 1.536713e+03	FES = 897100	Diversity = 2.3842e+00
Generation 8980:	Best = 1.536638e+03	FES = 898100	Diversity = 2.3772e+00
Generation 8990:	Best = 1.535911e+03	FES = 899100	Diversity = 2.3687e+00
Generation 9000:	Best = 1.535771e+03	FES = 900100	Diversity = 2.3642e+00
Generation 9010:	Best = 1.535451e+03	FES = 901100	Diversity = 2.3665e+00
Generation 9020:	Best = 1.535426e+03	FES = 902100	Diversity = 2.3714e+00
Generation 9030:	Best = 1.535139e+03	FES = 903100	Diversity = 2.3589e+00
Generation 9040:	Best = 1.535084e+03	FES = 904100	Diversity = 2.3588e+00
Generation 9050:	Best = 1.535026e+03	FES = 905100	Diversity = 2.3630e+00
Generation 9060:	Best = 1.534578e+03	FES = 906100	Diversity = 2.3575e+00
Generation 9070:	Best = 1.534401e+03	FES = 907100	Diversity = 2.3487e+00
Generation 9080:	Best = 1.533497e+03	FES = 908100	Diversity = 2.3487e+00

Generation 9090:	Best = 1.532865e+03	FES = 909100	Diversity = 2.3495e+00
Generation 9100:	Best = 1.532455e+03	FES = 910100	Diversity = 2.3461e+00
Generation 9110:	Best = 1.532349e+03	FES = 911100	Diversity = 2.3505e+00
Generation 9120:	Best = 1.532321e+03	FES = 912100	Diversity = 2.3400e+00
Generation 9130:	Best = 1.532287e+03	FES = 913100	Diversity = 2.3212e+00
Generation 9140:	Best = 1.532270e+03	FES = 914100	Diversity = 2.3144e+00
Generation 9150:	Best = 1.532260e+03	FES = 915100	Diversity = 2.3073e+00
Generation 9160:	Best = 1.532256e+03	FES = 916100	Diversity = 2.3072e+00
Generation 9170:	Best = 1.532240e+03	FES = 917100	Diversity = 2.3062e+00
Generation 9180:	Best = 1.532236e+03	FES = 918100	Diversity = 2.3071e+00
Generation 9190:	Best = 1.532226e+03	FES = 919100	Diversity = 2.3062e+00
Generation 9200:	Best = 1.532223e+03	FES = 920100	Diversity = 2.3059e+00
Generation 9210:	Best = 1.532220e+03	FES = 921100	Diversity = 2.2916e+00
Generation 9220:	Best = 1.532220e+03	FES = 922100	Diversity = 2.2787e+00
Generation 9230:	Best = 1.532220e+03	FES = 923100	Diversity = 2.2794e+00
Generation 9240:	Best = 1.532219e+03	FES = 924100	Diversity = 2.2791e+00
Generation 9250:	Best = 1.532219e+03	FES = 925100	Diversity = 2.2523e+00
Generation 9260:	Best = 1.532219e+03	FES = 926100	Diversity = 2.2294e+00
Generation 9270:	Best = 1.532219e+03	FES = 927100	Diversity = 2.2300e+00
Generation 9280:	Best = 1.532218e+03	FES = 928100	Diversity = 2.2063e+00
Generation 9290:	Best = 1.532218e+03	FES = 929100	Diversity = 2.1903e+00
Generation 9300:	Best = 1.532218e+03	FES = 930100	Diversity = 2.1884e+00
Generation 9310:	Best = 1.529366e+03	FES = 931100	Diversity = 2.1856e+00
Generation 9320:	Best = 1.529024e+03	FES = 932100	Diversity = 2.1733e+00
Generation 9330:	Best = 1.528602e+03	FES = 933100	Diversity = 2.1673e+00
Generation 9340:	Best = 1.528144e+03	FES = 934100	Diversity = 2.1533e+00
Generation 9350:	Best = 1.528023e+03	FES = 935100	Diversity = 2.1209e+00
Generation 9360:	Best = 1.525528e+03	FES = 936100	Diversity = 2.1109e+00
Generation 9370:	Best = 1.525511e+03	FES = 937100	Diversity = 2.1000e+00
Generation 9380:	Best = 1.524822e+03	FES = 938100	Diversity = 2.0855e+00
Generation 9390:	Best = 1.522120e+03	FES = 939100	Diversity = 2.0414e+00
Generation 9400:	Best = 1.519816e+03	FES = 940100	Diversity = 2.0268e+00
Generation 9410:	Best = 1.518851e+03	FES = 941100	Diversity = 1.9843e+00
Generation 9420:	Best = 1.518692e+03	FES = 942100	Diversity = 1.9720e+00
Generation 9430:	Best = 1.518456e+03	FES = 943100	Diversity = 1.9176e+00
Generation 9440:	Best = 1.518418e+03	FES = 944100	Diversity = 1.8233e+00
Generation 9450:	Best = 1.518368e+03	FES = 945100	Diversity = 1.7313e+00
Generation 9460:	Best = 1.518329e+03	FES = 946100	Diversity = 1.5123e+00
Generation 9470:	Best = 1.510571e+03	FES = 947100	Diversity = 1.3485e+00
Generation 9480:	Best = 1.510237e+03	FES = 948100	Diversity = 1.2610e+00
Generation 9490:	Best = 1.509643e+03	FES = 949100	Diversity = 1.0340e+00
Generation 9500:	Best = 1.501002e+03	FES = 950100	Diversity = 9.4034e-01
Generation 9510:	Best = 1.499495e+03	FES = 951100	Diversity = 8.2091e-01
Generation 9520:	Best = 1.495068e+03	FES = 952100	Diversity = 7.4978e-01
Generation 9530:	Best = 1.491943e+03	FES = 953100	Diversity = 7.3616e-01
Generation 9540:	Best = 1.490642e+03	FES = 954100	Diversity = 7.2410e-01
Generation 9550:	Best = 1.489677e+03	FES = 955100	Diversity = 7.1644e-01
Generation 9560:	Best = 1.488826e+03	FES = 956100	Diversity = 6.2656e-01
Generation 9570:	Best = 1.488112e+03	FES = 957100	Diversity = 6.1916e-01
Generation 9580:	Best = 1.486136e+03	FES = 958100	Diversity = 4.8802e-01
Generation 9590:	Best = 1.483176e+03	FES = 959100	Diversity = 3.7786e-01
Generation 9600:	Best = 1.479696e+03	FES = 960100	Diversity = 1.2644e-01
Generation 9610:	Best = 1.479099e+03	FES = 961100	Diversity = 1.1627e-01
Generation 9620:	Best = 1.478205e+03	FES = 962100	Diversity = 1.1129e-01

Generation 9630:	Best = 1.475487e+03	FES = 963100	Diversity = 8.0361e-02
Generation 9640:	Best = 1.472577e+03	FES = 964100	Diversity = 7.5991e-02
Generation 9650:	Best = 1.472474e+03	FES = 965100	Diversity = 6.5861e-02
Generation 9660:	Best = 1.472458e+03	FES = 966100	Diversity = 7.0302e-02
Generation 9670:	Best = 1.472180e+03	FES = 967100	Diversity = 7.7710e-02
Generation 9680:	Best = 1.469735e+03	FES = 968100	Diversity = 7.3042e-02
Generation 9690:	Best = 1.468659e+03	FES = 969100	Diversity = 7.8872e-02
Generation 9700:	Best = 1.468009e+03	FES = 970100	Diversity = 7.5756e-02
Generation 9710:	Best = 1.467733e+03	FES = 971100	Diversity = 7.2275e-02
Generation 9720:	Best = 1.467424e+03	FES = 972100	Diversity = 8.7544e-02
Generation 9730:	Best = 1.466995e+03	FES = 973100	Diversity = 9.1990e-02
Generation 9740:	Best = 1.466872e+03	FES = 974100	Diversity = 9.2429e-02
Generation 9750:	Best = 1.466728e+03	FES = 975100	Diversity = 9.5976e-02
Generation 9760:	Best = 1.466532e+03	FES = 976100	Diversity = 9.4746e-02
Generation 9770:	Best = 1.465580e+03	FES = 977100	Diversity = 1.0392e-01
Generation 9780:	Best = 1.465580e+03	FES = 978100	Diversity = 1.0475e-01
Generation 9790:	Best = 1.465172e+03	FES = 979100	Diversity = 8.8803e-02
Generation 9800:	Best = 1.464283e+03	FES = 980100	Diversity = 7.0851e-02
Generation 9810:	Best = 1.464261e+03	FES = 981100	Diversity = 6.0387e-02
Generation 9820:	Best = 1.464124e+03	FES = 982100	Diversity = 5.6545e-02
Generation 9830:	Best = 1.463916e+03	FES = 983100	Diversity = 5.3762e-02
Generation 9840:	Best = 1.463886e+03	FES = 984100	Diversity = 5.4525e-02
Generation 9850:	Best = 1.463845e+03	FES = 985100	Diversity = 5.1822e-02
Generation 9860:	Best = 1.463828e+03	FES = 986100	Diversity = 4.9137e-02
Generation 9870:	Best = 1.463822e+03	FES = 987100	Diversity = 4.9082e-02
Generation 9880:	Best = 1.463803e+03	FES = 988100	Diversity = 4.9107e-02
Generation 9890:	Best = 1.463796e+03	FES = 989100	Diversity = 4.7037e-02
Generation 9900:	Best = 1.463789e+03	FES = 990100	Diversity = 4.6792e-02
Generation 9910:	Best = 1.463697e+03	FES = 991100	Diversity = 4.9044e-02
Generation 9920:	Best = 1.463697e+03	FES = 992100	Diversity = 4.9046e-02
Generation 9930:	Best = 1.463684e+03	FES = 993100	Diversity = 5.1567e-02
Generation 9940:	Best = 1.463682e+03	FES = 994100	Diversity = 5.1502e-02
Generation 9950:	Best = 1.463665e+03	FES = 995100	Diversity = 5.1499e-02
Generation 9960:	Best = 1.463663e+03	FES = 996100	Diversity = 5.1493e-02
Generation 9970:	Best = 1.463659e+03	FES = 997100	Diversity = 4.8817e-02
Generation 9980:	Best = 1.463656e+03	FES = 998100	Diversity = 4.8746e-02
Generation 9990:	Best = 1.463655e+03	FES = 999100	Diversity = 4.8726e-02

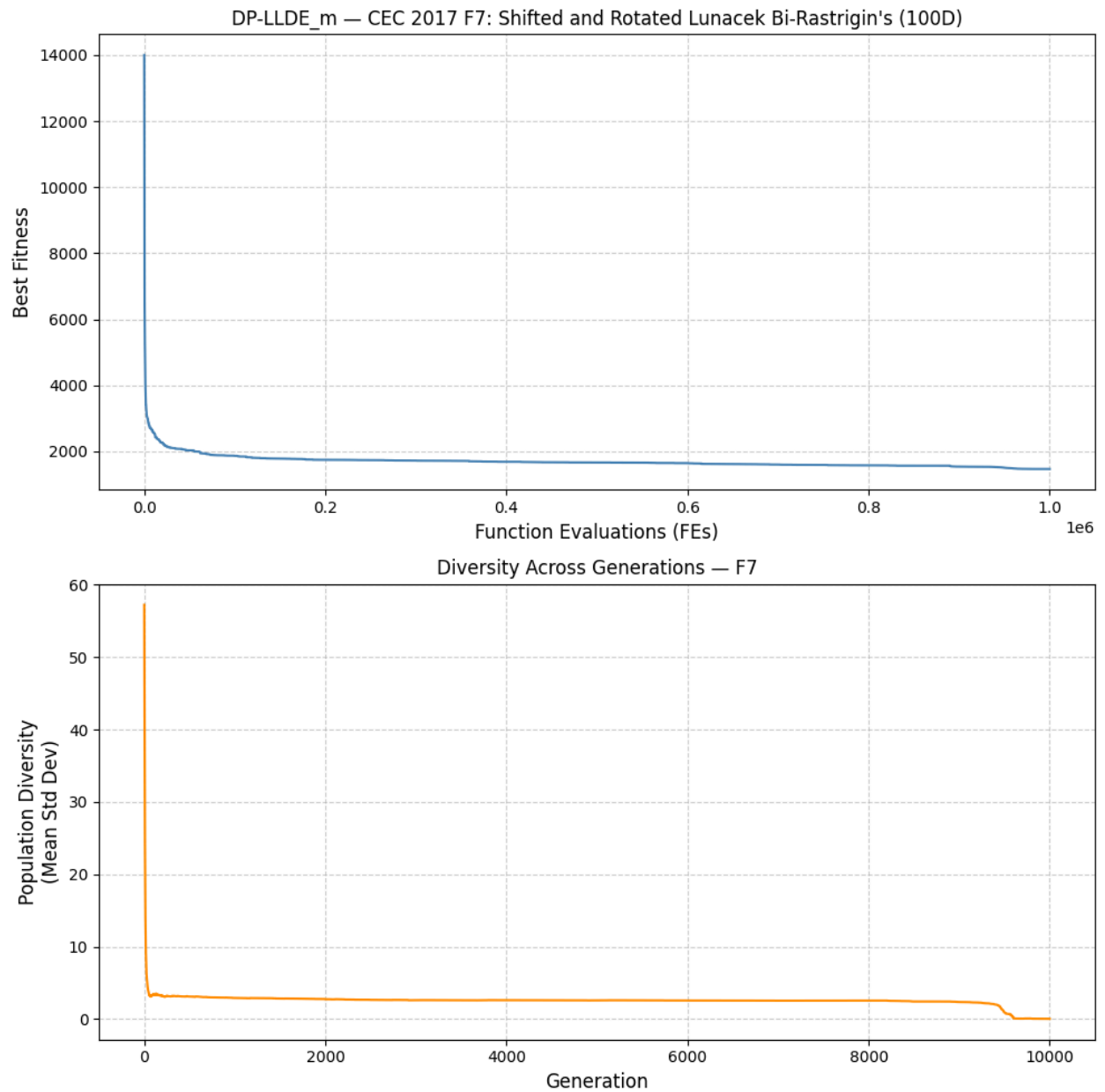
Optimization completed!

Final Best Fitness = 1.463655e+03

Total Function Evaluations = 1000000

```
=====
=
Results:
  Best Fitness           : 1.463655e+03
  Global Optimum         : 7.000000e+02
  Error (|best-optimum|) : 7.636549e+02
  Best Solution (5 dim)  : [-3.21962159e+01  3.77960164e+00 -1.11534667e+01
-2.84646928e-02
  1.99191869e+01]
  D^(0) (init. diversity): 5.7220e+01
=====
=
```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results_2/all_results_checkpoint.pkl



Graph saved as convergence_F7.png

✓ Plot saved to: /content/drive/My Drive/thesis_results_2/plots/convergence_F7.png

```
In [ ]: # =====  
# F8 — Shifted and Rotated Non-Continuous Rastrigin  
# =====  
run_and_plot('F8', make_wrapper(simple.f8), 800.0, "Shifted and Rotated Non-Co
```

```

=====
=
DP-LLDE_m on CEC 2017 F8 – Single Run
=====
=
Function      : Shifted and Rotated Non-Continuous Rastrigin's (100D)
Bounds       : [-100, 100]
Global optimum : 800.0
Initial D^(0) = 5.7220e+01 | Diversity threshold = 5.7220e+00
Generation 0: Best Fitness = 3.562373e+03 | Diversity = 5.7220e+01
Generation 10: Best = 2.457860e+03 FES = 1100 Diversity = 2.3419e+01
Generation 20: Best = 2.209997e+03 FES = 2100 Diversity = 1.4579e+01
Generation 30: Best = 2.108135e+03 FES = 3100 Diversity = 1.2185e+01
Generation 40: Best = 2.095524e+03 FES = 4100 Diversity = 1.1304e+01
Generation 50: Best = 2.060975e+03 FES = 5100 Diversity = 1.0578e+01
Generation 60: Best = 2.057877e+03 FES = 6100 Diversity = 9.7298e+00
Generation 70: Best = 2.025052e+03 FES = 7100 Diversity = 9.4991e+00
Generation 80: Best = 1.963235e+03 FES = 8100 Diversity = 9.0177e+00
Generation 90: Best = 1.963235e+03 FES = 9100 Diversity = 8.5523e+00
Generation 100: Best = 1.947416e+03 FES = 10100 Diversity = 8.1973e+00
Generation 110: Best = 1.915947e+03 FES = 11100 Diversity = 7.8202e+00
Generation 120: Best = 1.913171e+03 FES = 12100 Diversity = 7.6885e+00
Generation 130: Best = 1.888631e+03 FES = 13100 Diversity = 7.2903e+00
Generation 140: Best = 1.845615e+03 FES = 14100 Diversity = 7.1030e+00
Generation 150: Best = 1.832405e+03 FES = 15100 Diversity = 7.0018e+00
Generation 160: Best = 1.816084e+03 FES = 16100 Diversity = 6.8961e+00
Generation 170: Best = 1.811716e+03 FES = 17100 Diversity = 6.8600e+00
Generation 180: Best = 1.811716e+03 FES = 18100 Diversity = 6.7786e+00
Generation 190: Best = 1.806208e+03 FES = 19100 Diversity = 6.7504e+00
Generation 200: Best = 1.799852e+03 FES = 20100 Diversity = 6.5961e+00
Generation 210: Best = 1.792796e+03 FES = 21100 Diversity = 6.5516e+00
Generation 220: Best = 1.768769e+03 FES = 22100 Diversity = 6.4875e+00
Generation 230: Best = 1.768296e+03 FES = 23100 Diversity = 6.3719e+00
Generation 240: Best = 1.750880e+03 FES = 24100 Diversity = 6.3095e+00
Generation 250: Best = 1.750880e+03 FES = 25100 Diversity = 6.2309e+00
Generation 260: Best = 1.744920e+03 FES = 26100 Diversity = 6.1435e+00
Generation 270: Best = 1.736040e+03 FES = 27100 Diversity = 6.0475e+00
Generation 280: Best = 1.736040e+03 FES = 28100 Diversity = 5.9520e+00
Generation 290: Best = 1.736040e+03 FES = 29100 Diversity = 5.8615e+00
Generation 300: Best = 1.713432e+03 FES = 30100 Diversity = 5.8098e+00
Generation 310: Best = 1.707886e+03 FES = 31100 Diversity = 5.7324e+00
Generation 320: Best = 1.694710e+03 FES = 32100 Diversity = 5.6067e+00
Generation 330: Best = 1.678926e+03 FES = 33100 Diversity = 5.5755e+00
Generation 340: Best = 1.678926e+03 FES = 34100 Diversity = 5.6101e+00
Generation 350: Best = 1.668077e+03 FES = 35100 Diversity = 5.5614e+00
Generation 360: Best = 1.657883e+03 FES = 36100 Diversity = 5.4535e+00
Generation 370: Best = 1.654095e+03 FES = 37100 Diversity = 5.3779e+00
Generation 380: Best = 1.649832e+03 FES = 38100 Diversity = 5.3015e+00
Generation 390: Best = 1.637039e+03 FES = 39100 Diversity = 5.2121e+00
Generation 400: Best = 1.617487e+03 FES = 40100 Diversity = 5.1370e+00
Generation 410: Best = 1.598637e+03 FES = 41100 Diversity = 5.0892e+00
Generation 420: Best = 1.595986e+03 FES = 42100 Diversity = 4.9838e+00
Generation 430: Best = 1.591225e+03 FES = 43100 Diversity = 4.9360e+00
Generation 440: Best = 1.572896e+03 FES = 44100 Diversity = 4.8552e+00

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Generation	450:	Best = 1.570701e+03	FES = 45100	Diversity = 4.7979e+00
Generation	460:	Best = 1.564122e+03	FES = 46100	Diversity = 4.7344e+00
Generation	470:	Best = 1.561030e+03	FES = 47100	Diversity = 4.6029e+00
Generation	480:	Best = 1.560877e+03	FES = 48100	Diversity = 4.4871e+00
Generation	490:	Best = 1.549408e+03	FES = 49100	Diversity = 4.3382e+00
Generation	500:	Best = 1.536970e+03	FES = 50100	Diversity = 4.2906e+00
Generation	510:	Best = 1.526835e+03	FES = 51100	Diversity = 4.0973e+00
Generation	520:	Best = 1.496914e+03	FES = 52100	Diversity = 3.9435e+00
Generation	530:	Best = 1.489679e+03	FES = 53100	Diversity = 3.6244e+00
Generation	540:	Best = 1.464281e+03	FES = 54100	Diversity = 3.1230e+00
Generation	550:	Best = 1.461906e+03	FES = 55100	Diversity = 2.8074e+00
Generation	560:	Best = 1.447074e+03	FES = 56100	Diversity = 2.3185e+00
Generation	570:	Best = 1.416899e+03	FES = 57100	Diversity = 1.7778e+00
Generation	580:	Best = 1.407087e+03	FES = 58100	Diversity = 1.3415e+00
Generation	590:	Best = 1.380078e+03	FES = 59100	Diversity = 1.1119e+00
Generation	600:	Best = 1.348509e+03	FES = 60100	Diversity = 9.2275e-01
Generation	610:	Best = 1.329505e+03	FES = 61100	Diversity = 5.4466e-01
Generation	620:	Best = 1.317041e+03	FES = 62100	Diversity = 3.1751e-01
Generation	630:	Best = 1.309068e+03	FES = 63100	Diversity = 2.5927e-01
Generation	640:	Best = 1.303092e+03	FES = 64100	Diversity = 1.7806e-01
Generation	650:	Best = 1.299103e+03	FES = 65100	Diversity = 1.3683e-01
Generation	660:	Best = 1.295669e+03	FES = 66100	Diversity = 1.1161e-01
Generation	670:	Best = 1.293219e+03	FES = 67100	Diversity = 9.0582e-02
Generation	680:	Best = 1.290764e+03	FES = 68100	Diversity = 7.7729e-02
Generation	690:	Best = 1.288877e+03	FES = 69100	Diversity = 6.9806e-02
Generation	700:	Best = 1.287024e+03	FES = 70100	Diversity = 6.2142e-02
Generation	710:	Best = 1.285433e+03	FES = 71100	Diversity = 5.4854e-02
Generation	720:	Best = 1.284349e+03	FES = 72100	Diversity = 4.2409e-02
Generation	730:	Best = 1.283362e+03	FES = 73100	Diversity = 3.4793e-02
Generation	740:	Best = 1.282726e+03	FES = 74100	Diversity = 3.1769e-02
Generation	750:	Best = 1.282167e+03	FES = 75100	Diversity = 2.9328e-02
Generation	760:	Best = 1.281669e+03	FES = 76100	Diversity = 2.3423e-02
Generation	770:	Best = 1.281335e+03	FES = 77100	Diversity = 1.8152e-02
Generation	780:	Best = 1.280939e+03	FES = 78100	Diversity = 1.6057e-02
Generation	790:	Best = 1.280678e+03	FES = 79100	Diversity = 1.4788e-02
Generation	800:	Best = 1.280414e+03	FES = 80100	Diversity = 1.2235e-02
Generation	810:	Best = 1.280204e+03	FES = 81100	Diversity = 1.0123e-02
Generation	820:	Best = 1.279991e+03	FES = 82100	Diversity = 9.4321e-03
Generation	830:	Best = 1.279820e+03	FES = 83100	Diversity = 9.1320e-03
Generation	840:	Best = 1.279627e+03	FES = 84100	Diversity = 8.9715e-03
Generation	850:	Best = 1.279469e+03	FES = 85100	Diversity = 8.2976e-03
Generation	860:	Best = 1.279342e+03	FES = 86100	Diversity = 7.3720e-03
Generation	870:	Best = 1.279232e+03	FES = 87100	Diversity = 6.9360e-03
Generation	880:	Best = 1.279107e+03	FES = 88100	Diversity = 6.7336e-03
Generation	890:	Best = 1.278991e+03	FES = 89100	Diversity = 5.9138e-03
Generation	900:	Best = 1.278900e+03	FES = 90100	Diversity = 4.5874e-03
Generation	910:	Best = 1.278825e+03	FES = 91100	Diversity = 4.5190e-03
Generation	920:	Best = 1.278736e+03	FES = 92100	Diversity = 4.1552e-03
Generation	930:	Best = 1.278673e+03	FES = 93100	Diversity = 3.6318e-03
Generation	940:	Best = 1.278592e+03	FES = 94100	Diversity = 3.5079e-03
Generation	950:	Best = 1.278524e+03	FES = 95100	Diversity = 3.3091e-03
Generation	960:	Best = 1.278455e+03	FES = 96100	Diversity = 3.1835e-03
Generation	970:	Best = 1.278399e+03	FES = 97100	Diversity = 2.8596e-03
Generation	980:	Best = 1.278340e+03	FES = 98100	Diversity = 2.7667e-03

Generation 990:	Best = 1.278297e+03	FES = 99100	Diversity = 2.3100e-03
Generation 1000:	Best = 1.278252e+03	FES = 100100	Diversity = 2.1846e-03
Generation 1010:	Best = 1.278217e+03	FES = 101100	Diversity = 1.8927e-03
Generation 1020:	Best = 1.278183e+03	FES = 102100	Diversity = 1.8566e-03
Generation 1030:	Best = 1.278144e+03	FES = 103100	Diversity = 2.1549e-03
Generation 1040:	Best = 1.278108e+03	FES = 104100	Diversity = 2.1736e-03
Generation 1050:	Best = 1.278073e+03	FES = 105100	Diversity = 1.7978e-03
Generation 1060:	Best = 1.278052e+03	FES = 106100	Diversity = 1.3804e-03
Generation 1070:	Best = 1.278031e+03	FES = 107100	Diversity = 1.1998e-03
Generation 1080:	Best = 1.278015e+03	FES = 108100	Diversity = 1.1092e-03
Generation 1090:	Best = 1.278004e+03	FES = 109100	Diversity = 8.8379e-04
Generation 1100:	Best = 1.277990e+03	FES = 110100	Diversity = 9.0826e-04
Generation 1110:	Best = 1.277982e+03	FES = 111100	Diversity = 7.7120e-04
Generation 1120:	Best = 1.277972e+03	FES = 112100	Diversity = 8.0224e-04
Generation 1130:	Best = 1.277961e+03	FES = 113100	Diversity = 7.3089e-04
Generation 1140:	Best = 1.277954e+03	FES = 114100	Diversity = 6.4810e-04
Generation 1150:	Best = 1.277946e+03	FES = 115100	Diversity = 6.2417e-04
Generation 1160:	Best = 1.277940e+03	FES = 116100	Diversity = 4.8708e-04
Generation 1170:	Best = 1.277936e+03	FES = 117100	Diversity = 3.8619e-04
Generation 1180:	Best = 1.277934e+03	FES = 118100	Diversity = 2.6059e-04
Generation 1190:	Best = 1.277931e+03	FES = 119100	Diversity = 2.8752e-04
Generation 1200:	Best = 1.277929e+03	FES = 120100	Diversity = 2.4604e-04
Generation 1210:	Best = 1.277928e+03	FES = 121100	Diversity = 1.8970e-04
Generation 1220:	Best = 1.277927e+03	FES = 122100	Diversity = 1.9556e-04
Generation 1230:	Best = 1.277926e+03	FES = 123100	Diversity = 1.3475e-04
Generation 1240:	Best = 1.277926e+03	FES = 124100	Diversity = 1.3014e-04
Generation 1250:	Best = 1.277925e+03	FES = 125100	Diversity = 1.5040e-04
Generation 1260:	Best = 1.277924e+03	FES = 126100	Diversity = 1.1218e-04
Generation 1270:	Best = 1.277924e+03	FES = 127100	Diversity = 1.1123e-04
Generation 1280:	Best = 1.277923e+03	FES = 128100	Diversity = 9.5507e-05
Generation 1290:	Best = 1.277923e+03	FES = 129100	Diversity = 9.9499e-05
Generation 1300:	Best = 1.277923e+03	FES = 130100	Diversity = 7.6394e-05
Generation 1310:	Best = 1.277923e+03	FES = 131100	Diversity = 6.1292e-05
Generation 1320:	Best = 1.277922e+03	FES = 132100	Diversity = 6.3650e-05
Generation 1330:	Best = 1.277922e+03	FES = 133100	Diversity = 4.9789e-05
Generation 1340:	Best = 1.277922e+03	FES = 134100	Diversity = 3.9944e-05
Generation 1350:	Best = 1.277922e+03	FES = 135100	Diversity = 2.7350e-05
Generation 1360:	Best = 1.277922e+03	FES = 136100	Diversity = 2.7605e-05
Generation 1370:	Best = 1.277922e+03	FES = 137100	Diversity = 2.8569e-05
Generation 1380:	Best = 1.277922e+03	FES = 138100	Diversity = 2.3813e-05
Generation 1390:	Best = 1.277922e+03	FES = 139100	Diversity = 1.7791e-05
Generation 1400:	Best = 1.277922e+03	FES = 140100	Diversity = 1.3199e-05
Generation 1410:	Best = 1.277922e+03	FES = 141100	Diversity = 1.5359e-05
Generation 1420:	Best = 1.277922e+03	FES = 142100	Diversity = 1.0377e-05
Generation 1430:	Best = 1.277922e+03	FES = 143100	Diversity = 1.6151e-05
Generation 1440:	Best = 1.277922e+03	FES = 144100	Diversity = 1.0556e-05
Generation 1450:	Best = 1.277922e+03	FES = 145100	Diversity = 8.5969e-06
Generation 1460:	Best = 1.277922e+03	FES = 146100	Diversity = 7.3990e-06
Generation 1470:	Best = 1.277922e+03	FES = 147100	Diversity = 6.0300e-06
Generation 1480:	Best = 1.277922e+03	FES = 148100	Diversity = 4.3992e-06
Generation 1490:	Best = 1.277922e+03	FES = 149100	Diversity = 3.6932e-06
Generation 1500:	Best = 1.277922e+03	FES = 150100	Diversity = 3.5770e-06
Generation 1510:	Best = 1.277922e+03	FES = 151100	Diversity = 2.7676e-06
Generation 1520:	Best = 1.277922e+03	FES = 152100	Diversity = 2.7808e-06

Generation 1530:	Best = 1.277922e+03	FES = 153100	Diversity = 2.2149e-06
Generation 1540:	Best = 1.277922e+03	FES = 154100	Diversity = 1.2981e-06
Generation 1550:	Best = 1.277922e+03	FES = 155100	Diversity = 1.4635e-06
Generation 1560:	Best = 1.277922e+03	FES = 156100	Diversity = 1.1027e-06
Generation 1570:	Best = 1.277922e+03	FES = 157100	Diversity = 9.5308e-07
Generation 1580:	Best = 1.277922e+03	FES = 158100	Diversity = 8.4376e-07
Generation 1590:	Best = 1.277922e+03	FES = 159100	Diversity = 4.8543e-07
Generation 1600:	Best = 1.277922e+03	FES = 160100	Diversity = 4.1255e-07
Generation 1610:	Best = 1.277922e+03	FES = 161100	Diversity = 3.1624e-07
Generation 1620:	Best = 1.277922e+03	FES = 162100	Diversity = 2.8482e-07
Generation 1630:	Best = 1.277922e+03	FES = 163100	Diversity = 2.0184e-07
Generation 1640:	Best = 1.277922e+03	FES = 164100	Diversity = 2.0416e-07
Generation 1650:	Best = 1.277922e+03	FES = 165100	Diversity = 1.9485e-07
Generation 1660:	Best = 1.277922e+03	FES = 166100	Diversity = 1.2668e-07
Generation 1670:	Best = 1.277922e+03	FES = 167100	Diversity = 9.5221e-08
Generation 1680:	Best = 1.277922e+03	FES = 168100	Diversity = 9.2900e-08
Generation 1690:	Best = 1.277922e+03	FES = 169100	Diversity = 6.2709e-08
Generation 1700:	Best = 1.277922e+03	FES = 170100	Diversity = 4.3087e-08
Generation 1710:	Best = 1.277922e+03	FES = 171100	Diversity = 2.9357e-08
Generation 1720:	Best = 1.277922e+03	FES = 172100	Diversity = 3.5886e-08
Generation 1730:	Best = 1.277922e+03	FES = 173100	Diversity = 2.2523e-08
Generation 1740:	Best = 1.277922e+03	FES = 174100	Diversity = 1.7948e-08
Generation 1750:	Best = 1.277922e+03	FES = 175100	Diversity = 1.3569e-08
Generation 1760:	Best = 1.277922e+03	FES = 176100	Diversity = 1.1232e-08
Generation 1770:	Best = 1.277922e+03	FES = 177100	Diversity = 8.0099e-09
Generation 1780:	Best = 1.277922e+03	FES = 178100	Diversity = 6.1378e-09
Generation 1790:	Best = 1.277922e+03	FES = 179100	Diversity = 5.5410e-09
Generation 1800:	Best = 1.277922e+03	FES = 180100	Diversity = 2.8562e-09
Generation 1810:	Best = 1.277922e+03	FES = 181100	Diversity = 1.0680e-09
Generation 1820:	Best = 1.277922e+03	FES = 182100	Diversity = 2.0328e-11
Generation 1830:	Best = 1.277922e+03	FES = 183100	Diversity = 5.6108e-13
Generation 1840:	Best = 1.277922e+03	FES = 184100	Diversity = 3.1991e-13
Generation 1850:	Best = 1.277922e+03	FES = 185100	Diversity = 4.1782e-14
Generation 1860:	Best = 1.277922e+03	FES = 186100	Diversity = 2.6981e-14
Generation 1870:	Best = 1.277922e+03	FES = 187100	Diversity = 2.6981e-14
Generation 1880:	Best = 1.277922e+03	FES = 188100	Diversity = 2.6981e-14
Generation 1890:	Best = 1.277922e+03	FES = 189100	Diversity = 2.6981e-14
Generation 1900:	Best = 1.277922e+03	FES = 190100	Diversity = 2.6981e-14
Generation 1910:	Best = 1.277922e+03	FES = 191100	Diversity = 2.6981e-14
Generation 1920:	Best = 1.277922e+03	FES = 192100	Diversity = 2.6981e-14
Generation 1930:	Best = 1.277922e+03	FES = 193100	Diversity = 2.6981e-14
Generation 1940:	Best = 1.277922e+03	FES = 194100	Diversity = 2.6981e-14
Generation 1950:	Best = 1.277922e+03	FES = 195100	Diversity = 2.6981e-14
Generation 1960:	Best = 1.277922e+03	FES = 196100	Diversity = 2.6981e-14
Generation 1970:	Best = 1.277922e+03	FES = 197100	Diversity = 2.6981e-14
Generation 1980:	Best = 1.277922e+03	FES = 198100	Diversity = 2.6981e-14
Generation 1990:	Best = 1.277922e+03	FES = 199100	Diversity = 2.6981e-14
Generation 2000:	Best = 1.277922e+03	FES = 200100	Diversity = 2.6981e-14
Generation 2010:	Best = 1.277922e+03	FES = 201100	Diversity = 2.6981e-14
Generation 2020:	Best = 1.277922e+03	FES = 202100	Diversity = 2.6981e-14
Generation 2030:	Best = 1.277922e+03	FES = 203100	Diversity = 2.6981e-14
Generation 2040:	Best = 1.277922e+03	FES = 204100	Diversity = 2.6981e-14
Generation 2050:	Best = 1.277922e+03	FES = 205100	Diversity = 2.6981e-14
Generation 2060:	Best = 1.277922e+03	FES = 206100	Diversity = 2.6981e-14

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Generation 9630: Best = 1.277922e+03 FES = 963100 Diversity = 2.6981e-14
Generation 9640: Best = 1.277922e+03 FES = 964100 Diversity = 2.6981e-14
Generation 9650: Best = 1.277922e+03 FES = 965100 Diversity = 2.6981e-14
Generation 9660: Best = 1.277922e+03 FES = 966100 Diversity = 2.6981e-14
Generation 9670: Best = 1.277922e+03 FES = 967100 Diversity = 2.6981e-14
Generation 9680: Best = 1.277922e+03 FES = 968100 Diversity = 2.6981e-14
Generation 9690: Best = 1.277922e+03 FES = 969100 Diversity = 2.6981e-14
Generation 9700: Best = 1.277922e+03 FES = 970100 Diversity = 2.6981e-14
Generation 9710: Best = 1.277922e+03 FES = 971100 Diversity = 2.6981e-14
Generation 9720: Best = 1.277922e+03 FES = 972100 Diversity = 2.6981e-14
Generation 9730: Best = 1.277922e+03 FES = 973100 Diversity = 2.6981e-14
Generation 9740: Best = 1.277922e+03 FES = 974100 Diversity = 2.6981e-14
Generation 9750: Best = 1.277922e+03 FES = 975100 Diversity = 2.6981e-14
Generation 9760: Best = 1.277922e+03 FES = 976100 Diversity = 2.6981e-14
Generation 9770: Best = 1.277922e+03 FES = 977100 Diversity = 2.6981e-14
Generation 9780: Best = 1.277922e+03 FES = 978100 Diversity = 2.6981e-14
Generation 9790: Best = 1.277922e+03 FES = 979100 Diversity = 2.6981e-14
Generation 9800: Best = 1.277922e+03 FES = 980100 Diversity = 2.6981e-14
Generation 9810: Best = 1.277922e+03 FES = 981100 Diversity = 2.6981e-14
Generation 9820: Best = 1.277922e+03 FES = 982100 Diversity = 2.6981e-14
Generation 9830: Best = 1.277922e+03 FES = 983100 Diversity = 2.6981e-14
Generation 9840: Best = 1.277922e+03 FES = 984100 Diversity = 2.6981e-14
Generation 9850: Best = 1.277922e+03 FES = 985100 Diversity = 2.6981e-14
Generation 9860: Best = 1.277922e+03 FES = 986100 Diversity = 2.6981e-14
Generation 9870: Best = 1.277922e+03 FES = 987100 Diversity = 2.6981e-14
Generation 9880: Best = 1.277922e+03 FES = 988100 Diversity = 2.6981e-14
Generation 9890: Best = 1.277922e+03 FES = 989100 Diversity = 2.6981e-14
Generation 9900: Best = 1.277922e+03 FES = 990100 Diversity = 2.6981e-14
Generation 9910: Best = 1.277922e+03 FES = 991100 Diversity = 2.6981e-14
Generation 9920: Best = 1.277922e+03 FES = 992100 Diversity = 2.6981e-14
Generation 9930: Best = 1.277922e+03 FES = 993100 Diversity = 2.6981e-14
Generation 9940: Best = 1.277922e+03 FES = 994100 Diversity = 2.6981e-14
Generation 9950: Best = 1.277922e+03 FES = 995100 Diversity = 2.6981e-14
Generation 9960: Best = 1.277922e+03 FES = 996100 Diversity = 2.6981e-14
Generation 9970: Best = 1.277922e+03 FES = 997100 Diversity = 2.6981e-14
Generation 9980: Best = 1.277922e+03 FES = 998100 Diversity = 2.6981e-14
Generation 9990: Best = 1.277922e+03 FES = 999100 Diversity = 2.6981e-14

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Optimization completed!
Final Best Fitness = 1.277922e+03
Total Function Evaluations = 1000000

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=
Results:
  Best Fitness          : 1.277922e+03
  Global Optimum        : 8.000000e+02
  Error (|best-optimum|) : 4.779219e+02
  Best Solution (5 dim)  : [ 5.85777445 -40.69279014 13.03144174 2.31708892
-23.32770117]
  D^(0) (init. diversity): 5.7220e+01
=====
=

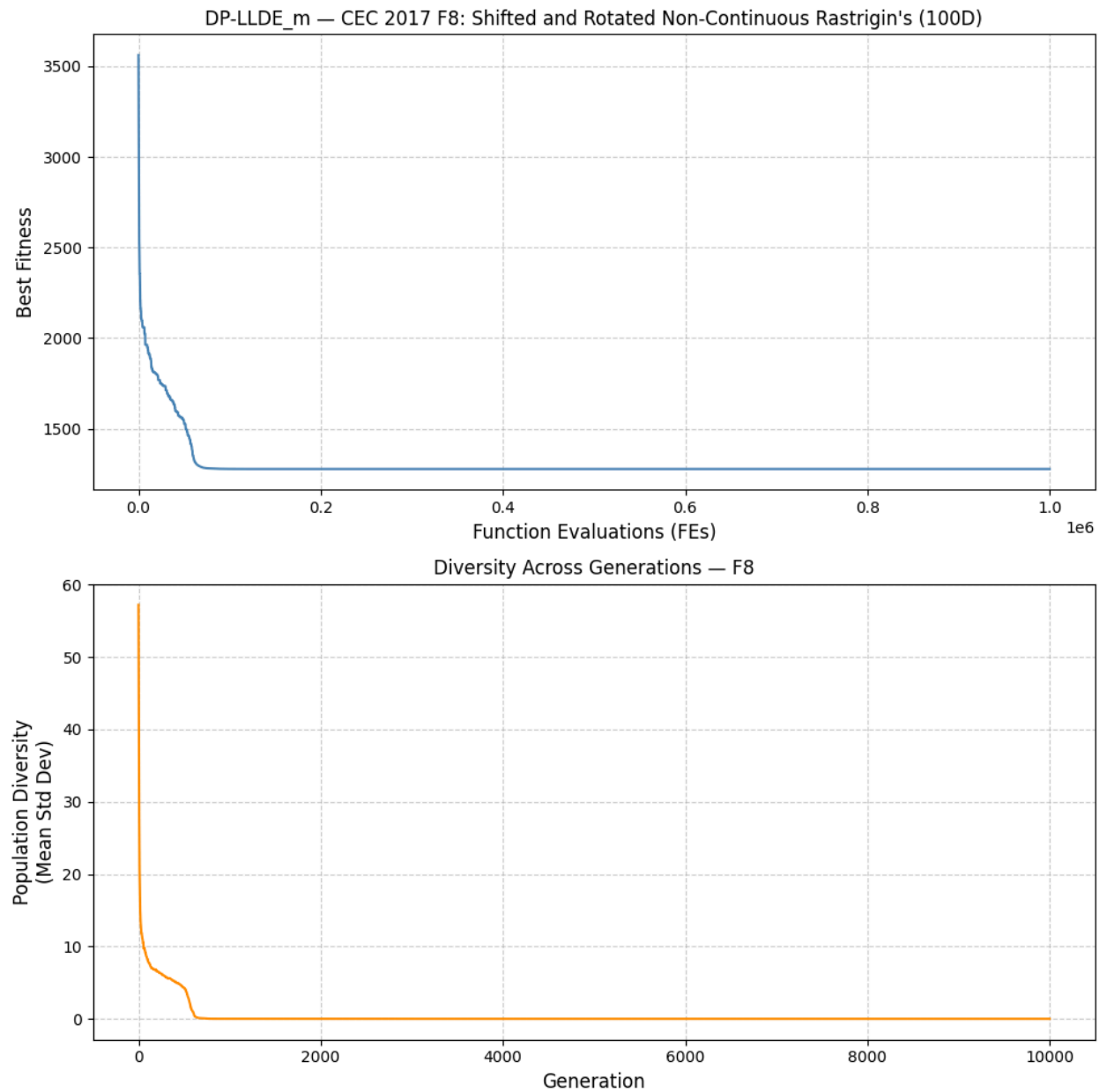
```

```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results_2/all_results_che

```

ckpoint.pkl



Graph saved as convergence_F8.png

✓ Plot saved to: /content/drive/My Drive/thesis_results_2/plots/convergence_F8.png

```
In [ ]: # =====  
# F9 — Shifted and Rotated Levy  
# =====  
run_and_plot('F9', make_wrapper(simple.f9), 900.0, 'Shifted and Rotated Levy')
```

```

=====
=
DP-LLDE_m on CEC 2017 F9 – Single Run
=====
=
Function      : Shifted and Rotated Levy (100D)
Bounds       : [-100, 100]
Global optimum : 900.0
Initial D^(0) = 5.7220e+01 | Diversity threshold = 5.7220e+00
Generation 0: Best Fitness = 2.313486e+05 | Diversity = 5.7220e+01
Generation 10: Best = 8.835712e+04 FES = 1100 Diversity = 2.8702e+01
Generation 20: Best = 5.844296e+04 FES = 2100 Diversity = 1.8714e+01
Generation 30: Best = 5.489564e+04 FES = 3100 Diversity = 1.4511e+01
Generation 40: Best = 5.028228e+04 FES = 4100 Diversity = 1.2785e+01
Generation 50: Best = 4.624467e+04 FES = 5100 Diversity = 1.2026e+01
Generation 60: Best = 4.323811e+04 FES = 6100 Diversity = 1.1264e+01
Generation 70: Best = 4.205328e+04 FES = 7100 Diversity = 1.1129e+01
Generation 80: Best = 3.616625e+04 FES = 8100 Diversity = 1.1048e+01
Generation 90: Best = 3.616625e+04 FES = 9100 Diversity = 1.0589e+01
Generation 100: Best = 3.358998e+04 FES = 10100 Diversity = 1.0271e+01
Generation 110: Best = 3.353542e+04 FES = 11100 Diversity = 9.5028e+00
Generation 120: Best = 3.183982e+04 FES = 12100 Diversity = 8.9433e+00
Generation 130: Best = 2.723154e+04 FES = 13100 Diversity = 8.8437e+00
Generation 140: Best = 2.701924e+04 FES = 14100 Diversity = 8.4241e+00
Generation 150: Best = 2.475849e+04 FES = 15100 Diversity = 8.6763e+00
Generation 160: Best = 2.391580e+04 FES = 16100 Diversity = 8.3180e+00
Generation 170: Best = 2.331182e+04 FES = 17100 Diversity = 7.9981e+00
Generation 180: Best = 2.029848e+04 FES = 18100 Diversity = 7.3531e+00
Generation 190: Best = 2.029848e+04 FES = 19100 Diversity = 7.1139e+00
Generation 200: Best = 1.950076e+04 FES = 20100 Diversity = 6.8381e+00
Generation 210: Best = 1.950076e+04 FES = 21100 Diversity = 6.7581e+00
Generation 220: Best = 1.677059e+04 FES = 22100 Diversity = 6.3762e+00
Generation 230: Best = 1.677059e+04 FES = 23100 Diversity = 6.1968e+00
Generation 240: Best = 1.677059e+04 FES = 24100 Diversity = 5.9368e+00
Generation 250: Best = 1.677059e+04 FES = 25100 Diversity = 5.7907e+00
Generation 260: Best = 1.677059e+04 FES = 26100 Diversity = 5.6932e+00
Generation 270: Best = 1.601262e+04 FES = 27100 Diversity = 5.5464e+00
Generation 280: Best = 1.546298e+04 FES = 28100 Diversity = 5.4823e+00
Generation 290: Best = 1.457711e+04 FES = 29100 Diversity = 5.5009e+00
Generation 300: Best = 1.422420e+04 FES = 30100 Diversity = 5.4916e+00
Generation 310: Best = 1.392244e+04 FES = 31100 Diversity = 5.4578e+00
Generation 320: Best = 1.392244e+04 FES = 32100 Diversity = 5.3465e+00
Generation 330: Best = 1.350058e+04 FES = 33100 Diversity = 5.3081e+00
Generation 340: Best = 1.350058e+04 FES = 34100 Diversity = 5.2375e+00
Generation 350: Best = 1.336331e+04 FES = 35100 Diversity = 5.2672e+00
Generation 360: Best = 1.336331e+04 FES = 36100 Diversity = 5.2509e+00
Generation 370: Best = 1.254299e+04 FES = 37100 Diversity = 5.2259e+00
Generation 380: Best = 1.254299e+04 FES = 38100 Diversity = 5.2644e+00
Generation 390: Best = 1.254299e+04 FES = 39100 Diversity = 5.3021e+00
Generation 400: Best = 1.183324e+04 FES = 40100 Diversity = 5.2933e+00
Generation 410: Best = 1.183324e+04 FES = 41100 Diversity = 5.2178e+00
Generation 420: Best = 1.168036e+04 FES = 42100 Diversity = 5.2189e+00
Generation 430: Best = 1.168036e+04 FES = 43100 Diversity = 5.1968e+00
Generation 440: Best = 1.168036e+04 FES = 44100 Diversity = 5.1970e+00

```

Generation	450:	Best = 1.168036e+04	FES = 45100	Diversity = 5.2133e+00
Generation	460:	Best = 1.168036e+04	FES = 46100	Diversity = 5.2110e+00
Generation	470:	Best = 1.168036e+04	FES = 47100	Diversity = 5.1866e+00
Generation	480:	Best = 1.168036e+04	FES = 48100	Diversity = 5.1706e+00
Generation	490:	Best = 1.168036e+04	FES = 49100	Diversity = 5.1614e+00
Generation	500:	Best = 1.168036e+04	FES = 50100	Diversity = 5.1604e+00
Generation	510:	Best = 1.168036e+04	FES = 51100	Diversity = 5.1232e+00
Generation	520:	Best = 1.168036e+04	FES = 52100	Diversity = 5.1173e+00
Generation	530:	Best = 1.168036e+04	FES = 53100	Diversity = 5.1032e+00
Generation	540:	Best = 1.168036e+04	FES = 54100	Diversity = 5.0992e+00
Generation	550:	Best = 1.168036e+04	FES = 55100	Diversity = 5.0954e+00
Generation	560:	Best = 1.166159e+04	FES = 56100	Diversity = 5.0404e+00
Generation	570:	Best = 1.166159e+04	FES = 57100	Diversity = 5.0200e+00
Generation	580:	Best = 1.166159e+04	FES = 58100	Diversity = 5.0141e+00
Generation	590:	Best = 1.155091e+04	FES = 59100	Diversity = 5.0095e+00
Generation	600:	Best = 1.155091e+04	FES = 60100	Diversity = 4.9663e+00
Generation	610:	Best = 1.155091e+04	FES = 61100	Diversity = 4.9634e+00
Generation	620:	Best = 1.122713e+04	FES = 62100	Diversity = 4.9433e+00
Generation	630:	Best = 1.098684e+04	FES = 63100	Diversity = 4.9057e+00
Generation	640:	Best = 1.098684e+04	FES = 64100	Diversity = 4.9008e+00
Generation	650:	Best = 1.098684e+04	FES = 65100	Diversity = 4.8970e+00
Generation	660:	Best = 1.098684e+04	FES = 66100	Diversity = 4.8993e+00
Generation	670:	Best = 1.098684e+04	FES = 67100	Diversity = 4.8917e+00
Generation	680:	Best = 1.098684e+04	FES = 68100	Diversity = 4.9075e+00
Generation	690:	Best = 1.095874e+04	FES = 69100	Diversity = 4.9046e+00
Generation	700:	Best = 1.095874e+04	FES = 70100	Diversity = 4.9124e+00
Generation	710:	Best = 1.095874e+04	FES = 71100	Diversity = 4.9157e+00
Generation	720:	Best = 1.080209e+04	FES = 72100	Diversity = 4.9123e+00
Generation	730:	Best = 1.080209e+04	FES = 73100	Diversity = 4.9028e+00
Generation	740:	Best = 1.023509e+04	FES = 74100	Diversity = 4.8797e+00
Generation	750:	Best = 1.023509e+04	FES = 75100	Diversity = 4.8787e+00
Generation	760:	Best = 1.023509e+04	FES = 76100	Diversity = 4.8783e+00
Generation	770:	Best = 1.023135e+04	FES = 77100	Diversity = 4.8812e+00
Generation	780:	Best = 1.023135e+04	FES = 78100	Diversity = 4.8779e+00
Generation	790:	Best = 1.023135e+04	FES = 79100	Diversity = 4.9029e+00
Generation	800:	Best = 1.023135e+04	FES = 80100	Diversity = 4.9011e+00
Generation	810:	Best = 1.023135e+04	FES = 81100	Diversity = 4.9027e+00
Generation	820:	Best = 1.023135e+04	FES = 82100	Diversity = 4.8918e+00
Generation	830:	Best = 1.023135e+04	FES = 83100	Diversity = 4.8717e+00
Generation	840:	Best = 1.023135e+04	FES = 84100	Diversity = 4.8912e+00
Generation	850:	Best = 1.023135e+04	FES = 85100	Diversity = 4.9230e+00
Generation	860:	Best = 1.023135e+04	FES = 86100	Diversity = 4.8900e+00
Generation	870:	Best = 1.010064e+04	FES = 87100	Diversity = 4.8887e+00
Generation	880:	Best = 1.010064e+04	FES = 88100	Diversity = 4.8838e+00
Generation	890:	Best = 9.891543e+03	FES = 89100	Diversity = 4.8829e+00
Generation	900:	Best = 9.818132e+03	FES = 90100	Diversity = 4.8804e+00
Generation	910:	Best = 9.818132e+03	FES = 91100	Diversity = 4.8788e+00
Generation	920:	Best = 9.818132e+03	FES = 92100	Diversity = 4.8626e+00
Generation	930:	Best = 9.818132e+03	FES = 93100	Diversity = 4.8617e+00
Generation	940:	Best = 9.818132e+03	FES = 94100	Diversity = 4.8597e+00
Generation	950:	Best = 9.818132e+03	FES = 95100	Diversity = 4.8586e+00
Generation	960:	Best = 9.818132e+03	FES = 96100	Diversity = 4.8564e+00
Generation	970:	Best = 9.818132e+03	FES = 97100	Diversity = 4.8561e+00
Generation	980:	Best = 9.785307e+03	FES = 98100	Diversity = 4.8529e+00

Generation 990:	Best = 9.785307e+03	FES = 99100	Diversity = 4.8516e+00
Generation 1000:	Best = 9.785307e+03	FES = 100100	Diversity = 4.8447e+00
Generation 1010:	Best = 9.726263e+03	FES = 101100	Diversity = 4.8432e+00
Generation 1020:	Best = 9.702724e+03	FES = 102100	Diversity = 4.8401e+00
Generation 1030:	Best = 9.702724e+03	FES = 103100	Diversity = 4.8401e+00
Generation 1040:	Best = 9.702724e+03	FES = 104100	Diversity = 4.8371e+00
Generation 1050:	Best = 9.598946e+03	FES = 105100	Diversity = 4.8417e+00
Generation 1060:	Best = 9.598946e+03	FES = 106100	Diversity = 4.8412e+00
Generation 1070:	Best = 9.598946e+03	FES = 107100	Diversity = 4.8240e+00
Generation 1080:	Best = 9.598946e+03	FES = 108100	Diversity = 4.8235e+00
Generation 1090:	Best = 9.598946e+03	FES = 109100	Diversity = 4.8102e+00
Generation 1100:	Best = 9.598946e+03	FES = 110100	Diversity = 4.8069e+00
Generation 1110:	Best = 9.598946e+03	FES = 111100	Diversity = 4.8053e+00
Generation 1120:	Best = 9.598946e+03	FES = 112100	Diversity = 4.8049e+00
Generation 1130:	Best = 9.598946e+03	FES = 113100	Diversity = 4.8043e+00
Generation 1140:	Best = 9.598946e+03	FES = 114100	Diversity = 4.8029e+00
Generation 1150:	Best = 9.598946e+03	FES = 115100	Diversity = 4.7913e+00
Generation 1160:	Best = 9.598946e+03	FES = 116100	Diversity = 4.7864e+00
Generation 1170:	Best = 9.598946e+03	FES = 117100	Diversity = 4.7843e+00
Generation 1180:	Best = 9.598946e+03	FES = 118100	Diversity = 4.7846e+00
Generation 1190:	Best = 9.598946e+03	FES = 119100	Diversity = 4.7871e+00
Generation 1200:	Best = 9.598946e+03	FES = 120100	Diversity = 4.7877e+00
Generation 1210:	Best = 9.598946e+03	FES = 121100	Diversity = 4.7857e+00
Generation 1220:	Best = 9.598946e+03	FES = 122100	Diversity = 4.7606e+00
Generation 1230:	Best = 9.598946e+03	FES = 123100	Diversity = 4.7496e+00
Generation 1240:	Best = 9.598946e+03	FES = 124100	Diversity = 4.7526e+00
Generation 1250:	Best = 9.531954e+03	FES = 125100	Diversity = 4.7441e+00
Generation 1260:	Best = 9.531954e+03	FES = 126100	Diversity = 4.7425e+00
Generation 1270:	Best = 9.426974e+03	FES = 127100	Diversity = 4.7424e+00
Generation 1280:	Best = 9.337024e+03	FES = 128100	Diversity = 4.7474e+00
Generation 1290:	Best = 9.337024e+03	FES = 129100	Diversity = 4.7609e+00
Generation 1300:	Best = 9.337024e+03	FES = 130100	Diversity = 4.7583e+00
Generation 1310:	Best = 9.337024e+03	FES = 131100	Diversity = 4.7562e+00
Generation 1320:	Best = 9.337024e+03	FES = 132100	Diversity = 4.7550e+00
Generation 1330:	Best = 9.332241e+03	FES = 133100	Diversity = 4.7406e+00
Generation 1340:	Best = 9.332241e+03	FES = 134100	Diversity = 4.7384e+00
Generation 1350:	Best = 9.332241e+03	FES = 135100	Diversity = 4.7225e+00
Generation 1360:	Best = 9.332241e+03	FES = 136100	Diversity = 4.7207e+00
Generation 1370:	Best = 9.332241e+03	FES = 137100	Diversity = 4.7182e+00
Generation 1380:	Best = 9.332241e+03	FES = 138100	Diversity = 4.7122e+00
Generation 1390:	Best = 9.281493e+03	FES = 139100	Diversity = 4.7038e+00
Generation 1400:	Best = 9.268720e+03	FES = 140100	Diversity = 4.7006e+00
Generation 1410:	Best = 9.268720e+03	FES = 141100	Diversity = 4.6776e+00
Generation 1420:	Best = 9.268720e+03	FES = 142100	Diversity = 4.6764e+00
Generation 1430:	Best = 9.268720e+03	FES = 143100	Diversity = 4.6745e+00
Generation 1440:	Best = 9.268720e+03	FES = 144100	Diversity = 4.6730e+00
Generation 1450:	Best = 9.078488e+03	FES = 145100	Diversity = 4.6722e+00
Generation 1460:	Best = 9.078488e+03	FES = 146100	Diversity = 4.6606e+00
Generation 1470:	Best = 9.078488e+03	FES = 147100	Diversity = 4.6495e+00
Generation 1480:	Best = 9.078488e+03	FES = 148100	Diversity = 4.6495e+00
Generation 1490:	Best = 9.078488e+03	FES = 149100	Diversity = 4.6480e+00
Generation 1500:	Best = 9.037708e+03	FES = 150100	Diversity = 4.6466e+00
Generation 1510:	Best = 9.037708e+03	FES = 151100	Diversity = 4.6429e+00
Generation 1520:	Best = 9.037097e+03	FES = 152100	Diversity = 4.6395e+00

Generation 1530:	Best = 8.972267e+03	FES = 153100	Diversity = 4.6383e+00
Generation 1540:	Best = 8.972267e+03	FES = 154100	Diversity = 4.6375e+00
Generation 1550:	Best = 8.829068e+03	FES = 155100	Diversity = 4.6367e+00
Generation 1560:	Best = 8.800072e+03	FES = 156100	Diversity = 4.6230e+00
Generation 1570:	Best = 8.800072e+03	FES = 157100	Diversity = 4.6217e+00
Generation 1580:	Best = 8.800072e+03	FES = 158100	Diversity = 4.6213e+00
Generation 1590:	Best = 8.800072e+03	FES = 159100	Diversity = 4.6203e+00
Generation 1600:	Best = 8.800072e+03	FES = 160100	Diversity = 4.6176e+00
Generation 1610:	Best = 8.800072e+03	FES = 161100	Diversity = 4.5934e+00
Generation 1620:	Best = 8.800072e+03	FES = 162100	Diversity = 4.5925e+00
Generation 1630:	Best = 8.800072e+03	FES = 163100	Diversity = 4.5849e+00
Generation 1640:	Best = 8.800072e+03	FES = 164100	Diversity = 4.5840e+00
Generation 1650:	Best = 8.800072e+03	FES = 165100	Diversity = 4.5907e+00
Generation 1660:	Best = 8.800072e+03	FES = 166100	Diversity = 4.5907e+00
Generation 1670:	Best = 8.800072e+03	FES = 167100	Diversity = 4.5906e+00
Generation 1680:	Best = 8.800072e+03	FES = 168100	Diversity = 4.5912e+00
Generation 1690:	Best = 8.800072e+03	FES = 169100	Diversity = 4.5904e+00
Generation 1700:	Best = 8.800072e+03	FES = 170100	Diversity = 4.5904e+00
Generation 1710:	Best = 8.800072e+03	FES = 171100	Diversity = 4.5903e+00
Generation 1720:	Best = 8.800072e+03	FES = 172100	Diversity = 4.5932e+00
Generation 1730:	Best = 8.672623e+03	FES = 173100	Diversity = 4.5914e+00
Generation 1740:	Best = 8.672623e+03	FES = 174100	Diversity = 4.5909e+00
Generation 1750:	Best = 8.672623e+03	FES = 175100	Diversity = 4.5905e+00
Generation 1760:	Best = 8.672623e+03	FES = 176100	Diversity = 4.5898e+00
Generation 1770:	Best = 8.672623e+03	FES = 177100	Diversity = 4.5899e+00
Generation 1780:	Best = 8.672623e+03	FES = 178100	Diversity = 4.5863e+00
Generation 1790:	Best = 8.672623e+03	FES = 179100	Diversity = 4.5857e+00
Generation 1800:	Best = 8.606154e+03	FES = 180100	Diversity = 4.5850e+00
Generation 1810:	Best = 8.606154e+03	FES = 181100	Diversity = 4.5850e+00
Generation 1820:	Best = 8.529798e+03	FES = 182100	Diversity = 4.5843e+00
Generation 1830:	Best = 8.529798e+03	FES = 183100	Diversity = 4.5834e+00
Generation 1840:	Best = 8.529798e+03	FES = 184100	Diversity = 4.5785e+00
Generation 1850:	Best = 8.529798e+03	FES = 185100	Diversity = 4.5729e+00
Generation 1860:	Best = 8.529798e+03	FES = 186100	Diversity = 4.5725e+00
Generation 1870:	Best = 8.529798e+03	FES = 187100	Diversity = 4.5720e+00
Generation 1880:	Best = 8.513438e+03	FES = 188100	Diversity = 4.5710e+00
Generation 1890:	Best = 8.513438e+03	FES = 189100	Diversity = 4.5708e+00
Generation 1900:	Best = 8.513438e+03	FES = 190100	Diversity = 4.5699e+00
Generation 1910:	Best = 8.513438e+03	FES = 191100	Diversity = 4.5696e+00
Generation 1920:	Best = 8.513438e+03	FES = 192100	Diversity = 4.5696e+00
Generation 1930:	Best = 8.513438e+03	FES = 193100	Diversity = 4.5696e+00
Generation 1940:	Best = 8.513438e+03	FES = 194100	Diversity = 4.5696e+00
Generation 1950:	Best = 8.498803e+03	FES = 195100	Diversity = 4.5696e+00
Generation 1960:	Best = 8.498803e+03	FES = 196100	Diversity = 4.5686e+00
Generation 1970:	Best = 8.498803e+03	FES = 197100	Diversity = 4.5657e+00
Generation 1980:	Best = 8.498803e+03	FES = 198100	Diversity = 4.5644e+00
Generation 1990:	Best = 8.498803e+03	FES = 199100	Diversity = 4.5646e+00
Generation 2000:	Best = 8.498803e+03	FES = 200100	Diversity = 4.5643e+00
Generation 2010:	Best = 8.498803e+03	FES = 201100	Diversity = 4.5648e+00
Generation 2020:	Best = 8.498803e+03	FES = 202100	Diversity = 4.5644e+00
Generation 2030:	Best = 8.498803e+03	FES = 203100	Diversity = 4.5643e+00
Generation 2040:	Best = 8.498803e+03	FES = 204100	Diversity = 4.5638e+00
Generation 2050:	Best = 8.498803e+03	FES = 205100	Diversity = 4.5644e+00
Generation 2060:	Best = 8.498803e+03	FES = 206100	Diversity = 4.5607e+00

Generation 2070:	Best = 8.498803e+03	FES = 207100	Diversity = 4.5596e+00
Generation 2080:	Best = 8.498803e+03	FES = 208100	Diversity = 4.5595e+00
Generation 2090:	Best = 8.308712e+03	FES = 209100	Diversity = 4.5595e+00
Generation 2100:	Best = 8.308712e+03	FES = 210100	Diversity = 4.5593e+00
Generation 2110:	Best = 8.308712e+03	FES = 211100	Diversity = 4.5510e+00
Generation 2120:	Best = 8.308712e+03	FES = 212100	Diversity = 4.5505e+00
Generation 2130:	Best = 8.308712e+03	FES = 213100	Diversity = 4.5502e+00
Generation 2140:	Best = 8.308712e+03	FES = 214100	Diversity = 4.5536e+00
Generation 2150:	Best = 8.308712e+03	FES = 215100	Diversity = 4.5534e+00
Generation 2160:	Best = 8.308712e+03	FES = 216100	Diversity = 4.5528e+00
Generation 2170:	Best = 8.308712e+03	FES = 217100	Diversity = 4.5525e+00
Generation 2180:	Best = 8.308712e+03	FES = 218100	Diversity = 4.5508e+00
Generation 2190:	Best = 8.308712e+03	FES = 219100	Diversity = 4.5507e+00
Generation 2200:	Best = 8.308712e+03	FES = 220100	Diversity = 4.5496e+00
Generation 2210:	Best = 8.308712e+03	FES = 221100	Diversity = 4.5496e+00
Generation 2220:	Best = 8.308712e+03	FES = 222100	Diversity = 4.5496e+00
Generation 2230:	Best = 8.308712e+03	FES = 223100	Diversity = 4.5489e+00
Generation 2240:	Best = 8.308712e+03	FES = 224100	Diversity = 4.5491e+00
Generation 2250:	Best = 8.308712e+03	FES = 225100	Diversity = 4.5490e+00
Generation 2260:	Best = 8.308712e+03	FES = 226100	Diversity = 4.5487e+00
Generation 2270:	Best = 8.308712e+03	FES = 227100	Diversity = 4.5485e+00
Generation 2280:	Best = 8.308712e+03	FES = 228100	Diversity = 4.5468e+00
Generation 2290:	Best = 8.308712e+03	FES = 229100	Diversity = 4.5467e+00
Generation 2300:	Best = 8.308712e+03	FES = 230100	Diversity = 4.5468e+00
Generation 2310:	Best = 8.308712e+03	FES = 231100	Diversity = 4.5461e+00
Generation 2320:	Best = 8.272900e+03	FES = 232100	Diversity = 4.5456e+00
Generation 2330:	Best = 8.271156e+03	FES = 233100	Diversity = 4.5456e+00
Generation 2340:	Best = 8.271156e+03	FES = 234100	Diversity = 4.5506e+00
Generation 2350:	Best = 8.271156e+03	FES = 235100	Diversity = 4.5501e+00
Generation 2360:	Best = 8.271156e+03	FES = 236100	Diversity = 4.5478e+00
Generation 2370:	Best = 8.271156e+03	FES = 237100	Diversity = 4.5457e+00
Generation 2380:	Best = 8.271156e+03	FES = 238100	Diversity = 4.5485e+00
Generation 2390:	Best = 8.271156e+03	FES = 239100	Diversity = 4.5485e+00
Generation 2400:	Best = 8.271156e+03	FES = 240100	Diversity = 4.5480e+00
Generation 2410:	Best = 8.271156e+03	FES = 241100	Diversity = 4.5480e+00
Generation 2420:	Best = 8.271156e+03	FES = 242100	Diversity = 4.5479e+00
Generation 2430:	Best = 8.263361e+03	FES = 243100	Diversity = 4.5479e+00
Generation 2440:	Best = 8.263361e+03	FES = 244100	Diversity = 4.5479e+00
Generation 2450:	Best = 8.263361e+03	FES = 245100	Diversity = 4.5473e+00
Generation 2460:	Best = 8.263361e+03	FES = 246100	Diversity = 4.5457e+00
Generation 2470:	Best = 8.263361e+03	FES = 247100	Diversity = 4.5454e+00
Generation 2480:	Best = 8.263361e+03	FES = 248100	Diversity = 4.5302e+00
Generation 2490:	Best = 8.263361e+03	FES = 249100	Diversity = 4.5304e+00
Generation 2500:	Best = 8.263361e+03	FES = 250100	Diversity = 4.5426e+00
Generation 2510:	Best = 8.262148e+03	FES = 251100	Diversity = 4.5416e+00
Generation 2520:	Best = 8.262148e+03	FES = 252100	Diversity = 4.5416e+00
Generation 2530:	Best = 8.262148e+03	FES = 253100	Diversity = 4.5413e+00
Generation 2540:	Best = 8.109571e+03	FES = 254100	Diversity = 4.5408e+00
Generation 2550:	Best = 8.109571e+03	FES = 255100	Diversity = 4.5408e+00
Generation 2560:	Best = 8.109571e+03	FES = 256100	Diversity = 4.5397e+00
Generation 2570:	Best = 8.109571e+03	FES = 257100	Diversity = 4.5395e+00
Generation 2580:	Best = 8.109571e+03	FES = 258100	Diversity = 4.5395e+00
Generation 2590:	Best = 8.109571e+03	FES = 259100	Diversity = 4.5378e+00
Generation 2600:	Best = 8.109571e+03	FES = 260100	Diversity = 4.5378e+00

Generation 2610:	Best = 8.109571e+03	FES = 261100	Diversity = 4.5376e+00
Generation 2620:	Best = 8.109571e+03	FES = 262100	Diversity = 4.5360e+00
Generation 2630:	Best = 8.109571e+03	FES = 263100	Diversity = 4.5358e+00
Generation 2640:	Best = 8.109571e+03	FES = 264100	Diversity = 4.5357e+00
Generation 2650:	Best = 8.109571e+03	FES = 265100	Diversity = 4.5351e+00
Generation 2660:	Best = 8.109571e+03	FES = 266100	Diversity = 4.5294e+00
Generation 2670:	Best = 8.109571e+03	FES = 267100	Diversity = 4.5179e+00
Generation 2680:	Best = 8.085475e+03	FES = 268100	Diversity = 4.5108e+00
Generation 2690:	Best = 8.085475e+03	FES = 269100	Diversity = 4.5104e+00
Generation 2700:	Best = 8.085475e+03	FES = 270100	Diversity = 4.5098e+00
Generation 2710:	Best = 8.085475e+03	FES = 271100	Diversity = 4.5078e+00
Generation 2720:	Best = 8.052590e+03	FES = 272100	Diversity = 4.5062e+00
Generation 2730:	Best = 8.052590e+03	FES = 273100	Diversity = 4.5058e+00
Generation 2740:	Best = 8.052590e+03	FES = 274100	Diversity = 4.5057e+00
Generation 2750:	Best = 8.052590e+03	FES = 275100	Diversity = 4.5042e+00
Generation 2760:	Best = 8.052590e+03	FES = 276100	Diversity = 4.5037e+00
Generation 2770:	Best = 8.052590e+03	FES = 277100	Diversity = 4.5014e+00
Generation 2780:	Best = 8.052590e+03	FES = 278100	Diversity = 4.5011e+00
Generation 2790:	Best = 8.052590e+03	FES = 279100	Diversity = 4.5019e+00
Generation 2800:	Best = 8.052590e+03	FES = 280100	Diversity = 4.5019e+00
Generation 2810:	Best = 8.052590e+03	FES = 281100	Diversity = 4.5008e+00
Generation 2820:	Best = 7.974611e+03	FES = 282100	Diversity = 4.5011e+00
Generation 2830:	Best = 7.974611e+03	FES = 283100	Diversity = 4.5003e+00
Generation 2840:	Best = 7.974611e+03	FES = 284100	Diversity = 4.5002e+00
Generation 2850:	Best = 7.974611e+03	FES = 285100	Diversity = 4.5001e+00
Generation 2860:	Best = 7.974611e+03	FES = 286100	Diversity = 4.4999e+00
Generation 2870:	Best = 7.654809e+03	FES = 287100	Diversity = 4.5000e+00
Generation 2880:	Best = 7.654809e+03	FES = 288100	Diversity = 4.5000e+00
Generation 2890:	Best = 7.616598e+03	FES = 289100	Diversity = 4.5000e+00
Generation 2900:	Best = 7.616598e+03	FES = 290100	Diversity = 4.4999e+00
Generation 2910:	Best = 7.616598e+03	FES = 291100	Diversity = 4.4999e+00
Generation 2920:	Best = 7.616598e+03	FES = 292100	Diversity = 4.4994e+00
Generation 2930:	Best = 7.616598e+03	FES = 293100	Diversity = 4.4993e+00
Generation 2940:	Best = 7.616598e+03	FES = 294100	Diversity = 4.4993e+00
Generation 2950:	Best = 7.616598e+03	FES = 295100	Diversity = 4.4958e+00
Generation 2960:	Best = 7.616598e+03	FES = 296100	Diversity = 4.4952e+00
Generation 2970:	Best = 7.616598e+03	FES = 297100	Diversity = 4.4952e+00
Generation 2980:	Best = 7.616598e+03	FES = 298100	Diversity = 4.4957e+00
Generation 2990:	Best = 7.616598e+03	FES = 299100	Diversity = 4.4945e+00
Generation 3000:	Best = 7.616598e+03	FES = 300100	Diversity = 4.4945e+00
Generation 3010:	Best = 7.616598e+03	FES = 301100	Diversity = 4.4945e+00
Generation 3020:	Best = 7.616598e+03	FES = 302100	Diversity = 4.4975e+00
Generation 3030:	Best = 7.616598e+03	FES = 303100	Diversity = 4.4973e+00
Generation 3040:	Best = 7.616598e+03	FES = 304100	Diversity = 4.4972e+00
Generation 3050:	Best = 7.616598e+03	FES = 305100	Diversity = 4.4972e+00
Generation 3060:	Best = 7.616598e+03	FES = 306100	Diversity = 4.4974e+00
Generation 3070:	Best = 7.616598e+03	FES = 307100	Diversity = 4.4974e+00
Generation 3080:	Best = 7.616598e+03	FES = 308100	Diversity = 4.4964e+00
Generation 3090:	Best = 7.616598e+03	FES = 309100	Diversity = 4.4902e+00
Generation 3100:	Best = 7.616598e+03	FES = 310100	Diversity = 4.4895e+00
Generation 3110:	Best = 7.616598e+03	FES = 311100	Diversity = 4.4887e+00
Generation 3120:	Best = 7.616598e+03	FES = 312100	Diversity = 4.4891e+00
Generation 3130:	Best = 7.616598e+03	FES = 313100	Diversity = 4.4889e+00
Generation 3140:	Best = 7.616598e+03	FES = 314100	Diversity = 4.4886e+00

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Generation 3690:	Best = 7.607017e+03	FES = 369100	Diversity = 4.4707e+00
Generation 3700:	Best = 7.607017e+03	FES = 370100	Diversity = 4.4707e+00
Generation 3710:	Best = 7.607017e+03	FES = 371100	Diversity = 4.4708e+00
Generation 3720:	Best = 7.607017e+03	FES = 372100	Diversity = 4.4708e+00
Generation 3730:	Best = 7.607017e+03	FES = 373100	Diversity = 4.4707e+00
Generation 3740:	Best = 7.607017e+03	FES = 374100	Diversity = 4.4706e+00
Generation 3750:	Best = 7.607017e+03	FES = 375100	Diversity = 4.4706e+00
Generation 3760:	Best = 7.607017e+03	FES = 376100	Diversity = 4.4694e+00
Generation 3770:	Best = 7.607017e+03	FES = 377100	Diversity = 4.4692e+00
Generation 3780:	Best = 7.607017e+03	FES = 378100	Diversity = 4.4686e+00
Generation 3790:	Best = 7.607017e+03	FES = 379100	Diversity = 4.4678e+00
Generation 3800:	Best = 7.607017e+03	FES = 380100	Diversity = 4.4679e+00
Generation 3810:	Best = 7.607017e+03	FES = 381100	Diversity = 4.4682e+00
Generation 3820:	Best = 7.597582e+03	FES = 382100	Diversity = 4.4681e+00
Generation 3830:	Best = 7.597582e+03	FES = 383100	Diversity = 4.4688e+00
Generation 3840:	Best = 7.597582e+03	FES = 384100	Diversity = 4.4687e+00
Generation 3850:	Best = 7.597582e+03	FES = 385100	Diversity = 4.4682e+00
Generation 3860:	Best = 7.597582e+03	FES = 386100	Diversity = 4.4678e+00
Generation 3870:	Best = 7.597582e+03	FES = 387100	Diversity = 4.4680e+00
Generation 3880:	Best = 7.597582e+03	FES = 388100	Diversity = 4.4680e+00
Generation 3890:	Best = 7.597582e+03	FES = 389100	Diversity = 4.4679e+00
Generation 3900:	Best = 7.597582e+03	FES = 390100	Diversity = 4.4679e+00
Generation 3910:	Best = 7.597582e+03	FES = 391100	Diversity = 4.4718e+00
Generation 3920:	Best = 7.597582e+03	FES = 392100	Diversity = 4.4718e+00
Generation 3930:	Best = 7.597582e+03	FES = 393100	Diversity = 4.4718e+00
Generation 3940:	Best = 7.597582e+03	FES = 394100	Diversity = 4.4708e+00
Generation 3950:	Best = 7.597582e+03	FES = 395100	Diversity = 4.4705e+00
Generation 3960:	Best = 7.597582e+03	FES = 396100	Diversity = 4.4705e+00
Generation 3970:	Best = 7.597582e+03	FES = 397100	Diversity = 4.4704e+00
Generation 3980:	Best = 7.597582e+03	FES = 398100	Diversity = 4.4702e+00
Generation 3990:	Best = 7.597582e+03	FES = 399100	Diversity = 4.4701e+00
Generation 4000:	Best = 7.597582e+03	FES = 400100	Diversity = 4.4701e+00
Generation 4010:	Best = 7.597582e+03	FES = 401100	Diversity = 4.4699e+00
Generation 4020:	Best = 7.597582e+03	FES = 402100	Diversity = 4.4700e+00
Generation 4030:	Best = 7.597582e+03	FES = 403100	Diversity = 4.4696e+00
Generation 4040:	Best = 7.597582e+03	FES = 404100	Diversity = 4.4696e+00
Generation 4050:	Best = 7.597582e+03	FES = 405100	Diversity = 4.4697e+00
Generation 4060:	Best = 7.597582e+03	FES = 406100	Diversity = 4.4690e+00
Generation 4070:	Best = 7.597582e+03	FES = 407100	Diversity = 4.4687e+00
Generation 4080:	Best = 7.597582e+03	FES = 408100	Diversity = 4.4649e+00
Generation 4090:	Best = 7.597582e+03	FES = 409100	Diversity = 4.4628e+00
Generation 4100:	Best = 7.597582e+03	FES = 410100	Diversity = 4.4623e+00
Generation 4110:	Best = 7.597582e+03	FES = 411100	Diversity = 4.4592e+00
Generation 4120:	Best = 7.544156e+03	FES = 412100	Diversity = 4.4583e+00
Generation 4130:	Best = 7.544156e+03	FES = 413100	Diversity = 4.4578e+00
Generation 4140:	Best = 7.544156e+03	FES = 414100	Diversity = 4.4555e+00
Generation 4150:	Best = 7.544156e+03	FES = 415100	Diversity = 4.4546e+00
Generation 4160:	Best = 7.544156e+03	FES = 416100	Diversity = 4.4539e+00
Generation 4170:	Best = 7.544156e+03	FES = 417100	Diversity = 4.4519e+00
Generation 4180:	Best = 7.544156e+03	FES = 418100	Diversity = 4.4516e+00
Generation 4190:	Best = 7.544156e+03	FES = 419100	Diversity = 4.4514e+00
Generation 4200:	Best = 7.544156e+03	FES = 420100	Diversity = 4.4506e+00
Generation 4210:	Best = 7.544156e+03	FES = 421100	Diversity = 4.4476e+00
Generation 4220:	Best = 7.544156e+03	FES = 422100	Diversity = 4.4471e+00

Generation 4230:	Best = 7.544156e+03	FES = 423100	Diversity = 4.4432e+00
Generation 4240:	Best = 7.544156e+03	FES = 424100	Diversity = 4.4432e+00
Generation 4250:	Best = 7.544156e+03	FES = 425100	Diversity = 4.4431e+00
Generation 4260:	Best = 7.544156e+03	FES = 426100	Diversity = 4.4428e+00
Generation 4270:	Best = 7.544156e+03	FES = 427100	Diversity = 4.4428e+00
Generation 4280:	Best = 7.544156e+03	FES = 428100	Diversity = 4.4408e+00
Generation 4290:	Best = 7.544156e+03	FES = 429100	Diversity = 4.4405e+00
Generation 4300:	Best = 7.544156e+03	FES = 430100	Diversity = 4.4397e+00
Generation 4310:	Best = 7.544156e+03	FES = 431100	Diversity = 4.4389e+00
Generation 4320:	Best = 7.544156e+03	FES = 432100	Diversity = 4.4388e+00
Generation 4330:	Best = 7.544156e+03	FES = 433100	Diversity = 4.4385e+00
Generation 4340:	Best = 7.544156e+03	FES = 434100	Diversity = 4.4380e+00
Generation 4350:	Best = 7.544156e+03	FES = 435100	Diversity = 4.4377e+00
Generation 4360:	Best = 7.544156e+03	FES = 436100	Diversity = 4.4378e+00
Generation 4370:	Best = 7.544156e+03	FES = 437100	Diversity = 4.4392e+00
Generation 4380:	Best = 7.544156e+03	FES = 438100	Diversity = 4.4392e+00
Generation 4390:	Best = 7.544156e+03	FES = 439100	Diversity = 4.4391e+00
Generation 4400:	Best = 7.544156e+03	FES = 440100	Diversity = 4.4391e+00
Generation 4410:	Best = 7.544156e+03	FES = 441100	Diversity = 4.4390e+00
Generation 4420:	Best = 7.544156e+03	FES = 442100	Diversity = 4.4389e+00
Generation 4430:	Best = 7.544156e+03	FES = 443100	Diversity = 4.4389e+00
Generation 4440:	Best = 7.544156e+03	FES = 444100	Diversity = 4.4389e+00
Generation 4450:	Best = 7.536105e+03	FES = 445100	Diversity = 4.4368e+00
Generation 4460:	Best = 7.536105e+03	FES = 446100	Diversity = 4.4367e+00
Generation 4470:	Best = 7.536105e+03	FES = 447100	Diversity = 4.4339e+00
Generation 4480:	Best = 7.484667e+03	FES = 448100	Diversity = 4.4339e+00
Generation 4490:	Best = 7.484667e+03	FES = 449100	Diversity = 4.4334e+00
Generation 4500:	Best = 7.484667e+03	FES = 450100	Diversity = 4.4334e+00
Generation 4510:	Best = 7.484667e+03	FES = 451100	Diversity = 4.4325e+00
Generation 4520:	Best = 7.484667e+03	FES = 452100	Diversity = 4.4307e+00
Generation 4530:	Best = 7.484667e+03	FES = 453100	Diversity = 4.4299e+00
Generation 4540:	Best = 7.484667e+03	FES = 454100	Diversity = 4.4300e+00
Generation 4550:	Best = 7.484667e+03	FES = 455100	Diversity = 4.4299e+00
Generation 4560:	Best = 7.484667e+03	FES = 456100	Diversity = 4.4295e+00
Generation 4570:	Best = 7.484667e+03	FES = 457100	Diversity = 4.4293e+00
Generation 4580:	Best = 7.484667e+03	FES = 458100	Diversity = 4.4376e+00
Generation 4590:	Best = 7.484667e+03	FES = 459100	Diversity = 4.4374e+00
Generation 4600:	Best = 7.484667e+03	FES = 460100	Diversity = 4.4364e+00
Generation 4610:	Best = 7.484667e+03	FES = 461100	Diversity = 4.4364e+00
Generation 4620:	Best = 7.484667e+03	FES = 462100	Diversity = 4.4365e+00
Generation 4630:	Best = 7.484667e+03	FES = 463100	Diversity = 4.4369e+00
Generation 4640:	Best = 7.484667e+03	FES = 464100	Diversity = 4.4371e+00
Generation 4650:	Best = 7.484667e+03	FES = 465100	Diversity = 4.4372e+00
Generation 4660:	Best = 7.484667e+03	FES = 466100	Diversity = 4.4341e+00
Generation 4670:	Best = 7.484667e+03	FES = 467100	Diversity = 4.4347e+00
Generation 4680:	Best = 7.484667e+03	FES = 468100	Diversity = 4.4349e+00
Generation 4690:	Best = 7.484667e+03	FES = 469100	Diversity = 4.4348e+00
Generation 4700:	Best = 7.484667e+03	FES = 470100	Diversity = 4.4351e+00
Generation 4710:	Best = 7.484667e+03	FES = 471100	Diversity = 4.4351e+00
Generation 4720:	Best = 7.484667e+03	FES = 472100	Diversity = 4.4351e+00
Generation 4730:	Best = 7.479692e+03	FES = 473100	Diversity = 4.4351e+00
Generation 4740:	Best = 7.479692e+03	FES = 474100	Diversity = 4.4351e+00
Generation 4750:	Best = 7.479692e+03	FES = 475100	Diversity = 4.4354e+00
Generation 4760:	Best = 7.479692e+03	FES = 476100	Diversity = 4.4346e+00

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[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

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Generation 9630: Best = 6.894113e+03 FES = 963100 Diversity = 4.4773e+00
Generation 9640: Best = 6.894113e+03 FES = 964100 Diversity = 4.4772e+00
Generation 9650: Best = 6.894113e+03 FES = 965100 Diversity = 4.4772e+00
Generation 9660: Best = 6.894113e+03 FES = 966100 Diversity = 4.4771e+00
Generation 9670: Best = 6.894113e+03 FES = 967100 Diversity = 4.4771e+00
Generation 9680: Best = 6.894113e+03 FES = 968100 Diversity = 4.4771e+00
Generation 9690: Best = 6.894113e+03 FES = 969100 Diversity = 4.4771e+00
Generation 9700: Best = 6.894113e+03 FES = 970100 Diversity = 4.4771e+00
Generation 9710: Best = 6.894113e+03 FES = 971100 Diversity = 4.4771e+00
Generation 9720: Best = 6.894113e+03 FES = 972100 Diversity = 4.4769e+00
Generation 9730: Best = 6.894113e+03 FES = 973100 Diversity = 4.4769e+00
Generation 9740: Best = 6.894113e+03 FES = 974100 Diversity = 4.4769e+00
Generation 9750: Best = 6.894113e+03 FES = 975100 Diversity = 4.4774e+00
Generation 9760: Best = 6.894113e+03 FES = 976100 Diversity = 4.4774e+00
Generation 9770: Best = 6.894113e+03 FES = 977100 Diversity = 4.4775e+00
Generation 9780: Best = 6.894113e+03 FES = 978100 Diversity = 4.4775e+00
Generation 9790: Best = 6.894113e+03 FES = 979100 Diversity = 4.4775e+00
Generation 9800: Best = 6.894113e+03 FES = 980100 Diversity = 4.4773e+00
Generation 9810: Best = 6.894113e+03 FES = 981100 Diversity = 4.4773e+00
Generation 9820: Best = 6.894113e+03 FES = 982100 Diversity = 4.4773e+00
Generation 9830: Best = 6.894113e+03 FES = 983100 Diversity = 4.4773e+00
Generation 9840: Best = 6.894113e+03 FES = 984100 Diversity = 4.4772e+00
Generation 9850: Best = 6.894113e+03 FES = 985100 Diversity = 4.4769e+00
Generation 9860: Best = 6.894113e+03 FES = 986100 Diversity = 4.4769e+00
Generation 9870: Best = 6.894113e+03 FES = 987100 Diversity = 4.4769e+00
Generation 9880: Best = 6.894113e+03 FES = 988100 Diversity = 4.4769e+00
Generation 9890: Best = 6.894113e+03 FES = 989100 Diversity = 4.4769e+00
Generation 9900: Best = 6.894113e+03 FES = 990100 Diversity = 4.4765e+00
Generation 9910: Best = 6.894113e+03 FES = 991100 Diversity = 4.4765e+00
Generation 9920: Best = 6.894113e+03 FES = 992100 Diversity = 4.4765e+00
Generation 9930: Best = 6.894113e+03 FES = 993100 Diversity = 4.4755e+00
Generation 9940: Best = 6.894113e+03 FES = 994100 Diversity = 4.4755e+00
Generation 9950: Best = 6.894113e+03 FES = 995100 Diversity = 4.4755e+00
Generation 9960: Best = 6.894113e+03 FES = 996100 Diversity = 4.4755e+00
Generation 9970: Best = 6.894113e+03 FES = 997100 Diversity = 4.4755e+00
Generation 9980: Best = 6.894113e+03 FES = 998100 Diversity = 4.4755e+00
Generation 9990: Best = 6.894113e+03 FES = 999100 Diversity = 4.4754e+00

```

Optimization completed!
 Final Best Fitness = 6.894113e+03
 Total Function Evaluations = 1000000

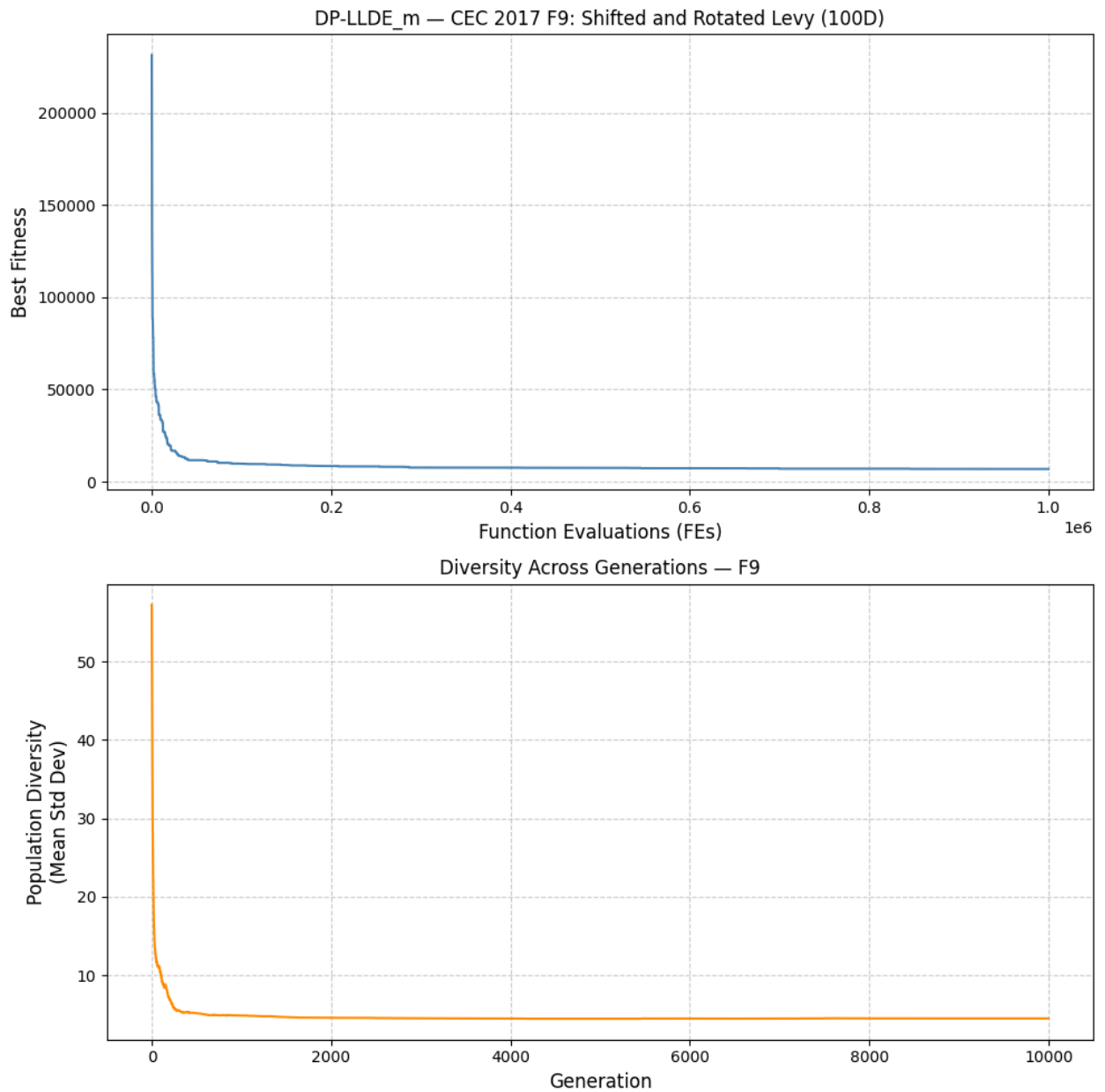
```

=====
=
Results:
  Best Fitness           : 6.894113e+03
  Global Optimum         : 9.000000e+02
  Error (|best-optimum|) : 5.994113e+03
  Best Solution (5 dim)  : [-11.36966753 -20.18362899 34.28206245 48.53351802
-24.18811285]
  D^(0) (init. diversity): 5.7220e+01
=====
=

```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results_2/all_results_che

ckpoint.pkl



Graph saved as convergence_F9.png

✓ Plot saved to: /content/drive/My Drive/thesis_results_2/plots/convergence_F9.png

```
In [ ]: # =====  
# F10 — Shifted and Rotated Schwefel  
# =====  
run_and_plot('F10', make_wrapper(simple.f10), 1000.0, "Shifted and Rotated Sch
```

=====

=

DP-LLDE_m on CEC 2017 F10 – Single Run

=====

=

Function : Shifted and Rotated Schwefel's (100D)

Bounds : [-100, 100]

Global optimum : 1000.0

Initial $D^{(0)} = 5.7220e+01$ | Diversity threshold = 5.7220e+00

Generation 0: Best Fitness = 3.685326e+04 | Diversity = 5.7220e+01

Generation	10:	Best = 3.395799e+04	FES = 1100	Diversity = 4.9727e+01
Generation	20:	Best = 3.395799e+04	FES = 2100	Diversity = 4.7521e+01
Generation	30:	Best = 3.275614e+04	FES = 3100	Diversity = 4.6833e+01
Generation	40:	Best = 3.275614e+04	FES = 4100	Diversity = 4.6413e+01
Generation	50:	Best = 3.275614e+04	FES = 5100	Diversity = 4.4874e+01
Generation	60:	Best = 3.275614e+04	FES = 6100	Diversity = 4.4246e+01
Generation	70:	Best = 3.236373e+04	FES = 7100	Diversity = 4.3896e+01
Generation	80:	Best = 3.162763e+04	FES = 8100	Diversity = 4.3612e+01
Generation	90:	Best = 3.152022e+04	FES = 9100	Diversity = 4.3674e+01
Generation	100:	Best = 3.149132e+04	FES = 10100	Diversity = 4.3587e+01
Generation	110:	Best = 2.968259e+04	FES = 11100	Diversity = 4.3701e+01
Generation	120:	Best = 2.968259e+04	FES = 12100	Diversity = 4.3525e+01
Generation	130:	Best = 2.968259e+04	FES = 13100	Diversity = 4.3864e+01
Generation	140:	Best = 2.968259e+04	FES = 14100	Diversity = 4.3557e+01
Generation	150:	Best = 2.968259e+04	FES = 15100	Diversity = 4.3331e+01
Generation	160:	Best = 2.968259e+04	FES = 16100	Diversity = 4.3249e+01
Generation	170:	Best = 2.968259e+04	FES = 17100	Diversity = 4.3141e+01
Generation	180:	Best = 2.968259e+04	FES = 18100	Diversity = 4.2671e+01
Generation	190:	Best = 2.890716e+04	FES = 19100	Diversity = 4.2580e+01
Generation	200:	Best = 2.774488e+04	FES = 20100	Diversity = 4.2599e+01
Generation	210:	Best = 2.581123e+04	FES = 21100	Diversity = 4.2467e+01
Generation	220:	Best = 2.581123e+04	FES = 22100	Diversity = 4.2164e+01
Generation	230:	Best = 2.581123e+04	FES = 23100	Diversity = 4.2196e+01
Generation	240:	Best = 2.581123e+04	FES = 24100	Diversity = 4.2340e+01
Generation	250:	Best = 2.581123e+04	FES = 25100	Diversity = 4.2091e+01
Generation	260:	Best = 2.581123e+04	FES = 26100	Diversity = 4.2043e+01
Generation	270:	Best = 2.581123e+04	FES = 27100	Diversity = 4.2102e+01
Generation	280:	Best = 2.581123e+04	FES = 28100	Diversity = 4.2058e+01
Generation	290:	Best = 2.581123e+04	FES = 29100	Diversity = 4.2022e+01
Generation	300:	Best = 2.581123e+04	FES = 30100	Diversity = 4.2232e+01
Generation	310:	Best = 2.581123e+04	FES = 31100	Diversity = 4.2225e+01
Generation	320:	Best = 2.581123e+04	FES = 32100	Diversity = 4.2263e+01
Generation	330:	Best = 2.581123e+04	FES = 33100	Diversity = 4.2242e+01
Generation	340:	Best = 2.581123e+04	FES = 34100	Diversity = 4.2260e+01
Generation	350:	Best = 2.581123e+04	FES = 35100	Diversity = 4.2228e+01
Generation	360:	Best = 2.581123e+04	FES = 36100	Diversity = 4.2245e+01
Generation	370:	Best = 2.581123e+04	FES = 37100	Diversity = 4.2241e+01
Generation	380:	Best = 2.581123e+04	FES = 38100	Diversity = 4.1925e+01
Generation	390:	Best = 2.581123e+04	FES = 39100	Diversity = 4.1803e+01
Generation	400:	Best = 2.581123e+04	FES = 40100	Diversity = 4.1792e+01
Generation	410:	Best = 2.581123e+04	FES = 41100	Diversity = 4.1693e+01
Generation	420:	Best = 2.581123e+04	FES = 42100	Diversity = 4.1683e+01
Generation	430:	Best = 2.581123e+04	FES = 43100	Diversity = 4.1407e+01
Generation	440:	Best = 2.581123e+04	FES = 44100	Diversity = 4.1263e+01

Generation	450:	Best = 2.581123e+04	FES = 45100	Diversity = 4.1241e+01
Generation	460:	Best = 2.572485e+04	FES = 46100	Diversity = 4.1212e+01
Generation	470:	Best = 2.572485e+04	FES = 47100	Diversity = 4.1193e+01
Generation	480:	Best = 2.478588e+04	FES = 48100	Diversity = 4.1157e+01
Generation	490:	Best = 2.478588e+04	FES = 49100	Diversity = 4.1146e+01
Generation	500:	Best = 2.478588e+04	FES = 50100	Diversity = 4.0826e+01
Generation	510:	Best = 2.478588e+04	FES = 51100	Diversity = 4.0863e+01
Generation	520:	Best = 2.478588e+04	FES = 52100	Diversity = 4.0851e+01
Generation	530:	Best = 2.478588e+04	FES = 53100	Diversity = 4.0844e+01
Generation	540:	Best = 2.478588e+04	FES = 54100	Diversity = 4.0870e+01
Generation	550:	Best = 2.478588e+04	FES = 55100	Diversity = 4.0780e+01
Generation	560:	Best = 2.471798e+04	FES = 56100	Diversity = 4.0742e+01
Generation	570:	Best = 2.471798e+04	FES = 57100	Diversity = 4.0646e+01
Generation	580:	Best = 2.471798e+04	FES = 58100	Diversity = 4.0573e+01
Generation	590:	Best = 2.446791e+04	FES = 59100	Diversity = 4.0559e+01
Generation	600:	Best = 2.446791e+04	FES = 60100	Diversity = 4.0521e+01
Generation	610:	Best = 2.446791e+04	FES = 61100	Diversity = 4.0273e+01
Generation	620:	Best = 2.446791e+04	FES = 62100	Diversity = 4.0255e+01
Generation	630:	Best = 2.446791e+04	FES = 63100	Diversity = 4.0246e+01
Generation	640:	Best = 2.446791e+04	FES = 64100	Diversity = 4.0440e+01
Generation	650:	Best = 2.446791e+04	FES = 65100	Diversity = 4.0444e+01
Generation	660:	Best = 2.446791e+04	FES = 66100	Diversity = 4.0350e+01
Generation	670:	Best = 2.446791e+04	FES = 67100	Diversity = 4.0384e+01
Generation	680:	Best = 2.439392e+04	FES = 68100	Diversity = 4.0354e+01
Generation	690:	Best = 2.419012e+04	FES = 69100	Diversity = 4.0290e+01
Generation	700:	Best = 2.419012e+04	FES = 70100	Diversity = 4.0283e+01
Generation	710:	Best = 2.419012e+04	FES = 71100	Diversity = 4.0264e+01
Generation	720:	Best = 2.419012e+04	FES = 72100	Diversity = 4.0245e+01
Generation	730:	Best = 2.419012e+04	FES = 73100	Diversity = 4.0223e+01
Generation	740:	Best = 2.419012e+04	FES = 74100	Diversity = 4.0177e+01
Generation	750:	Best = 2.416771e+04	FES = 75100	Diversity = 4.0174e+01
Generation	760:	Best = 2.416771e+04	FES = 76100	Diversity = 4.0107e+01
Generation	770:	Best = 2.416771e+04	FES = 77100	Diversity = 3.9901e+01
Generation	780:	Best = 2.416771e+04	FES = 78100	Diversity = 3.9767e+01
Generation	790:	Best = 2.416771e+04	FES = 79100	Diversity = 3.9767e+01
Generation	800:	Best = 2.416771e+04	FES = 80100	Diversity = 3.9912e+01
Generation	810:	Best = 2.416771e+04	FES = 81100	Diversity = 3.9912e+01
Generation	820:	Best = 2.416771e+04	FES = 82100	Diversity = 3.9909e+01
Generation	830:	Best = 2.416771e+04	FES = 83100	Diversity = 4.0049e+01
Generation	840:	Best = 2.416771e+04	FES = 84100	Diversity = 3.9898e+01
Generation	850:	Best = 2.416771e+04	FES = 85100	Diversity = 3.9908e+01
Generation	860:	Best = 2.416771e+04	FES = 86100	Diversity = 3.9583e+01
Generation	870:	Best = 2.414280e+04	FES = 87100	Diversity = 3.9590e+01
Generation	880:	Best = 2.401313e+04	FES = 88100	Diversity = 3.9571e+01
Generation	890:	Best = 2.401313e+04	FES = 89100	Diversity = 3.9480e+01
Generation	900:	Best = 2.401313e+04	FES = 90100	Diversity = 3.9467e+01
Generation	910:	Best = 2.401313e+04	FES = 91100	Diversity = 3.9464e+01
Generation	920:	Best = 2.401313e+04	FES = 92100	Diversity = 3.9444e+01
Generation	930:	Best = 2.401313e+04	FES = 93100	Diversity = 3.9399e+01
Generation	940:	Best = 2.401313e+04	FES = 94100	Diversity = 3.9394e+01
Generation	950:	Best = 2.401313e+04	FES = 95100	Diversity = 3.9384e+01
Generation	960:	Best = 2.401313e+04	FES = 96100	Diversity = 3.9393e+01
Generation	970:	Best = 2.401313e+04	FES = 97100	Diversity = 3.9387e+01
Generation	980:	Best = 2.401313e+04	FES = 98100	Diversity = 3.9359e+01

Generation 990:	Best = 2.401313e+04	FES = 99100	Diversity = 3.9347e+01
Generation 1000:	Best = 2.401313e+04	FES = 100100	Diversity = 3.9339e+01
Generation 1010:	Best = 2.401313e+04	FES = 101100	Diversity = 3.9337e+01
Generation 1020:	Best = 2.401313e+04	FES = 102100	Diversity = 3.9358e+01
Generation 1030:	Best = 2.401313e+04	FES = 103100	Diversity = 3.9365e+01
Generation 1040:	Best = 2.401313e+04	FES = 104100	Diversity = 3.9364e+01
Generation 1050:	Best = 2.401313e+04	FES = 105100	Diversity = 3.9358e+01
Generation 1060:	Best = 2.401313e+04	FES = 106100	Diversity = 3.9348e+01
Generation 1070:	Best = 2.401313e+04	FES = 107100	Diversity = 3.9348e+01
Generation 1080:	Best = 2.401313e+04	FES = 108100	Diversity = 3.9346e+01
Generation 1090:	Best = 2.401313e+04	FES = 109100	Diversity = 3.9347e+01
Generation 1100:	Best = 2.401313e+04	FES = 110100	Diversity = 3.9339e+01
Generation 1110:	Best = 2.401313e+04	FES = 111100	Diversity = 3.9339e+01
Generation 1120:	Best = 2.401313e+04	FES = 112100	Diversity = 3.9333e+01
Generation 1130:	Best = 2.401313e+04	FES = 113100	Diversity = 3.9338e+01
Generation 1140:	Best = 2.399348e+04	FES = 114100	Diversity = 3.9337e+01
Generation 1150:	Best = 2.399348e+04	FES = 115100	Diversity = 3.9333e+01
Generation 1160:	Best = 2.399348e+04	FES = 116100	Diversity = 3.9333e+01
Generation 1170:	Best = 2.399348e+04	FES = 117100	Diversity = 3.9334e+01
Generation 1180:	Best = 2.399348e+04	FES = 118100	Diversity = 3.9332e+01
Generation 1190:	Best = 2.399348e+04	FES = 119100	Diversity = 3.9323e+01
Generation 1200:	Best = 2.399348e+04	FES = 120100	Diversity = 3.9312e+01
Generation 1210:	Best = 2.399348e+04	FES = 121100	Diversity = 3.9310e+01
Generation 1220:	Best = 2.399348e+04	FES = 122100	Diversity = 3.9304e+01
Generation 1230:	Best = 2.399348e+04	FES = 123100	Diversity = 3.9302e+01
Generation 1240:	Best = 2.399348e+04	FES = 124100	Diversity = 3.9294e+01
Generation 1250:	Best = 2.399348e+04	FES = 125100	Diversity = 3.9283e+01
Generation 1260:	Best = 2.399348e+04	FES = 126100	Diversity = 3.9205e+01
Generation 1270:	Best = 2.399348e+04	FES = 127100	Diversity = 3.9201e+01
Generation 1280:	Best = 2.399348e+04	FES = 128100	Diversity = 3.9200e+01
Generation 1290:	Best = 2.399348e+04	FES = 129100	Diversity = 3.9199e+01
Generation 1300:	Best = 2.399348e+04	FES = 130100	Diversity = 3.9189e+01
Generation 1310:	Best = 2.399348e+04	FES = 131100	Diversity = 3.9190e+01
Generation 1320:	Best = 2.387791e+04	FES = 132100	Diversity = 3.9173e+01
Generation 1330:	Best = 2.387791e+04	FES = 133100	Diversity = 3.9173e+01
Generation 1340:	Best = 2.387791e+04	FES = 134100	Diversity = 3.9163e+01
Generation 1350:	Best = 2.387791e+04	FES = 135100	Diversity = 3.9153e+01
Generation 1360:	Best = 2.387791e+04	FES = 136100	Diversity = 3.9126e+01
Generation 1370:	Best = 2.387791e+04	FES = 137100	Diversity = 3.9137e+01
Generation 1380:	Best = 2.387791e+04	FES = 138100	Diversity = 3.9140e+01
Generation 1390:	Best = 2.387791e+04	FES = 139100	Diversity = 3.9129e+01
Generation 1400:	Best = 2.387791e+04	FES = 140100	Diversity = 3.9116e+01
Generation 1410:	Best = 2.387791e+04	FES = 141100	Diversity = 3.9116e+01
Generation 1420:	Best = 2.387791e+04	FES = 142100	Diversity = 3.9110e+01
Generation 1430:	Best = 2.387791e+04	FES = 143100	Diversity = 3.9106e+01
Generation 1440:	Best = 2.387791e+04	FES = 144100	Diversity = 3.9108e+01
Generation 1450:	Best = 2.387791e+04	FES = 145100	Diversity = 3.9094e+01
Generation 1460:	Best = 2.387791e+04	FES = 146100	Diversity = 3.9075e+01
Generation 1470:	Best = 2.387791e+04	FES = 147100	Diversity = 3.9081e+01
Generation 1480:	Best = 2.387791e+04	FES = 148100	Diversity = 3.8967e+01
Generation 1490:	Best = 2.387791e+04	FES = 149100	Diversity = 3.8963e+01
Generation 1500:	Best = 2.387791e+04	FES = 150100	Diversity = 3.8791e+01
Generation 1510:	Best = 2.387791e+04	FES = 151100	Diversity = 3.8764e+01
Generation 1520:	Best = 2.387791e+04	FES = 152100	Diversity = 3.8811e+01

Generation 1530:	Best = 2.387791e+04	FES = 153100	Diversity = 3.8977e+01
Generation 1540:	Best = 2.387791e+04	FES = 154100	Diversity = 3.8957e+01
Generation 1550:	Best = 2.387791e+04	FES = 155100	Diversity = 3.8995e+01
Generation 1560:	Best = 2.373372e+04	FES = 156100	Diversity = 3.8971e+01
Generation 1570:	Best = 2.373372e+04	FES = 157100	Diversity = 3.8976e+01
Generation 1580:	Best = 2.373372e+04	FES = 158100	Diversity = 3.8981e+01
Generation 1590:	Best = 2.373372e+04	FES = 159100	Diversity = 3.9004e+01
Generation 1600:	Best = 2.373372e+04	FES = 160100	Diversity = 3.8991e+01
Generation 1610:	Best = 2.373372e+04	FES = 161100	Diversity = 3.8988e+01
Generation 1620:	Best = 2.373372e+04	FES = 162100	Diversity = 3.9050e+01
Generation 1630:	Best = 2.373372e+04	FES = 163100	Diversity = 3.9038e+01
Generation 1640:	Best = 2.373372e+04	FES = 164100	Diversity = 3.9033e+01
Generation 1650:	Best = 2.373372e+04	FES = 165100	Diversity = 3.9033e+01
Generation 1660:	Best = 2.373372e+04	FES = 166100	Diversity = 3.9034e+01
Generation 1670:	Best = 2.373372e+04	FES = 167100	Diversity = 3.9021e+01
Generation 1680:	Best = 2.373372e+04	FES = 168100	Diversity = 3.9019e+01
Generation 1690:	Best = 2.373372e+04	FES = 169100	Diversity = 3.9017e+01
Generation 1700:	Best = 2.373372e+04	FES = 170100	Diversity = 3.9016e+01
Generation 1710:	Best = 2.373372e+04	FES = 171100	Diversity = 3.9016e+01
Generation 1720:	Best = 2.373372e+04	FES = 172100	Diversity = 3.9016e+01
Generation 1730:	Best = 2.373372e+04	FES = 173100	Diversity = 3.9016e+01
Generation 1740:	Best = 2.373372e+04	FES = 174100	Diversity = 3.9016e+01
Generation 1750:	Best = 2.373372e+04	FES = 175100	Diversity = 3.9016e+01
Generation 1760:	Best = 2.373372e+04	FES = 176100	Diversity = 3.9016e+01
Generation 1770:	Best = 2.373372e+04	FES = 177100	Diversity = 3.9014e+01
Generation 1780:	Best = 2.373372e+04	FES = 178100	Diversity = 3.9014e+01
Generation 1790:	Best = 2.373372e+04	FES = 179100	Diversity = 3.9014e+01
Generation 1800:	Best = 2.373372e+04	FES = 180100	Diversity = 3.9014e+01
Generation 1810:	Best = 2.373372e+04	FES = 181100	Diversity = 3.9011e+01
Generation 1820:	Best = 2.373372e+04	FES = 182100	Diversity = 3.9012e+01
Generation 1830:	Best = 2.373372e+04	FES = 183100	Diversity = 3.9013e+01
Generation 1840:	Best = 2.353962e+04	FES = 184100	Diversity = 3.8954e+01
Generation 1850:	Best = 2.353962e+04	FES = 185100	Diversity = 3.8957e+01
Generation 1860:	Best = 2.353962e+04	FES = 186100	Diversity = 3.8958e+01
Generation 1870:	Best = 2.353962e+04	FES = 187100	Diversity = 3.9079e+01
Generation 1880:	Best = 2.353962e+04	FES = 188100	Diversity = 3.9075e+01
Generation 1890:	Best = 2.353962e+04	FES = 189100	Diversity = 3.9074e+01
Generation 1900:	Best = 2.353962e+04	FES = 190100	Diversity = 3.9066e+01
Generation 1910:	Best = 2.353962e+04	FES = 191100	Diversity = 3.9064e+01
Generation 1920:	Best = 2.353962e+04	FES = 192100	Diversity = 3.9061e+01
Generation 1930:	Best = 2.353962e+04	FES = 193100	Diversity = 3.9048e+01
Generation 1940:	Best = 2.353962e+04	FES = 194100	Diversity = 3.9039e+01
Generation 1950:	Best = 2.353962e+04	FES = 195100	Diversity = 3.9031e+01
Generation 1960:	Best = 2.353962e+04	FES = 196100	Diversity = 3.9019e+01
Generation 1970:	Best = 2.353962e+04	FES = 197100	Diversity = 3.9018e+01
Generation 1980:	Best = 2.353962e+04	FES = 198100	Diversity = 3.9021e+01
Generation 1990:	Best = 2.353962e+04	FES = 199100	Diversity = 3.9058e+01
Generation 2000:	Best = 2.353962e+04	FES = 200100	Diversity = 3.9056e+01
Generation 2010:	Best = 2.353962e+04	FES = 201100	Diversity = 3.9044e+01
Generation 2020:	Best = 2.353962e+04	FES = 202100	Diversity = 3.9021e+01
Generation 2030:	Best = 2.353962e+04	FES = 203100	Diversity = 3.9010e+01
Generation 2040:	Best = 2.353962e+04	FES = 204100	Diversity = 3.9011e+01
Generation 2050:	Best = 2.353962e+04	FES = 205100	Diversity = 3.9004e+01
Generation 2060:	Best = 2.353962e+04	FES = 206100	Diversity = 3.9003e+01

Generation 2070:	Best = 2.353962e+04	FES = 207100	Diversity = 3.9000e+01
Generation 2080:	Best = 2.353962e+04	FES = 208100	Diversity = 3.8999e+01
Generation 2090:	Best = 2.353962e+04	FES = 209100	Diversity = 3.8996e+01
Generation 2100:	Best = 2.353962e+04	FES = 210100	Diversity = 3.8995e+01
Generation 2110:	Best = 2.353962e+04	FES = 211100	Diversity = 3.8984e+01
Generation 2120:	Best = 2.353962e+04	FES = 212100	Diversity = 3.8984e+01
Generation 2130:	Best = 2.353962e+04	FES = 213100	Diversity = 3.8938e+01
Generation 2140:	Best = 2.353962e+04	FES = 214100	Diversity = 3.8933e+01
Generation 2150:	Best = 2.353962e+04	FES = 215100	Diversity = 3.8932e+01
Generation 2160:	Best = 2.345141e+04	FES = 216100	Diversity = 3.8889e+01
Generation 2170:	Best = 2.345141e+04	FES = 217100	Diversity = 3.8895e+01
Generation 2180:	Best = 2.345141e+04	FES = 218100	Diversity = 3.8894e+01
Generation 2190:	Best = 2.345141e+04	FES = 219100	Diversity = 3.8876e+01
Generation 2200:	Best = 2.345141e+04	FES = 220100	Diversity = 3.8804e+01
Generation 2210:	Best = 2.345141e+04	FES = 221100	Diversity = 3.8792e+01
Generation 2220:	Best = 2.345141e+04	FES = 222100	Diversity = 3.8778e+01
Generation 2230:	Best = 2.345141e+04	FES = 223100	Diversity = 3.8778e+01
Generation 2240:	Best = 2.345141e+04	FES = 224100	Diversity = 3.8759e+01
Generation 2250:	Best = 2.345141e+04	FES = 225100	Diversity = 3.8759e+01
Generation 2260:	Best = 2.345141e+04	FES = 226100	Diversity = 3.8756e+01
Generation 2270:	Best = 2.345141e+04	FES = 227100	Diversity = 3.8758e+01
Generation 2280:	Best = 2.345141e+04	FES = 228100	Diversity = 3.8759e+01
Generation 2290:	Best = 2.345141e+04	FES = 229100	Diversity = 3.8759e+01
Generation 2300:	Best = 2.345141e+04	FES = 230100	Diversity = 3.8759e+01
Generation 2310:	Best = 2.345141e+04	FES = 231100	Diversity = 3.8753e+01
Generation 2320:	Best = 2.345141e+04	FES = 232100	Diversity = 3.8753e+01
Generation 2330:	Best = 2.345141e+04	FES = 233100	Diversity = 3.8753e+01
Generation 2340:	Best = 2.345141e+04	FES = 234100	Diversity = 3.8761e+01
Generation 2350:	Best = 2.345141e+04	FES = 235100	Diversity = 3.8761e+01
Generation 2360:	Best = 2.345141e+04	FES = 236100	Diversity = 3.8669e+01
Generation 2370:	Best = 2.345141e+04	FES = 237100	Diversity = 3.8667e+01
Generation 2380:	Best = 2.345141e+04	FES = 238100	Diversity = 3.8666e+01
Generation 2390:	Best = 2.345141e+04	FES = 239100	Diversity = 3.8665e+01
Generation 2400:	Best = 2.345141e+04	FES = 240100	Diversity = 3.8665e+01
Generation 2410:	Best = 2.345141e+04	FES = 241100	Diversity = 3.8657e+01
Generation 2420:	Best = 2.345141e+04	FES = 242100	Diversity = 3.8624e+01
Generation 2430:	Best = 2.345141e+04	FES = 243100	Diversity = 3.8621e+01
Generation 2440:	Best = 2.345141e+04	FES = 244100	Diversity = 3.8616e+01
Generation 2450:	Best = 2.345141e+04	FES = 245100	Diversity = 3.8622e+01
Generation 2460:	Best = 2.342021e+04	FES = 246100	Diversity = 3.8824e+01
Generation 2470:	Best = 2.342021e+04	FES = 247100	Diversity = 3.8824e+01
Generation 2480:	Best = 2.342021e+04	FES = 248100	Diversity = 3.8824e+01
Generation 2490:	Best = 2.342021e+04	FES = 249100	Diversity = 3.8821e+01
Generation 2500:	Best = 2.342021e+04	FES = 250100	Diversity = 3.8820e+01
Generation 2510:	Best = 2.342021e+04	FES = 251100	Diversity = 3.8824e+01
Generation 2520:	Best = 2.342021e+04	FES = 252100	Diversity = 3.8824e+01
Generation 2530:	Best = 2.342021e+04	FES = 253100	Diversity = 3.8821e+01
Generation 2540:	Best = 2.342021e+04	FES = 254100	Diversity = 3.8821e+01
Generation 2550:	Best = 2.342021e+04	FES = 255100	Diversity = 3.8821e+01
Generation 2560:	Best = 2.342021e+04	FES = 256100	Diversity = 3.8820e+01
Generation 2570:	Best = 2.342021e+04	FES = 257100	Diversity = 3.8799e+01
Generation 2580:	Best = 2.336462e+04	FES = 258100	Diversity = 3.8799e+01
Generation 2590:	Best = 2.336462e+04	FES = 259100	Diversity = 3.8801e+01
Generation 2600:	Best = 2.317959e+04	FES = 260100	Diversity = 3.8798e+01

Generation 3150:	Best = 2.297794e+04	FES = 315100	Diversity = 3.8345e+01
Generation 3160:	Best = 2.297794e+04	FES = 316100	Diversity = 3.8330e+01
Generation 3170:	Best = 2.297794e+04	FES = 317100	Diversity = 3.8331e+01
Generation 3180:	Best = 2.297794e+04	FES = 318100	Diversity = 3.8330e+01
Generation 3190:	Best = 2.297794e+04	FES = 319100	Diversity = 3.8329e+01
Generation 3200:	Best = 2.295315e+04	FES = 320100	Diversity = 3.8329e+01
Generation 3210:	Best = 2.295315e+04	FES = 321100	Diversity = 3.8326e+01
Generation 3220:	Best = 2.295315e+04	FES = 322100	Diversity = 3.8325e+01
Generation 3230:	Best = 2.295315e+04	FES = 323100	Diversity = 3.8315e+01
Generation 3240:	Best = 2.295315e+04	FES = 324100	Diversity = 3.8368e+01
Generation 3250:	Best = 2.295315e+04	FES = 325100	Diversity = 3.8363e+01
Generation 3260:	Best = 2.295315e+04	FES = 326100	Diversity = 3.8362e+01
Generation 3270:	Best = 2.295315e+04	FES = 327100	Diversity = 3.8352e+01
Generation 3280:	Best = 2.295315e+04	FES = 328100	Diversity = 3.8349e+01
Generation 3290:	Best = 2.295315e+04	FES = 329100	Diversity = 3.8351e+01
Generation 3300:	Best = 2.295315e+04	FES = 330100	Diversity = 3.8347e+01
Generation 3310:	Best = 2.295315e+04	FES = 331100	Diversity = 3.8345e+01
Generation 3320:	Best = 2.295315e+04	FES = 332100	Diversity = 3.8344e+01
Generation 3330:	Best = 2.288019e+04	FES = 333100	Diversity = 3.8343e+01
Generation 3340:	Best = 2.288019e+04	FES = 334100	Diversity = 3.8344e+01
Generation 3350:	Best = 2.288019e+04	FES = 335100	Diversity = 3.8338e+01
Generation 3360:	Best = 2.288019e+04	FES = 336100	Diversity = 3.8339e+01
Generation 3370:	Best = 2.288019e+04	FES = 337100	Diversity = 3.8339e+01
Generation 3380:	Best = 2.268116e+04	FES = 338100	Diversity = 3.8342e+01
Generation 3390:	Best = 2.268116e+04	FES = 339100	Diversity = 3.8343e+01
Generation 3400:	Best = 2.268116e+04	FES = 340100	Diversity = 3.8341e+01
Generation 3410:	Best = 2.268116e+04	FES = 341100	Diversity = 3.8341e+01
Generation 3420:	Best = 2.268116e+04	FES = 342100	Diversity = 3.8341e+01
Generation 3430:	Best = 2.268116e+04	FES = 343100	Diversity = 3.8341e+01
Generation 3440:	Best = 2.268116e+04	FES = 344100	Diversity = 3.8341e+01
Generation 3450:	Best = 2.268116e+04	FES = 345100	Diversity = 3.8341e+01
Generation 3460:	Best = 2.268116e+04	FES = 346100	Diversity = 3.8341e+01
Generation 3470:	Best = 2.268116e+04	FES = 347100	Diversity = 3.8330e+01
Generation 3480:	Best = 2.268116e+04	FES = 348100	Diversity = 3.8327e+01
Generation 3490:	Best = 2.268116e+04	FES = 349100	Diversity = 3.8317e+01
Generation 3500:	Best = 2.268116e+04	FES = 350100	Diversity = 3.8316e+01
Generation 3510:	Best = 2.268116e+04	FES = 351100	Diversity = 3.8316e+01
Generation 3520:	Best = 2.268116e+04	FES = 352100	Diversity = 3.8307e+01
Generation 3530:	Best = 2.268116e+04	FES = 353100	Diversity = 3.8307e+01
Generation 3540:	Best = 2.268116e+04	FES = 354100	Diversity = 3.8307e+01
Generation 3550:	Best = 2.268116e+04	FES = 355100	Diversity = 3.8299e+01
Generation 3560:	Best = 2.268116e+04	FES = 356100	Diversity = 3.8293e+01
Generation 3570:	Best = 2.268116e+04	FES = 357100	Diversity = 3.8293e+01
Generation 3580:	Best = 2.264335e+04	FES = 358100	Diversity = 3.8290e+01
Generation 3590:	Best = 2.264335e+04	FES = 359100	Diversity = 3.8280e+01
Generation 3600:	Best = 2.264335e+04	FES = 360100	Diversity = 3.8276e+01
Generation 3610:	Best = 2.264335e+04	FES = 361100	Diversity = 3.8271e+01
Generation 3620:	Best = 2.264335e+04	FES = 362100	Diversity = 3.8276e+01
Generation 3630:	Best = 2.214545e+04	FES = 363100	Diversity = 3.8277e+01
Generation 3640:	Best = 2.214545e+04	FES = 364100	Diversity = 3.8274e+01
Generation 3650:	Best = 2.214545e+04	FES = 365100	Diversity = 3.8270e+01
Generation 3660:	Best = 2.214545e+04	FES = 366100	Diversity = 3.8267e+01
Generation 3670:	Best = 2.214545e+04	FES = 367100	Diversity = 3.8269e+01
Generation 3680:	Best = 2.214545e+04	FES = 368100	Diversity = 3.8267e+01

Generation 3690:	Best = 2.214545e+04	FES = 369100	Diversity = 3.8263e+01
Generation 3700:	Best = 2.214545e+04	FES = 370100	Diversity = 3.8258e+01
Generation 3710:	Best = 2.214545e+04	FES = 371100	Diversity = 3.8256e+01
Generation 3720:	Best = 2.214545e+04	FES = 372100	Diversity = 3.8234e+01
Generation 3730:	Best = 2.214545e+04	FES = 373100	Diversity = 3.8230e+01
Generation 3740:	Best = 2.200348e+04	FES = 374100	Diversity = 3.8226e+01
Generation 3750:	Best = 2.200348e+04	FES = 375100	Diversity = 3.8221e+01
Generation 3760:	Best = 2.200348e+04	FES = 376100	Diversity = 3.8217e+01
Generation 3770:	Best = 2.190239e+04	FES = 377100	Diversity = 3.8216e+01
Generation 3780:	Best = 2.190239e+04	FES = 378100	Diversity = 3.8214e+01
Generation 3790:	Best = 2.190239e+04	FES = 379100	Diversity = 3.8213e+01
Generation 3800:	Best = 2.190239e+04	FES = 380100	Diversity = 3.8211e+01
Generation 3810:	Best = 2.190239e+04	FES = 381100	Diversity = 3.8207e+01
Generation 3820:	Best = 2.190239e+04	FES = 382100	Diversity = 3.8137e+01
Generation 3830:	Best = 2.190239e+04	FES = 383100	Diversity = 3.8136e+01
Generation 3840:	Best = 2.190239e+04	FES = 384100	Diversity = 3.8133e+01
Generation 3850:	Best = 2.190239e+04	FES = 385100	Diversity = 3.8111e+01
Generation 3860:	Best = 2.190239e+04	FES = 386100	Diversity = 3.8110e+01
Generation 3870:	Best = 2.190239e+04	FES = 387100	Diversity = 3.8109e+01
Generation 3880:	Best = 2.190239e+04	FES = 388100	Diversity = 3.8112e+01
Generation 3890:	Best = 2.190239e+04	FES = 389100	Diversity = 3.8108e+01
Generation 3900:	Best = 2.190239e+04	FES = 390100	Diversity = 3.8108e+01
Generation 3910:	Best = 2.190239e+04	FES = 391100	Diversity = 3.8058e+01
Generation 3920:	Best = 2.190239e+04	FES = 392100	Diversity = 3.8054e+01
Generation 3930:	Best = 2.190239e+04	FES = 393100	Diversity = 3.8052e+01
Generation 3940:	Best = 2.190239e+04	FES = 394100	Diversity = 3.8053e+01
Generation 3950:	Best = 2.190239e+04	FES = 395100	Diversity = 3.7981e+01
Generation 3960:	Best = 2.190239e+04	FES = 396100	Diversity = 3.7979e+01
Generation 3970:	Best = 2.190239e+04	FES = 397100	Diversity = 3.7973e+01
Generation 3980:	Best = 2.190239e+04	FES = 398100	Diversity = 3.7970e+01
Generation 3990:	Best = 2.190239e+04	FES = 399100	Diversity = 3.7956e+01
Generation 4000:	Best = 2.190239e+04	FES = 400100	Diversity = 3.7943e+01
Generation 4010:	Best = 2.190239e+04	FES = 401100	Diversity = 3.7943e+01
Generation 4020:	Best = 2.190239e+04	FES = 402100	Diversity = 3.7937e+01
Generation 4030:	Best = 2.190239e+04	FES = 403100	Diversity = 3.7931e+01
Generation 4040:	Best = 2.190239e+04	FES = 404100	Diversity = 3.7930e+01
Generation 4050:	Best = 2.190239e+04	FES = 405100	Diversity = 3.7930e+01
Generation 4060:	Best = 2.189572e+04	FES = 406100	Diversity = 3.7917e+01
Generation 4070:	Best = 2.189572e+04	FES = 407100	Diversity = 3.7904e+01
Generation 4080:	Best = 2.189572e+04	FES = 408100	Diversity = 3.7902e+01
Generation 4090:	Best = 2.189572e+04	FES = 409100	Diversity = 3.7895e+01
Generation 4100:	Best = 2.189572e+04	FES = 410100	Diversity = 3.7893e+01
Generation 4110:	Best = 2.189572e+04	FES = 411100	Diversity = 3.7918e+01
Generation 4120:	Best = 2.189572e+04	FES = 412100	Diversity = 3.7912e+01
Generation 4130:	Best = 2.189572e+04	FES = 413100	Diversity = 3.7936e+01
Generation 4140:	Best = 2.189572e+04	FES = 414100	Diversity = 3.7933e+01
Generation 4150:	Best = 2.189572e+04	FES = 415100	Diversity = 3.7927e+01
Generation 4160:	Best = 2.189572e+04	FES = 416100	Diversity = 3.7909e+01
Generation 4170:	Best = 2.189572e+04	FES = 417100	Diversity = 3.7910e+01
Generation 4180:	Best = 2.189572e+04	FES = 418100	Diversity = 3.7911e+01
Generation 4190:	Best = 2.189572e+04	FES = 419100	Diversity = 3.7909e+01
Generation 4200:	Best = 2.189572e+04	FES = 420100	Diversity = 3.7909e+01
Generation 4210:	Best = 2.189572e+04	FES = 421100	Diversity = 3.7908e+01
Generation 4220:	Best = 2.189572e+04	FES = 422100	Diversity = 3.7900e+01

Generation 4230:	Best = 2.189572e+04	FES = 423100	Diversity = 3.7896e+01
Generation 4240:	Best = 2.189572e+04	FES = 424100	Diversity = 3.7896e+01
Generation 4250:	Best = 2.189572e+04	FES = 425100	Diversity = 3.7897e+01
Generation 4260:	Best = 2.189572e+04	FES = 426100	Diversity = 3.7898e+01
Generation 4270:	Best = 2.187860e+04	FES = 427100	Diversity = 3.7867e+01
Generation 4280:	Best = 2.187860e+04	FES = 428100	Diversity = 3.7863e+01
Generation 4290:	Best = 2.187860e+04	FES = 429100	Diversity = 3.7865e+01
Generation 4300:	Best = 2.187860e+04	FES = 430100	Diversity = 3.7865e+01
Generation 4310:	Best = 2.187860e+04	FES = 431100	Diversity = 3.7861e+01
Generation 4320:	Best = 2.187860e+04	FES = 432100	Diversity = 3.7861e+01
Generation 4330:	Best = 2.187860e+04	FES = 433100	Diversity = 3.7860e+01
Generation 4340:	Best = 2.187860e+04	FES = 434100	Diversity = 3.7861e+01
Generation 4350:	Best = 2.187860e+04	FES = 435100	Diversity = 3.7861e+01
Generation 4360:	Best = 2.187860e+04	FES = 436100	Diversity = 3.7858e+01
Generation 4370:	Best = 2.187860e+04	FES = 437100	Diversity = 3.7858e+01
Generation 4380:	Best = 2.187860e+04	FES = 438100	Diversity = 3.7858e+01
Generation 4390:	Best = 2.187860e+04	FES = 439100	Diversity = 3.7853e+01
Generation 4400:	Best = 2.187860e+04	FES = 440100	Diversity = 3.7853e+01
Generation 4410:	Best = 2.187860e+04	FES = 441100	Diversity = 3.7852e+01
Generation 4420:	Best = 2.187860e+04	FES = 442100	Diversity = 3.7852e+01
Generation 4430:	Best = 2.187860e+04	FES = 443100	Diversity = 3.7852e+01
Generation 4440:	Best = 2.187860e+04	FES = 444100	Diversity = 3.7845e+01
Generation 4450:	Best = 2.187860e+04	FES = 445100	Diversity = 3.7843e+01
Generation 4460:	Best = 2.187860e+04	FES = 446100	Diversity = 3.7849e+01
Generation 4470:	Best = 2.187860e+04	FES = 447100	Diversity = 3.7847e+01
Generation 4480:	Best = 2.187860e+04	FES = 448100	Diversity = 3.7844e+01
Generation 4490:	Best = 2.187860e+04	FES = 449100	Diversity = 3.7844e+01
Generation 4500:	Best = 2.187860e+04	FES = 450100	Diversity = 3.7838e+01
Generation 4510:	Best = 2.187860e+04	FES = 451100	Diversity = 3.7839e+01
Generation 4520:	Best = 2.176926e+04	FES = 452100	Diversity = 3.7828e+01
Generation 4530:	Best = 2.174415e+04	FES = 453100	Diversity = 3.7825e+01
Generation 4540:	Best = 2.174415e+04	FES = 454100	Diversity = 3.7824e+01
Generation 4550:	Best = 2.174415e+04	FES = 455100	Diversity = 3.7823e+01
Generation 4560:	Best = 2.174415e+04	FES = 456100	Diversity = 3.7815e+01
Generation 4570:	Best = 2.174415e+04	FES = 457100	Diversity = 3.7814e+01
Generation 4580:	Best = 2.174415e+04	FES = 458100	Diversity = 3.7812e+01
Generation 4590:	Best = 2.174415e+04	FES = 459100	Diversity = 3.7807e+01
Generation 4600:	Best = 2.174415e+04	FES = 460100	Diversity = 3.7811e+01
Generation 4610:	Best = 2.174415e+04	FES = 461100	Diversity = 3.7813e+01
Generation 4620:	Best = 2.174415e+04	FES = 462100	Diversity = 3.7812e+01
Generation 4630:	Best = 2.162511e+04	FES = 463100	Diversity = 3.7813e+01
Generation 4640:	Best = 2.162511e+04	FES = 464100	Diversity = 3.7813e+01
Generation 4650:	Best = 2.162511e+04	FES = 465100	Diversity = 3.7812e+01
Generation 4660:	Best = 2.162511e+04	FES = 466100	Diversity = 3.7809e+01
Generation 4670:	Best = 2.162511e+04	FES = 467100	Diversity = 3.7809e+01
Generation 4680:	Best = 2.162511e+04	FES = 468100	Diversity = 3.7809e+01
Generation 4690:	Best = 2.162511e+04	FES = 469100	Diversity = 3.7809e+01
Generation 4700:	Best = 2.162511e+04	FES = 470100	Diversity = 3.7807e+01
Generation 4710:	Best = 2.162511e+04	FES = 471100	Diversity = 3.7803e+01
Generation 4720:	Best = 2.162511e+04	FES = 472100	Diversity = 3.7802e+01
Generation 4730:	Best = 2.162511e+04	FES = 473100	Diversity = 3.7798e+01
Generation 4740:	Best = 2.162511e+04	FES = 474100	Diversity = 3.7798e+01
Generation 4750:	Best = 2.162511e+04	FES = 475100	Diversity = 3.7797e+01
Generation 4760:	Best = 2.162511e+04	FES = 476100	Diversity = 3.7795e+01

[illegible]

[illegible]

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[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

```

Generation 9630: Best = 2.081448e+04 FES = 963100 Diversity = 3.7228e+01
Generation 9640: Best = 2.081448e+04 FES = 964100 Diversity = 3.7229e+01
Generation 9650: Best = 2.081448e+04 FES = 965100 Diversity = 3.7228e+01
Generation 9660: Best = 2.081448e+04 FES = 966100 Diversity = 3.7228e+01
Generation 9670: Best = 2.081448e+04 FES = 967100 Diversity = 3.7228e+01
Generation 9680: Best = 2.081448e+04 FES = 968100 Diversity = 3.7227e+01
Generation 9690: Best = 2.081448e+04 FES = 969100 Diversity = 3.7226e+01
Generation 9700: Best = 2.081448e+04 FES = 970100 Diversity = 3.7225e+01
Generation 9710: Best = 2.081448e+04 FES = 971100 Diversity = 3.7230e+01
Generation 9720: Best = 2.066255e+04 FES = 972100 Diversity = 3.7229e+01
Generation 9730: Best = 2.066255e+04 FES = 973100 Diversity = 3.7229e+01
Generation 9740: Best = 2.066255e+04 FES = 974100 Diversity = 3.7229e+01
Generation 9750: Best = 2.066255e+04 FES = 975100 Diversity = 3.7229e+01
Generation 9760: Best = 2.066255e+04 FES = 976100 Diversity = 3.7229e+01
Generation 9770: Best = 2.066255e+04 FES = 977100 Diversity = 3.7229e+01
Generation 9780: Best = 2.066255e+04 FES = 978100 Diversity = 3.7178e+01
Generation 9790: Best = 2.066255e+04 FES = 979100 Diversity = 3.7178e+01
Generation 9800: Best = 2.066255e+04 FES = 980100 Diversity = 3.7178e+01
Generation 9810: Best = 2.066255e+04 FES = 981100 Diversity = 3.7178e+01
Generation 9820: Best = 2.066255e+04 FES = 982100 Diversity = 3.7177e+01
Generation 9830: Best = 2.066255e+04 FES = 983100 Diversity = 3.7178e+01
Generation 9840: Best = 2.066255e+04 FES = 984100 Diversity = 3.7178e+01
Generation 9850: Best = 2.066255e+04 FES = 985100 Diversity = 3.7178e+01
Generation 9860: Best = 2.066255e+04 FES = 986100 Diversity = 3.7176e+01
Generation 9870: Best = 2.061254e+04 FES = 987100 Diversity = 3.7173e+01
Generation 9880: Best = 2.061254e+04 FES = 988100 Diversity = 3.7173e+01
Generation 9890: Best = 2.061254e+04 FES = 989100 Diversity = 3.7179e+01
Generation 9900: Best = 2.061254e+04 FES = 990100 Diversity = 3.7178e+01
Generation 9910: Best = 2.061254e+04 FES = 991100 Diversity = 3.7123e+01
Generation 9920: Best = 2.061254e+04 FES = 992100 Diversity = 3.7117e+01
Generation 9930: Best = 2.061254e+04 FES = 993100 Diversity = 3.7112e+01
Generation 9940: Best = 2.061254e+04 FES = 994100 Diversity = 3.7111e+01
Generation 9950: Best = 2.061254e+04 FES = 995100 Diversity = 3.7109e+01
Generation 9960: Best = 2.061254e+04 FES = 996100 Diversity = 3.7108e+01
Generation 9970: Best = 2.061254e+04 FES = 997100 Diversity = 3.7108e+01
Generation 9980: Best = 2.024832e+04 FES = 998100 Diversity = 3.7105e+01
Generation 9990: Best = 2.024832e+04 FES = 999100 Diversity = 3.7103e+01

```

Optimization completed!
 Final Best Fitness = 2.024832e+04
 Total Function Evaluations = 1000000

=====

=

Results:

```

  Best Fitness           : 2.024832e+04
  Global Optimum         : 1.000000e+03
  Error (|best-optimum|) : 1.924832e+04
  Best Solution (5 dim)  : [40.47902514  0.46131094 39.92867798 56.72518952
9.96819533]
  D^(0) (init. diversity): 5.7220e+01

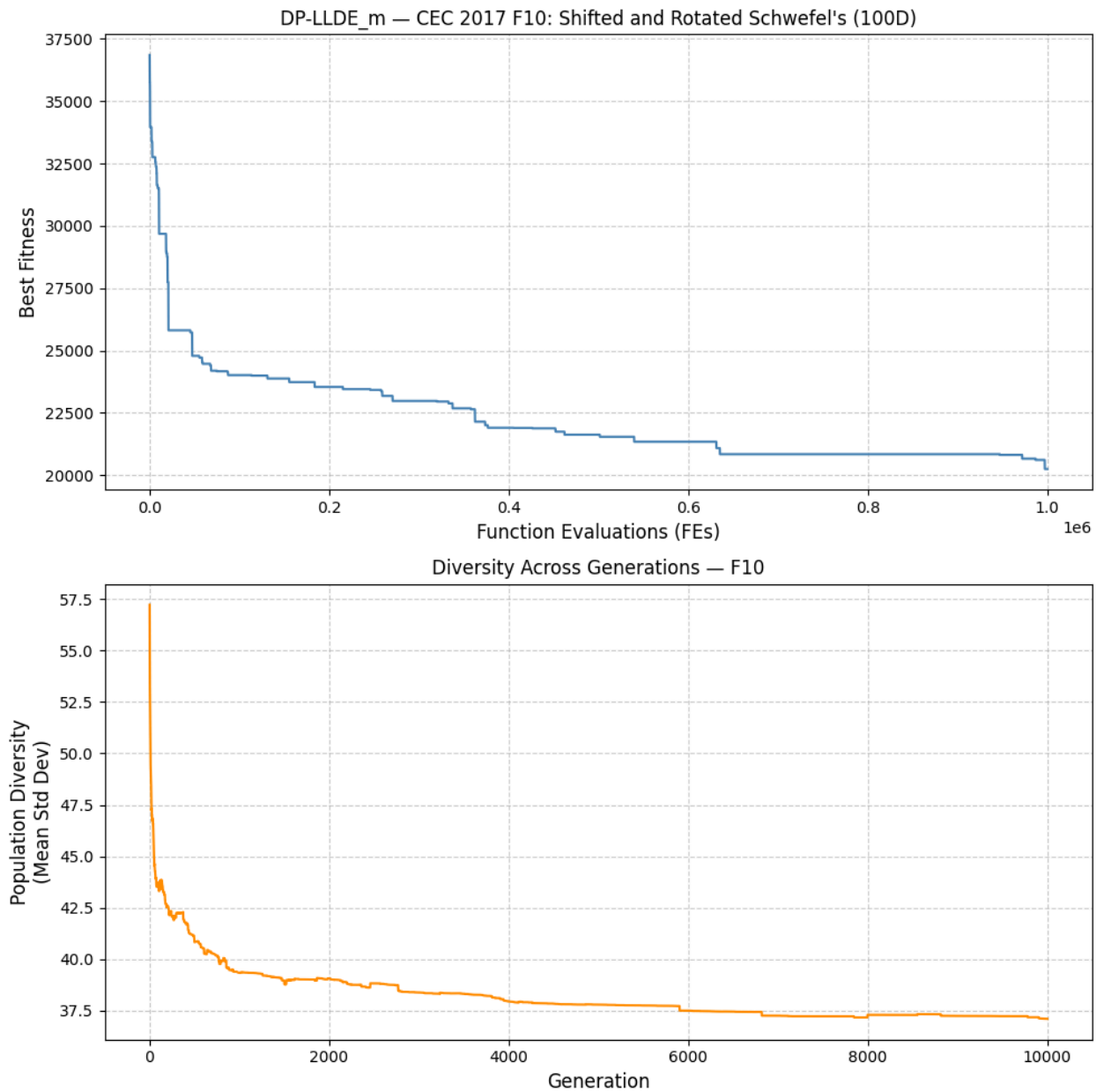
```

=====

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✓ Checkpoint saved to: /content/drive/My Drive/thesis_results_2/all_results_che

ckpoint.pkl



Graph saved as convergence_F10.png

✓ Plot saved to: /content/drive/My Drive/thesis_results_2/plots/convergence_F10.png

6.3 Hybrid Functions (F11-F20)

```
In [ ]: # =====  
# F11 – Hybrid Function 1 (Zakharov; Rosenbrock; Rastrigin)  
# =====  
run_and_plot('F11', make_wrapper(hybrid.f11), 1100.0, 'Hybrid: Zakharov; Roser
```

=====

=

DP-LLDE_m on CEC 2017 F11 – Single Run

=====

=

Function : Hybrid: Zakharov; Rosenbrock; Rastrigin (100D)

Bounds : [-100, 100]

Global optimum : 1100.0

Initial $D^{(0)} = 5.7220e+01$ | Diversity threshold = 5.7220e+00

Generation 0: Best Fitness = 7.262506e+05 | Diversity = 5.7220e+01

Generation	10:	Best = 4.028405e+05	FES = 1100	Diversity = 4.5941e+01
Generation	20:	Best = 2.119206e+05	FES = 2100	Diversity = 3.8023e+01
Generation	30:	Best = 1.464496e+05	FES = 3100	Diversity = 2.5493e+01
Generation	40:	Best = 1.331412e+05	FES = 4100	Diversity = 1.7074e+01
Generation	50:	Best = 1.070240e+05	FES = 5100	Diversity = 1.2672e+01
Generation	60:	Best = 1.012395e+05	FES = 6100	Diversity = 1.0633e+01
Generation	70:	Best = 9.810307e+04	FES = 7100	Diversity = 8.4744e+00
Generation	80:	Best = 9.016403e+04	FES = 8100	Diversity = 6.9474e+00
Generation	90:	Best = 8.531244e+04	FES = 9100	Diversity = 6.8341e+00
Generation	100:	Best = 7.821367e+04	FES = 10100	Diversity = 6.0209e+00
Generation	110:	Best = 7.575130e+04	FES = 11100	Diversity = 6.2958e+00
Generation	120:	Best = 7.382139e+04	FES = 12100	Diversity = 5.9267e+00
Generation	130:	Best = 6.850038e+04	FES = 13100	Diversity = 5.9525e+00
Generation	140:	Best = 6.766867e+04	FES = 14100	Diversity = 6.1137e+00
Generation	150:	Best = 6.327064e+04	FES = 15100	Diversity = 6.1242e+00
Generation	160:	Best = 5.971221e+04	FES = 16100	Diversity = 6.2838e+00
Generation	170:	Best = 5.937406e+04	FES = 17100	Diversity = 5.9822e+00
Generation	180:	Best = 5.726229e+04	FES = 18100	Diversity = 6.2641e+00
Generation	190:	Best = 5.500900e+04	FES = 19100	Diversity = 6.6372e+00
Generation	200:	Best = 5.140707e+04	FES = 20100	Diversity = 6.5428e+00
Generation	210:	Best = 5.034690e+04	FES = 21100	Diversity = 6.7664e+00
Generation	220:	Best = 4.958344e+04	FES = 22100	Diversity = 6.2455e+00
Generation	230:	Best = 4.725413e+04	FES = 23100	Diversity = 5.5631e+00
Generation	240:	Best = 4.594571e+04	FES = 24100	Diversity = 5.2567e+00
Generation	250:	Best = 4.454173e+04	FES = 25100	Diversity = 5.3609e+00
Generation	260:	Best = 4.413856e+04	FES = 26100	Diversity = 5.0113e+00
Generation	270:	Best = 4.286871e+04	FES = 27100	Diversity = 5.4770e+00
Generation	280:	Best = 4.176293e+04	FES = 28100	Diversity = 5.7523e+00
Generation	290:	Best = 4.047630e+04	FES = 29100	Diversity = 5.7529e+00
Generation	300:	Best = 3.976612e+04	FES = 30100	Diversity = 5.9920e+00
Generation	310:	Best = 3.864996e+04	FES = 31100	Diversity = 5.8509e+00
Generation	320:	Best = 3.633257e+04	FES = 32100	Diversity = 5.9479e+00
Generation	330:	Best = 3.512174e+04	FES = 33100	Diversity = 5.9727e+00
Generation	340:	Best = 3.423326e+04	FES = 34100	Diversity = 5.7995e+00
Generation	350:	Best = 3.326007e+04	FES = 35100	Diversity = 6.0488e+00
Generation	360:	Best = 3.250361e+04	FES = 36100	Diversity = 6.0555e+00
Generation	370:	Best = 3.199896e+04	FES = 37100	Diversity = 5.8100e+00
Generation	380:	Best = 3.095909e+04	FES = 38100	Diversity = 5.3498e+00
Generation	390:	Best = 3.045634e+04	FES = 39100	Diversity = 4.8257e+00
Generation	400:	Best = 2.946068e+04	FES = 40100	Diversity = 5.0776e+00
Generation	410:	Best = 2.844971e+04	FES = 41100	Diversity = 5.1444e+00
Generation	420:	Best = 2.834193e+04	FES = 42100	Diversity = 4.9345e+00
Generation	430:	Best = 2.755808e+04	FES = 43100	Diversity = 4.8642e+00
Generation	440:	Best = 2.669778e+04	FES = 44100	Diversity = 4.9673e+00

Generation	450:	Best = 2.610542e+04	FES = 45100	Diversity = 4.6277e+00
Generation	460:	Best = 2.559719e+04	FES = 46100	Diversity = 4.5257e+00
Generation	470:	Best = 2.519692e+04	FES = 47100	Diversity = 4.7703e+00
Generation	480:	Best = 2.460419e+04	FES = 48100	Diversity = 5.1305e+00
Generation	490:	Best = 2.378821e+04	FES = 49100	Diversity = 5.4156e+00
Generation	500:	Best = 2.346319e+04	FES = 50100	Diversity = 5.3715e+00
Generation	510:	Best = 2.283848e+04	FES = 51100	Diversity = 5.2547e+00
Generation	520:	Best = 2.216030e+04	FES = 52100	Diversity = 5.6931e+00
Generation	530:	Best = 2.129084e+04	FES = 53100	Diversity = 6.0563e+00
Generation	540:	Best = 2.113541e+04	FES = 54100	Diversity = 6.0255e+00
Generation	550:	Best = 2.037201e+04	FES = 55100	Diversity = 5.3822e+00
Generation	560:	Best = 2.025731e+04	FES = 56100	Diversity = 4.9569e+00
Generation	570:	Best = 1.981216e+04	FES = 57100	Diversity = 5.0912e+00
Generation	580:	Best = 1.931597e+04	FES = 58100	Diversity = 5.2792e+00
Generation	590:	Best = 1.881548e+04	FES = 59100	Diversity = 5.0334e+00
Generation	600:	Best = 1.841388e+04	FES = 60100	Diversity = 5.2030e+00
Generation	610:	Best = 1.806541e+04	FES = 61100	Diversity = 5.0401e+00
Generation	620:	Best = 1.762655e+04	FES = 62100	Diversity = 4.8682e+00
Generation	630:	Best = 1.726075e+04	FES = 63100	Diversity = 5.3927e+00
Generation	640:	Best = 1.693831e+04	FES = 64100	Diversity = 5.2375e+00
Generation	650:	Best = 1.671480e+04	FES = 65100	Diversity = 5.0044e+00
Generation	660:	Best = 1.615248e+04	FES = 66100	Diversity = 5.0854e+00
Generation	670:	Best = 1.592202e+04	FES = 67100	Diversity = 5.6158e+00
Generation	680:	Best = 1.571066e+04	FES = 68100	Diversity = 5.3996e+00
Generation	690:	Best = 1.524971e+04	FES = 69100	Diversity = 5.6631e+00
Generation	700:	Best = 1.491196e+04	FES = 70100	Diversity = 5.9085e+00
Generation	710:	Best = 1.465956e+04	FES = 71100	Diversity = 6.0464e+00
Generation	720:	Best = 1.453944e+04	FES = 72100	Diversity = 5.9836e+00
Generation	730:	Best = 1.400461e+04	FES = 73100	Diversity = 5.7359e+00
Generation	740:	Best = 1.382400e+04	FES = 74100	Diversity = 5.3966e+00
Generation	750:	Best = 1.353026e+04	FES = 75100	Diversity = 5.9381e+00
Generation	760:	Best = 1.323726e+04	FES = 76100	Diversity = 6.2100e+00
Generation	770:	Best = 1.276717e+04	FES = 77100	Diversity = 6.4486e+00
Generation	780:	Best = 1.241289e+04	FES = 78100	Diversity = 5.6145e+00
Generation	790:	Best = 1.222862e+04	FES = 79100	Diversity = 5.5669e+00
Generation	800:	Best = 1.193638e+04	FES = 80100	Diversity = 5.3198e+00
Generation	810:	Best = 1.185111e+04	FES = 81100	Diversity = 5.1024e+00
Generation	820:	Best = 1.170447e+04	FES = 82100	Diversity = 4.5124e+00
Generation	830:	Best = 1.143135e+04	FES = 83100	Diversity = 4.3375e+00
Generation	840:	Best = 1.134435e+04	FES = 84100	Diversity = 4.5329e+00
Generation	850:	Best = 1.122366e+04	FES = 85100	Diversity = 4.7144e+00
Generation	860:	Best = 1.092124e+04	FES = 86100	Diversity = 5.1225e+00
Generation	870:	Best = 1.075076e+04	FES = 87100	Diversity = 4.8007e+00
Generation	880:	Best = 1.067253e+04	FES = 88100	Diversity = 5.1772e+00
Generation	890:	Best = 1.041751e+04	FES = 89100	Diversity = 5.4735e+00
Generation	900:	Best = 1.026603e+04	FES = 90100	Diversity = 5.1053e+00
Generation	910:	Best = 1.004749e+04	FES = 91100	Diversity = 5.0906e+00
Generation	920:	Best = 9.906550e+03	FES = 92100	Diversity = 5.0504e+00
Generation	930:	Best = 9.861903e+03	FES = 93100	Diversity = 4.9077e+00
Generation	940:	Best = 9.623281e+03	FES = 94100	Diversity = 4.6261e+00
Generation	950:	Best = 9.409832e+03	FES = 95100	Diversity = 4.9507e+00
Generation	960:	Best = 9.174026e+03	FES = 96100	Diversity = 5.1700e+00
Generation	970:	Best = 8.991971e+03	FES = 97100	Diversity = 5.4413e+00
Generation	980:	Best = 8.794234e+03	FES = 98100	Diversity = 4.9491e+00

Generation 990:	Best = 8.686151e+03	FES = 99100	Diversity = 5.1553e+00
Generation 1000:	Best = 8.497343e+03	FES = 100100	Diversity = 5.7369e+00
Generation 1010:	Best = 8.377069e+03	FES = 101100	Diversity = 6.1226e+00
Generation 1020:	Best = 8.193615e+03	FES = 102100	Diversity = 5.9762e+00
Generation 1030:	Best = 7.973639e+03	FES = 103100	Diversity = 5.7951e+00
Generation 1040:	Best = 7.819945e+03	FES = 104100	Diversity = 5.4958e+00
Generation 1050:	Best = 7.737210e+03	FES = 105100	Diversity = 5.4222e+00
Generation 1060:	Best = 7.602678e+03	FES = 106100	Diversity = 5.1459e+00
Generation 1070:	Best = 7.372602e+03	FES = 107100	Diversity = 4.9274e+00
Generation 1080:	Best = 7.279477e+03	FES = 108100	Diversity = 4.6694e+00
Generation 1090:	Best = 7.170042e+03	FES = 109100	Diversity = 4.4814e+00
Generation 1100:	Best = 7.037819e+03	FES = 110100	Diversity = 4.5455e+00
Generation 1110:	Best = 6.893590e+03	FES = 111100	Diversity = 4.7360e+00
Generation 1120:	Best = 6.786777e+03	FES = 112100	Diversity = 4.7654e+00
Generation 1130:	Best = 6.579641e+03	FES = 113100	Diversity = 4.5618e+00
Generation 1140:	Best = 6.486759e+03	FES = 114100	Diversity = 4.6534e+00
Generation 1150:	Best = 6.253986e+03	FES = 115100	Diversity = 5.1364e+00
Generation 1160:	Best = 6.062349e+03	FES = 116100	Diversity = 5.3582e+00
Generation 1170:	Best = 5.951810e+03	FES = 117100	Diversity = 5.1189e+00
Generation 1180:	Best = 5.896822e+03	FES = 118100	Diversity = 4.9196e+00
Generation 1190:	Best = 5.717706e+03	FES = 119100	Diversity = 5.1320e+00
Generation 1200:	Best = 5.549055e+03	FES = 120100	Diversity = 5.2282e+00
Generation 1210:	Best = 5.378797e+03	FES = 121100	Diversity = 5.2565e+00
Generation 1220:	Best = 5.316531e+03	FES = 122100	Diversity = 5.3866e+00
Generation 1230:	Best = 5.244625e+03	FES = 123100	Diversity = 5.0522e+00
Generation 1240:	Best = 5.130072e+03	FES = 124100	Diversity = 4.7362e+00
Generation 1250:	Best = 5.050849e+03	FES = 125100	Diversity = 4.8058e+00
Generation 1260:	Best = 4.968248e+03	FES = 126100	Diversity = 4.7406e+00
Generation 1270:	Best = 4.882500e+03	FES = 127100	Diversity = 4.7520e+00
Generation 1280:	Best = 4.828664e+03	FES = 128100	Diversity = 4.3080e+00
Generation 1290:	Best = 4.811732e+03	FES = 129100	Diversity = 4.2242e+00
Generation 1300:	Best = 4.686910e+03	FES = 130100	Diversity = 4.1862e+00
Generation 1310:	Best = 4.681639e+03	FES = 131100	Diversity = 4.2130e+00
Generation 1320:	Best = 4.603963e+03	FES = 132100	Diversity = 4.1260e+00
Generation 1330:	Best = 4.563788e+03	FES = 133100	Diversity = 3.9794e+00
Generation 1340:	Best = 4.542469e+03	FES = 134100	Diversity = 4.2367e+00
Generation 1350:	Best = 4.508785e+03	FES = 135100	Diversity = 4.2084e+00
Generation 1360:	Best = 4.478647e+03	FES = 136100	Diversity = 4.0621e+00
Generation 1370:	Best = 4.399139e+03	FES = 137100	Diversity = 3.9150e+00
Generation 1380:	Best = 4.379412e+03	FES = 138100	Diversity = 3.7002e+00
Generation 1390:	Best = 4.305887e+03	FES = 139100	Diversity = 3.5235e+00
Generation 1400:	Best = 4.303118e+03	FES = 140100	Diversity = 3.3709e+00
Generation 1410:	Best = 4.232456e+03	FES = 141100	Diversity = 3.3258e+00
Generation 1420:	Best = 4.166775e+03	FES = 142100	Diversity = 3.4698e+00
Generation 1430:	Best = 4.156468e+03	FES = 143100	Diversity = 3.6556e+00
Generation 1440:	Best = 4.105521e+03	FES = 144100	Diversity = 3.6173e+00
Generation 1450:	Best = 4.035341e+03	FES = 145100	Diversity = 3.5958e+00
Generation 1460:	Best = 3.950840e+03	FES = 146100	Diversity = 3.8501e+00
Generation 1470:	Best = 3.922567e+03	FES = 147100	Diversity = 4.0247e+00
Generation 1480:	Best = 3.800579e+03	FES = 148100	Diversity = 4.1353e+00
Generation 1490:	Best = 3.784256e+03	FES = 149100	Diversity = 4.0825e+00
Generation 1500:	Best = 3.589091e+03	FES = 150100	Diversity = 4.0733e+00
Generation 1510:	Best = 3.521806e+03	FES = 151100	Diversity = 4.1265e+00
Generation 1520:	Best = 3.514846e+03	FES = 152100	Diversity = 4.0155e+00

Generation 1530:	Best = 3.458309e+03	FES = 153100	Diversity = 4.1012e+00
Generation 1540:	Best = 3.429943e+03	FES = 154100	Diversity = 3.9393e+00
Generation 1550:	Best = 3.365657e+03	FES = 155100	Diversity = 3.8578e+00
Generation 1560:	Best = 3.338855e+03	FES = 156100	Diversity = 3.9484e+00
Generation 1570:	Best = 3.253782e+03	FES = 157100	Diversity = 3.9039e+00
Generation 1580:	Best = 3.164758e+03	FES = 158100	Diversity = 3.9412e+00
Generation 1590:	Best = 3.137817e+03	FES = 159100	Diversity = 3.8506e+00
Generation 1600:	Best = 3.122784e+03	FES = 160100	Diversity = 3.7711e+00
Generation 1610:	Best = 3.107831e+03	FES = 161100	Diversity = 3.6520e+00
Generation 1620:	Best = 3.071007e+03	FES = 162100	Diversity = 3.6491e+00
Generation 1630:	Best = 3.047030e+03	FES = 163100	Diversity = 3.8590e+00
Generation 1640:	Best = 2.981207e+03	FES = 164100	Diversity = 3.7389e+00
Generation 1650:	Best = 2.948931e+03	FES = 165100	Diversity = 3.7379e+00
Generation 1660:	Best = 2.906326e+03	FES = 166100	Diversity = 3.6121e+00
Generation 1670:	Best = 2.906326e+03	FES = 167100	Diversity = 3.5091e+00
Generation 1680:	Best = 2.886568e+03	FES = 168100	Diversity = 3.5169e+00
Generation 1690:	Best = 2.858910e+03	FES = 169100	Diversity = 3.3935e+00
Generation 1700:	Best = 2.846761e+03	FES = 170100	Diversity = 3.3783e+00
Generation 1710:	Best = 2.814577e+03	FES = 171100	Diversity = 3.4145e+00
Generation 1720:	Best = 2.726333e+03	FES = 172100	Diversity = 3.3787e+00
Generation 1730:	Best = 2.724055e+03	FES = 173100	Diversity = 3.4716e+00
Generation 1740:	Best = 2.687311e+03	FES = 174100	Diversity = 3.6575e+00
Generation 1750:	Best = 2.642211e+03	FES = 175100	Diversity = 3.6541e+00
Generation 1760:	Best = 2.617282e+03	FES = 176100	Diversity = 3.8753e+00
Generation 1770:	Best = 2.566702e+03	FES = 177100	Diversity = 3.7586e+00
Generation 1780:	Best = 2.563265e+03	FES = 178100	Diversity = 3.7038e+00
Generation 1790:	Best = 2.498330e+03	FES = 179100	Diversity = 3.6803e+00
Generation 1800:	Best = 2.465768e+03	FES = 180100	Diversity = 3.5380e+00
Generation 1810:	Best = 2.448759e+03	FES = 181100	Diversity = 3.4143e+00
Generation 1820:	Best = 2.448759e+03	FES = 182100	Diversity = 3.2919e+00
Generation 1830:	Best = 2.448759e+03	FES = 183100	Diversity = 3.2049e+00
Generation 1840:	Best = 2.448318e+03	FES = 184100	Diversity = 3.0777e+00
Generation 1850:	Best = 2.448318e+03	FES = 185100	Diversity = 3.0037e+00
Generation 1860:	Best = 2.444746e+03	FES = 186100	Diversity = 2.9196e+00
Generation 1870:	Best = 2.444746e+03	FES = 187100	Diversity = 2.8911e+00
Generation 1880:	Best = 2.439735e+03	FES = 188100	Diversity = 2.7796e+00
Generation 1890:	Best = 2.424082e+03	FES = 189100	Diversity = 2.7140e+00
Generation 1900:	Best = 2.424082e+03	FES = 190100	Diversity = 2.7161e+00
Generation 1910:	Best = 2.413106e+03	FES = 191100	Diversity = 2.6117e+00
Generation 1920:	Best = 2.401276e+03	FES = 192100	Diversity = 2.5361e+00
Generation 1930:	Best = 2.382368e+03	FES = 193100	Diversity = 2.3417e+00
Generation 1940:	Best = 2.366874e+03	FES = 194100	Diversity = 2.2142e+00
Generation 1950:	Best = 2.332918e+03	FES = 195100	Diversity = 2.0729e+00
Generation 1960:	Best = 2.328180e+03	FES = 196100	Diversity = 2.0196e+00
Generation 1970:	Best = 2.323689e+03	FES = 197100	Diversity = 1.9267e+00
Generation 1980:	Best = 2.323355e+03	FES = 198100	Diversity = 1.7062e+00
Generation 1990:	Best = 2.285054e+03	FES = 199100	Diversity = 1.4747e+00
Generation 2000:	Best = 2.285054e+03	FES = 200100	Diversity = 1.0084e+00
Generation 2010:	Best = 2.268367e+03	FES = 201100	Diversity = 5.7759e-01
Generation 2020:	Best = 2.249975e+03	FES = 202100	Diversity = 4.4040e-01
Generation 2030:	Best = 2.228448e+03	FES = 203100	Diversity = 4.1117e-01
Generation 2040:	Best = 2.214564e+03	FES = 204100	Diversity = 3.7120e-01
Generation 2050:	Best = 2.201581e+03	FES = 205100	Diversity = 3.3921e-01
Generation 2060:	Best = 2.188822e+03	FES = 206100	Diversity = 3.0125e-01

Generation 2070:	Best = 2.183132e+03	FES = 207100	Diversity = 2.5934e-01
Generation 2080:	Best = 2.175923e+03	FES = 208100	Diversity = 2.3226e-01
Generation 2090:	Best = 2.171424e+03	FES = 209100	Diversity = 1.9367e-01
Generation 2100:	Best = 2.166454e+03	FES = 210100	Diversity = 1.5758e-01
Generation 2110:	Best = 2.164170e+03	FES = 211100	Diversity = 1.3299e-01
Generation 2120:	Best = 2.161173e+03	FES = 212100	Diversity = 1.2051e-01
Generation 2130:	Best = 2.159020e+03	FES = 213100	Diversity = 1.0056e-01
Generation 2140:	Best = 2.155673e+03	FES = 214100	Diversity = 9.4202e-02
Generation 2150:	Best = 2.152147e+03	FES = 215100	Diversity = 8.8559e-02
Generation 2160:	Best = 2.148144e+03	FES = 216100	Diversity = 9.6484e-02
Generation 2170:	Best = 2.146196e+03	FES = 217100	Diversity = 9.4679e-02
Generation 2180:	Best = 2.144346e+03	FES = 218100	Diversity = 8.6091e-02
Generation 2190:	Best = 2.142418e+03	FES = 219100	Diversity = 7.6062e-02
Generation 2200:	Best = 2.140660e+03	FES = 220100	Diversity = 6.6347e-02
Generation 2210:	Best = 2.138540e+03	FES = 221100	Diversity = 6.3943e-02
Generation 2220:	Best = 2.136596e+03	FES = 222100	Diversity = 6.3830e-02
Generation 2230:	Best = 2.135106e+03	FES = 223100	Diversity = 5.7022e-02
Generation 2240:	Best = 2.133737e+03	FES = 224100	Diversity = 5.1747e-02
Generation 2250:	Best = 2.132174e+03	FES = 225100	Diversity = 4.5172e-02
Generation 2260:	Best = 2.131024e+03	FES = 226100	Diversity = 4.1896e-02
Generation 2270:	Best = 2.130074e+03	FES = 227100	Diversity = 3.3842e-02
Generation 2280:	Best = 2.129198e+03	FES = 228100	Diversity = 2.8143e-02
Generation 2290:	Best = 2.128725e+03	FES = 229100	Diversity = 2.6638e-02
Generation 2300:	Best = 2.127937e+03	FES = 230100	Diversity = 2.7503e-02
Generation 2310:	Best = 2.127098e+03	FES = 231100	Diversity = 2.7295e-02
Generation 2320:	Best = 2.126434e+03	FES = 232100	Diversity = 2.5415e-02
Generation 2330:	Best = 2.125699e+03	FES = 233100	Diversity = 2.7190e-02
Generation 2340:	Best = 2.125197e+03	FES = 234100	Diversity = 2.3292e-02
Generation 2350:	Best = 2.124528e+03	FES = 235100	Diversity = 2.4592e-02
Generation 2360:	Best = 2.123992e+03	FES = 236100	Diversity = 2.2739e-02
Generation 2370:	Best = 2.123676e+03	FES = 237100	Diversity = 2.1024e-02
Generation 2380:	Best = 2.123381e+03	FES = 238100	Diversity = 1.8647e-02
Generation 2390:	Best = 2.123079e+03	FES = 239100	Diversity = 1.4804e-02
Generation 2400:	Best = 2.122892e+03	FES = 240100	Diversity = 1.3372e-02
Generation 2410:	Best = 2.122643e+03	FES = 241100	Diversity = 1.2778e-02
Generation 2420:	Best = 2.122468e+03	FES = 242100	Diversity = 1.1787e-02
Generation 2430:	Best = 2.122231e+03	FES = 243100	Diversity = 1.2716e-02
Generation 2440:	Best = 2.122046e+03	FES = 244100	Diversity = 1.3917e-02
Generation 2450:	Best = 2.121880e+03	FES = 245100	Diversity = 1.2094e-02
Generation 2460:	Best = 2.121739e+03	FES = 246100	Diversity = 1.0249e-02
Generation 2470:	Best = 2.121635e+03	FES = 247100	Diversity = 7.4886e-03
Generation 2480:	Best = 2.121517e+03	FES = 248100	Diversity = 7.8610e-03
Generation 2490:	Best = 2.121370e+03	FES = 249100	Diversity = 9.6780e-03
Generation 2500:	Best = 2.121187e+03	FES = 250100	Diversity = 1.1353e-02
Generation 2510:	Best = 2.121002e+03	FES = 251100	Diversity = 1.0592e-02
Generation 2520:	Best = 2.120841e+03	FES = 252100	Diversity = 1.0228e-02
Generation 2530:	Best = 2.120770e+03	FES = 253100	Diversity = 8.3382e-03
Generation 2540:	Best = 2.120675e+03	FES = 254100	Diversity = 7.9797e-03
Generation 2550:	Best = 2.120534e+03	FES = 255100	Diversity = 8.2501e-03
Generation 2560:	Best = 2.120450e+03	FES = 256100	Diversity = 7.0843e-03
Generation 2570:	Best = 2.120373e+03	FES = 257100	Diversity = 6.4492e-03
Generation 2580:	Best = 2.120297e+03	FES = 258100	Diversity = 5.7931e-03
Generation 2590:	Best = 2.120196e+03	FES = 259100	Diversity = 5.7050e-03
Generation 2600:	Best = 2.120117e+03	FES = 260100	Diversity = 5.7663e-03

Generation 2610:	Best = 2.120017e+03	FES = 261100	Diversity = 6.5239e-03
Generation 2620:	Best = 2.119949e+03	FES = 262100	Diversity = 6.7110e-03
Generation 2630:	Best = 2.119870e+03	FES = 263100	Diversity = 6.3625e-03
Generation 2640:	Best = 2.119787e+03	FES = 264100	Diversity = 7.3391e-03
Generation 2650:	Best = 2.119707e+03	FES = 265100	Diversity = 6.2777e-03
Generation 2660:	Best = 2.119585e+03	FES = 266100	Diversity = 5.6759e-03
Generation 2670:	Best = 2.119495e+03	FES = 267100	Diversity = 5.1551e-03
Generation 2680:	Best = 2.119378e+03	FES = 268100	Diversity = 6.5028e-03
Generation 2690:	Best = 2.119300e+03	FES = 269100	Diversity = 6.2387e-03
Generation 2700:	Best = 2.119204e+03	FES = 270100	Diversity = 5.8268e-03
Generation 2710:	Best = 2.119148e+03	FES = 271100	Diversity = 5.4992e-03
Generation 2720:	Best = 2.119103e+03	FES = 272100	Diversity = 5.2528e-03
Generation 2730:	Best = 2.119072e+03	FES = 273100	Diversity = 3.7737e-03
Generation 2740:	Best = 2.119047e+03	FES = 274100	Diversity = 2.7885e-03
Generation 2750:	Best = 2.119029e+03	FES = 275100	Diversity = 1.9749e-03
Generation 2760:	Best = 2.119009e+03	FES = 276100	Diversity = 2.0066e-03
Generation 2770:	Best = 2.118987e+03	FES = 277100	Diversity = 2.0739e-03
Generation 2780:	Best = 2.118962e+03	FES = 278100	Diversity = 2.0740e-03
Generation 2790:	Best = 2.118939e+03	FES = 279100	Diversity = 1.8764e-03
Generation 2800:	Best = 2.118923e+03	FES = 280100	Diversity = 1.7050e-03
Generation 2810:	Best = 2.118910e+03	FES = 281100	Diversity = 1.2503e-03
Generation 2820:	Best = 2.118895e+03	FES = 282100	Diversity = 1.1462e-03
Generation 2830:	Best = 2.118882e+03	FES = 283100	Diversity = 1.2624e-03
Generation 2840:	Best = 2.118871e+03	FES = 284100	Diversity = 8.6736e-04
Generation 2850:	Best = 2.118864e+03	FES = 285100	Diversity = 6.0002e-04
Generation 2860:	Best = 2.118861e+03	FES = 286100	Diversity = 4.4405e-04
Generation 2870:	Best = 2.118858e+03	FES = 287100	Diversity = 4.1438e-04
Generation 2880:	Best = 2.118855e+03	FES = 288100	Diversity = 3.4199e-04
Generation 2890:	Best = 2.118851e+03	FES = 289100	Diversity = 5.2204e-04
Generation 2900:	Best = 2.118849e+03	FES = 290100	Diversity = 3.2987e-04
Generation 2910:	Best = 2.118848e+03	FES = 291100	Diversity = 2.5538e-04
Generation 2920:	Best = 2.118846e+03	FES = 292100	Diversity = 2.6863e-04
Generation 2930:	Best = 2.118845e+03	FES = 293100	Diversity = 2.5542e-04
Generation 2940:	Best = 2.118843e+03	FES = 294100	Diversity = 1.9817e-04
Generation 2950:	Best = 2.118843e+03	FES = 295100	Diversity = 1.4330e-04
Generation 2960:	Best = 2.118843e+03	FES = 296100	Diversity = 1.0253e-04
Generation 2970:	Best = 2.118842e+03	FES = 297100	Diversity = 1.2239e-04
Generation 2980:	Best = 2.118842e+03	FES = 298100	Diversity = 1.4551e-04
Generation 2990:	Best = 2.118842e+03	FES = 299100	Diversity = 1.0246e-04
Generation 3000:	Best = 2.118841e+03	FES = 300100	Diversity = 1.0904e-04
Generation 3010:	Best = 2.118841e+03	FES = 301100	Diversity = 8.6128e-05
Generation 3020:	Best = 2.118841e+03	FES = 302100	Diversity = 9.5003e-05
Generation 3030:	Best = 2.118841e+03	FES = 303100	Diversity = 8.4709e-05
Generation 3040:	Best = 2.118841e+03	FES = 304100	Diversity = 6.3359e-05
Generation 3050:	Best = 2.118840e+03	FES = 305100	Diversity = 7.0205e-05
Generation 3060:	Best = 2.118840e+03	FES = 306100	Diversity = 4.4061e-05
Generation 3070:	Best = 2.118840e+03	FES = 307100	Diversity = 4.4619e-05
Generation 3080:	Best = 2.118840e+03	FES = 308100	Diversity = 3.6678e-05
Generation 3090:	Best = 2.118840e+03	FES = 309100	Diversity = 3.8590e-05
Generation 3100:	Best = 2.118840e+03	FES = 310100	Diversity = 3.1161e-05
Generation 3110:	Best = 2.118840e+03	FES = 311100	Diversity = 2.9471e-05
Generation 3120:	Best = 2.118840e+03	FES = 312100	Diversity = 1.5695e-05
Generation 3130:	Best = 2.118840e+03	FES = 313100	Diversity = 1.3521e-05
Generation 3140:	Best = 2.118840e+03	FES = 314100	Diversity = 1.2578e-05

Generation 3150:	Best = 2.118840e+03	FES = 315100	Diversity = 8.4286e-06
Generation 3160:	Best = 2.118840e+03	FES = 316100	Diversity = 8.1758e-06
Generation 3170:	Best = 2.118840e+03	FES = 317100	Diversity = 5.6911e-06
Generation 3180:	Best = 2.118840e+03	FES = 318100	Diversity = 5.3712e-06
Generation 3190:	Best = 2.118840e+03	FES = 319100	Diversity = 2.8751e-06
Generation 3200:	Best = 2.118840e+03	FES = 320100	Diversity = 2.7105e-06
Generation 3210:	Best = 2.118840e+03	FES = 321100	Diversity = 1.7771e-06
Generation 3220:	Best = 2.118840e+03	FES = 322100	Diversity = 1.5396e-06
Generation 3230:	Best = 2.118840e+03	FES = 323100	Diversity = 1.0237e-06
Generation 3240:	Best = 2.118840e+03	FES = 324100	Diversity = 1.1547e-06
Generation 3250:	Best = 2.118840e+03	FES = 325100	Diversity = 1.3579e-06
Generation 3260:	Best = 2.118840e+03	FES = 326100	Diversity = 1.0139e-06
Generation 3270:	Best = 2.118840e+03	FES = 327100	Diversity = 9.5087e-07
Generation 3280:	Best = 2.118840e+03	FES = 328100	Diversity = 6.3743e-07
Generation 3290:	Best = 2.118840e+03	FES = 329100	Diversity = 4.6619e-07
Generation 3300:	Best = 2.118840e+03	FES = 330100	Diversity = 3.0193e-07
Generation 3310:	Best = 2.118840e+03	FES = 331100	Diversity = 3.1295e-07
Generation 3320:	Best = 2.118840e+03	FES = 332100	Diversity = 2.9110e-07
Generation 3330:	Best = 2.118840e+03	FES = 333100	Diversity = 2.6011e-07
Generation 3340:	Best = 2.118840e+03	FES = 334100	Diversity = 2.5808e-07
Generation 3350:	Best = 2.118840e+03	FES = 335100	Diversity = 2.2936e-07
Generation 3360:	Best = 2.118840e+03	FES = 336100	Diversity = 1.4761e-07
Generation 3370:	Best = 2.118840e+03	FES = 337100	Diversity = 1.3462e-07
Generation 3380:	Best = 2.118840e+03	FES = 338100	Diversity = 1.0329e-07
Generation 3390:	Best = 2.118840e+03	FES = 339100	Diversity = 5.9338e-08
Generation 3400:	Best = 2.118840e+03	FES = 340100	Diversity = 5.1061e-08
Generation 3410:	Best = 2.118840e+03	FES = 341100	Diversity = 4.2269e-08
Generation 3420:	Best = 2.118840e+03	FES = 342100	Diversity = 2.3566e-08
Generation 3430:	Best = 2.118840e+03	FES = 343100	Diversity = 2.3992e-08
Generation 3440:	Best = 2.118840e+03	FES = 344100	Diversity = 2.0498e-08
Generation 3450:	Best = 2.118840e+03	FES = 345100	Diversity = 1.2086e-08
Generation 3460:	Best = 2.118840e+03	FES = 346100	Diversity = 6.2536e-09
Generation 3470:	Best = 2.118840e+03	FES = 347100	Diversity = 7.2744e-09
Generation 3480:	Best = 2.118840e+03	FES = 348100	Diversity = 3.9839e-09
Generation 3490:	Best = 2.118840e+03	FES = 349100	Diversity = 3.0625e-09
Generation 3500:	Best = 2.118840e+03	FES = 350100	Diversity = 2.1034e-09
Generation 3510:	Best = 2.118840e+03	FES = 351100	Diversity = 3.4050e-10
Generation 3520:	Best = 2.118840e+03	FES = 352100	Diversity = 2.1828e-12
Generation 3530:	Best = 2.118840e+03	FES = 353100	Diversity = 5.3349e-14
Generation 3540:	Best = 2.118840e+03	FES = 354100	Diversity = 4.7564e-14
Generation 3550:	Best = 2.118840e+03	FES = 355100	Diversity = 4.7273e-14
Generation 3560:	Best = 2.118840e+03	FES = 356100	Diversity = 4.7259e-14
Generation 3570:	Best = 2.118840e+03	FES = 357100	Diversity = 4.7258e-14
Generation 3580:	Best = 2.118840e+03	FES = 358100	Diversity = 4.7258e-14
Generation 3590:	Best = 2.118840e+03	FES = 359100	Diversity = 4.7258e-14
Generation 3600:	Best = 2.118840e+03	FES = 360100	Diversity = 4.7258e-14
Generation 3610:	Best = 2.118840e+03	FES = 361100	Diversity = 4.7258e-14
Generation 3620:	Best = 2.118840e+03	FES = 362100	Diversity = 4.7258e-14
Generation 3630:	Best = 2.118840e+03	FES = 363100	Diversity = 4.7258e-14
Generation 3640:	Best = 2.118840e+03	FES = 364100	Diversity = 4.7258e-14
Generation 3650:	Best = 2.118840e+03	FES = 365100	Diversity = 4.7258e-14
Generation 3660:	Best = 2.118840e+03	FES = 366100	Diversity = 4.7258e-14
Generation 3670:	Best = 2.118840e+03	FES = 367100	Diversity = 4.7258e-14
Generation 3680:	Best = 2.118840e+03	FES = 368100	Diversity = 4.7258e-14

[illegible]

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[illegible]

[illegible]

[illegible]

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[illegible]

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Generation 9630: Best = 2.118840e+03 FES = 963100 Diversity = 4.7258e-14
Generation 9640: Best = 2.118840e+03 FES = 964100 Diversity = 4.7258e-14
Generation 9650: Best = 2.118840e+03 FES = 965100 Diversity = 4.7258e-14
Generation 9660: Best = 2.118840e+03 FES = 966100 Diversity = 4.7258e-14
Generation 9670: Best = 2.118840e+03 FES = 967100 Diversity = 4.7258e-14
Generation 9680: Best = 2.118840e+03 FES = 968100 Diversity = 4.7258e-14
Generation 9690: Best = 2.118840e+03 FES = 969100 Diversity = 4.7258e-14
Generation 9700: Best = 2.118840e+03 FES = 970100 Diversity = 4.7258e-14
Generation 9710: Best = 2.118840e+03 FES = 971100 Diversity = 4.7258e-14
Generation 9720: Best = 2.118840e+03 FES = 972100 Diversity = 4.7258e-14
Generation 9730: Best = 2.118840e+03 FES = 973100 Diversity = 4.7258e-14
Generation 9740: Best = 2.118840e+03 FES = 974100 Diversity = 4.7258e-14
Generation 9750: Best = 2.118840e+03 FES = 975100 Diversity = 4.7258e-14
Generation 9760: Best = 2.118840e+03 FES = 976100 Diversity = 4.7258e-14
Generation 9770: Best = 2.118840e+03 FES = 977100 Diversity = 4.7258e-14
Generation 9780: Best = 2.118840e+03 FES = 978100 Diversity = 4.7258e-14
Generation 9790: Best = 2.118840e+03 FES = 979100 Diversity = 4.7258e-14
Generation 9800: Best = 2.118840e+03 FES = 980100 Diversity = 4.7258e-14
Generation 9810: Best = 2.118840e+03 FES = 981100 Diversity = 4.7258e-14
Generation 9820: Best = 2.118840e+03 FES = 982100 Diversity = 4.7258e-14
Generation 9830: Best = 2.118840e+03 FES = 983100 Diversity = 4.7258e-14
Generation 9840: Best = 2.118840e+03 FES = 984100 Diversity = 4.7258e-14
Generation 9850: Best = 2.118840e+03 FES = 985100 Diversity = 4.7258e-14
Generation 9860: Best = 2.118840e+03 FES = 986100 Diversity = 4.7258e-14
Generation 9870: Best = 2.118840e+03 FES = 987100 Diversity = 4.7258e-14
Generation 9880: Best = 2.118840e+03 FES = 988100 Diversity = 4.7258e-14
Generation 9890: Best = 2.118840e+03 FES = 989100 Diversity = 4.7258e-14
Generation 9900: Best = 2.118840e+03 FES = 990100 Diversity = 4.7258e-14
Generation 9910: Best = 2.118840e+03 FES = 991100 Diversity = 4.7258e-14
Generation 9920: Best = 2.118840e+03 FES = 992100 Diversity = 4.7258e-14
Generation 9930: Best = 2.118840e+03 FES = 993100 Diversity = 4.7258e-14
Generation 9940: Best = 2.118840e+03 FES = 994100 Diversity = 4.7258e-14
Generation 9950: Best = 2.118840e+03 FES = 995100 Diversity = 4.7258e-14
Generation 9960: Best = 2.118840e+03 FES = 996100 Diversity = 4.7258e-14
Generation 9970: Best = 2.118840e+03 FES = 997100 Diversity = 4.7258e-14
Generation 9980: Best = 2.118840e+03 FES = 998100 Diversity = 4.7258e-14
Generation 9990: Best = 2.118840e+03 FES = 999100 Diversity = 4.7258e-14

```

```

Optimization completed!
Final Best Fitness = 2.118840e+03
Total Function Evaluations = 1000000

```

```

=====
=
Results:
  Best Fitness           : 2.118840e+03
  Global Optimum         : 1.100000e+03
  Error (|best-optimum|) : 1.018840e+03
  Best Solution (5 dim)  : [-26.71931314 -59.07721078 -61.36888896 12.33953073
16.75058842]
  D^(0) (init. diversity): 5.7220e+01
=====
=

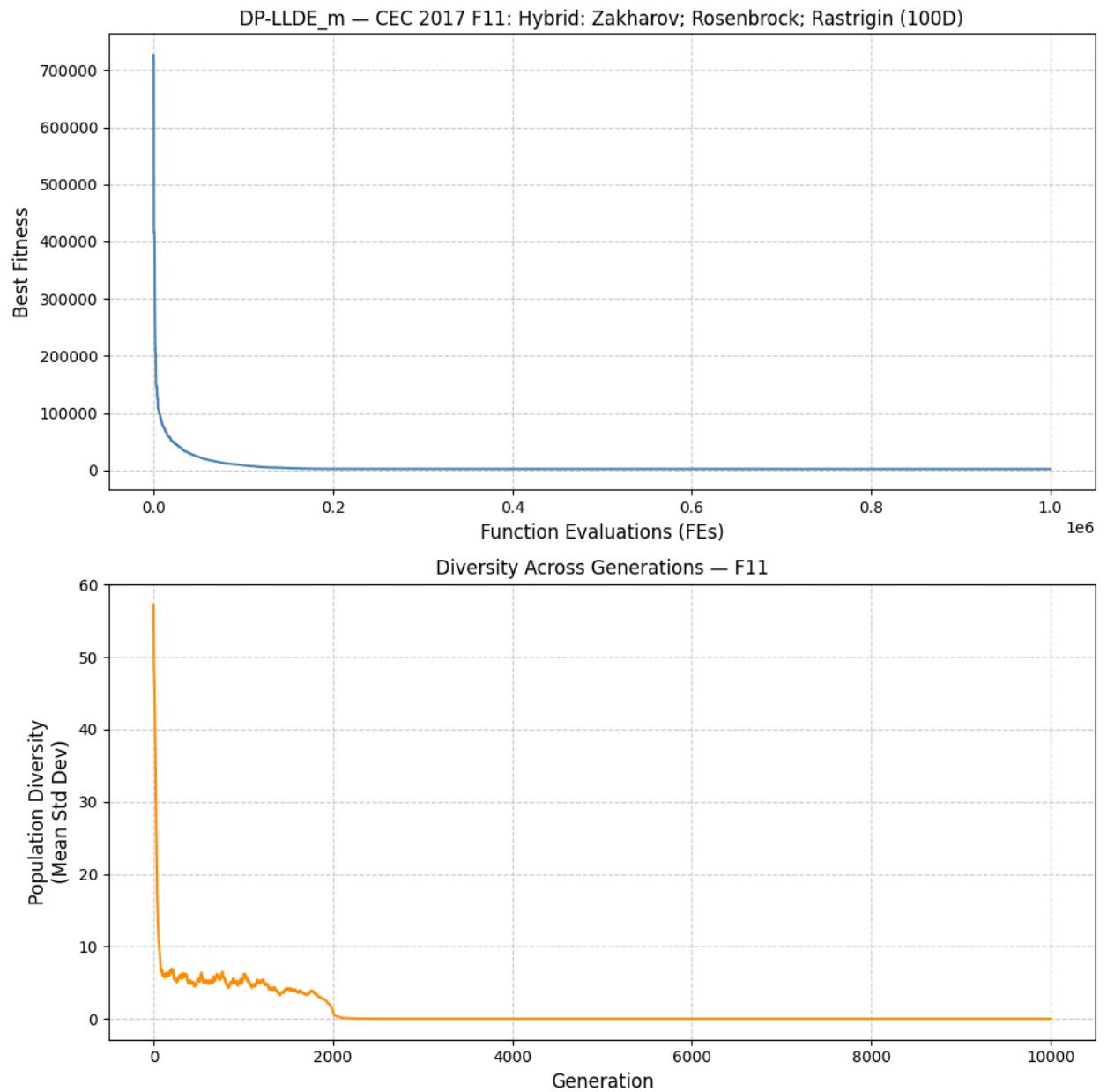
```

```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results_2/all_results_che

```

ckptpoint.pkl



Graph saved as convergence_F11.png

✓ Plot saved to: /content/drive/My Drive/thesis_results_2/plots/convergence_F11.png

```
In [ ]: # =====  
# F12 — Hybrid Function 2 (Elliptic; Schwefel; Bent Cigar)  
# =====  
run_and_plot('F12', make_wrapper(hybrid.f12), 1200.0, 'Hybrid: Elliptic; Schwe
```

```

=====
=
DP-LLDE_m on CEC 2017 F12 – Single Run
=====
=
Function      : Hybrid: Elliptic; Schwefel; Bent Cigar (100D)
Bounds       : [-100, 100]
Global optimum : 1200.0
Initial D^(0) = 5.7220e+01 | Diversity threshold = 5.7220e+00
Generation 0: Best Fitness = 3.383138e+12 | Diversity = 5.7220e+01
Generation 10: Best = 8.783886e+11 FES = 1100 Diversity = 2.2834e+01
Generation 20: Best = 5.474510e+11 FES = 2100 Diversity = 1.0071e+01
Generation 30: Best = 3.909716e+11 FES = 3100 Diversity = 6.3853e+00
Generation 40: Best = 3.136442e+11 FES = 4100 Diversity = 3.7789e+00
Generation 50: Best = 2.792893e+11 FES = 5100 Diversity = 2.5481e+00
Generation 60: Best = 2.641358e+11 FES = 6100 Diversity = 2.1510e+00
Generation 70: Best = 2.462022e+11 FES = 7100 Diversity = 1.8576e+00
Generation 80: Best = 2.329100e+11 FES = 8100 Diversity = 1.9674e+00
Generation 90: Best = 2.209979e+11 FES = 9100 Diversity = 1.8659e+00
Generation 100: Best = 2.078271e+11 FES = 10100 Diversity = 1.9273e+00
Generation 110: Best = 1.960287e+11 FES = 11100 Diversity = 2.3305e+00
Generation 120: Best = 1.880587e+11 FES = 12100 Diversity = 1.7843e+00
Generation 130: Best = 1.795005e+11 FES = 13100 Diversity = 1.6484e+00
Generation 140: Best = 1.719229e+11 FES = 14100 Diversity = 1.8590e+00
Generation 150: Best = 1.671138e+11 FES = 15100 Diversity = 1.7406e+00
Generation 160: Best = 1.618338e+11 FES = 16100 Diversity = 1.8488e+00
Generation 170: Best = 1.564979e+11 FES = 17100 Diversity = 1.9812e+00
Generation 180: Best = 1.522411e+11 FES = 18100 Diversity = 2.0293e+00
Generation 190: Best = 1.464741e+11 FES = 19100 Diversity = 2.3962e+00
Generation 200: Best = 1.414995e+11 FES = 20100 Diversity = 2.6581e+00
Generation 210: Best = 1.372198e+11 FES = 21100 Diversity = 2.8047e+00
Generation 220: Best = 1.336646e+11 FES = 22100 Diversity = 2.3130e+00
Generation 230: Best = 1.301806e+11 FES = 23100 Diversity = 2.3129e+00
Generation 240: Best = 1.272953e+11 FES = 24100 Diversity = 2.0115e+00
Generation 250: Best = 1.246685e+11 FES = 25100 Diversity = 2.0401e+00
Generation 260: Best = 1.224936e+11 FES = 26100 Diversity = 2.0298e+00
Generation 270: Best = 1.198494e+11 FES = 27100 Diversity = 1.8496e+00
Generation 280: Best = 1.176478e+11 FES = 28100 Diversity = 1.9210e+00
Generation 290: Best = 1.152609e+11 FES = 29100 Diversity = 2.0645e+00
Generation 300: Best = 1.137430e+11 FES = 30100 Diversity = 1.6801e+00
Generation 310: Best = 1.117374e+11 FES = 31100 Diversity = 1.2842e+00
Generation 320: Best = 1.100775e+11 FES = 32100 Diversity = 1.3078e+00
Generation 330: Best = 1.083070e+11 FES = 33100 Diversity = 1.3502e+00
Generation 340: Best = 1.062679e+11 FES = 34100 Diversity = 1.4924e+00
Generation 350: Best = 1.043888e+11 FES = 35100 Diversity = 1.4551e+00
Generation 360: Best = 1.028788e+11 FES = 36100 Diversity = 1.2828e+00
Generation 370: Best = 1.013846e+11 FES = 37100 Diversity = 1.2373e+00
Generation 380: Best = 1.003123e+11 FES = 38100 Diversity = 1.2210e+00
Generation 390: Best = 9.943283e+10 FES = 39100 Diversity = 1.1020e+00
Generation 400: Best = 9.879080e+10 FES = 40100 Diversity = 1.0168e+00
Generation 410: Best = 9.822000e+10 FES = 41100 Diversity = 9.4222e-01
Generation 420: Best = 9.756390e+10 FES = 42100 Diversity = 1.1665e+00
Generation 430: Best = 9.696952e+10 FES = 43100 Diversity = 1.0419e+00
Generation 440: Best = 9.635688e+10 FES = 44100 Diversity = 1.1877e+00

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Generation	450:	Best = 9.595042e+10	FES = 45100	Diversity = 9.3399e-01
Generation	460:	Best = 9.548067e+10	FES = 46100	Diversity = 8.5024e-01
Generation	470:	Best = 9.508925e+10	FES = 47100	Diversity = 1.1040e+00
Generation	480:	Best = 9.476653e+10	FES = 48100	Diversity = 8.1048e-01
Generation	490:	Best = 9.454930e+10	FES = 49100	Diversity = 6.4656e-01
Generation	500:	Best = 9.435652e+10	FES = 50100	Diversity = 5.6347e-01
Generation	510:	Best = 9.414614e+10	FES = 51100	Diversity = 6.3220e-01
Generation	520:	Best = 9.400550e+10	FES = 52100	Diversity = 5.3200e-01
Generation	530:	Best = 9.388397e+10	FES = 53100	Diversity = 7.1122e-01
Generation	540:	Best = 9.374379e+10	FES = 54100	Diversity = 9.4497e-01
Generation	550:	Best = 9.363948e+10	FES = 55100	Diversity = 5.2147e-01
Generation	560:	Best = 9.358576e+10	FES = 56100	Diversity = 5.6732e-01
Generation	570:	Best = 9.353404e+10	FES = 57100	Diversity = 5.4484e-01
Generation	580:	Best = 9.348388e+10	FES = 58100	Diversity = 4.2656e-01
Generation	590:	Best = 9.343892e+10	FES = 59100	Diversity = 5.7325e-01
Generation	600:	Best = 9.341141e+10	FES = 60100	Diversity = 7.0468e-01
Generation	610:	Best = 9.337826e+10	FES = 61100	Diversity = 4.6321e-01
Generation	620:	Best = 9.334628e+10	FES = 62100	Diversity = 3.2186e-01
Generation	630:	Best = 9.331962e+10	FES = 63100	Diversity = 4.2495e-01
Generation	640:	Best = 9.328976e+10	FES = 64100	Diversity = 4.1254e-01
Generation	650:	Best = 9.326742e+10	FES = 65100	Diversity = 3.7591e-01
Generation	660:	Best = 9.324126e+10	FES = 66100	Diversity = 5.5643e-01
Generation	670:	Best = 9.322185e+10	FES = 67100	Diversity = 4.2629e-01
Generation	680:	Best = 9.321085e+10	FES = 68100	Diversity = 5.5443e-01
Generation	690:	Best = 9.320130e+10	FES = 69100	Diversity = 4.2812e-01
Generation	700:	Best = 9.319172e+10	FES = 70100	Diversity = 2.6348e-01
Generation	710:	Best = 9.318379e+10	FES = 71100	Diversity = 9.4440e-02
Generation	720:	Best = 9.317969e+10	FES = 72100	Diversity = 1.1398e-01
Generation	730:	Best = 9.317316e+10	FES = 73100	Diversity = 1.4727e-01
Generation	740:	Best = 9.316481e+10	FES = 74100	Diversity = 2.1358e-01
Generation	750:	Best = 9.315976e+10	FES = 75100	Diversity = 2.3363e-01
Generation	760:	Best = 9.315364e+10	FES = 76100	Diversity = 3.1815e-01
Generation	770:	Best = 9.314762e+10	FES = 77100	Diversity = 1.7738e-01
Generation	780:	Best = 9.314300e+10	FES = 78100	Diversity = 1.8579e-01
Generation	790:	Best = 9.313662e+10	FES = 79100	Diversity = 1.5043e-01
Generation	800:	Best = 9.313380e+10	FES = 80100	Diversity = 1.2795e-01
Generation	810:	Best = 9.313058e+10	FES = 81100	Diversity = 1.6923e-01
Generation	820:	Best = 9.312724e+10	FES = 82100	Diversity = 2.1068e-01
Generation	830:	Best = 9.312486e+10	FES = 83100	Diversity = 1.7584e-01
Generation	840:	Best = 9.312230e+10	FES = 84100	Diversity = 8.6274e-02
Generation	850:	Best = 9.311940e+10	FES = 85100	Diversity = 1.0817e-01
Generation	860:	Best = 9.311667e+10	FES = 86100	Diversity = 2.0022e-01
Generation	870:	Best = 9.311444e+10	FES = 87100	Diversity = 1.7516e-01
Generation	880:	Best = 9.311308e+10	FES = 88100	Diversity = 7.0792e-02
Generation	890:	Best = 9.311147e+10	FES = 89100	Diversity = 8.2190e-02
Generation	900:	Best = 9.310991e+10	FES = 90100	Diversity = 5.3988e-02
Generation	910:	Best = 9.310844e+10	FES = 91100	Diversity = 4.7368e-02
Generation	920:	Best = 9.310725e+10	FES = 92100	Diversity = 1.2281e-01
Generation	930:	Best = 9.310612e+10	FES = 93100	Diversity = 1.7962e-01
Generation	940:	Best = 9.310551e+10	FES = 94100	Diversity = 5.4959e-02
Generation	950:	Best = 9.310487e+10	FES = 95100	Diversity = 8.7457e-03
Generation	960:	Best = 9.310456e+10	FES = 96100	Diversity = 7.5713e-03
Generation	970:	Best = 9.310398e+10	FES = 97100	Diversity = 9.9149e-03
Generation	980:	Best = 9.310328e+10	FES = 98100	Diversity = 9.5740e-03

Generation 990:	Best = 9.310263e+10	FES = 99100	Diversity = 2.7974e-02
Generation 1000:	Best = 9.310099e+10	FES = 100100	Diversity = 1.4671e-01
Generation 1010:	Best = 9.309902e+10	FES = 101100	Diversity = 8.6503e-02
Generation 1020:	Best = 9.309794e+10	FES = 102100	Diversity = 2.7227e-02
Generation 1030:	Best = 9.309644e+10	FES = 103100	Diversity = 4.0711e-02
Generation 1040:	Best = 9.309367e+10	FES = 104100	Diversity = 7.3375e-02
Generation 1050:	Best = 9.308610e+10	FES = 105100	Diversity = 2.5327e-01
Generation 1060:	Best = 9.307677e+10	FES = 106100	Diversity = 1.7383e-01
Generation 1070:	Best = 9.307131e+10	FES = 107100	Diversity = 1.7287e-01
Generation 1080:	Best = 9.306097e+10	FES = 108100	Diversity = 2.1829e-01
Generation 1090:	Best = 9.304789e+10	FES = 109100	Diversity = 3.4286e-01
Generation 1100:	Best = 9.304003e+10	FES = 110100	Diversity = 2.5627e-01
Generation 1110:	Best = 9.303510e+10	FES = 111100	Diversity = 2.3700e-01
Generation 1120:	Best = 9.303370e+10	FES = 112100	Diversity = 1.2259e-01
Generation 1130:	Best = 9.303254e+10	FES = 113100	Diversity = 3.0903e-02
Generation 1140:	Best = 9.302982e+10	FES = 114100	Diversity = 1.5353e-01
Generation 1150:	Best = 9.302779e+10	FES = 115100	Diversity = 1.9957e-01
Generation 1160:	Best = 9.302572e+10	FES = 116100	Diversity = 4.8905e-01
Generation 1170:	Best = 9.302450e+10	FES = 117100	Diversity = 1.1256e-01
Generation 1180:	Best = 9.302359e+10	FES = 118100	Diversity = 2.5820e-01
Generation 1190:	Best = 9.302250e+10	FES = 119100	Diversity = 2.9615e-02
Generation 1200:	Best = 9.302163e+10	FES = 120100	Diversity = 6.3827e-02
Generation 1210:	Best = 9.302057e+10	FES = 121100	Diversity = 2.3985e-01
Generation 1220:	Best = 9.301922e+10	FES = 122100	Diversity = 3.7254e-01
Generation 1230:	Best = 9.301859e+10	FES = 123100	Diversity = 3.7602e-03
Generation 1240:	Best = 9.301735e+10	FES = 124100	Diversity = 1.7594e-02
Generation 1250:	Best = 9.300165e+10	FES = 125100	Diversity = 7.6055e-02
Generation 1260:	Best = 9.297971e+10	FES = 126100	Diversity = 1.4490e-01
Generation 1270:	Best = 9.296339e+10	FES = 127100	Diversity = 2.0792e-01
Generation 1280:	Best = 9.295297e+10	FES = 128100	Diversity = 1.7519e-01
Generation 1290:	Best = 9.294457e+10	FES = 129100	Diversity = 1.7046e-01
Generation 1300:	Best = 9.293874e+10	FES = 130100	Diversity = 3.2127e-01
Generation 1310:	Best = 9.293230e+10	FES = 131100	Diversity = 2.2959e-01
Generation 1320:	Best = 9.292929e+10	FES = 132100	Diversity = 8.9769e-02
Generation 1330:	Best = 9.292789e+10	FES = 133100	Diversity = 6.3000e-03
Generation 1340:	Best = 9.292671e+10	FES = 134100	Diversity = 2.5897e-02
Generation 1350:	Best = 9.292473e+10	FES = 135100	Diversity = 8.5821e-02
Generation 1360:	Best = 9.292318e+10	FES = 136100	Diversity = 1.3719e-01
Generation 1370:	Best = 9.292150e+10	FES = 137100	Diversity = 4.9213e-03
Generation 1380:	Best = 9.292028e+10	FES = 138100	Diversity = 7.5294e-03
Generation 1390:	Best = 9.291987e+10	FES = 139100	Diversity = 4.4789e-03
Generation 1400:	Best = 9.291968e+10	FES = 140100	Diversity = 2.5808e-03
Generation 1410:	Best = 9.291961e+10	FES = 141100	Diversity = 1.5711e-03
Generation 1420:	Best = 9.291959e+10	FES = 142100	Diversity = 9.8277e-04
Generation 1430:	Best = 9.291958e+10	FES = 143100	Diversity = 7.2383e-04
Generation 1440:	Best = 9.291957e+10	FES = 144100	Diversity = 6.0631e-04
Generation 1450:	Best = 9.291957e+10	FES = 145100	Diversity = 1.7986e-04
Generation 1460:	Best = 9.291957e+10	FES = 146100	Diversity = 1.3547e-04
Generation 1470:	Best = 9.291957e+10	FES = 147100	Diversity = 7.5498e-05
Generation 1480:	Best = 9.291957e+10	FES = 148100	Diversity = 5.4048e-05
Generation 1490:	Best = 9.291957e+10	FES = 149100	Diversity = 7.8738e-05
Generation 1500:	Best = 9.291957e+10	FES = 150100	Diversity = 3.0453e-05
Generation 1510:	Best = 9.291957e+10	FES = 151100	Diversity = 1.8895e-05
Generation 1520:	Best = 9.291957e+10	FES = 152100	Diversity = 1.3959e-05

Generation 1530:	Best = 9.291957e+10	FES = 153100	Diversity = 5.7524e-06
Generation 1540:	Best = 9.291957e+10	FES = 154100	Diversity = 5.0320e-06
Generation 1550:	Best = 9.291957e+10	FES = 155100	Diversity = 1.8700e-06
Generation 1560:	Best = 9.291957e+10	FES = 156100	Diversity = 9.2092e-07
Generation 1570:	Best = 9.291957e+10	FES = 157100	Diversity = 7.7505e-07
Generation 1580:	Best = 9.291957e+10	FES = 158100	Diversity = 5.8889e-07
Generation 1590:	Best = 9.291957e+10	FES = 159100	Diversity = 4.2055e-07
Generation 1600:	Best = 9.291957e+10	FES = 160100	Diversity = 2.2881e-07
Generation 1610:	Best = 9.291957e+10	FES = 161100	Diversity = 1.3625e-07
Generation 1620:	Best = 9.291957e+10	FES = 162100	Diversity = 1.0033e-07
Generation 1630:	Best = 9.291957e+10	FES = 163100	Diversity = 7.5660e-08
Generation 1640:	Best = 9.291957e+10	FES = 164100	Diversity = 6.4662e-08
Generation 1650:	Best = 9.291957e+10	FES = 165100	Diversity = 5.9866e-08
Generation 1660:	Best = 9.291957e+10	FES = 166100	Diversity = 7.1789e-08
Generation 1670:	Best = 9.291957e+10	FES = 167100	Diversity = 5.6326e-08
Generation 1680:	Best = 9.291957e+10	FES = 168100	Diversity = 3.6906e-08
Generation 1690:	Best = 9.291957e+10	FES = 169100	Diversity = 2.9035e-08
Generation 1700:	Best = 9.291957e+10	FES = 170100	Diversity = 7.4350e-08
Generation 1710:	Best = 9.291957e+10	FES = 171100	Diversity = 4.4473e-08
Generation 1720:	Best = 9.291957e+10	FES = 172100	Diversity = 3.1903e-08
Generation 1730:	Best = 9.291957e+10	FES = 173100	Diversity = 1.7783e-08
Generation 1740:	Best = 9.291957e+10	FES = 174100	Diversity = 1.6013e-08
Generation 1750:	Best = 9.291957e+10	FES = 175100	Diversity = 1.0311e-08
Generation 1760:	Best = 9.291957e+10	FES = 176100	Diversity = 1.0610e-08
Generation 1770:	Best = 9.291957e+10	FES = 177100	Diversity = 2.6409e-09
Generation 1780:	Best = 9.291957e+10	FES = 178100	Diversity = 5.0257e-14
Generation 1790:	Best = 9.291957e+10	FES = 179100	Diversity = 5.0256e-14
Generation 1800:	Best = 9.291957e+10	FES = 180100	Diversity = 5.0256e-14
Generation 1810:	Best = 9.291957e+10	FES = 181100	Diversity = 5.0256e-14
Generation 1820:	Best = 9.291957e+10	FES = 182100	Diversity = 5.0256e-14
Generation 1830:	Best = 9.291957e+10	FES = 183100	Diversity = 5.0256e-14
Generation 1840:	Best = 9.291957e+10	FES = 184100	Diversity = 5.0256e-14
Generation 1850:	Best = 9.291957e+10	FES = 185100	Diversity = 5.0256e-14
Generation 1860:	Best = 9.291957e+10	FES = 186100	Diversity = 5.0256e-14
Generation 1870:	Best = 9.291957e+10	FES = 187100	Diversity = 5.0256e-14
Generation 1880:	Best = 9.291957e+10	FES = 188100	Diversity = 5.0256e-14
Generation 1890:	Best = 9.291957e+10	FES = 189100	Diversity = 5.0256e-14
Generation 1900:	Best = 9.291957e+10	FES = 190100	Diversity = 5.0256e-14
Generation 1910:	Best = 9.291957e+10	FES = 191100	Diversity = 5.0256e-14
Generation 1920:	Best = 9.291957e+10	FES = 192100	Diversity = 5.0256e-14
Generation 1930:	Best = 9.291957e+10	FES = 193100	Diversity = 5.0256e-14
Generation 1940:	Best = 9.291957e+10	FES = 194100	Diversity = 5.0256e-14
Generation 1950:	Best = 9.291957e+10	FES = 195100	Diversity = 5.0256e-14
Generation 1960:	Best = 9.291957e+10	FES = 196100	Diversity = 5.0256e-14
Generation 1970:	Best = 9.291957e+10	FES = 197100	Diversity = 5.0256e-14
Generation 1980:	Best = 9.291957e+10	FES = 198100	Diversity = 5.0256e-14
Generation 1990:	Best = 9.291957e+10	FES = 199100	Diversity = 5.0256e-14
Generation 2000:	Best = 9.291957e+10	FES = 200100	Diversity = 5.0256e-14
Generation 2010:	Best = 9.291957e+10	FES = 201100	Diversity = 5.0256e-14
Generation 2020:	Best = 9.291957e+10	FES = 202100	Diversity = 5.0256e-14
Generation 2030:	Best = 9.291957e+10	FES = 203100	Diversity = 5.0256e-14
Generation 2040:	Best = 9.291957e+10	FES = 204100	Diversity = 5.0256e-14
Generation 2050:	Best = 9.291957e+10	FES = 205100	Diversity = 5.0256e-14
Generation 2060:	Best = 9.291957e+10	FES = 206100	Diversity = 5.0256e-14

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Generation 9630: Best = 9.291957e+10 FES = 963100 Diversity = 5.0256e-14
Generation 9640: Best = 9.291957e+10 FES = 964100 Diversity = 5.0256e-14
Generation 9650: Best = 9.291957e+10 FES = 965100 Diversity = 5.0256e-14
Generation 9660: Best = 9.291957e+10 FES = 966100 Diversity = 5.0256e-14
Generation 9670: Best = 9.291957e+10 FES = 967100 Diversity = 5.0256e-14
Generation 9680: Best = 9.291957e+10 FES = 968100 Diversity = 5.0256e-14
Generation 9690: Best = 9.291957e+10 FES = 969100 Diversity = 5.0256e-14
Generation 9700: Best = 9.291957e+10 FES = 970100 Diversity = 5.0256e-14
Generation 9710: Best = 9.291957e+10 FES = 971100 Diversity = 5.0256e-14
Generation 9720: Best = 9.291957e+10 FES = 972100 Diversity = 5.0256e-14
Generation 9730: Best = 9.291957e+10 FES = 973100 Diversity = 5.0256e-14
Generation 9740: Best = 9.291957e+10 FES = 974100 Diversity = 5.0256e-14
Generation 9750: Best = 9.291957e+10 FES = 975100 Diversity = 5.0256e-14
Generation 9760: Best = 9.291957e+10 FES = 976100 Diversity = 5.0256e-14
Generation 9770: Best = 9.291957e+10 FES = 977100 Diversity = 5.0256e-14
Generation 9780: Best = 9.291957e+10 FES = 978100 Diversity = 5.0256e-14
Generation 9790: Best = 9.291957e+10 FES = 979100 Diversity = 5.0256e-14
Generation 9800: Best = 9.291957e+10 FES = 980100 Diversity = 5.0256e-14
Generation 9810: Best = 9.291957e+10 FES = 981100 Diversity = 5.0256e-14
Generation 9820: Best = 9.291957e+10 FES = 982100 Diversity = 5.0256e-14
Generation 9830: Best = 9.291957e+10 FES = 983100 Diversity = 5.0256e-14
Generation 9840: Best = 9.291957e+10 FES = 984100 Diversity = 5.0256e-14
Generation 9850: Best = 9.291957e+10 FES = 985100 Diversity = 5.0256e-14
Generation 9860: Best = 9.291957e+10 FES = 986100 Diversity = 5.0256e-14
Generation 9870: Best = 9.291957e+10 FES = 987100 Diversity = 5.0256e-14
Generation 9880: Best = 9.291957e+10 FES = 988100 Diversity = 5.0256e-14
Generation 9890: Best = 9.291957e+10 FES = 989100 Diversity = 5.0256e-14
Generation 9900: Best = 9.291957e+10 FES = 990100 Diversity = 5.0256e-14
Generation 9910: Best = 9.291957e+10 FES = 991100 Diversity = 5.0256e-14
Generation 9920: Best = 9.291957e+10 FES = 992100 Diversity = 5.0256e-14
Generation 9930: Best = 9.291957e+10 FES = 993100 Diversity = 5.0256e-14
Generation 9940: Best = 9.291957e+10 FES = 994100 Diversity = 5.0256e-14
Generation 9950: Best = 9.291957e+10 FES = 995100 Diversity = 5.0256e-14
Generation 9960: Best = 9.291957e+10 FES = 996100 Diversity = 5.0256e-14
Generation 9970: Best = 9.291957e+10 FES = 997100 Diversity = 5.0256e-14
Generation 9980: Best = 9.291957e+10 FES = 998100 Diversity = 5.0256e-14
Generation 9990: Best = 9.291957e+10 FES = 999100 Diversity = 5.0256e-14

```

Optimization completed!

Final Best Fitness = 9.291957e+10

Total Function Evaluations = 1000000

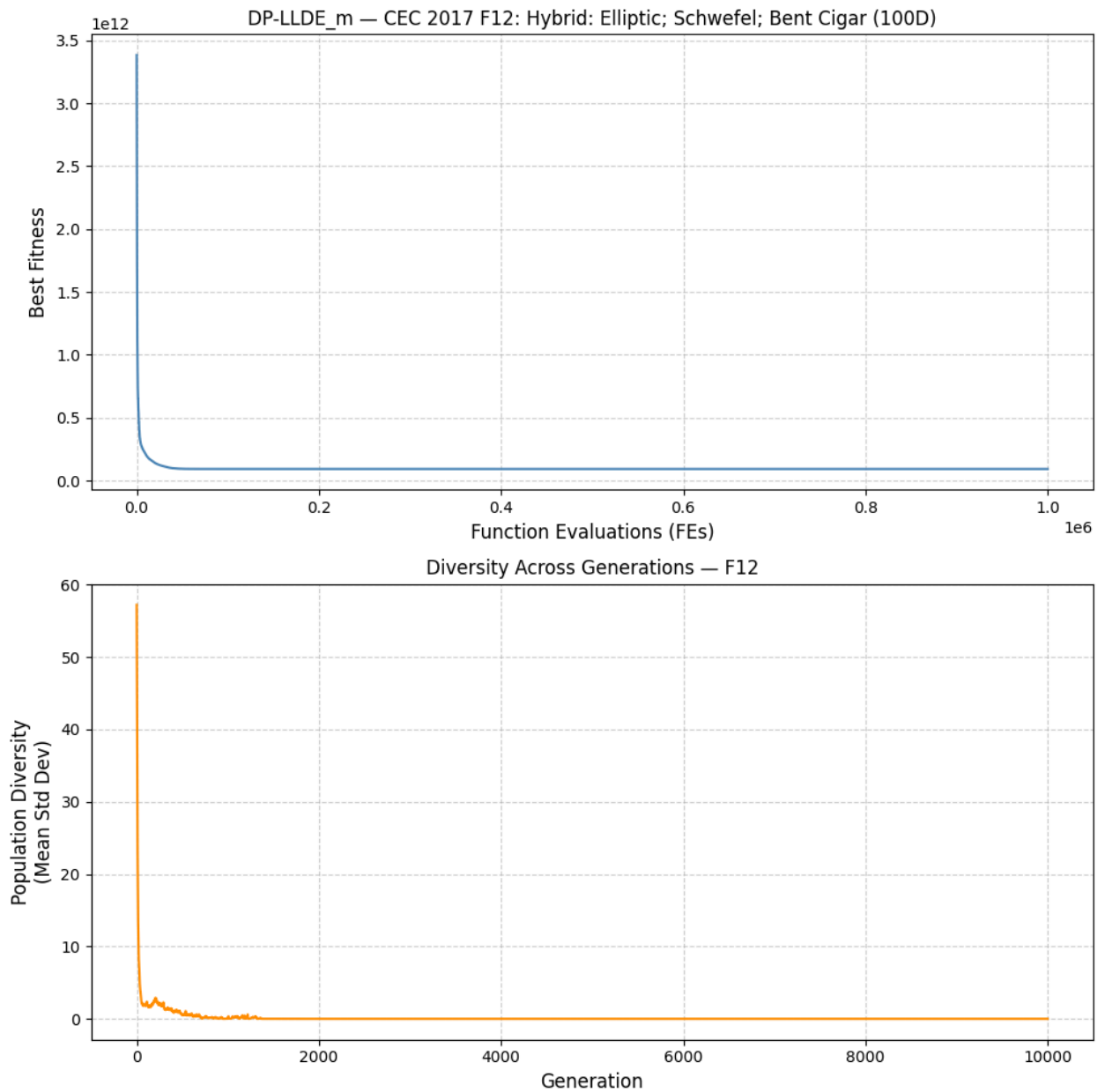
```

=====
=
Results:
  Best Fitness           : 9.291957e+10
  Global Optimum         : 1.200000e+03
  Error (|best-optimum|) : 9.291957e+10
  Best Solution (5 dim)  : [ 57.54576733  99.99591699   7.07035158  31.21126667
-96.40002748]
  D^(0) (init. diversity): 5.7220e+01
=====
=

```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results_2/all_results_che

ckpoint.pkl



Graph saved as convergence_F12.png

✓ Plot saved to: /content/drive/My Drive/thesis_results_2/plots/convergence_F12.png

```
In [ ]: # =====  
# F13 — Hybrid Function 3 (Bent Cigar; Rosenbrock; Bi-Rastrigin)  
# =====  
run_and_plot('F13', make_wrapper(hybrid.f13), 1300.0, 'Hybrid: Bent Cigar; Ros
```

=====

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DP-LLDE_m on CEC 2017 F13 – Single Run

=====

=

Function : Hybrid: Bent Cigar; Rosenbrock; Bi-Rastrigin (100D)

Bounds : [-100, 100]

Global optimum : 1300.0

Initial $D^{(0)} = 5.7220e+01$ | Diversity threshold = 5.7220e+00

Generation 0: Best Fitness = 9.446380e+11 | Diversity = 5.7220e+01

Generation	10:	Best = 2.200403e+11	FES = 1100	Diversity = 2.3108e+01
Generation	20:	Best = 1.124960e+11	FES = 2100	Diversity = 1.1330e+01
Generation	30:	Best = 7.797203e+10	FES = 3100	Diversity = 6.7453e+00
Generation	40:	Best = 6.143825e+10	FES = 4100	Diversity = 4.2492e+00
Generation	50:	Best = 5.438082e+10	FES = 5100	Diversity = 3.6249e+00
Generation	60:	Best = 4.970880e+10	FES = 6100	Diversity = 3.3440e+00
Generation	70:	Best = 4.608228e+10	FES = 7100	Diversity = 3.1843e+00
Generation	80:	Best = 4.227689e+10	FES = 8100	Diversity = 3.2115e+00
Generation	90:	Best = 3.959523e+10	FES = 9100	Diversity = 3.8264e+00
Generation	100:	Best = 3.780062e+10	FES = 10100	Diversity = 4.2578e+00
Generation	110:	Best = 3.545459e+10	FES = 11100	Diversity = 4.8617e+00
Generation	120:	Best = 3.345868e+10	FES = 12100	Diversity = 4.6978e+00
Generation	130:	Best = 3.189618e+10	FES = 13100	Diversity = 4.1580e+00
Generation	140:	Best = 3.046536e+10	FES = 14100	Diversity = 5.2628e+00
Generation	150:	Best = 2.896418e+10	FES = 15100	Diversity = 4.9755e+00
Generation	160:	Best = 2.803665e+10	FES = 16100	Diversity = 5.1478e+00
Generation	170:	Best = 2.731956e+10	FES = 17100	Diversity = 4.7386e+00
Generation	180:	Best = 2.643956e+10	FES = 18100	Diversity = 5.5112e+00
Generation	190:	Best = 2.555967e+10	FES = 19100	Diversity = 5.8966e+00
Generation	200:	Best = 2.492157e+10	FES = 20100	Diversity = 6.1613e+00
Generation	210:	Best = 2.447999e+10	FES = 21100	Diversity = 5.8502e+00
Generation	220:	Best = 2.407222e+10	FES = 22100	Diversity = 6.4911e+00
Generation	230:	Best = 2.373073e+10	FES = 23100	Diversity = 6.5341e+00
Generation	240:	Best = 2.342166e+10	FES = 24100	Diversity = 5.7212e+00
Generation	250:	Best = 2.304237e+10	FES = 25100	Diversity = 5.3129e+00
Generation	260:	Best = 2.281059e+10	FES = 26100	Diversity = 5.4928e+00
Generation	270:	Best = 2.263788e+10	FES = 27100	Diversity = 5.7069e+00
Generation	280:	Best = 2.242278e+10	FES = 28100	Diversity = 6.5600e+00
Generation	290:	Best = 2.230158e+10	FES = 29100	Diversity = 7.2397e+00
Generation	300:	Best = 2.219865e+10	FES = 30100	Diversity = 6.3866e+00
Generation	310:	Best = 2.211995e+10	FES = 31100	Diversity = 6.7481e+00
Generation	320:	Best = 2.206066e+10	FES = 32100	Diversity = 6.2497e+00
Generation	330:	Best = 2.200377e+10	FES = 33100	Diversity = 5.1889e+00
Generation	340:	Best = 2.195448e+10	FES = 34100	Diversity = 5.4222e+00
Generation	350:	Best = 2.191829e+10	FES = 35100	Diversity = 5.3850e+00
Generation	360:	Best = 2.187193e+10	FES = 36100	Diversity = 5.4835e+00
Generation	370:	Best = 2.184200e+10	FES = 37100	Diversity = 5.3385e+00
Generation	380:	Best = 2.180173e+10	FES = 38100	Diversity = 5.1565e+00
Generation	390:	Best = 2.176118e+10	FES = 39100	Diversity = 4.4604e+00
Generation	400:	Best = 2.172185e+10	FES = 40100	Diversity = 4.4686e+00
Generation	410:	Best = 2.169246e+10	FES = 41100	Diversity = 4.7588e+00
Generation	420:	Best = 2.167017e+10	FES = 42100	Diversity = 5.1051e+00
Generation	430:	Best = 2.165259e+10	FES = 43100	Diversity = 4.0686e+00
Generation	440:	Best = 2.163711e+10	FES = 44100	Diversity = 3.8122e+00

Generation	450:	Best = 2.162479e+10	FES = 45100	Diversity = 3.4446e+00
Generation	460:	Best = 2.161149e+10	FES = 46100	Diversity = 3.6866e+00
Generation	470:	Best = 2.160554e+10	FES = 47100	Diversity = 3.9028e+00
Generation	480:	Best = 2.160096e+10	FES = 48100	Diversity = 3.8707e+00
Generation	490:	Best = 2.159554e+10	FES = 49100	Diversity = 4.1315e+00
Generation	500:	Best = 2.159085e+10	FES = 50100	Diversity = 3.4529e+00
Generation	510:	Best = 2.158648e+10	FES = 51100	Diversity = 3.4870e+00
Generation	520:	Best = 2.158234e+10	FES = 52100	Diversity = 3.2380e+00
Generation	530:	Best = 2.157870e+10	FES = 53100	Diversity = 3.2090e+00
Generation	540:	Best = 2.157414e+10	FES = 54100	Diversity = 3.0932e+00
Generation	550:	Best = 2.157131e+10	FES = 55100	Diversity = 2.5329e+00
Generation	560:	Best = 2.156994e+10	FES = 56100	Diversity = 2.2272e+00
Generation	570:	Best = 2.156896e+10	FES = 57100	Diversity = 1.6012e+00
Generation	580:	Best = 2.156824e+10	FES = 58100	Diversity = 1.6309e+00
Generation	590:	Best = 2.156773e+10	FES = 59100	Diversity = 1.6348e+00
Generation	600:	Best = 2.156716e+10	FES = 60100	Diversity = 1.6242e+00
Generation	610:	Best = 2.156656e+10	FES = 61100	Diversity = 1.2664e+00
Generation	620:	Best = 2.156611e+10	FES = 62100	Diversity = 1.3968e+00
Generation	630:	Best = 2.156576e+10	FES = 63100	Diversity = 1.7152e+00
Generation	640:	Best = 2.156545e+10	FES = 64100	Diversity = 1.5917e+00
Generation	650:	Best = 2.156520e+10	FES = 65100	Diversity = 1.6372e+00
Generation	660:	Best = 2.156492e+10	FES = 66100	Diversity = 1.8836e+00
Generation	670:	Best = 2.156467e+10	FES = 67100	Diversity = 1.0867e+00
Generation	680:	Best = 2.156450e+10	FES = 68100	Diversity = 6.5366e-01
Generation	690:	Best = 2.156443e+10	FES = 69100	Diversity = 7.3491e-01
Generation	700:	Best = 2.156440e+10	FES = 70100	Diversity = 6.4065e-01
Generation	710:	Best = 2.156437e+10	FES = 71100	Diversity = 4.8455e-01
Generation	720:	Best = 2.156435e+10	FES = 72100	Diversity = 4.4219e-01
Generation	730:	Best = 2.156432e+10	FES = 73100	Diversity = 4.5011e-01
Generation	740:	Best = 2.156429e+10	FES = 74100	Diversity = 5.6645e-01
Generation	750:	Best = 2.156425e+10	FES = 75100	Diversity = 8.3364e-01
Generation	760:	Best = 2.156422e+10	FES = 76100	Diversity = 6.1408e-01
Generation	770:	Best = 2.156420e+10	FES = 77100	Diversity = 7.0372e-01
Generation	780:	Best = 2.156419e+10	FES = 78100	Diversity = 3.9082e-01
Generation	790:	Best = 2.156418e+10	FES = 79100	Diversity = 2.8069e-01
Generation	800:	Best = 2.156418e+10	FES = 80100	Diversity = 2.4688e-01
Generation	810:	Best = 2.156418e+10	FES = 81100	Diversity = 2.7659e-01
Generation	820:	Best = 2.156418e+10	FES = 82100	Diversity = 1.7406e-01
Generation	830:	Best = 2.156417e+10	FES = 83100	Diversity = 1.8211e-01
Generation	840:	Best = 2.156417e+10	FES = 84100	Diversity = 1.9864e-01
Generation	850:	Best = 2.156417e+10	FES = 85100	Diversity = 1.1520e-01
Generation	860:	Best = 2.156417e+10	FES = 86100	Diversity = 7.3952e-02
Generation	870:	Best = 2.156417e+10	FES = 87100	Diversity = 1.2319e-02
Generation	880:	Best = 2.156417e+10	FES = 88100	Diversity = 1.3058e-02
Generation	890:	Best = 2.156417e+10	FES = 89100	Diversity = 8.2424e-03
Generation	900:	Best = 2.156417e+10	FES = 90100	Diversity = 3.9074e-03
Generation	910:	Best = 2.156417e+10	FES = 91100	Diversity = 3.2643e-03
Generation	920:	Best = 2.156417e+10	FES = 92100	Diversity = 2.9122e-03
Generation	930:	Best = 2.156417e+10	FES = 93100	Diversity = 1.5413e-03
Generation	940:	Best = 2.156417e+10	FES = 94100	Diversity = 6.7510e-04
Generation	950:	Best = 2.156417e+10	FES = 95100	Diversity = 2.9404e-04
Generation	960:	Best = 2.156417e+10	FES = 96100	Diversity = 1.9214e-04
Generation	970:	Best = 2.156417e+10	FES = 97100	Diversity = 1.4145e-04
Generation	980:	Best = 2.156417e+10	FES = 98100	Diversity = 9.1460e-05

[illegible]

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[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]


```

Generation 9630: Best = 2.156417e+10 FES = 963100 Diversity = 5.6120e-14
Generation 9640: Best = 2.156417e+10 FES = 964100 Diversity = 5.6120e-14
Generation 9650: Best = 2.156417e+10 FES = 965100 Diversity = 5.6120e-14
Generation 9660: Best = 2.156417e+10 FES = 966100 Diversity = 5.6120e-14
Generation 9670: Best = 2.156417e+10 FES = 967100 Diversity = 5.6120e-14
Generation 9680: Best = 2.156417e+10 FES = 968100 Diversity = 5.6120e-14
Generation 9690: Best = 2.156417e+10 FES = 969100 Diversity = 5.6120e-14
Generation 9700: Best = 2.156417e+10 FES = 970100 Diversity = 5.6120e-14
Generation 9710: Best = 2.156417e+10 FES = 971100 Diversity = 5.6120e-14
Generation 9720: Best = 2.156417e+10 FES = 972100 Diversity = 5.6120e-14
Generation 9730: Best = 2.156417e+10 FES = 973100 Diversity = 5.6120e-14
Generation 9740: Best = 2.156417e+10 FES = 974100 Diversity = 5.6120e-14
Generation 9750: Best = 2.156417e+10 FES = 975100 Diversity = 5.6120e-14
Generation 9760: Best = 2.156417e+10 FES = 976100 Diversity = 5.6120e-14
Generation 9770: Best = 2.156417e+10 FES = 977100 Diversity = 5.6120e-14
Generation 9780: Best = 2.156417e+10 FES = 978100 Diversity = 5.6120e-14
Generation 9790: Best = 2.156417e+10 FES = 979100 Diversity = 5.6120e-14
Generation 9800: Best = 2.156417e+10 FES = 980100 Diversity = 5.6120e-14
Generation 9810: Best = 2.156417e+10 FES = 981100 Diversity = 5.6120e-14
Generation 9820: Best = 2.156417e+10 FES = 982100 Diversity = 5.6120e-14
Generation 9830: Best = 2.156417e+10 FES = 983100 Diversity = 5.6120e-14
Generation 9840: Best = 2.156417e+10 FES = 984100 Diversity = 5.6120e-14
Generation 9850: Best = 2.156417e+10 FES = 985100 Diversity = 5.6120e-14
Generation 9860: Best = 2.156417e+10 FES = 986100 Diversity = 5.6120e-14
Generation 9870: Best = 2.156417e+10 FES = 987100 Diversity = 5.6120e-14
Generation 9880: Best = 2.156417e+10 FES = 988100 Diversity = 5.6120e-14
Generation 9890: Best = 2.156417e+10 FES = 989100 Diversity = 5.6120e-14
Generation 9900: Best = 2.156417e+10 FES = 990100 Diversity = 5.6120e-14
Generation 9910: Best = 2.156417e+10 FES = 991100 Diversity = 5.6120e-14
Generation 9920: Best = 2.156417e+10 FES = 992100 Diversity = 5.6120e-14
Generation 9930: Best = 2.156417e+10 FES = 993100 Diversity = 5.6120e-14
Generation 9940: Best = 2.156417e+10 FES = 994100 Diversity = 5.6120e-14
Generation 9950: Best = 2.156417e+10 FES = 995100 Diversity = 5.6120e-14
Generation 9960: Best = 2.156417e+10 FES = 996100 Diversity = 5.6120e-14
Generation 9970: Best = 2.156417e+10 FES = 997100 Diversity = 5.6120e-14
Generation 9980: Best = 2.156417e+10 FES = 998100 Diversity = 5.6120e-14
Generation 9990: Best = 2.156417e+10 FES = 999100 Diversity = 5.6120e-14

```

Optimization completed!

Final Best Fitness = 2.156417e+10

Total Function Evaluations = 1000000

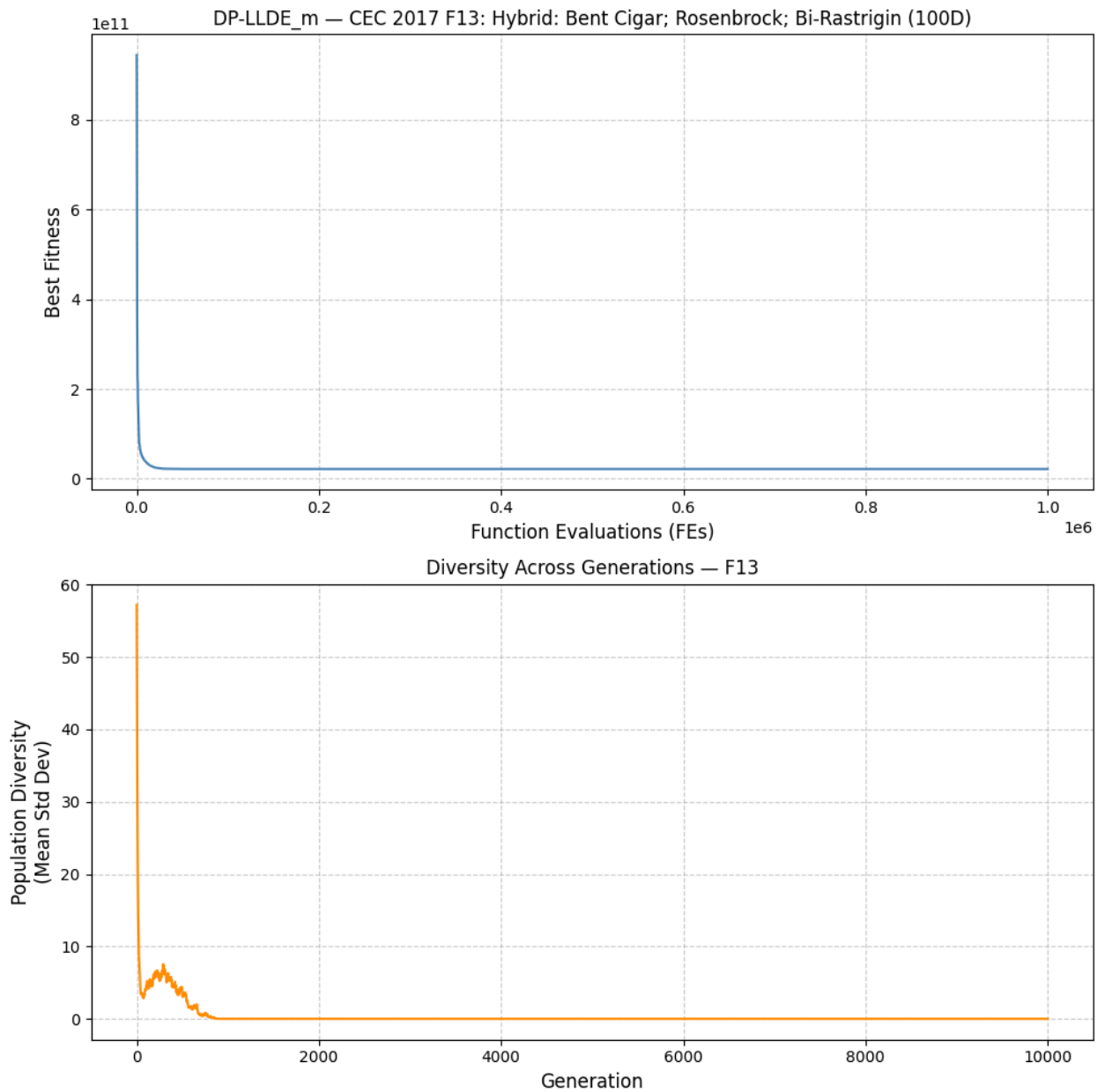
```

=====
=
Results:
  Best Fitness          : 2.156417e+10
  Global Optimum        : 1.300000e+03
  Error (|best-optimum|) : 2.156417e+10
  Best Solution (5 dim)  : [-83.50929478 -62.57831055 -36.92686325 -12.14156847
87.17847046]
  D^(0) (init. diversity): 5.7220e+01
=====
=

```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results_2/all_results_che

ckpoint.pkl



Graph saved as convergence_F13.png

✓ Plot saved to: /content/drive/My Drive/thesis_results_2/plots/convergence_F13.png

```
In [ ]: # =====  
# F14 — Hybrid Function 4 (Elliptic; Ackley; Schaffer; Rastrigin)  
# =====  
run_and_plot('F14', make_wrapper(hybrid.f14), 1400.0, 'Hybrid: Elliptic; Ackle
```

=====

=

DP-LLDE_m on CEC 2017 F14 – Single Run

=====

=

Function : Hybrid: Elliptic; Ackley; Schaffer; Rastrigin (100D)
Bounds : [-100, 100]
Global optimum : 1400.0
Initial $D^*(0) = 5.7220e+01$ | Diversity threshold = 5.7220e+00
Generation 0: Best Fitness = 7.040180e+08 | Diversity = 5.7220e+01

Generation	10:	Best = 4.519382e+07	FES = 1100	Diversity = 3.6592e+01
Generation	20:	Best = 1.300455e+07	FES = 2100	Diversity = 2.6623e+01
Generation	30:	Best = 2.975191e+06	FES = 3100	Diversity = 2.1730e+01
Generation	40:	Best = 2.224422e+06	FES = 4100	Diversity = 1.6865e+01
Generation	50:	Best = 1.552843e+06	FES = 5100	Diversity = 1.3523e+01
Generation	60:	Best = 1.018599e+06	FES = 6100	Diversity = 1.1108e+01
Generation	70:	Best = 7.486922e+05	FES = 7100	Diversity = 9.1998e+00
Generation	80:	Best = 5.731923e+05	FES = 8100	Diversity = 7.8583e+00
Generation	90:	Best = 5.219742e+05	FES = 9100	Diversity = 6.7584e+00
Generation	100:	Best = 5.219327e+05	FES = 10100	Diversity = 5.4485e+00
Generation	110:	Best = 4.867540e+05	FES = 11100	Diversity = 5.0582e+00
Generation	120:	Best = 4.648035e+05	FES = 12100	Diversity = 5.2947e+00
Generation	130:	Best = 4.532202e+05	FES = 13100	Diversity = 4.2312e+00
Generation	140:	Best = 4.435811e+05	FES = 14100	Diversity = 4.6706e+00
Generation	150:	Best = 4.294801e+05	FES = 15100	Diversity = 4.9223e+00
Generation	160:	Best = 4.139503e+05	FES = 16100	Diversity = 4.6900e+00
Generation	170:	Best = 4.042215e+05	FES = 17100	Diversity = 4.9120e+00
Generation	180:	Best = 4.036985e+05	FES = 18100	Diversity = 4.5336e+00
Generation	190:	Best = 3.964959e+05	FES = 19100	Diversity = 4.4482e+00
Generation	200:	Best = 3.924124e+05	FES = 20100	Diversity = 4.5849e+00
Generation	210:	Best = 3.841829e+05	FES = 21100	Diversity = 4.2666e+00
Generation	220:	Best = 3.806551e+05	FES = 22100	Diversity = 4.9253e+00
Generation	230:	Best = 3.725349e+05	FES = 23100	Diversity = 5.3557e+00
Generation	240:	Best = 3.725318e+05	FES = 24100	Diversity = 5.3539e+00
Generation	250:	Best = 3.693035e+05	FES = 25100	Diversity = 5.5263e+00
Generation	260:	Best = 3.661441e+05	FES = 26100	Diversity = 5.2503e+00
Generation	270:	Best = 3.634297e+05	FES = 27100	Diversity = 5.9057e+00
Generation	280:	Best = 3.577025e+05	FES = 28100	Diversity = 6.0698e+00
Generation	290:	Best = 3.527362e+05	FES = 29100	Diversity = 6.7612e+00
Generation	300:	Best = 3.480102e+05	FES = 30100	Diversity = 7.9601e+00
Generation	310:	Best = 3.457401e+05	FES = 31100	Diversity = 8.0764e+00
Generation	320:	Best = 3.418394e+05	FES = 32100	Diversity = 7.5002e+00
Generation	330:	Best = 3.407665e+05	FES = 33100	Diversity = 6.9405e+00
Generation	340:	Best = 3.387265e+05	FES = 34100	Diversity = 6.7966e+00
Generation	350:	Best = 3.364254e+05	FES = 35100	Diversity = 7.7588e+00
Generation	360:	Best = 3.347291e+05	FES = 36100	Diversity = 8.2817e+00
Generation	370:	Best = 3.287628e+05	FES = 37100	Diversity = 7.7860e+00
Generation	380:	Best = 3.283997e+05	FES = 38100	Diversity = 8.1704e+00
Generation	390:	Best = 3.242333e+05	FES = 39100	Diversity = 9.9019e+00
Generation	400:	Best = 3.174578e+05	FES = 40100	Diversity = 1.0020e+01
Generation	410:	Best = 3.120501e+05	FES = 41100	Diversity = 1.1249e+01
Generation	420:	Best = 3.062270e+05	FES = 42100	Diversity = 1.2842e+01
Generation	430:	Best = 3.045922e+05	FES = 43100	Diversity = 1.2854e+01
Generation	440:	Best = 3.023687e+05	FES = 44100	Diversity = 1.2592e+01

Generation	450:	Best = 3.006084e+05	FES = 45100	Diversity = 1.2156e+01
Generation	460:	Best = 2.981781e+05	FES = 46100	Diversity = 1.1873e+01
Generation	470:	Best = 2.962163e+05	FES = 47100	Diversity = 1.2458e+01
Generation	480:	Best = 2.955717e+05	FES = 48100	Diversity = 1.2534e+01
Generation	490:	Best = 2.911434e+05	FES = 49100	Diversity = 1.1926e+01
Generation	500:	Best = 2.892489e+05	FES = 50100	Diversity = 1.3523e+01
Generation	510:	Best = 2.874573e+05	FES = 51100	Diversity = 1.1223e+01
Generation	520:	Best = 2.857787e+05	FES = 52100	Diversity = 1.1722e+01
Generation	530:	Best = 2.806520e+05	FES = 53100	Diversity = 1.2566e+01
Generation	540:	Best = 2.789465e+05	FES = 54100	Diversity = 1.3793e+01
Generation	550:	Best = 2.745188e+05	FES = 55100	Diversity = 1.5460e+01
Generation	560:	Best = 2.719999e+05	FES = 56100	Diversity = 1.5179e+01
Generation	570:	Best = 2.685784e+05	FES = 57100	Diversity = 1.6136e+01
Generation	580:	Best = 2.646706e+05	FES = 58100	Diversity = 1.6630e+01
Generation	590:	Best = 2.592169e+05	FES = 59100	Diversity = 1.5190e+01
Generation	600:	Best = 2.586666e+05	FES = 60100	Diversity = 1.4233e+01
Generation	610:	Best = 2.579886e+05	FES = 61100	Diversity = 1.3281e+01
Generation	620:	Best = 2.570300e+05	FES = 62100	Diversity = 1.2909e+01
Generation	630:	Best = 2.559680e+05	FES = 63100	Diversity = 1.4020e+01
Generation	640:	Best = 2.535069e+05	FES = 64100	Diversity = 1.4743e+01
Generation	650:	Best = 2.503622e+05	FES = 65100	Diversity = 1.6348e+01
Generation	660:	Best = 2.479716e+05	FES = 66100	Diversity = 1.6966e+01
Generation	670:	Best = 2.472189e+05	FES = 67100	Diversity = 1.8239e+01
Generation	680:	Best = 2.467976e+05	FES = 68100	Diversity = 1.6496e+01
Generation	690:	Best = 2.448888e+05	FES = 69100	Diversity = 1.5669e+01
Generation	700:	Best = 2.444367e+05	FES = 70100	Diversity = 1.6315e+01
Generation	710:	Best = 2.430529e+05	FES = 71100	Diversity = 1.5427e+01
Generation	720:	Best = 2.425979e+05	FES = 72100	Diversity = 1.4568e+01
Generation	730:	Best = 2.405833e+05	FES = 73100	Diversity = 1.3712e+01
Generation	740:	Best = 2.387470e+05	FES = 74100	Diversity = 1.5341e+01
Generation	750:	Best = 2.373323e+05	FES = 75100	Diversity = 1.4503e+01
Generation	760:	Best = 2.362362e+05	FES = 76100	Diversity = 1.4546e+01
Generation	770:	Best = 2.349632e+05	FES = 77100	Diversity = 1.3903e+01
Generation	780:	Best = 2.343210e+05	FES = 78100	Diversity = 1.3469e+01
Generation	790:	Best = 2.330544e+05	FES = 79100	Diversity = 1.2731e+01
Generation	800:	Best = 2.313139e+05	FES = 80100	Diversity = 1.3676e+01
Generation	810:	Best = 2.297726e+05	FES = 81100	Diversity = 1.4442e+01
Generation	820:	Best = 2.278226e+05	FES = 82100	Diversity = 1.4355e+01
Generation	830:	Best = 2.274430e+05	FES = 83100	Diversity = 1.3424e+01
Generation	840:	Best = 2.263084e+05	FES = 84100	Diversity = 1.4522e+01
Generation	850:	Best = 2.258023e+05	FES = 85100	Diversity = 1.4350e+01
Generation	860:	Best = 2.251266e+05	FES = 86100	Diversity = 1.3111e+01
Generation	870:	Best = 2.243054e+05	FES = 87100	Diversity = 1.2433e+01
Generation	880:	Best = 2.234012e+05	FES = 88100	Diversity = 1.2543e+01
Generation	890:	Best = 2.222190e+05	FES = 89100	Diversity = 1.1173e+01
Generation	900:	Best = 2.213433e+05	FES = 90100	Diversity = 1.2262e+01
Generation	910:	Best = 2.192707e+05	FES = 91100	Diversity = 1.3882e+01
Generation	920:	Best = 2.181783e+05	FES = 92100	Diversity = 1.5032e+01
Generation	930:	Best = 2.173708e+05	FES = 93100	Diversity = 1.4335e+01
Generation	940:	Best = 2.169922e+05	FES = 94100	Diversity = 1.2864e+01
Generation	950:	Best = 2.150373e+05	FES = 95100	Diversity = 1.2468e+01
Generation	960:	Best = 2.136237e+05	FES = 96100	Diversity = 1.1970e+01
Generation	970:	Best = 2.127353e+05	FES = 97100	Diversity = 1.2000e+01
Generation	980:	Best = 2.118826e+05	FES = 98100	Diversity = 1.0992e+01

Generation 990:	Best = 2.113202e+05	FES = 99100	Diversity = 1.0749e+01
Generation 1000:	Best = 2.103551e+05	FES = 100100	Diversity = 1.0794e+01
Generation 1010:	Best = 2.098214e+05	FES = 101100	Diversity = 1.0287e+01
Generation 1020:	Best = 2.086791e+05	FES = 102100	Diversity = 9.3069e+00
Generation 1030:	Best = 2.073503e+05	FES = 103100	Diversity = 1.0043e+01
Generation 1040:	Best = 2.070505e+05	FES = 104100	Diversity = 1.0525e+01
Generation 1050:	Best = 2.059856e+05	FES = 105100	Diversity = 9.6038e+00
Generation 1060:	Best = 2.058784e+05	FES = 106100	Diversity = 9.6558e+00
Generation 1070:	Best = 2.055348e+05	FES = 107100	Diversity = 9.4390e+00
Generation 1080:	Best = 2.051051e+05	FES = 108100	Diversity = 8.3895e+00
Generation 1090:	Best = 2.045585e+05	FES = 109100	Diversity = 8.4322e+00
Generation 1100:	Best = 2.042691e+05	FES = 110100	Diversity = 8.8950e+00
Generation 1110:	Best = 2.034911e+05	FES = 111100	Diversity = 8.1972e+00
Generation 1120:	Best = 2.028860e+05	FES = 112100	Diversity = 8.8300e+00
Generation 1130:	Best = 2.021723e+05	FES = 113100	Diversity = 8.8965e+00
Generation 1140:	Best = 2.020625e+05	FES = 114100	Diversity = 8.2902e+00
Generation 1150:	Best = 2.016349e+05	FES = 115100	Diversity = 9.0367e+00
Generation 1160:	Best = 2.015469e+05	FES = 116100	Diversity = 8.8925e+00
Generation 1170:	Best = 2.011266e+05	FES = 117100	Diversity = 8.4196e+00
Generation 1180:	Best = 2.008501e+05	FES = 118100	Diversity = 8.4081e+00
Generation 1190:	Best = 2.004329e+05	FES = 119100	Diversity = 9.3520e+00
Generation 1200:	Best = 2.001478e+05	FES = 120100	Diversity = 9.6664e+00
Generation 1210:	Best = 1.996427e+05	FES = 121100	Diversity = 9.3856e+00
Generation 1220:	Best = 1.987388e+05	FES = 122100	Diversity = 9.8302e+00
Generation 1230:	Best = 1.979155e+05	FES = 123100	Diversity = 1.0004e+01
Generation 1240:	Best = 1.977008e+05	FES = 124100	Diversity = 1.0586e+01
Generation 1250:	Best = 1.973252e+05	FES = 125100	Diversity = 1.1147e+01
Generation 1260:	Best = 1.968967e+05	FES = 126100	Diversity = 1.0993e+01
Generation 1270:	Best = 1.967934e+05	FES = 127100	Diversity = 1.0781e+01
Generation 1280:	Best = 1.962322e+05	FES = 128100	Diversity = 1.0085e+01
Generation 1290:	Best = 1.959766e+05	FES = 129100	Diversity = 1.0694e+01
Generation 1300:	Best = 1.957538e+05	FES = 130100	Diversity = 9.7094e+00
Generation 1310:	Best = 1.954362e+05	FES = 131100	Diversity = 9.9332e+00
Generation 1320:	Best = 1.952350e+05	FES = 132100	Diversity = 9.7296e+00
Generation 1330:	Best = 1.952247e+05	FES = 133100	Diversity = 9.6234e+00
Generation 1340:	Best = 1.949705e+05	FES = 134100	Diversity = 8.5406e+00
Generation 1350:	Best = 1.947920e+05	FES = 135100	Diversity = 8.3046e+00
Generation 1360:	Best = 1.944325e+05	FES = 136100	Diversity = 8.7509e+00
Generation 1370:	Best = 1.936127e+05	FES = 137100	Diversity = 9.4057e+00
Generation 1380:	Best = 1.934787e+05	FES = 138100	Diversity = 1.0565e+01
Generation 1390:	Best = 1.930914e+05	FES = 139100	Diversity = 1.0695e+01
Generation 1400:	Best = 1.929408e+05	FES = 140100	Diversity = 1.0414e+01
Generation 1410:	Best = 1.924785e+05	FES = 141100	Diversity = 9.6641e+00
Generation 1420:	Best = 1.914180e+05	FES = 142100	Diversity = 1.0096e+01
Generation 1430:	Best = 1.906349e+05	FES = 143100	Diversity = 1.0825e+01
Generation 1440:	Best = 1.899020e+05	FES = 144100	Diversity = 1.0730e+01
Generation 1450:	Best = 1.895702e+05	FES = 145100	Diversity = 1.0874e+01
Generation 1460:	Best = 1.891668e+05	FES = 146100	Diversity = 1.1587e+01
Generation 1470:	Best = 1.889355e+05	FES = 147100	Diversity = 1.0657e+01
Generation 1480:	Best = 1.881613e+05	FES = 148100	Diversity = 1.1602e+01
Generation 1490:	Best = 1.879693e+05	FES = 149100	Diversity = 1.2420e+01
Generation 1500:	Best = 1.869593e+05	FES = 150100	Diversity = 1.3406e+01
Generation 1510:	Best = 1.867044e+05	FES = 151100	Diversity = 1.3106e+01
Generation 1520:	Best = 1.862114e+05	FES = 152100	Diversity = 1.1550e+01

Generation 1530:	Best = 1.860574e+05	FES = 153100	Diversity = 1.1332e+01
Generation 1540:	Best = 1.854672e+05	FES = 154100	Diversity = 1.2120e+01
Generation 1550:	Best = 1.852575e+05	FES = 155100	Diversity = 1.2134e+01
Generation 1560:	Best = 1.847609e+05	FES = 156100	Diversity = 1.0432e+01
Generation 1570:	Best = 1.845077e+05	FES = 157100	Diversity = 1.0688e+01
Generation 1580:	Best = 1.842834e+05	FES = 158100	Diversity = 1.1009e+01
Generation 1590:	Best = 1.840007e+05	FES = 159100	Diversity = 1.0535e+01
Generation 1600:	Best = 1.837647e+05	FES = 160100	Diversity = 1.1023e+01
Generation 1610:	Best = 1.835032e+05	FES = 161100	Diversity = 1.1752e+01
Generation 1620:	Best = 1.830040e+05	FES = 162100	Diversity = 1.1832e+01
Generation 1630:	Best = 1.826952e+05	FES = 163100	Diversity = 1.2411e+01
Generation 1640:	Best = 1.822585e+05	FES = 164100	Diversity = 1.2708e+01
Generation 1650:	Best = 1.819092e+05	FES = 165100	Diversity = 1.4347e+01
Generation 1660:	Best = 1.813349e+05	FES = 166100	Diversity = 1.3502e+01
Generation 1670:	Best = 1.812434e+05	FES = 167100	Diversity = 1.2919e+01
Generation 1680:	Best = 1.808328e+05	FES = 168100	Diversity = 1.1378e+01
Generation 1690:	Best = 1.806221e+05	FES = 169100	Diversity = 1.0479e+01
Generation 1700:	Best = 1.803661e+05	FES = 170100	Diversity = 9.9042e+00
Generation 1710:	Best = 1.799650e+05	FES = 171100	Diversity = 8.8122e+00
Generation 1720:	Best = 1.793850e+05	FES = 172100	Diversity = 9.6417e+00
Generation 1730:	Best = 1.791417e+05	FES = 173100	Diversity = 9.8260e+00
Generation 1740:	Best = 1.787320e+05	FES = 174100	Diversity = 1.0707e+01
Generation 1750:	Best = 1.784479e+05	FES = 175100	Diversity = 9.7288e+00
Generation 1760:	Best = 1.779069e+05	FES = 176100	Diversity = 9.0940e+00
Generation 1770:	Best = 1.777760e+05	FES = 177100	Diversity = 8.8843e+00
Generation 1780:	Best = 1.776378e+05	FES = 178100	Diversity = 9.0713e+00
Generation 1790:	Best = 1.772726e+05	FES = 179100	Diversity = 8.4694e+00
Generation 1800:	Best = 1.771673e+05	FES = 180100	Diversity = 9.9002e+00
Generation 1810:	Best = 1.766604e+05	FES = 181100	Diversity = 9.5698e+00
Generation 1820:	Best = 1.764240e+05	FES = 182100	Diversity = 1.0343e+01
Generation 1830:	Best = 1.761054e+05	FES = 183100	Diversity = 1.1051e+01
Generation 1840:	Best = 1.759160e+05	FES = 184100	Diversity = 1.0821e+01
Generation 1850:	Best = 1.757130e+05	FES = 185100	Diversity = 1.1034e+01
Generation 1860:	Best = 1.752087e+05	FES = 186100	Diversity = 1.1382e+01
Generation 1870:	Best = 1.749585e+05	FES = 187100	Diversity = 1.1744e+01
Generation 1880:	Best = 1.746964e+05	FES = 188100	Diversity = 1.1189e+01
Generation 1890:	Best = 1.742445e+05	FES = 189100	Diversity = 1.0864e+01
Generation 1900:	Best = 1.740113e+05	FES = 190100	Diversity = 1.1266e+01
Generation 1910:	Best = 1.738412e+05	FES = 191100	Diversity = 1.0641e+01
Generation 1920:	Best = 1.736026e+05	FES = 192100	Diversity = 9.8471e+00
Generation 1930:	Best = 1.733757e+05	FES = 193100	Diversity = 9.2199e+00
Generation 1940:	Best = 1.732523e+05	FES = 194100	Diversity = 9.0681e+00
Generation 1950:	Best = 1.730388e+05	FES = 195100	Diversity = 9.1666e+00
Generation 1960:	Best = 1.728566e+05	FES = 196100	Diversity = 9.5303e+00
Generation 1970:	Best = 1.726616e+05	FES = 197100	Diversity = 9.5773e+00
Generation 1980:	Best = 1.725049e+05	FES = 198100	Diversity = 1.0416e+01
Generation 1990:	Best = 1.723549e+05	FES = 199100	Diversity = 9.6229e+00
Generation 2000:	Best = 1.721736e+05	FES = 200100	Diversity = 9.3311e+00
Generation 2010:	Best = 1.721103e+05	FES = 201100	Diversity = 9.6447e+00
Generation 2020:	Best = 1.717744e+05	FES = 202100	Diversity = 8.1812e+00
Generation 2030:	Best = 1.716798e+05	FES = 203100	Diversity = 8.1333e+00
Generation 2040:	Best = 1.714483e+05	FES = 204100	Diversity = 8.9183e+00
Generation 2050:	Best = 1.712829e+05	FES = 205100	Diversity = 9.4477e+00
Generation 2060:	Best = 1.707508e+05	FES = 206100	Diversity = 1.1287e+01

Generation 2070:	Best = 1.704136e+05	FES = 207100	Diversity = 1.0981e+01
Generation 2080:	Best = 1.701107e+05	FES = 208100	Diversity = 1.1497e+01
Generation 2090:	Best = 1.698515e+05	FES = 209100	Diversity = 1.0949e+01
Generation 2100:	Best = 1.696350e+05	FES = 210100	Diversity = 1.0744e+01
Generation 2110:	Best = 1.694992e+05	FES = 211100	Diversity = 9.6043e+00
Generation 2120:	Best = 1.692756e+05	FES = 212100	Diversity = 9.9568e+00
Generation 2130:	Best = 1.691264e+05	FES = 213100	Diversity = 9.5255e+00
Generation 2140:	Best = 1.689577e+05	FES = 214100	Diversity = 9.6335e+00
Generation 2150:	Best = 1.687629e+05	FES = 215100	Diversity = 9.4517e+00
Generation 2160:	Best = 1.687103e+05	FES = 216100	Diversity = 8.7290e+00
Generation 2170:	Best = 1.685639e+05	FES = 217100	Diversity = 9.1400e+00
Generation 2180:	Best = 1.685259e+05	FES = 218100	Diversity = 8.6733e+00
Generation 2190:	Best = 1.682487e+05	FES = 219100	Diversity = 8.8300e+00
Generation 2200:	Best = 1.681306e+05	FES = 220100	Diversity = 9.0512e+00
Generation 2210:	Best = 1.679158e+05	FES = 221100	Diversity = 9.7612e+00
Generation 2220:	Best = 1.675911e+05	FES = 222100	Diversity = 1.0647e+01
Generation 2230:	Best = 1.673353e+05	FES = 223100	Diversity = 1.1372e+01
Generation 2240:	Best = 1.670851e+05	FES = 224100	Diversity = 1.1951e+01
Generation 2250:	Best = 1.669131e+05	FES = 225100	Diversity = 1.1641e+01
Generation 2260:	Best = 1.664085e+05	FES = 226100	Diversity = 1.0986e+01
Generation 2270:	Best = 1.662307e+05	FES = 227100	Diversity = 1.1332e+01
Generation 2280:	Best = 1.661774e+05	FES = 228100	Diversity = 1.1090e+01
Generation 2290:	Best = 1.659614e+05	FES = 229100	Diversity = 1.0615e+01
Generation 2300:	Best = 1.657767e+05	FES = 230100	Diversity = 1.0438e+01
Generation 2310:	Best = 1.653489e+05	FES = 231100	Diversity = 1.0031e+01
Generation 2320:	Best = 1.651450e+05	FES = 232100	Diversity = 1.1396e+01
Generation 2330:	Best = 1.649665e+05	FES = 233100	Diversity = 1.1068e+01
Generation 2340:	Best = 1.646611e+05	FES = 234100	Diversity = 1.0347e+01
Generation 2350:	Best = 1.645945e+05	FES = 235100	Diversity = 9.9601e+00
Generation 2360:	Best = 1.643096e+05	FES = 236100	Diversity = 9.2020e+00
Generation 2370:	Best = 1.642150e+05	FES = 237100	Diversity = 1.0558e+01
Generation 2380:	Best = 1.639126e+05	FES = 238100	Diversity = 1.0490e+01
Generation 2390:	Best = 1.638464e+05	FES = 239100	Diversity = 9.5175e+00
Generation 2400:	Best = 1.635645e+05	FES = 240100	Diversity = 9.9086e+00
Generation 2410:	Best = 1.634151e+05	FES = 241100	Diversity = 9.9440e+00
Generation 2420:	Best = 1.632883e+05	FES = 242100	Diversity = 9.1205e+00
Generation 2430:	Best = 1.631522e+05	FES = 243100	Diversity = 9.6344e+00
Generation 2440:	Best = 1.629522e+05	FES = 244100	Diversity = 9.3695e+00
Generation 2450:	Best = 1.626924e+05	FES = 245100	Diversity = 1.0197e+01
Generation 2460:	Best = 1.625760e+05	FES = 246100	Diversity = 1.0411e+01
Generation 2470:	Best = 1.621911e+05	FES = 247100	Diversity = 1.0744e+01
Generation 2480:	Best = 1.621027e+05	FES = 248100	Diversity = 1.0237e+01
Generation 2490:	Best = 1.618018e+05	FES = 249100	Diversity = 9.1639e+00
Generation 2500:	Best = 1.616774e+05	FES = 250100	Diversity = 9.1705e+00
Generation 2510:	Best = 1.615246e+05	FES = 251100	Diversity = 9.1827e+00
Generation 2520:	Best = 1.613951e+05	FES = 252100	Diversity = 9.0393e+00
Generation 2530:	Best = 1.613099e+05	FES = 253100	Diversity = 8.0835e+00
Generation 2540:	Best = 1.611564e+05	FES = 254100	Diversity = 7.9734e+00
Generation 2550:	Best = 1.608762e+05	FES = 255100	Diversity = 8.6059e+00
Generation 2560:	Best = 1.604806e+05	FES = 256100	Diversity = 8.4918e+00
Generation 2570:	Best = 1.602630e+05	FES = 257100	Diversity = 9.3234e+00
Generation 2580:	Best = 1.600752e+05	FES = 258100	Diversity = 9.1488e+00
Generation 2590:	Best = 1.599211e+05	FES = 259100	Diversity = 9.7000e+00
Generation 2600:	Best = 1.597802e+05	FES = 260100	Diversity = 9.8780e+00

Generation 2610:	Best = 1.596155e+05	FES = 261100	Diversity = 1.0344e+01
Generation 2620:	Best = 1.591952e+05	FES = 262100	Diversity = 1.0167e+01
Generation 2630:	Best = 1.591952e+05	FES = 263100	Diversity = 9.4371e+00
Generation 2640:	Best = 1.590508e+05	FES = 264100	Diversity = 9.1249e+00
Generation 2650:	Best = 1.589812e+05	FES = 265100	Diversity = 9.1590e+00
Generation 2660:	Best = 1.588772e+05	FES = 266100	Diversity = 9.3907e+00
Generation 2670:	Best = 1.587479e+05	FES = 267100	Diversity = 9.2397e+00
Generation 2680:	Best = 1.586629e+05	FES = 268100	Diversity = 9.6358e+00
Generation 2690:	Best = 1.585176e+05	FES = 269100	Diversity = 1.0279e+01
Generation 2700:	Best = 1.584057e+05	FES = 270100	Diversity = 1.0806e+01
Generation 2710:	Best = 1.581427e+05	FES = 271100	Diversity = 1.0769e+01
Generation 2720:	Best = 1.579300e+05	FES = 272100	Diversity = 1.0950e+01
Generation 2730:	Best = 1.576682e+05	FES = 273100	Diversity = 1.0294e+01
Generation 2740:	Best = 1.575859e+05	FES = 274100	Diversity = 9.6041e+00
Generation 2750:	Best = 1.573577e+05	FES = 275100	Diversity = 1.0180e+01
Generation 2760:	Best = 1.571112e+05	FES = 276100	Diversity = 1.1245e+01
Generation 2770:	Best = 1.570548e+05	FES = 277100	Diversity = 1.0615e+01
Generation 2780:	Best = 1.568286e+05	FES = 278100	Diversity = 1.0924e+01
Generation 2790:	Best = 1.567636e+05	FES = 279100	Diversity = 1.1111e+01
Generation 2800:	Best = 1.566002e+05	FES = 280100	Diversity = 1.1371e+01
Generation 2810:	Best = 1.563938e+05	FES = 281100	Diversity = 1.1472e+01
Generation 2820:	Best = 1.562563e+05	FES = 282100	Diversity = 1.1657e+01
Generation 2830:	Best = 1.560891e+05	FES = 283100	Diversity = 1.0882e+01
Generation 2840:	Best = 1.559692e+05	FES = 284100	Diversity = 1.1272e+01
Generation 2850:	Best = 1.557222e+05	FES = 285100	Diversity = 1.1508e+01
Generation 2860:	Best = 1.555924e+05	FES = 286100	Diversity = 1.1229e+01
Generation 2870:	Best = 1.555225e+05	FES = 287100	Diversity = 1.1995e+01
Generation 2880:	Best = 1.553379e+05	FES = 288100	Diversity = 1.2643e+01
Generation 2890:	Best = 1.551933e+05	FES = 289100	Diversity = 1.2359e+01
Generation 2900:	Best = 1.550791e+05	FES = 290100	Diversity = 1.1293e+01
Generation 2910:	Best = 1.549470e+05	FES = 291100	Diversity = 1.0017e+01
Generation 2920:	Best = 1.548332e+05	FES = 292100	Diversity = 1.0034e+01
Generation 2930:	Best = 1.546148e+05	FES = 293100	Diversity = 1.0244e+01
Generation 2940:	Best = 1.542717e+05	FES = 294100	Diversity = 1.0280e+01
Generation 2950:	Best = 1.540898e+05	FES = 295100	Diversity = 1.1395e+01
Generation 2960:	Best = 1.537318e+05	FES = 296100	Diversity = 1.0844e+01
Generation 2970:	Best = 1.533100e+05	FES = 297100	Diversity = 1.1015e+01
Generation 2980:	Best = 1.532141e+05	FES = 298100	Diversity = 1.1598e+01
Generation 2990:	Best = 1.526444e+05	FES = 299100	Diversity = 1.0839e+01
Generation 3000:	Best = 1.526339e+05	FES = 300100	Diversity = 1.0508e+01
Generation 3010:	Best = 1.525168e+05	FES = 301100	Diversity = 1.0934e+01
Generation 3020:	Best = 1.520147e+05	FES = 302100	Diversity = 1.1839e+01
Generation 3030:	Best = 1.515764e+05	FES = 303100	Diversity = 1.0955e+01
Generation 3040:	Best = 1.513874e+05	FES = 304100	Diversity = 1.1268e+01
Generation 3050:	Best = 1.512515e+05	FES = 305100	Diversity = 1.0874e+01
Generation 3060:	Best = 1.509658e+05	FES = 306100	Diversity = 1.1618e+01
Generation 3070:	Best = 1.505021e+05	FES = 307100	Diversity = 1.1862e+01
Generation 3080:	Best = 1.502924e+05	FES = 308100	Diversity = 1.1265e+01
Generation 3090:	Best = 1.500603e+05	FES = 309100	Diversity = 1.0226e+01
Generation 3100:	Best = 1.498112e+05	FES = 310100	Diversity = 9.0792e+00
Generation 3110:	Best = 1.495807e+05	FES = 311100	Diversity = 8.8294e+00
Generation 3120:	Best = 1.492034e+05	FES = 312100	Diversity = 9.1823e+00
Generation 3130:	Best = 1.491182e+05	FES = 313100	Diversity = 9.5098e+00
Generation 3140:	Best = 1.489468e+05	FES = 314100	Diversity = 9.7089e+00

Generation 3150:	Best = 1.487048e+05	FES = 315100	Diversity = 8.7658e+00
Generation 3160:	Best = 1.485712e+05	FES = 316100	Diversity = 9.2545e+00
Generation 3170:	Best = 1.483532e+05	FES = 317100	Diversity = 8.8063e+00
Generation 3180:	Best = 1.481738e+05	FES = 318100	Diversity = 7.9501e+00
Generation 3190:	Best = 1.480841e+05	FES = 319100	Diversity = 8.2055e+00
Generation 3200:	Best = 1.478785e+05	FES = 320100	Diversity = 8.2030e+00
Generation 3210:	Best = 1.477254e+05	FES = 321100	Diversity = 7.7517e+00
Generation 3220:	Best = 1.475689e+05	FES = 322100	Diversity = 7.2212e+00
Generation 3230:	Best = 1.473965e+05	FES = 323100	Diversity = 7.0955e+00
Generation 3240:	Best = 1.472609e+05	FES = 324100	Diversity = 6.4973e+00
Generation 3250:	Best = 1.471769e+05	FES = 325100	Diversity = 5.9928e+00
Generation 3260:	Best = 1.470289e+05	FES = 326100	Diversity = 6.5772e+00
Generation 3270:	Best = 1.468350e+05	FES = 327100	Diversity = 7.0994e+00
Generation 3280:	Best = 1.466885e+05	FES = 328100	Diversity = 6.9136e+00
Generation 3290:	Best = 1.464100e+05	FES = 329100	Diversity = 6.6931e+00
Generation 3300:	Best = 1.462515e+05	FES = 330100	Diversity = 7.2564e+00
Generation 3310:	Best = 1.461251e+05	FES = 331100	Diversity = 7.0692e+00
Generation 3320:	Best = 1.459477e+05	FES = 332100	Diversity = 7.4506e+00
Generation 3330:	Best = 1.457474e+05	FES = 333100	Diversity = 7.4856e+00
Generation 3340:	Best = 1.454804e+05	FES = 334100	Diversity = 7.4611e+00
Generation 3350:	Best = 1.450433e+05	FES = 335100	Diversity = 8.1189e+00
Generation 3360:	Best = 1.448800e+05	FES = 336100	Diversity = 8.2252e+00
Generation 3370:	Best = 1.445744e+05	FES = 337100	Diversity = 8.1907e+00
Generation 3380:	Best = 1.443151e+05	FES = 338100	Diversity = 8.5321e+00
Generation 3390:	Best = 1.441157e+05	FES = 339100	Diversity = 9.1396e+00
Generation 3400:	Best = 1.438418e+05	FES = 340100	Diversity = 9.1871e+00
Generation 3410:	Best = 1.437275e+05	FES = 341100	Diversity = 8.3458e+00
Generation 3420:	Best = 1.435978e+05	FES = 342100	Diversity = 8.4378e+00
Generation 3430:	Best = 1.434755e+05	FES = 343100	Diversity = 8.2861e+00
Generation 3440:	Best = 1.432879e+05	FES = 344100	Diversity = 8.2251e+00
Generation 3450:	Best = 1.431981e+05	FES = 345100	Diversity = 7.5428e+00
Generation 3460:	Best = 1.430105e+05	FES = 346100	Diversity = 7.2640e+00
Generation 3470:	Best = 1.428695e+05	FES = 347100	Diversity = 7.2849e+00
Generation 3480:	Best = 1.426957e+05	FES = 348100	Diversity = 7.0785e+00
Generation 3490:	Best = 1.425326e+05	FES = 349100	Diversity = 6.9453e+00
Generation 3500:	Best = 1.423087e+05	FES = 350100	Diversity = 6.6585e+00
Generation 3510:	Best = 1.421740e+05	FES = 351100	Diversity = 5.9819e+00
Generation 3520:	Best = 1.418651e+05	FES = 352100	Diversity = 6.1991e+00
Generation 3530:	Best = 1.418035e+05	FES = 353100	Diversity = 6.4103e+00
Generation 3540:	Best = 1.414616e+05	FES = 354100	Diversity = 6.8114e+00
Generation 3550:	Best = 1.413693e+05	FES = 355100	Diversity = 6.4850e+00
Generation 3560:	Best = 1.412330e+05	FES = 356100	Diversity = 6.6297e+00
Generation 3570:	Best = 1.410890e+05	FES = 357100	Diversity = 7.0538e+00
Generation 3580:	Best = 1.407686e+05	FES = 358100	Diversity = 6.7527e+00
Generation 3590:	Best = 1.406510e+05	FES = 359100	Diversity = 6.9028e+00
Generation 3600:	Best = 1.403663e+05	FES = 360100	Diversity = 8.1453e+00
Generation 3610:	Best = 1.401395e+05	FES = 361100	Diversity = 8.0789e+00
Generation 3620:	Best = 1.398403e+05	FES = 362100	Diversity = 7.7750e+00
Generation 3630:	Best = 1.394730e+05	FES = 363100	Diversity = 9.2122e+00
Generation 3640:	Best = 1.393501e+05	FES = 364100	Diversity = 9.1237e+00
Generation 3650:	Best = 1.392115e+05	FES = 365100	Diversity = 8.3791e+00
Generation 3660:	Best = 1.389642e+05	FES = 366100	Diversity = 8.5675e+00
Generation 3670:	Best = 1.388302e+05	FES = 367100	Diversity = 8.9428e+00
Generation 3680:	Best = 1.384483e+05	FES = 368100	Diversity = 9.3056e+00

Generation 3690:	Best = 1.384125e+05	FES = 369100	Diversity = 8.8662e+00
Generation 3700:	Best = 1.382716e+05	FES = 370100	Diversity = 8.5695e+00
Generation 3710:	Best = 1.380857e+05	FES = 371100	Diversity = 8.2915e+00
Generation 3720:	Best = 1.380308e+05	FES = 372100	Diversity = 7.9238e+00
Generation 3730:	Best = 1.379640e+05	FES = 373100	Diversity = 7.9845e+00
Generation 3740:	Best = 1.378143e+05	FES = 374100	Diversity = 7.8966e+00
Generation 3750:	Best = 1.376971e+05	FES = 375100	Diversity = 6.9201e+00
Generation 3760:	Best = 1.374349e+05	FES = 376100	Diversity = 8.0341e+00
Generation 3770:	Best = 1.372364e+05	FES = 377100	Diversity = 8.2499e+00
Generation 3780:	Best = 1.370268e+05	FES = 378100	Diversity = 8.7115e+00
Generation 3790:	Best = 1.368998e+05	FES = 379100	Diversity = 9.3749e+00
Generation 3800:	Best = 1.366196e+05	FES = 380100	Diversity = 9.9693e+00
Generation 3810:	Best = 1.364491e+05	FES = 381100	Diversity = 9.9941e+00
Generation 3820:	Best = 1.363054e+05	FES = 382100	Diversity = 9.2342e+00
Generation 3830:	Best = 1.360948e+05	FES = 383100	Diversity = 9.1789e+00
Generation 3840:	Best = 1.357358e+05	FES = 384100	Diversity = 8.3001e+00
Generation 3850:	Best = 1.353820e+05	FES = 385100	Diversity = 8.6569e+00
Generation 3860:	Best = 1.352685e+05	FES = 386100	Diversity = 8.0128e+00
Generation 3870:	Best = 1.350285e+05	FES = 387100	Diversity = 8.3354e+00
Generation 3880:	Best = 1.348173e+05	FES = 388100	Diversity = 7.8673e+00
Generation 3890:	Best = 1.345814e+05	FES = 389100	Diversity = 8.6740e+00
Generation 3900:	Best = 1.344983e+05	FES = 390100	Diversity = 8.8211e+00
Generation 3910:	Best = 1.343162e+05	FES = 391100	Diversity = 8.5190e+00
Generation 3920:	Best = 1.342478e+05	FES = 392100	Diversity = 7.9139e+00
Generation 3930:	Best = 1.341165e+05	FES = 393100	Diversity = 7.6911e+00
Generation 3940:	Best = 1.340210e+05	FES = 394100	Diversity = 7.9062e+00
Generation 3950:	Best = 1.336877e+05	FES = 395100	Diversity = 8.9373e+00
Generation 3960:	Best = 1.336356e+05	FES = 396100	Diversity = 7.6819e+00
Generation 3970:	Best = 1.334808e+05	FES = 397100	Diversity = 8.1869e+00
Generation 3980:	Best = 1.332593e+05	FES = 398100	Diversity = 8.3942e+00
Generation 3990:	Best = 1.328388e+05	FES = 399100	Diversity = 8.5415e+00
Generation 4000:	Best = 1.328182e+05	FES = 400100	Diversity = 8.9520e+00
Generation 4010:	Best = 1.326301e+05	FES = 401100	Diversity = 8.9797e+00
Generation 4020:	Best = 1.325043e+05	FES = 402100	Diversity = 9.1788e+00
Generation 4030:	Best = 1.321519e+05	FES = 403100	Diversity = 9.3259e+00
Generation 4040:	Best = 1.320167e+05	FES = 404100	Diversity = 1.0075e+01
Generation 4050:	Best = 1.318227e+05	FES = 405100	Diversity = 1.0111e+01
Generation 4060:	Best = 1.316368e+05	FES = 406100	Diversity = 1.0699e+01
Generation 4070:	Best = 1.314339e+05	FES = 407100	Diversity = 1.1044e+01
Generation 4080:	Best = 1.312143e+05	FES = 408100	Diversity = 1.0624e+01
Generation 4090:	Best = 1.310481e+05	FES = 409100	Diversity = 1.0877e+01
Generation 4100:	Best = 1.308323e+05	FES = 410100	Diversity = 1.0648e+01
Generation 4110:	Best = 1.307513e+05	FES = 411100	Diversity = 9.1929e+00
Generation 4120:	Best = 1.305111e+05	FES = 412100	Diversity = 8.7318e+00
Generation 4130:	Best = 1.302606e+05	FES = 413100	Diversity = 9.3651e+00
Generation 4140:	Best = 1.301301e+05	FES = 414100	Diversity = 9.4003e+00
Generation 4150:	Best = 1.298493e+05	FES = 415100	Diversity = 9.2364e+00
Generation 4160:	Best = 1.295823e+05	FES = 416100	Diversity = 9.5016e+00
Generation 4170:	Best = 1.294157e+05	FES = 417100	Diversity = 9.4504e+00
Generation 4180:	Best = 1.290900e+05	FES = 418100	Diversity = 1.0457e+01
Generation 4190:	Best = 1.287160e+05	FES = 419100	Diversity = 1.0418e+01
Generation 4200:	Best = 1.285722e+05	FES = 420100	Diversity = 9.2992e+00
Generation 4210:	Best = 1.283930e+05	FES = 421100	Diversity = 8.1735e+00
Generation 4220:	Best = 1.282423e+05	FES = 422100	Diversity = 8.8264e+00

Generation 4230:	Best = 1.280643e+05	FES = 423100	Diversity = 8.6853e+00
Generation 4240:	Best = 1.279793e+05	FES = 424100	Diversity = 9.2287e+00
Generation 4250:	Best = 1.277982e+05	FES = 425100	Diversity = 7.9314e+00
Generation 4260:	Best = 1.275738e+05	FES = 426100	Diversity = 7.2821e+00
Generation 4270:	Best = 1.274318e+05	FES = 427100	Diversity = 7.5816e+00
Generation 4280:	Best = 1.272340e+05	FES = 428100	Diversity = 7.8098e+00
Generation 4290:	Best = 1.271280e+05	FES = 429100	Diversity = 8.3925e+00
Generation 4300:	Best = 1.269026e+05	FES = 430100	Diversity = 8.8397e+00
Generation 4310:	Best = 1.267192e+05	FES = 431100	Diversity = 9.1506e+00
Generation 4320:	Best = 1.263251e+05	FES = 432100	Diversity = 9.5032e+00
Generation 4330:	Best = 1.261592e+05	FES = 433100	Diversity = 8.8567e+00
Generation 4340:	Best = 1.260424e+05	FES = 434100	Diversity = 8.2974e+00
Generation 4350:	Best = 1.258937e+05	FES = 435100	Diversity = 7.2887e+00
Generation 4360:	Best = 1.257819e+05	FES = 436100	Diversity = 7.1867e+00
Generation 4370:	Best = 1.256058e+05	FES = 437100	Diversity = 6.9249e+00
Generation 4380:	Best = 1.254808e+05	FES = 438100	Diversity = 7.1055e+00
Generation 4390:	Best = 1.252740e+05	FES = 439100	Diversity = 7.4977e+00
Generation 4400:	Best = 1.250022e+05	FES = 440100	Diversity = 7.5898e+00
Generation 4410:	Best = 1.249424e+05	FES = 441100	Diversity = 7.7361e+00
Generation 4420:	Best = 1.246594e+05	FES = 442100	Diversity = 8.1382e+00
Generation 4430:	Best = 1.246487e+05	FES = 443100	Diversity = 8.2895e+00
Generation 4440:	Best = 1.245141e+05	FES = 444100	Diversity = 8.3742e+00
Generation 4450:	Best = 1.244234e+05	FES = 445100	Diversity = 7.7599e+00
Generation 4460:	Best = 1.242588e+05	FES = 446100	Diversity = 7.1565e+00
Generation 4470:	Best = 1.241756e+05	FES = 447100	Diversity = 6.8208e+00
Generation 4480:	Best = 1.240401e+05	FES = 448100	Diversity = 7.0560e+00
Generation 4490:	Best = 1.238528e+05	FES = 449100	Diversity = 7.1912e+00
Generation 4500:	Best = 1.237313e+05	FES = 450100	Diversity = 7.2339e+00
Generation 4510:	Best = 1.234516e+05	FES = 451100	Diversity = 7.4209e+00
Generation 4520:	Best = 1.233682e+05	FES = 452100	Diversity = 7.0302e+00
Generation 4530:	Best = 1.232477e+05	FES = 453100	Diversity = 6.7045e+00
Generation 4540:	Best = 1.231017e+05	FES = 454100	Diversity = 6.4700e+00
Generation 4550:	Best = 1.229140e+05	FES = 455100	Diversity = 7.3252e+00
Generation 4560:	Best = 1.226909e+05	FES = 456100	Diversity = 7.7440e+00
Generation 4570:	Best = 1.225341e+05	FES = 457100	Diversity = 8.0859e+00
Generation 4580:	Best = 1.223123e+05	FES = 458100	Diversity = 8.5844e+00
Generation 4590:	Best = 1.222331e+05	FES = 459100	Diversity = 8.3462e+00
Generation 4600:	Best = 1.221292e+05	FES = 460100	Diversity = 8.0342e+00
Generation 4610:	Best = 1.219682e+05	FES = 461100	Diversity = 7.5971e+00
Generation 4620:	Best = 1.216173e+05	FES = 462100	Diversity = 7.3512e+00
Generation 4630:	Best = 1.214636e+05	FES = 463100	Diversity = 7.1505e+00
Generation 4640:	Best = 1.211385e+05	FES = 464100	Diversity = 8.0132e+00
Generation 4650:	Best = 1.210444e+05	FES = 465100	Diversity = 7.8177e+00
Generation 4660:	Best = 1.209607e+05	FES = 466100	Diversity = 8.4712e+00
Generation 4670:	Best = 1.207664e+05	FES = 467100	Diversity = 8.4064e+00
Generation 4680:	Best = 1.205784e+05	FES = 468100	Diversity = 8.1585e+00
Generation 4690:	Best = 1.204956e+05	FES = 469100	Diversity = 8.3778e+00
Generation 4700:	Best = 1.204345e+05	FES = 470100	Diversity = 7.7965e+00
Generation 4710:	Best = 1.203139e+05	FES = 471100	Diversity = 7.6666e+00
Generation 4720:	Best = 1.202164e+05	FES = 472100	Diversity = 7.8264e+00
Generation 4730:	Best = 1.200553e+05	FES = 473100	Diversity = 7.9193e+00
Generation 4740:	Best = 1.199824e+05	FES = 474100	Diversity = 8.2336e+00
Generation 4750:	Best = 1.198267e+05	FES = 475100	Diversity = 7.7998e+00
Generation 4760:	Best = 1.197563e+05	FES = 476100	Diversity = 7.6280e+00

Generation 4770:	Best = 1.195460e+05	FES = 477100	Diversity = 8.5340e+00
Generation 4780:	Best = 1.194746e+05	FES = 478100	Diversity = 8.0987e+00
Generation 4790:	Best = 1.193963e+05	FES = 479100	Diversity = 7.8103e+00
Generation 4800:	Best = 1.192545e+05	FES = 480100	Diversity = 7.5692e+00
Generation 4810:	Best = 1.190951e+05	FES = 481100	Diversity = 7.5252e+00
Generation 4820:	Best = 1.190645e+05	FES = 482100	Diversity = 7.6475e+00
Generation 4830:	Best = 1.189824e+05	FES = 483100	Diversity = 7.5323e+00
Generation 4840:	Best = 1.188719e+05	FES = 484100	Diversity = 7.4483e+00
Generation 4850:	Best = 1.187964e+05	FES = 485100	Diversity = 7.4391e+00
Generation 4860:	Best = 1.186740e+05	FES = 486100	Diversity = 7.8671e+00
Generation 4870:	Best = 1.185265e+05	FES = 487100	Diversity = 8.4817e+00
Generation 4880:	Best = 1.183937e+05	FES = 488100	Diversity = 8.9946e+00
Generation 4890:	Best = 1.182899e+05	FES = 489100	Diversity = 8.5590e+00
Generation 4900:	Best = 1.182526e+05	FES = 490100	Diversity = 7.9618e+00
Generation 4910:	Best = 1.180607e+05	FES = 491100	Diversity = 8.1305e+00
Generation 4920:	Best = 1.179090e+05	FES = 492100	Diversity = 8.8081e+00
Generation 4930:	Best = 1.177365e+05	FES = 493100	Diversity = 8.8235e+00
Generation 4940:	Best = 1.173600e+05	FES = 494100	Diversity = 9.6748e+00
Generation 4950:	Best = 1.173196e+05	FES = 495100	Diversity = 9.2619e+00
Generation 4960:	Best = 1.169901e+05	FES = 496100	Diversity = 9.2084e+00
Generation 4970:	Best = 1.168392e+05	FES = 497100	Diversity = 9.8085e+00
Generation 4980:	Best = 1.166952e+05	FES = 498100	Diversity = 1.0274e+01
Generation 4990:	Best = 1.165545e+05	FES = 499100	Diversity = 9.5428e+00
Generation 5000:	Best = 1.164399e+05	FES = 500100	Diversity = 8.9639e+00
Generation 5010:	Best = 1.163159e+05	FES = 501100	Diversity = 8.6092e+00
Generation 5020:	Best = 1.162381e+05	FES = 502100	Diversity = 8.9487e+00
Generation 5030:	Best = 1.160692e+05	FES = 503100	Diversity = 9.5269e+00
Generation 5040:	Best = 1.157764e+05	FES = 504100	Diversity = 1.0246e+01
Generation 5050:	Best = 1.156551e+05	FES = 505100	Diversity = 9.8807e+00
Generation 5060:	Best = 1.153742e+05	FES = 506100	Diversity = 9.4287e+00
Generation 5070:	Best = 1.152869e+05	FES = 507100	Diversity = 9.2453e+00
Generation 5080:	Best = 1.150411e+05	FES = 508100	Diversity = 9.1643e+00
Generation 5090:	Best = 1.150153e+05	FES = 509100	Diversity = 9.2819e+00
Generation 5100:	Best = 1.148848e+05	FES = 510100	Diversity = 9.1582e+00
Generation 5110:	Best = 1.148067e+05	FES = 511100	Diversity = 8.8340e+00
Generation 5120:	Best = 1.146683e+05	FES = 512100	Diversity = 8.4772e+00
Generation 5130:	Best = 1.145693e+05	FES = 513100	Diversity = 8.6135e+00
Generation 5140:	Best = 1.144010e+05	FES = 514100	Diversity = 9.1545e+00
Generation 5150:	Best = 1.142434e+05	FES = 515100	Diversity = 8.9342e+00
Generation 5160:	Best = 1.141716e+05	FES = 516100	Diversity = 8.6109e+00
Generation 5170:	Best = 1.139745e+05	FES = 517100	Diversity = 8.5634e+00
Generation 5180:	Best = 1.137351e+05	FES = 518100	Diversity = 8.6147e+00
Generation 5190:	Best = 1.137072e+05	FES = 519100	Diversity = 8.6339e+00
Generation 5200:	Best = 1.135512e+05	FES = 520100	Diversity = 8.4402e+00
Generation 5210:	Best = 1.134216e+05	FES = 521100	Diversity = 8.3028e+00
Generation 5220:	Best = 1.132797e+05	FES = 522100	Diversity = 8.5663e+00
Generation 5230:	Best = 1.131087e+05	FES = 523100	Diversity = 8.1683e+00
Generation 5240:	Best = 1.129362e+05	FES = 524100	Diversity = 8.8215e+00
Generation 5250:	Best = 1.126963e+05	FES = 525100	Diversity = 8.1979e+00
Generation 5260:	Best = 1.124604e+05	FES = 526100	Diversity = 8.5874e+00
Generation 5270:	Best = 1.124110e+05	FES = 527100	Diversity = 8.5401e+00
Generation 5280:	Best = 1.122366e+05	FES = 528100	Diversity = 8.0998e+00
Generation 5290:	Best = 1.120739e+05	FES = 529100	Diversity = 7.7880e+00
Generation 5300:	Best = 1.119278e+05	FES = 530100	Diversity = 8.3038e+00

Generation 5310:	Best = 1.116427e+05	FES = 531100	Diversity = 8.7275e+00
Generation 5320:	Best = 1.115879e+05	FES = 532100	Diversity = 8.5533e+00
Generation 5330:	Best = 1.114282e+05	FES = 533100	Diversity = 8.1577e+00
Generation 5340:	Best = 1.113978e+05	FES = 534100	Diversity = 8.1730e+00
Generation 5350:	Best = 1.111154e+05	FES = 535100	Diversity = 7.9926e+00
Generation 5360:	Best = 1.111154e+05	FES = 536100	Diversity = 8.0644e+00
Generation 5370:	Best = 1.109957e+05	FES = 537100	Diversity = 8.4845e+00
Generation 5380:	Best = 1.106876e+05	FES = 538100	Diversity = 8.1929e+00
Generation 5390:	Best = 1.106544e+05	FES = 539100	Diversity = 8.8896e+00
Generation 5400:	Best = 1.104650e+05	FES = 540100	Diversity = 8.1834e+00
Generation 5410:	Best = 1.103966e+05	FES = 541100	Diversity = 8.2842e+00
Generation 5420:	Best = 1.102246e+05	FES = 542100	Diversity = 8.8432e+00
Generation 5430:	Best = 1.100756e+05	FES = 543100	Diversity = 8.3585e+00
Generation 5440:	Best = 1.099512e+05	FES = 544100	Diversity = 8.1454e+00
Generation 5450:	Best = 1.096628e+05	FES = 545100	Diversity = 8.7790e+00
Generation 5460:	Best = 1.096628e+05	FES = 546100	Diversity = 9.2947e+00
Generation 5470:	Best = 1.094579e+05	FES = 547100	Diversity = 8.8435e+00
Generation 5480:	Best = 1.092588e+05	FES = 548100	Diversity = 8.7834e+00
Generation 5490:	Best = 1.091543e+05	FES = 549100	Diversity = 9.1752e+00
Generation 5500:	Best = 1.090255e+05	FES = 550100	Diversity = 9.2314e+00
Generation 5510:	Best = 1.089247e+05	FES = 551100	Diversity = 8.6432e+00
Generation 5520:	Best = 1.087501e+05	FES = 552100	Diversity = 8.8225e+00
Generation 5530:	Best = 1.086278e+05	FES = 553100	Diversity = 9.2529e+00
Generation 5540:	Best = 1.085690e+05	FES = 554100	Diversity = 8.8642e+00
Generation 5550:	Best = 1.084617e+05	FES = 555100	Diversity = 8.2485e+00
Generation 5560:	Best = 1.083548e+05	FES = 556100	Diversity = 8.3961e+00
Generation 5570:	Best = 1.081735e+05	FES = 557100	Diversity = 9.4200e+00
Generation 5580:	Best = 1.081605e+05	FES = 558100	Diversity = 8.9848e+00
Generation 5590:	Best = 1.080263e+05	FES = 559100	Diversity = 8.3803e+00
Generation 5600:	Best = 1.079021e+05	FES = 560100	Diversity = 8.2643e+00
Generation 5610:	Best = 1.076390e+05	FES = 561100	Diversity = 7.7472e+00
Generation 5620:	Best = 1.075337e+05	FES = 562100	Diversity = 7.8200e+00
Generation 5630:	Best = 1.074197e+05	FES = 563100	Diversity = 7.1855e+00
Generation 5640:	Best = 1.072432e+05	FES = 564100	Diversity = 7.4446e+00
Generation 5650:	Best = 1.070144e+05	FES = 565100	Diversity = 7.5672e+00
Generation 5660:	Best = 1.068203e+05	FES = 566100	Diversity = 8.2278e+00
Generation 5670:	Best = 1.067877e+05	FES = 567100	Diversity = 8.1539e+00
Generation 5680:	Best = 1.065850e+05	FES = 568100	Diversity = 7.6056e+00
Generation 5690:	Best = 1.064620e+05	FES = 569100	Diversity = 7.8418e+00
Generation 5700:	Best = 1.061286e+05	FES = 570100	Diversity = 9.1926e+00
Generation 5710:	Best = 1.058618e+05	FES = 571100	Diversity = 9.9956e+00
Generation 5720:	Best = 1.056446e+05	FES = 572100	Diversity = 9.0258e+00
Generation 5730:	Best = 1.052239e+05	FES = 573100	Diversity = 9.6121e+00
Generation 5740:	Best = 1.051320e+05	FES = 574100	Diversity = 1.0562e+01
Generation 5750:	Best = 1.049217e+05	FES = 575100	Diversity = 1.0804e+01
Generation 5760:	Best = 1.048048e+05	FES = 576100	Diversity = 1.0068e+01
Generation 5770:	Best = 1.045852e+05	FES = 577100	Diversity = 1.0078e+01
Generation 5780:	Best = 1.045524e+05	FES = 578100	Diversity = 1.0095e+01
Generation 5790:	Best = 1.042801e+05	FES = 579100	Diversity = 1.0922e+01
Generation 5800:	Best = 1.040631e+05	FES = 580100	Diversity = 1.0519e+01
Generation 5810:	Best = 1.038884e+05	FES = 581100	Diversity = 9.6446e+00
Generation 5820:	Best = 1.037369e+05	FES = 582100	Diversity = 9.6924e+00
Generation 5830:	Best = 1.035686e+05	FES = 583100	Diversity = 9.3077e+00
Generation 5840:	Best = 1.034188e+05	FES = 584100	Diversity = 9.5192e+00

Generation 5850:	Best = 1.033375e+05	FES = 585100	Diversity = 9.4387e+00
Generation 5860:	Best = 1.031387e+05	FES = 586100	Diversity = 9.5434e+00
Generation 5870:	Best = 1.029536e+05	FES = 587100	Diversity = 9.2844e+00
Generation 5880:	Best = 1.028993e+05	FES = 588100	Diversity = 8.9589e+00
Generation 5890:	Best = 1.026870e+05	FES = 589100	Diversity = 9.4983e+00
Generation 5900:	Best = 1.024263e+05	FES = 590100	Diversity = 8.9858e+00
Generation 5910:	Best = 1.023275e+05	FES = 591100	Diversity = 9.2356e+00
Generation 5920:	Best = 1.021986e+05	FES = 592100	Diversity = 9.2991e+00
Generation 5930:	Best = 1.020204e+05	FES = 593100	Diversity = 8.5216e+00
Generation 5940:	Best = 1.018521e+05	FES = 594100	Diversity = 8.0276e+00
Generation 5950:	Best = 1.017402e+05	FES = 595100	Diversity = 9.2316e+00
Generation 5960:	Best = 1.014860e+05	FES = 596100	Diversity = 9.9614e+00
Generation 5970:	Best = 1.013588e+05	FES = 597100	Diversity = 9.3316e+00
Generation 5980:	Best = 1.012181e+05	FES = 598100	Diversity = 9.4031e+00
Generation 5990:	Best = 1.010429e+05	FES = 599100	Diversity = 8.7639e+00
Generation 6000:	Best = 1.008820e+05	FES = 600100	Diversity = 8.2527e+00
Generation 6010:	Best = 1.008100e+05	FES = 601100	Diversity = 8.1409e+00
Generation 6020:	Best = 1.007246e+05	FES = 602100	Diversity = 8.0957e+00
Generation 6030:	Best = 1.003649e+05	FES = 603100	Diversity = 8.8489e+00
Generation 6040:	Best = 1.001660e+05	FES = 604100	Diversity = 7.9377e+00
Generation 6050:	Best = 9.983883e+04	FES = 605100	Diversity = 7.5146e+00
Generation 6060:	Best = 9.956317e+04	FES = 606100	Diversity = 7.9901e+00
Generation 6070:	Best = 9.923911e+04	FES = 607100	Diversity = 8.1937e+00
Generation 6080:	Best = 9.904047e+04	FES = 608100	Diversity = 8.5884e+00
Generation 6090:	Best = 9.887484e+04	FES = 609100	Diversity = 9.2443e+00
Generation 6100:	Best = 9.877595e+04	FES = 610100	Diversity = 9.2182e+00
Generation 6110:	Best = 9.860429e+04	FES = 611100	Diversity = 9.3598e+00
Generation 6120:	Best = 9.844659e+04	FES = 612100	Diversity = 8.2388e+00
Generation 6130:	Best = 9.839575e+04	FES = 613100	Diversity = 8.0540e+00
Generation 6140:	Best = 9.830066e+04	FES = 614100	Diversity = 7.9137e+00
Generation 6150:	Best = 9.816615e+04	FES = 615100	Diversity = 8.1715e+00
Generation 6160:	Best = 9.800429e+04	FES = 616100	Diversity = 8.8001e+00
Generation 6170:	Best = 9.791880e+04	FES = 617100	Diversity = 8.6406e+00
Generation 6180:	Best = 9.776849e+04	FES = 618100	Diversity = 8.1982e+00
Generation 6190:	Best = 9.740322e+04	FES = 619100	Diversity = 9.0674e+00
Generation 6200:	Best = 9.739817e+04	FES = 620100	Diversity = 8.3828e+00
Generation 6210:	Best = 9.703720e+04	FES = 621100	Diversity = 8.2492e+00
Generation 6220:	Best = 9.681095e+04	FES = 622100	Diversity = 8.4156e+00
Generation 6230:	Best = 9.668854e+04	FES = 623100	Diversity = 9.3894e+00
Generation 6240:	Best = 9.646953e+04	FES = 624100	Diversity = 9.6996e+00
Generation 6250:	Best = 9.628694e+04	FES = 625100	Diversity = 9.2756e+00
Generation 6260:	Best = 9.616876e+04	FES = 626100	Diversity = 8.7992e+00
Generation 6270:	Best = 9.595411e+04	FES = 627100	Diversity = 9.0288e+00
Generation 6280:	Best = 9.576324e+04	FES = 628100	Diversity = 8.8068e+00
Generation 6290:	Best = 9.567083e+04	FES = 629100	Diversity = 9.1238e+00
Generation 6300:	Best = 9.534452e+04	FES = 630100	Diversity = 9.1767e+00
Generation 6310:	Best = 9.529119e+04	FES = 631100	Diversity = 8.6444e+00
Generation 6320:	Best = 9.522095e+04	FES = 632100	Diversity = 7.8934e+00
Generation 6330:	Best = 9.515772e+04	FES = 633100	Diversity = 7.6637e+00
Generation 6340:	Best = 9.509774e+04	FES = 634100	Diversity = 7.2437e+00
Generation 6350:	Best = 9.501992e+04	FES = 635100	Diversity = 7.1168e+00
Generation 6360:	Best = 9.482803e+04	FES = 636100	Diversity = 7.0462e+00
Generation 6370:	Best = 9.469594e+04	FES = 637100	Diversity = 7.0497e+00
Generation 6380:	Best = 9.463147e+04	FES = 638100	Diversity = 6.6984e+00

Generation 6390:	Best = 9.449658e+04	FES = 639100	Diversity = 5.9635e+00
Generation 6400:	Best = 9.441012e+04	FES = 640100	Diversity = 6.0869e+00
Generation 6410:	Best = 9.428348e+04	FES = 641100	Diversity = 6.2800e+00
Generation 6420:	Best = 9.412498e+04	FES = 642100	Diversity = 6.2578e+00
Generation 6430:	Best = 9.396542e+04	FES = 643100	Diversity = 6.4456e+00
Generation 6440:	Best = 9.377406e+04	FES = 644100	Diversity = 6.3861e+00
Generation 6450:	Best = 9.371088e+04	FES = 645100	Diversity = 6.7651e+00
Generation 6460:	Best = 9.353161e+04	FES = 646100	Diversity = 7.2667e+00
Generation 6470:	Best = 9.335699e+04	FES = 647100	Diversity = 7.3492e+00
Generation 6480:	Best = 9.322855e+04	FES = 648100	Diversity = 7.5118e+00
Generation 6490:	Best = 9.303601e+04	FES = 649100	Diversity = 7.0740e+00
Generation 6500:	Best = 9.279115e+04	FES = 650100	Diversity = 7.3096e+00
Generation 6510:	Best = 9.275329e+04	FES = 651100	Diversity = 7.6857e+00
Generation 6520:	Best = 9.259081e+04	FES = 652100	Diversity = 7.8447e+00
Generation 6530:	Best = 9.237950e+04	FES = 653100	Diversity = 8.3809e+00
Generation 6540:	Best = 9.221344e+04	FES = 654100	Diversity = 8.6651e+00
Generation 6550:	Best = 9.209979e+04	FES = 655100	Diversity = 8.9850e+00
Generation 6560:	Best = 9.196691e+04	FES = 656100	Diversity = 8.2227e+00
Generation 6570:	Best = 9.183269e+04	FES = 657100	Diversity = 8.4637e+00
Generation 6580:	Best = 9.165781e+04	FES = 658100	Diversity = 8.5513e+00
Generation 6590:	Best = 9.135385e+04	FES = 659100	Diversity = 8.2698e+00
Generation 6600:	Best = 9.118563e+04	FES = 660100	Diversity = 8.4637e+00
Generation 6610:	Best = 9.099104e+04	FES = 661100	Diversity = 8.2768e+00
Generation 6620:	Best = 9.083949e+04	FES = 662100	Diversity = 8.4044e+00
Generation 6630:	Best = 9.072453e+04	FES = 663100	Diversity = 7.7353e+00
Generation 6640:	Best = 9.055565e+04	FES = 664100	Diversity = 8.4314e+00
Generation 6650:	Best = 9.048796e+04	FES = 665100	Diversity = 8.6415e+00
Generation 6660:	Best = 9.041149e+04	FES = 666100	Diversity = 8.9013e+00
Generation 6670:	Best = 9.028179e+04	FES = 667100	Diversity = 8.6356e+00
Generation 6680:	Best = 9.018981e+04	FES = 668100	Diversity = 8.0926e+00
Generation 6690:	Best = 9.011149e+04	FES = 669100	Diversity = 8.4910e+00
Generation 6700:	Best = 9.004776e+04	FES = 670100	Diversity = 7.6039e+00
Generation 6710:	Best = 8.986948e+04	FES = 671100	Diversity = 8.2054e+00
Generation 6720:	Best = 8.985704e+04	FES = 672100	Diversity = 8.4324e+00
Generation 6730:	Best = 8.976786e+04	FES = 673100	Diversity = 8.3440e+00
Generation 6740:	Best = 8.964926e+04	FES = 674100	Diversity = 8.1903e+00
Generation 6750:	Best = 8.961558e+04	FES = 675100	Diversity = 8.0975e+00
Generation 6760:	Best = 8.949237e+04	FES = 676100	Diversity = 8.5667e+00
Generation 6770:	Best = 8.945907e+04	FES = 677100	Diversity = 8.5506e+00
Generation 6780:	Best = 8.937718e+04	FES = 678100	Diversity = 8.9029e+00
Generation 6790:	Best = 8.924064e+04	FES = 679100	Diversity = 9.3257e+00
Generation 6800:	Best = 8.910054e+04	FES = 680100	Diversity = 9.2630e+00
Generation 6810:	Best = 8.906148e+04	FES = 681100	Diversity = 9.0854e+00
Generation 6820:	Best = 8.899503e+04	FES = 682100	Diversity = 9.0670e+00
Generation 6830:	Best = 8.893725e+04	FES = 683100	Diversity = 8.1785e+00
Generation 6840:	Best = 8.884032e+04	FES = 684100	Diversity = 8.5584e+00
Generation 6850:	Best = 8.877217e+04	FES = 685100	Diversity = 8.9847e+00
Generation 6860:	Best = 8.865339e+04	FES = 686100	Diversity = 8.9163e+00
Generation 6870:	Best = 8.863278e+04	FES = 687100	Diversity = 9.0487e+00
Generation 6880:	Best = 8.849007e+04	FES = 688100	Diversity = 8.4182e+00
Generation 6890:	Best = 8.834940e+04	FES = 689100	Diversity = 8.3237e+00
Generation 6900:	Best = 8.821598e+04	FES = 690100	Diversity = 7.8532e+00
Generation 6910:	Best = 8.817441e+04	FES = 691100	Diversity = 8.4098e+00
Generation 6920:	Best = 8.802353e+04	FES = 692100	Diversity = 8.6713e+00

Generation 6930:	Best = 8.778283e+04	FES = 693100	Diversity = 8.5394e+00
Generation 6940:	Best = 8.776874e+04	FES = 694100	Diversity = 8.8299e+00
Generation 6950:	Best = 8.752054e+04	FES = 695100	Diversity = 8.5194e+00
Generation 6960:	Best = 8.741067e+04	FES = 696100	Diversity = 8.7563e+00
Generation 6970:	Best = 8.736539e+04	FES = 697100	Diversity = 9.3947e+00
Generation 6980:	Best = 8.720344e+04	FES = 698100	Diversity = 9.0306e+00
Generation 6990:	Best = 8.704644e+04	FES = 699100	Diversity = 9.5894e+00
Generation 7000:	Best = 8.688845e+04	FES = 700100	Diversity = 1.0106e+01
Generation 7010:	Best = 8.674428e+04	FES = 701100	Diversity = 9.8381e+00
Generation 7020:	Best = 8.657150e+04	FES = 702100	Diversity = 1.0005e+01
Generation 7030:	Best = 8.636551e+04	FES = 703100	Diversity = 9.8858e+00
Generation 7040:	Best = 8.636150e+04	FES = 704100	Diversity = 1.0042e+01
Generation 7050:	Best = 8.613244e+04	FES = 705100	Diversity = 1.0164e+01
Generation 7060:	Best = 8.595811e+04	FES = 706100	Diversity = 9.4188e+00
Generation 7070:	Best = 8.593566e+04	FES = 707100	Diversity = 9.0524e+00
Generation 7080:	Best = 8.585320e+04	FES = 708100	Diversity = 8.9917e+00
Generation 7090:	Best = 8.575545e+04	FES = 709100	Diversity = 9.2550e+00
Generation 7100:	Best = 8.556480e+04	FES = 710100	Diversity = 9.0893e+00
Generation 7110:	Best = 8.542790e+04	FES = 711100	Diversity = 8.7776e+00
Generation 7120:	Best = 8.528743e+04	FES = 712100	Diversity = 8.4472e+00
Generation 7130:	Best = 8.523834e+04	FES = 713100	Diversity = 8.3019e+00
Generation 7140:	Best = 8.509285e+04	FES = 714100	Diversity = 8.2505e+00
Generation 7150:	Best = 8.495628e+04	FES = 715100	Diversity = 8.0187e+00
Generation 7160:	Best = 8.488980e+04	FES = 716100	Diversity = 7.7546e+00
Generation 7170:	Best = 8.468976e+04	FES = 717100	Diversity = 8.9034e+00
Generation 7180:	Best = 8.453395e+04	FES = 718100	Diversity = 1.0834e+01
Generation 7190:	Best = 8.441522e+04	FES = 719100	Diversity = 1.0834e+01
Generation 7200:	Best = 8.423616e+04	FES = 720100	Diversity = 1.1743e+01
Generation 7210:	Best = 8.401823e+04	FES = 721100	Diversity = 1.1928e+01
Generation 7220:	Best = 8.372066e+04	FES = 722100	Diversity = 1.3137e+01
Generation 7230:	Best = 8.326980e+04	FES = 723100	Diversity = 1.2855e+01
Generation 7240:	Best = 8.291503e+04	FES = 724100	Diversity = 1.2606e+01
Generation 7250:	Best = 8.255097e+04	FES = 725100	Diversity = 1.3522e+01
Generation 7260:	Best = 8.220619e+04	FES = 726100	Diversity = 1.3236e+01
Generation 7270:	Best = 8.176184e+04	FES = 727100	Diversity = 1.3074e+01
Generation 7280:	Best = 8.152709e+04	FES = 728100	Diversity = 1.3391e+01
Generation 7290:	Best = 8.125031e+04	FES = 729100	Diversity = 1.3164e+01
Generation 7300:	Best = 8.076121e+04	FES = 730100	Diversity = 1.2501e+01
Generation 7310:	Best = 8.047868e+04	FES = 731100	Diversity = 1.1206e+01
Generation 7320:	Best = 8.000677e+04	FES = 732100	Diversity = 1.2493e+01
Generation 7330:	Best = 7.991543e+04	FES = 733100	Diversity = 1.2790e+01
Generation 7340:	Best = 7.963303e+04	FES = 734100	Diversity = 1.1911e+01
Generation 7350:	Best = 7.920390e+04	FES = 735100	Diversity = 1.2166e+01
Generation 7360:	Best = 7.901166e+04	FES = 736100	Diversity = 1.1464e+01
Generation 7370:	Best = 7.871131e+04	FES = 737100	Diversity = 1.1051e+01
Generation 7380:	Best = 7.847016e+04	FES = 738100	Diversity = 1.1728e+01
Generation 7390:	Best = 7.838861e+04	FES = 739100	Diversity = 1.1405e+01
Generation 7400:	Best = 7.811598e+04	FES = 740100	Diversity = 1.2677e+01
Generation 7410:	Best = 7.796920e+04	FES = 741100	Diversity = 1.2595e+01
Generation 7420:	Best = 7.769228e+04	FES = 742100	Diversity = 1.2054e+01
Generation 7430:	Best = 7.769228e+04	FES = 743100	Diversity = 1.1586e+01
Generation 7440:	Best = 7.747135e+04	FES = 744100	Diversity = 1.1762e+01
Generation 7450:	Best = 7.727941e+04	FES = 745100	Diversity = 1.2007e+01
Generation 7460:	Best = 7.708763e+04	FES = 746100	Diversity = 1.0818e+01

Generation 7470:	Best = 7.693107e+04	FES = 747100	Diversity = 1.0413e+01
Generation 7480:	Best = 7.669926e+04	FES = 748100	Diversity = 9.9806e+00
Generation 7490:	Best = 7.649917e+04	FES = 749100	Diversity = 1.1708e+01
Generation 7500:	Best = 7.627402e+04	FES = 750100	Diversity = 1.1408e+01
Generation 7510:	Best = 7.614874e+04	FES = 751100	Diversity = 1.0888e+01
Generation 7520:	Best = 7.601371e+04	FES = 752100	Diversity = 1.0109e+01
Generation 7530:	Best = 7.588030e+04	FES = 753100	Diversity = 1.0119e+01
Generation 7540:	Best = 7.578316e+04	FES = 754100	Diversity = 9.3465e+00
Generation 7550:	Best = 7.572793e+04	FES = 755100	Diversity = 9.5188e+00
Generation 7560:	Best = 7.550162e+04	FES = 756100	Diversity = 9.8770e+00
Generation 7570:	Best = 7.537467e+04	FES = 757100	Diversity = 9.3841e+00
Generation 7580:	Best = 7.529973e+04	FES = 758100	Diversity = 9.4980e+00
Generation 7590:	Best = 7.497658e+04	FES = 759100	Diversity = 1.0527e+01
Generation 7600:	Best = 7.488318e+04	FES = 760100	Diversity = 1.0952e+01
Generation 7610:	Best = 7.467142e+04	FES = 761100	Diversity = 1.0888e+01
Generation 7620:	Best = 7.458873e+04	FES = 762100	Diversity = 1.0861e+01
Generation 7630:	Best = 7.444877e+04	FES = 763100	Diversity = 1.0382e+01
Generation 7640:	Best = 7.436805e+04	FES = 764100	Diversity = 9.9107e+00
Generation 7650:	Best = 7.430744e+04	FES = 765100	Diversity = 1.0186e+01
Generation 7660:	Best = 7.424071e+04	FES = 766100	Diversity = 1.0041e+01
Generation 7670:	Best = 7.413663e+04	FES = 767100	Diversity = 9.4810e+00
Generation 7680:	Best = 7.408143e+04	FES = 768100	Diversity = 9.5283e+00
Generation 7690:	Best = 7.389684e+04	FES = 769100	Diversity = 1.0513e+01
Generation 7700:	Best = 7.377346e+04	FES = 770100	Diversity = 1.0586e+01
Generation 7710:	Best = 7.362138e+04	FES = 771100	Diversity = 1.2201e+01
Generation 7720:	Best = 7.344118e+04	FES = 772100	Diversity = 1.1951e+01
Generation 7730:	Best = 7.327123e+04	FES = 773100	Diversity = 1.1373e+01
Generation 7740:	Best = 7.310392e+04	FES = 774100	Diversity = 1.1282e+01
Generation 7750:	Best = 7.295279e+04	FES = 775100	Diversity = 1.0591e+01
Generation 7760:	Best = 7.278187e+04	FES = 776100	Diversity = 1.0222e+01
Generation 7770:	Best = 7.258670e+04	FES = 777100	Diversity = 1.0146e+01
Generation 7780:	Best = 7.236874e+04	FES = 778100	Diversity = 9.7876e+00
Generation 7790:	Best = 7.216316e+04	FES = 779100	Diversity = 9.8715e+00
Generation 7800:	Best = 7.183988e+04	FES = 780100	Diversity = 1.1746e+01
Generation 7810:	Best = 7.162459e+04	FES = 781100	Diversity = 1.2973e+01
Generation 7820:	Best = 7.122691e+04	FES = 782100	Diversity = 1.2294e+01
Generation 7830:	Best = 7.112859e+04	FES = 783100	Diversity = 1.1916e+01
Generation 7840:	Best = 7.095864e+04	FES = 784100	Diversity = 1.2109e+01
Generation 7850:	Best = 7.081576e+04	FES = 785100	Diversity = 1.1537e+01
Generation 7860:	Best = 7.068961e+04	FES = 786100	Diversity = 1.0987e+01
Generation 7870:	Best = 7.056765e+04	FES = 787100	Diversity = 1.1197e+01
Generation 7880:	Best = 7.035185e+04	FES = 788100	Diversity = 1.0784e+01
Generation 7890:	Best = 7.033062e+04	FES = 789100	Diversity = 1.0729e+01
Generation 7900:	Best = 7.011000e+04	FES = 790100	Diversity = 1.0250e+01
Generation 7910:	Best = 6.992275e+04	FES = 791100	Diversity = 1.0714e+01
Generation 7920:	Best = 6.977199e+04	FES = 792100	Diversity = 1.1085e+01
Generation 7930:	Best = 6.960044e+04	FES = 793100	Diversity = 9.7665e+00
Generation 7940:	Best = 6.951595e+04	FES = 794100	Diversity = 8.9180e+00
Generation 7950:	Best = 6.938735e+04	FES = 795100	Diversity = 8.6367e+00
Generation 7960:	Best = 6.919342e+04	FES = 796100	Diversity = 1.0133e+01
Generation 7970:	Best = 6.900163e+04	FES = 797100	Diversity = 1.0935e+01
Generation 7980:	Best = 6.874927e+04	FES = 798100	Diversity = 1.1206e+01
Generation 7990:	Best = 6.845784e+04	FES = 799100	Diversity = 1.1609e+01
Generation 8000:	Best = 6.830353e+04	FES = 800100	Diversity = 1.2389e+01

Generation 8010:	Best = 6.809927e+04	FES = 801100	Diversity = 1.1858e+01
Generation 8020:	Best = 6.785726e+04	FES = 802100	Diversity = 1.1713e+01
Generation 8030:	Best = 6.777772e+04	FES = 803100	Diversity = 1.1757e+01
Generation 8040:	Best = 6.765458e+04	FES = 804100	Diversity = 1.0895e+01
Generation 8050:	Best = 6.752471e+04	FES = 805100	Diversity = 1.0459e+01
Generation 8060:	Best = 6.733849e+04	FES = 806100	Diversity = 1.0586e+01
Generation 8070:	Best = 6.716528e+04	FES = 807100	Diversity = 1.0968e+01
Generation 8080:	Best = 6.706309e+04	FES = 808100	Diversity = 1.1471e+01
Generation 8090:	Best = 6.688546e+04	FES = 809100	Diversity = 1.1084e+01
Generation 8100:	Best = 6.682988e+04	FES = 810100	Diversity = 1.1402e+01
Generation 8110:	Best = 6.672717e+04	FES = 811100	Diversity = 1.1543e+01
Generation 8120:	Best = 6.663216e+04	FES = 812100	Diversity = 1.0912e+01
Generation 8130:	Best = 6.652194e+04	FES = 813100	Diversity = 1.0511e+01
Generation 8140:	Best = 6.649447e+04	FES = 814100	Diversity = 1.0197e+01
Generation 8150:	Best = 6.648731e+04	FES = 815100	Diversity = 9.7319e+00
Generation 8160:	Best = 6.643822e+04	FES = 816100	Diversity = 9.4523e+00
Generation 8170:	Best = 6.637864e+04	FES = 817100	Diversity = 1.0242e+01
Generation 8180:	Best = 6.631211e+04	FES = 818100	Diversity = 1.0345e+01
Generation 8190:	Best = 6.627569e+04	FES = 819100	Diversity = 9.9885e+00
Generation 8200:	Best = 6.621424e+04	FES = 820100	Diversity = 9.7071e+00
Generation 8210:	Best = 6.616148e+04	FES = 821100	Diversity = 9.1920e+00
Generation 8220:	Best = 6.609303e+04	FES = 822100	Diversity = 9.2546e+00
Generation 8230:	Best = 6.601596e+04	FES = 823100	Diversity = 9.3669e+00
Generation 8240:	Best = 6.593339e+04	FES = 824100	Diversity = 9.5433e+00
Generation 8250:	Best = 6.582476e+04	FES = 825100	Diversity = 9.4694e+00
Generation 8260:	Best = 6.579856e+04	FES = 826100	Diversity = 9.9307e+00
Generation 8270:	Best = 6.571425e+04	FES = 827100	Diversity = 9.3238e+00
Generation 8280:	Best = 6.557081e+04	FES = 828100	Diversity = 9.4529e+00
Generation 8290:	Best = 6.542560e+04	FES = 829100	Diversity = 9.0278e+00
Generation 8300:	Best = 6.531632e+04	FES = 830100	Diversity = 9.3642e+00
Generation 8310:	Best = 6.523005e+04	FES = 831100	Diversity = 9.2026e+00
Generation 8320:	Best = 6.516589e+04	FES = 832100	Diversity = 9.3025e+00
Generation 8330:	Best = 6.505721e+04	FES = 833100	Diversity = 1.0544e+01
Generation 8340:	Best = 6.489503e+04	FES = 834100	Diversity = 9.8341e+00
Generation 8350:	Best = 6.477358e+04	FES = 835100	Diversity = 9.3895e+00
Generation 8360:	Best = 6.455512e+04	FES = 836100	Diversity = 1.0701e+01
Generation 8370:	Best = 6.445789e+04	FES = 837100	Diversity = 1.1157e+01
Generation 8380:	Best = 6.433845e+04	FES = 838100	Diversity = 1.0798e+01
Generation 8390:	Best = 6.412857e+04	FES = 839100	Diversity = 1.0705e+01
Generation 8400:	Best = 6.411713e+04	FES = 840100	Diversity = 1.0754e+01
Generation 8410:	Best = 6.400216e+04	FES = 841100	Diversity = 9.7958e+00
Generation 8420:	Best = 6.389312e+04	FES = 842100	Diversity = 9.1759e+00
Generation 8430:	Best = 6.364261e+04	FES = 843100	Diversity = 8.3994e+00
Generation 8440:	Best = 6.362961e+04	FES = 844100	Diversity = 8.5632e+00
Generation 8450:	Best = 6.358348e+04	FES = 845100	Diversity = 8.6663e+00
Generation 8460:	Best = 6.344525e+04	FES = 846100	Diversity = 9.2253e+00
Generation 8470:	Best = 6.330526e+04	FES = 847100	Diversity = 9.0917e+00
Generation 8480:	Best = 6.321972e+04	FES = 848100	Diversity = 9.5080e+00
Generation 8490:	Best = 6.311031e+04	FES = 849100	Diversity = 1.0090e+01
Generation 8500:	Best = 6.297722e+04	FES = 850100	Diversity = 1.0741e+01
Generation 8510:	Best = 6.286331e+04	FES = 851100	Diversity = 9.8214e+00
Generation 8520:	Best = 6.273747e+04	FES = 852100	Diversity = 1.0153e+01
Generation 8530:	Best = 6.264120e+04	FES = 853100	Diversity = 9.5712e+00
Generation 8540:	Best = 6.261729e+04	FES = 854100	Diversity = 9.1771e+00

Generation 8550:	Best = 6.253383e+04	FES = 855100	Diversity = 8.7605e+00
Generation 8560:	Best = 6.240052e+04	FES = 856100	Diversity = 9.4232e+00
Generation 8570:	Best = 6.226232e+04	FES = 857100	Diversity = 9.8183e+00
Generation 8580:	Best = 6.225820e+04	FES = 858100	Diversity = 9.3777e+00
Generation 8590:	Best = 6.220361e+04	FES = 859100	Diversity = 9.7350e+00
Generation 8600:	Best = 6.215391e+04	FES = 860100	Diversity = 9.0605e+00
Generation 8610:	Best = 6.203205e+04	FES = 861100	Diversity = 8.3936e+00
Generation 8620:	Best = 6.186156e+04	FES = 862100	Diversity = 8.0596e+00
Generation 8630:	Best = 6.184274e+04	FES = 863100	Diversity = 8.3179e+00
Generation 8640:	Best = 6.171020e+04	FES = 864100	Diversity = 8.1473e+00
Generation 8650:	Best = 6.170129e+04	FES = 865100	Diversity = 8.5111e+00
Generation 8660:	Best = 6.167659e+04	FES = 866100	Diversity = 8.8827e+00
Generation 8670:	Best = 6.159265e+04	FES = 867100	Diversity = 9.8694e+00
Generation 8680:	Best = 6.151509e+04	FES = 868100	Diversity = 1.0236e+01
Generation 8690:	Best = 6.143855e+04	FES = 869100	Diversity = 1.0405e+01
Generation 8700:	Best = 6.137456e+04	FES = 870100	Diversity = 1.0163e+01
Generation 8710:	Best = 6.132166e+04	FES = 871100	Diversity = 1.0145e+01
Generation 8720:	Best = 6.123021e+04	FES = 872100	Diversity = 9.5339e+00
Generation 8730:	Best = 6.116873e+04	FES = 873100	Diversity = 9.6810e+00
Generation 8740:	Best = 6.113498e+04	FES = 874100	Diversity = 9.2578e+00
Generation 8750:	Best = 6.107464e+04	FES = 875100	Diversity = 9.1539e+00
Generation 8760:	Best = 6.098558e+04	FES = 876100	Diversity = 9.6904e+00
Generation 8770:	Best = 6.083299e+04	FES = 877100	Diversity = 1.0150e+01
Generation 8780:	Best = 6.080908e+04	FES = 878100	Diversity = 9.7603e+00
Generation 8790:	Best = 6.066733e+04	FES = 879100	Diversity = 9.2892e+00
Generation 8800:	Best = 6.056277e+04	FES = 880100	Diversity = 1.0113e+01
Generation 8810:	Best = 6.041768e+04	FES = 881100	Diversity = 9.9617e+00
Generation 8820:	Best = 6.027963e+04	FES = 882100	Diversity = 1.0277e+01
Generation 8830:	Best = 6.027540e+04	FES = 883100	Diversity = 1.1096e+01
Generation 8840:	Best = 6.007620e+04	FES = 884100	Diversity = 1.0875e+01
Generation 8850:	Best = 5.988313e+04	FES = 885100	Diversity = 1.1477e+01
Generation 8860:	Best = 5.984295e+04	FES = 886100	Diversity = 1.0955e+01
Generation 8870:	Best = 5.969765e+04	FES = 887100	Diversity = 1.0948e+01
Generation 8880:	Best = 5.964693e+04	FES = 888100	Diversity = 1.0486e+01
Generation 8890:	Best = 5.940858e+04	FES = 889100	Diversity = 1.1609e+01
Generation 8900:	Best = 5.915909e+04	FES = 890100	Diversity = 1.1959e+01
Generation 8910:	Best = 5.912634e+04	FES = 891100	Diversity = 1.2423e+01
Generation 8920:	Best = 5.903646e+04	FES = 892100	Diversity = 1.2974e+01
Generation 8930:	Best = 5.887954e+04	FES = 893100	Diversity = 1.2222e+01
Generation 8940:	Best = 5.873633e+04	FES = 894100	Diversity = 1.2263e+01
Generation 8950:	Best = 5.856011e+04	FES = 895100	Diversity = 1.2359e+01
Generation 8960:	Best = 5.852390e+04	FES = 896100	Diversity = 1.2079e+01
Generation 8970:	Best = 5.847035e+04	FES = 897100	Diversity = 1.1955e+01
Generation 8980:	Best = 5.838248e+04	FES = 898100	Diversity = 1.0327e+01
Generation 8990:	Best = 5.824844e+04	FES = 899100	Diversity = 1.0018e+01
Generation 9000:	Best = 5.815801e+04	FES = 900100	Diversity = 1.0082e+01
Generation 9010:	Best = 5.802233e+04	FES = 901100	Diversity = 1.0287e+01
Generation 9020:	Best = 5.788169e+04	FES = 902100	Diversity = 1.0817e+01
Generation 9030:	Best = 5.758179e+04	FES = 903100	Diversity = 1.1510e+01
Generation 9040:	Best = 5.737569e+04	FES = 904100	Diversity = 1.1678e+01
Generation 9050:	Best = 5.726662e+04	FES = 905100	Diversity = 1.2178e+01
Generation 9060:	Best = 5.702050e+04	FES = 906100	Diversity = 1.2581e+01
Generation 9070:	Best = 5.686025e+04	FES = 907100	Diversity = 1.2423e+01
Generation 9080:	Best = 5.671053e+04	FES = 908100	Diversity = 1.0940e+01

Generation 9090:	Best = 5.652874e+04	FES = 909100	Diversity = 1.0367e+01
Generation 9100:	Best = 5.636048e+04	FES = 910100	Diversity = 1.0538e+01
Generation 9110:	Best = 5.625904e+04	FES = 911100	Diversity = 9.7277e+00
Generation 9120:	Best = 5.597711e+04	FES = 912100	Diversity = 1.0394e+01
Generation 9130:	Best = 5.580612e+04	FES = 913100	Diversity = 1.0033e+01
Generation 9140:	Best = 5.572176e+04	FES = 914100	Diversity = 1.0225e+01
Generation 9150:	Best = 5.552673e+04	FES = 915100	Diversity = 9.7016e+00
Generation 9160:	Best = 5.542930e+04	FES = 916100	Diversity = 9.3064e+00
Generation 9170:	Best = 5.523406e+04	FES = 917100	Diversity = 1.0076e+01
Generation 9180:	Best = 5.510681e+04	FES = 918100	Diversity = 9.3186e+00
Generation 9190:	Best = 5.496389e+04	FES = 919100	Diversity = 8.1323e+00
Generation 9200:	Best = 5.488215e+04	FES = 920100	Diversity = 8.5085e+00
Generation 9210:	Best = 5.477842e+04	FES = 921100	Diversity = 8.9820e+00
Generation 9220:	Best = 5.465589e+04	FES = 922100	Diversity = 8.2382e+00
Generation 9230:	Best = 5.463226e+04	FES = 923100	Diversity = 7.9277e+00
Generation 9240:	Best = 5.456859e+04	FES = 924100	Diversity = 7.7307e+00
Generation 9250:	Best = 5.449493e+04	FES = 925100	Diversity = 8.1567e+00
Generation 9260:	Best = 5.444386e+04	FES = 926100	Diversity = 8.7300e+00
Generation 9270:	Best = 5.441964e+04	FES = 927100	Diversity = 8.6389e+00
Generation 9280:	Best = 5.439711e+04	FES = 928100	Diversity = 8.5967e+00
Generation 9290:	Best = 5.427058e+04	FES = 929100	Diversity = 8.1142e+00
Generation 9300:	Best = 5.418668e+04	FES = 930100	Diversity = 8.1008e+00
Generation 9310:	Best = 5.408138e+04	FES = 931100	Diversity = 8.3156e+00
Generation 9320:	Best = 5.398427e+04	FES = 932100	Diversity = 7.7390e+00
Generation 9330:	Best = 5.394378e+04	FES = 933100	Diversity = 7.6797e+00
Generation 9340:	Best = 5.382708e+04	FES = 934100	Diversity = 8.0665e+00
Generation 9350:	Best = 5.380046e+04	FES = 935100	Diversity = 8.3895e+00
Generation 9360:	Best = 5.374784e+04	FES = 936100	Diversity = 8.4637e+00
Generation 9370:	Best = 5.363683e+04	FES = 937100	Diversity = 8.0740e+00
Generation 9380:	Best = 5.360827e+04	FES = 938100	Diversity = 7.3757e+00
Generation 9390:	Best = 5.358104e+04	FES = 939100	Diversity = 7.3304e+00
Generation 9400:	Best = 5.354537e+04	FES = 940100	Diversity = 6.9211e+00
Generation 9410:	Best = 5.351379e+04	FES = 941100	Diversity = 6.8233e+00
Generation 9420:	Best = 5.344049e+04	FES = 942100	Diversity = 6.3736e+00
Generation 9430:	Best = 5.338537e+04	FES = 943100	Diversity = 5.8900e+00
Generation 9440:	Best = 5.338537e+04	FES = 944100	Diversity = 5.9032e+00
Generation 9450:	Best = 5.338537e+04	FES = 945100	Diversity = 6.0010e+00
Generation 9460:	Best = 5.326762e+04	FES = 946100	Diversity = 5.8228e+00
Generation 9470:	Best = 5.326577e+04	FES = 947100	Diversity = 6.1035e+00
Generation 9480:	Best = 5.319503e+04	FES = 948100	Diversity = 6.6532e+00
Generation 9490:	Best = 5.312205e+04	FES = 949100	Diversity = 7.4285e+00
Generation 9500:	Best = 5.301099e+04	FES = 950100	Diversity = 6.7442e+00
Generation 9510:	Best = 5.288500e+04	FES = 951100	Diversity = 7.3048e+00
Generation 9520:	Best = 5.280420e+04	FES = 952100	Diversity = 7.8556e+00
Generation 9530:	Best = 5.262477e+04	FES = 953100	Diversity = 7.6240e+00
Generation 9540:	Best = 5.250713e+04	FES = 954100	Diversity = 8.0783e+00
Generation 9550:	Best = 5.231277e+04	FES = 955100	Diversity = 9.1632e+00
Generation 9560:	Best = 5.217391e+04	FES = 956100	Diversity = 8.8312e+00
Generation 9570:	Best = 5.196166e+04	FES = 957100	Diversity = 8.4757e+00
Generation 9580:	Best = 5.175744e+04	FES = 958100	Diversity = 8.8494e+00
Generation 9590:	Best = 5.151876e+04	FES = 959100	Diversity = 8.7788e+00
Generation 9600:	Best = 5.140425e+04	FES = 960100	Diversity = 9.0131e+00
Generation 9610:	Best = 5.129237e+04	FES = 961100	Diversity = 8.3660e+00
Generation 9620:	Best = 5.112233e+04	FES = 962100	Diversity = 7.5854e+00

Generation 9630:	Best = 5.099336e+04	FES = 963100	Diversity = 8.1289e+00
Generation 9640:	Best = 5.099336e+04	FES = 964100	Diversity = 8.2261e+00
Generation 9650:	Best = 5.087900e+04	FES = 965100	Diversity = 8.5164e+00
Generation 9660:	Best = 5.074652e+04	FES = 966100	Diversity = 8.8888e+00
Generation 9670:	Best = 5.066667e+04	FES = 967100	Diversity = 8.9445e+00
Generation 9680:	Best = 5.047516e+04	FES = 968100	Diversity = 7.9805e+00
Generation 9690:	Best = 5.031381e+04	FES = 969100	Diversity = 7.6702e+00
Generation 9700:	Best = 5.005578e+04	FES = 970100	Diversity = 8.1028e+00
Generation 9710:	Best = 5.000814e+04	FES = 971100	Diversity = 8.4557e+00
Generation 9720:	Best = 4.993528e+04	FES = 972100	Diversity = 7.7127e+00
Generation 9730:	Best = 4.993528e+04	FES = 973100	Diversity = 7.3265e+00
Generation 9740:	Best = 4.989887e+04	FES = 974100	Diversity = 7.5887e+00
Generation 9750:	Best = 4.984191e+04	FES = 975100	Diversity = 7.3529e+00
Generation 9760:	Best = 4.980066e+04	FES = 976100	Diversity = 7.2850e+00
Generation 9770:	Best = 4.971356e+04	FES = 977100	Diversity = 7.5810e+00
Generation 9780:	Best = 4.969666e+04	FES = 978100	Diversity = 7.8976e+00
Generation 9790:	Best = 4.951050e+04	FES = 979100	Diversity = 8.7119e+00
Generation 9800:	Best = 4.938370e+04	FES = 980100	Diversity = 9.0843e+00
Generation 9810:	Best = 4.925218e+04	FES = 981100	Diversity = 9.4696e+00
Generation 9820:	Best = 4.916166e+04	FES = 982100	Diversity = 9.1558e+00
Generation 9830:	Best = 4.908170e+04	FES = 983100	Diversity = 9.4198e+00
Generation 9840:	Best = 4.906316e+04	FES = 984100	Diversity = 9.4612e+00
Generation 9850:	Best = 4.898080e+04	FES = 985100	Diversity = 9.6609e+00
Generation 9860:	Best = 4.887452e+04	FES = 986100	Diversity = 9.1985e+00
Generation 9870:	Best = 4.879905e+04	FES = 987100	Diversity = 8.9806e+00
Generation 9880:	Best = 4.871519e+04	FES = 988100	Diversity = 8.9668e+00
Generation 9890:	Best = 4.865582e+04	FES = 989100	Diversity = 8.4247e+00
Generation 9900:	Best = 4.860024e+04	FES = 990100	Diversity = 8.3122e+00
Generation 9910:	Best = 4.856103e+04	FES = 991100	Diversity = 8.3200e+00
Generation 9920:	Best = 4.851108e+04	FES = 992100	Diversity = 7.9430e+00
Generation 9930:	Best = 4.848681e+04	FES = 993100	Diversity = 7.6552e+00
Generation 9940:	Best = 4.842016e+04	FES = 994100	Diversity = 7.5986e+00
Generation 9950:	Best = 4.838846e+04	FES = 995100	Diversity = 7.0945e+00
Generation 9960:	Best = 4.837001e+04	FES = 996100	Diversity = 7.0248e+00
Generation 9970:	Best = 4.831552e+04	FES = 997100	Diversity = 6.6881e+00
Generation 9980:	Best = 4.829000e+04	FES = 998100	Diversity = 6.6587e+00
Generation 9990:	Best = 4.819654e+04	FES = 999100	Diversity = 6.2919e+00

Optimization completed!

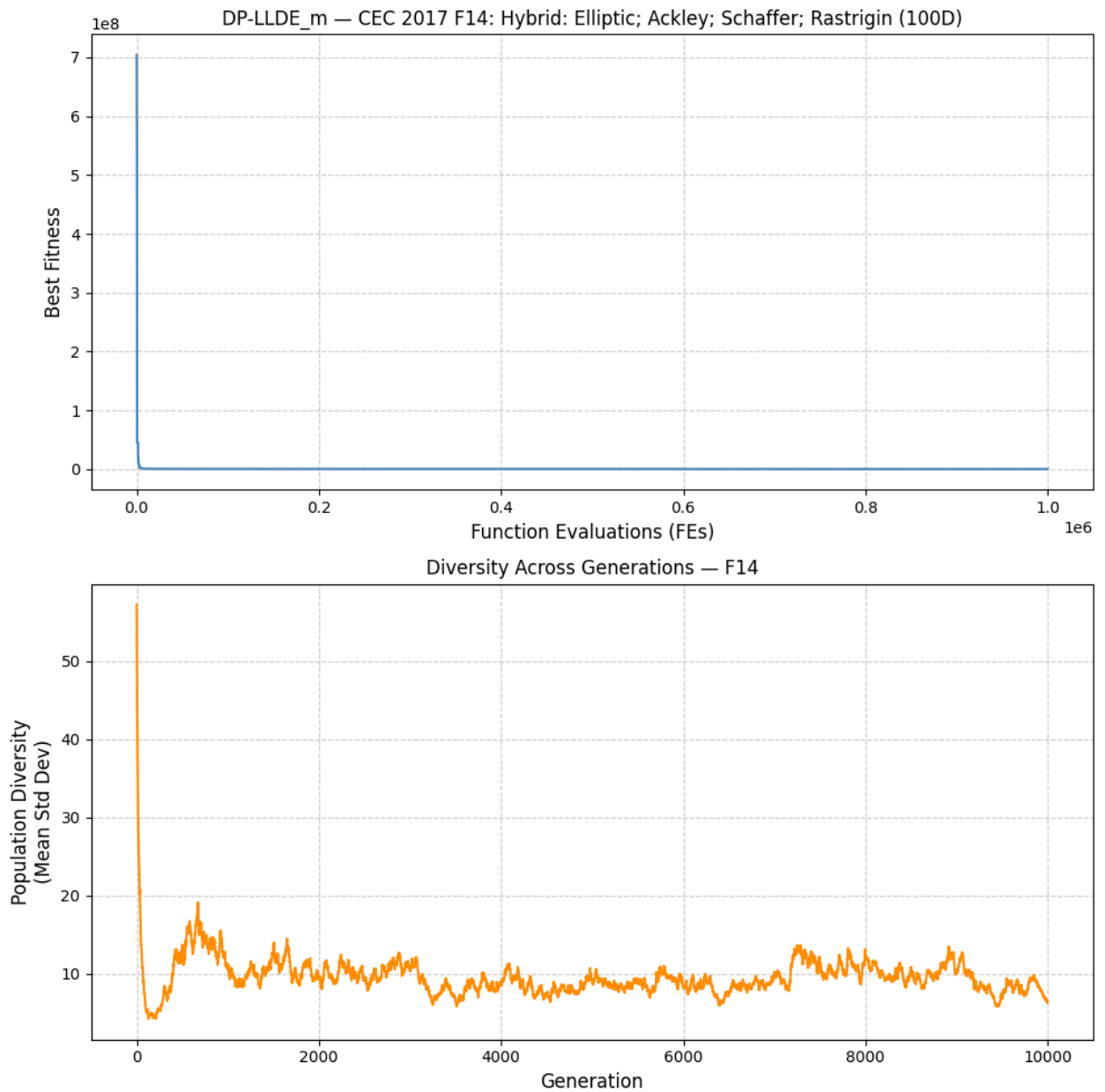
Final Best Fitness = 4.807895e+04

Total Function Evaluations = 1000000

```
=====
=
Results:
  Best Fitness           : 4.807895e+04
  Global Optimum         : 1.400000e+03
  Error (|best-optimum|) : 4.667895e+04
  Best Solution (5 dim)  : [ 87.67084275 -55.31941574 -94.50061238  85.05044751
-95.80647486]
  D^(0) (init. diversity): 5.7220e+01
=====
```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results_2/all_results_che

ckpoint.pkl



Graph saved as convergence_F14.png

✓ Plot saved to: /content/drive/My Drive/thesis_results_2/plots/convergence_F14.png

```
In [ ]: # =====  
# F15 — Hybrid Function 5 (Bent Cigar; HGBat; Rastrigin; Rosenbrock)  
# =====  
run_and_plot('F15', make_wrapper(hybrid.f15), 1500.0, 'Hybrid: Bent Cigar; HGE
```

=====

=

DP-LLDE_m on CEC 2017 F15 – Single Run

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Function : Hybrid: Bent Cigar; HGBat; Rastrigin; Rosenbrock (100D)

Bounds : [-100, 100]

Global optimum : 1500.0

Initial $D^{(0)} = 5.7220e+01$ | Diversity threshold = 5.7220e+00

Generation 0: Best Fitness = 3.781522e+11 | Diversity = 5.7220e+01

Generation	10:	Best = 7.877121e+10	FES = 1100	Diversity = 2.4281e+01
Generation	20:	Best = 3.637890e+10	FES = 2100	Diversity = 1.1493e+01
Generation	30:	Best = 2.705117e+10	FES = 3100	Diversity = 7.0533e+00
Generation	40:	Best = 2.170759e+10	FES = 4100	Diversity = 5.1639e+00
Generation	50:	Best = 1.962685e+10	FES = 5100	Diversity = 4.3119e+00
Generation	60:	Best = 1.838191e+10	FES = 6100	Diversity = 3.6552e+00
Generation	70:	Best = 1.748687e+10	FES = 7100	Diversity = 3.7556e+00
Generation	80:	Best = 1.672610e+10	FES = 8100	Diversity = 3.6377e+00
Generation	90:	Best = 1.617060e+10	FES = 9100	Diversity = 3.8506e+00
Generation	100:	Best = 1.582780e+10	FES = 10100	Diversity = 4.6581e+00
Generation	110:	Best = 1.546348e+10	FES = 11100	Diversity = 6.7931e+00
Generation	120:	Best = 1.525589e+10	FES = 12100	Diversity = 6.8176e+00
Generation	130:	Best = 1.499881e+10	FES = 13100	Diversity = 8.1470e+00
Generation	140:	Best = 1.479342e+10	FES = 14100	Diversity = 1.0399e+01
Generation	150:	Best = 1.461632e+10	FES = 15100	Diversity = 1.0808e+01
Generation	160:	Best = 1.445194e+10	FES = 16100	Diversity = 1.1413e+01
Generation	170:	Best = 1.429223e+10	FES = 17100	Diversity = 1.2624e+01
Generation	180:	Best = 1.416129e+10	FES = 18100	Diversity = 1.2138e+01
Generation	190:	Best = 1.405908e+10	FES = 19100	Diversity = 1.4184e+01
Generation	200:	Best = 1.392234e+10	FES = 20100	Diversity = 1.5760e+01
Generation	210:	Best = 1.383252e+10	FES = 21100	Diversity = 1.5348e+01
Generation	220:	Best = 1.376922e+10	FES = 22100	Diversity = 1.5142e+01
Generation	230:	Best = 1.372933e+10	FES = 23100	Diversity = 1.3472e+01
Generation	240:	Best = 1.368108e+10	FES = 24100	Diversity = 1.2854e+01
Generation	250:	Best = 1.364758e+10	FES = 25100	Diversity = 1.1623e+01
Generation	260:	Best = 1.363065e+10	FES = 26100	Diversity = 1.1300e+01
Generation	270:	Best = 1.361848e+10	FES = 27100	Diversity = 1.1405e+01
Generation	280:	Best = 1.361008e+10	FES = 28100	Diversity = 1.0863e+01
Generation	290:	Best = 1.360123e+10	FES = 29100	Diversity = 1.0940e+01
Generation	300:	Best = 1.359551e+10	FES = 30100	Diversity = 9.8603e+00
Generation	310:	Best = 1.358967e+10	FES = 31100	Diversity = 9.5069e+00
Generation	320:	Best = 1.358496e+10	FES = 32100	Diversity = 1.0652e+01
Generation	330:	Best = 1.357931e+10	FES = 33100	Diversity = 9.9315e+00
Generation	340:	Best = 1.357452e+10	FES = 34100	Diversity = 1.1435e+01
Generation	350:	Best = 1.357044e+10	FES = 35100	Diversity = 1.0246e+01
Generation	360:	Best = 1.356727e+10	FES = 36100	Diversity = 1.0572e+01
Generation	370:	Best = 1.356456e+10	FES = 37100	Diversity = 9.3052e+00
Generation	380:	Best = 1.356206e+10	FES = 38100	Diversity = 7.6158e+00
Generation	390:	Best = 1.356018e+10	FES = 39100	Diversity = 7.9919e+00
Generation	400:	Best = 1.355905e+10	FES = 40100	Diversity = 7.6116e+00
Generation	410:	Best = 1.355811e+10	FES = 41100	Diversity = 7.9843e+00
Generation	420:	Best = 1.355707e+10	FES = 42100	Diversity = 7.3531e+00
Generation	430:	Best = 1.355601e+10	FES = 43100	Diversity = 6.7375e+00
Generation	440:	Best = 1.355527e+10	FES = 44100	Diversity = 4.9230e+00

Generation	450:	Best = 1.355485e+10	FES = 45100	Diversity = 3.8359e+00
Generation	460:	Best = 1.355445e+10	FES = 46100	Diversity = 4.2198e+00
Generation	470:	Best = 1.355396e+10	FES = 47100	Diversity = 4.4748e+00
Generation	480:	Best = 1.355355e+10	FES = 48100	Diversity = 4.2128e+00
Generation	490:	Best = 1.355329e+10	FES = 49100	Diversity = 4.1269e+00
Generation	500:	Best = 1.355306e+10	FES = 50100	Diversity = 3.7605e+00
Generation	510:	Best = 1.355289e+10	FES = 51100	Diversity = 3.6355e+00
Generation	520:	Best = 1.355272e+10	FES = 52100	Diversity = 2.3143e+00
Generation	530:	Best = 1.355256e+10	FES = 53100	Diversity = 2.1735e+00
Generation	540:	Best = 1.355242e+10	FES = 54100	Diversity = 2.6254e+00
Generation	550:	Best = 1.355224e+10	FES = 55100	Diversity = 3.0028e+00
Generation	560:	Best = 1.355215e+10	FES = 56100	Diversity = 2.6567e+00
Generation	570:	Best = 1.355208e+10	FES = 57100	Diversity = 2.5853e+00
Generation	580:	Best = 1.355202e+10	FES = 58100	Diversity = 2.1531e+00
Generation	590:	Best = 1.355198e+10	FES = 59100	Diversity = 2.1098e+00
Generation	600:	Best = 1.355195e+10	FES = 60100	Diversity = 2.8009e+00
Generation	610:	Best = 1.355192e+10	FES = 61100	Diversity = 2.3765e+00
Generation	620:	Best = 1.355189e+10	FES = 62100	Diversity = 1.5968e+00
Generation	630:	Best = 1.355187e+10	FES = 63100	Diversity = 1.6068e+00
Generation	640:	Best = 1.355186e+10	FES = 64100	Diversity = 1.6424e+00
Generation	650:	Best = 1.355185e+10	FES = 65100	Diversity = 1.2348e+00
Generation	660:	Best = 1.355185e+10	FES = 66100	Diversity = 9.3783e-01
Generation	670:	Best = 1.355184e+10	FES = 67100	Diversity = 1.0318e+00
Generation	680:	Best = 1.355184e+10	FES = 68100	Diversity = 7.2580e-01
Generation	690:	Best = 1.355183e+10	FES = 69100	Diversity = 6.8854e-01
Generation	700:	Best = 1.355183e+10	FES = 70100	Diversity = 6.4115e-01
Generation	710:	Best = 1.355183e+10	FES = 71100	Diversity = 3.5604e-01
Generation	720:	Best = 1.355183e+10	FES = 72100	Diversity = 3.1901e-01
Generation	730:	Best = 1.355183e+10	FES = 73100	Diversity = 3.4399e-01
Generation	740:	Best = 1.355183e+10	FES = 74100	Diversity = 3.6119e-01
Generation	750:	Best = 1.355183e+10	FES = 75100	Diversity = 6.0528e-01
Generation	760:	Best = 1.355183e+10	FES = 76100	Diversity = 6.4574e-01
Generation	770:	Best = 1.355182e+10	FES = 77100	Diversity = 1.5937e-01
Generation	780:	Best = 1.355182e+10	FES = 78100	Diversity = 1.1880e-01
Generation	790:	Best = 1.355182e+10	FES = 79100	Diversity = 1.3214e-01
Generation	800:	Best = 1.355182e+10	FES = 80100	Diversity = 1.7971e-01
Generation	810:	Best = 1.355182e+10	FES = 81100	Diversity = 1.7898e-01
Generation	820:	Best = 1.355182e+10	FES = 82100	Diversity = 1.6368e-01
Generation	830:	Best = 1.355182e+10	FES = 83100	Diversity = 1.1368e-01
Generation	840:	Best = 1.355182e+10	FES = 84100	Diversity = 1.0789e-01
Generation	850:	Best = 1.355182e+10	FES = 85100	Diversity = 1.0257e-01
Generation	860:	Best = 1.355182e+10	FES = 86100	Diversity = 6.1795e-02
Generation	870:	Best = 1.355182e+10	FES = 87100	Diversity = 3.6126e-02
Generation	880:	Best = 1.355182e+10	FES = 88100	Diversity = 3.1681e-02
Generation	890:	Best = 1.355182e+10	FES = 89100	Diversity = 2.8291e-02
Generation	900:	Best = 1.355182e+10	FES = 90100	Diversity = 2.1691e-02
Generation	910:	Best = 1.355182e+10	FES = 91100	Diversity = 1.5195e-02
Generation	920:	Best = 1.355182e+10	FES = 92100	Diversity = 1.3036e-02
Generation	930:	Best = 1.355182e+10	FES = 93100	Diversity = 1.6896e-02
Generation	940:	Best = 1.355182e+10	FES = 94100	Diversity = 9.8141e-03
Generation	950:	Best = 1.355182e+10	FES = 95100	Diversity = 1.0248e-02
Generation	960:	Best = 1.355182e+10	FES = 96100	Diversity = 4.5675e-03
Generation	970:	Best = 1.355182e+10	FES = 97100	Diversity = 1.6703e-03
Generation	980:	Best = 1.355182e+10	FES = 98100	Diversity = 1.8831e-03

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[illegible]

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```

Generation 9630: Best = 1.355182e+10 FES = 963100 Diversity = 5.6255e-14
Generation 9640: Best = 1.355182e+10 FES = 964100 Diversity = 5.6255e-14
Generation 9650: Best = 1.355182e+10 FES = 965100 Diversity = 5.6255e-14
Generation 9660: Best = 1.355182e+10 FES = 966100 Diversity = 5.6255e-14
Generation 9670: Best = 1.355182e+10 FES = 967100 Diversity = 5.6255e-14
Generation 9680: Best = 1.355182e+10 FES = 968100 Diversity = 5.6255e-14
Generation 9690: Best = 1.355182e+10 FES = 969100 Diversity = 5.6255e-14
Generation 9700: Best = 1.355182e+10 FES = 970100 Diversity = 5.6255e-14
Generation 9710: Best = 1.355182e+10 FES = 971100 Diversity = 5.6255e-14
Generation 9720: Best = 1.355182e+10 FES = 972100 Diversity = 5.6255e-14
Generation 9730: Best = 1.355182e+10 FES = 973100 Diversity = 5.6255e-14
Generation 9740: Best = 1.355182e+10 FES = 974100 Diversity = 5.6255e-14
Generation 9750: Best = 1.355182e+10 FES = 975100 Diversity = 5.6255e-14
Generation 9760: Best = 1.355182e+10 FES = 976100 Diversity = 5.6255e-14
Generation 9770: Best = 1.355182e+10 FES = 977100 Diversity = 5.6255e-14
Generation 9780: Best = 1.355182e+10 FES = 978100 Diversity = 5.6255e-14
Generation 9790: Best = 1.355182e+10 FES = 979100 Diversity = 5.6255e-14
Generation 9800: Best = 1.355182e+10 FES = 980100 Diversity = 5.6255e-14
Generation 9810: Best = 1.355182e+10 FES = 981100 Diversity = 5.6255e-14
Generation 9820: Best = 1.355182e+10 FES = 982100 Diversity = 5.6255e-14
Generation 9830: Best = 1.355182e+10 FES = 983100 Diversity = 5.6255e-14
Generation 9840: Best = 1.355182e+10 FES = 984100 Diversity = 5.6255e-14
Generation 9850: Best = 1.355182e+10 FES = 985100 Diversity = 5.6255e-14
Generation 9860: Best = 1.355182e+10 FES = 986100 Diversity = 5.6255e-14
Generation 9870: Best = 1.355182e+10 FES = 987100 Diversity = 5.6255e-14
Generation 9880: Best = 1.355182e+10 FES = 988100 Diversity = 5.6255e-14
Generation 9890: Best = 1.355182e+10 FES = 989100 Diversity = 5.6255e-14
Generation 9900: Best = 1.355182e+10 FES = 990100 Diversity = 5.6255e-14
Generation 9910: Best = 1.355182e+10 FES = 991100 Diversity = 5.6255e-14
Generation 9920: Best = 1.355182e+10 FES = 992100 Diversity = 5.6255e-14
Generation 9930: Best = 1.355182e+10 FES = 993100 Diversity = 5.6255e-14
Generation 9940: Best = 1.355182e+10 FES = 994100 Diversity = 5.6255e-14
Generation 9950: Best = 1.355182e+10 FES = 995100 Diversity = 5.6255e-14
Generation 9960: Best = 1.355182e+10 FES = 996100 Diversity = 5.6255e-14
Generation 9970: Best = 1.355182e+10 FES = 997100 Diversity = 5.6255e-14
Generation 9980: Best = 1.355182e+10 FES = 998100 Diversity = 5.6255e-14
Generation 9990: Best = 1.355182e+10 FES = 999100 Diversity = 5.6255e-14

```

Optimization completed!

Final Best Fitness = 1.355182e+10

Total Function Evaluations = 1000000

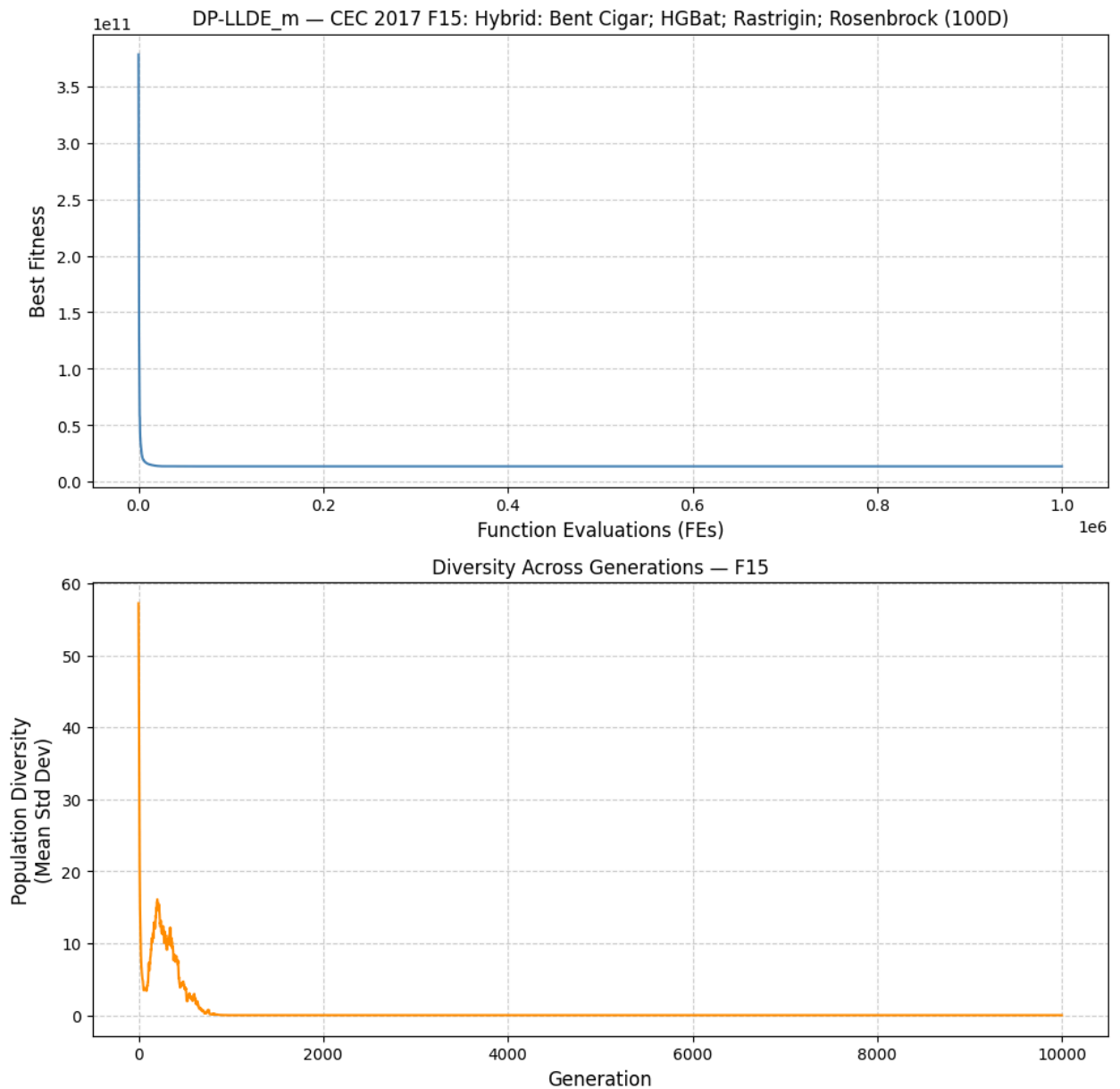
```

=====
=
Results:
  Best Fitness          : 1.355182e+10
  Global Optimum        : 1.500000e+03
  Error (|best-optimum|) : 1.355182e+10
  Best Solution (5 dim)  : [-71.71286508 -12.26856821 -86.07259518 -86.17874233
-37.54895879]
  D^(0) (init. diversity): 5.7220e+01
=====
=

```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results_2/all_results_che

ckpoint.pkl



Graph saved as convergence_F15.png

✓ Plot saved to: /content/drive/My Drive/thesis_results_2/plots/convergence_F15.png

```
In [ ]: # =====  
# F16 – Hybrid Function 6 (Schaffer; HGBat; Rosenbrock; Schwefel)  
# =====  
run_and_plot('F16', make_wrapper(hybrid.f16), 1600.0, 'Hybrid: Schaffer; HGBat
```

=====

=

DP-LLDE_m on CEC 2017 F16 – Single Run

=====

=

Function : Hybrid: Schaffer; HGBat; Rosenbrock; Schwefel (100D)

Bounds : [-100, 100]

Global optimum : 1600.0

Initial $D^{(0)} = 5.7220e+01$ | Diversity threshold = 5.7220e+00

Generation 0: Best Fitness = 3.124977e+04 | Diversity = 5.7220e+01

Generation	10:	Best = 1.583665e+04	FES = 1100	Diversity = 2.6570e+01
Generation	20:	Best = 1.263720e+04	FES = 2100	Diversity = 1.8636e+01
Generation	30:	Best = 1.205862e+04	FES = 3100	Diversity = 1.5877e+01
Generation	40:	Best = 1.205862e+04	FES = 4100	Diversity = 1.4148e+01
Generation	50:	Best = 1.095285e+04	FES = 5100	Diversity = 1.3497e+01
Generation	60:	Best = 1.054430e+04	FES = 6100	Diversity = 1.2668e+01
Generation	70:	Best = 1.054430e+04	FES = 7100	Diversity = 1.2169e+01
Generation	80:	Best = 1.025054e+04	FES = 8100	Diversity = 1.1800e+01
Generation	90:	Best = 1.017669e+04	FES = 9100	Diversity = 1.1348e+01
Generation	100:	Best = 9.695362e+03	FES = 10100	Diversity = 1.0824e+01
Generation	110:	Best = 9.569081e+03	FES = 11100	Diversity = 1.0417e+01
Generation	120:	Best = 9.567487e+03	FES = 12100	Diversity = 9.7459e+00
Generation	130:	Best = 8.982370e+03	FES = 13100	Diversity = 9.5939e+00
Generation	140:	Best = 8.965203e+03	FES = 14100	Diversity = 9.1811e+00
Generation	150:	Best = 8.930107e+03	FES = 15100	Diversity = 8.8787e+00
Generation	160:	Best = 8.392953e+03	FES = 16100	Diversity = 8.8059e+00
Generation	170:	Best = 8.392953e+03	FES = 17100	Diversity = 8.6814e+00
Generation	180:	Best = 8.323881e+03	FES = 18100	Diversity = 8.4927e+00
Generation	190:	Best = 8.053784e+03	FES = 19100	Diversity = 8.4329e+00
Generation	200:	Best = 8.013902e+03	FES = 20100	Diversity = 8.0908e+00
Generation	210:	Best = 7.694281e+03	FES = 21100	Diversity = 7.9855e+00
Generation	220:	Best = 7.656837e+03	FES = 22100	Diversity = 7.8399e+00
Generation	230:	Best = 7.656837e+03	FES = 23100	Diversity = 7.6870e+00
Generation	240:	Best = 7.653595e+03	FES = 24100	Diversity = 7.6485e+00
Generation	250:	Best = 7.360233e+03	FES = 25100	Diversity = 7.5680e+00
Generation	260:	Best = 7.204196e+03	FES = 26100	Diversity = 7.3008e+00
Generation	270:	Best = 7.120659e+03	FES = 27100	Diversity = 7.0131e+00
Generation	280:	Best = 6.996361e+03	FES = 28100	Diversity = 6.8690e+00
Generation	290:	Best = 6.890501e+03	FES = 29100	Diversity = 6.6472e+00
Generation	300:	Best = 6.811811e+03	FES = 30100	Diversity = 6.5908e+00
Generation	310:	Best = 6.754215e+03	FES = 31100	Diversity = 6.4932e+00
Generation	320:	Best = 6.673168e+03	FES = 32100	Diversity = 6.3830e+00
Generation	330:	Best = 6.531335e+03	FES = 33100	Diversity = 6.0295e+00
Generation	340:	Best = 6.294380e+03	FES = 34100	Diversity = 5.7678e+00
Generation	350:	Best = 6.085190e+03	FES = 35100	Diversity = 5.3177e+00
Generation	360:	Best = 5.973070e+03	FES = 36100	Diversity = 4.5443e+00
Generation	370:	Best = 5.750621e+03	FES = 37100	Diversity = 1.6620e+00
Generation	380:	Best = 5.543708e+03	FES = 38100	Diversity = 8.9719e-01
Generation	390:	Best = 5.324954e+03	FES = 39100	Diversity = 9.0399e-01
Generation	400:	Best = 5.252279e+03	FES = 40100	Diversity = 7.6145e-01
Generation	410:	Best = 5.154520e+03	FES = 41100	Diversity = 6.9452e-01
Generation	420:	Best = 5.088955e+03	FES = 42100	Diversity = 7.9113e-01
Generation	430:	Best = 5.037373e+03	FES = 43100	Diversity = 8.3848e-01
Generation	440:	Best = 4.986064e+03	FES = 44100	Diversity = 1.0343e+00

Generation	450:	Best = 4.953760e+03	FES = 45100	Diversity = 8.9341e-01
Generation	460:	Best = 4.914550e+03	FES = 46100	Diversity = 8.1472e-01
Generation	470:	Best = 4.886013e+03	FES = 47100	Diversity = 7.9835e-01
Generation	480:	Best = 4.841957e+03	FES = 48100	Diversity = 8.9049e-01
Generation	490:	Best = 4.803647e+03	FES = 49100	Diversity = 1.0152e+00
Generation	500:	Best = 4.759667e+03	FES = 50100	Diversity = 1.0459e+00
Generation	510:	Best = 4.723525e+03	FES = 51100	Diversity = 1.1395e+00
Generation	520:	Best = 4.683159e+03	FES = 52100	Diversity = 1.1923e+00
Generation	530:	Best = 4.646209e+03	FES = 53100	Diversity = 1.3497e+00
Generation	540:	Best = 4.624320e+03	FES = 54100	Diversity = 1.1676e+00
Generation	550:	Best = 4.595183e+03	FES = 55100	Diversity = 1.2412e+00
Generation	560:	Best = 4.569852e+03	FES = 56100	Diversity = 1.5361e+00
Generation	570:	Best = 4.546635e+03	FES = 57100	Diversity = 1.7279e+00
Generation	580:	Best = 4.523707e+03	FES = 58100	Diversity = 1.6838e+00
Generation	590:	Best = 4.497618e+03	FES = 59100	Diversity = 1.7926e+00
Generation	600:	Best = 4.477623e+03	FES = 60100	Diversity = 1.7746e+00
Generation	610:	Best = 4.454737e+03	FES = 61100	Diversity = 1.6046e+00
Generation	620:	Best = 4.438031e+03	FES = 62100	Diversity = 1.5388e+00
Generation	630:	Best = 4.419177e+03	FES = 63100	Diversity = 1.6060e+00
Generation	640:	Best = 4.405387e+03	FES = 64100	Diversity = 1.4468e+00
Generation	650:	Best = 4.392841e+03	FES = 65100	Diversity = 1.2133e+00
Generation	660:	Best = 4.381285e+03	FES = 66100	Diversity = 1.3387e+00
Generation	670:	Best = 4.370190e+03	FES = 67100	Diversity = 1.4075e+00
Generation	680:	Best = 4.361199e+03	FES = 68100	Diversity = 1.4049e+00
Generation	690:	Best = 4.352795e+03	FES = 69100	Diversity = 1.2493e+00
Generation	700:	Best = 4.344959e+03	FES = 70100	Diversity = 1.1111e+00
Generation	710:	Best = 4.337510e+03	FES = 71100	Diversity = 1.1390e+00
Generation	720:	Best = 4.329534e+03	FES = 72100	Diversity = 1.0879e+00
Generation	730:	Best = 4.323513e+03	FES = 73100	Diversity = 9.9981e-01
Generation	740:	Best = 4.317815e+03	FES = 74100	Diversity = 1.0074e+00
Generation	750:	Best = 4.312236e+03	FES = 75100	Diversity = 1.0727e+00
Generation	760:	Best = 4.308336e+03	FES = 76100	Diversity = 9.1345e-01
Generation	770:	Best = 4.303504e+03	FES = 77100	Diversity = 1.0942e+00
Generation	780:	Best = 4.299072e+03	FES = 78100	Diversity = 1.0670e+00
Generation	790:	Best = 4.294929e+03	FES = 79100	Diversity = 9.3849e-01
Generation	800:	Best = 4.289469e+03	FES = 80100	Diversity = 1.1908e+00
Generation	810:	Best = 4.284218e+03	FES = 81100	Diversity = 1.1702e+00
Generation	820:	Best = 4.279196e+03	FES = 82100	Diversity = 1.1325e+00
Generation	830:	Best = 4.276355e+03	FES = 83100	Diversity = 9.9844e-01
Generation	840:	Best = 4.273041e+03	FES = 84100	Diversity = 1.0893e+00
Generation	850:	Best = 4.269315e+03	FES = 85100	Diversity = 1.0660e+00
Generation	860:	Best = 4.266604e+03	FES = 86100	Diversity = 1.1479e+00
Generation	870:	Best = 4.262857e+03	FES = 87100	Diversity = 1.1071e+00
Generation	880:	Best = 4.259036e+03	FES = 88100	Diversity = 1.3100e+00
Generation	890:	Best = 4.255339e+03	FES = 89100	Diversity = 1.5476e+00
Generation	900:	Best = 4.252239e+03	FES = 90100	Diversity = 1.9407e+00
Generation	910:	Best = 4.248193e+03	FES = 91100	Diversity = 1.6356e+00
Generation	920:	Best = 4.243739e+03	FES = 92100	Diversity = 1.9134e+00
Generation	930:	Best = 4.239138e+03	FES = 93100	Diversity = 1.9417e+00
Generation	940:	Best = 4.232802e+03	FES = 94100	Diversity = 2.0216e+00
Generation	950:	Best = 4.228086e+03	FES = 95100	Diversity = 1.8525e+00
Generation	960:	Best = 4.222542e+03	FES = 96100	Diversity = 2.1921e+00
Generation	970:	Best = 4.218864e+03	FES = 97100	Diversity = 2.2664e+00
Generation	980:	Best = 4.214828e+03	FES = 98100	Diversity = 2.2570e+00

Generation 990:	Best = 4.211076e+03	FES = 99100	Diversity = 2.1752e+00
Generation 1000:	Best = 4.206681e+03	FES = 100100	Diversity = 2.4417e+00
Generation 1010:	Best = 4.203653e+03	FES = 101100	Diversity = 2.4963e+00
Generation 1020:	Best = 4.199294e+03	FES = 102100	Diversity = 2.6938e+00
Generation 1030:	Best = 4.195453e+03	FES = 103100	Diversity = 2.7292e+00
Generation 1040:	Best = 4.192301e+03	FES = 104100	Diversity = 2.8448e+00
Generation 1050:	Best = 4.189198e+03	FES = 105100	Diversity = 3.2147e+00
Generation 1060:	Best = 4.187282e+03	FES = 106100	Diversity = 3.2041e+00
Generation 1070:	Best = 4.184994e+03	FES = 107100	Diversity = 2.9660e+00
Generation 1080:	Best = 4.182471e+03	FES = 108100	Diversity = 2.6521e+00
Generation 1090:	Best = 4.180437e+03	FES = 109100	Diversity = 2.6550e+00
Generation 1100:	Best = 4.177683e+03	FES = 110100	Diversity = 2.5456e+00
Generation 1110:	Best = 4.175860e+03	FES = 111100	Diversity = 2.7173e+00
Generation 1120:	Best = 4.173814e+03	FES = 112100	Diversity = 2.1780e+00
Generation 1130:	Best = 4.172882e+03	FES = 113100	Diversity = 2.0447e+00
Generation 1140:	Best = 4.171513e+03	FES = 114100	Diversity = 2.2768e+00
Generation 1150:	Best = 4.170487e+03	FES = 115100	Diversity = 2.9715e+00
Generation 1160:	Best = 4.168662e+03	FES = 116100	Diversity = 2.9105e+00
Generation 1170:	Best = 4.166820e+03	FES = 117100	Diversity = 2.7368e+00
Generation 1180:	Best = 4.163946e+03	FES = 118100	Diversity = 2.1897e+00
Generation 1190:	Best = 4.162324e+03	FES = 119100	Diversity = 1.7220e+00
Generation 1200:	Best = 4.160233e+03	FES = 120100	Diversity = 1.5575e+00
Generation 1210:	Best = 4.158083e+03	FES = 121100	Diversity = 1.6220e+00
Generation 1220:	Best = 4.154000e+03	FES = 122100	Diversity = 1.3898e+00
Generation 1230:	Best = 4.150963e+03	FES = 123100	Diversity = 1.7724e+00
Generation 1240:	Best = 4.148451e+03	FES = 124100	Diversity = 1.2227e+00
Generation 1250:	Best = 4.146944e+03	FES = 125100	Diversity = 1.1360e+00
Generation 1260:	Best = 4.146180e+03	FES = 126100	Diversity = 1.3317e+00
Generation 1270:	Best = 4.145802e+03	FES = 127100	Diversity = 1.0737e+00
Generation 1280:	Best = 4.145330e+03	FES = 128100	Diversity = 1.0599e+00
Generation 1290:	Best = 4.144962e+03	FES = 129100	Diversity = 1.1605e+00
Generation 1300:	Best = 4.144362e+03	FES = 130100	Diversity = 1.0133e+00
Generation 1310:	Best = 4.143330e+03	FES = 131100	Diversity = 1.2613e+00
Generation 1320:	Best = 4.141938e+03	FES = 132100	Diversity = 7.1794e-01
Generation 1330:	Best = 4.141445e+03	FES = 133100	Diversity = 6.6998e-01
Generation 1340:	Best = 4.140668e+03	FES = 134100	Diversity = 7.1490e-01
Generation 1350:	Best = 4.140100e+03	FES = 135100	Diversity = 5.3726e-01
Generation 1360:	Best = 4.138947e+03	FES = 136100	Diversity = 6.8360e-01
Generation 1370:	Best = 4.137051e+03	FES = 137100	Diversity = 8.3522e-01
Generation 1380:	Best = 4.136631e+03	FES = 138100	Diversity = 4.9279e-01
Generation 1390:	Best = 4.136536e+03	FES = 139100	Diversity = 3.4495e-01
Generation 1400:	Best = 4.136318e+03	FES = 140100	Diversity = 1.8839e-01
Generation 1410:	Best = 4.136181e+03	FES = 141100	Diversity = 1.4966e-02
Generation 1420:	Best = 4.136144e+03	FES = 142100	Diversity = 9.2713e-03
Generation 1430:	Best = 4.136135e+03	FES = 143100	Diversity = 2.3936e-03
Generation 1440:	Best = 4.136132e+03	FES = 144100	Diversity = 1.3156e-03
Generation 1450:	Best = 4.136131e+03	FES = 145100	Diversity = 9.2394e-04
Generation 1460:	Best = 4.136131e+03	FES = 146100	Diversity = 7.8823e-04
Generation 1470:	Best = 4.136131e+03	FES = 147100	Diversity = 3.2822e-04
Generation 1480:	Best = 4.136130e+03	FES = 148100	Diversity = 2.6820e-04
Generation 1490:	Best = 4.136130e+03	FES = 149100	Diversity = 1.9921e-04
Generation 1500:	Best = 4.136130e+03	FES = 150100	Diversity = 1.2023e-04
Generation 1510:	Best = 4.136130e+03	FES = 151100	Diversity = 6.2675e-05
Generation 1520:	Best = 4.136130e+03	FES = 152100	Diversity = 4.5795e-05

[illegible]

[illegible]

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Generation 9630: Best = 4.136130e+03 FES = 963100 Diversity = 5.0342e-14
Generation 9640: Best = 4.136130e+03 FES = 964100 Diversity = 5.0342e-14
Generation 9650: Best = 4.136130e+03 FES = 965100 Diversity = 5.0342e-14
Generation 9660: Best = 4.136130e+03 FES = 966100 Diversity = 5.0342e-14
Generation 9670: Best = 4.136130e+03 FES = 967100 Diversity = 5.0342e-14
Generation 9680: Best = 4.136130e+03 FES = 968100 Diversity = 5.0342e-14
Generation 9690: Best = 4.136130e+03 FES = 969100 Diversity = 5.0342e-14
Generation 9700: Best = 4.136130e+03 FES = 970100 Diversity = 5.0342e-14
Generation 9710: Best = 4.136130e+03 FES = 971100 Diversity = 5.0342e-14
Generation 9720: Best = 4.136130e+03 FES = 972100 Diversity = 5.0342e-14
Generation 9730: Best = 4.136130e+03 FES = 973100 Diversity = 5.0342e-14
Generation 9740: Best = 4.136130e+03 FES = 974100 Diversity = 5.0342e-14
Generation 9750: Best = 4.136130e+03 FES = 975100 Diversity = 5.0342e-14
Generation 9760: Best = 4.136130e+03 FES = 976100 Diversity = 5.0342e-14
Generation 9770: Best = 4.136130e+03 FES = 977100 Diversity = 5.0342e-14
Generation 9780: Best = 4.136130e+03 FES = 978100 Diversity = 5.0342e-14
Generation 9790: Best = 4.136130e+03 FES = 979100 Diversity = 5.0342e-14
Generation 9800: Best = 4.136130e+03 FES = 980100 Diversity = 5.0342e-14
Generation 9810: Best = 4.136130e+03 FES = 981100 Diversity = 5.0342e-14
Generation 9820: Best = 4.136130e+03 FES = 982100 Diversity = 5.0342e-14
Generation 9830: Best = 4.136130e+03 FES = 983100 Diversity = 5.0342e-14
Generation 9840: Best = 4.136130e+03 FES = 984100 Diversity = 5.0342e-14
Generation 9850: Best = 4.136130e+03 FES = 985100 Diversity = 5.0342e-14
Generation 9860: Best = 4.136130e+03 FES = 986100 Diversity = 5.0342e-14
Generation 9870: Best = 4.136130e+03 FES = 987100 Diversity = 5.0342e-14
Generation 9880: Best = 4.136130e+03 FES = 988100 Diversity = 5.0342e-14
Generation 9890: Best = 4.136130e+03 FES = 989100 Diversity = 5.0342e-14
Generation 9900: Best = 4.136130e+03 FES = 990100 Diversity = 5.0342e-14
Generation 9910: Best = 4.136130e+03 FES = 991100 Diversity = 5.0342e-14
Generation 9920: Best = 4.136130e+03 FES = 992100 Diversity = 5.0342e-14
Generation 9930: Best = 4.136130e+03 FES = 993100 Diversity = 5.0342e-14
Generation 9940: Best = 4.136130e+03 FES = 994100 Diversity = 5.0342e-14
Generation 9950: Best = 4.136130e+03 FES = 995100 Diversity = 5.0342e-14
Generation 9960: Best = 4.136130e+03 FES = 996100 Diversity = 5.0342e-14
Generation 9970: Best = 4.136130e+03 FES = 997100 Diversity = 5.0342e-14
Generation 9980: Best = 4.136130e+03 FES = 998100 Diversity = 5.0342e-14
Generation 9990: Best = 4.136130e+03 FES = 999100 Diversity = 5.0342e-14

```

```

Optimization completed!
Final Best Fitness = 4.136130e+03
Total Function Evaluations = 1000000

```

```

=====
=
Results:
  Best Fitness          : 4.136130e+03
  Global Optimum        : 1.600000e+03
  Error (|best-optimum|) : 2.536130e+03
  Best Solution (5 dim)  : [ 21.02353731  36.19739732  47.89953852 -12.48409606
94.1201577 ]
  D^(0) (init. diversity): 5.7220e+01
=====
=

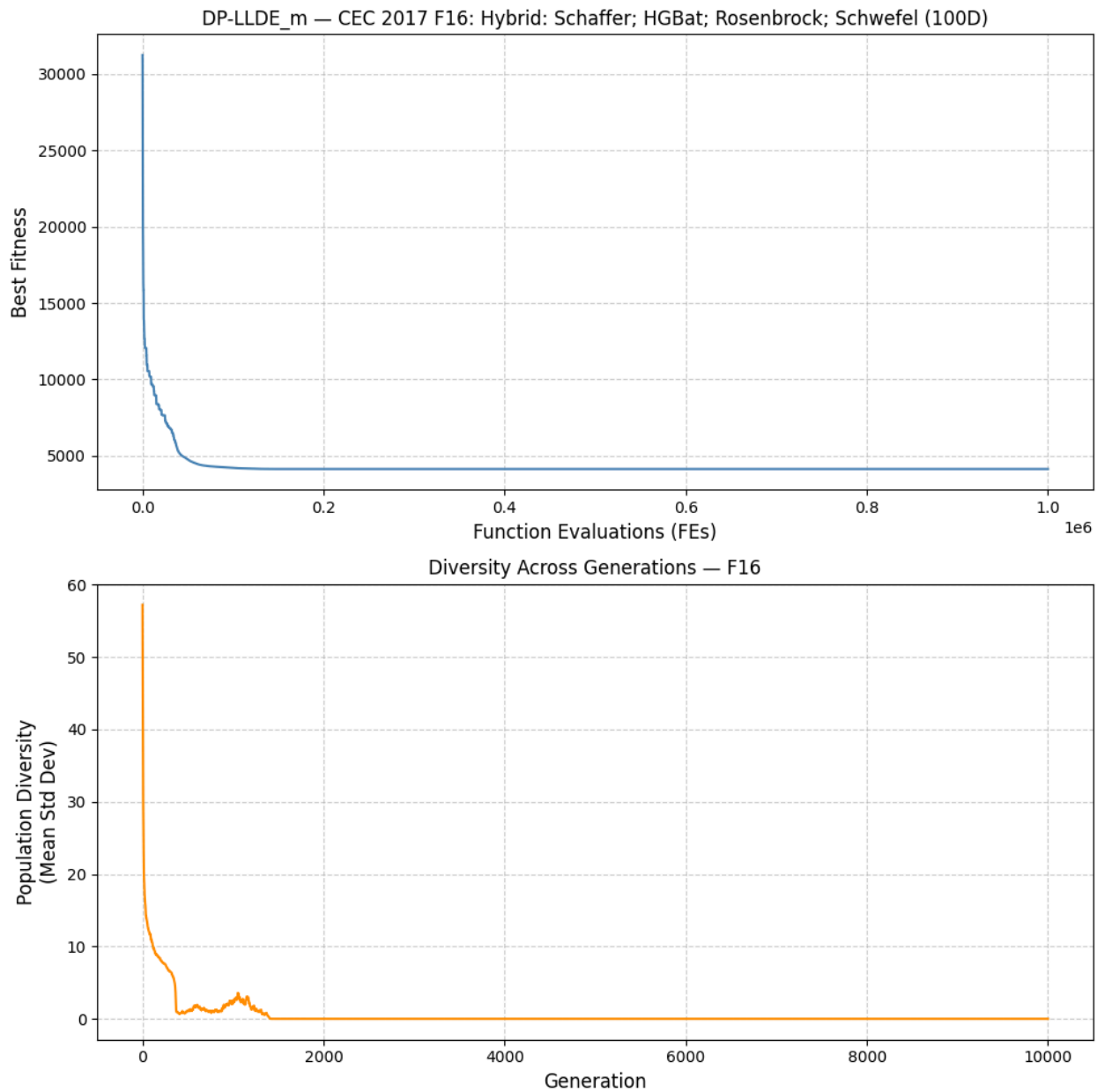
```

```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results_2/all_results_che

```

ckpoint.pkl



Graph saved as convergence_F16.png

✓ Plot saved to: /content/drive/My Drive/thesis_results_2/plots/convergence_F16.png

```
In [ ]: # =====  
# F17 — Hybrid Function 7 (Katsuura; Ackley; Griewank+Rosenbrock; Schwefel; Ra  
# =====  
run_and_plot('F17', make_wrapper(hybrid.f17), 1700.0, 'Hybrid: Katsuura; Ackle
```

=====

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DP-LLDE_m on CEC 2017 F17 – Single Run

=====

=

Function : Hybrid: Katsuura; Ackley; Griewank+Rosenbrock; Schwefel; Rastrigin (100D)

Bounds : [-100, 100]

Global optimum : 1700.0

Initial $D^{(0)} = 5.7220e+01$ | Diversity threshold = 5.7220e+00

Generation 0: Best Fitness = 3.179716e+07 | Diversity = 5.7220e+01

Generation 10: Best = 5.246809e+04 FES = 1100 Diversity = 2.6224e+01

Generation 20: Best = 1.059534e+04 FES = 2100 Diversity = 1.3391e+01

Generation 30: Best = 8.425846e+03 FES = 3100 Diversity = 9.8655e+00

Generation 40: Best = 7.868939e+03 FES = 4100 Diversity = 8.8144e+00

Generation 50: Best = 7.868939e+03 FES = 5100 Diversity = 7.9645e+00

Generation 60: Best = 7.384714e+03 FES = 6100 Diversity = 7.5980e+00

Generation 70: Best = 7.088299e+03 FES = 7100 Diversity = 7.5790e+00

Generation 80: Best = 6.896787e+03 FES = 8100 Diversity = 7.4022e+00

Generation 90: Best = 6.885118e+03 FES = 9100 Diversity = 7.2271e+00

Generation 100: Best = 6.712739e+03 FES = 10100 Diversity = 7.0775e+00

Generation 110: Best = 6.348023e+03 FES = 11100 Diversity = 7.0286e+00

Generation 120: Best = 6.192725e+03 FES = 12100 Diversity = 6.9663e+00

Generation 130: Best = 6.192725e+03 FES = 13100 Diversity = 6.9336e+00

Generation 140: Best = 6.192352e+03 FES = 14100 Diversity = 6.8326e+00

Generation 150: Best = 6.112728e+03 FES = 15100 Diversity = 6.8611e+00

Generation 160: Best = 6.112728e+03 FES = 16100 Diversity = 6.8130e+00

Generation 170: Best = 6.041400e+03 FES = 17100 Diversity = 6.8850e+00

Generation 180: Best = 5.792597e+03 FES = 18100 Diversity = 6.6978e+00

Generation 190: Best = 5.738261e+03 FES = 19100 Diversity = 6.6788e+00

Generation 200: Best = 5.729309e+03 FES = 20100 Diversity = 6.6294e+00

Generation 210: Best = 5.728995e+03 FES = 21100 Diversity = 6.6288e+00

Generation 220: Best = 5.728995e+03 FES = 22100 Diversity = 6.6000e+00

Generation 230: Best = 5.728995e+03 FES = 23100 Diversity = 6.5355e+00

Generation 240: Best = 5.726689e+03 FES = 24100 Diversity = 6.5012e+00

Generation 250: Best = 5.724261e+03 FES = 25100 Diversity = 6.4723e+00

Generation 260: Best = 5.680345e+03 FES = 26100 Diversity = 6.3774e+00

Generation 270: Best = 5.593855e+03 FES = 27100 Diversity = 6.2384e+00

Generation 280: Best = 5.559369e+03 FES = 28100 Diversity = 6.2651e+00

Generation 290: Best = 5.535312e+03 FES = 29100 Diversity = 6.2598e+00

Generation 300: Best = 5.535312e+03 FES = 30100 Diversity = 6.1976e+00

Generation 310: Best = 5.530012e+03 FES = 31100 Diversity = 6.1820e+00

Generation 320: Best = 5.530012e+03 FES = 32100 Diversity = 6.1590e+00

Generation 330: Best = 5.530012e+03 FES = 33100 Diversity = 6.1424e+00

Generation 340: Best = 5.530012e+03 FES = 34100 Diversity = 6.1131e+00

Generation 350: Best = 5.530012e+03 FES = 35100 Diversity = 6.0953e+00

Generation 360: Best = 5.530012e+03 FES = 36100 Diversity = 6.0810e+00

Generation 370: Best = 5.526948e+03 FES = 37100 Diversity = 6.0413e+00

Generation 380: Best = 5.522688e+03 FES = 38100 Diversity = 5.9814e+00

Generation 390: Best = 5.522688e+03 FES = 39100 Diversity = 5.9584e+00

Generation 400: Best = 5.522633e+03 FES = 40100 Diversity = 5.9638e+00

Generation 410: Best = 5.461263e+03 FES = 41100 Diversity = 5.9251e+00

Generation 420: Best = 5.347870e+03 FES = 42100 Diversity = 5.9098e+00

Generation 430: Best = 5.151235e+03 FES = 43100 Diversity = 5.8551e+00

Generation	440:	Best = 5.144376e+03	FES = 44100	Diversity = 5.7908e+00
Generation	450:	Best = 5.144376e+03	FES = 45100	Diversity = 5.7805e+00
Generation	460:	Best = 5.144342e+03	FES = 46100	Diversity = 5.7431e+00
Generation	470:	Best = 5.078490e+03	FES = 47100	Diversity = 5.6780e+00
Generation	480:	Best = 5.013735e+03	FES = 48100	Diversity = 5.5944e+00
Generation	490:	Best = 5.011238e+03	FES = 49100	Diversity = 5.5545e+00
Generation	500:	Best = 5.010916e+03	FES = 50100	Diversity = 5.5279e+00
Generation	510:	Best = 4.858717e+03	FES = 51100	Diversity = 5.4839e+00
Generation	520:	Best = 4.858717e+03	FES = 52100	Diversity = 5.4161e+00
Generation	530:	Best = 4.858717e+03	FES = 53100	Diversity = 5.3873e+00
Generation	540:	Best = 4.848263e+03	FES = 54100	Diversity = 5.4002e+00
Generation	550:	Best = 4.848263e+03	FES = 55100	Diversity = 5.3947e+00
Generation	560:	Best = 4.809419e+03	FES = 56100	Diversity = 5.3817e+00
Generation	570:	Best = 4.808539e+03	FES = 57100	Diversity = 5.3631e+00
Generation	580:	Best = 4.756939e+03	FES = 58100	Diversity = 5.3121e+00
Generation	590:	Best = 4.741072e+03	FES = 59100	Diversity = 5.3053e+00
Generation	600:	Best = 4.741072e+03	FES = 60100	Diversity = 5.2780e+00
Generation	610:	Best = 4.739728e+03	FES = 61100	Diversity = 5.2025e+00
Generation	620:	Best = 4.739728e+03	FES = 62100	Diversity = 5.1676e+00
Generation	630:	Best = 4.739728e+03	FES = 63100	Diversity = 5.1720e+00
Generation	640:	Best = 4.739728e+03	FES = 64100	Diversity = 5.1407e+00
Generation	650:	Best = 4.739395e+03	FES = 65100	Diversity = 5.0926e+00
Generation	660:	Best = 4.739317e+03	FES = 66100	Diversity = 5.0630e+00
Generation	670:	Best = 4.739280e+03	FES = 67100	Diversity = 5.0311e+00
Generation	680:	Best = 4.727711e+03	FES = 68100	Diversity = 5.0573e+00
Generation	690:	Best = 4.720996e+03	FES = 69100	Diversity = 5.0508e+00
Generation	700:	Best = 4.708796e+03	FES = 70100	Diversity = 5.0174e+00
Generation	710:	Best = 4.708796e+03	FES = 71100	Diversity = 5.0132e+00
Generation	720:	Best = 4.708796e+03	FES = 72100	Diversity = 5.0008e+00
Generation	730:	Best = 4.669181e+03	FES = 73100	Diversity = 4.9735e+00
Generation	740:	Best = 4.669045e+03	FES = 74100	Diversity = 4.9926e+00
Generation	750:	Best = 4.669045e+03	FES = 75100	Diversity = 5.0170e+00
Generation	760:	Best = 4.669045e+03	FES = 76100	Diversity = 4.9748e+00
Generation	770:	Best = 4.669045e+03	FES = 77100	Diversity = 4.9365e+00
Generation	780:	Best = 4.669045e+03	FES = 78100	Diversity = 4.9292e+00
Generation	790:	Best = 4.660336e+03	FES = 79100	Diversity = 4.8906e+00
Generation	800:	Best = 4.660336e+03	FES = 80100	Diversity = 4.8949e+00
Generation	810:	Best = 4.660336e+03	FES = 81100	Diversity = 4.8776e+00
Generation	820:	Best = 4.659043e+03	FES = 82100	Diversity = 4.8343e+00
Generation	830:	Best = 4.659043e+03	FES = 83100	Diversity = 4.8147e+00
Generation	840:	Best = 4.659043e+03	FES = 84100	Diversity = 4.8084e+00
Generation	850:	Best = 4.659043e+03	FES = 85100	Diversity = 4.7592e+00
Generation	860:	Best = 4.646257e+03	FES = 86100	Diversity = 4.7122e+00
Generation	870:	Best = 4.646257e+03	FES = 87100	Diversity = 4.7022e+00
Generation	880:	Best = 4.646257e+03	FES = 88100	Diversity = 4.6900e+00
Generation	890:	Best = 4.646257e+03	FES = 89100	Diversity = 4.6620e+00
Generation	900:	Best = 4.646257e+03	FES = 90100	Diversity = 4.6539e+00
Generation	910:	Best = 4.646257e+03	FES = 91100	Diversity = 4.6351e+00
Generation	920:	Best = 4.646257e+03	FES = 92100	Diversity = 4.5890e+00
Generation	930:	Best = 4.646257e+03	FES = 93100	Diversity = 4.5827e+00
Generation	940:	Best = 4.645734e+03	FES = 94100	Diversity = 4.5828e+00
Generation	950:	Best = 4.643002e+03	FES = 95100	Diversity = 4.5796e+00
Generation	960:	Best = 4.643002e+03	FES = 96100	Diversity = 4.6007e+00
Generation	970:	Best = 4.642936e+03	FES = 97100	Diversity = 4.6022e+00

Generation 980:	Best = 4.642936e+03	FES = 98100	Diversity = 4.5881e+00
Generation 990:	Best = 4.642936e+03	FES = 99100	Diversity = 4.5841e+00
Generation 1000:	Best = 4.642936e+03	FES = 100100	Diversity = 4.5935e+00
Generation 1010:	Best = 4.642361e+03	FES = 101100	Diversity = 4.5888e+00
Generation 1020:	Best = 4.641579e+03	FES = 102100	Diversity = 4.6048e+00
Generation 1030:	Best = 4.641435e+03	FES = 103100	Diversity = 4.5954e+00
Generation 1040:	Best = 4.641435e+03	FES = 104100	Diversity = 4.5688e+00
Generation 1050:	Best = 4.641417e+03	FES = 105100	Diversity = 4.5363e+00
Generation 1060:	Best = 4.641417e+03	FES = 106100	Diversity = 4.5249e+00
Generation 1070:	Best = 4.639992e+03	FES = 107100	Diversity = 4.4861e+00
Generation 1080:	Best = 4.639992e+03	FES = 108100	Diversity = 4.4785e+00
Generation 1090:	Best = 4.639992e+03	FES = 109100	Diversity = 4.4335e+00
Generation 1100:	Best = 4.639992e+03	FES = 110100	Diversity = 4.4308e+00
Generation 1110:	Best = 4.639992e+03	FES = 111100	Diversity = 4.4238e+00
Generation 1120:	Best = 4.639309e+03	FES = 112100	Diversity = 4.4154e+00
Generation 1130:	Best = 4.639161e+03	FES = 113100	Diversity = 4.4081e+00
Generation 1140:	Best = 4.639161e+03	FES = 114100	Diversity = 4.3991e+00
Generation 1150:	Best = 4.639161e+03	FES = 115100	Diversity = 4.3750e+00
Generation 1160:	Best = 4.636543e+03	FES = 116100	Diversity = 4.3507e+00
Generation 1170:	Best = 4.636543e+03	FES = 117100	Diversity = 4.3342e+00
Generation 1180:	Best = 4.636543e+03	FES = 118100	Diversity = 4.3174e+00
Generation 1190:	Best = 4.636543e+03	FES = 119100	Diversity = 4.3155e+00
Generation 1200:	Best = 4.636543e+03	FES = 120100	Diversity = 4.3055e+00
Generation 1210:	Best = 4.636543e+03	FES = 121100	Diversity = 4.2337e+00
Generation 1220:	Best = 4.635156e+03	FES = 122100	Diversity = 4.2353e+00
Generation 1230:	Best = 4.576257e+03	FES = 123100	Diversity = 4.2306e+00
Generation 1240:	Best = 4.576257e+03	FES = 124100	Diversity = 4.2228e+00
Generation 1250:	Best = 4.576257e+03	FES = 125100	Diversity = 4.2038e+00
Generation 1260:	Best = 4.573816e+03	FES = 126100	Diversity = 4.1859e+00
Generation 1270:	Best = 4.573816e+03	FES = 127100	Diversity = 4.1692e+00
Generation 1280:	Best = 4.396622e+03	FES = 128100	Diversity = 4.1725e+00
Generation 1290:	Best = 4.393482e+03	FES = 129100	Diversity = 4.1353e+00
Generation 1300:	Best = 4.386608e+03	FES = 130100	Diversity = 4.1331e+00
Generation 1310:	Best = 4.349674e+03	FES = 131100	Diversity = 4.1380e+00
Generation 1320:	Best = 4.318255e+03	FES = 132100	Diversity = 4.1340e+00
Generation 1330:	Best = 4.318255e+03	FES = 133100	Diversity = 4.1174e+00
Generation 1340:	Best = 4.318255e+03	FES = 134100	Diversity = 4.1177e+00
Generation 1350:	Best = 4.313453e+03	FES = 135100	Diversity = 4.0774e+00
Generation 1360:	Best = 4.313186e+03	FES = 136100	Diversity = 4.0835e+00
Generation 1370:	Best = 4.313186e+03	FES = 137100	Diversity = 4.0759e+00
Generation 1380:	Best = 4.302786e+03	FES = 138100	Diversity = 3.9676e+00
Generation 1390:	Best = 4.302777e+03	FES = 139100	Diversity = 3.9588e+00
Generation 1400:	Best = 4.282747e+03	FES = 140100	Diversity = 3.9651e+00
Generation 1410:	Best = 4.213004e+03	FES = 141100	Diversity = 3.9149e+00
Generation 1420:	Best = 4.213004e+03	FES = 142100	Diversity = 3.8898e+00
Generation 1430:	Best = 4.182111e+03	FES = 143100	Diversity = 3.7881e+00
Generation 1440:	Best = 4.178384e+03	FES = 144100	Diversity = 3.7753e+00
Generation 1450:	Best = 4.149805e+03	FES = 145100	Diversity = 3.7220e+00
Generation 1460:	Best = 4.105072e+03	FES = 146100	Diversity = 3.7166e+00
Generation 1470:	Best = 4.094582e+03	FES = 147100	Diversity = 3.6771e+00
Generation 1480:	Best = 4.092670e+03	FES = 148100	Diversity = 3.5477e+00
Generation 1490:	Best = 4.092463e+03	FES = 149100	Diversity = 3.5409e+00
Generation 1500:	Best = 4.077011e+03	FES = 150100	Diversity = 3.5295e+00
Generation 1510:	Best = 4.069339e+03	FES = 151100	Diversity = 3.4646e+00

Generation 1520:	Best = 4.067710e+03	FES = 152100	Diversity = 3.3487e+00
Generation 1530:	Best = 4.052014e+03	FES = 153100	Diversity = 3.1055e+00
Generation 1540:	Best = 4.048765e+03	FES = 154100	Diversity = 2.9537e+00
Generation 1550:	Best = 4.043362e+03	FES = 155100	Diversity = 2.5492e+00
Generation 1560:	Best = 4.034939e+03	FES = 156100	Diversity = 1.0703e+00
Generation 1570:	Best = 4.021475e+03	FES = 157100	Diversity = 4.2697e-01
Generation 1580:	Best = 3.998444e+03	FES = 158100	Diversity = 4.1295e-01
Generation 1590:	Best = 3.954918e+03	FES = 159100	Diversity = 3.9320e-01
Generation 1600:	Best = 3.945170e+03	FES = 160100	Diversity = 3.5428e-01
Generation 1610:	Best = 3.935253e+03	FES = 161100	Diversity = 3.3410e-01
Generation 1620:	Best = 3.916385e+03	FES = 162100	Diversity = 3.1631e-01
Generation 1630:	Best = 3.915888e+03	FES = 163100	Diversity = 2.8724e-01
Generation 1640:	Best = 3.904715e+03	FES = 164100	Diversity = 2.6599e-01
Generation 1650:	Best = 3.902819e+03	FES = 165100	Diversity = 2.2944e-01
Generation 1660:	Best = 3.898070e+03	FES = 166100	Diversity = 2.1058e-01
Generation 1670:	Best = 3.894514e+03	FES = 167100	Diversity = 2.1235e-01
Generation 1680:	Best = 3.892463e+03	FES = 168100	Diversity = 2.0131e-01
Generation 1690:	Best = 3.884417e+03	FES = 169100	Diversity = 2.0959e-01
Generation 1700:	Best = 3.884395e+03	FES = 170100	Diversity = 2.1537e-01
Generation 1710:	Best = 3.883396e+03	FES = 171100	Diversity = 2.1012e-01
Generation 1720:	Best = 3.882598e+03	FES = 172100	Diversity = 2.1226e-01
Generation 1730:	Best = 3.879299e+03	FES = 173100	Diversity = 1.9010e-01
Generation 1740:	Best = 3.877479e+03	FES = 174100	Diversity = 1.8402e-01
Generation 1750:	Best = 3.874478e+03	FES = 175100	Diversity = 2.2066e-01
Generation 1760:	Best = 3.873212e+03	FES = 176100	Diversity = 2.2724e-01
Generation 1770:	Best = 3.872111e+03	FES = 177100	Diversity = 2.4366e-01
Generation 1780:	Best = 3.870299e+03	FES = 178100	Diversity = 2.8487e-01
Generation 1790:	Best = 3.869003e+03	FES = 179100	Diversity = 2.5593e-01
Generation 1800:	Best = 3.867021e+03	FES = 180100	Diversity = 2.5603e-01
Generation 1810:	Best = 3.865341e+03	FES = 181100	Diversity = 2.6384e-01
Generation 1820:	Best = 3.862272e+03	FES = 182100	Diversity = 2.9689e-01
Generation 1830:	Best = 3.861613e+03	FES = 183100	Diversity = 2.8625e-01
Generation 1840:	Best = 3.860230e+03	FES = 184100	Diversity = 3.1861e-01
Generation 1850:	Best = 3.858526e+03	FES = 185100	Diversity = 3.4798e-01
Generation 1860:	Best = 3.857087e+03	FES = 186100	Diversity = 3.2498e-01
Generation 1870:	Best = 3.855902e+03	FES = 187100	Diversity = 3.1574e-01
Generation 1880:	Best = 3.855058e+03	FES = 188100	Diversity = 2.6938e-01
Generation 1890:	Best = 3.854494e+03	FES = 189100	Diversity = 2.7330e-01
Generation 1900:	Best = 3.853621e+03	FES = 190100	Diversity = 3.0570e-01
Generation 1910:	Best = 3.852657e+03	FES = 191100	Diversity = 2.8090e-01
Generation 1920:	Best = 3.852024e+03	FES = 192100	Diversity = 2.8459e-01
Generation 1930:	Best = 3.851454e+03	FES = 193100	Diversity = 1.8583e-01
Generation 1940:	Best = 3.850401e+03	FES = 194100	Diversity = 1.6675e-01
Generation 1950:	Best = 3.850107e+03	FES = 195100	Diversity = 1.5987e-01
Generation 1960:	Best = 3.849303e+03	FES = 196100	Diversity = 1.6919e-01
Generation 1970:	Best = 3.848804e+03	FES = 197100	Diversity = 1.6096e-01
Generation 1980:	Best = 3.848087e+03	FES = 198100	Diversity = 1.6253e-01
Generation 1990:	Best = 3.847663e+03	FES = 199100	Diversity = 1.4064e-01
Generation 2000:	Best = 3.846943e+03	FES = 200100	Diversity = 1.2964e-01
Generation 2010:	Best = 3.846440e+03	FES = 201100	Diversity = 9.2525e-02
Generation 2020:	Best = 3.846035e+03	FES = 202100	Diversity = 5.6182e-02
Generation 2030:	Best = 3.845492e+03	FES = 203100	Diversity = 4.3165e-02
Generation 2040:	Best = 3.845044e+03	FES = 204100	Diversity = 3.8564e-02
Generation 2050:	Best = 3.844605e+03	FES = 205100	Diversity = 3.5978e-02

Generation 2060:	Best = 3.844423e+03	FES = 206100	Diversity = 3.5896e-02
Generation 2070:	Best = 3.844225e+03	FES = 207100	Diversity = 3.4631e-02
Generation 2080:	Best = 3.844108e+03	FES = 208100	Diversity = 3.7342e-02
Generation 2090:	Best = 3.843913e+03	FES = 209100	Diversity = 3.4106e-02
Generation 2100:	Best = 3.843859e+03	FES = 210100	Diversity = 3.5247e-02
Generation 2110:	Best = 3.843681e+03	FES = 211100	Diversity = 3.7514e-02
Generation 2120:	Best = 3.843665e+03	FES = 212100	Diversity = 3.6034e-02
Generation 2130:	Best = 3.843424e+03	FES = 213100	Diversity = 3.6505e-02
Generation 2140:	Best = 3.843244e+03	FES = 214100	Diversity = 3.2842e-02
Generation 2150:	Best = 3.843223e+03	FES = 215100	Diversity = 2.8847e-02
Generation 2160:	Best = 3.843177e+03	FES = 216100	Diversity = 2.7799e-02
Generation 2170:	Best = 3.843107e+03	FES = 217100	Diversity = 2.7156e-02
Generation 2180:	Best = 3.843039e+03	FES = 218100	Diversity = 2.8618e-02
Generation 2190:	Best = 3.842925e+03	FES = 219100	Diversity = 2.6422e-02
Generation 2200:	Best = 3.842803e+03	FES = 220100	Diversity = 2.7070e-02
Generation 2210:	Best = 3.842715e+03	FES = 221100	Diversity = 2.7579e-02
Generation 2220:	Best = 3.842460e+03	FES = 222100	Diversity = 2.5179e-02
Generation 2230:	Best = 3.842434e+03	FES = 223100	Diversity = 2.4982e-02
Generation 2240:	Best = 3.842416e+03	FES = 224100	Diversity = 1.9281e-02
Generation 2250:	Best = 3.842414e+03	FES = 225100	Diversity = 1.6951e-02
Generation 2260:	Best = 3.842338e+03	FES = 226100	Diversity = 1.6324e-02
Generation 2270:	Best = 3.842200e+03	FES = 227100	Diversity = 1.2828e-02
Generation 2280:	Best = 3.842145e+03	FES = 228100	Diversity = 1.1508e-02
Generation 2290:	Best = 3.842112e+03	FES = 229100	Diversity = 1.0669e-02
Generation 2300:	Best = 3.842027e+03	FES = 230100	Diversity = 9.9028e-03
Generation 2310:	Best = 3.841975e+03	FES = 231100	Diversity = 9.0810e-03
Generation 2320:	Best = 3.841967e+03	FES = 232100	Diversity = 8.6800e-03
Generation 2330:	Best = 3.841941e+03	FES = 233100	Diversity = 8.2343e-03
Generation 2340:	Best = 3.841878e+03	FES = 234100	Diversity = 7.5483e-03
Generation 2350:	Best = 3.841874e+03	FES = 235100	Diversity = 7.4008e-03
Generation 2360:	Best = 3.841848e+03	FES = 236100	Diversity = 7.0263e-03
Generation 2370:	Best = 3.841826e+03	FES = 237100	Diversity = 6.4503e-03
Generation 2380:	Best = 3.841782e+03	FES = 238100	Diversity = 6.0723e-03
Generation 2390:	Best = 3.841769e+03	FES = 239100	Diversity = 5.8910e-03
Generation 2400:	Best = 3.841725e+03	FES = 240100	Diversity = 5.7268e-03
Generation 2410:	Best = 3.841691e+03	FES = 241100	Diversity = 5.4732e-03
Generation 2420:	Best = 3.841691e+03	FES = 242100	Diversity = 5.2424e-03
Generation 2430:	Best = 3.841688e+03	FES = 243100	Diversity = 5.0539e-03
Generation 2440:	Best = 3.841688e+03	FES = 244100	Diversity = 4.5115e-03
Generation 2450:	Best = 3.841678e+03	FES = 245100	Diversity = 4.1658e-03
Generation 2460:	Best = 3.841661e+03	FES = 246100	Diversity = 4.2023e-03
Generation 2470:	Best = 3.841655e+03	FES = 247100	Diversity = 4.2048e-03
Generation 2480:	Best = 3.841643e+03	FES = 248100	Diversity = 4.2243e-03
Generation 2490:	Best = 3.841602e+03	FES = 249100	Diversity = 4.3042e-03
Generation 2500:	Best = 3.841590e+03	FES = 250100	Diversity = 4.1930e-03
Generation 2510:	Best = 3.841580e+03	FES = 251100	Diversity = 4.5072e-03
Generation 2520:	Best = 3.841529e+03	FES = 252100	Diversity = 4.6878e-03
Generation 2530:	Best = 3.841515e+03	FES = 253100	Diversity = 4.8449e-03
Generation 2540:	Best = 3.841500e+03	FES = 254100	Diversity = 4.9704e-03
Generation 2550:	Best = 3.841469e+03	FES = 255100	Diversity = 5.1239e-03
Generation 2560:	Best = 3.841447e+03	FES = 256100	Diversity = 5.3004e-03
Generation 2570:	Best = 3.841433e+03	FES = 257100	Diversity = 5.2543e-03
Generation 2580:	Best = 3.841389e+03	FES = 258100	Diversity = 5.2877e-03
Generation 2590:	Best = 3.841336e+03	FES = 259100	Diversity = 5.3076e-03

Generation 2600:	Best = 3.841326e+03	FES = 260100	Diversity = 5.6999e-03
Generation 2610:	Best = 3.841308e+03	FES = 261100	Diversity = 5.6468e-03
Generation 2620:	Best = 3.841256e+03	FES = 262100	Diversity = 5.4860e-03
Generation 2630:	Best = 3.841247e+03	FES = 263100	Diversity = 5.2775e-03
Generation 2640:	Best = 3.841221e+03	FES = 264100	Diversity = 5.0226e-03
Generation 2650:	Best = 3.841209e+03	FES = 265100	Diversity = 4.5089e-03
Generation 2660:	Best = 3.841180e+03	FES = 266100	Diversity = 4.3441e-03
Generation 2670:	Best = 3.841170e+03	FES = 267100	Diversity = 4.4140e-03
Generation 2680:	Best = 3.841126e+03	FES = 268100	Diversity = 4.4743e-03
Generation 2690:	Best = 3.841078e+03	FES = 269100	Diversity = 4.2276e-03
Generation 2700:	Best = 3.841058e+03	FES = 270100	Diversity = 4.2092e-03
Generation 2710:	Best = 3.841016e+03	FES = 271100	Diversity = 4.2887e-03
Generation 2720:	Best = 3.841010e+03	FES = 272100	Diversity = 4.0337e-03
Generation 2730:	Best = 3.840992e+03	FES = 273100	Diversity = 4.0818e-03
Generation 2740:	Best = 3.840976e+03	FES = 274100	Diversity = 3.7890e-03
Generation 2750:	Best = 3.840951e+03	FES = 275100	Diversity = 3.9679e-03
Generation 2760:	Best = 3.840924e+03	FES = 276100	Diversity = 3.9499e-03
Generation 2770:	Best = 3.840907e+03	FES = 277100	Diversity = 3.8828e-03
Generation 2780:	Best = 3.840889e+03	FES = 278100	Diversity = 3.8108e-03
Generation 2790:	Best = 3.840886e+03	FES = 279100	Diversity = 3.7052e-03
Generation 2800:	Best = 3.840862e+03	FES = 280100	Diversity = 3.7493e-03
Generation 2810:	Best = 3.840817e+03	FES = 281100	Diversity = 3.7327e-03
Generation 2820:	Best = 3.840810e+03	FES = 282100	Diversity = 3.6307e-03
Generation 2830:	Best = 3.840790e+03	FES = 283100	Diversity = 3.6805e-03
Generation 2840:	Best = 3.840751e+03	FES = 284100	Diversity = 3.7016e-03
Generation 2850:	Best = 3.840702e+03	FES = 285100	Diversity = 3.4447e-03
Generation 2860:	Best = 3.840692e+03	FES = 286100	Diversity = 3.3860e-03
Generation 2870:	Best = 3.840686e+03	FES = 287100	Diversity = 3.5954e-03
Generation 2880:	Best = 3.840639e+03	FES = 288100	Diversity = 3.7528e-03
Generation 2890:	Best = 3.840616e+03	FES = 289100	Diversity = 3.6135e-03
Generation 2900:	Best = 3.840603e+03	FES = 290100	Diversity = 3.5211e-03
Generation 2910:	Best = 3.840593e+03	FES = 291100	Diversity = 3.5654e-03
Generation 2920:	Best = 3.840583e+03	FES = 292100	Diversity = 3.5394e-03
Generation 2930:	Best = 3.840563e+03	FES = 293100	Diversity = 3.5303e-03
Generation 2940:	Best = 3.840541e+03	FES = 294100	Diversity = 3.3262e-03
Generation 2950:	Best = 3.840466e+03	FES = 295100	Diversity = 3.3291e-03
Generation 2960:	Best = 3.840434e+03	FES = 296100	Diversity = 3.6564e-03
Generation 2970:	Best = 3.840410e+03	FES = 297100	Diversity = 3.8949e-03
Generation 2980:	Best = 3.840399e+03	FES = 298100	Diversity = 4.1115e-03
Generation 2990:	Best = 3.840395e+03	FES = 299100	Diversity = 3.7739e-03
Generation 3000:	Best = 3.840374e+03	FES = 300100	Diversity = 3.6117e-03
Generation 3010:	Best = 3.840323e+03	FES = 301100	Diversity = 3.6366e-03
Generation 3020:	Best = 3.840306e+03	FES = 302100	Diversity = 3.8401e-03
Generation 3030:	Best = 3.840288e+03	FES = 303100	Diversity = 4.0098e-03
Generation 3040:	Best = 3.840226e+03	FES = 304100	Diversity = 4.3038e-03
Generation 3050:	Best = 3.840215e+03	FES = 305100	Diversity = 4.3818e-03
Generation 3060:	Best = 3.840204e+03	FES = 306100	Diversity = 4.6059e-03
Generation 3070:	Best = 3.840185e+03	FES = 307100	Diversity = 4.5850e-03
Generation 3080:	Best = 3.840159e+03	FES = 308100	Diversity = 4.4269e-03
Generation 3090:	Best = 3.840137e+03	FES = 309100	Diversity = 4.1924e-03
Generation 3100:	Best = 3.840116e+03	FES = 310100	Diversity = 4.1365e-03
Generation 3110:	Best = 3.840077e+03	FES = 311100	Diversity = 3.7859e-03
Generation 3120:	Best = 3.840043e+03	FES = 312100	Diversity = 3.6992e-03
Generation 3130:	Best = 3.840015e+03	FES = 313100	Diversity = 3.6784e-03

Generation 3140:	Best = 3.839977e+03	FES = 314100	Diversity = 4.1247e-03
Generation 3150:	Best = 3.839954e+03	FES = 315100	Diversity = 3.9595e-03
Generation 3160:	Best = 3.839938e+03	FES = 316100	Diversity = 3.7905e-03
Generation 3170:	Best = 3.839934e+03	FES = 317100	Diversity = 3.5827e-03
Generation 3180:	Best = 3.839907e+03	FES = 318100	Diversity = 3.4576e-03
Generation 3190:	Best = 3.839886e+03	FES = 319100	Diversity = 3.4250e-03
Generation 3200:	Best = 3.839874e+03	FES = 320100	Diversity = 3.3055e-03
Generation 3210:	Best = 3.839874e+03	FES = 321100	Diversity = 3.3364e-03
Generation 3220:	Best = 3.839869e+03	FES = 322100	Diversity = 3.0268e-03
Generation 3230:	Best = 3.839847e+03	FES = 323100	Diversity = 2.7780e-03
Generation 3240:	Best = 3.839835e+03	FES = 324100	Diversity = 2.9604e-03
Generation 3250:	Best = 3.839829e+03	FES = 325100	Diversity = 2.6685e-03
Generation 3260:	Best = 3.839820e+03	FES = 326100	Diversity = 2.6782e-03
Generation 3270:	Best = 3.839775e+03	FES = 327100	Diversity = 2.8421e-03
Generation 3280:	Best = 3.839760e+03	FES = 328100	Diversity = 3.0398e-03
Generation 3290:	Best = 3.839741e+03	FES = 329100	Diversity = 3.1477e-03
Generation 3300:	Best = 3.839729e+03	FES = 330100	Diversity = 3.0844e-03
Generation 3310:	Best = 3.839713e+03	FES = 331100	Diversity = 3.1735e-03
Generation 3320:	Best = 3.839706e+03	FES = 332100	Diversity = 3.0597e-03
Generation 3330:	Best = 3.839680e+03	FES = 333100	Diversity = 3.0186e-03
Generation 3340:	Best = 3.839663e+03	FES = 334100	Diversity = 3.2602e-03
Generation 3350:	Best = 3.839631e+03	FES = 335100	Diversity = 3.1453e-03
Generation 3360:	Best = 3.839618e+03	FES = 336100	Diversity = 3.2387e-03
Generation 3370:	Best = 3.839594e+03	FES = 337100	Diversity = 3.1535e-03
Generation 3380:	Best = 3.839584e+03	FES = 338100	Diversity = 3.0413e-03
Generation 3390:	Best = 3.839572e+03	FES = 339100	Diversity = 2.9170e-03
Generation 3400:	Best = 3.839549e+03	FES = 340100	Diversity = 2.8033e-03
Generation 3410:	Best = 3.839549e+03	FES = 341100	Diversity = 2.7692e-03
Generation 3420:	Best = 3.839540e+03	FES = 342100	Diversity = 2.8347e-03
Generation 3430:	Best = 3.839530e+03	FES = 343100	Diversity = 2.5933e-03
Generation 3440:	Best = 3.839516e+03	FES = 344100	Diversity = 2.4683e-03
Generation 3450:	Best = 3.839497e+03	FES = 345100	Diversity = 2.4308e-03
Generation 3460:	Best = 3.839492e+03	FES = 346100	Diversity = 2.3880e-03
Generation 3470:	Best = 3.839485e+03	FES = 347100	Diversity = 2.1740e-03
Generation 3480:	Best = 3.839481e+03	FES = 348100	Diversity = 2.1119e-03
Generation 3490:	Best = 3.839463e+03	FES = 349100	Diversity = 2.1312e-03
Generation 3500:	Best = 3.839463e+03	FES = 350100	Diversity = 2.0588e-03
Generation 3510:	Best = 3.839457e+03	FES = 351100	Diversity = 1.9869e-03
Generation 3520:	Best = 3.839446e+03	FES = 352100	Diversity = 1.9491e-03
Generation 3530:	Best = 3.839433e+03	FES = 353100	Diversity = 2.1965e-03
Generation 3540:	Best = 3.839425e+03	FES = 354100	Diversity = 2.4490e-03
Generation 3550:	Best = 3.839407e+03	FES = 355100	Diversity = 2.5590e-03
Generation 3560:	Best = 3.839392e+03	FES = 356100	Diversity = 2.6425e-03
Generation 3570:	Best = 3.839391e+03	FES = 357100	Diversity = 2.7173e-03
Generation 3580:	Best = 3.839381e+03	FES = 358100	Diversity = 2.7539e-03
Generation 3590:	Best = 3.839377e+03	FES = 359100	Diversity = 2.6806e-03
Generation 3600:	Best = 3.839365e+03	FES = 360100	Diversity = 2.6378e-03
Generation 3610:	Best = 3.839334e+03	FES = 361100	Diversity = 2.7030e-03
Generation 3620:	Best = 3.839320e+03	FES = 362100	Diversity = 2.8810e-03
Generation 3630:	Best = 3.839303e+03	FES = 363100	Diversity = 2.7522e-03
Generation 3640:	Best = 3.839285e+03	FES = 364100	Diversity = 2.8163e-03
Generation 3650:	Best = 3.839268e+03	FES = 365100	Diversity = 2.9040e-03
Generation 3660:	Best = 3.839261e+03	FES = 366100	Diversity = 3.0237e-03
Generation 3670:	Best = 3.839242e+03	FES = 367100	Diversity = 3.1981e-03

Generation 3680:	Best = 3.839202e+03	FES = 368100	Diversity = 3.1331e-03
Generation 3690:	Best = 3.839199e+03	FES = 369100	Diversity = 3.2221e-03
Generation 3700:	Best = 3.839169e+03	FES = 370100	Diversity = 3.3627e-03
Generation 3710:	Best = 3.839158e+03	FES = 371100	Diversity = 3.6311e-03
Generation 3720:	Best = 3.839102e+03	FES = 372100	Diversity = 3.6103e-03
Generation 3730:	Best = 3.839094e+03	FES = 373100	Diversity = 3.6560e-03
Generation 3740:	Best = 3.839081e+03	FES = 374100	Diversity = 3.4018e-03
Generation 3750:	Best = 3.839058e+03	FES = 375100	Diversity = 3.3658e-03
Generation 3760:	Best = 3.839038e+03	FES = 376100	Diversity = 3.2652e-03
Generation 3770:	Best = 3.839036e+03	FES = 377100	Diversity = 3.0385e-03
Generation 3780:	Best = 3.839005e+03	FES = 378100	Diversity = 2.8186e-03
Generation 3790:	Best = 3.838999e+03	FES = 379100	Diversity = 2.9480e-03
Generation 3800:	Best = 3.838994e+03	FES = 380100	Diversity = 2.9306e-03
Generation 3810:	Best = 3.838982e+03	FES = 381100	Diversity = 2.7566e-03
Generation 3820:	Best = 3.838974e+03	FES = 382100	Diversity = 2.7014e-03
Generation 3830:	Best = 3.838952e+03	FES = 383100	Diversity = 2.7002e-03
Generation 3840:	Best = 3.838952e+03	FES = 384100	Diversity = 2.5555e-03
Generation 3850:	Best = 3.838930e+03	FES = 385100	Diversity = 2.8820e-03
Generation 3860:	Best = 3.838927e+03	FES = 386100	Diversity = 2.7576e-03
Generation 3870:	Best = 3.838919e+03	FES = 387100	Diversity = 2.7470e-03
Generation 3880:	Best = 3.838900e+03	FES = 388100	Diversity = 2.5982e-03
Generation 3890:	Best = 3.838894e+03	FES = 389100	Diversity = 2.6477e-03
Generation 3900:	Best = 3.838886e+03	FES = 390100	Diversity = 2.5720e-03
Generation 3910:	Best = 3.838883e+03	FES = 391100	Diversity = 2.5171e-03
Generation 3920:	Best = 3.838877e+03	FES = 392100	Diversity = 2.4712e-03
Generation 3930:	Best = 3.838867e+03	FES = 393100	Diversity = 2.4790e-03
Generation 3940:	Best = 3.838858e+03	FES = 394100	Diversity = 2.4612e-03
Generation 3950:	Best = 3.838853e+03	FES = 395100	Diversity = 2.4070e-03
Generation 3960:	Best = 3.838837e+03	FES = 396100	Diversity = 2.5056e-03
Generation 3970:	Best = 3.838831e+03	FES = 397100	Diversity = 2.5091e-03
Generation 3980:	Best = 3.838826e+03	FES = 398100	Diversity = 2.4282e-03
Generation 3990:	Best = 3.838823e+03	FES = 399100	Diversity = 2.3706e-03
Generation 4000:	Best = 3.838815e+03	FES = 400100	Diversity = 2.4043e-03
Generation 4010:	Best = 3.838806e+03	FES = 401100	Diversity = 2.3183e-03
Generation 4020:	Best = 3.838788e+03	FES = 402100	Diversity = 2.3404e-03
Generation 4030:	Best = 3.838785e+03	FES = 403100	Diversity = 2.1673e-03
Generation 4040:	Best = 3.838779e+03	FES = 404100	Diversity = 2.0286e-03
Generation 4050:	Best = 3.838775e+03	FES = 405100	Diversity = 2.0639e-03
Generation 4060:	Best = 3.838762e+03	FES = 406100	Diversity = 2.0025e-03
Generation 4070:	Best = 3.838762e+03	FES = 407100	Diversity = 2.0736e-03
Generation 4080:	Best = 3.838754e+03	FES = 408100	Diversity = 2.0630e-03
Generation 4090:	Best = 3.838746e+03	FES = 409100	Diversity = 2.0567e-03
Generation 4100:	Best = 3.838744e+03	FES = 410100	Diversity = 1.9562e-03
Generation 4110:	Best = 3.838743e+03	FES = 411100	Diversity = 1.9617e-03
Generation 4120:	Best = 3.838741e+03	FES = 412100	Diversity = 1.8831e-03
Generation 4130:	Best = 3.838738e+03	FES = 413100	Diversity = 1.8232e-03
Generation 4140:	Best = 3.838736e+03	FES = 414100	Diversity = 1.6915e-03
Generation 4150:	Best = 3.838733e+03	FES = 415100	Diversity = 1.6972e-03
Generation 4160:	Best = 3.838726e+03	FES = 416100	Diversity = 1.5923e-03
Generation 4170:	Best = 3.838720e+03	FES = 417100	Diversity = 1.5560e-03
Generation 4180:	Best = 3.838719e+03	FES = 418100	Diversity = 1.5333e-03
Generation 4190:	Best = 3.838710e+03	FES = 419100	Diversity = 1.5554e-03
Generation 4200:	Best = 3.838709e+03	FES = 420100	Diversity = 1.5660e-03
Generation 4210:	Best = 3.838704e+03	FES = 421100	Diversity = 1.6281e-03

Generation 4220:	Best = 3.838702e+03	FES = 422100	Diversity = 1.6149e-03
Generation 4230:	Best = 3.838697e+03	FES = 423100	Diversity = 1.7139e-03
Generation 4240:	Best = 3.838684e+03	FES = 424100	Diversity = 1.8143e-03
Generation 4250:	Best = 3.838678e+03	FES = 425100	Diversity = 1.8720e-03
Generation 4260:	Best = 3.838670e+03	FES = 426100	Diversity = 1.8698e-03
Generation 4270:	Best = 3.838669e+03	FES = 427100	Diversity = 1.8204e-03
Generation 4280:	Best = 3.838665e+03	FES = 428100	Diversity = 1.7789e-03
Generation 4290:	Best = 3.838658e+03	FES = 429100	Diversity = 1.8438e-03
Generation 4300:	Best = 3.838653e+03	FES = 430100	Diversity = 1.8537e-03
Generation 4310:	Best = 3.838649e+03	FES = 431100	Diversity = 1.7828e-03
Generation 4320:	Best = 3.838646e+03	FES = 432100	Diversity = 1.7279e-03
Generation 4330:	Best = 3.838643e+03	FES = 433100	Diversity = 1.7081e-03
Generation 4340:	Best = 3.838639e+03	FES = 434100	Diversity = 1.7208e-03
Generation 4350:	Best = 3.838632e+03	FES = 435100	Diversity = 1.7371e-03
Generation 4360:	Best = 3.838631e+03	FES = 436100	Diversity = 1.6690e-03
Generation 4370:	Best = 3.838624e+03	FES = 437100	Diversity = 1.5170e-03
Generation 4380:	Best = 3.838621e+03	FES = 438100	Diversity = 1.4164e-03
Generation 4390:	Best = 3.838621e+03	FES = 439100	Diversity = 1.5544e-03
Generation 4400:	Best = 3.838618e+03	FES = 440100	Diversity = 1.5111e-03
Generation 4410:	Best = 3.838608e+03	FES = 441100	Diversity = 1.4157e-03
Generation 4420:	Best = 3.838606e+03	FES = 442100	Diversity = 1.5032e-03
Generation 4430:	Best = 3.838598e+03	FES = 443100	Diversity = 1.5958e-03
Generation 4440:	Best = 3.838593e+03	FES = 444100	Diversity = 1.5610e-03
Generation 4450:	Best = 3.838581e+03	FES = 445100	Diversity = 1.4518e-03
Generation 4460:	Best = 3.838581e+03	FES = 446100	Diversity = 1.6202e-03
Generation 4470:	Best = 3.838577e+03	FES = 447100	Diversity = 1.5824e-03
Generation 4480:	Best = 3.838573e+03	FES = 448100	Diversity = 1.5755e-03
Generation 4490:	Best = 3.838567e+03	FES = 449100	Diversity = 1.4802e-03
Generation 4500:	Best = 3.838561e+03	FES = 450100	Diversity = 1.5434e-03
Generation 4510:	Best = 3.838554e+03	FES = 451100	Diversity = 1.7540e-03
Generation 4520:	Best = 3.838546e+03	FES = 452100	Diversity = 1.7462e-03
Generation 4530:	Best = 3.838546e+03	FES = 453100	Diversity = 1.6821e-03
Generation 4540:	Best = 3.838540e+03	FES = 454100	Diversity = 1.7223e-03
Generation 4550:	Best = 3.838534e+03	FES = 455100	Diversity = 1.7852e-03
Generation 4560:	Best = 3.838518e+03	FES = 456100	Diversity = 1.8844e-03
Generation 4570:	Best = 3.838515e+03	FES = 457100	Diversity = 2.0103e-03
Generation 4580:	Best = 3.838510e+03	FES = 458100	Diversity = 1.9785e-03
Generation 4590:	Best = 3.838505e+03	FES = 459100	Diversity = 2.0059e-03
Generation 4600:	Best = 3.838499e+03	FES = 460100	Diversity = 1.8926e-03
Generation 4610:	Best = 3.838496e+03	FES = 461100	Diversity = 1.7821e-03
Generation 4620:	Best = 3.838490e+03	FES = 462100	Diversity = 2.0587e-03
Generation 4630:	Best = 3.838482e+03	FES = 463100	Diversity = 2.0310e-03
Generation 4640:	Best = 3.838479e+03	FES = 464100	Diversity = 1.9179e-03
Generation 4650:	Best = 3.838468e+03	FES = 465100	Diversity = 1.9265e-03
Generation 4660:	Best = 3.838465e+03	FES = 466100	Diversity = 1.9210e-03
Generation 4670:	Best = 3.838453e+03	FES = 467100	Diversity = 1.9207e-03
Generation 4680:	Best = 3.838450e+03	FES = 468100	Diversity = 2.0816e-03
Generation 4690:	Best = 3.838429e+03	FES = 469100	Diversity = 2.0505e-03
Generation 4700:	Best = 3.838428e+03	FES = 470100	Diversity = 2.1132e-03
Generation 4710:	Best = 3.838428e+03	FES = 471100	Diversity = 2.1534e-03
Generation 4720:	Best = 3.838425e+03	FES = 472100	Diversity = 2.1355e-03
Generation 4730:	Best = 3.838418e+03	FES = 473100	Diversity = 2.0379e-03
Generation 4740:	Best = 3.838411e+03	FES = 474100	Diversity = 1.9490e-03
Generation 4750:	Best = 3.838411e+03	FES = 475100	Diversity = 1.9925e-03

Generation 4760:	Best = 3.838406e+03	FES = 476100	Diversity = 1.8709e-03
Generation 4770:	Best = 3.838402e+03	FES = 477100	Diversity = 1.7567e-03
Generation 4780:	Best = 3.838395e+03	FES = 478100	Diversity = 1.7480e-03
Generation 4790:	Best = 3.838395e+03	FES = 479100	Diversity = 1.6805e-03
Generation 4800:	Best = 3.838390e+03	FES = 480100	Diversity = 1.5971e-03
Generation 4810:	Best = 3.838388e+03	FES = 481100	Diversity = 1.5899e-03
Generation 4820:	Best = 3.838386e+03	FES = 482100	Diversity = 1.6070e-03
Generation 4830:	Best = 3.838380e+03	FES = 483100	Diversity = 1.6644e-03
Generation 4840:	Best = 3.838367e+03	FES = 484100	Diversity = 1.8311e-03
Generation 4850:	Best = 3.838364e+03	FES = 485100	Diversity = 1.9281e-03
Generation 4860:	Best = 3.838349e+03	FES = 486100	Diversity = 2.1119e-03
Generation 4870:	Best = 3.838332e+03	FES = 487100	Diversity = 2.6345e-03
Generation 4880:	Best = 3.838312e+03	FES = 488100	Diversity = 3.1615e-03
Generation 4890:	Best = 3.838259e+03	FES = 489100	Diversity = 3.6194e-03
Generation 4900:	Best = 3.838246e+03	FES = 490100	Diversity = 3.6000e-03
Generation 4910:	Best = 3.838242e+03	FES = 491100	Diversity = 3.4333e-03
Generation 4920:	Best = 3.838230e+03	FES = 492100	Diversity = 3.2532e-03
Generation 4930:	Best = 3.838215e+03	FES = 493100	Diversity = 3.0403e-03
Generation 4940:	Best = 3.838214e+03	FES = 494100	Diversity = 2.9841e-03
Generation 4950:	Best = 3.838201e+03	FES = 495100	Diversity = 2.9876e-03
Generation 4960:	Best = 3.838183e+03	FES = 496100	Diversity = 2.9784e-03
Generation 4970:	Best = 3.838181e+03	FES = 497100	Diversity = 2.9691e-03
Generation 4980:	Best = 3.838171e+03	FES = 498100	Diversity = 2.9642e-03
Generation 4990:	Best = 3.838163e+03	FES = 499100	Diversity = 3.0188e-03
Generation 5000:	Best = 3.838146e+03	FES = 500100	Diversity = 2.9646e-03
Generation 5010:	Best = 3.838135e+03	FES = 501100	Diversity = 3.0176e-03
Generation 5020:	Best = 3.838122e+03	FES = 502100	Diversity = 2.9912e-03
Generation 5030:	Best = 3.838119e+03	FES = 503100	Diversity = 3.1388e-03
Generation 5040:	Best = 3.838107e+03	FES = 504100	Diversity = 3.2404e-03
Generation 5050:	Best = 3.838102e+03	FES = 505100	Diversity = 3.3117e-03
Generation 5060:	Best = 3.838093e+03	FES = 506100	Diversity = 3.1819e-03
Generation 5070:	Best = 3.838089e+03	FES = 507100	Diversity = 3.0121e-03
Generation 5080:	Best = 3.838078e+03	FES = 508100	Diversity = 3.0197e-03
Generation 5090:	Best = 3.838066e+03	FES = 509100	Diversity = 2.9346e-03
Generation 5100:	Best = 3.838053e+03	FES = 510100	Diversity = 2.9256e-03
Generation 5110:	Best = 3.838039e+03	FES = 511100	Diversity = 2.9198e-03
Generation 5120:	Best = 3.838029e+03	FES = 512100	Diversity = 3.1202e-03
Generation 5130:	Best = 3.838015e+03	FES = 513100	Diversity = 3.1404e-03
Generation 5140:	Best = 3.838006e+03	FES = 514100	Diversity = 3.1533e-03
Generation 5150:	Best = 3.837994e+03	FES = 515100	Diversity = 2.9659e-03
Generation 5160:	Best = 3.837991e+03	FES = 516100	Diversity = 2.7248e-03
Generation 5170:	Best = 3.837979e+03	FES = 517100	Diversity = 2.5743e-03
Generation 5180:	Best = 3.837971e+03	FES = 518100	Diversity = 2.6422e-03
Generation 5190:	Best = 3.837967e+03	FES = 519100	Diversity = 2.5525e-03
Generation 5200:	Best = 3.837961e+03	FES = 520100	Diversity = 2.3932e-03
Generation 5210:	Best = 3.837953e+03	FES = 521100	Diversity = 2.5778e-03
Generation 5220:	Best = 3.837951e+03	FES = 522100	Diversity = 2.4426e-03
Generation 5230:	Best = 3.837942e+03	FES = 523100	Diversity = 2.3214e-03
Generation 5240:	Best = 3.837930e+03	FES = 524100	Diversity = 2.2988e-03
Generation 5250:	Best = 3.837930e+03	FES = 525100	Diversity = 2.5389e-03
Generation 5260:	Best = 3.837925e+03	FES = 526100	Diversity = 2.4594e-03
Generation 5270:	Best = 3.837921e+03	FES = 527100	Diversity = 2.3881e-03
Generation 5280:	Best = 3.837917e+03	FES = 528100	Diversity = 2.5091e-03
Generation 5290:	Best = 3.837910e+03	FES = 529100	Diversity = 2.4843e-03

Generation 5300:	Best = 3.837900e+03	FES = 530100	Diversity = 2.5344e-03
Generation 5310:	Best = 3.837896e+03	FES = 531100	Diversity = 2.6031e-03
Generation 5320:	Best = 3.837883e+03	FES = 532100	Diversity = 2.6224e-03
Generation 5330:	Best = 3.837882e+03	FES = 533100	Diversity = 2.5798e-03
Generation 5340:	Best = 3.837880e+03	FES = 534100	Diversity = 2.5060e-03
Generation 5350:	Best = 3.837874e+03	FES = 535100	Diversity = 2.4296e-03
Generation 5360:	Best = 3.837862e+03	FES = 536100	Diversity = 2.4238e-03
Generation 5370:	Best = 3.837856e+03	FES = 537100	Diversity = 2.3773e-03
Generation 5380:	Best = 3.837852e+03	FES = 538100	Diversity = 2.3720e-03
Generation 5390:	Best = 3.837841e+03	FES = 539100	Diversity = 2.5052e-03
Generation 5400:	Best = 3.837841e+03	FES = 540100	Diversity = 2.3997e-03
Generation 5410:	Best = 3.837840e+03	FES = 541100	Diversity = 2.2953e-03
Generation 5420:	Best = 3.837825e+03	FES = 542100	Diversity = 2.2155e-03
Generation 5430:	Best = 3.837822e+03	FES = 543100	Diversity = 2.2760e-03
Generation 5440:	Best = 3.837819e+03	FES = 544100	Diversity = 2.3152e-03
Generation 5450:	Best = 3.837812e+03	FES = 545100	Diversity = 2.2946e-03
Generation 5460:	Best = 3.837809e+03	FES = 546100	Diversity = 2.3732e-03
Generation 5470:	Best = 3.837796e+03	FES = 547100	Diversity = 2.4433e-03
Generation 5480:	Best = 3.837774e+03	FES = 548100	Diversity = 2.6133e-03
Generation 5490:	Best = 3.837762e+03	FES = 549100	Diversity = 2.9579e-03
Generation 5500:	Best = 3.837751e+03	FES = 550100	Diversity = 2.9959e-03
Generation 5510:	Best = 3.837751e+03	FES = 551100	Diversity = 2.9492e-03
Generation 5520:	Best = 3.837733e+03	FES = 552100	Diversity = 3.0649e-03
Generation 5530:	Best = 3.837710e+03	FES = 553100	Diversity = 3.4124e-03
Generation 5540:	Best = 3.837709e+03	FES = 554100	Diversity = 3.4141e-03
Generation 5550:	Best = 3.837697e+03	FES = 555100	Diversity = 3.4770e-03
Generation 5560:	Best = 3.837690e+03	FES = 556100	Diversity = 3.1144e-03
Generation 5570:	Best = 3.837679e+03	FES = 557100	Diversity = 3.1449e-03
Generation 5580:	Best = 3.837671e+03	FES = 558100	Diversity = 3.0952e-03
Generation 5590:	Best = 3.837667e+03	FES = 559100	Diversity = 2.9398e-03
Generation 5600:	Best = 3.837662e+03	FES = 560100	Diversity = 2.9697e-03
Generation 5610:	Best = 3.837655e+03	FES = 561100	Diversity = 2.8499e-03
Generation 5620:	Best = 3.837654e+03	FES = 562100	Diversity = 2.6389e-03
Generation 5630:	Best = 3.837643e+03	FES = 563100	Diversity = 2.6125e-03
Generation 5640:	Best = 3.837632e+03	FES = 564100	Diversity = 2.5697e-03
Generation 5650:	Best = 3.837629e+03	FES = 565100	Diversity = 2.5020e-03
Generation 5660:	Best = 3.837629e+03	FES = 566100	Diversity = 2.4756e-03
Generation 5670:	Best = 3.837620e+03	FES = 567100	Diversity = 2.4740e-03
Generation 5680:	Best = 3.837613e+03	FES = 568100	Diversity = 2.3483e-03
Generation 5690:	Best = 3.837610e+03	FES = 569100	Diversity = 2.2625e-03
Generation 5700:	Best = 3.837605e+03	FES = 570100	Diversity = 2.1831e-03
Generation 5710:	Best = 3.837600e+03	FES = 571100	Diversity = 2.2863e-03
Generation 5720:	Best = 3.837594e+03	FES = 572100	Diversity = 2.1739e-03
Generation 5730:	Best = 3.837593e+03	FES = 573100	Diversity = 2.1494e-03
Generation 5740:	Best = 3.837593e+03	FES = 574100	Diversity = 2.0731e-03
Generation 5750:	Best = 3.837587e+03	FES = 575100	Diversity = 1.9655e-03
Generation 5760:	Best = 3.837582e+03	FES = 576100	Diversity = 1.8184e-03
Generation 5770:	Best = 3.837582e+03	FES = 577100	Diversity = 1.8273e-03
Generation 5780:	Best = 3.837582e+03	FES = 578100	Diversity = 1.7480e-03
Generation 5790:	Best = 3.837579e+03	FES = 579100	Diversity = 1.7038e-03
Generation 5800:	Best = 3.837576e+03	FES = 580100	Diversity = 1.5634e-03
Generation 5810:	Best = 3.837574e+03	FES = 581100	Diversity = 1.5099e-03
Generation 5820:	Best = 3.837569e+03	FES = 582100	Diversity = 1.4610e-03
Generation 5830:	Best = 3.837569e+03	FES = 583100	Diversity = 1.3898e-03

Generation 5840:	Best = 3.837567e+03	FES = 584100	Diversity = 1.3961e-03
Generation 5850:	Best = 3.837565e+03	FES = 585100	Diversity = 1.4671e-03
Generation 5860:	Best = 3.837564e+03	FES = 586100	Diversity = 1.3951e-03
Generation 5870:	Best = 3.837559e+03	FES = 587100	Diversity = 1.4062e-03
Generation 5880:	Best = 3.837557e+03	FES = 588100	Diversity = 1.4246e-03
Generation 5890:	Best = 3.837550e+03	FES = 589100	Diversity = 1.5496e-03
Generation 5900:	Best = 3.837549e+03	FES = 590100	Diversity = 1.5059e-03
Generation 5910:	Best = 3.837547e+03	FES = 591100	Diversity = 1.5617e-03
Generation 5920:	Best = 3.837544e+03	FES = 592100	Diversity = 1.5982e-03
Generation 5930:	Best = 3.837543e+03	FES = 593100	Diversity = 1.5661e-03
Generation 5940:	Best = 3.837536e+03	FES = 594100	Diversity = 1.6058e-03
Generation 5950:	Best = 3.837527e+03	FES = 595100	Diversity = 1.6202e-03
Generation 5960:	Best = 3.837527e+03	FES = 596100	Diversity = 1.6540e-03
Generation 5970:	Best = 3.837525e+03	FES = 597100	Diversity = 1.7073e-03
Generation 5980:	Best = 3.837524e+03	FES = 598100	Diversity = 1.5900e-03
Generation 5990:	Best = 3.837523e+03	FES = 599100	Diversity = 1.5240e-03
Generation 6000:	Best = 3.837521e+03	FES = 600100	Diversity = 1.4432e-03
Generation 6010:	Best = 3.837514e+03	FES = 601100	Diversity = 1.4786e-03
Generation 6020:	Best = 3.837514e+03	FES = 602100	Diversity = 1.5442e-03
Generation 6030:	Best = 3.837514e+03	FES = 603100	Diversity = 1.4406e-03
Generation 6040:	Best = 3.837513e+03	FES = 604100	Diversity = 1.4397e-03
Generation 6050:	Best = 3.837511e+03	FES = 605100	Diversity = 1.2775e-03
Generation 6060:	Best = 3.837509e+03	FES = 606100	Diversity = 1.3011e-03
Generation 6070:	Best = 3.837501e+03	FES = 607100	Diversity = 1.2719e-03
Generation 6080:	Best = 3.837499e+03	FES = 608100	Diversity = 1.2340e-03
Generation 6090:	Best = 3.837498e+03	FES = 609100	Diversity = 1.1744e-03
Generation 6100:	Best = 3.837497e+03	FES = 610100	Diversity = 1.1460e-03
Generation 6110:	Best = 3.837497e+03	FES = 611100	Diversity = 1.0753e-03
Generation 6120:	Best = 3.837496e+03	FES = 612100	Diversity = 1.1272e-03
Generation 6130:	Best = 3.837490e+03	FES = 613100	Diversity = 1.2122e-03
Generation 6140:	Best = 3.837490e+03	FES = 614100	Diversity = 1.2384e-03
Generation 6150:	Best = 3.837489e+03	FES = 615100	Diversity = 1.2336e-03
Generation 6160:	Best = 3.837482e+03	FES = 616100	Diversity = 1.2722e-03
Generation 6170:	Best = 3.837481e+03	FES = 617100	Diversity = 1.3750e-03
Generation 6180:	Best = 3.837480e+03	FES = 618100	Diversity = 1.3584e-03
Generation 6190:	Best = 3.837480e+03	FES = 619100	Diversity = 1.4176e-03
Generation 6200:	Best = 3.837478e+03	FES = 620100	Diversity = 1.3795e-03
Generation 6210:	Best = 3.837475e+03	FES = 621100	Diversity = 1.2838e-03
Generation 6220:	Best = 3.837473e+03	FES = 622100	Diversity = 1.2593e-03
Generation 6230:	Best = 3.837473e+03	FES = 623100	Diversity = 1.2188e-03
Generation 6240:	Best = 3.837471e+03	FES = 624100	Diversity = 1.1217e-03
Generation 6250:	Best = 3.837470e+03	FES = 625100	Diversity = 1.1131e-03
Generation 6260:	Best = 3.837468e+03	FES = 626100	Diversity = 1.0312e-03
Generation 6270:	Best = 3.837466e+03	FES = 627100	Diversity = 9.2065e-04
Generation 6280:	Best = 3.837465e+03	FES = 628100	Diversity = 9.1202e-04
Generation 6290:	Best = 3.837463e+03	FES = 629100	Diversity = 8.5852e-04
Generation 6300:	Best = 3.837462e+03	FES = 630100	Diversity = 8.6622e-04
Generation 6310:	Best = 3.837462e+03	FES = 631100	Diversity = 8.2852e-04
Generation 6320:	Best = 3.837461e+03	FES = 632100	Diversity = 7.8889e-04
Generation 6330:	Best = 3.837460e+03	FES = 633100	Diversity = 7.7497e-04
Generation 6340:	Best = 3.837457e+03	FES = 634100	Diversity = 7.4795e-04
Generation 6350:	Best = 3.837457e+03	FES = 635100	Diversity = 7.7835e-04
Generation 6360:	Best = 3.837455e+03	FES = 636100	Diversity = 8.1249e-04
Generation 6370:	Best = 3.837455e+03	FES = 637100	Diversity = 8.7284e-04

Generation 6380:	Best = 3.837452e+03	FES = 638100	Diversity = 9.0402e-04
Generation 6390:	Best = 3.837450e+03	FES = 639100	Diversity = 8.5017e-04
Generation 6400:	Best = 3.837449e+03	FES = 640100	Diversity = 8.3422e-04
Generation 6410:	Best = 3.837447e+03	FES = 641100	Diversity = 9.1253e-04
Generation 6420:	Best = 3.837444e+03	FES = 642100	Diversity = 8.9297e-04
Generation 6430:	Best = 3.837443e+03	FES = 643100	Diversity = 8.5478e-04
Generation 6440:	Best = 3.837442e+03	FES = 644100	Diversity = 7.7766e-04
Generation 6450:	Best = 3.837441e+03	FES = 645100	Diversity = 7.8841e-04
Generation 6460:	Best = 3.837441e+03	FES = 646100	Diversity = 7.7837e-04
Generation 6470:	Best = 3.837439e+03	FES = 647100	Diversity = 7.6027e-04
Generation 6480:	Best = 3.837437e+03	FES = 648100	Diversity = 7.7719e-04
Generation 6490:	Best = 3.837432e+03	FES = 649100	Diversity = 8.2441e-04
Generation 6500:	Best = 3.837431e+03	FES = 650100	Diversity = 8.4436e-04
Generation 6510:	Best = 3.837431e+03	FES = 651100	Diversity = 8.0960e-04
Generation 6520:	Best = 3.837430e+03	FES = 652100	Diversity = 7.6895e-04
Generation 6530:	Best = 3.837429e+03	FES = 653100	Diversity = 7.4940e-04
Generation 6540:	Best = 3.837428e+03	FES = 654100	Diversity = 7.3291e-04
Generation 6550:	Best = 3.837427e+03	FES = 655100	Diversity = 7.3345e-04
Generation 6560:	Best = 3.837422e+03	FES = 656100	Diversity = 8.2999e-04
Generation 6570:	Best = 3.837421e+03	FES = 657100	Diversity = 8.4029e-04
Generation 6580:	Best = 3.837421e+03	FES = 658100	Diversity = 8.1195e-04
Generation 6590:	Best = 3.837421e+03	FES = 659100	Diversity = 7.9560e-04
Generation 6600:	Best = 3.837420e+03	FES = 660100	Diversity = 7.9027e-04
Generation 6610:	Best = 3.837417e+03	FES = 661100	Diversity = 7.5521e-04
Generation 6620:	Best = 3.837417e+03	FES = 662100	Diversity = 7.5113e-04
Generation 6630:	Best = 3.837414e+03	FES = 663100	Diversity = 7.9642e-04
Generation 6640:	Best = 3.837413e+03	FES = 664100	Diversity = 7.2462e-04
Generation 6650:	Best = 3.837413e+03	FES = 665100	Diversity = 7.2858e-04
Generation 6660:	Best = 3.837412e+03	FES = 666100	Diversity = 7.6094e-04
Generation 6670:	Best = 3.837410e+03	FES = 667100	Diversity = 7.8791e-04
Generation 6680:	Best = 3.837407e+03	FES = 668100	Diversity = 7.4921e-04
Generation 6690:	Best = 3.837405e+03	FES = 669100	Diversity = 8.1015e-04
Generation 6700:	Best = 3.837402e+03	FES = 670100	Diversity = 7.3922e-04
Generation 6710:	Best = 3.837402e+03	FES = 671100	Diversity = 7.5293e-04
Generation 6720:	Best = 3.837401e+03	FES = 672100	Diversity = 7.1230e-04
Generation 6730:	Best = 3.837401e+03	FES = 673100	Diversity = 7.6717e-04
Generation 6740:	Best = 3.837398e+03	FES = 674100	Diversity = 7.5575e-04
Generation 6750:	Best = 3.837397e+03	FES = 675100	Diversity = 7.5560e-04
Generation 6760:	Best = 3.837395e+03	FES = 676100	Diversity = 7.4224e-04
Generation 6770:	Best = 3.837394e+03	FES = 677100	Diversity = 7.5449e-04
Generation 6780:	Best = 3.837391e+03	FES = 678100	Diversity = 7.5597e-04
Generation 6790:	Best = 3.837390e+03	FES = 679100	Diversity = 7.7714e-04
Generation 6800:	Best = 3.837388e+03	FES = 680100	Diversity = 8.2178e-04
Generation 6810:	Best = 3.837388e+03	FES = 681100	Diversity = 7.8843e-04
Generation 6820:	Best = 3.837386e+03	FES = 682100	Diversity = 7.6695e-04
Generation 6830:	Best = 3.837384e+03	FES = 683100	Diversity = 8.6329e-04
Generation 6840:	Best = 3.837384e+03	FES = 684100	Diversity = 8.7812e-04
Generation 6850:	Best = 3.837382e+03	FES = 685100	Diversity = 9.1521e-04
Generation 6860:	Best = 3.837380e+03	FES = 686100	Diversity = 8.5626e-04
Generation 6870:	Best = 3.837379e+03	FES = 687100	Diversity = 8.7535e-04
Generation 6880:	Best = 3.837378e+03	FES = 688100	Diversity = 8.6095e-04
Generation 6890:	Best = 3.837376e+03	FES = 689100	Diversity = 8.5794e-04
Generation 6900:	Best = 3.837375e+03	FES = 690100	Diversity = 8.4754e-04
Generation 6910:	Best = 3.837375e+03	FES = 691100	Diversity = 8.5995e-04

Generation 6920:	Best = 3.837369e+03	FES = 692100	Diversity = 9.1620e-04
Generation 6930:	Best = 3.837369e+03	FES = 693100	Diversity = 9.5816e-04
Generation 6940:	Best = 3.837366e+03	FES = 694100	Diversity = 1.0805e-03
Generation 6950:	Best = 3.837365e+03	FES = 695100	Diversity = 1.0656e-03
Generation 6960:	Best = 3.837359e+03	FES = 696100	Diversity = 1.0873e-03
Generation 6970:	Best = 3.837359e+03	FES = 697100	Diversity = 1.1984e-03
Generation 6980:	Best = 3.837358e+03	FES = 698100	Diversity = 1.1799e-03
Generation 6990:	Best = 3.837353e+03	FES = 699100	Diversity = 1.1694e-03
Generation 7000:	Best = 3.837353e+03	FES = 700100	Diversity = 1.2106e-03
Generation 7010:	Best = 3.837351e+03	FES = 701100	Diversity = 1.2119e-03
Generation 7020:	Best = 3.837346e+03	FES = 702100	Diversity = 1.2005e-03
Generation 7030:	Best = 3.837341e+03	FES = 703100	Diversity = 1.2972e-03
Generation 7040:	Best = 3.837338e+03	FES = 704100	Diversity = 1.3897e-03
Generation 7050:	Best = 3.837334e+03	FES = 705100	Diversity = 1.4682e-03
Generation 7060:	Best = 3.837323e+03	FES = 706100	Diversity = 1.5814e-03
Generation 7070:	Best = 3.837320e+03	FES = 707100	Diversity = 1.6102e-03
Generation 7080:	Best = 3.837320e+03	FES = 708100	Diversity = 1.6444e-03
Generation 7090:	Best = 3.837316e+03	FES = 709100	Diversity = 1.7005e-03
Generation 7100:	Best = 3.837315e+03	FES = 710100	Diversity = 1.6354e-03
Generation 7110:	Best = 3.837312e+03	FES = 711100	Diversity = 1.6089e-03
Generation 7120:	Best = 3.837309e+03	FES = 712100	Diversity = 1.5291e-03
Generation 7130:	Best = 3.837309e+03	FES = 713100	Diversity = 1.4014e-03
Generation 7140:	Best = 3.837308e+03	FES = 714100	Diversity = 1.2943e-03
Generation 7150:	Best = 3.837306e+03	FES = 715100	Diversity = 1.2348e-03
Generation 7160:	Best = 3.837304e+03	FES = 716100	Diversity = 1.2134e-03
Generation 7170:	Best = 3.837304e+03	FES = 717100	Diversity = 1.1679e-03
Generation 7180:	Best = 3.837302e+03	FES = 718100	Diversity = 1.1473e-03
Generation 7190:	Best = 3.837300e+03	FES = 719100	Diversity = 1.1041e-03
Generation 7200:	Best = 3.837298e+03	FES = 720100	Diversity = 1.0405e-03
Generation 7210:	Best = 3.837297e+03	FES = 721100	Diversity = 1.0581e-03
Generation 7220:	Best = 3.837295e+03	FES = 722100	Diversity = 1.0253e-03
Generation 7230:	Best = 3.837294e+03	FES = 723100	Diversity = 9.4256e-04
Generation 7240:	Best = 3.837291e+03	FES = 724100	Diversity = 8.9519e-04
Generation 7250:	Best = 3.837289e+03	FES = 725100	Diversity = 9.4734e-04
Generation 7260:	Best = 3.837289e+03	FES = 726100	Diversity = 9.7641e-04
Generation 7270:	Best = 3.837287e+03	FES = 727100	Diversity = 1.0168e-03
Generation 7280:	Best = 3.837286e+03	FES = 728100	Diversity = 1.0436e-03
Generation 7290:	Best = 3.837286e+03	FES = 729100	Diversity = 1.0305e-03
Generation 7300:	Best = 3.837285e+03	FES = 730100	Diversity = 9.6311e-04
Generation 7310:	Best = 3.837284e+03	FES = 731100	Diversity = 9.6860e-04
Generation 7320:	Best = 3.837283e+03	FES = 732100	Diversity = 9.2632e-04
Generation 7330:	Best = 3.837282e+03	FES = 733100	Diversity = 8.9847e-04
Generation 7340:	Best = 3.837281e+03	FES = 734100	Diversity = 9.0456e-04
Generation 7350:	Best = 3.837281e+03	FES = 735100	Diversity = 8.6677e-04
Generation 7360:	Best = 3.837280e+03	FES = 736100	Diversity = 8.9028e-04
Generation 7370:	Best = 3.837280e+03	FES = 737100	Diversity = 8.6754e-04
Generation 7380:	Best = 3.837279e+03	FES = 738100	Diversity = 7.8975e-04
Generation 7390:	Best = 3.837277e+03	FES = 739100	Diversity = 8.5131e-04
Generation 7400:	Best = 3.837275e+03	FES = 740100	Diversity = 8.5345e-04
Generation 7410:	Best = 3.837275e+03	FES = 741100	Diversity = 8.3705e-04
Generation 7420:	Best = 3.837275e+03	FES = 742100	Diversity = 8.2877e-04
Generation 7430:	Best = 3.837274e+03	FES = 743100	Diversity = 7.9364e-04
Generation 7440:	Best = 3.837274e+03	FES = 744100	Diversity = 7.8838e-04
Generation 7450:	Best = 3.837272e+03	FES = 745100	Diversity = 8.0824e-04

Generation 7460:	Best = 3.837272e+03	FES = 746100	Diversity = 7.4965e-04
Generation 7470:	Best = 3.837271e+03	FES = 747100	Diversity = 7.5301e-04
Generation 7480:	Best = 3.837271e+03	FES = 748100	Diversity = 7.0470e-04
Generation 7490:	Best = 3.837271e+03	FES = 749100	Diversity = 6.9963e-04
Generation 7500:	Best = 3.837270e+03	FES = 750100	Diversity = 6.0701e-04
Generation 7510:	Best = 3.837270e+03	FES = 751100	Diversity = 5.8511e-04
Generation 7520:	Best = 3.837269e+03	FES = 752100	Diversity = 5.8375e-04
Generation 7530:	Best = 3.837269e+03	FES = 753100	Diversity = 6.0083e-04
Generation 7540:	Best = 3.837268e+03	FES = 754100	Diversity = 5.9002e-04
Generation 7550:	Best = 3.837267e+03	FES = 755100	Diversity = 6.2233e-04
Generation 7560:	Best = 3.837267e+03	FES = 756100	Diversity = 5.9043e-04
Generation 7570:	Best = 3.837267e+03	FES = 757100	Diversity = 5.9092e-04
Generation 7580:	Best = 3.837265e+03	FES = 758100	Diversity = 6.4129e-04
Generation 7590:	Best = 3.837264e+03	FES = 759100	Diversity = 6.1146e-04
Generation 7600:	Best = 3.837264e+03	FES = 760100	Diversity = 5.5581e-04
Generation 7610:	Best = 3.837263e+03	FES = 761100	Diversity = 5.9385e-04
Generation 7620:	Best = 3.837263e+03	FES = 762100	Diversity = 6.1663e-04
Generation 7630:	Best = 3.837262e+03	FES = 763100	Diversity = 6.2147e-04
Generation 7640:	Best = 3.837261e+03	FES = 764100	Diversity = 6.2524e-04
Generation 7650:	Best = 3.837260e+03	FES = 765100	Diversity = 6.3451e-04
Generation 7660:	Best = 3.837258e+03	FES = 766100	Diversity = 7.2426e-04
Generation 7670:	Best = 3.837258e+03	FES = 767100	Diversity = 7.4866e-04
Generation 7680:	Best = 3.837257e+03	FES = 768100	Diversity = 7.5708e-04
Generation 7690:	Best = 3.837256e+03	FES = 769100	Diversity = 8.1189e-04
Generation 7700:	Best = 3.837256e+03	FES = 770100	Diversity = 8.5496e-04
Generation 7710:	Best = 3.837253e+03	FES = 771100	Diversity = 8.3209e-04
Generation 7720:	Best = 3.837253e+03	FES = 772100	Diversity = 8.2636e-04
Generation 7730:	Best = 3.837251e+03	FES = 773100	Diversity = 8.4754e-04
Generation 7740:	Best = 3.837250e+03	FES = 774100	Diversity = 9.4979e-04
Generation 7750:	Best = 3.837249e+03	FES = 775100	Diversity = 9.8100e-04
Generation 7760:	Best = 3.837248e+03	FES = 776100	Diversity = 9.4852e-04
Generation 7770:	Best = 3.837248e+03	FES = 777100	Diversity = 9.5103e-04
Generation 7780:	Best = 3.837247e+03	FES = 778100	Diversity = 8.9828e-04
Generation 7790:	Best = 3.837247e+03	FES = 779100	Diversity = 8.3632e-04
Generation 7800:	Best = 3.837246e+03	FES = 780100	Diversity = 7.5885e-04
Generation 7810:	Best = 3.837245e+03	FES = 781100	Diversity = 7.6621e-04
Generation 7820:	Best = 3.837244e+03	FES = 782100	Diversity = 7.5595e-04
Generation 7830:	Best = 3.837242e+03	FES = 783100	Diversity = 7.8090e-04
Generation 7840:	Best = 3.837241e+03	FES = 784100	Diversity = 7.0151e-04
Generation 7850:	Best = 3.837240e+03	FES = 785100	Diversity = 6.6934e-04
Generation 7860:	Best = 3.837239e+03	FES = 786100	Diversity = 6.9930e-04
Generation 7870:	Best = 3.837239e+03	FES = 787100	Diversity = 6.7050e-04
Generation 7880:	Best = 3.837238e+03	FES = 788100	Diversity = 6.5921e-04
Generation 7890:	Best = 3.837238e+03	FES = 789100	Diversity = 6.2648e-04
Generation 7900:	Best = 3.837236e+03	FES = 790100	Diversity = 6.0858e-04
Generation 7910:	Best = 3.837235e+03	FES = 791100	Diversity = 6.6722e-04
Generation 7920:	Best = 3.837234e+03	FES = 792100	Diversity = 6.9417e-04
Generation 7930:	Best = 3.837230e+03	FES = 793100	Diversity = 7.5304e-04
Generation 7940:	Best = 3.837229e+03	FES = 794100	Diversity = 7.1841e-04
Generation 7950:	Best = 3.837229e+03	FES = 795100	Diversity = 7.8009e-04
Generation 7960:	Best = 3.837226e+03	FES = 796100	Diversity = 7.9534e-04
Generation 7970:	Best = 3.837225e+03	FES = 797100	Diversity = 8.0443e-04
Generation 7980:	Best = 3.837222e+03	FES = 798100	Diversity = 8.6992e-04
Generation 7990:	Best = 3.837221e+03	FES = 799100	Diversity = 9.3433e-04

Generation 8000:	Best = 3.837219e+03	FES = 800100	Diversity = 9.0664e-04
Generation 8010:	Best = 3.837219e+03	FES = 801100	Diversity = 9.2058e-04
Generation 8020:	Best = 3.837217e+03	FES = 802100	Diversity = 9.3759e-04
Generation 8030:	Best = 3.837217e+03	FES = 803100	Diversity = 9.7444e-04
Generation 8040:	Best = 3.837217e+03	FES = 804100	Diversity = 9.5193e-04
Generation 8050:	Best = 3.837217e+03	FES = 805100	Diversity = 9.1101e-04
Generation 8060:	Best = 3.837213e+03	FES = 806100	Diversity = 8.8273e-04
Generation 8070:	Best = 3.837212e+03	FES = 807100	Diversity = 8.5422e-04
Generation 8080:	Best = 3.837212e+03	FES = 808100	Diversity = 8.2388e-04
Generation 8090:	Best = 3.837212e+03	FES = 809100	Diversity = 8.1869e-04
Generation 8100:	Best = 3.837211e+03	FES = 810100	Diversity = 7.6840e-04
Generation 8110:	Best = 3.837211e+03	FES = 811100	Diversity = 7.5726e-04
Generation 8120:	Best = 3.837210e+03	FES = 812100	Diversity = 7.0814e-04
Generation 8130:	Best = 3.837209e+03	FES = 813100	Diversity = 7.3105e-04
Generation 8140:	Best = 3.837208e+03	FES = 814100	Diversity = 7.4289e-04
Generation 8150:	Best = 3.837207e+03	FES = 815100	Diversity = 7.5806e-04
Generation 8160:	Best = 3.837206e+03	FES = 816100	Diversity = 7.0832e-04
Generation 8170:	Best = 3.837205e+03	FES = 817100	Diversity = 6.6147e-04
Generation 8180:	Best = 3.837204e+03	FES = 818100	Diversity = 6.9992e-04
Generation 8190:	Best = 3.837204e+03	FES = 819100	Diversity = 7.3281e-04
Generation 8200:	Best = 3.837202e+03	FES = 820100	Diversity = 7.2620e-04
Generation 8210:	Best = 3.837202e+03	FES = 821100	Diversity = 7.0880e-04
Generation 8220:	Best = 3.837199e+03	FES = 822100	Diversity = 7.3799e-04
Generation 8230:	Best = 3.837198e+03	FES = 823100	Diversity = 7.5938e-04
Generation 8240:	Best = 3.837198e+03	FES = 824100	Diversity = 7.6961e-04
Generation 8250:	Best = 3.837196e+03	FES = 825100	Diversity = 7.1909e-04
Generation 8260:	Best = 3.837196e+03	FES = 826100	Diversity = 7.4646e-04
Generation 8270:	Best = 3.837196e+03	FES = 827100	Diversity = 7.0680e-04
Generation 8280:	Best = 3.837194e+03	FES = 828100	Diversity = 6.4887e-04
Generation 8290:	Best = 3.837194e+03	FES = 829100	Diversity = 5.9008e-04
Generation 8300:	Best = 3.837193e+03	FES = 830100	Diversity = 5.8169e-04
Generation 8310:	Best = 3.837193e+03	FES = 831100	Diversity = 5.8662e-04
Generation 8320:	Best = 3.837192e+03	FES = 832100	Diversity = 5.5716e-04
Generation 8330:	Best = 3.837192e+03	FES = 833100	Diversity = 5.4919e-04
Generation 8340:	Best = 3.837192e+03	FES = 834100	Diversity = 5.4629e-04
Generation 8350:	Best = 3.837191e+03	FES = 835100	Diversity = 5.4626e-04
Generation 8360:	Best = 3.837191e+03	FES = 836100	Diversity = 5.2442e-04
Generation 8370:	Best = 3.837190e+03	FES = 837100	Diversity = 5.5067e-04
Generation 8380:	Best = 3.837190e+03	FES = 838100	Diversity = 5.6058e-04
Generation 8390:	Best = 3.837189e+03	FES = 839100	Diversity = 5.4695e-04
Generation 8400:	Best = 3.837188e+03	FES = 840100	Diversity = 5.3018e-04
Generation 8410:	Best = 3.837187e+03	FES = 841100	Diversity = 5.7967e-04
Generation 8420:	Best = 3.837186e+03	FES = 842100	Diversity = 6.2266e-04
Generation 8430:	Best = 3.837186e+03	FES = 843100	Diversity = 5.8885e-04
Generation 8440:	Best = 3.837186e+03	FES = 844100	Diversity = 5.7302e-04
Generation 8450:	Best = 3.837186e+03	FES = 845100	Diversity = 5.4739e-04
Generation 8460:	Best = 3.837185e+03	FES = 846100	Diversity = 5.3568e-04
Generation 8470:	Best = 3.837184e+03	FES = 847100	Diversity = 5.1930e-04
Generation 8480:	Best = 3.837183e+03	FES = 848100	Diversity = 4.9244e-04
Generation 8490:	Best = 3.837183e+03	FES = 849100	Diversity = 4.9853e-04
Generation 8500:	Best = 3.837182e+03	FES = 850100	Diversity = 4.9112e-04
Generation 8510:	Best = 3.837181e+03	FES = 851100	Diversity = 4.7616e-04
Generation 8520:	Best = 3.837181e+03	FES = 852100	Diversity = 5.3886e-04
Generation 8530:	Best = 3.837180e+03	FES = 853100	Diversity = 5.4551e-04

Generation 8540:	Best = 3.837180e+03	FES = 854100	Diversity = 5.1570e-04
Generation 8550:	Best = 3.837178e+03	FES = 855100	Diversity = 4.8684e-04
Generation 8560:	Best = 3.837178e+03	FES = 856100	Diversity = 5.0550e-04
Generation 8570:	Best = 3.837177e+03	FES = 857100	Diversity = 5.5755e-04
Generation 8580:	Best = 3.837176e+03	FES = 858100	Diversity = 6.1957e-04
Generation 8590:	Best = 3.837175e+03	FES = 859100	Diversity = 6.4019e-04
Generation 8600:	Best = 3.837174e+03	FES = 860100	Diversity = 6.5573e-04
Generation 8610:	Best = 3.837174e+03	FES = 861100	Diversity = 6.2598e-04
Generation 8620:	Best = 3.837172e+03	FES = 862100	Diversity = 6.0941e-04
Generation 8630:	Best = 3.837171e+03	FES = 863100	Diversity = 6.1763e-04
Generation 8640:	Best = 3.837171e+03	FES = 864100	Diversity = 6.4129e-04
Generation 8650:	Best = 3.837170e+03	FES = 865100	Diversity = 6.5216e-04
Generation 8660:	Best = 3.837169e+03	FES = 866100	Diversity = 6.5932e-04
Generation 8670:	Best = 3.837168e+03	FES = 867100	Diversity = 6.6310e-04
Generation 8680:	Best = 3.837168e+03	FES = 868100	Diversity = 6.9001e-04
Generation 8690:	Best = 3.837166e+03	FES = 869100	Diversity = 6.7394e-04
Generation 8700:	Best = 3.837165e+03	FES = 870100	Diversity = 7.2232e-04
Generation 8710:	Best = 3.837164e+03	FES = 871100	Diversity = 7.6065e-04
Generation 8720:	Best = 3.837163e+03	FES = 872100	Diversity = 7.4071e-04
Generation 8730:	Best = 3.837161e+03	FES = 873100	Diversity = 7.3733e-04
Generation 8740:	Best = 3.837161e+03	FES = 874100	Diversity = 6.5980e-04
Generation 8750:	Best = 3.837161e+03	FES = 875100	Diversity = 6.4592e-04
Generation 8760:	Best = 3.837161e+03	FES = 876100	Diversity = 6.1697e-04
Generation 8770:	Best = 3.837160e+03	FES = 877100	Diversity = 5.5889e-04
Generation 8780:	Best = 3.837160e+03	FES = 878100	Diversity = 5.6869e-04
Generation 8790:	Best = 3.837158e+03	FES = 879100	Diversity = 5.6629e-04
Generation 8800:	Best = 3.837157e+03	FES = 880100	Diversity = 6.0447e-04
Generation 8810:	Best = 3.837157e+03	FES = 881100	Diversity = 6.0824e-04
Generation 8820:	Best = 3.837156e+03	FES = 882100	Diversity = 6.5199e-04
Generation 8830:	Best = 3.837154e+03	FES = 883100	Diversity = 6.7080e-04
Generation 8840:	Best = 3.837153e+03	FES = 884100	Diversity = 6.6670e-04
Generation 8850:	Best = 3.837152e+03	FES = 885100	Diversity = 6.7677e-04
Generation 8860:	Best = 3.837151e+03	FES = 886100	Diversity = 6.7395e-04
Generation 8870:	Best = 3.837151e+03	FES = 887100	Diversity = 6.5279e-04
Generation 8880:	Best = 3.837148e+03	FES = 888100	Diversity = 7.2644e-04
Generation 8890:	Best = 3.837148e+03	FES = 889100	Diversity = 7.6662e-04
Generation 8900:	Best = 3.837148e+03	FES = 890100	Diversity = 7.4443e-04
Generation 8910:	Best = 3.837147e+03	FES = 891100	Diversity = 7.1979e-04
Generation 8920:	Best = 3.837147e+03	FES = 892100	Diversity = 6.6590e-04
Generation 8930:	Best = 3.837146e+03	FES = 893100	Diversity = 6.0909e-04
Generation 8940:	Best = 3.837146e+03	FES = 894100	Diversity = 5.7174e-04
Generation 8950:	Best = 3.837145e+03	FES = 895100	Diversity = 5.7730e-04
Generation 8960:	Best = 3.837144e+03	FES = 896100	Diversity = 6.1248e-04
Generation 8970:	Best = 3.837144e+03	FES = 897100	Diversity = 6.2925e-04
Generation 8980:	Best = 3.837143e+03	FES = 898100	Diversity = 6.1574e-04
Generation 8990:	Best = 3.837141e+03	FES = 899100	Diversity = 6.0115e-04
Generation 9000:	Best = 3.837140e+03	FES = 900100	Diversity = 6.6268e-04
Generation 9010:	Best = 3.837139e+03	FES = 901100	Diversity = 7.6057e-04
Generation 9020:	Best = 3.837138e+03	FES = 902100	Diversity = 7.5673e-04
Generation 9030:	Best = 3.837135e+03	FES = 903100	Diversity = 7.8622e-04
Generation 9040:	Best = 3.837135e+03	FES = 904100	Diversity = 8.4324e-04
Generation 9050:	Best = 3.837135e+03	FES = 905100	Diversity = 7.9680e-04
Generation 9060:	Best = 3.837134e+03	FES = 906100	Diversity = 7.5070e-04
Generation 9070:	Best = 3.837132e+03	FES = 907100	Diversity = 7.3335e-04

Generation 9080:	Best = 3.837132e+03	FES = 908100	Diversity = 7.8999e-04
Generation 9090:	Best = 3.837131e+03	FES = 909100	Diversity = 7.9270e-04
Generation 9100:	Best = 3.837129e+03	FES = 910100	Diversity = 7.8942e-04
Generation 9110:	Best = 3.837128e+03	FES = 911100	Diversity = 8.2633e-04
Generation 9120:	Best = 3.837127e+03	FES = 912100	Diversity = 8.1009e-04
Generation 9130:	Best = 3.837126e+03	FES = 913100	Diversity = 8.4632e-04
Generation 9140:	Best = 3.837125e+03	FES = 914100	Diversity = 9.0478e-04
Generation 9150:	Best = 3.837124e+03	FES = 915100	Diversity = 8.9525e-04
Generation 9160:	Best = 3.837123e+03	FES = 916100	Diversity = 8.2916e-04
Generation 9170:	Best = 3.837122e+03	FES = 917100	Diversity = 7.8461e-04
Generation 9180:	Best = 3.837120e+03	FES = 918100	Diversity = 8.2164e-04
Generation 9190:	Best = 3.837120e+03	FES = 919100	Diversity = 8.3303e-04
Generation 9200:	Best = 3.837118e+03	FES = 920100	Diversity = 8.1139e-04
Generation 9210:	Best = 3.837116e+03	FES = 921100	Diversity = 7.9929e-04
Generation 9220:	Best = 3.837115e+03	FES = 922100	Diversity = 7.9505e-04
Generation 9230:	Best = 3.837115e+03	FES = 923100	Diversity = 8.3443e-04
Generation 9240:	Best = 3.837112e+03	FES = 924100	Diversity = 8.6881e-04
Generation 9250:	Best = 3.837109e+03	FES = 925100	Diversity = 8.5117e-04
Generation 9260:	Best = 3.837109e+03	FES = 926100	Diversity = 9.3500e-04
Generation 9270:	Best = 3.837109e+03	FES = 927100	Diversity = 8.3883e-04
Generation 9280:	Best = 3.837107e+03	FES = 928100	Diversity = 8.2786e-04
Generation 9290:	Best = 3.837106e+03	FES = 929100	Diversity = 8.0503e-04
Generation 9300:	Best = 3.837104e+03	FES = 930100	Diversity = 7.3236e-04
Generation 9310:	Best = 3.837103e+03	FES = 931100	Diversity = 8.0585e-04
Generation 9320:	Best = 3.837102e+03	FES = 932100	Diversity = 8.3245e-04
Generation 9330:	Best = 3.837099e+03	FES = 933100	Diversity = 8.2814e-04
Generation 9340:	Best = 3.837097e+03	FES = 934100	Diversity = 8.5583e-04
Generation 9350:	Best = 3.837095e+03	FES = 935100	Diversity = 8.3206e-04
Generation 9360:	Best = 3.837093e+03	FES = 936100	Diversity = 8.5468e-04
Generation 9370:	Best = 3.837092e+03	FES = 937100	Diversity = 8.4019e-04
Generation 9380:	Best = 3.837088e+03	FES = 938100	Diversity = 8.4413e-04
Generation 9390:	Best = 3.837087e+03	FES = 939100	Diversity = 9.1375e-04
Generation 9400:	Best = 3.837086e+03	FES = 940100	Diversity = 9.1000e-04
Generation 9410:	Best = 3.837085e+03	FES = 941100	Diversity = 9.0331e-04
Generation 9420:	Best = 3.837084e+03	FES = 942100	Diversity = 8.6722e-04
Generation 9430:	Best = 3.837081e+03	FES = 943100	Diversity = 8.1302e-04
Generation 9440:	Best = 3.837081e+03	FES = 944100	Diversity = 8.9393e-04
Generation 9450:	Best = 3.837079e+03	FES = 945100	Diversity = 9.1844e-04
Generation 9460:	Best = 3.837079e+03	FES = 946100	Diversity = 8.6183e-04
Generation 9470:	Best = 3.837075e+03	FES = 947100	Diversity = 8.6519e-04
Generation 9480:	Best = 3.837073e+03	FES = 948100	Diversity = 8.4267e-04
Generation 9490:	Best = 3.837073e+03	FES = 949100	Diversity = 8.2693e-04
Generation 9500:	Best = 3.837071e+03	FES = 950100	Diversity = 7.9660e-04
Generation 9510:	Best = 3.837071e+03	FES = 951100	Diversity = 8.0403e-04
Generation 9520:	Best = 3.837069e+03	FES = 952100	Diversity = 7.9103e-04
Generation 9530:	Best = 3.837069e+03	FES = 953100	Diversity = 8.7581e-04
Generation 9540:	Best = 3.837068e+03	FES = 954100	Diversity = 8.3114e-04
Generation 9550:	Best = 3.837063e+03	FES = 955100	Diversity = 8.4477e-04
Generation 9560:	Best = 3.837063e+03	FES = 956100	Diversity = 9.3325e-04
Generation 9570:	Best = 3.837063e+03	FES = 957100	Diversity = 8.4374e-04
Generation 9580:	Best = 3.837063e+03	FES = 958100	Diversity = 8.5426e-04
Generation 9590:	Best = 3.837062e+03	FES = 959100	Diversity = 7.5813e-04
Generation 9600:	Best = 3.837058e+03	FES = 960100	Diversity = 7.3506e-04
Generation 9610:	Best = 3.837058e+03	FES = 961100	Diversity = 7.6460e-04

Generation 9620:	Best = 3.837056e+03	FES = 962100	Diversity = 7.8860e-04
Generation 9630:	Best = 3.837055e+03	FES = 963100	Diversity = 8.4799e-04
Generation 9640:	Best = 3.837053e+03	FES = 964100	Diversity = 8.5465e-04
Generation 9650:	Best = 3.837053e+03	FES = 965100	Diversity = 7.6691e-04
Generation 9660:	Best = 3.837049e+03	FES = 966100	Diversity = 7.0153e-04
Generation 9670:	Best = 3.837049e+03	FES = 967100	Diversity = 7.2862e-04
Generation 9680:	Best = 3.837047e+03	FES = 968100	Diversity = 7.3966e-04
Generation 9690:	Best = 3.837047e+03	FES = 969100	Diversity = 7.0627e-04
Generation 9700:	Best = 3.837045e+03	FES = 970100	Diversity = 7.0758e-04
Generation 9710:	Best = 3.837045e+03	FES = 971100	Diversity = 7.0961e-04
Generation 9720:	Best = 3.837045e+03	FES = 972100	Diversity = 6.9524e-04
Generation 9730:	Best = 3.837043e+03	FES = 973100	Diversity = 6.6843e-04
Generation 9740:	Best = 3.837040e+03	FES = 974100	Diversity = 6.4705e-04
Generation 9750:	Best = 3.837040e+03	FES = 975100	Diversity = 6.6257e-04
Generation 9760:	Best = 3.837038e+03	FES = 976100	Diversity = 6.5875e-04
Generation 9770:	Best = 3.837038e+03	FES = 977100	Diversity = 6.4261e-04
Generation 9780:	Best = 3.837037e+03	FES = 978100	Diversity = 6.4927e-04
Generation 9790:	Best = 3.837034e+03	FES = 979100	Diversity = 6.4264e-04
Generation 9800:	Best = 3.837032e+03	FES = 980100	Diversity = 6.1749e-04
Generation 9810:	Best = 3.837031e+03	FES = 981100	Diversity = 6.0206e-04
Generation 9820:	Best = 3.837031e+03	FES = 982100	Diversity = 6.7397e-04
Generation 9830:	Best = 3.837030e+03	FES = 983100	Diversity = 6.6652e-04
Generation 9840:	Best = 3.837030e+03	FES = 984100	Diversity = 6.6652e-04
Generation 9850:	Best = 3.837029e+03	FES = 985100	Diversity = 6.5582e-04
Generation 9860:	Best = 3.837028e+03	FES = 986100	Diversity = 6.3519e-04
Generation 9870:	Best = 3.837025e+03	FES = 987100	Diversity = 7.1144e-04
Generation 9880:	Best = 3.837024e+03	FES = 988100	Diversity = 6.9203e-04
Generation 9890:	Best = 3.837023e+03	FES = 989100	Diversity = 7.3995e-04
Generation 9900:	Best = 3.837020e+03	FES = 990100	Diversity = 7.5559e-04
Generation 9910:	Best = 3.837020e+03	FES = 991100	Diversity = 7.3984e-04
Generation 9920:	Best = 3.837019e+03	FES = 992100	Diversity = 7.1634e-04
Generation 9930:	Best = 3.837017e+03	FES = 993100	Diversity = 7.5107e-04
Generation 9940:	Best = 3.837016e+03	FES = 994100	Diversity = 8.1904e-04
Generation 9950:	Best = 3.837015e+03	FES = 995100	Diversity = 8.2733e-04
Generation 9960:	Best = 3.837013e+03	FES = 996100	Diversity = 8.1513e-04
Generation 9970:	Best = 3.837013e+03	FES = 997100	Diversity = 8.2498e-04
Generation 9980:	Best = 3.837009e+03	FES = 998100	Diversity = 8.6459e-04
Generation 9990:	Best = 3.837007e+03	FES = 999100	Diversity = 8.4687e-04

Optimization completed!

Final Best Fitness = 3.837005e+03

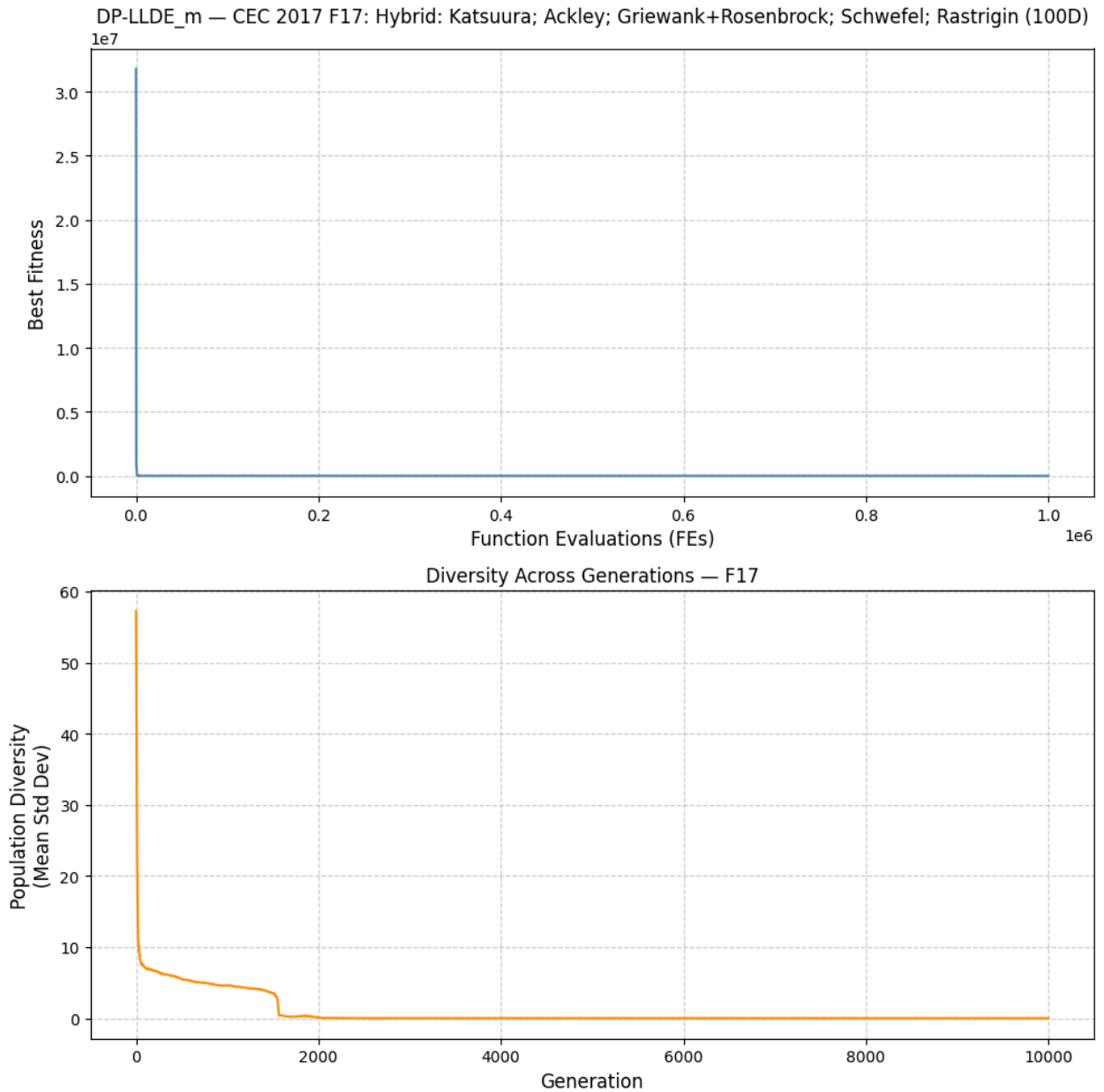
Total Function Evaluations = 1000000

```

=====
=
Results:
  Best Fitness           : 3.837005e+03
  Global Optimum         : 1.700000e+03
  Error (|best-optimum|) : 2.137005e+03
  Best Solution (5 dim)  : [-34.71413082  -4.71069519  16.13555616  15.81956618
6.95863875]
  D^(0) (init. diversity): 5.7220e+01
=====
=

```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results_2/all_results_checkpoint.pkl



Graph saved as convergence_F17.png

✓ Plot saved to: /content/drive/My Drive/thesis_results_2/plots/convergence_F17.png

```
In [ ]: # =====  
# F18 — Hybrid Function 8 (Elliptic; Ackley; Rastrigin; HGBat; Discus)  
# =====  
run_and_plot('F18', make_wrapper(hybrid.f18), 1800.0, 'Hybrid: Elliptic; Ackle
```

=====

=

DP-LLDE_m on CEC 2017 F18 – Single Run

=====

=

Function : Hybrid: Elliptic; Ackley; Rastrigin; HGBat; Discus (100D)

Bounds : [-100, 100]

Global optimum : 1800.0

Initial $D^{(0)} = 5.7220e+01$ | Diversity threshold = 5.7220e+00

Generation 0: Best Fitness = 1.395496e+09 | Diversity = 5.7220e+01

Generation 10: Best = 9.172835e+07 FES = 1100 Diversity = 3.4582e+01

Generation 20: Best = 2.864048e+07 FES = 2100 Diversity = 2.7741e+01

Generation 30: Best = 1.047296e+07 FES = 3100 Diversity = 1.9009e+01

Generation 40: Best = 5.553631e+06 FES = 4100 Diversity = 1.3963e+01

Generation 50: Best = 3.314962e+06 FES = 5100 Diversity = 9.7459e+00

Generation 60: Best = 2.742439e+06 FES = 6100 Diversity = 8.0387e+00

Generation 70: Best = 2.356076e+06 FES = 7100 Diversity = 7.3119e+00

Generation 80: Best = 1.875143e+06 FES = 8100 Diversity = 6.9614e+00

Generation 90: Best = 1.687895e+06 FES = 9100 Diversity = 5.6949e+00

Generation 100: Best = 1.593721e+06 FES = 10100 Diversity = 4.5174e+00

Generation 110: Best = 1.496641e+06 FES = 11100 Diversity = 4.0589e+00

Generation 120: Best = 1.434726e+06 FES = 12100 Diversity = 4.0481e+00

Generation 130: Best = 1.386570e+06 FES = 13100 Diversity = 3.7896e+00

Generation 140: Best = 1.348786e+06 FES = 14100 Diversity = 3.3259e+00

Generation 150: Best = 1.302995e+06 FES = 15100 Diversity = 3.2491e+00

Generation 160: Best = 1.282197e+06 FES = 16100 Diversity = 3.4799e+00

Generation 170: Best = 1.268608e+06 FES = 17100 Diversity = 2.9079e+00

Generation 180: Best = 1.254601e+06 FES = 18100 Diversity = 2.5980e+00

Generation 190: Best = 1.241162e+06 FES = 19100 Diversity = 2.2571e+00

Generation 200: Best = 1.218110e+06 FES = 20100 Diversity = 2.5066e+00

Generation 210: Best = 1.199783e+06 FES = 21100 Diversity = 2.8970e+00

Generation 220: Best = 1.193893e+06 FES = 22100 Diversity = 2.3354e+00

Generation 230: Best = 1.183501e+06 FES = 23100 Diversity = 2.1428e+00

Generation 240: Best = 1.175361e+06 FES = 24100 Diversity = 2.0876e+00

Generation 250: Best = 1.172557e+06 FES = 25100 Diversity = 2.2545e+00

Generation 260: Best = 1.166292e+06 FES = 26100 Diversity = 2.2080e+00

Generation 270: Best = 1.162571e+06 FES = 27100 Diversity = 1.9347e+00

Generation 280: Best = 1.155091e+06 FES = 28100 Diversity = 2.0894e+00

Generation 290: Best = 1.147740e+06 FES = 29100 Diversity = 2.0252e+00

Generation 300: Best = 1.143029e+06 FES = 30100 Diversity = 2.2329e+00

Generation 310: Best = 1.138572e+06 FES = 31100 Diversity = 2.2288e+00

Generation 320: Best = 1.129211e+06 FES = 32100 Diversity = 2.0025e+00

Generation 330: Best = 1.118260e+06 FES = 33100 Diversity = 2.3532e+00

Generation 340: Best = 1.109417e+06 FES = 34100 Diversity = 3.0284e+00

Generation 350: Best = 1.103893e+06 FES = 35100 Diversity = 2.3043e+00

Generation 360: Best = 1.098546e+06 FES = 36100 Diversity = 2.2323e+00

Generation 370: Best = 1.092002e+06 FES = 37100 Diversity = 2.1129e+00

Generation 380: Best = 1.088326e+06 FES = 38100 Diversity = 2.1014e+00

Generation 390: Best = 1.082593e+06 FES = 39100 Diversity = 1.8410e+00

Generation 400: Best = 1.080576e+06 FES = 40100 Diversity = 2.1135e+00

Generation 410: Best = 1.077636e+06 FES = 41100 Diversity = 1.9955e+00

Generation 420: Best = 1.072409e+06 FES = 42100 Diversity = 2.5921e+00

Generation 430: Best = 1.069184e+06 FES = 43100 Diversity = 3.0943e+00

Generation 440: Best = 1.059122e+06 FES = 44100 Diversity = 3.2624e+00

Generation	450:	Best = 1.052300e+06	FES = 45100	Diversity = 4.0531e+00
Generation	460:	Best = 1.044039e+06	FES = 46100	Diversity = 4.5629e+00
Generation	470:	Best = 1.038457e+06	FES = 47100	Diversity = 4.8341e+00
Generation	480:	Best = 1.034048e+06	FES = 48100	Diversity = 4.8506e+00
Generation	490:	Best = 1.024290e+06	FES = 49100	Diversity = 4.8020e+00
Generation	500:	Best = 1.021443e+06	FES = 50100	Diversity = 4.7238e+00
Generation	510:	Best = 1.012985e+06	FES = 51100	Diversity = 4.2863e+00
Generation	520:	Best = 1.011616e+06	FES = 52100	Diversity = 4.3396e+00
Generation	530:	Best = 1.009291e+06	FES = 53100	Diversity = 3.9554e+00
Generation	540:	Best = 1.004276e+06	FES = 54100	Diversity = 3.9476e+00
Generation	550:	Best = 1.000116e+06	FES = 55100	Diversity = 4.1300e+00
Generation	560:	Best = 9.989107e+05	FES = 56100	Diversity = 3.8413e+00
Generation	570:	Best = 9.954172e+05	FES = 57100	Diversity = 3.0028e+00
Generation	580:	Best = 9.881508e+05	FES = 58100	Diversity = 3.6908e+00
Generation	590:	Best = 9.858304e+05	FES = 59100	Diversity = 4.1797e+00
Generation	600:	Best = 9.810794e+05	FES = 60100	Diversity = 4.5755e+00
Generation	610:	Best = 9.789058e+05	FES = 61100	Diversity = 4.8993e+00
Generation	620:	Best = 9.739999e+05	FES = 62100	Diversity = 4.7936e+00
Generation	630:	Best = 9.727231e+05	FES = 63100	Diversity = 4.7682e+00
Generation	640:	Best = 9.714246e+05	FES = 64100	Diversity = 4.5313e+00
Generation	650:	Best = 9.681571e+05	FES = 65100	Diversity = 5.2069e+00
Generation	660:	Best = 9.656886e+05	FES = 66100	Diversity = 5.2345e+00
Generation	670:	Best = 9.585997e+05	FES = 67100	Diversity = 5.2324e+00
Generation	680:	Best = 9.568485e+05	FES = 68100	Diversity = 5.3764e+00
Generation	690:	Best = 9.535183e+05	FES = 69100	Diversity = 5.7399e+00
Generation	700:	Best = 9.487920e+05	FES = 70100	Diversity = 6.0701e+00
Generation	710:	Best = 9.467235e+05	FES = 71100	Diversity = 5.9398e+00
Generation	720:	Best = 9.426017e+05	FES = 72100	Diversity = 6.3659e+00
Generation	730:	Best = 9.361773e+05	FES = 73100	Diversity = 6.8891e+00
Generation	740:	Best = 9.361192e+05	FES = 74100	Diversity = 7.5833e+00
Generation	750:	Best = 9.327875e+05	FES = 75100	Diversity = 7.6127e+00
Generation	760:	Best = 9.283633e+05	FES = 76100	Diversity = 7.2389e+00
Generation	770:	Best = 9.249168e+05	FES = 77100	Diversity = 7.4080e+00
Generation	780:	Best = 9.230232e+05	FES = 78100	Diversity = 7.0801e+00
Generation	790:	Best = 9.219057e+05	FES = 79100	Diversity = 6.3082e+00
Generation	800:	Best = 9.175822e+05	FES = 80100	Diversity = 6.3080e+00
Generation	810:	Best = 9.157804e+05	FES = 81100	Diversity = 6.1430e+00
Generation	820:	Best = 9.142911e+05	FES = 82100	Diversity = 6.5802e+00
Generation	830:	Best = 9.118817e+05	FES = 83100	Diversity = 6.9772e+00
Generation	840:	Best = 9.084946e+05	FES = 84100	Diversity = 6.7418e+00
Generation	850:	Best = 9.064421e+05	FES = 85100	Diversity = 7.0732e+00
Generation	860:	Best = 9.034281e+05	FES = 86100	Diversity = 6.6896e+00
Generation	870:	Best = 8.969877e+05	FES = 87100	Diversity = 6.0150e+00
Generation	880:	Best = 8.966834e+05	FES = 88100	Diversity = 5.9546e+00
Generation	890:	Best = 8.922109e+05	FES = 89100	Diversity = 5.2290e+00
Generation	900:	Best = 8.869787e+05	FES = 90100	Diversity = 5.2521e+00
Generation	910:	Best = 8.853927e+05	FES = 91100	Diversity = 5.9596e+00
Generation	920:	Best = 8.825713e+05	FES = 92100	Diversity = 5.7913e+00
Generation	930:	Best = 8.800874e+05	FES = 93100	Diversity = 5.8386e+00
Generation	940:	Best = 8.785904e+05	FES = 94100	Diversity = 6.5432e+00
Generation	950:	Best = 8.776278e+05	FES = 95100	Diversity = 6.2847e+00
Generation	960:	Best = 8.756874e+05	FES = 96100	Diversity = 6.6188e+00
Generation	970:	Best = 8.736817e+05	FES = 97100	Diversity = 7.1169e+00
Generation	980:	Best = 8.691013e+05	FES = 98100	Diversity = 6.8210e+00

Generation 990:	Best = 8.678831e+05	FES = 99100	Diversity = 6.7715e+00
Generation 1000:	Best = 8.660311e+05	FES = 100100	Diversity = 5.6809e+00
Generation 1010:	Best = 8.648086e+05	FES = 101100	Diversity = 5.5752e+00
Generation 1020:	Best = 8.636128e+05	FES = 102100	Diversity = 6.0782e+00
Generation 1030:	Best = 8.620383e+05	FES = 103100	Diversity = 5.6903e+00
Generation 1040:	Best = 8.609477e+05	FES = 104100	Diversity = 5.8023e+00
Generation 1050:	Best = 8.598149e+05	FES = 105100	Diversity = 5.2573e+00
Generation 1060:	Best = 8.580428e+05	FES = 106100	Diversity = 5.1384e+00
Generation 1070:	Best = 8.555763e+05	FES = 107100	Diversity = 5.6851e+00
Generation 1080:	Best = 8.542535e+05	FES = 108100	Diversity = 5.5645e+00
Generation 1090:	Best = 8.535592e+05	FES = 109100	Diversity = 5.4094e+00
Generation 1100:	Best = 8.506549e+05	FES = 110100	Diversity = 5.9046e+00
Generation 1110:	Best = 8.502206e+05	FES = 111100	Diversity = 6.7991e+00
Generation 1120:	Best = 8.491190e+05	FES = 112100	Diversity = 6.4240e+00
Generation 1130:	Best = 8.476738e+05	FES = 113100	Diversity = 6.4959e+00
Generation 1140:	Best = 8.462534e+05	FES = 114100	Diversity = 6.4740e+00
Generation 1150:	Best = 8.458909e+05	FES = 115100	Diversity = 6.6163e+00
Generation 1160:	Best = 8.436841e+05	FES = 116100	Diversity = 6.1717e+00
Generation 1170:	Best = 8.435515e+05	FES = 117100	Diversity = 7.0137e+00
Generation 1180:	Best = 8.422224e+05	FES = 118100	Diversity = 7.4620e+00
Generation 1190:	Best = 8.409074e+05	FES = 119100	Diversity = 6.4658e+00
Generation 1200:	Best = 8.399374e+05	FES = 120100	Diversity = 7.4402e+00
Generation 1210:	Best = 8.376564e+05	FES = 121100	Diversity = 7.8202e+00
Generation 1220:	Best = 8.364893e+05	FES = 122100	Diversity = 9.1207e+00
Generation 1230:	Best = 8.352195e+05	FES = 123100	Diversity = 7.8450e+00
Generation 1240:	Best = 8.324920e+05	FES = 124100	Diversity = 7.9400e+00
Generation 1250:	Best = 8.309805e+05	FES = 125100	Diversity = 7.3389e+00
Generation 1260:	Best = 8.304980e+05	FES = 126100	Diversity = 6.7360e+00
Generation 1270:	Best = 8.289180e+05	FES = 127100	Diversity = 6.5202e+00
Generation 1280:	Best = 8.266662e+05	FES = 128100	Diversity = 6.6106e+00
Generation 1290:	Best = 8.263896e+05	FES = 129100	Diversity = 6.9720e+00
Generation 1300:	Best = 8.233153e+05	FES = 130100	Diversity = 6.2547e+00
Generation 1310:	Best = 8.229464e+05	FES = 131100	Diversity = 7.0326e+00
Generation 1320:	Best = 8.219579e+05	FES = 132100	Diversity = 6.4908e+00
Generation 1330:	Best = 8.203679e+05	FES = 133100	Diversity = 6.5753e+00
Generation 1340:	Best = 8.179936e+05	FES = 134100	Diversity = 6.7997e+00
Generation 1350:	Best = 8.169059e+05	FES = 135100	Diversity = 6.8969e+00
Generation 1360:	Best = 8.153400e+05	FES = 136100	Diversity = 6.1567e+00
Generation 1370:	Best = 8.142046e+05	FES = 137100	Diversity = 6.3742e+00
Generation 1380:	Best = 8.133056e+05	FES = 138100	Diversity = 7.3779e+00
Generation 1390:	Best = 8.111377e+05	FES = 139100	Diversity = 7.5822e+00
Generation 1400:	Best = 8.091413e+05	FES = 140100	Diversity = 7.3324e+00
Generation 1410:	Best = 8.083638e+05	FES = 141100	Diversity = 6.6710e+00
Generation 1420:	Best = 8.070875e+05	FES = 142100	Diversity = 7.0416e+00
Generation 1430:	Best = 8.059288e+05	FES = 143100	Diversity = 6.4017e+00
Generation 1440:	Best = 8.051791e+05	FES = 144100	Diversity = 6.4596e+00
Generation 1450:	Best = 8.032428e+05	FES = 145100	Diversity = 7.6550e+00
Generation 1460:	Best = 8.022814e+05	FES = 146100	Diversity = 7.5831e+00
Generation 1470:	Best = 8.008515e+05	FES = 147100	Diversity = 6.3858e+00
Generation 1480:	Best = 7.998216e+05	FES = 148100	Diversity = 6.3454e+00
Generation 1490:	Best = 7.986337e+05	FES = 149100	Diversity = 6.4277e+00
Generation 1500:	Best = 7.972772e+05	FES = 150100	Diversity = 6.0989e+00
Generation 1510:	Best = 7.965501e+05	FES = 151100	Diversity = 6.2978e+00
Generation 1520:	Best = 7.944124e+05	FES = 152100	Diversity = 7.0486e+00

Generation 1530:	Best = 7.930891e+05	FES = 153100	Diversity = 6.6726e+00
Generation 1540:	Best = 7.911644e+05	FES = 154100	Diversity = 7.7487e+00
Generation 1550:	Best = 7.902148e+05	FES = 155100	Diversity = 7.3908e+00
Generation 1560:	Best = 7.880061e+05	FES = 156100	Diversity = 8.7903e+00
Generation 1570:	Best = 7.852818e+05	FES = 157100	Diversity = 8.1220e+00
Generation 1580:	Best = 7.841872e+05	FES = 158100	Diversity = 7.7001e+00
Generation 1590:	Best = 7.829129e+05	FES = 159100	Diversity = 7.0403e+00
Generation 1600:	Best = 7.813110e+05	FES = 160100	Diversity = 8.0303e+00
Generation 1610:	Best = 7.796176e+05	FES = 161100	Diversity = 7.4456e+00
Generation 1620:	Best = 7.787264e+05	FES = 162100	Diversity = 6.5656e+00
Generation 1630:	Best = 7.771036e+05	FES = 163100	Diversity = 6.5342e+00
Generation 1640:	Best = 7.760792e+05	FES = 164100	Diversity = 6.1025e+00
Generation 1650:	Best = 7.748082e+05	FES = 165100	Diversity = 7.6934e+00
Generation 1660:	Best = 7.738938e+05	FES = 166100	Diversity = 7.5568e+00
Generation 1670:	Best = 7.704465e+05	FES = 167100	Diversity = 7.3986e+00
Generation 1680:	Best = 7.676059e+05	FES = 168100	Diversity = 7.3497e+00
Generation 1690:	Best = 7.661601e+05	FES = 169100	Diversity = 8.5800e+00
Generation 1700:	Best = 7.656810e+05	FES = 170100	Diversity = 9.3223e+00
Generation 1710:	Best = 7.650222e+05	FES = 171100	Diversity = 8.8388e+00
Generation 1720:	Best = 7.635181e+05	FES = 172100	Diversity = 8.2230e+00
Generation 1730:	Best = 7.624632e+05	FES = 173100	Diversity = 7.5909e+00
Generation 1740:	Best = 7.619875e+05	FES = 174100	Diversity = 7.2518e+00
Generation 1750:	Best = 7.610055e+05	FES = 175100	Diversity = 7.2113e+00
Generation 1760:	Best = 7.603041e+05	FES = 176100	Diversity = 6.7895e+00
Generation 1770:	Best = 7.600682e+05	FES = 177100	Diversity = 5.5829e+00
Generation 1780:	Best = 7.591698e+05	FES = 178100	Diversity = 5.9422e+00
Generation 1790:	Best = 7.585498e+05	FES = 179100	Diversity = 5.8257e+00
Generation 1800:	Best = 7.577705e+05	FES = 180100	Diversity = 5.7287e+00
Generation 1810:	Best = 7.567780e+05	FES = 181100	Diversity = 6.0174e+00
Generation 1820:	Best = 7.554301e+05	FES = 182100	Diversity = 5.8794e+00
Generation 1830:	Best = 7.544874e+05	FES = 183100	Diversity = 6.2506e+00
Generation 1840:	Best = 7.540717e+05	FES = 184100	Diversity = 5.9011e+00
Generation 1850:	Best = 7.534567e+05	FES = 185100	Diversity = 5.9642e+00
Generation 1860:	Best = 7.526522e+05	FES = 186100	Diversity = 6.1265e+00
Generation 1870:	Best = 7.523301e+05	FES = 187100	Diversity = 5.5048e+00
Generation 1880:	Best = 7.511395e+05	FES = 188100	Diversity = 5.2210e+00
Generation 1890:	Best = 7.504988e+05	FES = 189100	Diversity = 5.3121e+00
Generation 1900:	Best = 7.496190e+05	FES = 190100	Diversity = 5.7748e+00
Generation 1910:	Best = 7.485969e+05	FES = 191100	Diversity = 6.0373e+00
Generation 1920:	Best = 7.476057e+05	FES = 192100	Diversity = 6.1585e+00
Generation 1930:	Best = 7.466350e+05	FES = 193100	Diversity = 6.7349e+00
Generation 1940:	Best = 7.452638e+05	FES = 194100	Diversity = 6.7200e+00
Generation 1950:	Best = 7.445279e+05	FES = 195100	Diversity = 7.1372e+00
Generation 1960:	Best = 7.423363e+05	FES = 196100	Diversity = 8.3930e+00
Generation 1970:	Best = 7.401687e+05	FES = 197100	Diversity = 8.9554e+00
Generation 1980:	Best = 7.395436e+05	FES = 198100	Diversity = 8.3320e+00
Generation 1990:	Best = 7.386371e+05	FES = 199100	Diversity = 7.8752e+00
Generation 2000:	Best = 7.381137e+05	FES = 200100	Diversity = 6.6211e+00
Generation 2010:	Best = 7.374932e+05	FES = 201100	Diversity = 6.5549e+00
Generation 2020:	Best = 7.365777e+05	FES = 202100	Diversity = 5.7306e+00
Generation 2030:	Best = 7.360260e+05	FES = 203100	Diversity = 5.8565e+00
Generation 2040:	Best = 7.353268e+05	FES = 204100	Diversity = 6.3718e+00
Generation 2050:	Best = 7.350620e+05	FES = 205100	Diversity = 6.8264e+00
Generation 2060:	Best = 7.346956e+05	FES = 206100	Diversity = 6.4389e+00

Generation 2070:	Best = 7.338442e+05	FES = 207100	Diversity = 5.4533e+00
Generation 2080:	Best = 7.328295e+05	FES = 208100	Diversity = 6.0595e+00
Generation 2090:	Best = 7.322216e+05	FES = 209100	Diversity = 6.3245e+00
Generation 2100:	Best = 7.303153e+05	FES = 210100	Diversity = 6.9549e+00
Generation 2110:	Best = 7.301292e+05	FES = 211100	Diversity = 7.5879e+00
Generation 2120:	Best = 7.293324e+05	FES = 212100	Diversity = 7.6141e+00
Generation 2130:	Best = 7.286698e+05	FES = 213100	Diversity = 7.9249e+00
Generation 2140:	Best = 7.277229e+05	FES = 214100	Diversity = 7.8729e+00
Generation 2150:	Best = 7.266285e+05	FES = 215100	Diversity = 7.5920e+00
Generation 2160:	Best = 7.260595e+05	FES = 216100	Diversity = 7.9554e+00
Generation 2170:	Best = 7.247528e+05	FES = 217100	Diversity = 7.7844e+00
Generation 2180:	Best = 7.238951e+05	FES = 218100	Diversity = 8.5727e+00
Generation 2190:	Best = 7.230348e+05	FES = 219100	Diversity = 7.7912e+00
Generation 2200:	Best = 7.221409e+05	FES = 220100	Diversity = 7.1268e+00
Generation 2210:	Best = 7.218876e+05	FES = 221100	Diversity = 6.1161e+00
Generation 2220:	Best = 7.213814e+05	FES = 222100	Diversity = 5.3777e+00
Generation 2230:	Best = 7.204935e+05	FES = 223100	Diversity = 5.2223e+00
Generation 2240:	Best = 7.200245e+05	FES = 224100	Diversity = 5.0544e+00
Generation 2250:	Best = 7.194612e+05	FES = 225100	Diversity = 5.3509e+00
Generation 2260:	Best = 7.187617e+05	FES = 226100	Diversity = 5.9519e+00
Generation 2270:	Best = 7.179430e+05	FES = 227100	Diversity = 6.4434e+00
Generation 2280:	Best = 7.172196e+05	FES = 228100	Diversity = 6.2450e+00
Generation 2290:	Best = 7.166892e+05	FES = 229100	Diversity = 7.0320e+00
Generation 2300:	Best = 7.153802e+05	FES = 230100	Diversity = 7.6364e+00
Generation 2310:	Best = 7.143936e+05	FES = 231100	Diversity = 7.4712e+00
Generation 2320:	Best = 7.137178e+05	FES = 232100	Diversity = 7.6077e+00
Generation 2330:	Best = 7.123544e+05	FES = 233100	Diversity = 8.2951e+00
Generation 2340:	Best = 7.111343e+05	FES = 234100	Diversity = 7.3430e+00
Generation 2350:	Best = 7.100470e+05	FES = 235100	Diversity = 7.4485e+00
Generation 2360:	Best = 7.090916e+05	FES = 236100	Diversity = 7.0212e+00
Generation 2370:	Best = 7.083904e+05	FES = 237100	Diversity = 6.5524e+00
Generation 2380:	Best = 7.072990e+05	FES = 238100	Diversity = 6.8282e+00
Generation 2390:	Best = 7.065804e+05	FES = 239100	Diversity = 7.3023e+00
Generation 2400:	Best = 7.058489e+05	FES = 240100	Diversity = 7.5967e+00
Generation 2410:	Best = 7.048256e+05	FES = 241100	Diversity = 6.9479e+00
Generation 2420:	Best = 7.035453e+05	FES = 242100	Diversity = 6.9046e+00
Generation 2430:	Best = 7.029508e+05	FES = 243100	Diversity = 7.1809e+00
Generation 2440:	Best = 7.021430e+05	FES = 244100	Diversity = 7.2479e+00
Generation 2450:	Best = 7.015854e+05	FES = 245100	Diversity = 7.5305e+00
Generation 2460:	Best = 7.007060e+05	FES = 246100	Diversity = 7.5956e+00
Generation 2470:	Best = 7.002566e+05	FES = 247100	Diversity = 7.3641e+00
Generation 2480:	Best = 6.987433e+05	FES = 248100	Diversity = 7.1412e+00
Generation 2490:	Best = 6.984803e+05	FES = 249100	Diversity = 7.3942e+00
Generation 2500:	Best = 6.982206e+05	FES = 250100	Diversity = 7.0740e+00
Generation 2510:	Best = 6.975952e+05	FES = 251100	Diversity = 6.6485e+00
Generation 2520:	Best = 6.967573e+05	FES = 252100	Diversity = 6.1405e+00
Generation 2530:	Best = 6.956425e+05	FES = 253100	Diversity = 6.0134e+00
Generation 2540:	Best = 6.949657e+05	FES = 254100	Diversity = 6.6126e+00
Generation 2550:	Best = 6.943059e+05	FES = 255100	Diversity = 7.2932e+00
Generation 2560:	Best = 6.933423e+05	FES = 256100	Diversity = 6.8368e+00
Generation 2570:	Best = 6.924978e+05	FES = 257100	Diversity = 6.6857e+00
Generation 2580:	Best = 6.918309e+05	FES = 258100	Diversity = 5.5563e+00
Generation 2590:	Best = 6.914237e+05	FES = 259100	Diversity = 5.5888e+00
Generation 2600:	Best = 6.905440e+05	FES = 260100	Diversity = 5.7768e+00

Generation 2610:	Best = 6.890702e+05	FES = 261100	Diversity = 5.8879e+00
Generation 2620:	Best = 6.880168e+05	FES = 262100	Diversity = 6.6707e+00
Generation 2630:	Best = 6.871697e+05	FES = 263100	Diversity = 6.1548e+00
Generation 2640:	Best = 6.859104e+05	FES = 264100	Diversity = 5.8963e+00
Generation 2650:	Best = 6.850068e+05	FES = 265100	Diversity = 6.3091e+00
Generation 2660:	Best = 6.839617e+05	FES = 266100	Diversity = 6.1653e+00
Generation 2670:	Best = 6.832563e+05	FES = 267100	Diversity = 5.8393e+00
Generation 2680:	Best = 6.827561e+05	FES = 268100	Diversity = 4.9784e+00
Generation 2690:	Best = 6.822789e+05	FES = 269100	Diversity = 5.4311e+00
Generation 2700:	Best = 6.818887e+05	FES = 270100	Diversity = 4.8857e+00
Generation 2710:	Best = 6.816060e+05	FES = 271100	Diversity = 4.7920e+00
Generation 2720:	Best = 6.813973e+05	FES = 272100	Diversity = 4.9002e+00
Generation 2730:	Best = 6.809693e+05	FES = 273100	Diversity = 4.6221e+00
Generation 2740:	Best = 6.805048e+05	FES = 274100	Diversity = 4.7328e+00
Generation 2750:	Best = 6.800068e+05	FES = 275100	Diversity = 4.9617e+00
Generation 2760:	Best = 6.794091e+05	FES = 276100	Diversity = 5.0700e+00
Generation 2770:	Best = 6.786776e+05	FES = 277100	Diversity = 4.8822e+00
Generation 2780:	Best = 6.785398e+05	FES = 278100	Diversity = 5.3777e+00
Generation 2790:	Best = 6.776371e+05	FES = 279100	Diversity = 5.7065e+00
Generation 2800:	Best = 6.768431e+05	FES = 280100	Diversity = 6.2878e+00
Generation 2810:	Best = 6.762486e+05	FES = 281100	Diversity = 6.1169e+00
Generation 2820:	Best = 6.757163e+05	FES = 282100	Diversity = 5.3658e+00
Generation 2830:	Best = 6.750074e+05	FES = 283100	Diversity = 5.2029e+00
Generation 2840:	Best = 6.747307e+05	FES = 284100	Diversity = 5.8659e+00
Generation 2850:	Best = 6.739807e+05	FES = 285100	Diversity = 6.0298e+00
Generation 2860:	Best = 6.729598e+05	FES = 286100	Diversity = 5.8184e+00
Generation 2870:	Best = 6.728254e+05	FES = 287100	Diversity = 6.1448e+00
Generation 2880:	Best = 6.722162e+05	FES = 288100	Diversity = 5.7849e+00
Generation 2890:	Best = 6.718199e+05	FES = 289100	Diversity = 6.1657e+00
Generation 2900:	Best = 6.713623e+05	FES = 290100	Diversity = 5.9891e+00
Generation 2910:	Best = 6.711642e+05	FES = 291100	Diversity = 5.6958e+00
Generation 2920:	Best = 6.705559e+05	FES = 292100	Diversity = 5.9355e+00
Generation 2930:	Best = 6.700290e+05	FES = 293100	Diversity = 5.8559e+00
Generation 2940:	Best = 6.696132e+05	FES = 294100	Diversity = 5.6777e+00
Generation 2950:	Best = 6.682683e+05	FES = 295100	Diversity = 6.3129e+00
Generation 2960:	Best = 6.681679e+05	FES = 296100	Diversity = 6.5249e+00
Generation 2970:	Best = 6.674105e+05	FES = 297100	Diversity = 6.3529e+00
Generation 2980:	Best = 6.666027e+05	FES = 298100	Diversity = 7.0780e+00
Generation 2990:	Best = 6.663366e+05	FES = 299100	Diversity = 6.6228e+00
Generation 3000:	Best = 6.659311e+05	FES = 300100	Diversity = 5.8069e+00
Generation 3010:	Best = 6.653467e+05	FES = 301100	Diversity = 5.5120e+00
Generation 3020:	Best = 6.649679e+05	FES = 302100	Diversity = 5.1661e+00
Generation 3030:	Best = 6.645242e+05	FES = 303100	Diversity = 4.8805e+00
Generation 3040:	Best = 6.638038e+05	FES = 304100	Diversity = 5.4580e+00
Generation 3050:	Best = 6.632599e+05	FES = 305100	Diversity = 6.2293e+00
Generation 3060:	Best = 6.629015e+05	FES = 306100	Diversity = 5.8612e+00
Generation 3070:	Best = 6.625587e+05	FES = 307100	Diversity = 5.2076e+00
Generation 3080:	Best = 6.617470e+05	FES = 308100	Diversity = 4.9169e+00
Generation 3090:	Best = 6.617470e+05	FES = 309100	Diversity = 5.1511e+00
Generation 3100:	Best = 6.613916e+05	FES = 310100	Diversity = 4.6594e+00
Generation 3110:	Best = 6.608031e+05	FES = 311100	Diversity = 4.8153e+00
Generation 3120:	Best = 6.606728e+05	FES = 312100	Diversity = 5.3793e+00
Generation 3130:	Best = 6.600870e+05	FES = 313100	Diversity = 5.6871e+00
Generation 3140:	Best = 6.598283e+05	FES = 314100	Diversity = 5.9559e+00

Generation 3150:	Best = 6.591493e+05	FES = 315100	Diversity = 6.0320e+00
Generation 3160:	Best = 6.585925e+05	FES = 316100	Diversity = 5.8948e+00
Generation 3170:	Best = 6.579873e+05	FES = 317100	Diversity = 6.5819e+00
Generation 3180:	Best = 6.578498e+05	FES = 318100	Diversity = 6.8953e+00
Generation 3190:	Best = 6.569093e+05	FES = 319100	Diversity = 6.8110e+00
Generation 3200:	Best = 6.566511e+05	FES = 320100	Diversity = 6.1312e+00
Generation 3210:	Best = 6.559033e+05	FES = 321100	Diversity = 5.8524e+00
Generation 3220:	Best = 6.553513e+05	FES = 322100	Diversity = 6.0370e+00
Generation 3230:	Best = 6.551136e+05	FES = 323100	Diversity = 6.2039e+00
Generation 3240:	Best = 6.547311e+05	FES = 324100	Diversity = 5.0921e+00
Generation 3250:	Best = 6.544094e+05	FES = 325100	Diversity = 4.9639e+00
Generation 3260:	Best = 6.541962e+05	FES = 326100	Diversity = 5.1934e+00
Generation 3270:	Best = 6.538101e+05	FES = 327100	Diversity = 5.4389e+00
Generation 3280:	Best = 6.531958e+05	FES = 328100	Diversity = 5.3387e+00
Generation 3290:	Best = 6.522994e+05	FES = 329100	Diversity = 4.9115e+00
Generation 3300:	Best = 6.516611e+05	FES = 330100	Diversity = 5.6782e+00
Generation 3310:	Best = 6.510294e+05	FES = 331100	Diversity = 5.4016e+00
Generation 3320:	Best = 6.506592e+05	FES = 332100	Diversity = 5.4279e+00
Generation 3330:	Best = 6.498731e+05	FES = 333100	Diversity = 5.3732e+00
Generation 3340:	Best = 6.496121e+05	FES = 334100	Diversity = 5.5448e+00
Generation 3350:	Best = 6.489056e+05	FES = 335100	Diversity = 4.8519e+00
Generation 3360:	Best = 6.484026e+05	FES = 336100	Diversity = 5.1424e+00
Generation 3370:	Best = 6.480879e+05	FES = 337100	Diversity = 5.5023e+00
Generation 3380:	Best = 6.476462e+05	FES = 338100	Diversity = 5.0941e+00
Generation 3390:	Best = 6.473026e+05	FES = 339100	Diversity = 5.3253e+00
Generation 3400:	Best = 6.463409e+05	FES = 340100	Diversity = 4.8975e+00
Generation 3410:	Best = 6.460588e+05	FES = 341100	Diversity = 4.8271e+00
Generation 3420:	Best = 6.456937e+05	FES = 342100	Diversity = 4.7599e+00
Generation 3430:	Best = 6.452422e+05	FES = 343100	Diversity = 5.1101e+00
Generation 3440:	Best = 6.449485e+05	FES = 344100	Diversity = 5.0769e+00
Generation 3450:	Best = 6.446236e+05	FES = 345100	Diversity = 4.8445e+00
Generation 3460:	Best = 6.434927e+05	FES = 346100	Diversity = 4.5803e+00
Generation 3470:	Best = 6.431929e+05	FES = 347100	Diversity = 4.7558e+00
Generation 3480:	Best = 6.424942e+05	FES = 348100	Diversity = 4.9346e+00
Generation 3490:	Best = 6.417125e+05	FES = 349100	Diversity = 5.9767e+00
Generation 3500:	Best = 6.399217e+05	FES = 350100	Diversity = 5.5883e+00
Generation 3510:	Best = 6.395988e+05	FES = 351100	Diversity = 5.7632e+00
Generation 3520:	Best = 6.386386e+05	FES = 352100	Diversity = 5.7613e+00
Generation 3530:	Best = 6.380827e+05	FES = 353100	Diversity = 5.9202e+00
Generation 3540:	Best = 6.379409e+05	FES = 354100	Diversity = 5.4183e+00
Generation 3550:	Best = 6.372773e+05	FES = 355100	Diversity = 5.2517e+00
Generation 3560:	Best = 6.368170e+05	FES = 356100	Diversity = 5.4492e+00
Generation 3570:	Best = 6.365356e+05	FES = 357100	Diversity = 5.2059e+00
Generation 3580:	Best = 6.349115e+05	FES = 358100	Diversity = 5.2648e+00
Generation 3590:	Best = 6.346215e+05	FES = 359100	Diversity = 5.3359e+00
Generation 3600:	Best = 6.343001e+05	FES = 360100	Diversity = 5.1628e+00
Generation 3610:	Best = 6.337336e+05	FES = 361100	Diversity = 4.7542e+00
Generation 3620:	Best = 6.329060e+05	FES = 362100	Diversity = 4.8765e+00
Generation 3630:	Best = 6.322571e+05	FES = 363100	Diversity = 5.0810e+00
Generation 3640:	Best = 6.317053e+05	FES = 364100	Diversity = 5.3379e+00
Generation 3650:	Best = 6.315185e+05	FES = 365100	Diversity = 5.9207e+00
Generation 3660:	Best = 6.307956e+05	FES = 366100	Diversity = 5.5768e+00
Generation 3670:	Best = 6.302444e+05	FES = 367100	Diversity = 5.7471e+00
Generation 3680:	Best = 6.298596e+05	FES = 368100	Diversity = 5.5188e+00

Generation 3690:	Best = 6.291748e+05	FES = 369100	Diversity = 4.7686e+00
Generation 3700:	Best = 6.284250e+05	FES = 370100	Diversity = 5.2484e+00
Generation 3710:	Best = 6.280056e+05	FES = 371100	Diversity = 5.6607e+00
Generation 3720:	Best = 6.272716e+05	FES = 372100	Diversity = 5.6551e+00
Generation 3730:	Best = 6.269105e+05	FES = 373100	Diversity = 5.7366e+00
Generation 3740:	Best = 6.263647e+05	FES = 374100	Diversity = 5.6869e+00
Generation 3750:	Best = 6.260116e+05	FES = 375100	Diversity = 4.8023e+00
Generation 3760:	Best = 6.255772e+05	FES = 376100	Diversity = 4.6413e+00
Generation 3770:	Best = 6.251575e+05	FES = 377100	Diversity = 4.7286e+00
Generation 3780:	Best = 6.247774e+05	FES = 378100	Diversity = 5.1802e+00
Generation 3790:	Best = 6.246401e+05	FES = 379100	Diversity = 4.7948e+00
Generation 3800:	Best = 6.242077e+05	FES = 380100	Diversity = 4.8429e+00
Generation 3810:	Best = 6.237498e+05	FES = 381100	Diversity = 5.0563e+00
Generation 3820:	Best = 6.233972e+05	FES = 382100	Diversity = 5.0405e+00
Generation 3830:	Best = 6.228432e+05	FES = 383100	Diversity = 4.9266e+00
Generation 3840:	Best = 6.225824e+05	FES = 384100	Diversity = 5.1808e+00
Generation 3850:	Best = 6.223242e+05	FES = 385100	Diversity = 5.4589e+00
Generation 3860:	Best = 6.219701e+05	FES = 386100	Diversity = 5.6950e+00
Generation 3870:	Best = 6.215062e+05	FES = 387100	Diversity = 5.5550e+00
Generation 3880:	Best = 6.212926e+05	FES = 388100	Diversity = 5.2143e+00
Generation 3890:	Best = 6.208437e+05	FES = 389100	Diversity = 5.1034e+00
Generation 3900:	Best = 6.203778e+05	FES = 390100	Diversity = 5.5332e+00
Generation 3910:	Best = 6.201659e+05	FES = 391100	Diversity = 5.5288e+00
Generation 3920:	Best = 6.198953e+05	FES = 392100	Diversity = 5.2658e+00
Generation 3930:	Best = 6.196122e+05	FES = 393100	Diversity = 5.1368e+00
Generation 3940:	Best = 6.193818e+05	FES = 394100	Diversity = 4.6137e+00
Generation 3950:	Best = 6.191448e+05	FES = 395100	Diversity = 4.5710e+00
Generation 3960:	Best = 6.188133e+05	FES = 396100	Diversity = 5.1194e+00
Generation 3970:	Best = 6.185580e+05	FES = 397100	Diversity = 5.2535e+00
Generation 3980:	Best = 6.182041e+05	FES = 398100	Diversity = 5.2220e+00
Generation 3990:	Best = 6.177621e+05	FES = 399100	Diversity = 4.7778e+00
Generation 4000:	Best = 6.175051e+05	FES = 400100	Diversity = 4.7973e+00
Generation 4010:	Best = 6.173302e+05	FES = 401100	Diversity = 4.7669e+00
Generation 4020:	Best = 6.169872e+05	FES = 402100	Diversity = 5.0559e+00
Generation 4030:	Best = 6.168228e+05	FES = 403100	Diversity = 4.8961e+00
Generation 4040:	Best = 6.165920e+05	FES = 404100	Diversity = 4.7285e+00
Generation 4050:	Best = 6.161275e+05	FES = 405100	Diversity = 4.9112e+00
Generation 4060:	Best = 6.158232e+05	FES = 406100	Diversity = 5.2301e+00
Generation 4070:	Best = 6.155763e+05	FES = 407100	Diversity = 5.4824e+00
Generation 4080:	Best = 6.150243e+05	FES = 408100	Diversity = 5.5946e+00
Generation 4090:	Best = 6.147651e+05	FES = 409100	Diversity = 5.3882e+00
Generation 4100:	Best = 6.142641e+05	FES = 410100	Diversity = 5.2122e+00
Generation 4110:	Best = 6.134215e+05	FES = 411100	Diversity = 5.3137e+00
Generation 4120:	Best = 6.131881e+05	FES = 412100	Diversity = 5.3514e+00
Generation 4130:	Best = 6.130719e+05	FES = 413100	Diversity = 5.1330e+00
Generation 4140:	Best = 6.125352e+05	FES = 414100	Diversity = 5.0558e+00
Generation 4150:	Best = 6.123344e+05	FES = 415100	Diversity = 4.9940e+00
Generation 4160:	Best = 6.118212e+05	FES = 416100	Diversity = 4.5412e+00
Generation 4170:	Best = 6.112733e+05	FES = 417100	Diversity = 5.2453e+00
Generation 4180:	Best = 6.109110e+05	FES = 418100	Diversity = 5.1903e+00
Generation 4190:	Best = 6.102464e+05	FES = 419100	Diversity = 5.2593e+00
Generation 4200:	Best = 6.097943e+05	FES = 420100	Diversity = 6.1565e+00
Generation 4210:	Best = 6.091547e+05	FES = 421100	Diversity = 5.9026e+00
Generation 4220:	Best = 6.084017e+05	FES = 422100	Diversity = 6.1943e+00

Generation 4230:	Best = 6.076807e+05	FES = 423100	Diversity = 5.9422e+00
Generation 4240:	Best = 6.070785e+05	FES = 424100	Diversity = 5.7473e+00
Generation 4250:	Best = 6.061392e+05	FES = 425100	Diversity = 6.2507e+00
Generation 4260:	Best = 6.054836e+05	FES = 426100	Diversity = 6.4316e+00
Generation 4270:	Best = 6.048364e+05	FES = 427100	Diversity = 5.6142e+00
Generation 4280:	Best = 6.041910e+05	FES = 428100	Diversity = 5.7068e+00
Generation 4290:	Best = 6.035636e+05	FES = 429100	Diversity = 5.8808e+00
Generation 4300:	Best = 6.031622e+05	FES = 430100	Diversity = 5.4029e+00
Generation 4310:	Best = 6.025001e+05	FES = 431100	Diversity = 5.1923e+00
Generation 4320:	Best = 6.022471e+05	FES = 432100	Diversity = 4.9757e+00
Generation 4330:	Best = 6.016825e+05	FES = 433100	Diversity = 4.8877e+00
Generation 4340:	Best = 6.013294e+05	FES = 434100	Diversity = 4.5751e+00
Generation 4350:	Best = 6.009464e+05	FES = 435100	Diversity = 4.3055e+00
Generation 4360:	Best = 6.005412e+05	FES = 436100	Diversity = 4.5468e+00
Generation 4370:	Best = 6.001828e+05	FES = 437100	Diversity = 4.2872e+00
Generation 4380:	Best = 5.995504e+05	FES = 438100	Diversity = 4.2234e+00
Generation 4390:	Best = 5.991413e+05	FES = 439100	Diversity = 4.4255e+00
Generation 4400:	Best = 5.986316e+05	FES = 440100	Diversity = 4.4741e+00
Generation 4410:	Best = 5.981211e+05	FES = 441100	Diversity = 4.5186e+00
Generation 4420:	Best = 5.977457e+05	FES = 442100	Diversity = 4.7971e+00
Generation 4430:	Best = 5.972550e+05	FES = 443100	Diversity = 4.5507e+00
Generation 4440:	Best = 5.966532e+05	FES = 444100	Diversity = 4.8120e+00
Generation 4450:	Best = 5.962015e+05	FES = 445100	Diversity = 4.3663e+00
Generation 4460:	Best = 5.958486e+05	FES = 446100	Diversity = 4.7353e+00
Generation 4470:	Best = 5.955535e+05	FES = 447100	Diversity = 5.2020e+00
Generation 4480:	Best = 5.947978e+05	FES = 448100	Diversity = 5.0995e+00
Generation 4490:	Best = 5.942634e+05	FES = 449100	Diversity = 4.9609e+00
Generation 4500:	Best = 5.930306e+05	FES = 450100	Diversity = 5.5533e+00
Generation 4510:	Best = 5.924946e+05	FES = 451100	Diversity = 6.0672e+00
Generation 4520:	Best = 5.920961e+05	FES = 452100	Diversity = 5.9813e+00
Generation 4530:	Best = 5.916408e+05	FES = 453100	Diversity = 5.4917e+00
Generation 4540:	Best = 5.914767e+05	FES = 454100	Diversity = 5.2276e+00
Generation 4550:	Best = 5.908601e+05	FES = 455100	Diversity = 5.3658e+00
Generation 4560:	Best = 5.903878e+05	FES = 456100	Diversity = 5.4285e+00
Generation 4570:	Best = 5.899778e+05	FES = 457100	Diversity = 5.1769e+00
Generation 4580:	Best = 5.894730e+05	FES = 458100	Diversity = 5.5031e+00
Generation 4590:	Best = 5.886446e+05	FES = 459100	Diversity = 6.1128e+00
Generation 4600:	Best = 5.882432e+05	FES = 460100	Diversity = 5.5995e+00
Generation 4610:	Best = 5.877041e+05	FES = 461100	Diversity = 5.7297e+00
Generation 4620:	Best = 5.872147e+05	FES = 462100	Diversity = 5.3391e+00
Generation 4630:	Best = 5.868610e+05	FES = 463100	Diversity = 4.5900e+00
Generation 4640:	Best = 5.864613e+05	FES = 464100	Diversity = 4.3983e+00
Generation 4650:	Best = 5.860752e+05	FES = 465100	Diversity = 4.0696e+00
Generation 4660:	Best = 5.857598e+05	FES = 466100	Diversity = 4.9332e+00
Generation 4670:	Best = 5.852598e+05	FES = 467100	Diversity = 5.0914e+00
Generation 4680:	Best = 5.847135e+05	FES = 468100	Diversity = 5.7949e+00
Generation 4690:	Best = 5.845537e+05	FES = 469100	Diversity = 5.7081e+00
Generation 4700:	Best = 5.839753e+05	FES = 470100	Diversity = 5.7555e+00
Generation 4710:	Best = 5.834478e+05	FES = 471100	Diversity = 5.4821e+00
Generation 4720:	Best = 5.828374e+05	FES = 472100	Diversity = 5.7056e+00
Generation 4730:	Best = 5.814526e+05	FES = 473100	Diversity = 5.9628e+00
Generation 4740:	Best = 5.809390e+05	FES = 474100	Diversity = 5.6560e+00
Generation 4750:	Best = 5.803807e+05	FES = 475100	Diversity = 5.2131e+00
Generation 4760:	Best = 5.799616e+05	FES = 476100	Diversity = 4.5628e+00

Generation 4770:	Best = 5.795167e+05	FES = 477100	Diversity = 3.8475e+00
Generation 4780:	Best = 5.788272e+05	FES = 478100	Diversity = 3.4731e+00
Generation 4790:	Best = 5.784211e+05	FES = 479100	Diversity = 3.5575e+00
Generation 4800:	Best = 5.776579e+05	FES = 480100	Diversity = 3.4358e+00
Generation 4810:	Best = 5.771947e+05	FES = 481100	Diversity = 3.7612e+00
Generation 4820:	Best = 5.766930e+05	FES = 482100	Diversity = 4.1399e+00
Generation 4830:	Best = 5.761631e+05	FES = 483100	Diversity = 4.3523e+00
Generation 4840:	Best = 5.758739e+05	FES = 484100	Diversity = 4.2969e+00
Generation 4850:	Best = 5.754300e+05	FES = 485100	Diversity = 4.7326e+00
Generation 4860:	Best = 5.745504e+05	FES = 486100	Diversity = 4.4685e+00
Generation 4870:	Best = 5.743883e+05	FES = 487100	Diversity = 4.4570e+00
Generation 4880:	Best = 5.739204e+05	FES = 488100	Diversity = 4.1913e+00
Generation 4890:	Best = 5.728055e+05	FES = 489100	Diversity = 4.6123e+00
Generation 4900:	Best = 5.725091e+05	FES = 490100	Diversity = 4.5062e+00
Generation 4910:	Best = 5.718033e+05	FES = 491100	Diversity = 3.9217e+00
Generation 4920:	Best = 5.713120e+05	FES = 492100	Diversity = 4.1817e+00
Generation 4930:	Best = 5.709108e+05	FES = 493100	Diversity = 4.0756e+00
Generation 4940:	Best = 5.704052e+05	FES = 494100	Diversity = 4.1385e+00
Generation 4950:	Best = 5.699856e+05	FES = 495100	Diversity = 4.1198e+00
Generation 4960:	Best = 5.698482e+05	FES = 496100	Diversity = 3.9488e+00
Generation 4970:	Best = 5.694683e+05	FES = 497100	Diversity = 4.0751e+00
Generation 4980:	Best = 5.689621e+05	FES = 498100	Diversity = 4.1568e+00
Generation 4990:	Best = 5.685786e+05	FES = 499100	Diversity = 4.2452e+00
Generation 5000:	Best = 5.678990e+05	FES = 500100	Diversity = 4.0405e+00
Generation 5010:	Best = 5.676212e+05	FES = 501100	Diversity = 3.8775e+00
Generation 5020:	Best = 5.674269e+05	FES = 502100	Diversity = 3.5354e+00
Generation 5030:	Best = 5.672369e+05	FES = 503100	Diversity = 3.3547e+00
Generation 5040:	Best = 5.668837e+05	FES = 504100	Diversity = 3.6609e+00
Generation 5050:	Best = 5.666739e+05	FES = 505100	Diversity = 3.6512e+00
Generation 5060:	Best = 5.664424e+05	FES = 506100	Diversity = 3.6726e+00
Generation 5070:	Best = 5.662115e+05	FES = 507100	Diversity = 3.8808e+00
Generation 5080:	Best = 5.660234e+05	FES = 508100	Diversity = 3.7825e+00
Generation 5090:	Best = 5.657745e+05	FES = 509100	Diversity = 4.0574e+00
Generation 5100:	Best = 5.648203e+05	FES = 510100	Diversity = 4.1387e+00
Generation 5110:	Best = 5.646696e+05	FES = 511100	Diversity = 4.6190e+00
Generation 5120:	Best = 5.643282e+05	FES = 512100	Diversity = 4.5809e+00
Generation 5130:	Best = 5.639257e+05	FES = 513100	Diversity = 4.5505e+00
Generation 5140:	Best = 5.635441e+05	FES = 514100	Diversity = 4.5969e+00
Generation 5150:	Best = 5.630567e+05	FES = 515100	Diversity = 4.1662e+00
Generation 5160:	Best = 5.627802e+05	FES = 516100	Diversity = 3.8508e+00
Generation 5170:	Best = 5.624417e+05	FES = 517100	Diversity = 3.6950e+00
Generation 5180:	Best = 5.616714e+05	FES = 518100	Diversity = 3.6689e+00
Generation 5190:	Best = 5.615182e+05	FES = 519100	Diversity = 3.4765e+00
Generation 5200:	Best = 5.612733e+05	FES = 520100	Diversity = 3.3509e+00
Generation 5210:	Best = 5.608062e+05	FES = 521100	Diversity = 3.5951e+00
Generation 5220:	Best = 5.606117e+05	FES = 522100	Diversity = 3.6160e+00
Generation 5230:	Best = 5.602008e+05	FES = 523100	Diversity = 3.6029e+00
Generation 5240:	Best = 5.600203e+05	FES = 524100	Diversity = 3.9049e+00
Generation 5250:	Best = 5.596308e+05	FES = 525100	Diversity = 3.7058e+00
Generation 5260:	Best = 5.591478e+05	FES = 526100	Diversity = 3.8001e+00
Generation 5270:	Best = 5.585998e+05	FES = 527100	Diversity = 4.6450e+00
Generation 5280:	Best = 5.578417e+05	FES = 528100	Diversity = 4.6552e+00
Generation 5290:	Best = 5.577215e+05	FES = 529100	Diversity = 4.4814e+00
Generation 5300:	Best = 5.572759e+05	FES = 530100	Diversity = 4.4951e+00

Generation 5310:	Best = 5.566405e+05	FES = 531100	Diversity = 4.2326e+00
Generation 5320:	Best = 5.562159e+05	FES = 532100	Diversity = 4.2827e+00
Generation 5330:	Best = 5.557310e+05	FES = 533100	Diversity = 4.1862e+00
Generation 5340:	Best = 5.550367e+05	FES = 534100	Diversity = 4.3244e+00
Generation 5350:	Best = 5.547297e+05	FES = 535100	Diversity = 4.0073e+00
Generation 5360:	Best = 5.545570e+05	FES = 536100	Diversity = 4.2960e+00
Generation 5370:	Best = 5.542325e+05	FES = 537100	Diversity = 4.4712e+00
Generation 5380:	Best = 5.538192e+05	FES = 538100	Diversity = 4.8470e+00
Generation 5390:	Best = 5.536238e+05	FES = 539100	Diversity = 5.1333e+00
Generation 5400:	Best = 5.533695e+05	FES = 540100	Diversity = 4.6091e+00
Generation 5410:	Best = 5.531375e+05	FES = 541100	Diversity = 4.8860e+00
Generation 5420:	Best = 5.527869e+05	FES = 542100	Diversity = 4.6091e+00
Generation 5430:	Best = 5.523013e+05	FES = 543100	Diversity = 4.9519e+00
Generation 5440:	Best = 5.520546e+05	FES = 544100	Diversity = 5.1652e+00
Generation 5450:	Best = 5.517520e+05	FES = 545100	Diversity = 4.9419e+00
Generation 5460:	Best = 5.514684e+05	FES = 546100	Diversity = 4.7208e+00
Generation 5470:	Best = 5.509807e+05	FES = 547100	Diversity = 5.6249e+00
Generation 5480:	Best = 5.506565e+05	FES = 548100	Diversity = 5.8014e+00
Generation 5490:	Best = 5.502186e+05	FES = 549100	Diversity = 5.5095e+00
Generation 5500:	Best = 5.494295e+05	FES = 550100	Diversity = 6.3556e+00
Generation 5510:	Best = 5.490940e+05	FES = 551100	Diversity = 5.7642e+00
Generation 5520:	Best = 5.488133e+05	FES = 552100	Diversity = 5.4949e+00
Generation 5530:	Best = 5.484570e+05	FES = 553100	Diversity = 5.6256e+00
Generation 5540:	Best = 5.482928e+05	FES = 554100	Diversity = 5.9394e+00
Generation 5550:	Best = 5.475429e+05	FES = 555100	Diversity = 5.8704e+00
Generation 5560:	Best = 5.468240e+05	FES = 556100	Diversity = 6.1691e+00
Generation 5570:	Best = 5.459970e+05	FES = 557100	Diversity = 5.8996e+00
Generation 5580:	Best = 5.455634e+05	FES = 558100	Diversity = 6.6242e+00
Generation 5590:	Best = 5.450636e+05	FES = 559100	Diversity = 6.1301e+00
Generation 5600:	Best = 5.448260e+05	FES = 560100	Diversity = 5.3524e+00
Generation 5610:	Best = 5.445579e+05	FES = 561100	Diversity = 4.7650e+00
Generation 5620:	Best = 5.441033e+05	FES = 562100	Diversity = 5.0918e+00
Generation 5630:	Best = 5.437975e+05	FES = 563100	Diversity = 5.6976e+00
Generation 5640:	Best = 5.434659e+05	FES = 564100	Diversity = 5.4078e+00
Generation 5650:	Best = 5.431693e+05	FES = 565100	Diversity = 5.3865e+00
Generation 5660:	Best = 5.425208e+05	FES = 566100	Diversity = 5.4338e+00
Generation 5670:	Best = 5.422023e+05	FES = 567100	Diversity = 5.3394e+00
Generation 5680:	Best = 5.417015e+05	FES = 568100	Diversity = 4.9720e+00
Generation 5690:	Best = 5.416092e+05	FES = 569100	Diversity = 4.9043e+00
Generation 5700:	Best = 5.410550e+05	FES = 570100	Diversity = 6.1726e+00
Generation 5710:	Best = 5.406731e+05	FES = 571100	Diversity = 6.0810e+00
Generation 5720:	Best = 5.398102e+05	FES = 572100	Diversity = 6.0562e+00
Generation 5730:	Best = 5.391328e+05	FES = 573100	Diversity = 5.9614e+00
Generation 5740:	Best = 5.385383e+05	FES = 574100	Diversity = 6.5034e+00
Generation 5750:	Best = 5.379147e+05	FES = 575100	Diversity = 6.0491e+00
Generation 5760:	Best = 5.372950e+05	FES = 576100	Diversity = 5.5057e+00
Generation 5770:	Best = 5.367848e+05	FES = 577100	Diversity = 5.6199e+00
Generation 5780:	Best = 5.364139e+05	FES = 578100	Diversity = 5.4239e+00
Generation 5790:	Best = 5.359826e+05	FES = 579100	Diversity = 5.5925e+00
Generation 5800:	Best = 5.353675e+05	FES = 580100	Diversity = 6.0583e+00
Generation 5810:	Best = 5.351113e+05	FES = 581100	Diversity = 5.9199e+00
Generation 5820:	Best = 5.348083e+05	FES = 582100	Diversity = 5.3855e+00
Generation 5830:	Best = 5.344052e+05	FES = 583100	Diversity = 4.8145e+00
Generation 5840:	Best = 5.338142e+05	FES = 584100	Diversity = 5.0451e+00

Generation 5850:	Best = 5.335104e+05	FES = 585100	Diversity = 5.7205e+00
Generation 5860:	Best = 5.332114e+05	FES = 586100	Diversity = 5.4217e+00
Generation 5870:	Best = 5.328219e+05	FES = 587100	Diversity = 5.4397e+00
Generation 5880:	Best = 5.326197e+05	FES = 588100	Diversity = 4.9782e+00
Generation 5890:	Best = 5.321796e+05	FES = 589100	Diversity = 5.2643e+00
Generation 5900:	Best = 5.312349e+05	FES = 590100	Diversity = 6.5644e+00
Generation 5910:	Best = 5.308342e+05	FES = 591100	Diversity = 6.5888e+00
Generation 5920:	Best = 5.299691e+05	FES = 592100	Diversity = 5.9905e+00
Generation 5930:	Best = 5.294801e+05	FES = 593100	Diversity = 6.2527e+00
Generation 5940:	Best = 5.284780e+05	FES = 594100	Diversity = 6.0322e+00
Generation 5950:	Best = 5.284780e+05	FES = 595100	Diversity = 5.9182e+00
Generation 5960:	Best = 5.281949e+05	FES = 596100	Diversity = 6.2471e+00
Generation 5970:	Best = 5.269535e+05	FES = 597100	Diversity = 5.6638e+00
Generation 5980:	Best = 5.267227e+05	FES = 598100	Diversity = 5.9997e+00
Generation 5990:	Best = 5.264161e+05	FES = 599100	Diversity = 5.3035e+00
Generation 6000:	Best = 5.256651e+05	FES = 600100	Diversity = 4.9836e+00
Generation 6010:	Best = 5.254262e+05	FES = 601100	Diversity = 5.2816e+00
Generation 6020:	Best = 5.246547e+05	FES = 602100	Diversity = 5.4630e+00
Generation 6030:	Best = 5.240007e+05	FES = 603100	Diversity = 5.7444e+00
Generation 6040:	Best = 5.234685e+05	FES = 604100	Diversity = 5.7810e+00
Generation 6050:	Best = 5.229555e+05	FES = 605100	Diversity = 5.4412e+00
Generation 6060:	Best = 5.225435e+05	FES = 606100	Diversity = 4.8867e+00
Generation 6070:	Best = 5.222910e+05	FES = 607100	Diversity = 4.1706e+00
Generation 6080:	Best = 5.219045e+05	FES = 608100	Diversity = 3.7961e+00
Generation 6090:	Best = 5.214905e+05	FES = 609100	Diversity = 3.5645e+00
Generation 6100:	Best = 5.211595e+05	FES = 610100	Diversity = 3.2893e+00
Generation 6110:	Best = 5.207394e+05	FES = 611100	Diversity = 3.3712e+00
Generation 6120:	Best = 5.198104e+05	FES = 612100	Diversity = 3.4036e+00
Generation 6130:	Best = 5.193972e+05	FES = 613100	Diversity = 3.8817e+00
Generation 6140:	Best = 5.188102e+05	FES = 614100	Diversity = 4.2165e+00
Generation 6150:	Best = 5.180447e+05	FES = 615100	Diversity = 4.2136e+00
Generation 6160:	Best = 5.173095e+05	FES = 616100	Diversity = 4.2193e+00
Generation 6170:	Best = 5.167396e+05	FES = 617100	Diversity = 3.8709e+00
Generation 6180:	Best = 5.164201e+05	FES = 618100	Diversity = 4.1752e+00
Generation 6190:	Best = 5.162862e+05	FES = 619100	Diversity = 4.5525e+00
Generation 6200:	Best = 5.160063e+05	FES = 620100	Diversity = 4.3564e+00
Generation 6210:	Best = 5.157028e+05	FES = 621100	Diversity = 4.2701e+00
Generation 6220:	Best = 5.153845e+05	FES = 622100	Diversity = 3.9133e+00
Generation 6230:	Best = 5.144010e+05	FES = 623100	Diversity = 4.0392e+00
Generation 6240:	Best = 5.140006e+05	FES = 624100	Diversity = 4.4738e+00
Generation 6250:	Best = 5.134238e+05	FES = 625100	Diversity = 4.5736e+00
Generation 6260:	Best = 5.130772e+05	FES = 626100	Diversity = 4.1669e+00
Generation 6270:	Best = 5.125242e+05	FES = 627100	Diversity = 4.0149e+00
Generation 6280:	Best = 5.119298e+05	FES = 628100	Diversity = 3.7411e+00
Generation 6290:	Best = 5.117569e+05	FES = 629100	Diversity = 3.9211e+00
Generation 6300:	Best = 5.113569e+05	FES = 630100	Diversity = 4.1280e+00
Generation 6310:	Best = 5.106293e+05	FES = 631100	Diversity = 3.7155e+00
Generation 6320:	Best = 5.100987e+05	FES = 632100	Diversity = 3.6001e+00
Generation 6330:	Best = 5.096256e+05	FES = 633100	Diversity = 4.1337e+00
Generation 6340:	Best = 5.093722e+05	FES = 634100	Diversity = 3.8764e+00
Generation 6350:	Best = 5.091687e+05	FES = 635100	Diversity = 3.8361e+00
Generation 6360:	Best = 5.087693e+05	FES = 636100	Diversity = 4.3162e+00
Generation 6370:	Best = 5.084362e+05	FES = 637100	Diversity = 5.3347e+00
Generation 6380:	Best = 5.078125e+05	FES = 638100	Diversity = 4.6173e+00

Generation 6390:	Best = 5.075902e+05	FES = 639100	Diversity = 4.6194e+00
Generation 6400:	Best = 5.071266e+05	FES = 640100	Diversity = 4.6206e+00
Generation 6410:	Best = 5.068316e+05	FES = 641100	Diversity = 4.2568e+00
Generation 6420:	Best = 5.065335e+05	FES = 642100	Diversity = 3.8384e+00
Generation 6430:	Best = 5.058765e+05	FES = 643100	Diversity = 3.6360e+00
Generation 6440:	Best = 5.055257e+05	FES = 644100	Diversity = 3.5569e+00
Generation 6450:	Best = 5.050628e+05	FES = 645100	Diversity = 3.5702e+00
Generation 6460:	Best = 5.047589e+05	FES = 646100	Diversity = 3.4305e+00
Generation 6470:	Best = 5.041109e+05	FES = 647100	Diversity = 3.4730e+00
Generation 6480:	Best = 5.037423e+05	FES = 648100	Diversity = 3.5666e+00
Generation 6490:	Best = 5.033693e+05	FES = 649100	Diversity = 3.6685e+00
Generation 6500:	Best = 5.026493e+05	FES = 650100	Diversity = 3.4814e+00
Generation 6510:	Best = 5.025917e+05	FES = 651100	Diversity = 3.3143e+00
Generation 6520:	Best = 5.019905e+05	FES = 652100	Diversity = 3.5474e+00
Generation 6530:	Best = 5.014039e+05	FES = 653100	Diversity = 3.6188e+00
Generation 6540:	Best = 5.009817e+05	FES = 654100	Diversity = 4.2883e+00
Generation 6550:	Best = 5.005840e+05	FES = 655100	Diversity = 4.2674e+00
Generation 6560:	Best = 5.001567e+05	FES = 656100	Diversity = 4.0937e+00
Generation 6570:	Best = 4.995075e+05	FES = 657100	Diversity = 4.2298e+00
Generation 6580:	Best = 4.995075e+05	FES = 658100	Diversity = 4.2914e+00
Generation 6590:	Best = 4.993844e+05	FES = 659100	Diversity = 4.4358e+00
Generation 6600:	Best = 4.989663e+05	FES = 660100	Diversity = 4.5316e+00
Generation 6610:	Best = 4.987582e+05	FES = 661100	Diversity = 4.9106e+00
Generation 6620:	Best = 4.978438e+05	FES = 662100	Diversity = 5.1769e+00
Generation 6630:	Best = 4.977227e+05	FES = 663100	Diversity = 5.0441e+00
Generation 6640:	Best = 4.972984e+05	FES = 664100	Diversity = 4.9725e+00
Generation 6650:	Best = 4.966328e+05	FES = 665100	Diversity = 5.1132e+00
Generation 6660:	Best = 4.965655e+05	FES = 666100	Diversity = 5.4821e+00
Generation 6670:	Best = 4.961504e+05	FES = 667100	Diversity = 5.0449e+00
Generation 6680:	Best = 4.957618e+05	FES = 668100	Diversity = 4.8822e+00
Generation 6690:	Best = 4.953828e+05	FES = 669100	Diversity = 4.8690e+00
Generation 6700:	Best = 4.953828e+05	FES = 670100	Diversity = 4.8819e+00
Generation 6710:	Best = 4.951454e+05	FES = 671100	Diversity = 4.7693e+00
Generation 6720:	Best = 4.949334e+05	FES = 672100	Diversity = 4.4705e+00
Generation 6730:	Best = 4.947341e+05	FES = 673100	Diversity = 4.2364e+00
Generation 6740:	Best = 4.944846e+05	FES = 674100	Diversity = 4.2346e+00
Generation 6750:	Best = 4.941253e+05	FES = 675100	Diversity = 4.4457e+00
Generation 6760:	Best = 4.940137e+05	FES = 676100	Diversity = 4.5700e+00
Generation 6770:	Best = 4.936356e+05	FES = 677100	Diversity = 4.0711e+00
Generation 6780:	Best = 4.934264e+05	FES = 678100	Diversity = 3.8400e+00
Generation 6790:	Best = 4.932422e+05	FES = 679100	Diversity = 3.9701e+00
Generation 6800:	Best = 4.927531e+05	FES = 680100	Diversity = 4.2080e+00
Generation 6810:	Best = 4.926821e+05	FES = 681100	Diversity = 4.6295e+00
Generation 6820:	Best = 4.923491e+05	FES = 682100	Diversity = 4.5918e+00
Generation 6830:	Best = 4.918162e+05	FES = 683100	Diversity = 5.1441e+00
Generation 6840:	Best = 4.912888e+05	FES = 684100	Diversity = 4.7133e+00
Generation 6850:	Best = 4.906250e+05	FES = 685100	Diversity = 4.4747e+00
Generation 6860:	Best = 4.903274e+05	FES = 686100	Diversity = 4.5943e+00
Generation 6870:	Best = 4.898620e+05	FES = 687100	Diversity = 4.5508e+00
Generation 6880:	Best = 4.897284e+05	FES = 688100	Diversity = 4.4152e+00
Generation 6890:	Best = 4.894385e+05	FES = 689100	Diversity = 3.5842e+00
Generation 6900:	Best = 4.891192e+05	FES = 690100	Diversity = 3.4201e+00
Generation 6910:	Best = 4.880446e+05	FES = 691100	Diversity = 3.5596e+00
Generation 6920:	Best = 4.869114e+05	FES = 692100	Diversity = 4.8039e+00

Generation 6930:	Best = 4.863370e+05	FES = 693100	Diversity = 5.0132e+00
Generation 6940:	Best = 4.858961e+05	FES = 694100	Diversity = 5.1632e+00
Generation 6950:	Best = 4.852734e+05	FES = 695100	Diversity = 5.8707e+00
Generation 6960:	Best = 4.844445e+05	FES = 696100	Diversity = 5.2872e+00
Generation 6970:	Best = 4.842215e+05	FES = 697100	Diversity = 4.8271e+00
Generation 6980:	Best = 4.837014e+05	FES = 698100	Diversity = 4.4776e+00
Generation 6990:	Best = 4.829218e+05	FES = 699100	Diversity = 4.8290e+00
Generation 7000:	Best = 4.816413e+05	FES = 700100	Diversity = 5.4539e+00
Generation 7010:	Best = 4.816385e+05	FES = 701100	Diversity = 5.0378e+00
Generation 7020:	Best = 4.809353e+05	FES = 702100	Diversity = 5.1104e+00
Generation 7030:	Best = 4.802079e+05	FES = 703100	Diversity = 5.1503e+00
Generation 7040:	Best = 4.791822e+05	FES = 704100	Diversity = 5.0613e+00
Generation 7050:	Best = 4.788595e+05	FES = 705100	Diversity = 4.7172e+00
Generation 7060:	Best = 4.782467e+05	FES = 706100	Diversity = 4.6927e+00
Generation 7070:	Best = 4.779209e+05	FES = 707100	Diversity = 5.2503e+00
Generation 7080:	Best = 4.772751e+05	FES = 708100	Diversity = 4.7414e+00
Generation 7090:	Best = 4.769970e+05	FES = 709100	Diversity = 4.0019e+00
Generation 7100:	Best = 4.766540e+05	FES = 710100	Diversity = 3.9336e+00
Generation 7110:	Best = 4.761290e+05	FES = 711100	Diversity = 3.5197e+00
Generation 7120:	Best = 4.761290e+05	FES = 712100	Diversity = 3.5229e+00
Generation 7130:	Best = 4.757719e+05	FES = 713100	Diversity = 3.0824e+00
Generation 7140:	Best = 4.756691e+05	FES = 714100	Diversity = 3.5394e+00
Generation 7150:	Best = 4.753005e+05	FES = 715100	Diversity = 3.4159e+00
Generation 7160:	Best = 4.750443e+05	FES = 716100	Diversity = 3.5415e+00
Generation 7170:	Best = 4.747018e+05	FES = 717100	Diversity = 3.8513e+00
Generation 7180:	Best = 4.742173e+05	FES = 718100	Diversity = 4.2346e+00
Generation 7190:	Best = 4.740285e+05	FES = 719100	Diversity = 3.7912e+00
Generation 7200:	Best = 4.730917e+05	FES = 720100	Diversity = 3.8470e+00
Generation 7210:	Best = 4.730232e+05	FES = 721100	Diversity = 3.9443e+00
Generation 7220:	Best = 4.724159e+05	FES = 722100	Diversity = 3.6518e+00
Generation 7230:	Best = 4.716599e+05	FES = 723100	Diversity = 4.0182e+00
Generation 7240:	Best = 4.712637e+05	FES = 724100	Diversity = 4.1202e+00
Generation 7250:	Best = 4.707590e+05	FES = 725100	Diversity = 3.8778e+00
Generation 7260:	Best = 4.701800e+05	FES = 726100	Diversity = 3.5266e+00
Generation 7270:	Best = 4.699893e+05	FES = 727100	Diversity = 3.7007e+00
Generation 7280:	Best = 4.694298e+05	FES = 728100	Diversity = 3.4398e+00
Generation 7290:	Best = 4.693180e+05	FES = 729100	Diversity = 3.4310e+00
Generation 7300:	Best = 4.689584e+05	FES = 730100	Diversity = 3.3792e+00
Generation 7310:	Best = 4.686425e+05	FES = 731100	Diversity = 3.4089e+00
Generation 7320:	Best = 4.684057e+05	FES = 732100	Diversity = 3.2741e+00
Generation 7330:	Best = 4.678856e+05	FES = 733100	Diversity = 3.0519e+00
Generation 7340:	Best = 4.673497e+05	FES = 734100	Diversity = 3.1628e+00
Generation 7350:	Best = 4.672316e+05	FES = 735100	Diversity = 3.2995e+00
Generation 7360:	Best = 4.667373e+05	FES = 736100	Diversity = 3.4470e+00
Generation 7370:	Best = 4.663408e+05	FES = 737100	Diversity = 3.5015e+00
Generation 7380:	Best = 4.660716e+05	FES = 738100	Diversity = 3.7511e+00
Generation 7390:	Best = 4.655446e+05	FES = 739100	Diversity = 3.8036e+00
Generation 7400:	Best = 4.653181e+05	FES = 740100	Diversity = 3.7166e+00
Generation 7410:	Best = 4.650270e+05	FES = 741100	Diversity = 3.6526e+00
Generation 7420:	Best = 4.646174e+05	FES = 742100	Diversity = 3.4432e+00
Generation 7430:	Best = 4.638666e+05	FES = 743100	Diversity = 3.4985e+00
Generation 7440:	Best = 4.634346e+05	FES = 744100	Diversity = 3.9144e+00
Generation 7450:	Best = 4.629043e+05	FES = 745100	Diversity = 3.7204e+00
Generation 7460:	Best = 4.627051e+05	FES = 746100	Diversity = 3.7242e+00

Generation 7470:	Best = 4.621128e+05	FES = 747100	Diversity = 3.4412e+00
Generation 7480:	Best = 4.615615e+05	FES = 748100	Diversity = 3.2990e+00
Generation 7490:	Best = 4.611041e+05	FES = 749100	Diversity = 3.3857e+00
Generation 7500:	Best = 4.608149e+05	FES = 750100	Diversity = 3.5196e+00
Generation 7510:	Best = 4.602117e+05	FES = 751100	Diversity = 3.9027e+00
Generation 7520:	Best = 4.599311e+05	FES = 752100	Diversity = 3.8613e+00
Generation 7530:	Best = 4.591203e+05	FES = 753100	Diversity = 3.2831e+00
Generation 7540:	Best = 4.585527e+05	FES = 754100	Diversity = 3.6126e+00
Generation 7550:	Best = 4.579321e+05	FES = 755100	Diversity = 3.4110e+00
Generation 7560:	Best = 4.577589e+05	FES = 756100	Diversity = 2.9194e+00
Generation 7570:	Best = 4.572121e+05	FES = 757100	Diversity = 3.1665e+00
Generation 7580:	Best = 4.568001e+05	FES = 758100	Diversity = 3.1645e+00
Generation 7590:	Best = 4.564356e+05	FES = 759100	Diversity = 2.8248e+00
Generation 7600:	Best = 4.561879e+05	FES = 760100	Diversity = 3.2563e+00
Generation 7610:	Best = 4.560085e+05	FES = 761100	Diversity = 3.2764e+00
Generation 7620:	Best = 4.558056e+05	FES = 762100	Diversity = 3.0317e+00
Generation 7630:	Best = 4.552917e+05	FES = 763100	Diversity = 2.6632e+00
Generation 7640:	Best = 4.550720e+05	FES = 764100	Diversity = 2.9886e+00
Generation 7650:	Best = 4.546969e+05	FES = 765100	Diversity = 3.0885e+00
Generation 7660:	Best = 4.543972e+05	FES = 766100	Diversity = 3.6508e+00
Generation 7670:	Best = 4.539495e+05	FES = 767100	Diversity = 3.9370e+00
Generation 7680:	Best = 4.537397e+05	FES = 768100	Diversity = 3.7881e+00
Generation 7690:	Best = 4.532456e+05	FES = 769100	Diversity = 3.7004e+00
Generation 7700:	Best = 4.526635e+05	FES = 770100	Diversity = 4.1310e+00
Generation 7710:	Best = 4.522562e+05	FES = 771100	Diversity = 4.4967e+00
Generation 7720:	Best = 4.518626e+05	FES = 772100	Diversity = 4.6284e+00
Generation 7730:	Best = 4.516091e+05	FES = 773100	Diversity = 4.3128e+00
Generation 7740:	Best = 4.511937e+05	FES = 774100	Diversity = 4.6694e+00
Generation 7750:	Best = 4.499479e+05	FES = 775100	Diversity = 4.7908e+00
Generation 7760:	Best = 4.494854e+05	FES = 776100	Diversity = 5.9302e+00
Generation 7770:	Best = 4.489602e+05	FES = 777100	Diversity = 6.1522e+00
Generation 7780:	Best = 4.487843e+05	FES = 778100	Diversity = 5.9823e+00
Generation 7790:	Best = 4.482070e+05	FES = 779100	Diversity = 5.7050e+00
Generation 7800:	Best = 4.478877e+05	FES = 780100	Diversity = 5.6760e+00
Generation 7810:	Best = 4.474631e+05	FES = 781100	Diversity = 5.2317e+00
Generation 7820:	Best = 4.470965e+05	FES = 782100	Diversity = 5.0673e+00
Generation 7830:	Best = 4.467833e+05	FES = 783100	Diversity = 4.4450e+00
Generation 7840:	Best = 4.462939e+05	FES = 784100	Diversity = 4.5077e+00
Generation 7850:	Best = 4.459472e+05	FES = 785100	Diversity = 4.9780e+00
Generation 7860:	Best = 4.453183e+05	FES = 786100	Diversity = 4.8184e+00
Generation 7870:	Best = 4.450196e+05	FES = 787100	Diversity = 4.6525e+00
Generation 7880:	Best = 4.444853e+05	FES = 788100	Diversity = 4.8043e+00
Generation 7890:	Best = 4.442369e+05	FES = 789100	Diversity = 4.7952e+00
Generation 7900:	Best = 4.439208e+05	FES = 790100	Diversity = 4.5734e+00
Generation 7910:	Best = 4.435856e+05	FES = 791100	Diversity = 4.5056e+00
Generation 7920:	Best = 4.433327e+05	FES = 792100	Diversity = 4.5256e+00
Generation 7930:	Best = 4.429975e+05	FES = 793100	Diversity = 3.9577e+00
Generation 7940:	Best = 4.427188e+05	FES = 794100	Diversity = 3.7751e+00
Generation 7950:	Best = 4.424792e+05	FES = 795100	Diversity = 3.6672e+00
Generation 7960:	Best = 4.423586e+05	FES = 796100	Diversity = 3.8028e+00
Generation 7970:	Best = 4.421053e+05	FES = 797100	Diversity = 3.7769e+00
Generation 7980:	Best = 4.419251e+05	FES = 798100	Diversity = 3.7150e+00
Generation 7990:	Best = 4.416910e+05	FES = 799100	Diversity = 3.8619e+00
Generation 8000:	Best = 4.413521e+05	FES = 800100	Diversity = 3.4860e+00

Generation 8010:	Best = 4.412746e+05	FES = 801100	Diversity = 3.5629e+00
Generation 8020:	Best = 4.409982e+05	FES = 802100	Diversity = 3.4245e+00
Generation 8030:	Best = 4.408906e+05	FES = 803100	Diversity = 3.0736e+00
Generation 8040:	Best = 4.407550e+05	FES = 804100	Diversity = 3.1948e+00
Generation 8050:	Best = 4.404977e+05	FES = 805100	Diversity = 3.4236e+00
Generation 8060:	Best = 4.402453e+05	FES = 806100	Diversity = 3.3201e+00
Generation 8070:	Best = 4.400550e+05	FES = 807100	Diversity = 3.3226e+00
Generation 8080:	Best = 4.395963e+05	FES = 808100	Diversity = 3.5109e+00
Generation 8090:	Best = 4.394100e+05	FES = 809100	Diversity = 3.8584e+00
Generation 8100:	Best = 4.390813e+05	FES = 810100	Diversity = 3.6929e+00
Generation 8110:	Best = 4.389483e+05	FES = 811100	Diversity = 3.8985e+00
Generation 8120:	Best = 4.386398e+05	FES = 812100	Diversity = 4.3018e+00
Generation 8130:	Best = 4.383549e+05	FES = 813100	Diversity = 4.7837e+00
Generation 8140:	Best = 4.380061e+05	FES = 814100	Diversity = 4.6629e+00
Generation 8150:	Best = 4.377717e+05	FES = 815100	Diversity = 5.0966e+00
Generation 8160:	Best = 4.376107e+05	FES = 816100	Diversity = 5.1204e+00
Generation 8170:	Best = 4.370719e+05	FES = 817100	Diversity = 5.6233e+00
Generation 8180:	Best = 4.368661e+05	FES = 818100	Diversity = 5.8720e+00
Generation 8190:	Best = 4.363493e+05	FES = 819100	Diversity = 5.3953e+00
Generation 8200:	Best = 4.361099e+05	FES = 820100	Diversity = 4.7273e+00
Generation 8210:	Best = 4.360983e+05	FES = 821100	Diversity = 4.9370e+00
Generation 8220:	Best = 4.358585e+05	FES = 822100	Diversity = 4.7634e+00
Generation 8230:	Best = 4.356401e+05	FES = 823100	Diversity = 4.8310e+00
Generation 8240:	Best = 4.352293e+05	FES = 824100	Diversity = 4.7316e+00
Generation 8250:	Best = 4.351453e+05	FES = 825100	Diversity = 4.9398e+00
Generation 8260:	Best = 4.345858e+05	FES = 826100	Diversity = 5.3579e+00
Generation 8270:	Best = 4.341202e+05	FES = 827100	Diversity = 5.5559e+00
Generation 8280:	Best = 4.339806e+05	FES = 828100	Diversity = 6.1390e+00
Generation 8290:	Best = 4.336523e+05	FES = 829100	Diversity = 6.2641e+00
Generation 8300:	Best = 4.332608e+05	FES = 830100	Diversity = 5.8943e+00
Generation 8310:	Best = 4.329459e+05	FES = 831100	Diversity = 5.3754e+00
Generation 8320:	Best = 4.325555e+05	FES = 832100	Diversity = 5.4032e+00
Generation 8330:	Best = 4.321420e+05	FES = 833100	Diversity = 5.8526e+00
Generation 8340:	Best = 4.320327e+05	FES = 834100	Diversity = 5.5375e+00
Generation 8350:	Best = 4.317563e+05	FES = 835100	Diversity = 5.5126e+00
Generation 8360:	Best = 4.313372e+05	FES = 836100	Diversity = 5.2373e+00
Generation 8370:	Best = 4.313219e+05	FES = 837100	Diversity = 5.2969e+00
Generation 8380:	Best = 4.310402e+05	FES = 838100	Diversity = 4.8113e+00
Generation 8390:	Best = 4.309048e+05	FES = 839100	Diversity = 4.4150e+00
Generation 8400:	Best = 4.305738e+05	FES = 840100	Diversity = 3.9474e+00
Generation 8410:	Best = 4.305451e+05	FES = 841100	Diversity = 3.8518e+00
Generation 8420:	Best = 4.303441e+05	FES = 842100	Diversity = 3.6354e+00
Generation 8430:	Best = 4.302487e+05	FES = 843100	Diversity = 3.5151e+00
Generation 8440:	Best = 4.300793e+05	FES = 844100	Diversity = 3.6722e+00
Generation 8450:	Best = 4.299590e+05	FES = 845100	Diversity = 3.2882e+00
Generation 8460:	Best = 4.298682e+05	FES = 846100	Diversity = 3.2433e+00
Generation 8470:	Best = 4.297008e+05	FES = 847100	Diversity = 3.4449e+00
Generation 8480:	Best = 4.295100e+05	FES = 848100	Diversity = 4.4508e+00
Generation 8490:	Best = 4.293311e+05	FES = 849100	Diversity = 4.8003e+00
Generation 8500:	Best = 4.290615e+05	FES = 850100	Diversity = 5.4361e+00
Generation 8510:	Best = 4.289245e+05	FES = 851100	Diversity = 5.0312e+00
Generation 8520:	Best = 4.287003e+05	FES = 852100	Diversity = 4.8093e+00
Generation 8530:	Best = 4.285914e+05	FES = 853100	Diversity = 5.1283e+00
Generation 8540:	Best = 4.283335e+05	FES = 854100	Diversity = 5.0263e+00

Generation 8550:	Best = 4.282481e+05	FES = 855100	Diversity = 4.9915e+00
Generation 8560:	Best = 4.280546e+05	FES = 856100	Diversity = 4.9219e+00
Generation 8570:	Best = 4.277659e+05	FES = 857100	Diversity = 5.1941e+00
Generation 8580:	Best = 4.275205e+05	FES = 858100	Diversity = 4.7267e+00
Generation 8590:	Best = 4.274680e+05	FES = 859100	Diversity = 4.9461e+00
Generation 8600:	Best = 4.273165e+05	FES = 860100	Diversity = 4.6610e+00
Generation 8610:	Best = 4.271568e+05	FES = 861100	Diversity = 4.5143e+00
Generation 8620:	Best = 4.270018e+05	FES = 862100	Diversity = 4.2427e+00
Generation 8630:	Best = 4.269237e+05	FES = 863100	Diversity = 4.4696e+00
Generation 8640:	Best = 4.267841e+05	FES = 864100	Diversity = 4.1047e+00
Generation 8650:	Best = 4.266899e+05	FES = 865100	Diversity = 3.9008e+00
Generation 8660:	Best = 4.265679e+05	FES = 866100	Diversity = 4.1950e+00
Generation 8670:	Best = 4.264702e+05	FES = 867100	Diversity = 4.3079e+00
Generation 8680:	Best = 4.261872e+05	FES = 868100	Diversity = 4.3594e+00
Generation 8690:	Best = 4.260542e+05	FES = 869100	Diversity = 4.6686e+00
Generation 8700:	Best = 4.258690e+05	FES = 870100	Diversity = 4.6053e+00
Generation 8710:	Best = 4.257165e+05	FES = 871100	Diversity = 4.0412e+00
Generation 8720:	Best = 4.255168e+05	FES = 872100	Diversity = 4.8884e+00
Generation 8730:	Best = 4.252873e+05	FES = 873100	Diversity = 5.5197e+00
Generation 8740:	Best = 4.251424e+05	FES = 874100	Diversity = 5.7852e+00
Generation 8750:	Best = 4.249375e+05	FES = 875100	Diversity = 5.4265e+00
Generation 8760:	Best = 4.247837e+05	FES = 876100	Diversity = 5.0673e+00
Generation 8770:	Best = 4.246105e+05	FES = 877100	Diversity = 4.6630e+00
Generation 8780:	Best = 4.244067e+05	FES = 878100	Diversity = 4.8851e+00
Generation 8790:	Best = 4.243359e+05	FES = 879100	Diversity = 4.7189e+00
Generation 8800:	Best = 4.242028e+05	FES = 880100	Diversity = 4.4870e+00
Generation 8810:	Best = 4.240331e+05	FES = 881100	Diversity = 4.4022e+00
Generation 8820:	Best = 4.240272e+05	FES = 882100	Diversity = 3.9052e+00
Generation 8830:	Best = 4.239909e+05	FES = 883100	Diversity = 3.7012e+00
Generation 8840:	Best = 4.238395e+05	FES = 884100	Diversity = 3.5309e+00
Generation 8850:	Best = 4.236675e+05	FES = 885100	Diversity = 3.5055e+00
Generation 8860:	Best = 4.236561e+05	FES = 886100	Diversity = 3.3126e+00
Generation 8870:	Best = 4.235869e+05	FES = 887100	Diversity = 2.6492e+00
Generation 8880:	Best = 4.235145e+05	FES = 888100	Diversity = 2.8888e+00
Generation 8890:	Best = 4.234423e+05	FES = 889100	Diversity = 2.8318e+00
Generation 8900:	Best = 4.233396e+05	FES = 890100	Diversity = 2.7295e+00
Generation 8910:	Best = 4.231427e+05	FES = 891100	Diversity = 3.0663e+00
Generation 8920:	Best = 4.229817e+05	FES = 892100	Diversity = 3.5695e+00
Generation 8930:	Best = 4.229523e+05	FES = 893100	Diversity = 3.6205e+00
Generation 8940:	Best = 4.227953e+05	FES = 894100	Diversity = 3.7185e+00
Generation 8950:	Best = 4.226902e+05	FES = 895100	Diversity = 3.8935e+00
Generation 8960:	Best = 4.226128e+05	FES = 896100	Diversity = 3.9761e+00
Generation 8970:	Best = 4.224224e+05	FES = 897100	Diversity = 4.3142e+00
Generation 8980:	Best = 4.223451e+05	FES = 898100	Diversity = 4.3091e+00
Generation 8990:	Best = 4.222793e+05	FES = 899100	Diversity = 4.4436e+00
Generation 9000:	Best = 4.221127e+05	FES = 900100	Diversity = 4.6686e+00
Generation 9010:	Best = 4.218751e+05	FES = 901100	Diversity = 4.6313e+00
Generation 9020:	Best = 4.216680e+05	FES = 902100	Diversity = 5.0718e+00
Generation 9030:	Best = 4.214684e+05	FES = 903100	Diversity = 5.1668e+00
Generation 9040:	Best = 4.212885e+05	FES = 904100	Diversity = 5.0630e+00
Generation 9050:	Best = 4.211587e+05	FES = 905100	Diversity = 5.3503e+00
Generation 9060:	Best = 4.208054e+05	FES = 906100	Diversity = 5.5791e+00
Generation 9070:	Best = 4.202447e+05	FES = 907100	Diversity = 5.7533e+00
Generation 9080:	Best = 4.199576e+05	FES = 908100	Diversity = 6.8427e+00

Generation 9090:	Best = 4.197462e+05	FES = 909100	Diversity = 5.8563e+00
Generation 9100:	Best = 4.194180e+05	FES = 910100	Diversity = 5.6347e+00
Generation 9110:	Best = 4.190685e+05	FES = 911100	Diversity = 5.8513e+00
Generation 9120:	Best = 4.188083e+05	FES = 912100	Diversity = 5.5743e+00
Generation 9130:	Best = 4.185207e+05	FES = 913100	Diversity = 5.4913e+00
Generation 9140:	Best = 4.182178e+05	FES = 914100	Diversity = 5.6276e+00
Generation 9150:	Best = 4.179820e+05	FES = 915100	Diversity = 5.7913e+00
Generation 9160:	Best = 4.177985e+05	FES = 916100	Diversity = 5.3449e+00
Generation 9170:	Best = 4.176575e+05	FES = 917100	Diversity = 5.1445e+00
Generation 9180:	Best = 4.174509e+05	FES = 918100	Diversity = 4.6749e+00
Generation 9190:	Best = 4.172398e+05	FES = 919100	Diversity = 4.9794e+00
Generation 9200:	Best = 4.171556e+05	FES = 920100	Diversity = 5.0249e+00
Generation 9210:	Best = 4.168684e+05	FES = 921100	Diversity = 4.6978e+00
Generation 9220:	Best = 4.167731e+05	FES = 922100	Diversity = 4.9558e+00
Generation 9230:	Best = 4.167098e+05	FES = 923100	Diversity = 4.9258e+00
Generation 9240:	Best = 4.166162e+05	FES = 924100	Diversity = 4.4306e+00
Generation 9250:	Best = 4.163581e+05	FES = 925100	Diversity = 4.8072e+00
Generation 9260:	Best = 4.160480e+05	FES = 926100	Diversity = 4.9433e+00
Generation 9270:	Best = 4.159052e+05	FES = 927100	Diversity = 5.1494e+00
Generation 9280:	Best = 4.156424e+05	FES = 928100	Diversity = 5.4561e+00
Generation 9290:	Best = 4.155493e+05	FES = 929100	Diversity = 5.8299e+00
Generation 9300:	Best = 4.153447e+05	FES = 930100	Diversity = 5.8467e+00
Generation 9310:	Best = 4.151120e+05	FES = 931100	Diversity = 5.7242e+00
Generation 9320:	Best = 4.147731e+05	FES = 932100	Diversity = 5.4494e+00
Generation 9330:	Best = 4.146880e+05	FES = 933100	Diversity = 5.4212e+00
Generation 9340:	Best = 4.145339e+05	FES = 934100	Diversity = 5.2365e+00
Generation 9350:	Best = 4.142883e+05	FES = 935100	Diversity = 5.2730e+00
Generation 9360:	Best = 4.142327e+05	FES = 936100	Diversity = 4.8823e+00
Generation 9370:	Best = 4.140148e+05	FES = 937100	Diversity = 5.3707e+00
Generation 9380:	Best = 4.138965e+05	FES = 938100	Diversity = 5.2660e+00
Generation 9390:	Best = 4.137593e+05	FES = 939100	Diversity = 4.4569e+00
Generation 9400:	Best = 4.135995e+05	FES = 940100	Diversity = 5.0534e+00
Generation 9410:	Best = 4.133368e+05	FES = 941100	Diversity = 4.7470e+00
Generation 9420:	Best = 4.132186e+05	FES = 942100	Diversity = 4.8402e+00
Generation 9430:	Best = 4.130592e+05	FES = 943100	Diversity = 4.6884e+00
Generation 9440:	Best = 4.127904e+05	FES = 944100	Diversity = 4.7218e+00
Generation 9450:	Best = 4.125755e+05	FES = 945100	Diversity = 5.0595e+00
Generation 9460:	Best = 4.124256e+05	FES = 946100	Diversity = 5.0852e+00
Generation 9470:	Best = 4.122610e+05	FES = 947100	Diversity = 4.9021e+00
Generation 9480:	Best = 4.118467e+05	FES = 948100	Diversity = 5.1723e+00
Generation 9490:	Best = 4.118165e+05	FES = 949100	Diversity = 5.8281e+00
Generation 9500:	Best = 4.116726e+05	FES = 950100	Diversity = 5.0671e+00
Generation 9510:	Best = 4.115579e+05	FES = 951100	Diversity = 5.0099e+00
Generation 9520:	Best = 4.114521e+05	FES = 952100	Diversity = 5.3486e+00
Generation 9530:	Best = 4.110450e+05	FES = 953100	Diversity = 5.4330e+00
Generation 9540:	Best = 4.106269e+05	FES = 954100	Diversity = 5.8709e+00
Generation 9550:	Best = 4.103549e+05	FES = 955100	Diversity = 6.4677e+00
Generation 9560:	Best = 4.099428e+05	FES = 956100	Diversity = 6.0542e+00
Generation 9570:	Best = 4.095139e+05	FES = 957100	Diversity = 6.1850e+00
Generation 9580:	Best = 4.093038e+05	FES = 958100	Diversity = 5.8476e+00
Generation 9590:	Best = 4.090849e+05	FES = 959100	Diversity = 5.7700e+00
Generation 9600:	Best = 4.083990e+05	FES = 960100	Diversity = 5.8844e+00
Generation 9610:	Best = 4.081079e+05	FES = 961100	Diversity = 6.3641e+00
Generation 9620:	Best = 4.078241e+05	FES = 962100	Diversity = 6.2816e+00

Generation 9630:	Best = 4.075843e+05	FES = 963100	Diversity = 6.4210e+00
Generation 9640:	Best = 4.074289e+05	FES = 964100	Diversity = 5.7740e+00
Generation 9650:	Best = 4.070801e+05	FES = 965100	Diversity = 5.8342e+00
Generation 9660:	Best = 4.069281e+05	FES = 966100	Diversity = 5.9125e+00
Generation 9670:	Best = 4.067021e+05	FES = 967100	Diversity = 5.1888e+00
Generation 9680:	Best = 4.063219e+05	FES = 968100	Diversity = 5.2737e+00
Generation 9690:	Best = 4.062277e+05	FES = 969100	Diversity = 5.8962e+00
Generation 9700:	Best = 4.061353e+05	FES = 970100	Diversity = 5.5943e+00
Generation 9710:	Best = 4.060588e+05	FES = 971100	Diversity = 5.2380e+00
Generation 9720:	Best = 4.059205e+05	FES = 972100	Diversity = 4.9882e+00
Generation 9730:	Best = 4.055850e+05	FES = 973100	Diversity = 5.0348e+00
Generation 9740:	Best = 4.055423e+05	FES = 974100	Diversity = 5.1097e+00
Generation 9750:	Best = 4.052686e+05	FES = 975100	Diversity = 5.2194e+00
Generation 9760:	Best = 4.049997e+05	FES = 976100	Diversity = 5.3325e+00
Generation 9770:	Best = 4.048001e+05	FES = 977100	Diversity = 5.5550e+00
Generation 9780:	Best = 4.046437e+05	FES = 978100	Diversity = 5.0854e+00
Generation 9790:	Best = 4.044798e+05	FES = 979100	Diversity = 5.1068e+00
Generation 9800:	Best = 4.043930e+05	FES = 980100	Diversity = 5.1114e+00
Generation 9810:	Best = 4.041253e+05	FES = 981100	Diversity = 5.3229e+00
Generation 9820:	Best = 4.039482e+05	FES = 982100	Diversity = 5.4793e+00
Generation 9830:	Best = 4.038564e+05	FES = 983100	Diversity = 5.2973e+00
Generation 9840:	Best = 4.037367e+05	FES = 984100	Diversity = 5.0909e+00
Generation 9850:	Best = 4.035145e+05	FES = 985100	Diversity = 4.8134e+00
Generation 9860:	Best = 4.034459e+05	FES = 986100	Diversity = 5.1101e+00
Generation 9870:	Best = 4.032951e+05	FES = 987100	Diversity = 5.1111e+00
Generation 9880:	Best = 4.031145e+05	FES = 988100	Diversity = 5.2931e+00
Generation 9890:	Best = 4.030050e+05	FES = 989100	Diversity = 5.7415e+00
Generation 9900:	Best = 4.028655e+05	FES = 990100	Diversity = 5.7104e+00
Generation 9910:	Best = 4.026948e+05	FES = 991100	Diversity = 6.0256e+00
Generation 9920:	Best = 4.023695e+05	FES = 992100	Diversity = 6.5625e+00
Generation 9930:	Best = 4.022299e+05	FES = 993100	Diversity = 6.3614e+00
Generation 9940:	Best = 4.020685e+05	FES = 994100	Diversity = 6.0049e+00
Generation 9950:	Best = 4.019481e+05	FES = 995100	Diversity = 5.5562e+00
Generation 9960:	Best = 4.017453e+05	FES = 996100	Diversity = 5.7580e+00
Generation 9970:	Best = 4.016419e+05	FES = 997100	Diversity = 5.2504e+00
Generation 9980:	Best = 4.015256e+05	FES = 998100	Diversity = 5.3107e+00
Generation 9990:	Best = 4.012792e+05	FES = 999100	Diversity = 4.9792e+00

Optimization completed!

Final Best Fitness = 4.010990e+05

Total Function Evaluations = 1000000

=====

=

Results:

Best Fitness : 4.010990e+05

Global Optimum : 1.800000e+03

Error (|best-optimum|) : 3.992990e+05

Best Solution (5 dim) : [99.99993722 15.68198631 -1.02649212 -8.03536847
-8.21463587]

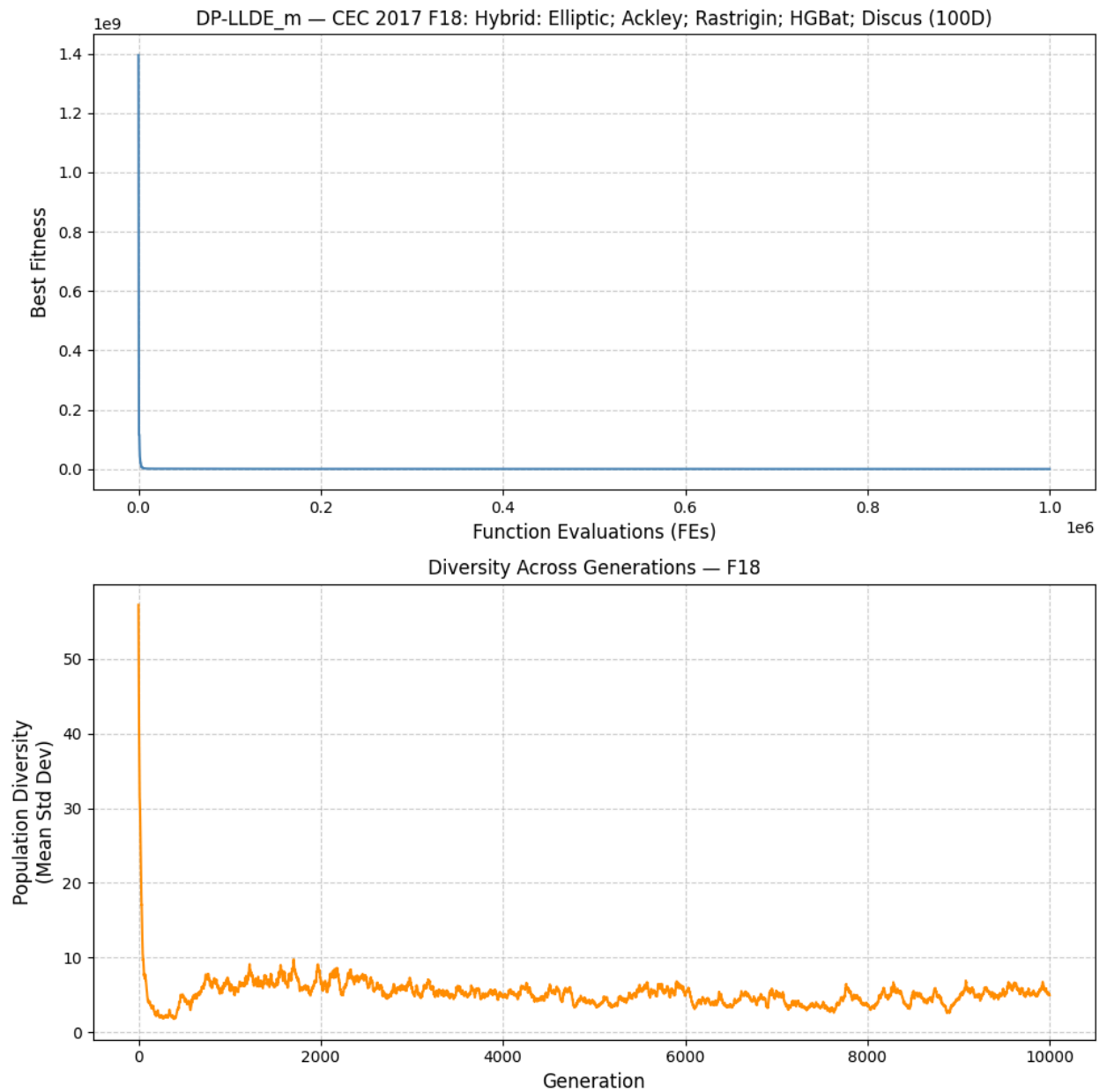
D^(0) (init. diversity): 5.7220e+01

=====

=

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results_2/all_results_che

ckpoint.pkl



Graph saved as convergence_F18.png

✓ Plot saved to: /content/drive/My Drive/thesis_results_2/plots/convergence_F18.png

```
In [ ]: # =====  
# F19 — Hybrid Function 9 (Bent Cigar; Rastrigin; Griewank+Rosenbrock; Weierst  
# =====  
run_and_plot('F19', make_wrapper(hybrid.f19), 1900.0, 'Hybrid: Bent Cigar; Ras
```

```

=====
=
DP-LLDE_m on CEC 2017 F19 – Single Run
=====
=
Function      : Hybrid: Bent Cigar; Rastrigin; Griewank+Rosenbrock; Weierstras
s; Schaffer (100D)
Bounds       : [-100, 100]
Global optimum : 1900.0
Initial  $D^*(0) = 5.7220e+01$  | Diversity threshold = 5.7220e+00
Generation 0: Best Fitness = 4.192174e+11 | Diversity = 5.7220e+01
Generation 10: Best = 8.830749e+10 FES = 1100 Diversity = 2.3857e+01
Generation 20: Best = 3.242801e+10 FES = 2100 Diversity = 1.1191e+01
Generation 30: Best = 1.711397e+10 FES = 3100 Diversity = 6.6065e+00
Generation 40: Best = 1.199087e+10 FES = 4100 Diversity = 4.4787e+00
Generation 50: Best = 9.627274e+09 FES = 5100 Diversity = 4.0919e+00
Generation 60: Best = 8.348780e+09 FES = 6100 Diversity = 4.0227e+00
Generation 70: Best = 7.377823e+09 FES = 7100 Diversity = 4.1411e+00
Generation 80: Best = 6.482109e+09 FES = 8100 Diversity = 5.1844e+00
Generation 90: Best = 5.874465e+09 FES = 9100 Diversity = 5.7912e+00
Generation 100: Best = 5.444957e+09 FES = 10100 Diversity = 5.6316e+00
Generation 110: Best = 4.937414e+09 FES = 11100 Diversity = 6.5101e+00
Generation 120: Best = 4.584907e+09 FES = 12100 Diversity = 6.4786e+00
Generation 130: Best = 4.377423e+09 FES = 13100 Diversity = 6.4551e+00
Generation 140: Best = 4.170493e+09 FES = 14100 Diversity = 7.1763e+00
Generation 150: Best = 4.073630e+09 FES = 15100 Diversity = 7.2845e+00
Generation 160: Best = 3.969485e+09 FES = 16100 Diversity = 8.5803e+00
Generation 170: Best = 3.887319e+09 FES = 17100 Diversity = 9.2079e+00
Generation 180: Best = 3.837077e+09 FES = 18100 Diversity = 7.7793e+00
Generation 190: Best = 3.776171e+09 FES = 19100 Diversity = 7.0962e+00
Generation 200: Best = 3.716418e+09 FES = 20100 Diversity = 8.9661e+00
Generation 210: Best = 3.662082e+09 FES = 21100 Diversity = 8.9551e+00
Generation 220: Best = 3.614099e+09 FES = 22100 Diversity = 9.6504e+00
Generation 230: Best = 3.574110e+09 FES = 23100 Diversity = 1.0772e+01
Generation 240: Best = 3.544111e+09 FES = 24100 Diversity = 9.9498e+00
Generation 250: Best = 3.526608e+09 FES = 25100 Diversity = 8.2649e+00
Generation 260: Best = 3.514160e+09 FES = 26100 Diversity = 6.8664e+00
Generation 270: Best = 3.505615e+09 FES = 27100 Diversity = 6.2844e+00
Generation 280: Best = 3.497478e+09 FES = 28100 Diversity = 5.6179e+00
Generation 290: Best = 3.488438e+09 FES = 29100 Diversity = 5.8821e+00
Generation 300: Best = 3.481938e+09 FES = 30100 Diversity = 5.9444e+00
Generation 310: Best = 3.477342e+09 FES = 31100 Diversity = 5.6305e+00
Generation 320: Best = 3.474162e+09 FES = 32100 Diversity = 5.9444e+00
Generation 330: Best = 3.470797e+09 FES = 33100 Diversity = 5.0497e+00
Generation 340: Best = 3.468547e+09 FES = 34100 Diversity = 5.5205e+00
Generation 350: Best = 3.466651e+09 FES = 35100 Diversity = 5.1593e+00
Generation 360: Best = 3.464922e+09 FES = 36100 Diversity = 5.1126e+00
Generation 370: Best = 3.463539e+09 FES = 37100 Diversity = 5.9736e+00
Generation 380: Best = 3.462379e+09 FES = 38100 Diversity = 5.1402e+00
Generation 390: Best = 3.461195e+09 FES = 39100 Diversity = 5.3308e+00
Generation 400: Best = 3.460323e+09 FES = 40100 Diversity = 4.8188e+00
Generation 410: Best = 3.459620e+09 FES = 41100 Diversity = 5.8633e+00
Generation 420: Best = 3.458964e+09 FES = 42100 Diversity = 5.9577e+00
Generation 430: Best = 3.458288e+09 FES = 43100 Diversity = 6.1643e+00

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Generation	440:	Best = 3.457681e+09	FES = 44100	Diversity = 5.4795e+00
Generation	450:	Best = 3.457282e+09	FES = 45100	Diversity = 4.9699e+00
Generation	460:	Best = 3.456956e+09	FES = 46100	Diversity = 4.4808e+00
Generation	470:	Best = 3.456761e+09	FES = 47100	Diversity = 3.4917e+00
Generation	480:	Best = 3.456608e+09	FES = 48100	Diversity = 2.8147e+00
Generation	490:	Best = 3.456503e+09	FES = 49100	Diversity = 2.2980e+00
Generation	500:	Best = 3.456425e+09	FES = 50100	Diversity = 2.1731e+00
Generation	510:	Best = 3.456344e+09	FES = 51100	Diversity = 1.8935e+00
Generation	520:	Best = 3.456287e+09	FES = 52100	Diversity = 1.6422e+00
Generation	530:	Best = 3.456241e+09	FES = 53100	Diversity = 1.5658e+00
Generation	540:	Best = 3.456213e+09	FES = 54100	Diversity = 1.2399e+00
Generation	550:	Best = 3.456192e+09	FES = 55100	Diversity = 1.3577e+00
Generation	560:	Best = 3.456176e+09	FES = 56100	Diversity = 7.9367e-01
Generation	570:	Best = 3.456166e+09	FES = 57100	Diversity = 8.5792e-01
Generation	580:	Best = 3.456158e+09	FES = 58100	Diversity = 5.7448e-01
Generation	590:	Best = 3.456151e+09	FES = 59100	Diversity = 7.4587e-01
Generation	600:	Best = 3.456145e+09	FES = 60100	Diversity = 7.1513e-01
Generation	610:	Best = 3.456141e+09	FES = 61100	Diversity = 5.4356e-01
Generation	620:	Best = 3.456138e+09	FES = 62100	Diversity = 3.2381e-01
Generation	630:	Best = 3.456136e+09	FES = 63100	Diversity = 3.9531e-01
Generation	640:	Best = 3.456135e+09	FES = 64100	Diversity = 3.3024e-01
Generation	650:	Best = 3.456133e+09	FES = 65100	Diversity = 8.2676e-01
Generation	660:	Best = 3.456132e+09	FES = 66100	Diversity = 5.8818e-01
Generation	670:	Best = 3.456131e+09	FES = 67100	Diversity = 4.2907e-01
Generation	680:	Best = 3.456130e+09	FES = 68100	Diversity = 4.1162e-01
Generation	690:	Best = 3.456130e+09	FES = 69100	Diversity = 5.0239e-01
Generation	700:	Best = 3.456129e+09	FES = 70100	Diversity = 5.3747e-01
Generation	710:	Best = 3.456129e+09	FES = 71100	Diversity = 5.3635e-01
Generation	720:	Best = 3.456129e+09	FES = 72100	Diversity = 4.4555e-01
Generation	730:	Best = 3.456128e+09	FES = 73100	Diversity = 4.4499e-01
Generation	740:	Best = 3.456128e+09	FES = 74100	Diversity = 6.4585e-01
Generation	750:	Best = 3.456128e+09	FES = 75100	Diversity = 1.8034e-01
Generation	760:	Best = 3.456128e+09	FES = 76100	Diversity = 6.3039e-02
Generation	770:	Best = 3.456128e+09	FES = 77100	Diversity = 8.1196e-02
Generation	780:	Best = 3.456128e+09	FES = 78100	Diversity = 1.5835e-01
Generation	790:	Best = 3.456128e+09	FES = 79100	Diversity = 2.5367e-01
Generation	800:	Best = 3.456128e+09	FES = 80100	Diversity = 4.4133e-01
Generation	810:	Best = 3.456128e+09	FES = 81100	Diversity = 3.9999e-01
Generation	820:	Best = 3.456128e+09	FES = 82100	Diversity = 3.3044e-01
Generation	830:	Best = 3.456128e+09	FES = 83100	Diversity = 2.5790e-01
Generation	840:	Best = 3.456128e+09	FES = 84100	Diversity = 3.2773e-01
Generation	850:	Best = 3.456128e+09	FES = 85100	Diversity = 4.5329e-01
Generation	860:	Best = 3.456128e+09	FES = 86100	Diversity = 4.3581e-01
Generation	870:	Best = 3.456128e+09	FES = 87100	Diversity = 4.3227e-01
Generation	880:	Best = 3.456128e+09	FES = 88100	Diversity = 5.2930e-01
Generation	890:	Best = 3.456128e+09	FES = 89100	Diversity = 1.8556e-01
Generation	900:	Best = 3.456128e+09	FES = 90100	Diversity = 2.0192e-02
Generation	910:	Best = 3.456128e+09	FES = 91100	Diversity = 4.2045e-02
Generation	920:	Best = 3.456128e+09	FES = 92100	Diversity = 2.1075e-01
Generation	930:	Best = 3.456128e+09	FES = 93100	Diversity = 2.6023e-01
Generation	940:	Best = 3.456128e+09	FES = 94100	Diversity = 1.2257e-01
Generation	950:	Best = 3.456128e+09	FES = 95100	Diversity = 1.6572e-01
Generation	960:	Best = 3.456128e+09	FES = 96100	Diversity = 8.3306e-02
Generation	970:	Best = 3.456128e+09	FES = 97100	Diversity = 8.9686e-02

Generation 980:	Best = 3.456128e+09	FES = 98100	Diversity = 2.0323e-01
Generation 990:	Best = 3.456128e+09	FES = 99100	Diversity = 6.0941e-01
Generation 1000:	Best = 3.456128e+09	FES = 100100	Diversity = 1.9059e-01
Generation 1010:	Best = 3.456128e+09	FES = 101100	Diversity = 4.0826e-01
Generation 1020:	Best = 3.456128e+09	FES = 102100	Diversity = 4.8104e-02
Generation 1030:	Best = 3.456128e+09	FES = 103100	Diversity = 1.4440e-02
Generation 1040:	Best = 3.456128e+09	FES = 104100	Diversity = 1.1834e-02
Generation 1050:	Best = 3.456128e+09	FES = 105100	Diversity = 4.1228e-03
Generation 1060:	Best = 3.456128e+09	FES = 106100	Diversity = 2.1294e-03
Generation 1070:	Best = 3.456128e+09	FES = 107100	Diversity = 6.2916e-04
Generation 1080:	Best = 3.456128e+09	FES = 108100	Diversity = 3.4031e-04
Generation 1090:	Best = 3.456128e+09	FES = 109100	Diversity = 1.9641e-04
Generation 1100:	Best = 3.456128e+09	FES = 110100	Diversity = 1.1143e-04
Generation 1110:	Best = 3.456128e+09	FES = 111100	Diversity = 1.1367e-04
Generation 1120:	Best = 3.456128e+09	FES = 112100	Diversity = 5.6376e-05
Generation 1130:	Best = 3.456128e+09	FES = 113100	Diversity = 1.6517e-05
Generation 1140:	Best = 3.456128e+09	FES = 114100	Diversity = 1.1232e-05
Generation 1150:	Best = 3.456128e+09	FES = 115100	Diversity = 1.2112e-05
Generation 1160:	Best = 3.456128e+09	FES = 116100	Diversity = 6.3266e-06
Generation 1170:	Best = 3.456128e+09	FES = 117100	Diversity = 1.1390e-05
Generation 1180:	Best = 3.456128e+09	FES = 118100	Diversity = 1.1441e-05
Generation 1190:	Best = 3.456128e+09	FES = 119100	Diversity = 7.3788e-06
Generation 1200:	Best = 3.456128e+09	FES = 120100	Diversity = 4.4238e-06
Generation 1210:	Best = 3.456128e+09	FES = 121100	Diversity = 3.8150e-06
Generation 1220:	Best = 3.456128e+09	FES = 122100	Diversity = 5.8914e-06
Generation 1230:	Best = 3.456128e+09	FES = 123100	Diversity = 3.0057e-06
Generation 1240:	Best = 3.456128e+09	FES = 124100	Diversity = 4.6623e-06
Generation 1250:	Best = 3.456128e+09	FES = 125100	Diversity = 1.1192e-06
Generation 1260:	Best = 3.456128e+09	FES = 126100	Diversity = 8.8010e-09
Generation 1270:	Best = 3.456128e+09	FES = 127100	Diversity = 6.6211e-10
Generation 1280:	Best = 3.456128e+09	FES = 128100	Diversity = 5.1306e-12
Generation 1290:	Best = 3.456128e+09	FES = 129100	Diversity = 6.4866e-14
Generation 1300:	Best = 3.456128e+09	FES = 130100	Diversity = 5.1441e-14
Generation 1310:	Best = 3.456128e+09	FES = 131100	Diversity = 5.0557e-14
Generation 1320:	Best = 3.456128e+09	FES = 132100	Diversity = 5.1439e-14
Generation 1330:	Best = 3.456128e+09	FES = 133100	Diversity = 5.1439e-14
Generation 1340:	Best = 3.456128e+09	FES = 134100	Diversity = 5.1439e-14
Generation 1350:	Best = 3.456128e+09	FES = 135100	Diversity = 5.1439e-14
Generation 1360:	Best = 3.456128e+09	FES = 136100	Diversity = 5.1439e-14
Generation 1370:	Best = 3.456128e+09	FES = 137100	Diversity = 5.1439e-14
Generation 1380:	Best = 3.456128e+09	FES = 138100	Diversity = 5.1439e-14
Generation 1390:	Best = 3.456128e+09	FES = 139100	Diversity = 5.1439e-14
Generation 1400:	Best = 3.456128e+09	FES = 140100	Diversity = 5.1439e-14
Generation 1410:	Best = 3.456128e+09	FES = 141100	Diversity = 5.1439e-14
Generation 1420:	Best = 3.456128e+09	FES = 142100	Diversity = 5.1439e-14
Generation 1430:	Best = 3.456128e+09	FES = 143100	Diversity = 5.1439e-14
Generation 1440:	Best = 3.456128e+09	FES = 144100	Diversity = 5.1439e-14
Generation 1450:	Best = 3.456128e+09	FES = 145100	Diversity = 5.1439e-14
Generation 1460:	Best = 3.456128e+09	FES = 146100	Diversity = 5.1439e-14
Generation 1470:	Best = 3.456128e+09	FES = 147100	Diversity = 5.1439e-14
Generation 1480:	Best = 3.456128e+09	FES = 148100	Diversity = 5.1439e-14
Generation 1490:	Best = 3.456128e+09	FES = 149100	Diversity = 5.1439e-14
Generation 1500:	Best = 3.456128e+09	FES = 150100	Diversity = 5.1439e-14
Generation 1510:	Best = 3.456128e+09	FES = 151100	Diversity = 5.1439e-14

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[illegible]

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[illegible]


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Generation 9620: Best = 3.456128e+09 FES = 962100 Diversity = 5.1439e-14
Generation 9630: Best = 3.456128e+09 FES = 963100 Diversity = 5.1439e-14
Generation 9640: Best = 3.456128e+09 FES = 964100 Diversity = 5.1439e-14
Generation 9650: Best = 3.456128e+09 FES = 965100 Diversity = 5.1439e-14
Generation 9660: Best = 3.456128e+09 FES = 966100 Diversity = 5.1439e-14
Generation 9670: Best = 3.456128e+09 FES = 967100 Diversity = 5.1439e-14
Generation 9680: Best = 3.456128e+09 FES = 968100 Diversity = 5.1439e-14
Generation 9690: Best = 3.456128e+09 FES = 969100 Diversity = 5.1439e-14
Generation 9700: Best = 3.456128e+09 FES = 970100 Diversity = 5.1439e-14
Generation 9710: Best = 3.456128e+09 FES = 971100 Diversity = 5.1439e-14
Generation 9720: Best = 3.456128e+09 FES = 972100 Diversity = 5.1439e-14
Generation 9730: Best = 3.456128e+09 FES = 973100 Diversity = 5.1439e-14
Generation 9740: Best = 3.456128e+09 FES = 974100 Diversity = 5.1439e-14
Generation 9750: Best = 3.456128e+09 FES = 975100 Diversity = 5.1439e-14
Generation 9760: Best = 3.456128e+09 FES = 976100 Diversity = 5.1439e-14
Generation 9770: Best = 3.456128e+09 FES = 977100 Diversity = 5.1439e-14
Generation 9780: Best = 3.456128e+09 FES = 978100 Diversity = 5.1439e-14
Generation 9790: Best = 3.456128e+09 FES = 979100 Diversity = 5.1439e-14
Generation 9800: Best = 3.456128e+09 FES = 980100 Diversity = 5.1439e-14
Generation 9810: Best = 3.456128e+09 FES = 981100 Diversity = 5.1439e-14
Generation 9820: Best = 3.456128e+09 FES = 982100 Diversity = 5.1439e-14
Generation 9830: Best = 3.456128e+09 FES = 983100 Diversity = 5.1439e-14
Generation 9840: Best = 3.456128e+09 FES = 984100 Diversity = 5.1439e-14
Generation 9850: Best = 3.456128e+09 FES = 985100 Diversity = 5.1439e-14
Generation 9860: Best = 3.456128e+09 FES = 986100 Diversity = 5.1439e-14
Generation 9870: Best = 3.456128e+09 FES = 987100 Diversity = 5.1439e-14
Generation 9880: Best = 3.456128e+09 FES = 988100 Diversity = 5.1439e-14
Generation 9890: Best = 3.456128e+09 FES = 989100 Diversity = 5.1439e-14
Generation 9900: Best = 3.456128e+09 FES = 990100 Diversity = 5.1439e-14
Generation 9910: Best = 3.456128e+09 FES = 991100 Diversity = 5.1439e-14
Generation 9920: Best = 3.456128e+09 FES = 992100 Diversity = 5.1439e-14
Generation 9930: Best = 3.456128e+09 FES = 993100 Diversity = 5.1439e-14
Generation 9940: Best = 3.456128e+09 FES = 994100 Diversity = 5.1439e-14
Generation 9950: Best = 3.456128e+09 FES = 995100 Diversity = 5.1439e-14
Generation 9960: Best = 3.456128e+09 FES = 996100 Diversity = 5.1439e-14
Generation 9970: Best = 3.456128e+09 FES = 997100 Diversity = 5.1439e-14
Generation 9980: Best = 3.456128e+09 FES = 998100 Diversity = 5.1439e-14
Generation 9990: Best = 3.456128e+09 FES = 999100 Diversity = 5.1439e-14

```

Optimization completed!

Final Best Fitness = 3.456128e+09

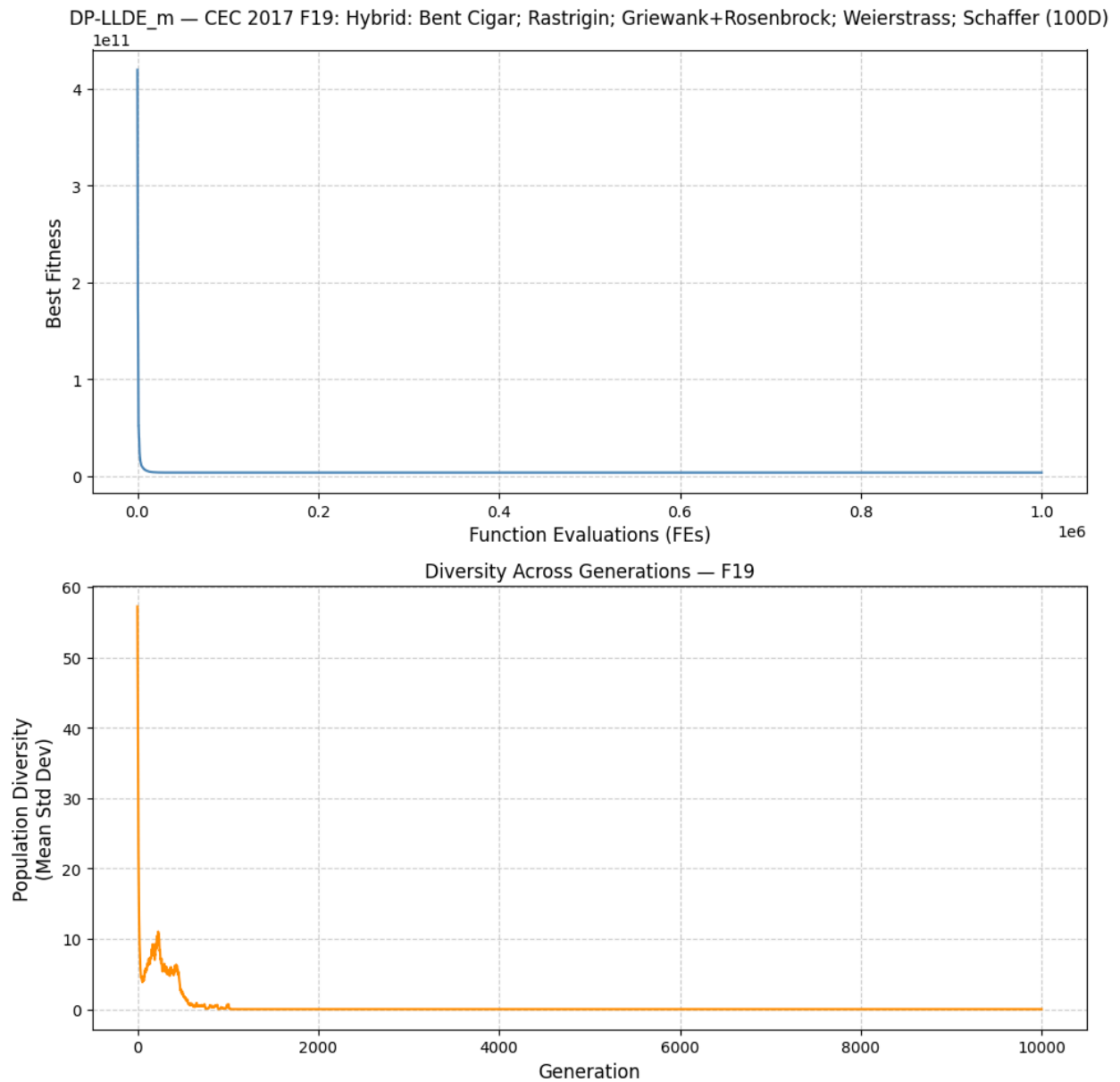
Total Function Evaluations = 1000000

```

=====
=
Results:
  Best Fitness           : 3.456128e+09
  Global Optimum         : 1.900000e+03
  Error (|best-optimum|) : 3.456126e+09
  Best Solution (5 dim)  : [ 99.05516427  11.84980868 -17.06334761 -34.59002363
-98.80295406]
  D^(0) (init. diversity): 5.7220e+01
=====
=

```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results_2/all_results_checkpoint.pkl



Graph saved as convergence_F19.png

✓ Plot saved to: /content/drive/My Drive/thesis_results_2/plots/convergence_F19.png

```
In [ ]: # =====  
# F20 — Hybrid Function 10 (HappyCat; Katsuura; Ackley; Rastrigin; Schwefel; S  
# =====  
run_and_plot('F20', make_wrapper(hybrid.f20), 2000.0, 'Hybrid: HappyCat; Katsu
```

=====

=

DP-LLDE_m on CEC 2017 F20 – Single Run

=====

=

Function : Hybrid: HappyCat; Katsuura; Ackley; Rastrigin; Schwefel; Schaffer (100D)

Bounds : [-100, 100]

Global optimum : 2000.0

Initial $D^{(0)} = 5.7220e+01$ | Diversity threshold = 5.7220e+00

Generation 0: Best Fitness = 8.995901e+03 | Diversity = 5.7220e+01

Generation 10: Best = 8.241507e+03 FES = 1100 Diversity = 4.7236e+01

Generation 20: Best = 8.096625e+03 FES = 2100 Diversity = 4.2724e+01

Generation 30: Best = 7.588296e+03 FES = 3100 Diversity = 4.1937e+01

Generation 40: Best = 7.588296e+03 FES = 4100 Diversity = 4.1559e+01

Generation 50: Best = 7.447472e+03 FES = 5100 Diversity = 4.0845e+01

Generation 60: Best = 7.447472e+03 FES = 6100 Diversity = 4.0238e+01

Generation 70: Best = 7.447472e+03 FES = 7100 Diversity = 4.0164e+01

Generation 80: Best = 7.312866e+03 FES = 8100 Diversity = 3.9964e+01

Generation 90: Best = 7.312866e+03 FES = 9100 Diversity = 3.9613e+01

Generation 100: Best = 7.311638e+03 FES = 10100 Diversity = 3.9410e+01

Generation 110: Best = 6.972480e+03 FES = 11100 Diversity = 3.9320e+01

Generation 120: Best = 6.972480e+03 FES = 12100 Diversity = 3.9143e+01

Generation 130: Best = 6.972480e+03 FES = 13100 Diversity = 3.9107e+01

Generation 140: Best = 6.972480e+03 FES = 14100 Diversity = 3.9023e+01

Generation 150: Best = 6.972480e+03 FES = 15100 Diversity = 3.8626e+01

Generation 160: Best = 6.972480e+03 FES = 16100 Diversity = 3.7905e+01

Generation 170: Best = 6.783038e+03 FES = 17100 Diversity = 3.7766e+01

Generation 180: Best = 6.783038e+03 FES = 18100 Diversity = 3.7596e+01

Generation 190: Best = 6.735575e+03 FES = 19100 Diversity = 3.7726e+01

Generation 200: Best = 6.735575e+03 FES = 20100 Diversity = 3.7574e+01

Generation 210: Best = 6.735575e+03 FES = 21100 Diversity = 3.7308e+01

Generation 220: Best = 6.735575e+03 FES = 22100 Diversity = 3.7315e+01

Generation 230: Best = 6.735575e+03 FES = 23100 Diversity = 3.7244e+01

Generation 240: Best = 6.735575e+03 FES = 24100 Diversity = 3.7282e+01

Generation 250: Best = 6.735575e+03 FES = 25100 Diversity = 3.7402e+01

Generation 260: Best = 6.460821e+03 FES = 26100 Diversity = 3.6853e+01

Generation 270: Best = 6.460821e+03 FES = 27100 Diversity = 3.7011e+01

Generation 280: Best = 6.460821e+03 FES = 28100 Diversity = 3.6843e+01

Generation 290: Best = 6.460821e+03 FES = 29100 Diversity = 3.6715e+01

Generation 300: Best = 6.456298e+03 FES = 30100 Diversity = 3.6646e+01

Generation 310: Best = 6.456298e+03 FES = 31100 Diversity = 3.6562e+01

Generation 320: Best = 6.401729e+03 FES = 32100 Diversity = 3.6528e+01

Generation 330: Best = 6.370066e+03 FES = 33100 Diversity = 3.6302e+01

Generation 340: Best = 6.370066e+03 FES = 34100 Diversity = 3.6017e+01

Generation 350: Best = 6.370066e+03 FES = 35100 Diversity = 3.5937e+01

Generation 360: Best = 6.370066e+03 FES = 36100 Diversity = 3.6038e+01

Generation 370: Best = 6.370066e+03 FES = 37100 Diversity = 3.5877e+01

Generation 380: Best = 6.362559e+03 FES = 38100 Diversity = 3.5878e+01

Generation 390: Best = 6.301935e+03 FES = 39100 Diversity = 3.5817e+01

Generation 400: Best = 6.301935e+03 FES = 40100 Diversity = 3.5586e+01

Generation 410: Best = 6.286271e+03 FES = 41100 Diversity = 3.5417e+01

Generation 420: Best = 6.254816e+03 FES = 42100 Diversity = 3.5299e+01

Generation 430: Best = 6.254816e+03 FES = 43100 Diversity = 3.5228e+01

Generation	440:	Best = 6.219227e+03	FES = 44100	Diversity = 3.5061e+01
Generation	450:	Best = 6.198148e+03	FES = 45100	Diversity = 3.4981e+01
Generation	460:	Best = 6.198148e+03	FES = 46100	Diversity = 3.4916e+01
Generation	470:	Best = 6.198148e+03	FES = 47100	Diversity = 3.4892e+01
Generation	480:	Best = 5.891106e+03	FES = 48100	Diversity = 3.4825e+01
Generation	490:	Best = 5.891106e+03	FES = 49100	Diversity = 3.5064e+01
Generation	500:	Best = 5.891106e+03	FES = 50100	Diversity = 3.5000e+01
Generation	510:	Best = 5.891106e+03	FES = 51100	Diversity = 3.4696e+01
Generation	520:	Best = 5.891106e+03	FES = 52100	Diversity = 3.4664e+01
Generation	530:	Best = 5.891106e+03	FES = 53100	Diversity = 3.4590e+01
Generation	540:	Best = 5.891106e+03	FES = 54100	Diversity = 3.4575e+01
Generation	550:	Best = 5.891099e+03	FES = 55100	Diversity = 3.4531e+01
Generation	560:	Best = 5.872992e+03	FES = 56100	Diversity = 3.4429e+01
Generation	570:	Best = 5.872926e+03	FES = 57100	Diversity = 3.4171e+01
Generation	580:	Best = 5.872926e+03	FES = 58100	Diversity = 3.4163e+01
Generation	590:	Best = 5.861754e+03	FES = 59100	Diversity = 3.4290e+01
Generation	600:	Best = 5.857014e+03	FES = 60100	Diversity = 3.4211e+01
Generation	610:	Best = 5.857007e+03	FES = 61100	Diversity = 3.4144e+01
Generation	620:	Best = 5.857007e+03	FES = 62100	Diversity = 3.3894e+01
Generation	630:	Best = 5.857007e+03	FES = 63100	Diversity = 3.3840e+01
Generation	640:	Best = 5.856393e+03	FES = 64100	Diversity = 3.3792e+01
Generation	650:	Best = 5.856020e+03	FES = 65100	Diversity = 3.3778e+01
Generation	660:	Best = 5.855705e+03	FES = 66100	Diversity = 3.3701e+01
Generation	670:	Best = 5.820048e+03	FES = 67100	Diversity = 3.3654e+01
Generation	680:	Best = 5.819987e+03	FES = 68100	Diversity = 3.3549e+01
Generation	690:	Best = 5.819987e+03	FES = 69100	Diversity = 3.3469e+01
Generation	700:	Best = 5.819987e+03	FES = 70100	Diversity = 3.3445e+01
Generation	710:	Best = 5.812647e+03	FES = 71100	Diversity = 3.3377e+01
Generation	720:	Best = 5.803369e+03	FES = 72100	Diversity = 3.3496e+01
Generation	730:	Best = 5.803369e+03	FES = 73100	Diversity = 3.3467e+01
Generation	740:	Best = 5.803369e+03	FES = 74100	Diversity = 3.3365e+01
Generation	750:	Best = 5.644888e+03	FES = 75100	Diversity = 3.3308e+01
Generation	760:	Best = 5.644888e+03	FES = 76100	Diversity = 3.3176e+01
Generation	770:	Best = 5.644888e+03	FES = 77100	Diversity = 3.3095e+01
Generation	780:	Best = 5.644888e+03	FES = 78100	Diversity = 3.2895e+01
Generation	790:	Best = 5.644888e+03	FES = 79100	Diversity = 3.2826e+01
Generation	800:	Best = 5.644888e+03	FES = 80100	Diversity = 3.2796e+01
Generation	810:	Best = 5.644888e+03	FES = 81100	Diversity = 3.2762e+01
Generation	820:	Best = 5.644888e+03	FES = 82100	Diversity = 3.2644e+01
Generation	830:	Best = 5.644326e+03	FES = 83100	Diversity = 3.2626e+01
Generation	840:	Best = 5.644326e+03	FES = 84100	Diversity = 3.2603e+01
Generation	850:	Best = 5.632565e+03	FES = 85100	Diversity = 3.2561e+01
Generation	860:	Best = 5.632565e+03	FES = 86100	Diversity = 3.2528e+01
Generation	870:	Best = 5.632472e+03	FES = 87100	Diversity = 3.2475e+01
Generation	880:	Best = 5.632472e+03	FES = 88100	Diversity = 3.2403e+01
Generation	890:	Best = 5.632472e+03	FES = 89100	Diversity = 3.2367e+01
Generation	900:	Best = 5.632472e+03	FES = 90100	Diversity = 3.2321e+01
Generation	910:	Best = 5.625106e+03	FES = 91100	Diversity = 3.2254e+01
Generation	920:	Best = 5.623141e+03	FES = 92100	Diversity = 3.2217e+01
Generation	930:	Best = 5.623141e+03	FES = 93100	Diversity = 3.2304e+01
Generation	940:	Best = 5.623141e+03	FES = 94100	Diversity = 3.2362e+01
Generation	950:	Best = 5.623141e+03	FES = 95100	Diversity = 3.2350e+01
Generation	960:	Best = 5.621724e+03	FES = 96100	Diversity = 3.2306e+01
Generation	970:	Best = 5.621724e+03	FES = 97100	Diversity = 3.2286e+01

Generation 980:	Best = 5.613505e+03	FES = 98100	Diversity = 3.2144e+01
Generation 990:	Best = 5.613505e+03	FES = 99100	Diversity = 3.1995e+01
Generation 1000:	Best = 5.613505e+03	FES = 100100	Diversity = 3.2054e+01
Generation 1010:	Best = 5.613505e+03	FES = 101100	Diversity = 3.2108e+01
Generation 1020:	Best = 5.595655e+03	FES = 102100	Diversity = 3.1953e+01
Generation 1030:	Best = 5.595112e+03	FES = 103100	Diversity = 3.1954e+01
Generation 1040:	Best = 5.595112e+03	FES = 104100	Diversity = 3.1871e+01
Generation 1050:	Best = 5.595112e+03	FES = 105100	Diversity = 3.1837e+01
Generation 1060:	Best = 5.595112e+03	FES = 106100	Diversity = 3.1356e+01
Generation 1070:	Best = 5.595112e+03	FES = 107100	Diversity = 3.1256e+01
Generation 1080:	Best = 5.594119e+03	FES = 108100	Diversity = 3.1220e+01
Generation 1090:	Best = 5.594119e+03	FES = 109100	Diversity = 3.1199e+01
Generation 1100:	Best = 5.594119e+03	FES = 110100	Diversity = 3.1179e+01
Generation 1110:	Best = 5.594119e+03	FES = 111100	Diversity = 3.1145e+01
Generation 1120:	Best = 5.594119e+03	FES = 112100	Diversity = 3.1097e+01
Generation 1130:	Best = 5.594119e+03	FES = 113100	Diversity = 3.0864e+01
Generation 1140:	Best = 5.594119e+03	FES = 114100	Diversity = 3.0833e+01
Generation 1150:	Best = 5.594084e+03	FES = 115100	Diversity = 3.0809e+01
Generation 1160:	Best = 5.594084e+03	FES = 116100	Diversity = 3.0759e+01
Generation 1170:	Best = 5.594084e+03	FES = 117100	Diversity = 3.0745e+01
Generation 1180:	Best = 5.571739e+03	FES = 118100	Diversity = 3.0710e+01
Generation 1190:	Best = 5.571739e+03	FES = 119100	Diversity = 3.0681e+01
Generation 1200:	Best = 5.571739e+03	FES = 120100	Diversity = 3.0691e+01
Generation 1210:	Best = 5.571739e+03	FES = 121100	Diversity = 3.0672e+01
Generation 1220:	Best = 5.571739e+03	FES = 122100	Diversity = 3.0731e+01
Generation 1230:	Best = 5.571739e+03	FES = 123100	Diversity = 3.0727e+01
Generation 1240:	Best = 5.571739e+03	FES = 124100	Diversity = 3.0707e+01
Generation 1250:	Best = 5.571739e+03	FES = 125100	Diversity = 3.0741e+01
Generation 1260:	Best = 5.571739e+03	FES = 126100	Diversity = 3.0516e+01
Generation 1270:	Best = 5.571739e+03	FES = 127100	Diversity = 3.0439e+01
Generation 1280:	Best = 5.571739e+03	FES = 128100	Diversity = 3.0417e+01
Generation 1290:	Best = 5.571739e+03	FES = 129100	Diversity = 3.0322e+01
Generation 1300:	Best = 5.571739e+03	FES = 130100	Diversity = 3.0292e+01
Generation 1310:	Best = 5.571739e+03	FES = 131100	Diversity = 3.0116e+01
Generation 1320:	Best = 5.571739e+03	FES = 132100	Diversity = 3.0081e+01
Generation 1330:	Best = 5.571739e+03	FES = 133100	Diversity = 3.0077e+01
Generation 1340:	Best = 5.571739e+03	FES = 134100	Diversity = 3.0022e+01
Generation 1350:	Best = 5.571739e+03	FES = 135100	Diversity = 2.9978e+01
Generation 1360:	Best = 5.571739e+03	FES = 136100	Diversity = 2.9967e+01
Generation 1370:	Best = 5.571515e+03	FES = 137100	Diversity = 2.9945e+01
Generation 1380:	Best = 5.571515e+03	FES = 138100	Diversity = 2.9948e+01
Generation 1390:	Best = 5.571515e+03	FES = 139100	Diversity = 2.9941e+01
Generation 1400:	Best = 5.568746e+03	FES = 140100	Diversity = 2.9913e+01
Generation 1410:	Best = 5.562999e+03	FES = 141100	Diversity = 2.9910e+01
Generation 1420:	Best = 5.562999e+03	FES = 142100	Diversity = 2.9882e+01
Generation 1430:	Best = 5.562999e+03	FES = 143100	Diversity = 2.9876e+01
Generation 1440:	Best = 5.558681e+03	FES = 144100	Diversity = 2.9831e+01
Generation 1450:	Best = 5.558681e+03	FES = 145100	Diversity = 2.9819e+01
Generation 1460:	Best = 5.558681e+03	FES = 146100	Diversity = 2.9793e+01
Generation 1470:	Best = 5.558681e+03	FES = 147100	Diversity = 2.9787e+01
Generation 1480:	Best = 5.558681e+03	FES = 148100	Diversity = 2.9739e+01
Generation 1490:	Best = 5.558681e+03	FES = 149100	Diversity = 2.9754e+01
Generation 1500:	Best = 5.541983e+03	FES = 150100	Diversity = 2.9717e+01
Generation 1510:	Best = 5.541305e+03	FES = 151100	Diversity = 2.9653e+01

Generation 1520:	Best = 5.539222e+03	FES = 152100	Diversity = 2.9649e+01
Generation 1530:	Best = 5.539222e+03	FES = 153100	Diversity = 2.9579e+01
Generation 1540:	Best = 5.539222e+03	FES = 154100	Diversity = 2.9573e+01
Generation 1550:	Best = 5.539222e+03	FES = 155100	Diversity = 2.9559e+01
Generation 1560:	Best = 5.539222e+03	FES = 156100	Diversity = 2.9544e+01
Generation 1570:	Best = 5.539071e+03	FES = 157100	Diversity = 2.9543e+01
Generation 1580:	Best = 5.539011e+03	FES = 158100	Diversity = 2.9540e+01
Generation 1590:	Best = 5.538587e+03	FES = 159100	Diversity = 2.9541e+01
Generation 1600:	Best = 5.538587e+03	FES = 160100	Diversity = 2.9486e+01
Generation 1610:	Best = 5.538386e+03	FES = 161100	Diversity = 2.9479e+01
Generation 1620:	Best = 5.538386e+03	FES = 162100	Diversity = 2.9450e+01
Generation 1630:	Best = 5.533203e+03	FES = 163100	Diversity = 2.9419e+01
Generation 1640:	Best = 5.533203e+03	FES = 164100	Diversity = 2.9330e+01
Generation 1650:	Best = 5.533154e+03	FES = 165100	Diversity = 2.9308e+01
Generation 1660:	Best = 5.533154e+03	FES = 166100	Diversity = 2.9314e+01
Generation 1670:	Best = 5.533154e+03	FES = 167100	Diversity = 2.9317e+01
Generation 1680:	Best = 5.533154e+03	FES = 168100	Diversity = 2.9307e+01
Generation 1690:	Best = 5.526730e+03	FES = 169100	Diversity = 2.9299e+01
Generation 1700:	Best = 5.526730e+03	FES = 170100	Diversity = 2.9296e+01
Generation 1710:	Best = 5.526730e+03	FES = 171100	Diversity = 2.9280e+01
Generation 1720:	Best = 5.526489e+03	FES = 172100	Diversity = 2.9241e+01
Generation 1730:	Best = 5.526489e+03	FES = 173100	Diversity = 2.9239e+01
Generation 1740:	Best = 5.526489e+03	FES = 174100	Diversity = 2.9230e+01
Generation 1750:	Best = 5.525340e+03	FES = 175100	Diversity = 2.9199e+01
Generation 1760:	Best = 5.522895e+03	FES = 176100	Diversity = 2.9175e+01
Generation 1770:	Best = 5.522895e+03	FES = 177100	Diversity = 2.9164e+01
Generation 1780:	Best = 5.522895e+03	FES = 178100	Diversity = 2.9054e+01
Generation 1790:	Best = 5.522895e+03	FES = 179100	Diversity = 2.9043e+01
Generation 1800:	Best = 5.522895e+03	FES = 180100	Diversity = 2.9037e+01
Generation 1810:	Best = 5.522207e+03	FES = 181100	Diversity = 2.9014e+01
Generation 1820:	Best = 5.522207e+03	FES = 182100	Diversity = 2.9001e+01
Generation 1830:	Best = 5.521385e+03	FES = 183100	Diversity = 2.8864e+01
Generation 1840:	Best = 5.521385e+03	FES = 184100	Diversity = 2.8837e+01
Generation 1850:	Best = 5.521385e+03	FES = 185100	Diversity = 2.8825e+01
Generation 1860:	Best = 5.483453e+03	FES = 186100	Diversity = 2.8799e+01
Generation 1870:	Best = 5.483453e+03	FES = 187100	Diversity = 2.8803e+01
Generation 1880:	Best = 5.483453e+03	FES = 188100	Diversity = 2.8782e+01
Generation 1890:	Best = 5.310382e+03	FES = 189100	Diversity = 2.8744e+01
Generation 1900:	Best = 5.310382e+03	FES = 190100	Diversity = 2.8732e+01
Generation 1910:	Best = 5.285122e+03	FES = 191100	Diversity = 2.8694e+01
Generation 1920:	Best = 5.285122e+03	FES = 192100	Diversity = 2.8697e+01
Generation 1930:	Best = 5.285122e+03	FES = 193100	Diversity = 2.8638e+01
Generation 1940:	Best = 5.285122e+03	FES = 194100	Diversity = 2.8593e+01
Generation 1950:	Best = 5.285122e+03	FES = 195100	Diversity = 2.8559e+01
Generation 1960:	Best = 5.285122e+03	FES = 196100	Diversity = 2.8542e+01
Generation 1970:	Best = 5.285122e+03	FES = 197100	Diversity = 2.8267e+01
Generation 1980:	Best = 5.285122e+03	FES = 198100	Diversity = 2.8254e+01
Generation 1990:	Best = 5.284917e+03	FES = 199100	Diversity = 2.8219e+01
Generation 2000:	Best = 5.284917e+03	FES = 200100	Diversity = 2.8195e+01
Generation 2010:	Best = 4.998091e+03	FES = 201100	Diversity = 2.8176e+01
Generation 2020:	Best = 4.998091e+03	FES = 202100	Diversity = 2.8057e+01
Generation 2030:	Best = 4.998091e+03	FES = 203100	Diversity = 2.8033e+01
Generation 2040:	Best = 4.998091e+03	FES = 204100	Diversity = 2.8010e+01
Generation 2050:	Best = 4.998091e+03	FES = 205100	Diversity = 2.8016e+01

Generation 2060:	Best = 4.998091e+03	FES = 206100	Diversity = 2.7991e+01
Generation 2070:	Best = 4.998091e+03	FES = 207100	Diversity = 2.7995e+01
Generation 2080:	Best = 4.998091e+03	FES = 208100	Diversity = 2.7971e+01
Generation 2090:	Best = 4.998091e+03	FES = 209100	Diversity = 2.7953e+01
Generation 2100:	Best = 4.979040e+03	FES = 210100	Diversity = 2.7933e+01
Generation 2110:	Best = 4.979040e+03	FES = 211100	Diversity = 2.7929e+01
Generation 2120:	Best = 4.979040e+03	FES = 212100	Diversity = 2.7918e+01
Generation 2130:	Best = 4.979040e+03	FES = 213100	Diversity = 2.7906e+01
Generation 2140:	Best = 4.978954e+03	FES = 214100	Diversity = 2.7881e+01
Generation 2150:	Best = 4.978934e+03	FES = 215100	Diversity = 2.7794e+01
Generation 2160:	Best = 4.978934e+03	FES = 216100	Diversity = 2.7758e+01
Generation 2170:	Best = 4.978934e+03	FES = 217100	Diversity = 2.7767e+01
Generation 2180:	Best = 4.895490e+03	FES = 218100	Diversity = 2.7761e+01
Generation 2190:	Best = 4.895490e+03	FES = 219100	Diversity = 2.7639e+01
Generation 2200:	Best = 4.895490e+03	FES = 220100	Diversity = 2.7612e+01
Generation 2210:	Best = 4.895490e+03	FES = 221100	Diversity = 2.7560e+01
Generation 2220:	Best = 4.895490e+03	FES = 222100	Diversity = 2.7527e+01
Generation 2230:	Best = 4.895490e+03	FES = 223100	Diversity = 2.7506e+01
Generation 2240:	Best = 4.895490e+03	FES = 224100	Diversity = 2.7480e+01
Generation 2250:	Best = 4.895490e+03	FES = 225100	Diversity = 2.7479e+01
Generation 2260:	Best = 4.895490e+03	FES = 226100	Diversity = 2.7322e+01
Generation 2270:	Best = 4.895490e+03	FES = 227100	Diversity = 2.7304e+01
Generation 2280:	Best = 4.895490e+03	FES = 228100	Diversity = 2.7294e+01
Generation 2290:	Best = 4.895490e+03	FES = 229100	Diversity = 2.7266e+01
Generation 2300:	Best = 4.895182e+03	FES = 230100	Diversity = 2.7254e+01
Generation 2310:	Best = 4.895182e+03	FES = 231100	Diversity = 2.7256e+01
Generation 2320:	Best = 4.895182e+03	FES = 232100	Diversity = 2.7255e+01
Generation 2330:	Best = 4.895182e+03	FES = 233100	Diversity = 2.7244e+01
Generation 2340:	Best = 4.895182e+03	FES = 234100	Diversity = 2.7235e+01
Generation 2350:	Best = 4.895182e+03	FES = 235100	Diversity = 2.7237e+01
Generation 2360:	Best = 4.895182e+03	FES = 236100	Diversity = 2.7292e+01
Generation 2370:	Best = 4.895182e+03	FES = 237100	Diversity = 2.7273e+01
Generation 2380:	Best = 4.895182e+03	FES = 238100	Diversity = 2.7256e+01
Generation 2390:	Best = 4.895182e+03	FES = 239100	Diversity = 2.7202e+01
Generation 2400:	Best = 4.895182e+03	FES = 240100	Diversity = 2.7212e+01
Generation 2410:	Best = 4.895182e+03	FES = 241100	Diversity = 2.7193e+01
Generation 2420:	Best = 4.895182e+03	FES = 242100	Diversity = 2.7156e+01
Generation 2430:	Best = 4.895182e+03	FES = 243100	Diversity = 2.7134e+01
Generation 2440:	Best = 4.895182e+03	FES = 244100	Diversity = 2.7134e+01
Generation 2450:	Best = 4.895182e+03	FES = 245100	Diversity = 2.7131e+01
Generation 2460:	Best = 4.895182e+03	FES = 246100	Diversity = 2.7129e+01
Generation 2470:	Best = 4.895182e+03	FES = 247100	Diversity = 2.7111e+01
Generation 2480:	Best = 4.895182e+03	FES = 248100	Diversity = 2.7103e+01
Generation 2490:	Best = 4.895182e+03	FES = 249100	Diversity = 2.7071e+01
Generation 2500:	Best = 4.895182e+03	FES = 250100	Diversity = 2.7063e+01
Generation 2510:	Best = 4.895182e+03	FES = 251100	Diversity = 2.7058e+01
Generation 2520:	Best = 4.895182e+03	FES = 252100	Diversity = 2.7059e+01
Generation 2530:	Best = 4.895182e+03	FES = 253100	Diversity = 2.7055e+01
Generation 2540:	Best = 4.895182e+03	FES = 254100	Diversity = 2.7045e+01
Generation 2550:	Best = 4.895182e+03	FES = 255100	Diversity = 2.7036e+01
Generation 2560:	Best = 4.895182e+03	FES = 256100	Diversity = 2.7034e+01
Generation 2570:	Best = 4.895182e+03	FES = 257100	Diversity = 2.7019e+01
Generation 2580:	Best = 4.895182e+03	FES = 258100	Diversity = 2.7021e+01
Generation 2590:	Best = 4.895182e+03	FES = 259100	Diversity = 2.7033e+01

Generation 2600:	Best = 4.895182e+03	FES = 260100	Diversity = 2.7034e+01
Generation 2610:	Best = 4.895182e+03	FES = 261100	Diversity = 2.7020e+01
Generation 2620:	Best = 4.895182e+03	FES = 262100	Diversity = 2.7015e+01
Generation 2630:	Best = 4.895106e+03	FES = 263100	Diversity = 2.6978e+01
Generation 2640:	Best = 4.895106e+03	FES = 264100	Diversity = 2.6905e+01
Generation 2650:	Best = 4.895106e+03	FES = 265100	Diversity = 2.6882e+01
Generation 2660:	Best = 4.895106e+03	FES = 266100	Diversity = 2.6854e+01
Generation 2670:	Best = 4.895106e+03	FES = 267100	Diversity = 2.6849e+01
Generation 2680:	Best = 4.895106e+03	FES = 268100	Diversity = 2.6841e+01
Generation 2690:	Best = 4.895106e+03	FES = 269100	Diversity = 2.6842e+01
Generation 2700:	Best = 4.895106e+03	FES = 270100	Diversity = 2.6842e+01
Generation 2710:	Best = 4.895106e+03	FES = 271100	Diversity = 2.6828e+01
Generation 2720:	Best = 4.895031e+03	FES = 272100	Diversity = 2.6821e+01
Generation 2730:	Best = 4.895031e+03	FES = 273100	Diversity = 2.6605e+01
Generation 2740:	Best = 4.895031e+03	FES = 274100	Diversity = 2.6587e+01
Generation 2750:	Best = 4.895031e+03	FES = 275100	Diversity = 2.6574e+01
Generation 2760:	Best = 4.892541e+03	FES = 276100	Diversity = 2.6574e+01
Generation 2770:	Best = 4.889080e+03	FES = 277100	Diversity = 2.6546e+01
Generation 2780:	Best = 4.889080e+03	FES = 278100	Diversity = 2.6547e+01
Generation 2790:	Best = 4.889080e+03	FES = 279100	Diversity = 2.6517e+01
Generation 2800:	Best = 4.889080e+03	FES = 280100	Diversity = 2.6508e+01
Generation 2810:	Best = 4.889080e+03	FES = 281100	Diversity = 2.6485e+01
Generation 2820:	Best = 4.889080e+03	FES = 282100	Diversity = 2.6496e+01
Generation 2830:	Best = 4.889080e+03	FES = 283100	Diversity = 2.6491e+01
Generation 2840:	Best = 4.889080e+03	FES = 284100	Diversity = 2.6477e+01
Generation 2850:	Best = 4.888783e+03	FES = 285100	Diversity = 2.6465e+01
Generation 2860:	Best = 4.888783e+03	FES = 286100	Diversity = 2.6457e+01
Generation 2870:	Best = 4.888783e+03	FES = 287100	Diversity = 2.6410e+01
Generation 2880:	Best = 4.888783e+03	FES = 288100	Diversity = 2.6399e+01
Generation 2890:	Best = 4.888783e+03	FES = 289100	Diversity = 2.6390e+01
Generation 2900:	Best = 4.888783e+03	FES = 290100	Diversity = 2.6352e+01
Generation 2910:	Best = 4.888783e+03	FES = 291100	Diversity = 2.6336e+01
Generation 2920:	Best = 4.888471e+03	FES = 292100	Diversity = 2.6323e+01
Generation 2930:	Best = 4.888471e+03	FES = 293100	Diversity = 2.6319e+01
Generation 2940:	Best = 4.887409e+03	FES = 294100	Diversity = 2.6300e+01
Generation 2950:	Best = 4.887409e+03	FES = 295100	Diversity = 2.6262e+01
Generation 2960:	Best = 4.887409e+03	FES = 296100	Diversity = 2.6246e+01
Generation 2970:	Best = 4.887409e+03	FES = 297100	Diversity = 2.6154e+01
Generation 2980:	Best = 4.887409e+03	FES = 298100	Diversity = 2.6150e+01
Generation 2990:	Best = 4.887409e+03	FES = 299100	Diversity = 2.6141e+01
Generation 3000:	Best = 4.887409e+03	FES = 300100	Diversity = 2.6124e+01
Generation 3010:	Best = 4.887409e+03	FES = 301100	Diversity = 2.6099e+01
Generation 3020:	Best = 4.887409e+03	FES = 302100	Diversity = 2.6081e+01
Generation 3030:	Best = 4.887409e+03	FES = 303100	Diversity = 2.6074e+01
Generation 3040:	Best = 4.887409e+03	FES = 304100	Diversity = 2.6071e+01
Generation 3050:	Best = 4.887398e+03	FES = 305100	Diversity = 2.6068e+01
Generation 3060:	Best = 4.887398e+03	FES = 306100	Diversity = 2.6050e+01
Generation 3070:	Best = 4.887398e+03	FES = 307100	Diversity = 2.6041e+01
Generation 3080:	Best = 4.887398e+03	FES = 308100	Diversity = 2.6030e+01
Generation 3090:	Best = 4.887278e+03	FES = 309100	Diversity = 2.6014e+01
Generation 3100:	Best = 4.887278e+03	FES = 310100	Diversity = 2.6002e+01
Generation 3110:	Best = 4.887278e+03	FES = 311100	Diversity = 2.5992e+01
Generation 3120:	Best = 4.887278e+03	FES = 312100	Diversity = 2.5989e+01
Generation 3130:	Best = 4.887278e+03	FES = 313100	Diversity = 2.5977e+01

Generation 3140:	Best = 4.681203e+03	FES = 314100	Diversity = 2.5971e+01
Generation 3150:	Best = 4.681203e+03	FES = 315100	Diversity = 2.5964e+01
Generation 3160:	Best = 4.681203e+03	FES = 316100	Diversity = 2.5952e+01
Generation 3170:	Best = 4.681203e+03	FES = 317100	Diversity = 2.5945e+01
Generation 3180:	Best = 4.681203e+03	FES = 318100	Diversity = 2.5916e+01
Generation 3190:	Best = 4.681203e+03	FES = 319100	Diversity = 2.5917e+01
Generation 3200:	Best = 4.681203e+03	FES = 320100	Diversity = 2.5905e+01
Generation 3210:	Best = 4.681203e+03	FES = 321100	Diversity = 2.5896e+01
Generation 3220:	Best = 4.679483e+03	FES = 322100	Diversity = 2.5886e+01
Generation 3230:	Best = 4.679483e+03	FES = 323100	Diversity = 2.5885e+01
Generation 3240:	Best = 4.679483e+03	FES = 324100	Diversity = 2.5887e+01
Generation 3250:	Best = 4.679483e+03	FES = 325100	Diversity = 2.5869e+01
Generation 3260:	Best = 4.679483e+03	FES = 326100	Diversity = 2.5838e+01
Generation 3270:	Best = 4.679483e+03	FES = 327100	Diversity = 2.5997e+01
Generation 3280:	Best = 4.679483e+03	FES = 328100	Diversity = 2.5995e+01
Generation 3290:	Best = 4.679483e+03	FES = 329100	Diversity = 2.5989e+01
Generation 3300:	Best = 4.676978e+03	FES = 330100	Diversity = 2.5967e+01
Generation 3310:	Best = 4.676978e+03	FES = 331100	Diversity = 2.5954e+01
Generation 3320:	Best = 4.676978e+03	FES = 332100	Diversity = 2.5925e+01
Generation 3330:	Best = 4.676978e+03	FES = 333100	Diversity = 2.5921e+01
Generation 3340:	Best = 4.676978e+03	FES = 334100	Diversity = 2.5804e+01
Generation 3350:	Best = 4.676978e+03	FES = 335100	Diversity = 2.5798e+01
Generation 3360:	Best = 4.676978e+03	FES = 336100	Diversity = 2.5789e+01
Generation 3370:	Best = 4.676978e+03	FES = 337100	Diversity = 2.5788e+01
Generation 3380:	Best = 4.676978e+03	FES = 338100	Diversity = 2.5785e+01
Generation 3390:	Best = 4.676978e+03	FES = 339100	Diversity = 2.5776e+01
Generation 3400:	Best = 4.676978e+03	FES = 340100	Diversity = 2.5749e+01
Generation 3410:	Best = 4.676978e+03	FES = 341100	Diversity = 2.5742e+01
Generation 3420:	Best = 4.676978e+03	FES = 342100	Diversity = 2.5702e+01
Generation 3430:	Best = 4.676978e+03	FES = 343100	Diversity = 2.5676e+01
Generation 3440:	Best = 4.676978e+03	FES = 344100	Diversity = 2.5674e+01
Generation 3450:	Best = 4.676978e+03	FES = 345100	Diversity = 2.5648e+01
Generation 3460:	Best = 4.676978e+03	FES = 346100	Diversity = 2.5637e+01
Generation 3470:	Best = 4.676978e+03	FES = 347100	Diversity = 2.5616e+01
Generation 3480:	Best = 4.676978e+03	FES = 348100	Diversity = 2.5606e+01
Generation 3490:	Best = 4.676978e+03	FES = 349100	Diversity = 2.5371e+01
Generation 3500:	Best = 4.676978e+03	FES = 350100	Diversity = 2.5354e+01
Generation 3510:	Best = 4.676978e+03	FES = 351100	Diversity = 2.5345e+01
Generation 3520:	Best = 4.676978e+03	FES = 352100	Diversity = 2.5225e+01
Generation 3530:	Best = 4.676978e+03	FES = 353100	Diversity = 2.5220e+01
Generation 3540:	Best = 4.676978e+03	FES = 354100	Diversity = 2.5213e+01
Generation 3550:	Best = 4.676978e+03	FES = 355100	Diversity = 2.5195e+01
Generation 3560:	Best = 4.676974e+03	FES = 356100	Diversity = 2.5162e+01
Generation 3570:	Best = 4.676974e+03	FES = 357100	Diversity = 2.5141e+01
Generation 3580:	Best = 4.676974e+03	FES = 358100	Diversity = 2.5140e+01
Generation 3590:	Best = 4.676974e+03	FES = 359100	Diversity = 2.5088e+01
Generation 3600:	Best = 4.676974e+03	FES = 360100	Diversity = 2.5090e+01
Generation 3610:	Best = 4.676974e+03	FES = 361100	Diversity = 2.5085e+01
Generation 3620:	Best = 4.676712e+03	FES = 362100	Diversity = 2.5077e+01
Generation 3630:	Best = 4.676712e+03	FES = 363100	Diversity = 2.5073e+01
Generation 3640:	Best = 4.676712e+03	FES = 364100	Diversity = 2.5067e+01
Generation 3650:	Best = 4.676712e+03	FES = 365100	Diversity = 2.5063e+01
Generation 3660:	Best = 4.676712e+03	FES = 366100	Diversity = 2.5045e+01
Generation 3670:	Best = 4.676712e+03	FES = 367100	Diversity = 2.5008e+01

Generation 3680:	Best = 4.676712e+03	FES = 368100	Diversity = 2.5004e+01
Generation 3690:	Best = 4.676712e+03	FES = 369100	Diversity = 2.4993e+01
Generation 3700:	Best = 4.676712e+03	FES = 370100	Diversity = 2.4975e+01
Generation 3710:	Best = 4.676011e+03	FES = 371100	Diversity = 2.4978e+01
Generation 3720:	Best = 4.675690e+03	FES = 372100	Diversity = 2.4972e+01
Generation 3730:	Best = 4.675690e+03	FES = 373100	Diversity = 2.4967e+01
Generation 3740:	Best = 4.675690e+03	FES = 374100	Diversity = 2.4967e+01
Generation 3750:	Best = 4.675678e+03	FES = 375100	Diversity = 2.4955e+01
Generation 3760:	Best = 4.675678e+03	FES = 376100	Diversity = 2.4954e+01
Generation 3770:	Best = 4.675678e+03	FES = 377100	Diversity = 2.4943e+01
Generation 3780:	Best = 4.675678e+03	FES = 378100	Diversity = 2.4927e+01
Generation 3790:	Best = 4.675678e+03	FES = 379100	Diversity = 2.4923e+01
Generation 3800:	Best = 4.675678e+03	FES = 380100	Diversity = 2.4912e+01
Generation 3810:	Best = 4.675678e+03	FES = 381100	Diversity = 2.4916e+01
Generation 3820:	Best = 4.675678e+03	FES = 382100	Diversity = 2.4904e+01
Generation 3830:	Best = 4.675678e+03	FES = 383100	Diversity = 2.4875e+01
Generation 3840:	Best = 4.675678e+03	FES = 384100	Diversity = 2.4872e+01
Generation 3850:	Best = 4.675678e+03	FES = 385100	Diversity = 2.4852e+01
Generation 3860:	Best = 4.675678e+03	FES = 386100	Diversity = 2.4839e+01
Generation 3870:	Best = 4.675678e+03	FES = 387100	Diversity = 2.4840e+01
Generation 3880:	Best = 4.675678e+03	FES = 388100	Diversity = 2.4820e+01
Generation 3890:	Best = 4.675678e+03	FES = 389100	Diversity = 2.4818e+01
Generation 3900:	Best = 4.675678e+03	FES = 390100	Diversity = 2.4817e+01
Generation 3910:	Best = 4.675678e+03	FES = 391100	Diversity = 2.4813e+01
Generation 3920:	Best = 4.675678e+03	FES = 392100	Diversity = 2.4807e+01
Generation 3930:	Best = 4.675678e+03	FES = 393100	Diversity = 2.4806e+01
Generation 3940:	Best = 4.675678e+03	FES = 394100	Diversity = 2.4800e+01
Generation 3950:	Best = 4.675664e+03	FES = 395100	Diversity = 2.4789e+01
Generation 3960:	Best = 4.675664e+03	FES = 396100	Diversity = 2.4780e+01
Generation 3970:	Best = 4.675664e+03	FES = 397100	Diversity = 2.4777e+01
Generation 3980:	Best = 4.675664e+03	FES = 398100	Diversity = 2.4769e+01
Generation 3990:	Best = 4.674665e+03	FES = 399100	Diversity = 2.4749e+01
Generation 4000:	Best = 4.674665e+03	FES = 400100	Diversity = 2.4744e+01
Generation 4010:	Best = 4.674665e+03	FES = 401100	Diversity = 2.4737e+01
Generation 4020:	Best = 4.674665e+03	FES = 402100	Diversity = 2.4737e+01
Generation 4030:	Best = 4.674665e+03	FES = 403100	Diversity = 2.4735e+01
Generation 4040:	Best = 4.674665e+03	FES = 404100	Diversity = 2.4733e+01
Generation 4050:	Best = 4.674665e+03	FES = 405100	Diversity = 2.4718e+01
Generation 4060:	Best = 4.674641e+03	FES = 406100	Diversity = 2.4712e+01
Generation 4070:	Best = 4.674641e+03	FES = 407100	Diversity = 2.4707e+01
Generation 4080:	Best = 4.674641e+03	FES = 408100	Diversity = 2.4699e+01
Generation 4090:	Best = 4.674641e+03	FES = 409100	Diversity = 2.4700e+01
Generation 4100:	Best = 4.674641e+03	FES = 410100	Diversity = 2.4704e+01
Generation 4110:	Best = 4.674641e+03	FES = 411100	Diversity = 2.4704e+01
Generation 4120:	Best = 4.674641e+03	FES = 412100	Diversity = 2.4698e+01
Generation 4130:	Best = 4.674641e+03	FES = 413100	Diversity = 2.4692e+01
Generation 4140:	Best = 4.674641e+03	FES = 414100	Diversity = 2.4690e+01
Generation 4150:	Best = 4.674641e+03	FES = 415100	Diversity = 2.4669e+01
Generation 4160:	Best = 4.674641e+03	FES = 416100	Diversity = 2.4667e+01
Generation 4170:	Best = 4.674641e+03	FES = 417100	Diversity = 2.4655e+01
Generation 4180:	Best = 4.674641e+03	FES = 418100	Diversity = 2.4637e+01
Generation 4190:	Best = 4.674641e+03	FES = 419100	Diversity = 2.4594e+01
Generation 4200:	Best = 4.674641e+03	FES = 420100	Diversity = 2.4582e+01
Generation 4210:	Best = 4.672552e+03	FES = 421100	Diversity = 2.4584e+01

Generation 4220:	Best = 4.672552e+03	FES = 422100	Diversity = 2.4574e+01
Generation 4230:	Best = 4.672552e+03	FES = 423100	Diversity = 2.4568e+01
Generation 4240:	Best = 4.672552e+03	FES = 424100	Diversity = 2.4568e+01
Generation 4250:	Best = 4.672552e+03	FES = 425100	Diversity = 2.4565e+01
Generation 4260:	Best = 4.672552e+03	FES = 426100	Diversity = 2.4563e+01
Generation 4270:	Best = 4.672552e+03	FES = 427100	Diversity = 2.4563e+01
Generation 4280:	Best = 4.672552e+03	FES = 428100	Diversity = 2.4560e+01
Generation 4290:	Best = 4.672552e+03	FES = 429100	Diversity = 2.4547e+01
Generation 4300:	Best = 4.672552e+03	FES = 430100	Diversity = 2.4298e+01
Generation 4310:	Best = 4.672552e+03	FES = 431100	Diversity = 2.4276e+01
Generation 4320:	Best = 4.672552e+03	FES = 432100	Diversity = 2.4274e+01
Generation 4330:	Best = 4.672552e+03	FES = 433100	Diversity = 2.4263e+01
Generation 4340:	Best = 4.672552e+03	FES = 434100	Diversity = 2.4258e+01
Generation 4350:	Best = 4.672552e+03	FES = 435100	Diversity = 2.4254e+01
Generation 4360:	Best = 4.672552e+03	FES = 436100	Diversity = 2.4249e+01
Generation 4370:	Best = 4.672495e+03	FES = 437100	Diversity = 2.4237e+01
Generation 4380:	Best = 4.672495e+03	FES = 438100	Diversity = 2.4233e+01
Generation 4390:	Best = 4.672495e+03	FES = 439100	Diversity = 2.4231e+01
Generation 4400:	Best = 4.672495e+03	FES = 440100	Diversity = 2.4098e+01
Generation 4410:	Best = 4.672495e+03	FES = 441100	Diversity = 2.4092e+01
Generation 4420:	Best = 4.672495e+03	FES = 442100	Diversity = 2.4095e+01
Generation 4430:	Best = 4.672495e+03	FES = 443100	Diversity = 2.4095e+01
Generation 4440:	Best = 4.672495e+03	FES = 444100	Diversity = 2.4086e+01
Generation 4450:	Best = 4.672392e+03	FES = 445100	Diversity = 2.4069e+01
Generation 4460:	Best = 4.672392e+03	FES = 446100	Diversity = 2.4059e+01
Generation 4470:	Best = 4.672392e+03	FES = 447100	Diversity = 2.4060e+01
Generation 4480:	Best = 4.672392e+03	FES = 448100	Diversity = 2.4062e+01
Generation 4490:	Best = 4.672392e+03	FES = 449100	Diversity = 2.4052e+01
Generation 4500:	Best = 4.672372e+03	FES = 450100	Diversity = 2.4032e+01
Generation 4510:	Best = 4.672372e+03	FES = 451100	Diversity = 2.4025e+01
Generation 4520:	Best = 4.672343e+03	FES = 452100	Diversity = 2.4020e+01
Generation 4530:	Best = 4.672343e+03	FES = 453100	Diversity = 2.4011e+01
Generation 4540:	Best = 4.672343e+03	FES = 454100	Diversity = 2.3995e+01
Generation 4550:	Best = 4.672343e+03	FES = 455100	Diversity = 2.3984e+01
Generation 4560:	Best = 4.672343e+03	FES = 456100	Diversity = 2.3975e+01
Generation 4570:	Best = 4.672343e+03	FES = 457100	Diversity = 2.3972e+01
Generation 4580:	Best = 4.672343e+03	FES = 458100	Diversity = 2.3966e+01
Generation 4590:	Best = 4.672343e+03	FES = 459100	Diversity = 2.3957e+01
Generation 4600:	Best = 4.672343e+03	FES = 460100	Diversity = 2.3908e+01
Generation 4610:	Best = 4.672343e+03	FES = 461100	Diversity = 2.3904e+01
Generation 4620:	Best = 4.672343e+03	FES = 462100	Diversity = 2.3901e+01
Generation 4630:	Best = 4.672343e+03	FES = 463100	Diversity = 2.3891e+01
Generation 4640:	Best = 4.672343e+03	FES = 464100	Diversity = 2.3889e+01
Generation 4650:	Best = 4.672343e+03	FES = 465100	Diversity = 2.3884e+01
Generation 4660:	Best = 4.672343e+03	FES = 466100	Diversity = 2.3885e+01
Generation 4670:	Best = 4.672343e+03	FES = 467100	Diversity = 2.3876e+01
Generation 4680:	Best = 4.672343e+03	FES = 468100	Diversity = 2.3874e+01
Generation 4690:	Best = 4.672343e+03	FES = 469100	Diversity = 2.3870e+01
Generation 4700:	Best = 4.672343e+03	FES = 470100	Diversity = 2.3867e+01
Generation 4710:	Best = 4.672343e+03	FES = 471100	Diversity = 2.3866e+01
Generation 4720:	Best = 4.672343e+03	FES = 472100	Diversity = 2.3866e+01
Generation 4730:	Best = 4.672343e+03	FES = 473100	Diversity = 2.3860e+01
Generation 4740:	Best = 4.672343e+03	FES = 474100	Diversity = 2.3839e+01
Generation 4750:	Best = 4.672343e+03	FES = 475100	Diversity = 2.3832e+01

Generation 4760:	Best = 4.672343e+03	FES = 476100	Diversity = 2.3937e+01
Generation 4770:	Best = 4.672213e+03	FES = 477100	Diversity = 2.3927e+01
Generation 4780:	Best = 4.669684e+03	FES = 478100	Diversity = 2.3926e+01
Generation 4790:	Best = 4.669684e+03	FES = 479100	Diversity = 2.3916e+01
Generation 4800:	Best = 4.669684e+03	FES = 480100	Diversity = 2.3913e+01
Generation 4810:	Best = 4.669684e+03	FES = 481100	Diversity = 2.3884e+01
Generation 4820:	Best = 4.669684e+03	FES = 482100	Diversity = 2.3876e+01
Generation 4830:	Best = 4.666684e+03	FES = 483100	Diversity = 2.3869e+01
Generation 4840:	Best = 4.666684e+03	FES = 484100	Diversity = 2.3857e+01
Generation 4850:	Best = 4.666684e+03	FES = 485100	Diversity = 2.3855e+01
Generation 4860:	Best = 4.666684e+03	FES = 486100	Diversity = 2.3853e+01
Generation 4870:	Best = 4.666684e+03	FES = 487100	Diversity = 2.3851e+01
Generation 4880:	Best = 4.666684e+03	FES = 488100	Diversity = 2.3844e+01
Generation 4890:	Best = 4.666684e+03	FES = 489100	Diversity = 2.3837e+01
Generation 4900:	Best = 4.666684e+03	FES = 490100	Diversity = 2.3842e+01
Generation 4910:	Best = 4.666684e+03	FES = 491100	Diversity = 2.3838e+01
Generation 4920:	Best = 4.666684e+03	FES = 492100	Diversity = 2.3837e+01
Generation 4930:	Best = 4.666684e+03	FES = 493100	Diversity = 2.3835e+01
Generation 4940:	Best = 4.633000e+03	FES = 494100	Diversity = 2.3818e+01
Generation 4950:	Best = 4.633000e+03	FES = 495100	Diversity = 2.3756e+01
Generation 4960:	Best = 4.633000e+03	FES = 496100	Diversity = 2.3755e+01
Generation 4970:	Best = 4.633000e+03	FES = 497100	Diversity = 2.3643e+01
Generation 4980:	Best = 4.633000e+03	FES = 498100	Diversity = 2.3638e+01
Generation 4990:	Best = 4.633000e+03	FES = 499100	Diversity = 2.3639e+01
Generation 5000:	Best = 4.632773e+03	FES = 500100	Diversity = 2.3617e+01
Generation 5010:	Best = 4.632773e+03	FES = 501100	Diversity = 2.3612e+01
Generation 5020:	Best = 4.632773e+03	FES = 502100	Diversity = 2.3611e+01
Generation 5030:	Best = 4.632773e+03	FES = 503100	Diversity = 2.3609e+01
Generation 5040:	Best = 4.632573e+03	FES = 504100	Diversity = 2.3610e+01
Generation 5050:	Best = 4.632573e+03	FES = 505100	Diversity = 2.3611e+01
Generation 5060:	Best = 4.632573e+03	FES = 506100	Diversity = 2.3611e+01
Generation 5070:	Best = 4.632573e+03	FES = 507100	Diversity = 2.3609e+01
Generation 5080:	Best = 4.435376e+03	FES = 508100	Diversity = 2.3593e+01
Generation 5090:	Best = 4.435376e+03	FES = 509100	Diversity = 2.3583e+01
Generation 5100:	Best = 4.435376e+03	FES = 510100	Diversity = 2.3576e+01
Generation 5110:	Best = 4.434360e+03	FES = 511100	Diversity = 2.3558e+01
Generation 5120:	Best = 4.434360e+03	FES = 512100	Diversity = 2.3560e+01
Generation 5130:	Best = 4.434360e+03	FES = 513100	Diversity = 2.3554e+01
Generation 5140:	Best = 4.434360e+03	FES = 514100	Diversity = 2.3551e+01
Generation 5150:	Best = 4.434360e+03	FES = 515100	Diversity = 2.3548e+01
Generation 5160:	Best = 4.426569e+03	FES = 516100	Diversity = 2.3545e+01
Generation 5170:	Best = 4.426569e+03	FES = 517100	Diversity = 2.3534e+01
Generation 5180:	Best = 4.426569e+03	FES = 518100	Diversity = 2.3523e+01
Generation 5190:	Best = 4.426569e+03	FES = 519100	Diversity = 2.3519e+01
Generation 5200:	Best = 4.426569e+03	FES = 520100	Diversity = 2.3520e+01
Generation 5210:	Best = 4.426569e+03	FES = 521100	Diversity = 2.3494e+01
Generation 5220:	Best = 4.426569e+03	FES = 522100	Diversity = 2.3475e+01
Generation 5230:	Best = 4.426569e+03	FES = 523100	Diversity = 2.3468e+01
Generation 5240:	Best = 4.426569e+03	FES = 524100	Diversity = 2.3474e+01
Generation 5250:	Best = 4.426569e+03	FES = 525100	Diversity = 2.3474e+01
Generation 5260:	Best = 4.426569e+03	FES = 526100	Diversity = 2.3466e+01
Generation 5270:	Best = 4.426569e+03	FES = 527100	Diversity = 2.3457e+01
Generation 5280:	Best = 4.426569e+03	FES = 528100	Diversity = 2.3451e+01
Generation 5290:	Best = 4.426569e+03	FES = 529100	Diversity = 2.3450e+01

Generation 5300:	Best = 4.426569e+03	FES = 530100	Diversity = 2.3443e+01
Generation 5310:	Best = 4.426569e+03	FES = 531100	Diversity = 2.3440e+01
Generation 5320:	Best = 4.426527e+03	FES = 532100	Diversity = 2.3429e+01
Generation 5330:	Best = 4.426527e+03	FES = 533100	Diversity = 2.3429e+01
Generation 5340:	Best = 4.426527e+03	FES = 534100	Diversity = 2.3419e+01
Generation 5350:	Best = 4.426527e+03	FES = 535100	Diversity = 2.3418e+01
Generation 5360:	Best = 4.426527e+03	FES = 536100	Diversity = 2.3418e+01
Generation 5370:	Best = 4.426527e+03	FES = 537100	Diversity = 2.3418e+01
Generation 5380:	Best = 4.426527e+03	FES = 538100	Diversity = 2.3415e+01
Generation 5390:	Best = 4.426527e+03	FES = 539100	Diversity = 2.3416e+01
Generation 5400:	Best = 4.426527e+03	FES = 540100	Diversity = 2.3406e+01
Generation 5410:	Best = 4.426527e+03	FES = 541100	Diversity = 2.3404e+01
Generation 5420:	Best = 4.426527e+03	FES = 542100	Diversity = 2.3406e+01
Generation 5430:	Best = 4.407078e+03	FES = 543100	Diversity = 2.3403e+01
Generation 5440:	Best = 4.407078e+03	FES = 544100	Diversity = 2.3403e+01
Generation 5450:	Best = 4.407066e+03	FES = 545100	Diversity = 2.3414e+01
Generation 5460:	Best = 4.407066e+03	FES = 546100	Diversity = 2.3395e+01
Generation 5470:	Best = 4.407066e+03	FES = 547100	Diversity = 2.3395e+01
Generation 5480:	Best = 4.407066e+03	FES = 548100	Diversity = 2.3400e+01
Generation 5490:	Best = 4.407066e+03	FES = 549100	Diversity = 2.3392e+01
Generation 5500:	Best = 4.407066e+03	FES = 550100	Diversity = 2.3398e+01
Generation 5510:	Best = 4.407066e+03	FES = 551100	Diversity = 2.3398e+01
Generation 5520:	Best = 4.407066e+03	FES = 552100	Diversity = 2.3396e+01
Generation 5530:	Best = 4.407066e+03	FES = 553100	Diversity = 2.3390e+01
Generation 5540:	Best = 4.407066e+03	FES = 554100	Diversity = 2.3367e+01
Generation 5550:	Best = 4.407066e+03	FES = 555100	Diversity = 2.3419e+01
Generation 5560:	Best = 4.407066e+03	FES = 556100	Diversity = 2.3420e+01
Generation 5570:	Best = 4.407036e+03	FES = 557100	Diversity = 2.3419e+01
Generation 5580:	Best = 4.406898e+03	FES = 558100	Diversity = 2.3411e+01
Generation 5590:	Best = 4.406898e+03	FES = 559100	Diversity = 2.3396e+01
Generation 5600:	Best = 4.406747e+03	FES = 560100	Diversity = 2.3402e+01
Generation 5610:	Best = 4.406747e+03	FES = 561100	Diversity = 2.3269e+01
Generation 5620:	Best = 4.406747e+03	FES = 562100	Diversity = 2.3261e+01
Generation 5630:	Best = 4.406747e+03	FES = 563100	Diversity = 2.3252e+01
Generation 5640:	Best = 4.406747e+03	FES = 564100	Diversity = 2.3246e+01
Generation 5650:	Best = 4.406736e+03	FES = 565100	Diversity = 2.3239e+01
Generation 5660:	Best = 4.406736e+03	FES = 566100	Diversity = 2.3232e+01
Generation 5670:	Best = 4.406736e+03	FES = 567100	Diversity = 2.3232e+01
Generation 5680:	Best = 4.406736e+03	FES = 568100	Diversity = 2.3229e+01
Generation 5690:	Best = 4.406652e+03	FES = 569100	Diversity = 2.3227e+01
Generation 5700:	Best = 4.406652e+03	FES = 570100	Diversity = 2.3272e+01
Generation 5710:	Best = 4.406531e+03	FES = 571100	Diversity = 2.3270e+01
Generation 5720:	Best = 4.406531e+03	FES = 572100	Diversity = 2.3265e+01
Generation 5730:	Best = 4.406531e+03	FES = 573100	Diversity = 2.3267e+01
Generation 5740:	Best = 4.406531e+03	FES = 574100	Diversity = 2.3261e+01
Generation 5750:	Best = 4.406531e+03	FES = 575100	Diversity = 2.3254e+01
Generation 5760:	Best = 4.406531e+03	FES = 576100	Diversity = 2.3247e+01
Generation 5770:	Best = 4.403364e+03	FES = 577100	Diversity = 2.3246e+01
Generation 5780:	Best = 4.403364e+03	FES = 578100	Diversity = 2.3058e+01
Generation 5790:	Best = 4.403364e+03	FES = 579100	Diversity = 2.3054e+01
Generation 5800:	Best = 4.403364e+03	FES = 580100	Diversity = 2.3045e+01
Generation 5810:	Best = 4.403364e+03	FES = 581100	Diversity = 2.3047e+01
Generation 5820:	Best = 4.403364e+03	FES = 582100	Diversity = 2.3044e+01
Generation 5830:	Best = 4.403364e+03	FES = 583100	Diversity = 2.3044e+01

Generation 5840:	Best = 4.403364e+03	FES = 584100	Diversity = 2.3042e+01
Generation 5850:	Best = 4.403208e+03	FES = 585100	Diversity = 2.3029e+01
Generation 5860:	Best = 4.403208e+03	FES = 586100	Diversity = 2.3024e+01
Generation 5870:	Best = 4.403208e+03	FES = 587100	Diversity = 2.3002e+01
Generation 5880:	Best = 4.403208e+03	FES = 588100	Diversity = 2.3003e+01
Generation 5890:	Best = 4.403208e+03	FES = 589100	Diversity = 2.2996e+01
Generation 5900:	Best = 4.403208e+03	FES = 590100	Diversity = 2.2990e+01
Generation 5910:	Best = 4.403208e+03	FES = 591100	Diversity = 2.2987e+01
Generation 5920:	Best = 4.403208e+03	FES = 592100	Diversity = 2.2986e+01
Generation 5930:	Best = 4.403208e+03	FES = 593100	Diversity = 2.2984e+01
Generation 5940:	Best = 4.403208e+03	FES = 594100	Diversity = 2.2982e+01
Generation 5950:	Best = 4.403050e+03	FES = 595100	Diversity = 2.2929e+01
Generation 5960:	Best = 4.403050e+03	FES = 596100	Diversity = 2.2906e+01
Generation 5970:	Best = 4.403050e+03	FES = 597100	Diversity = 2.2905e+01
Generation 5980:	Best = 4.403050e+03	FES = 598100	Diversity = 2.2893e+01
Generation 5990:	Best = 4.403050e+03	FES = 599100	Diversity = 2.2884e+01
Generation 6000:	Best = 4.403050e+03	FES = 600100	Diversity = 2.2885e+01
Generation 6010:	Best = 4.403050e+03	FES = 601100	Diversity = 2.2878e+01
Generation 6020:	Best = 4.403050e+03	FES = 602100	Diversity = 2.2877e+01
Generation 6030:	Best = 4.403050e+03	FES = 603100	Diversity = 2.2880e+01
Generation 6040:	Best = 4.402922e+03	FES = 604100	Diversity = 2.2873e+01
Generation 6050:	Best = 4.402922e+03	FES = 605100	Diversity = 2.2860e+01
Generation 6060:	Best = 4.402922e+03	FES = 606100	Diversity = 2.2854e+01
Generation 6070:	Best = 4.402922e+03	FES = 607100	Diversity = 2.2851e+01
Generation 6080:	Best = 4.402922e+03	FES = 608100	Diversity = 2.2838e+01
Generation 6090:	Best = 4.402922e+03	FES = 609100	Diversity = 2.2835e+01
Generation 6100:	Best = 4.402922e+03	FES = 610100	Diversity = 2.2827e+01
Generation 6110:	Best = 4.402922e+03	FES = 611100	Diversity = 2.2827e+01
Generation 6120:	Best = 4.402922e+03	FES = 612100	Diversity = 2.2819e+01
Generation 6130:	Best = 4.402922e+03	FES = 613100	Diversity = 2.2805e+01
Generation 6140:	Best = 4.402829e+03	FES = 614100	Diversity = 2.2799e+01
Generation 6150:	Best = 4.402829e+03	FES = 615100	Diversity = 2.2795e+01
Generation 6160:	Best = 4.402829e+03	FES = 616100	Diversity = 2.2785e+01
Generation 6170:	Best = 4.402829e+03	FES = 617100	Diversity = 2.2773e+01
Generation 6180:	Best = 4.402829e+03	FES = 618100	Diversity = 2.2774e+01
Generation 6190:	Best = 4.402829e+03	FES = 619100	Diversity = 2.2773e+01
Generation 6200:	Best = 4.402829e+03	FES = 620100	Diversity = 2.2767e+01
Generation 6210:	Best = 4.402829e+03	FES = 621100	Diversity = 2.2759e+01
Generation 6220:	Best = 4.402829e+03	FES = 622100	Diversity = 2.2757e+01
Generation 6230:	Best = 4.402829e+03	FES = 623100	Diversity = 2.2746e+01
Generation 6240:	Best = 4.402790e+03	FES = 624100	Diversity = 2.2738e+01
Generation 6250:	Best = 4.402790e+03	FES = 625100	Diversity = 2.2732e+01
Generation 6260:	Best = 4.402790e+03	FES = 626100	Diversity = 2.2731e+01
Generation 6270:	Best = 4.402790e+03	FES = 627100	Diversity = 2.2732e+01
Generation 6280:	Best = 4.402790e+03	FES = 628100	Diversity = 2.2724e+01
Generation 6290:	Best = 4.402779e+03	FES = 629100	Diversity = 2.2724e+01
Generation 6300:	Best = 4.402779e+03	FES = 630100	Diversity = 2.2718e+01
Generation 6310:	Best = 4.402779e+03	FES = 631100	Diversity = 2.2710e+01
Generation 6320:	Best = 4.402779e+03	FES = 632100	Diversity = 2.2709e+01
Generation 6330:	Best = 4.402779e+03	FES = 633100	Diversity = 2.2708e+01
Generation 6340:	Best = 4.402779e+03	FES = 634100	Diversity = 2.2708e+01
Generation 6350:	Best = 4.402779e+03	FES = 635100	Diversity = 2.2709e+01
Generation 6360:	Best = 4.402779e+03	FES = 636100	Diversity = 2.2708e+01
Generation 6370:	Best = 4.402779e+03	FES = 637100	Diversity = 2.2711e+01

Generation 6380:	Best = 4.402779e+03	FES = 638100	Diversity = 2.2711e+01
Generation 6390:	Best = 4.402779e+03	FES = 639100	Diversity = 2.2702e+01
Generation 6400:	Best = 4.402779e+03	FES = 640100	Diversity = 2.2703e+01
Generation 6410:	Best = 4.402779e+03	FES = 641100	Diversity = 2.2704e+01
Generation 6420:	Best = 4.402779e+03	FES = 642100	Diversity = 2.2700e+01
Generation 6430:	Best = 4.402779e+03	FES = 643100	Diversity = 2.2696e+01
Generation 6440:	Best = 4.402779e+03	FES = 644100	Diversity = 2.2680e+01
Generation 6450:	Best = 4.402779e+03	FES = 645100	Diversity = 2.2678e+01
Generation 6460:	Best = 4.402779e+03	FES = 646100	Diversity = 2.2666e+01
Generation 6470:	Best = 4.402779e+03	FES = 647100	Diversity = 2.2666e+01
Generation 6480:	Best = 4.402779e+03	FES = 648100	Diversity = 2.2662e+01
Generation 6490:	Best = 4.402779e+03	FES = 649100	Diversity = 2.2657e+01
Generation 6500:	Best = 4.401944e+03	FES = 650100	Diversity = 2.2655e+01
Generation 6510:	Best = 4.401944e+03	FES = 651100	Diversity = 2.2653e+01
Generation 6520:	Best = 4.401944e+03	FES = 652100	Diversity = 2.2651e+01
Generation 6530:	Best = 4.401589e+03	FES = 653100	Diversity = 2.2651e+01
Generation 6540:	Best = 4.401541e+03	FES = 654100	Diversity = 2.2644e+01
Generation 6550:	Best = 4.401541e+03	FES = 655100	Diversity = 2.2634e+01
Generation 6560:	Best = 4.399999e+03	FES = 656100	Diversity = 2.2624e+01
Generation 6570:	Best = 4.399868e+03	FES = 657100	Diversity = 2.2619e+01
Generation 6580:	Best = 4.399868e+03	FES = 658100	Diversity = 2.2612e+01
Generation 6590:	Best = 4.399868e+03	FES = 659100	Diversity = 2.2578e+01
Generation 6600:	Best = 4.399790e+03	FES = 660100	Diversity = 2.2576e+01
Generation 6610:	Best = 4.383876e+03	FES = 661100	Diversity = 2.2503e+01
Generation 6620:	Best = 4.383876e+03	FES = 662100	Diversity = 2.2502e+01
Generation 6630:	Best = 4.383876e+03	FES = 663100	Diversity = 2.2492e+01
Generation 6640:	Best = 4.383790e+03	FES = 664100	Diversity = 2.2478e+01
Generation 6650:	Best = 4.383790e+03	FES = 665100	Diversity = 2.2472e+01
Generation 6660:	Best = 4.383790e+03	FES = 666100	Diversity = 2.2472e+01
Generation 6670:	Best = 4.383739e+03	FES = 667100	Diversity = 2.2468e+01
Generation 6680:	Best = 4.383505e+03	FES = 668100	Diversity = 2.2458e+01
Generation 6690:	Best = 4.382784e+03	FES = 669100	Diversity = 2.2458e+01
Generation 6700:	Best = 4.382784e+03	FES = 670100	Diversity = 2.2454e+01
Generation 6710:	Best = 4.382784e+03	FES = 671100	Diversity = 2.2448e+01
Generation 6720:	Best = 4.379882e+03	FES = 672100	Diversity = 2.2423e+01
Generation 6730:	Best = 4.379882e+03	FES = 673100	Diversity = 2.2417e+01
Generation 6740:	Best = 4.379882e+03	FES = 674100	Diversity = 2.2405e+01
Generation 6750:	Best = 4.379882e+03	FES = 675100	Diversity = 2.2406e+01
Generation 6760:	Best = 4.379882e+03	FES = 676100	Diversity = 2.2401e+01
Generation 6770:	Best = 4.379882e+03	FES = 677100	Diversity = 2.2392e+01
Generation 6780:	Best = 4.379882e+03	FES = 678100	Diversity = 2.2392e+01
Generation 6790:	Best = 4.369387e+03	FES = 679100	Diversity = 2.2390e+01
Generation 6800:	Best = 4.369387e+03	FES = 680100	Diversity = 2.2386e+01
Generation 6810:	Best = 4.369387e+03	FES = 681100	Diversity = 2.2386e+01
Generation 6820:	Best = 4.369387e+03	FES = 682100	Diversity = 2.2386e+01
Generation 6830:	Best = 4.369387e+03	FES = 683100	Diversity = 2.2382e+01
Generation 6840:	Best = 4.369387e+03	FES = 684100	Diversity = 2.2381e+01
Generation 6850:	Best = 4.369326e+03	FES = 685100	Diversity = 2.2380e+01
Generation 6860:	Best = 4.369326e+03	FES = 686100	Diversity = 2.2375e+01
Generation 6870:	Best = 4.369326e+03	FES = 687100	Diversity = 2.2375e+01
Generation 6880:	Best = 4.369326e+03	FES = 688100	Diversity = 2.2373e+01
Generation 6890:	Best = 4.369326e+03	FES = 689100	Diversity = 2.2365e+01
Generation 6900:	Best = 4.369326e+03	FES = 690100	Diversity = 2.2361e+01
Generation 6910:	Best = 4.369326e+03	FES = 691100	Diversity = 2.2354e+01

Generation 6920:	Best = 4.369326e+03	FES = 692100	Diversity = 2.2353e+01
Generation 6930:	Best = 4.364796e+03	FES = 693100	Diversity = 2.2348e+01
Generation 6940:	Best = 4.364796e+03	FES = 694100	Diversity = 2.2345e+01
Generation 6950:	Best = 4.364796e+03	FES = 695100	Diversity = 2.2341e+01
Generation 6960:	Best = 4.364796e+03	FES = 696100	Diversity = 2.2340e+01
Generation 6970:	Best = 4.364796e+03	FES = 697100	Diversity = 2.2338e+01
Generation 6980:	Best = 4.364796e+03	FES = 698100	Diversity = 2.2328e+01
Generation 6990:	Best = 4.364796e+03	FES = 699100	Diversity = 2.2322e+01
Generation 7000:	Best = 4.364796e+03	FES = 700100	Diversity = 2.2321e+01
Generation 7010:	Best = 4.364796e+03	FES = 701100	Diversity = 2.2320e+01
Generation 7020:	Best = 4.364796e+03	FES = 702100	Diversity = 2.2316e+01
Generation 7030:	Best = 4.362092e+03	FES = 703100	Diversity = 2.2310e+01
Generation 7040:	Best = 4.361946e+03	FES = 704100	Diversity = 2.2296e+01
Generation 7050:	Best = 4.361946e+03	FES = 705100	Diversity = 2.2285e+01
Generation 7060:	Best = 4.361946e+03	FES = 706100	Diversity = 2.2273e+01
Generation 7070:	Best = 4.361946e+03	FES = 707100	Diversity = 2.2261e+01
Generation 7080:	Best = 4.361946e+03	FES = 708100	Diversity = 2.2261e+01
Generation 7090:	Best = 4.361946e+03	FES = 709100	Diversity = 2.2249e+01
Generation 7100:	Best = 4.361946e+03	FES = 710100	Diversity = 2.2231e+01
Generation 7110:	Best = 4.361946e+03	FES = 711100	Diversity = 2.2222e+01
Generation 7120:	Best = 4.361946e+03	FES = 712100	Diversity = 2.2222e+01
Generation 7130:	Best = 4.361946e+03	FES = 713100	Diversity = 2.2167e+01
Generation 7140:	Best = 4.361946e+03	FES = 714100	Diversity = 2.2167e+01
Generation 7150:	Best = 4.361946e+03	FES = 715100	Diversity = 2.2167e+01
Generation 7160:	Best = 4.361946e+03	FES = 716100	Diversity = 2.2167e+01
Generation 7170:	Best = 4.361946e+03	FES = 717100	Diversity = 2.2167e+01
Generation 7180:	Best = 4.361946e+03	FES = 718100	Diversity = 2.2165e+01
Generation 7190:	Best = 4.361946e+03	FES = 719100	Diversity = 2.2163e+01
Generation 7200:	Best = 4.361946e+03	FES = 720100	Diversity = 2.2160e+01
Generation 7210:	Best = 4.361946e+03	FES = 721100	Diversity = 2.2160e+01
Generation 7220:	Best = 4.361946e+03	FES = 722100	Diversity = 2.2155e+01
Generation 7230:	Best = 4.361946e+03	FES = 723100	Diversity = 2.2154e+01
Generation 7240:	Best = 4.361946e+03	FES = 724100	Diversity = 2.2152e+01
Generation 7250:	Best = 4.361946e+03	FES = 725100	Diversity = 2.2142e+01
Generation 7260:	Best = 4.361922e+03	FES = 726100	Diversity = 2.2143e+01
Generation 7270:	Best = 4.361922e+03	FES = 727100	Diversity = 2.2142e+01
Generation 7280:	Best = 4.361922e+03	FES = 728100	Diversity = 2.2139e+01
Generation 7290:	Best = 4.361922e+03	FES = 729100	Diversity = 2.1912e+01
Generation 7300:	Best = 4.361922e+03	FES = 730100	Diversity = 2.1913e+01
Generation 7310:	Best = 4.361922e+03	FES = 731100	Diversity = 2.1913e+01
Generation 7320:	Best = 4.361922e+03	FES = 732100	Diversity = 2.1902e+01
Generation 7330:	Best = 4.361922e+03	FES = 733100	Diversity = 2.1890e+01
Generation 7340:	Best = 4.361922e+03	FES = 734100	Diversity = 2.1884e+01
Generation 7350:	Best = 4.361922e+03	FES = 735100	Diversity = 2.1881e+01
Generation 7360:	Best = 4.361922e+03	FES = 736100	Diversity = 2.1884e+01
Generation 7370:	Best = 4.361922e+03	FES = 737100	Diversity = 2.1885e+01
Generation 7380:	Best = 4.361922e+03	FES = 738100	Diversity = 2.1885e+01
Generation 7390:	Best = 4.361922e+03	FES = 739100	Diversity = 2.1886e+01
Generation 7400:	Best = 4.361922e+03	FES = 740100	Diversity = 2.1879e+01
Generation 7410:	Best = 4.361922e+03	FES = 741100	Diversity = 2.1881e+01
Generation 7420:	Best = 4.359368e+03	FES = 742100	Diversity = 2.1882e+01
Generation 7430:	Best = 4.359368e+03	FES = 743100	Diversity = 2.1870e+01
Generation 7440:	Best = 4.359321e+03	FES = 744100	Diversity = 2.1862e+01
Generation 7450:	Best = 4.359321e+03	FES = 745100	Diversity = 2.1864e+01

Generation 7460:	Best = 4.359321e+03	FES = 746100	Diversity = 2.1861e+01
Generation 7470:	Best = 4.359321e+03	FES = 747100	Diversity = 2.1860e+01
Generation 7480:	Best = 4.359321e+03	FES = 748100	Diversity = 2.1849e+01
Generation 7490:	Best = 4.359321e+03	FES = 749100	Diversity = 2.1713e+01
Generation 7500:	Best = 4.359321e+03	FES = 750100	Diversity = 2.1713e+01
Generation 7510:	Best = 4.359321e+03	FES = 751100	Diversity = 2.1698e+01
Generation 7520:	Best = 4.359321e+03	FES = 752100	Diversity = 2.1691e+01
Generation 7530:	Best = 4.359321e+03	FES = 753100	Diversity = 2.1688e+01
Generation 7540:	Best = 4.359321e+03	FES = 754100	Diversity = 2.1688e+01
Generation 7550:	Best = 4.359321e+03	FES = 755100	Diversity = 2.1688e+01
Generation 7560:	Best = 4.359180e+03	FES = 756100	Diversity = 2.1689e+01
Generation 7570:	Best = 4.359180e+03	FES = 757100	Diversity = 2.1689e+01
Generation 7580:	Best = 4.359180e+03	FES = 758100	Diversity = 2.1689e+01
Generation 7590:	Best = 4.359180e+03	FES = 759100	Diversity = 2.1678e+01
Generation 7600:	Best = 4.359123e+03	FES = 760100	Diversity = 2.1677e+01
Generation 7610:	Best = 4.359123e+03	FES = 761100	Diversity = 2.1676e+01
Generation 7620:	Best = 4.359123e+03	FES = 762100	Diversity = 2.1666e+01
Generation 7630:	Best = 4.359123e+03	FES = 763100	Diversity = 2.1663e+01
Generation 7640:	Best = 4.359123e+03	FES = 764100	Diversity = 2.1659e+01
Generation 7650:	Best = 4.359123e+03	FES = 765100	Diversity = 2.1659e+01
Generation 7660:	Best = 4.359123e+03	FES = 766100	Diversity = 2.1656e+01
Generation 7670:	Best = 4.359123e+03	FES = 767100	Diversity = 2.1659e+01
Generation 7680:	Best = 4.359123e+03	FES = 768100	Diversity = 2.1659e+01
Generation 7690:	Best = 4.359123e+03	FES = 769100	Diversity = 2.1656e+01
Generation 7700:	Best = 4.359123e+03	FES = 770100	Diversity = 2.1656e+01
Generation 7710:	Best = 4.359123e+03	FES = 771100	Diversity = 2.1652e+01
Generation 7720:	Best = 4.359123e+03	FES = 772100	Diversity = 2.1651e+01
Generation 7730:	Best = 4.359123e+03	FES = 773100	Diversity = 2.1652e+01
Generation 7740:	Best = 4.359123e+03	FES = 774100	Diversity = 2.1651e+01
Generation 7750:	Best = 4.351537e+03	FES = 775100	Diversity = 2.1651e+01
Generation 7760:	Best = 4.351537e+03	FES = 776100	Diversity = 2.1651e+01
Generation 7770:	Best = 4.351537e+03	FES = 777100	Diversity = 2.1648e+01
Generation 7780:	Best = 4.351537e+03	FES = 778100	Diversity = 2.1645e+01
Generation 7790:	Best = 4.351537e+03	FES = 779100	Diversity = 2.1646e+01
Generation 7800:	Best = 4.351537e+03	FES = 780100	Diversity = 2.1645e+01
Generation 7810:	Best = 4.351240e+03	FES = 781100	Diversity = 2.1634e+01
Generation 7820:	Best = 4.351240e+03	FES = 782100	Diversity = 2.1615e+01
Generation 7830:	Best = 4.351240e+03	FES = 783100	Diversity = 2.1609e+01
Generation 7840:	Best = 4.347458e+03	FES = 784100	Diversity = 2.1604e+01
Generation 7850:	Best = 4.347458e+03	FES = 785100	Diversity = 2.1599e+01
Generation 7860:	Best = 4.347458e+03	FES = 786100	Diversity = 2.1594e+01
Generation 7870:	Best = 4.347458e+03	FES = 787100	Diversity = 2.1590e+01
Generation 7880:	Best = 4.347458e+03	FES = 788100	Diversity = 2.1587e+01
Generation 7890:	Best = 4.347458e+03	FES = 789100	Diversity = 2.1581e+01
Generation 7900:	Best = 4.347458e+03	FES = 790100	Diversity = 2.1579e+01
Generation 7910:	Best = 4.347458e+03	FES = 791100	Diversity = 2.1581e+01
Generation 7920:	Best = 4.347458e+03	FES = 792100	Diversity = 2.1575e+01
Generation 7930:	Best = 4.347458e+03	FES = 793100	Diversity = 2.1575e+01
Generation 7940:	Best = 4.347458e+03	FES = 794100	Diversity = 2.1568e+01
Generation 7950:	Best = 4.347458e+03	FES = 795100	Diversity = 2.1564e+01
Generation 7960:	Best = 4.347458e+03	FES = 796100	Diversity = 2.1563e+01
Generation 7970:	Best = 4.343896e+03	FES = 797100	Diversity = 2.1565e+01
Generation 7980:	Best = 4.343896e+03	FES = 798100	Diversity = 2.1561e+01
Generation 7990:	Best = 4.343577e+03	FES = 799100	Diversity = 2.1557e+01

Generation 8000:	Best = 4.343577e+03	FES = 800100	Diversity = 2.1559e+01
Generation 8010:	Best = 4.343577e+03	FES = 801100	Diversity = 2.1552e+01
Generation 8020:	Best = 4.343577e+03	FES = 802100	Diversity = 2.1546e+01
Generation 8030:	Best = 4.343556e+03	FES = 803100	Diversity = 2.1547e+01
Generation 8040:	Best = 4.343556e+03	FES = 804100	Diversity = 2.1542e+01
Generation 8050:	Best = 4.343556e+03	FES = 805100	Diversity = 2.1539e+01
Generation 8060:	Best = 4.343556e+03	FES = 806100	Diversity = 2.1539e+01
Generation 8070:	Best = 4.343556e+03	FES = 807100	Diversity = 2.1530e+01
Generation 8080:	Best = 4.343556e+03	FES = 808100	Diversity = 2.1529e+01
Generation 8090:	Best = 4.343556e+03	FES = 809100	Diversity = 2.1518e+01
Generation 8100:	Best = 4.343556e+03	FES = 810100	Diversity = 2.1517e+01
Generation 8110:	Best = 4.343556e+03	FES = 811100	Diversity = 2.1517e+01
Generation 8120:	Best = 4.343556e+03	FES = 812100	Diversity = 2.1513e+01
Generation 8130:	Best = 4.343556e+03	FES = 813100	Diversity = 2.1513e+01
Generation 8140:	Best = 4.343556e+03	FES = 814100	Diversity = 2.1510e+01
Generation 8150:	Best = 4.343556e+03	FES = 815100	Diversity = 2.1504e+01
Generation 8160:	Best = 4.343556e+03	FES = 816100	Diversity = 2.1501e+01
Generation 8170:	Best = 4.343556e+03	FES = 817100	Diversity = 2.1497e+01
Generation 8180:	Best = 4.343556e+03	FES = 818100	Diversity = 2.1494e+01
Generation 8190:	Best = 4.343556e+03	FES = 819100	Diversity = 2.1488e+01
Generation 8200:	Best = 4.343556e+03	FES = 820100	Diversity = 2.1487e+01
Generation 8210:	Best = 4.343458e+03	FES = 821100	Diversity = 2.1486e+01
Generation 8220:	Best = 4.343458e+03	FES = 822100	Diversity = 2.1487e+01
Generation 8230:	Best = 4.343458e+03	FES = 823100	Diversity = 2.1482e+01
Generation 8240:	Best = 4.343458e+03	FES = 824100	Diversity = 2.1483e+01
Generation 8250:	Best = 4.343458e+03	FES = 825100	Diversity = 2.1480e+01
Generation 8260:	Best = 4.343458e+03	FES = 826100	Diversity = 2.1481e+01
Generation 8270:	Best = 4.343458e+03	FES = 827100	Diversity = 2.1482e+01
Generation 8280:	Best = 4.343458e+03	FES = 828100	Diversity = 2.1481e+01
Generation 8290:	Best = 4.343320e+03	FES = 829100	Diversity = 2.1481e+01
Generation 8300:	Best = 4.343320e+03	FES = 830100	Diversity = 2.1474e+01
Generation 8310:	Best = 4.343320e+03	FES = 831100	Diversity = 2.1479e+01
Generation 8320:	Best = 4.343157e+03	FES = 832100	Diversity = 2.1467e+01
Generation 8330:	Best = 4.343157e+03	FES = 833100	Diversity = 2.1466e+01
Generation 8340:	Best = 4.343157e+03	FES = 834100	Diversity = 2.1455e+01
Generation 8350:	Best = 4.343157e+03	FES = 835100	Diversity = 2.1459e+01
Generation 8360:	Best = 4.343157e+03	FES = 836100	Diversity = 2.1458e+01
Generation 8370:	Best = 4.343157e+03	FES = 837100	Diversity = 2.1452e+01
Generation 8380:	Best = 4.343157e+03	FES = 838100	Diversity = 2.1452e+01
Generation 8390:	Best = 4.343157e+03	FES = 839100	Diversity = 2.1445e+01
Generation 8400:	Best = 4.343157e+03	FES = 840100	Diversity = 2.1441e+01
Generation 8410:	Best = 4.343157e+03	FES = 841100	Diversity = 2.1438e+01
Generation 8420:	Best = 4.343157e+03	FES = 842100	Diversity = 2.1436e+01
Generation 8430:	Best = 4.343157e+03	FES = 843100	Diversity = 2.1437e+01
Generation 8440:	Best = 4.343157e+03	FES = 844100	Diversity = 2.1436e+01
Generation 8450:	Best = 4.343157e+03	FES = 845100	Diversity = 2.1429e+01
Generation 8460:	Best = 4.343157e+03	FES = 846100	Diversity = 2.1429e+01
Generation 8470:	Best = 4.343157e+03	FES = 847100	Diversity = 2.1427e+01
Generation 8480:	Best = 4.343157e+03	FES = 848100	Diversity = 2.1425e+01
Generation 8490:	Best = 4.342270e+03	FES = 849100	Diversity = 2.1425e+01
Generation 8500:	Best = 4.341878e+03	FES = 850100	Diversity = 2.1424e+01
Generation 8510:	Best = 4.341810e+03	FES = 851100	Diversity = 2.1423e+01
Generation 8520:	Best = 4.341810e+03	FES = 852100	Diversity = 2.1422e+01
Generation 8530:	Best = 4.341810e+03	FES = 853100	Diversity = 2.1416e+01

Generation 8540:	Best = 4.341810e+03	FES = 854100	Diversity = 2.1417e+01
Generation 8550:	Best = 4.341810e+03	FES = 855100	Diversity = 2.1415e+01
Generation 8560:	Best = 4.341810e+03	FES = 856100	Diversity = 2.1415e+01
Generation 8570:	Best = 4.341810e+03	FES = 857100	Diversity = 2.1416e+01
Generation 8580:	Best = 4.341810e+03	FES = 858100	Diversity = 2.1416e+01
Generation 8590:	Best = 4.341810e+03	FES = 859100	Diversity = 2.1414e+01
Generation 8600:	Best = 4.341810e+03	FES = 860100	Diversity = 2.1414e+01
Generation 8610:	Best = 4.341810e+03	FES = 861100	Diversity = 2.1414e+01
Generation 8620:	Best = 4.341810e+03	FES = 862100	Diversity = 2.1414e+01
Generation 8630:	Best = 4.341810e+03	FES = 863100	Diversity = 2.1414e+01
Generation 8640:	Best = 4.341810e+03	FES = 864100	Diversity = 2.1413e+01
Generation 8650:	Best = 4.341810e+03	FES = 865100	Diversity = 2.1413e+01
Generation 8660:	Best = 4.341810e+03	FES = 866100	Diversity = 2.1412e+01
Generation 8670:	Best = 4.341810e+03	FES = 867100	Diversity = 2.1423e+01
Generation 8680:	Best = 4.341810e+03	FES = 868100	Diversity = 2.1421e+01
Generation 8690:	Best = 4.341810e+03	FES = 869100	Diversity = 2.1421e+01
Generation 8700:	Best = 4.341810e+03	FES = 870100	Diversity = 2.1419e+01
Generation 8710:	Best = 4.341788e+03	FES = 871100	Diversity = 2.1418e+01
Generation 8720:	Best = 4.341788e+03	FES = 872100	Diversity = 2.1417e+01
Generation 8730:	Best = 4.341788e+03	FES = 873100	Diversity = 2.1406e+01
Generation 8740:	Best = 4.341788e+03	FES = 874100	Diversity = 2.1406e+01
Generation 8750:	Best = 4.341784e+03	FES = 875100	Diversity = 2.1406e+01
Generation 8760:	Best = 4.341715e+03	FES = 876100	Diversity = 2.1405e+01
Generation 8770:	Best = 4.341715e+03	FES = 877100	Diversity = 2.1405e+01
Generation 8780:	Best = 4.341715e+03	FES = 878100	Diversity = 2.1401e+01
Generation 8790:	Best = 4.341715e+03	FES = 879100	Diversity = 2.1399e+01
Generation 8800:	Best = 4.341715e+03	FES = 880100	Diversity = 2.1393e+01
Generation 8810:	Best = 4.341715e+03	FES = 881100	Diversity = 2.1394e+01
Generation 8820:	Best = 4.341715e+03	FES = 882100	Diversity = 2.1392e+01
Generation 8830:	Best = 4.341715e+03	FES = 883100	Diversity = 2.1391e+01
Generation 8840:	Best = 4.339431e+03	FES = 884100	Diversity = 2.1386e+01
Generation 8850:	Best = 4.339431e+03	FES = 885100	Diversity = 2.1385e+01
Generation 8860:	Best = 4.339431e+03	FES = 886100	Diversity = 2.1383e+01
Generation 8870:	Best = 4.339431e+03	FES = 887100	Diversity = 2.1383e+01
Generation 8880:	Best = 4.339431e+03	FES = 888100	Diversity = 2.1378e+01
Generation 8890:	Best = 4.339431e+03	FES = 889100	Diversity = 2.1374e+01
Generation 8900:	Best = 4.339423e+03	FES = 890100	Diversity = 2.1369e+01
Generation 8910:	Best = 4.339423e+03	FES = 891100	Diversity = 2.1368e+01
Generation 8920:	Best = 4.339423e+03	FES = 892100	Diversity = 2.1367e+01
Generation 8930:	Best = 4.339423e+03	FES = 893100	Diversity = 2.1367e+01
Generation 8940:	Best = 4.339423e+03	FES = 894100	Diversity = 2.1360e+01
Generation 8950:	Best = 4.339423e+03	FES = 895100	Diversity = 2.1360e+01
Generation 8960:	Best = 4.339423e+03	FES = 896100	Diversity = 2.1361e+01
Generation 8970:	Best = 4.339423e+03	FES = 897100	Diversity = 2.1359e+01
Generation 8980:	Best = 4.332337e+03	FES = 898100	Diversity = 2.1359e+01
Generation 8990:	Best = 4.332337e+03	FES = 899100	Diversity = 2.1359e+01
Generation 9000:	Best = 4.332337e+03	FES = 900100	Diversity = 2.1355e+01
Generation 9010:	Best = 4.332337e+03	FES = 901100	Diversity = 2.1355e+01
Generation 9020:	Best = 4.332337e+03	FES = 902100	Diversity = 2.1355e+01
Generation 9030:	Best = 4.332337e+03	FES = 903100	Diversity = 2.1353e+01
Generation 9040:	Best = 4.332322e+03	FES = 904100	Diversity = 2.1347e+01
Generation 9050:	Best = 4.332322e+03	FES = 905100	Diversity = 2.1339e+01
Generation 9060:	Best = 4.332322e+03	FES = 906100	Diversity = 2.1332e+01
Generation 9070:	Best = 4.332322e+03	FES = 907100	Diversity = 2.1332e+01

Generation 9080:	Best = 4.332322e+03	FES = 908100	Diversity = 2.1329e+01
Generation 9090:	Best = 4.332322e+03	FES = 909100	Diversity = 2.1329e+01
Generation 9100:	Best = 4.332322e+03	FES = 910100	Diversity = 2.1327e+01
Generation 9110:	Best = 4.332322e+03	FES = 911100	Diversity = 2.1324e+01
Generation 9120:	Best = 4.332322e+03	FES = 912100	Diversity = 2.1322e+01
Generation 9130:	Best = 4.332322e+03	FES = 913100	Diversity = 2.1321e+01
Generation 9140:	Best = 4.332322e+03	FES = 914100	Diversity = 2.1316e+01
Generation 9150:	Best = 4.332322e+03	FES = 915100	Diversity = 2.1310e+01
Generation 9160:	Best = 4.332322e+03	FES = 916100	Diversity = 2.1252e+01
Generation 9170:	Best = 4.332322e+03	FES = 917100	Diversity = 2.1249e+01
Generation 9180:	Best = 4.332322e+03	FES = 918100	Diversity = 2.1246e+01
Generation 9190:	Best = 4.332322e+03	FES = 919100	Diversity = 2.1245e+01
Generation 9200:	Best = 4.332322e+03	FES = 920100	Diversity = 2.1243e+01
Generation 9210:	Best = 4.332322e+03	FES = 921100	Diversity = 2.1241e+01
Generation 9220:	Best = 4.332322e+03	FES = 922100	Diversity = 2.1237e+01
Generation 9230:	Best = 4.332322e+03	FES = 923100	Diversity = 2.1233e+01
Generation 9240:	Best = 4.331692e+03	FES = 924100	Diversity = 2.1232e+01
Generation 9250:	Best = 4.331566e+03	FES = 925100	Diversity = 2.1232e+01
Generation 9260:	Best = 4.331566e+03	FES = 926100	Diversity = 2.1228e+01
Generation 9270:	Best = 4.331527e+03	FES = 927100	Diversity = 2.1237e+01
Generation 9280:	Best = 4.331527e+03	FES = 928100	Diversity = 2.1235e+01
Generation 9290:	Best = 4.331527e+03	FES = 929100	Diversity = 2.1234e+01
Generation 9300:	Best = 4.331527e+03	FES = 930100	Diversity = 2.1231e+01
Generation 9310:	Best = 4.331527e+03	FES = 931100	Diversity = 2.1230e+01
Generation 9320:	Best = 4.331527e+03	FES = 932100	Diversity = 2.1226e+01
Generation 9330:	Best = 4.331527e+03	FES = 933100	Diversity = 2.1225e+01
Generation 9340:	Best = 4.331527e+03	FES = 934100	Diversity = 2.1225e+01
Generation 9350:	Best = 4.331527e+03	FES = 935100	Diversity = 2.1221e+01
Generation 9360:	Best = 4.331527e+03	FES = 936100	Diversity = 2.1221e+01
Generation 9370:	Best = 4.331527e+03	FES = 937100	Diversity = 2.1221e+01
Generation 9380:	Best = 4.331527e+03	FES = 938100	Diversity = 2.1213e+01
Generation 9390:	Best = 4.331527e+03	FES = 939100	Diversity = 2.1207e+01
Generation 9400:	Best = 4.331527e+03	FES = 940100	Diversity = 2.1204e+01
Generation 9410:	Best = 4.331527e+03	FES = 941100	Diversity = 2.1204e+01
Generation 9420:	Best = 4.331527e+03	FES = 942100	Diversity = 2.1204e+01
Generation 9430:	Best = 4.331527e+03	FES = 943100	Diversity = 2.1201e+01
Generation 9440:	Best = 4.331527e+03	FES = 944100	Diversity = 2.1198e+01
Generation 9450:	Best = 4.331527e+03	FES = 945100	Diversity = 2.1198e+01
Generation 9460:	Best = 4.331527e+03	FES = 946100	Diversity = 2.1196e+01
Generation 9470:	Best = 4.331527e+03	FES = 947100	Diversity = 2.1197e+01
Generation 9480:	Best = 4.331527e+03	FES = 948100	Diversity = 2.1196e+01
Generation 9490:	Best = 4.331527e+03	FES = 949100	Diversity = 2.1197e+01
Generation 9500:	Best = 4.331527e+03	FES = 950100	Diversity = 2.1196e+01
Generation 9510:	Best = 4.331446e+03	FES = 951100	Diversity = 2.1195e+01
Generation 9520:	Best = 4.331446e+03	FES = 952100	Diversity = 2.1195e+01
Generation 9530:	Best = 4.331446e+03	FES = 953100	Diversity = 2.1190e+01
Generation 9540:	Best = 4.331446e+03	FES = 954100	Diversity = 2.1189e+01
Generation 9550:	Best = 4.331446e+03	FES = 955100	Diversity = 2.1186e+01
Generation 9560:	Best = 4.331446e+03	FES = 956100	Diversity = 2.1184e+01
Generation 9570:	Best = 4.331446e+03	FES = 957100	Diversity = 2.1181e+01
Generation 9580:	Best = 4.331446e+03	FES = 958100	Diversity = 2.1159e+01
Generation 9590:	Best = 4.331446e+03	FES = 959100	Diversity = 2.1159e+01
Generation 9600:	Best = 4.330970e+03	FES = 960100	Diversity = 2.1157e+01
Generation 9610:	Best = 4.330970e+03	FES = 961100	Diversity = 2.1159e+01

```

Generation 9620: Best = 4.330970e+03 FES = 962100 Diversity = 2.1158e+01
Generation 9630: Best = 4.330970e+03 FES = 963100 Diversity = 2.1156e+01
Generation 9640: Best = 4.330970e+03 FES = 964100 Diversity = 2.1155e+01
Generation 9650: Best = 4.330970e+03 FES = 965100 Diversity = 2.1150e+01
Generation 9660: Best = 4.330970e+03 FES = 966100 Diversity = 2.1149e+01
Generation 9670: Best = 4.330970e+03 FES = 967100 Diversity = 2.1149e+01
Generation 9680: Best = 4.330970e+03 FES = 968100 Diversity = 2.1150e+01
Generation 9690: Best = 4.330970e+03 FES = 969100 Diversity = 2.1151e+01
Generation 9700: Best = 4.330505e+03 FES = 970100 Diversity = 2.1148e+01
Generation 9710: Best = 4.330505e+03 FES = 971100 Diversity = 2.1147e+01
Generation 9720: Best = 4.330505e+03 FES = 972100 Diversity = 2.1144e+01
Generation 9730: Best = 4.330505e+03 FES = 973100 Diversity = 2.1143e+01
Generation 9740: Best = 4.330150e+03 FES = 974100 Diversity = 2.1143e+01
Generation 9750: Best = 4.330150e+03 FES = 975100 Diversity = 2.1144e+01
Generation 9760: Best = 4.330150e+03 FES = 976100 Diversity = 2.1142e+01
Generation 9770: Best = 4.330150e+03 FES = 977100 Diversity = 2.1143e+01
Generation 9780: Best = 4.330150e+03 FES = 978100 Diversity = 2.1141e+01
Generation 9790: Best = 4.330150e+03 FES = 979100 Diversity = 2.1140e+01
Generation 9800: Best = 4.330150e+03 FES = 980100 Diversity = 2.1141e+01
Generation 9810: Best = 4.330150e+03 FES = 981100 Diversity = 2.1141e+01
Generation 9820: Best = 4.330150e+03 FES = 982100 Diversity = 2.1142e+01
Generation 9830: Best = 4.330150e+03 FES = 983100 Diversity = 2.1143e+01
Generation 9840: Best = 4.330150e+03 FES = 984100 Diversity = 2.1144e+01
Generation 9850: Best = 4.330150e+03 FES = 985100 Diversity = 2.1141e+01
Generation 9860: Best = 4.330150e+03 FES = 986100 Diversity = 2.1136e+01
Generation 9870: Best = 4.330113e+03 FES = 987100 Diversity = 2.1133e+01
Generation 9880: Best = 4.330113e+03 FES = 988100 Diversity = 2.1129e+01
Generation 9890: Best = 4.330113e+03 FES = 989100 Diversity = 2.1133e+01
Generation 9900: Best = 4.330113e+03 FES = 990100 Diversity = 2.1133e+01
Generation 9910: Best = 4.330100e+03 FES = 991100 Diversity = 2.1134e+01
Generation 9920: Best = 4.330100e+03 FES = 992100 Diversity = 2.1134e+01
Generation 9930: Best = 4.330100e+03 FES = 993100 Diversity = 2.1135e+01
Generation 9940: Best = 4.330100e+03 FES = 994100 Diversity = 2.1135e+01
Generation 9950: Best = 4.330100e+03 FES = 995100 Diversity = 2.1119e+01
Generation 9960: Best = 4.330100e+03 FES = 996100 Diversity = 2.1120e+01
Generation 9970: Best = 4.330100e+03 FES = 997100 Diversity = 2.1119e+01
Generation 9980: Best = 4.330100e+03 FES = 998100 Diversity = 2.1118e+01
Generation 9990: Best = 4.330100e+03 FES = 999100 Diversity = 2.1120e+01

```

Optimization completed!

Final Best Fitness = 4.330084e+03

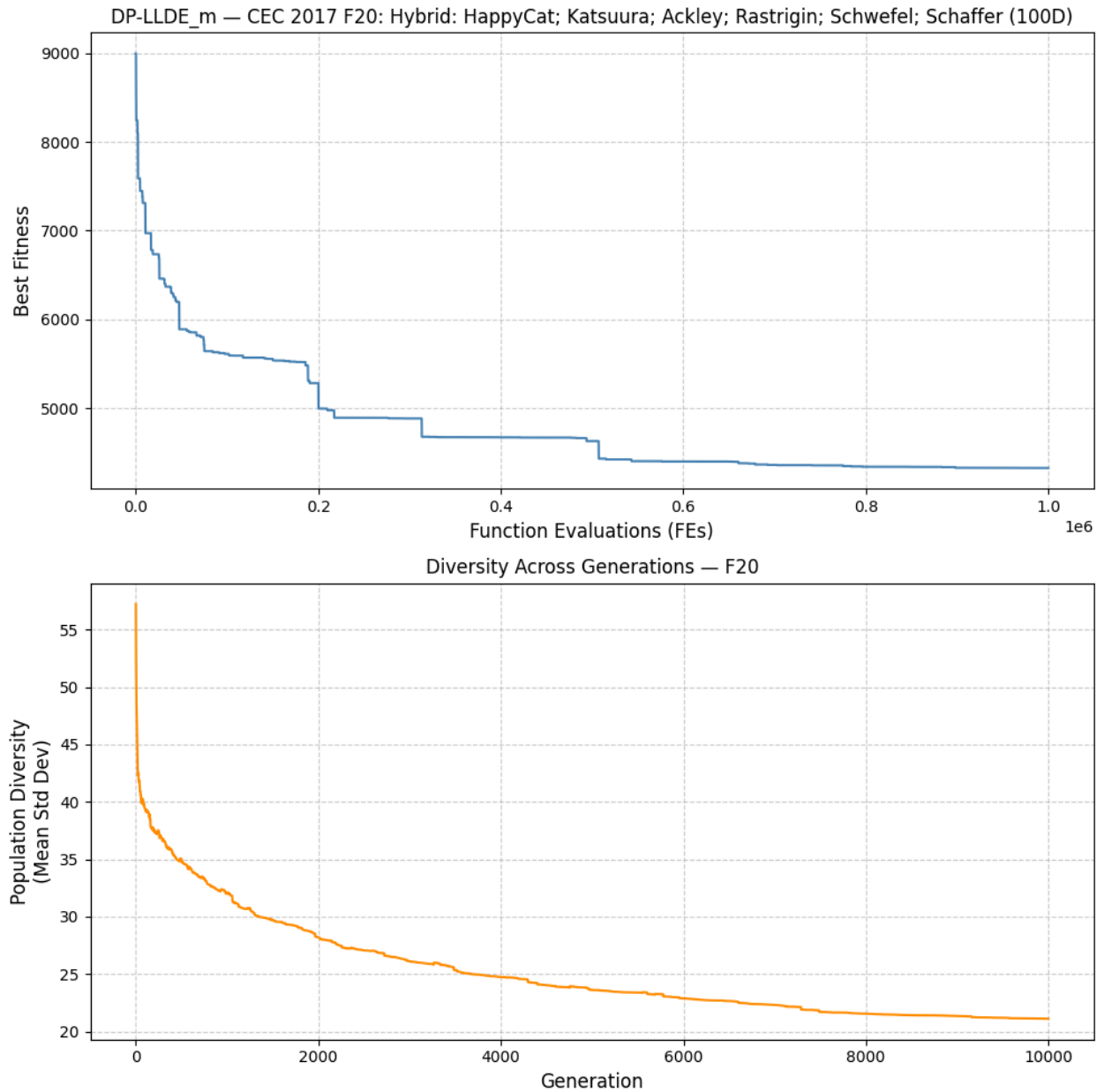
Total Function Evaluations = 1000000

```

=====
=
Results:
  Best Fitness           : 4.330084e+03
  Global Optimum         : 2.000000e+03
  Error (|best-optimum|) : 2.330084e+03
  Best Solution (5 dim)  : [ 9.62889071 48.67688585 -46.88433508 40.79311055
51.18823841]
  D^(0) (init. diversity): 5.7220e+01
=====
=

```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results_2/all_results_checkpoint.pkl



Graph saved as convergence_F20.png

✓ Plot saved to: /content/drive/My Drive/thesis_results_2/plots/convergence_F20.png

6.4 Composite Functions (F21-F30)

```
In [ ]: # =====  
# F21 — Composition Function 1 (Rosenbrock; Elliptic; Rastrigin)  
# =====  
run_and_plot('F21', make_wrapper(composition.f21), 2100.0, 'Composition: Roser
```

=====

=

DP-LLDE_m on CEC 2017 F21 – Single Run

=====

=

Function : Composition: Rosenbrock; Elliptic; Rastrigin (100D)

Bounds : [-100, 100]

Global optimum : 2100.0

Initial $D^{(0)} = 5.7220e+01$ | Diversity threshold = 5.7220e+00

Generation 0: Best Fitness = 5.273291e+03 | Diversity = 5.7220e+01

Generation	10:	Best = 4.233778e+03	FES = 1100	Diversity = 3.1406e+01
Generation	20:	Best = 3.801906e+03	FES = 2100	Diversity = 1.8468e+01
Generation	30:	Best = 3.636505e+03	FES = 3100	Diversity = 1.4811e+01
Generation	40:	Best = 3.510152e+03	FES = 4100	Diversity = 1.2780e+01
Generation	50:	Best = 3.492517e+03	FES = 5100	Diversity = 1.1588e+01
Generation	60:	Best = 3.406030e+03	FES = 6100	Diversity = 1.0672e+01
Generation	70:	Best = 3.363257e+03	FES = 7100	Diversity = 1.0492e+01
Generation	80:	Best = 3.363257e+03	FES = 8100	Diversity = 1.0261e+01
Generation	90:	Best = 3.348481e+03	FES = 9100	Diversity = 9.7769e+00
Generation	100:	Best = 3.330108e+03	FES = 10100	Diversity = 9.6474e+00
Generation	110:	Best = 3.317492e+03	FES = 11100	Diversity = 9.3884e+00
Generation	120:	Best = 3.271040e+03	FES = 12100	Diversity = 9.4522e+00
Generation	130:	Best = 3.271040e+03	FES = 13100	Diversity = 9.3427e+00
Generation	140:	Best = 3.255462e+03	FES = 14100	Diversity = 9.1662e+00
Generation	150:	Best = 3.241459e+03	FES = 15100	Diversity = 8.9980e+00
Generation	160:	Best = 3.240227e+03	FES = 16100	Diversity = 8.9295e+00
Generation	170:	Best = 3.235115e+03	FES = 17100	Diversity = 8.8714e+00
Generation	180:	Best = 3.232035e+03	FES = 18100	Diversity = 8.7718e+00
Generation	190:	Best = 3.229660e+03	FES = 19100	Diversity = 8.7093e+00
Generation	200:	Best = 3.227876e+03	FES = 20100	Diversity = 8.6018e+00
Generation	210:	Best = 3.213993e+03	FES = 21100	Diversity = 8.4741e+00
Generation	220:	Best = 3.202567e+03	FES = 22100	Diversity = 8.3927e+00
Generation	230:	Best = 3.200210e+03	FES = 23100	Diversity = 8.2566e+00
Generation	240:	Best = 3.183944e+03	FES = 24100	Diversity = 8.1113e+00
Generation	250:	Best = 3.183944e+03	FES = 25100	Diversity = 8.1378e+00
Generation	260:	Best = 3.183944e+03	FES = 26100	Diversity = 8.1127e+00
Generation	270:	Best = 3.183944e+03	FES = 27100	Diversity = 8.0151e+00
Generation	280:	Best = 3.174988e+03	FES = 28100	Diversity = 7.9929e+00
Generation	290:	Best = 3.160393e+03	FES = 29100	Diversity = 7.9607e+00
Generation	300:	Best = 3.156057e+03	FES = 30100	Diversity = 7.9367e+00
Generation	310:	Best = 3.146445e+03	FES = 31100	Diversity = 7.9034e+00
Generation	320:	Best = 3.146445e+03	FES = 32100	Diversity = 7.8847e+00
Generation	330:	Best = 3.121489e+03	FES = 33100	Diversity = 7.8024e+00
Generation	340:	Best = 3.121489e+03	FES = 34100	Diversity = 7.8246e+00
Generation	350:	Best = 3.121489e+03	FES = 35100	Diversity = 7.7365e+00
Generation	360:	Best = 3.121051e+03	FES = 36100	Diversity = 7.7181e+00
Generation	370:	Best = 3.121051e+03	FES = 37100	Diversity = 7.6991e+00
Generation	380:	Best = 3.112682e+03	FES = 38100	Diversity = 7.6467e+00
Generation	390:	Best = 3.106460e+03	FES = 39100	Diversity = 7.6342e+00
Generation	400:	Best = 3.106460e+03	FES = 40100	Diversity = 7.6483e+00
Generation	410:	Best = 3.092075e+03	FES = 41100	Diversity = 7.5996e+00
Generation	420:	Best = 3.090264e+03	FES = 42100	Diversity = 7.5368e+00
Generation	430:	Best = 3.087043e+03	FES = 43100	Diversity = 7.4801e+00
Generation	440:	Best = 3.058245e+03	FES = 44100	Diversity = 7.4697e+00

Generation	450:	Best = 3.047662e+03	FES = 45100	Diversity = 7.4718e+00
Generation	460:	Best = 3.035262e+03	FES = 46100	Diversity = 7.4498e+00
Generation	470:	Best = 3.018566e+03	FES = 47100	Diversity = 7.4629e+00
Generation	480:	Best = 3.018566e+03	FES = 48100	Diversity = 7.4227e+00
Generation	490:	Best = 3.018566e+03	FES = 49100	Diversity = 7.4251e+00
Generation	500:	Best = 3.004940e+03	FES = 50100	Diversity = 7.4151e+00
Generation	510:	Best = 3.004568e+03	FES = 51100	Diversity = 7.3911e+00
Generation	520:	Best = 2.994437e+03	FES = 52100	Diversity = 7.3783e+00
Generation	530:	Best = 2.994292e+03	FES = 53100	Diversity = 7.3505e+00
Generation	540:	Best = 2.994292e+03	FES = 54100	Diversity = 7.3095e+00
Generation	550:	Best = 2.993864e+03	FES = 55100	Diversity = 7.3116e+00
Generation	560:	Best = 2.993718e+03	FES = 56100	Diversity = 7.3079e+00
Generation	570:	Best = 2.993718e+03	FES = 57100	Diversity = 7.3098e+00
Generation	580:	Best = 2.993718e+03	FES = 58100	Diversity = 7.2939e+00
Generation	590:	Best = 2.993718e+03	FES = 59100	Diversity = 7.2731e+00
Generation	600:	Best = 2.990064e+03	FES = 60100	Diversity = 7.2618e+00
Generation	610:	Best = 2.985301e+03	FES = 61100	Diversity = 7.2524e+00
Generation	620:	Best = 2.971065e+03	FES = 62100	Diversity = 7.2423e+00
Generation	630:	Best = 2.971065e+03	FES = 63100	Diversity = 7.2353e+00
Generation	640:	Best = 2.971065e+03	FES = 64100	Diversity = 7.2905e+00
Generation	650:	Best = 2.963204e+03	FES = 65100	Diversity = 7.1683e+00
Generation	660:	Best = 2.962001e+03	FES = 66100	Diversity = 7.1747e+00
Generation	670:	Best = 2.958637e+03	FES = 67100	Diversity = 7.1738e+00
Generation	680:	Best = 2.958448e+03	FES = 68100	Diversity = 7.1349e+00
Generation	690:	Best = 2.958448e+03	FES = 69100	Diversity = 7.1180e+00
Generation	700:	Best = 2.947148e+03	FES = 70100	Diversity = 7.0860e+00
Generation	710:	Best = 2.947148e+03	FES = 71100	Diversity = 7.0743e+00
Generation	720:	Best = 2.946282e+03	FES = 72100	Diversity = 7.0647e+00
Generation	730:	Best = 2.946282e+03	FES = 73100	Diversity = 7.0491e+00
Generation	740:	Best = 2.943420e+03	FES = 74100	Diversity = 7.0434e+00
Generation	750:	Best = 2.943402e+03	FES = 75100	Diversity = 7.0118e+00
Generation	760:	Best = 2.937146e+03	FES = 76100	Diversity = 6.9867e+00
Generation	770:	Best = 2.936079e+03	FES = 77100	Diversity = 6.9685e+00
Generation	780:	Best = 2.934916e+03	FES = 78100	Diversity = 6.9628e+00
Generation	790:	Best = 2.934764e+03	FES = 79100	Diversity = 6.9225e+00
Generation	800:	Best = 2.929922e+03	FES = 80100	Diversity = 6.9209e+00
Generation	810:	Best = 2.925784e+03	FES = 81100	Diversity = 6.9274e+00
Generation	820:	Best = 2.920607e+03	FES = 82100	Diversity = 6.9079e+00
Generation	830:	Best = 2.919782e+03	FES = 83100	Diversity = 6.9178e+00
Generation	840:	Best = 2.916678e+03	FES = 84100	Diversity = 6.8986e+00
Generation	850:	Best = 2.916678e+03	FES = 85100	Diversity = 6.8972e+00
Generation	860:	Best = 2.914200e+03	FES = 86100	Diversity = 6.8806e+00
Generation	870:	Best = 2.911822e+03	FES = 87100	Diversity = 6.8785e+00
Generation	880:	Best = 2.907975e+03	FES = 88100	Diversity = 6.8965e+00
Generation	890:	Best = 2.902595e+03	FES = 89100	Diversity = 6.7525e+00
Generation	900:	Best = 2.902595e+03	FES = 90100	Diversity = 6.7362e+00
Generation	910:	Best = 2.901287e+03	FES = 91100	Diversity = 6.7369e+00
Generation	920:	Best = 2.899642e+03	FES = 92100	Diversity = 6.7533e+00
Generation	930:	Best = 2.899306e+03	FES = 93100	Diversity = 6.7305e+00
Generation	940:	Best = 2.896510e+03	FES = 94100	Diversity = 6.7277e+00
Generation	950:	Best = 2.895103e+03	FES = 95100	Diversity = 6.7549e+00
Generation	960:	Best = 2.895074e+03	FES = 96100	Diversity = 6.7603e+00
Generation	970:	Best = 2.894874e+03	FES = 97100	Diversity = 6.7168e+00
Generation	980:	Best = 2.894874e+03	FES = 98100	Diversity = 6.7130e+00

Generation 990:	Best = 2.894801e+03	FES = 99100	Diversity = 6.7047e+00
Generation 1000:	Best = 2.892726e+03	FES = 100100	Diversity = 6.7038e+00
Generation 1010:	Best = 2.891441e+03	FES = 101100	Diversity = 6.7303e+00
Generation 1020:	Best = 2.890572e+03	FES = 102100	Diversity = 6.7321e+00
Generation 1030:	Best = 2.889965e+03	FES = 103100	Diversity = 6.6996e+00
Generation 1040:	Best = 2.887842e+03	FES = 104100	Diversity = 6.6747e+00
Generation 1050:	Best = 2.887831e+03	FES = 105100	Diversity = 6.6694e+00
Generation 1060:	Best = 2.886766e+03	FES = 106100	Diversity = 6.6666e+00
Generation 1070:	Best = 2.870947e+03	FES = 107100	Diversity = 6.6359e+00
Generation 1080:	Best = 2.867902e+03	FES = 108100	Diversity = 6.4313e+00
Generation 1090:	Best = 2.860930e+03	FES = 109100	Diversity = 6.2817e+00
Generation 1100:	Best = 2.860219e+03	FES = 110100	Diversity = 6.2585e+00
Generation 1110:	Best = 2.855394e+03	FES = 111100	Diversity = 6.2145e+00
Generation 1120:	Best = 2.854347e+03	FES = 112100	Diversity = 6.1228e+00
Generation 1130:	Best = 2.853404e+03	FES = 113100	Diversity = 6.0394e+00
Generation 1140:	Best = 2.851437e+03	FES = 114100	Diversity = 6.0278e+00
Generation 1150:	Best = 2.850423e+03	FES = 115100	Diversity = 6.0814e+00
Generation 1160:	Best = 2.846755e+03	FES = 116100	Diversity = 6.0245e+00
Generation 1170:	Best = 2.834977e+03	FES = 117100	Diversity = 5.9550e+00
Generation 1180:	Best = 2.828715e+03	FES = 118100	Diversity = 5.9374e+00
Generation 1190:	Best = 2.826109e+03	FES = 119100	Diversity = 5.9293e+00
Generation 1200:	Best = 2.824554e+03	FES = 120100	Diversity = 5.8818e+00
Generation 1210:	Best = 2.824146e+03	FES = 121100	Diversity = 5.8359e+00
Generation 1220:	Best = 2.818293e+03	FES = 122100	Diversity = 5.7092e+00
Generation 1230:	Best = 2.810640e+03	FES = 123100	Diversity = 5.4516e+00
Generation 1240:	Best = 2.801075e+03	FES = 124100	Diversity = 5.1198e+00
Generation 1250:	Best = 2.797088e+03	FES = 125100	Diversity = 4.2940e+00
Generation 1260:	Best = 2.780003e+03	FES = 126100	Diversity = 8.0240e-01
Generation 1270:	Best = 2.764187e+03	FES = 127100	Diversity = 3.1042e-01
Generation 1280:	Best = 2.750948e+03	FES = 128100	Diversity = 2.2846e-01
Generation 1290:	Best = 2.740415e+03	FES = 129100	Diversity = 1.6936e-01
Generation 1300:	Best = 2.734465e+03	FES = 130100	Diversity = 1.2483e-01
Generation 1310:	Best = 2.728526e+03	FES = 131100	Diversity = 1.1818e-01
Generation 1320:	Best = 2.723970e+03	FES = 132100	Diversity = 1.0090e-01
Generation 1330:	Best = 2.721503e+03	FES = 133100	Diversity = 7.4151e-02
Generation 1340:	Best = 2.719673e+03	FES = 134100	Diversity = 6.0776e-02
Generation 1350:	Best = 2.717857e+03	FES = 135100	Diversity = 5.1440e-02
Generation 1360:	Best = 2.716212e+03	FES = 136100	Diversity = 4.8166e-02
Generation 1370:	Best = 2.714950e+03	FES = 137100	Diversity = 4.3384e-02
Generation 1380:	Best = 2.713905e+03	FES = 138100	Diversity = 3.4583e-02
Generation 1390:	Best = 2.712830e+03	FES = 139100	Diversity = 3.3646e-02
Generation 1400:	Best = 2.712061e+03	FES = 140100	Diversity = 2.7135e-02
Generation 1410:	Best = 2.711540e+03	FES = 141100	Diversity = 2.0656e-02
Generation 1420:	Best = 2.711101e+03	FES = 142100	Diversity = 1.6722e-02
Generation 1430:	Best = 2.710610e+03	FES = 143100	Diversity = 1.6032e-02
Generation 1440:	Best = 2.710148e+03	FES = 144100	Diversity = 1.7115e-02
Generation 1450:	Best = 2.709658e+03	FES = 145100	Diversity = 1.6224e-02
Generation 1460:	Best = 2.709185e+03	FES = 146100	Diversity = 1.4692e-02
Generation 1470:	Best = 2.708949e+03	FES = 147100	Diversity = 1.2931e-02
Generation 1480:	Best = 2.708641e+03	FES = 148100	Diversity = 1.0396e-02
Generation 1490:	Best = 2.708369e+03	FES = 149100	Diversity = 9.9065e-03
Generation 1500:	Best = 2.708244e+03	FES = 150100	Diversity = 8.8808e-03
Generation 1510:	Best = 2.708035e+03	FES = 151100	Diversity = 8.7966e-03
Generation 1520:	Best = 2.707906e+03	FES = 152100	Diversity = 7.4423e-03

Generation 1530:	Best = 2.707741e+03	FES = 153100	Diversity = 8.3510e-03
Generation 1540:	Best = 2.707555e+03	FES = 154100	Diversity = 8.2149e-03
Generation 1550:	Best = 2.707396e+03	FES = 155100	Diversity = 7.1385e-03
Generation 1560:	Best = 2.707233e+03	FES = 156100	Diversity = 6.6320e-03
Generation 1570:	Best = 2.707046e+03	FES = 157100	Diversity = 6.6954e-03
Generation 1580:	Best = 2.706933e+03	FES = 158100	Diversity = 4.8939e-03
Generation 1590:	Best = 2.706843e+03	FES = 159100	Diversity = 3.9937e-03
Generation 1600:	Best = 2.706796e+03	FES = 160100	Diversity = 3.6343e-03
Generation 1610:	Best = 2.706739e+03	FES = 161100	Diversity = 3.1281e-03
Generation 1620:	Best = 2.706678e+03	FES = 162100	Diversity = 2.9520e-03
Generation 1630:	Best = 2.706618e+03	FES = 163100	Diversity = 2.8269e-03
Generation 1640:	Best = 2.706564e+03	FES = 164100	Diversity = 2.6176e-03
Generation 1650:	Best = 2.706519e+03	FES = 165100	Diversity = 2.1529e-03
Generation 1660:	Best = 2.706488e+03	FES = 166100	Diversity = 1.5697e-03
Generation 1670:	Best = 2.706468e+03	FES = 167100	Diversity = 1.2640e-03
Generation 1680:	Best = 2.706442e+03	FES = 168100	Diversity = 1.6356e-03
Generation 1690:	Best = 2.706421e+03	FES = 169100	Diversity = 1.6777e-03
Generation 1700:	Best = 2.706399e+03	FES = 170100	Diversity = 1.3427e-03
Generation 1710:	Best = 2.706371e+03	FES = 171100	Diversity = 1.4077e-03
Generation 1720:	Best = 2.706352e+03	FES = 172100	Diversity = 1.3152e-03
Generation 1730:	Best = 2.706340e+03	FES = 173100	Diversity = 1.0703e-03
Generation 1740:	Best = 2.706327e+03	FES = 174100	Diversity = 9.6230e-04
Generation 1750:	Best = 2.706319e+03	FES = 175100	Diversity = 7.0616e-04
Generation 1760:	Best = 2.706312e+03	FES = 176100	Diversity = 5.7381e-04
Generation 1770:	Best = 2.706306e+03	FES = 177100	Diversity = 4.9645e-04
Generation 1780:	Best = 2.706301e+03	FES = 178100	Diversity = 5.3159e-04
Generation 1790:	Best = 2.706296e+03	FES = 179100	Diversity = 5.5171e-04
Generation 1800:	Best = 2.706293e+03	FES = 180100	Diversity = 3.7873e-04
Generation 1810:	Best = 2.706291e+03	FES = 181100	Diversity = 3.1529e-04
Generation 1820:	Best = 2.706290e+03	FES = 182100	Diversity = 2.0037e-04
Generation 1830:	Best = 2.706289e+03	FES = 183100	Diversity = 2.1560e-04
Generation 1840:	Best = 2.706288e+03	FES = 184100	Diversity = 1.6618e-04
Generation 1850:	Best = 2.706287e+03	FES = 185100	Diversity = 1.2423e-04
Generation 1860:	Best = 2.706287e+03	FES = 186100	Diversity = 1.2902e-04
Generation 1870:	Best = 2.706286e+03	FES = 187100	Diversity = 1.0328e-04
Generation 1880:	Best = 2.706286e+03	FES = 188100	Diversity = 7.9999e-05
Generation 1890:	Best = 2.706286e+03	FES = 189100	Diversity = 9.0949e-05
Generation 1900:	Best = 2.706285e+03	FES = 190100	Diversity = 9.4672e-05
Generation 1910:	Best = 2.706285e+03	FES = 191100	Diversity = 6.5495e-05
Generation 1920:	Best = 2.706285e+03	FES = 192100	Diversity = 4.9115e-05
Generation 1930:	Best = 2.706285e+03	FES = 193100	Diversity = 3.3796e-05
Generation 1940:	Best = 2.706285e+03	FES = 194100	Diversity = 2.6400e-05
Generation 1950:	Best = 2.706285e+03	FES = 195100	Diversity = 2.5688e-05
Generation 1960:	Best = 2.706285e+03	FES = 196100	Diversity = 2.1630e-05
Generation 1970:	Best = 2.706285e+03	FES = 197100	Diversity = 1.9754e-05
Generation 1980:	Best = 2.706285e+03	FES = 198100	Diversity = 1.3611e-05
Generation 1990:	Best = 2.706285e+03	FES = 199100	Diversity = 1.0739e-05
Generation 2000:	Best = 2.706285e+03	FES = 200100	Diversity = 8.6922e-06
Generation 2010:	Best = 2.706285e+03	FES = 201100	Diversity = 5.6521e-06
Generation 2020:	Best = 2.706285e+03	FES = 202100	Diversity = 5.3928e-06
Generation 2030:	Best = 2.706285e+03	FES = 203100	Diversity = 3.9797e-06
Generation 2040:	Best = 2.706285e+03	FES = 204100	Diversity = 3.5229e-06
Generation 2050:	Best = 2.706285e+03	FES = 205100	Diversity = 2.0615e-06
Generation 2060:	Best = 2.706285e+03	FES = 206100	Diversity = 1.7642e-06

Generation 2070:	Best = 2.706285e+03	FES = 207100	Diversity = 1.3132e-06
Generation 2080:	Best = 2.706285e+03	FES = 208100	Diversity = 1.0236e-06
Generation 2090:	Best = 2.706285e+03	FES = 209100	Diversity = 7.9899e-07
Generation 2100:	Best = 2.706285e+03	FES = 210100	Diversity = 7.5983e-07
Generation 2110:	Best = 2.706285e+03	FES = 211100	Diversity = 6.9015e-07
Generation 2120:	Best = 2.706285e+03	FES = 212100	Diversity = 5.8984e-07
Generation 2130:	Best = 2.706285e+03	FES = 213100	Diversity = 4.9035e-07
Generation 2140:	Best = 2.706285e+03	FES = 214100	Diversity = 2.5792e-07
Generation 2150:	Best = 2.706285e+03	FES = 215100	Diversity = 2.1496e-07
Generation 2160:	Best = 2.706285e+03	FES = 216100	Diversity = 2.0808e-07
Generation 2170:	Best = 2.706285e+03	FES = 217100	Diversity = 1.4963e-07
Generation 2180:	Best = 2.706285e+03	FES = 218100	Diversity = 1.4065e-07
Generation 2190:	Best = 2.706285e+03	FES = 219100	Diversity = 1.2922e-07
Generation 2200:	Best = 2.706285e+03	FES = 220100	Diversity = 9.8237e-08
Generation 2210:	Best = 2.706285e+03	FES = 221100	Diversity = 5.3764e-08
Generation 2220:	Best = 2.706285e+03	FES = 222100	Diversity = 4.9763e-08
Generation 2230:	Best = 2.706285e+03	FES = 223100	Diversity = 2.4607e-08
Generation 2240:	Best = 2.706285e+03	FES = 224100	Diversity = 2.7777e-08
Generation 2250:	Best = 2.706285e+03	FES = 225100	Diversity = 2.6210e-08
Generation 2260:	Best = 2.706285e+03	FES = 226100	Diversity = 1.6508e-08
Generation 2270:	Best = 2.706285e+03	FES = 227100	Diversity = 1.3744e-08
Generation 2280:	Best = 2.706285e+03	FES = 228100	Diversity = 8.6866e-09
Generation 2290:	Best = 2.706285e+03	FES = 229100	Diversity = 7.1115e-09
Generation 2300:	Best = 2.706285e+03	FES = 230100	Diversity = 5.2472e-09
Generation 2310:	Best = 2.706285e+03	FES = 231100	Diversity = 3.0934e-09
Generation 2320:	Best = 2.706285e+03	FES = 232100	Diversity = 1.8464e-10
Generation 2330:	Best = 2.706285e+03	FES = 233100	Diversity = 9.5312e-13
Generation 2340:	Best = 2.706285e+03	FES = 234100	Diversity = 6.1928e-14
Generation 2350:	Best = 2.706285e+03	FES = 235100	Diversity = 3.7845e-14
Generation 2360:	Best = 2.706285e+03	FES = 236100	Diversity = 3.7820e-14
Generation 2370:	Best = 2.706285e+03	FES = 237100	Diversity = 3.7820e-14
Generation 2380:	Best = 2.706285e+03	FES = 238100	Diversity = 3.7820e-14
Generation 2390:	Best = 2.706285e+03	FES = 239100	Diversity = 3.7820e-14
Generation 2400:	Best = 2.706285e+03	FES = 240100	Diversity = 3.7820e-14
Generation 2410:	Best = 2.706285e+03	FES = 241100	Diversity = 3.7820e-14
Generation 2420:	Best = 2.706285e+03	FES = 242100	Diversity = 3.7820e-14
Generation 2430:	Best = 2.706285e+03	FES = 243100	Diversity = 3.7820e-14
Generation 2440:	Best = 2.706285e+03	FES = 244100	Diversity = 3.7820e-14
Generation 2450:	Best = 2.706285e+03	FES = 245100	Diversity = 3.7820e-14
Generation 2460:	Best = 2.706285e+03	FES = 246100	Diversity = 3.7820e-14
Generation 2470:	Best = 2.706285e+03	FES = 247100	Diversity = 3.7820e-14
Generation 2480:	Best = 2.706285e+03	FES = 248100	Diversity = 3.7820e-14
Generation 2490:	Best = 2.706285e+03	FES = 249100	Diversity = 3.7820e-14
Generation 2500:	Best = 2.706285e+03	FES = 250100	Diversity = 3.7820e-14
Generation 2510:	Best = 2.706285e+03	FES = 251100	Diversity = 3.7820e-14
Generation 2520:	Best = 2.706285e+03	FES = 252100	Diversity = 3.7820e-14
Generation 2530:	Best = 2.706285e+03	FES = 253100	Diversity = 3.7820e-14
Generation 2540:	Best = 2.706285e+03	FES = 254100	Diversity = 3.7820e-14
Generation 2550:	Best = 2.706285e+03	FES = 255100	Diversity = 3.7820e-14
Generation 2560:	Best = 2.706285e+03	FES = 256100	Diversity = 3.7820e-14
Generation 2570:	Best = 2.706285e+03	FES = 257100	Diversity = 3.7820e-14
Generation 2580:	Best = 2.706285e+03	FES = 258100	Diversity = 3.7820e-14
Generation 2590:	Best = 2.706285e+03	FES = 259100	Diversity = 3.7820e-14
Generation 2600:	Best = 2.706285e+03	FES = 260100	Diversity = 3.7820e-14

[illegible]

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Generation 9630: Best = 2.706285e+03 FES = 963100 Diversity = 3.7820e-14
Generation 9640: Best = 2.706285e+03 FES = 964100 Diversity = 3.7820e-14
Generation 9650: Best = 2.706285e+03 FES = 965100 Diversity = 3.7820e-14
Generation 9660: Best = 2.706285e+03 FES = 966100 Diversity = 3.7820e-14
Generation 9670: Best = 2.706285e+03 FES = 967100 Diversity = 3.7820e-14
Generation 9680: Best = 2.706285e+03 FES = 968100 Diversity = 3.7820e-14
Generation 9690: Best = 2.706285e+03 FES = 969100 Diversity = 3.7820e-14
Generation 9700: Best = 2.706285e+03 FES = 970100 Diversity = 3.7820e-14
Generation 9710: Best = 2.706285e+03 FES = 971100 Diversity = 3.7820e-14
Generation 9720: Best = 2.706285e+03 FES = 972100 Diversity = 3.7820e-14
Generation 9730: Best = 2.706285e+03 FES = 973100 Diversity = 3.7820e-14
Generation 9740: Best = 2.706285e+03 FES = 974100 Diversity = 3.7820e-14
Generation 9750: Best = 2.706285e+03 FES = 975100 Diversity = 3.7820e-14
Generation 9760: Best = 2.706285e+03 FES = 976100 Diversity = 3.7820e-14
Generation 9770: Best = 2.706285e+03 FES = 977100 Diversity = 3.7820e-14
Generation 9780: Best = 2.706285e+03 FES = 978100 Diversity = 3.7820e-14
Generation 9790: Best = 2.706285e+03 FES = 979100 Diversity = 3.7820e-14
Generation 9800: Best = 2.706285e+03 FES = 980100 Diversity = 3.7820e-14
Generation 9810: Best = 2.706285e+03 FES = 981100 Diversity = 3.7820e-14
Generation 9820: Best = 2.706285e+03 FES = 982100 Diversity = 3.7820e-14
Generation 9830: Best = 2.706285e+03 FES = 983100 Diversity = 3.7820e-14
Generation 9840: Best = 2.706285e+03 FES = 984100 Diversity = 3.7820e-14
Generation 9850: Best = 2.706285e+03 FES = 985100 Diversity = 3.7820e-14
Generation 9860: Best = 2.706285e+03 FES = 986100 Diversity = 3.7820e-14
Generation 9870: Best = 2.706285e+03 FES = 987100 Diversity = 3.7820e-14
Generation 9880: Best = 2.706285e+03 FES = 988100 Diversity = 3.7820e-14
Generation 9890: Best = 2.706285e+03 FES = 989100 Diversity = 3.7820e-14
Generation 9900: Best = 2.706285e+03 FES = 990100 Diversity = 3.7820e-14
Generation 9910: Best = 2.706285e+03 FES = 991100 Diversity = 3.7820e-14
Generation 9920: Best = 2.706285e+03 FES = 992100 Diversity = 3.7820e-14
Generation 9930: Best = 2.706285e+03 FES = 993100 Diversity = 3.7820e-14
Generation 9940: Best = 2.706285e+03 FES = 994100 Diversity = 3.7820e-14
Generation 9950: Best = 2.706285e+03 FES = 995100 Diversity = 3.7820e-14
Generation 9960: Best = 2.706285e+03 FES = 996100 Diversity = 3.7820e-14
Generation 9970: Best = 2.706285e+03 FES = 997100 Diversity = 3.7820e-14
Generation 9980: Best = 2.706285e+03 FES = 998100 Diversity = 3.7820e-14
Generation 9990: Best = 2.706285e+03 FES = 999100 Diversity = 3.7820e-14

```

```

Optimization completed!
Final Best Fitness = 2.706285e+03
Total Function Evaluations = 1000000

```

```

=====
=
Results:
  Best Fitness          : 2.706285e+03
  Global Optimum        : 2.100000e+03
  Error (|best-optimum|) : 6.062849e+02
  Best Solution (5 dim)  : [-17.76366169   6.972554   63.20718915  54.98734521
39.76699237]
  D^(0) (init. diversity): 5.7220e+01
=====
=

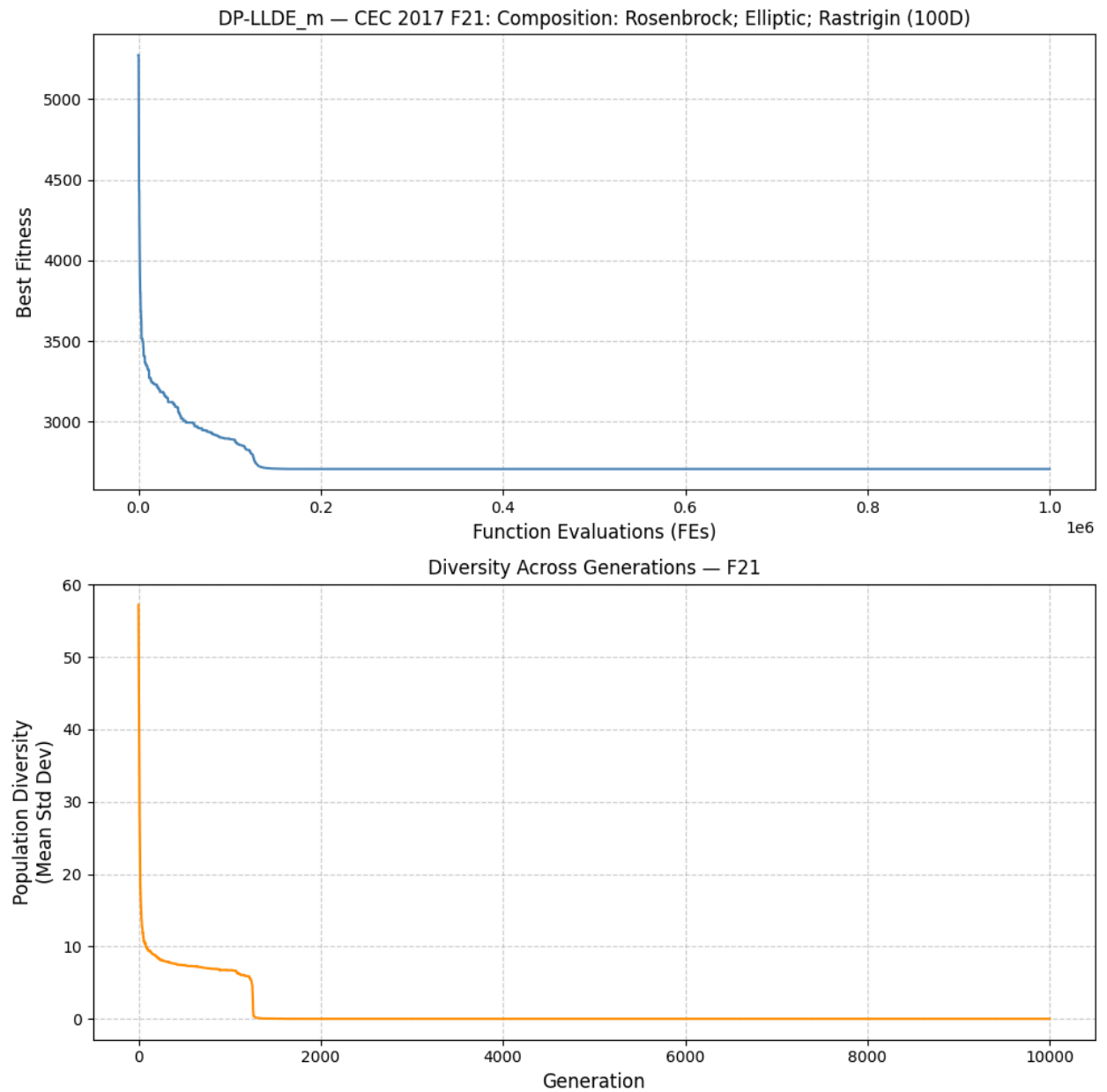
```

```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results_2/all_results_che

```

ckpoint.pkl



Graph saved as convergence_F21.png

✓ Plot saved to: /content/drive/My Drive/thesis_results_2/plots/convergence_F21.png

```
In [ ]: # =====  
# F22 — Composition Function 2 (Rastrigin; Griewank; Schwefel)  
# =====  
run_and_plot('F22', make_wrapper(composition.f22), 2200.0, 'Composition: Rastr
```

=====

=

DP-LLDE_m on CEC 2017 F22 – Single Run

=====

=

Function : Composition: Rastrigin; Griewank; Schwefel (100D)

Bounds : [-100, 100]

Global optimum : 2200.0

Initial $D^*(0) = 5.7220e+01$ | Diversity threshold = 5.7220e+00

Generation 0: Best Fitness = 4.002781e+04 | Diversity = 5.7220e+01

Generation	10:	Best = 3.719342e+04	FES = 1100	Diversity = 4.8855e+01
Generation	20:	Best = 3.603505e+04	FES = 2100	Diversity = 4.7267e+01
Generation	30:	Best = 3.603505e+04	FES = 3100	Diversity = 4.6187e+01
Generation	40:	Best = 3.510166e+04	FES = 4100	Diversity = 4.5279e+01
Generation	50:	Best = 3.500723e+04	FES = 5100	Diversity = 4.4796e+01
Generation	60:	Best = 3.389887e+04	FES = 6100	Diversity = 4.4989e+01
Generation	70:	Best = 3.389887e+04	FES = 7100	Diversity = 4.4614e+01
Generation	80:	Best = 3.389887e+04	FES = 8100	Diversity = 4.4569e+01
Generation	90:	Best = 3.389887e+04	FES = 9100	Diversity = 4.4188e+01
Generation	100:	Best = 3.389887e+04	FES = 10100	Diversity = 4.3811e+01
Generation	110:	Best = 3.389887e+04	FES = 11100	Diversity = 4.3787e+01
Generation	120:	Best = 3.369733e+04	FES = 12100	Diversity = 4.4048e+01
Generation	130:	Best = 3.369733e+04	FES = 13100	Diversity = 4.3903e+01
Generation	140:	Best = 3.185608e+04	FES = 14100	Diversity = 4.3821e+01
Generation	150:	Best = 3.185608e+04	FES = 15100	Diversity = 4.3924e+01
Generation	160:	Best = 3.185608e+04	FES = 16100	Diversity = 4.4118e+01
Generation	170:	Best = 3.183160e+04	FES = 17100	Diversity = 4.4219e+01
Generation	180:	Best = 3.083532e+04	FES = 18100	Diversity = 4.4062e+01
Generation	190:	Best = 3.083532e+04	FES = 19100	Diversity = 4.4121e+01
Generation	200:	Best = 3.083532e+04	FES = 20100	Diversity = 4.4073e+01
Generation	210:	Best = 3.083532e+04	FES = 21100	Diversity = 4.3765e+01
Generation	220:	Best = 3.083532e+04	FES = 22100	Diversity = 4.3505e+01
Generation	230:	Best = 3.083532e+04	FES = 23100	Diversity = 4.3499e+01
Generation	240:	Best = 3.083532e+04	FES = 24100	Diversity = 4.3289e+01
Generation	250:	Best = 3.083532e+04	FES = 25100	Diversity = 4.3050e+01
Generation	260:	Best = 3.083532e+04	FES = 26100	Diversity = 4.2769e+01
Generation	270:	Best = 3.083532e+04	FES = 27100	Diversity = 4.2779e+01
Generation	280:	Best = 3.083532e+04	FES = 28100	Diversity = 4.3132e+01
Generation	290:	Best = 3.083532e+04	FES = 29100	Diversity = 4.3148e+01
Generation	300:	Best = 3.069600e+04	FES = 30100	Diversity = 4.3109e+01
Generation	310:	Best = 3.069600e+04	FES = 31100	Diversity = 4.2926e+01
Generation	320:	Best = 3.069600e+04	FES = 32100	Diversity = 4.3164e+01
Generation	330:	Best = 3.069600e+04	FES = 33100	Diversity = 4.3110e+01
Generation	340:	Best = 3.069600e+04	FES = 34100	Diversity = 4.2897e+01
Generation	350:	Best = 3.069600e+04	FES = 35100	Diversity = 4.2873e+01
Generation	360:	Best = 3.066455e+04	FES = 36100	Diversity = 4.2732e+01
Generation	370:	Best = 3.066455e+04	FES = 37100	Diversity = 4.2684e+01
Generation	380:	Best = 3.066455e+04	FES = 38100	Diversity = 4.2515e+01
Generation	390:	Best = 3.057743e+04	FES = 39100	Diversity = 4.2635e+01
Generation	400:	Best = 3.057743e+04	FES = 40100	Diversity = 4.2635e+01
Generation	410:	Best = 3.057743e+04	FES = 41100	Diversity = 4.2663e+01
Generation	420:	Best = 3.057743e+04	FES = 42100	Diversity = 4.2645e+01
Generation	430:	Best = 3.053880e+04	FES = 43100	Diversity = 4.2262e+01
Generation	440:	Best = 2.994804e+04	FES = 44100	Diversity = 4.2199e+01

Generation	450:	Best = 2.994804e+04	FES = 45100	Diversity = 4.2183e+01
Generation	460:	Best = 2.994804e+04	FES = 46100	Diversity = 4.2607e+01
Generation	470:	Best = 2.994804e+04	FES = 47100	Diversity = 4.2658e+01
Generation	480:	Best = 2.994804e+04	FES = 48100	Diversity = 4.2652e+01
Generation	490:	Best = 2.994804e+04	FES = 49100	Diversity = 4.2645e+01
Generation	500:	Best = 2.994804e+04	FES = 50100	Diversity = 4.2803e+01
Generation	510:	Best = 2.994804e+04	FES = 51100	Diversity = 4.2803e+01
Generation	520:	Best = 2.955683e+04	FES = 52100	Diversity = 4.2707e+01
Generation	530:	Best = 2.955683e+04	FES = 53100	Diversity = 4.2990e+01
Generation	540:	Best = 2.955683e+04	FES = 54100	Diversity = 4.2926e+01
Generation	550:	Best = 2.940961e+04	FES = 55100	Diversity = 4.2804e+01
Generation	560:	Best = 2.940961e+04	FES = 56100	Diversity = 4.2775e+01
Generation	570:	Best = 2.940961e+04	FES = 57100	Diversity = 4.2751e+01
Generation	580:	Best = 2.900641e+04	FES = 58100	Diversity = 4.2747e+01
Generation	590:	Best = 2.900641e+04	FES = 59100	Diversity = 4.2725e+01
Generation	600:	Best = 2.900641e+04	FES = 60100	Diversity = 4.2774e+01
Generation	610:	Best = 2.857371e+04	FES = 61100	Diversity = 4.2715e+01
Generation	620:	Best = 2.857371e+04	FES = 62100	Diversity = 4.2796e+01
Generation	630:	Best = 2.857371e+04	FES = 63100	Diversity = 4.2676e+01
Generation	640:	Best = 2.857371e+04	FES = 64100	Diversity = 4.2666e+01
Generation	650:	Best = 2.857371e+04	FES = 65100	Diversity = 4.2668e+01
Generation	660:	Best = 2.857371e+04	FES = 66100	Diversity = 4.2550e+01
Generation	670:	Best = 2.857371e+04	FES = 67100	Diversity = 4.2472e+01
Generation	680:	Best = 2.857371e+04	FES = 68100	Diversity = 4.2467e+01
Generation	690:	Best = 2.857371e+04	FES = 69100	Diversity = 4.2387e+01
Generation	700:	Best = 2.857371e+04	FES = 70100	Diversity = 4.2373e+01
Generation	710:	Best = 2.857371e+04	FES = 71100	Diversity = 4.2379e+01
Generation	720:	Best = 2.857371e+04	FES = 72100	Diversity = 4.2355e+01
Generation	730:	Best = 2.857371e+04	FES = 73100	Diversity = 4.2353e+01
Generation	740:	Best = 2.857371e+04	FES = 74100	Diversity = 4.2345e+01
Generation	750:	Best = 2.857371e+04	FES = 75100	Diversity = 4.2137e+01
Generation	760:	Best = 2.857371e+04	FES = 76100	Diversity = 4.2132e+01
Generation	770:	Best = 2.839008e+04	FES = 77100	Diversity = 4.2132e+01
Generation	780:	Best = 2.839008e+04	FES = 78100	Diversity = 4.2117e+01
Generation	790:	Best = 2.839008e+04	FES = 79100	Diversity = 4.2094e+01
Generation	800:	Best = 2.839008e+04	FES = 80100	Diversity = 4.2063e+01
Generation	810:	Best = 2.839008e+04	FES = 81100	Diversity = 4.2062e+01
Generation	820:	Best = 2.839008e+04	FES = 82100	Diversity = 4.2191e+01
Generation	830:	Best = 2.839008e+04	FES = 83100	Diversity = 4.2189e+01
Generation	840:	Best = 2.812454e+04	FES = 84100	Diversity = 4.2147e+01
Generation	850:	Best = 2.812454e+04	FES = 85100	Diversity = 4.2148e+01
Generation	860:	Best = 2.812454e+04	FES = 86100	Diversity = 4.1788e+01
Generation	870:	Best = 2.812454e+04	FES = 87100	Diversity = 4.1822e+01
Generation	880:	Best = 2.812454e+04	FES = 88100	Diversity = 4.1803e+01
Generation	890:	Best = 2.812454e+04	FES = 89100	Diversity = 4.1803e+01
Generation	900:	Best = 2.812454e+04	FES = 90100	Diversity = 4.1658e+01
Generation	910:	Best = 2.812454e+04	FES = 91100	Diversity = 4.1660e+01
Generation	920:	Best = 2.804975e+04	FES = 92100	Diversity = 4.1674e+01
Generation	930:	Best = 2.804975e+04	FES = 93100	Diversity = 4.1674e+01
Generation	940:	Best = 2.804975e+04	FES = 94100	Diversity = 4.1657e+01
Generation	950:	Best = 2.804975e+04	FES = 95100	Diversity = 4.1648e+01
Generation	960:	Best = 2.804975e+04	FES = 96100	Diversity = 4.1679e+01
Generation	970:	Best = 2.804975e+04	FES = 97100	Diversity = 4.1677e+01
Generation	980:	Best = 2.804975e+04	FES = 98100	Diversity = 4.1665e+01

Generation 990:	Best = 2.804975e+04	FES = 99100	Diversity = 4.1649e+01
Generation 1000:	Best = 2.804975e+04	FES = 100100	Diversity = 4.1595e+01
Generation 1010:	Best = 2.804975e+04	FES = 101100	Diversity = 4.1508e+01
Generation 1020:	Best = 2.804975e+04	FES = 102100	Diversity = 4.1456e+01
Generation 1030:	Best = 2.804975e+04	FES = 103100	Diversity = 4.1450e+01
Generation 1040:	Best = 2.795585e+04	FES = 104100	Diversity = 4.1215e+01
Generation 1050:	Best = 2.795585e+04	FES = 105100	Diversity = 4.1210e+01
Generation 1060:	Best = 2.795585e+04	FES = 106100	Diversity = 4.1209e+01
Generation 1070:	Best = 2.795585e+04	FES = 107100	Diversity = 4.0902e+01
Generation 1080:	Best = 2.795585e+04	FES = 108100	Diversity = 4.0885e+01
Generation 1090:	Best = 2.781277e+04	FES = 109100	Diversity = 4.0867e+01
Generation 1100:	Best = 2.754119e+04	FES = 110100	Diversity = 4.0834e+01
Generation 1110:	Best = 2.754119e+04	FES = 111100	Diversity = 4.0796e+01
Generation 1120:	Best = 2.751942e+04	FES = 112100	Diversity = 4.0794e+01
Generation 1130:	Best = 2.751942e+04	FES = 113100	Diversity = 4.0616e+01
Generation 1140:	Best = 2.751942e+04	FES = 114100	Diversity = 4.0610e+01
Generation 1150:	Best = 2.751942e+04	FES = 115100	Diversity = 4.0603e+01
Generation 1160:	Best = 2.751942e+04	FES = 116100	Diversity = 4.0561e+01
Generation 1170:	Best = 2.751942e+04	FES = 117100	Diversity = 4.0551e+01
Generation 1180:	Best = 2.751942e+04	FES = 118100	Diversity = 4.0547e+01
Generation 1190:	Best = 2.751942e+04	FES = 119100	Diversity = 4.0519e+01
Generation 1200:	Best = 2.751942e+04	FES = 120100	Diversity = 4.0519e+01
Generation 1210:	Best = 2.744052e+04	FES = 121100	Diversity = 4.0499e+01
Generation 1220:	Best = 2.744052e+04	FES = 122100	Diversity = 4.0681e+01
Generation 1230:	Best = 2.744052e+04	FES = 123100	Diversity = 4.0680e+01
Generation 1240:	Best = 2.744052e+04	FES = 124100	Diversity = 4.0754e+01
Generation 1250:	Best = 2.744052e+04	FES = 125100	Diversity = 4.0754e+01
Generation 1260:	Best = 2.744052e+04	FES = 126100	Diversity = 4.0755e+01
Generation 1270:	Best = 2.744052e+04	FES = 127100	Diversity = 4.0744e+01
Generation 1280:	Best = 2.744052e+04	FES = 128100	Diversity = 4.0744e+01
Generation 1290:	Best = 2.744052e+04	FES = 129100	Diversity = 4.0728e+01
Generation 1300:	Best = 2.744052e+04	FES = 130100	Diversity = 4.0684e+01
Generation 1310:	Best = 2.744052e+04	FES = 131100	Diversity = 4.0493e+01
Generation 1320:	Best = 2.719436e+04	FES = 132100	Diversity = 4.0456e+01
Generation 1330:	Best = 2.719436e+04	FES = 133100	Diversity = 4.0403e+01
Generation 1340:	Best = 2.719436e+04	FES = 134100	Diversity = 4.0402e+01
Generation 1350:	Best = 2.719436e+04	FES = 135100	Diversity = 4.0318e+01
Generation 1360:	Best = 2.719436e+04	FES = 136100	Diversity = 4.0302e+01
Generation 1370:	Best = 2.719436e+04	FES = 137100	Diversity = 4.0283e+01
Generation 1380:	Best = 2.719436e+04	FES = 138100	Diversity = 4.0284e+01
Generation 1390:	Best = 2.719436e+04	FES = 139100	Diversity = 4.0277e+01
Generation 1400:	Best = 2.695990e+04	FES = 140100	Diversity = 4.0266e+01
Generation 1410:	Best = 2.695990e+04	FES = 141100	Diversity = 4.0260e+01
Generation 1420:	Best = 2.695990e+04	FES = 142100	Diversity = 4.0182e+01
Generation 1430:	Best = 2.695990e+04	FES = 143100	Diversity = 4.0149e+01
Generation 1440:	Best = 2.683256e+04	FES = 144100	Diversity = 4.0139e+01
Generation 1450:	Best = 2.683256e+04	FES = 145100	Diversity = 4.0000e+01
Generation 1460:	Best = 2.683256e+04	FES = 146100	Diversity = 3.9986e+01
Generation 1470:	Best = 2.683256e+04	FES = 147100	Diversity = 3.9982e+01
Generation 1480:	Best = 2.683256e+04	FES = 148100	Diversity = 3.9979e+01
Generation 1490:	Best = 2.683256e+04	FES = 149100	Diversity = 3.9976e+01
Generation 1500:	Best = 2.671952e+04	FES = 150100	Diversity = 3.9965e+01
Generation 1510:	Best = 2.671952e+04	FES = 151100	Diversity = 3.9964e+01
Generation 1520:	Best = 2.671952e+04	FES = 152100	Diversity = 3.9951e+01

Generation 1530:	Best = 2.671952e+04	FES = 153100	Diversity = 3.9939e+01
Generation 1540:	Best = 2.671952e+04	FES = 154100	Diversity = 3.9928e+01
Generation 1550:	Best = 2.671952e+04	FES = 155100	Diversity = 3.9921e+01
Generation 1560:	Best = 2.671952e+04	FES = 156100	Diversity = 4.0015e+01
Generation 1570:	Best = 2.671952e+04	FES = 157100	Diversity = 4.0009e+01
Generation 1580:	Best = 2.671952e+04	FES = 158100	Diversity = 3.9903e+01
Generation 1590:	Best = 2.671952e+04	FES = 159100	Diversity = 3.9904e+01
Generation 1600:	Best = 2.671952e+04	FES = 160100	Diversity = 3.9875e+01
Generation 1610:	Best = 2.671952e+04	FES = 161100	Diversity = 3.9875e+01
Generation 1620:	Best = 2.671952e+04	FES = 162100	Diversity = 3.9667e+01
Generation 1630:	Best = 2.671952e+04	FES = 163100	Diversity = 3.9652e+01
Generation 1640:	Best = 2.671952e+04	FES = 164100	Diversity = 3.9657e+01
Generation 1650:	Best = 2.671952e+04	FES = 165100	Diversity = 3.9656e+01
Generation 1660:	Best = 2.670638e+04	FES = 166100	Diversity = 3.9653e+01
Generation 1670:	Best = 2.670638e+04	FES = 167100	Diversity = 3.9640e+01
Generation 1680:	Best = 2.670638e+04	FES = 168100	Diversity = 3.9792e+01
Generation 1690:	Best = 2.670638e+04	FES = 169100	Diversity = 3.9793e+01
Generation 1700:	Best = 2.603989e+04	FES = 170100	Diversity = 3.9767e+01
Generation 1710:	Best = 2.603989e+04	FES = 171100	Diversity = 3.9766e+01
Generation 1720:	Best = 2.603989e+04	FES = 172100	Diversity = 3.9752e+01
Generation 1730:	Best = 2.603989e+04	FES = 173100	Diversity = 3.9751e+01
Generation 1740:	Best = 2.603989e+04	FES = 174100	Diversity = 3.9731e+01
Generation 1750:	Best = 2.603989e+04	FES = 175100	Diversity = 3.9728e+01
Generation 1760:	Best = 2.603989e+04	FES = 176100	Diversity = 3.9677e+01
Generation 1770:	Best = 2.603989e+04	FES = 177100	Diversity = 3.9675e+01
Generation 1780:	Best = 2.603989e+04	FES = 178100	Diversity = 3.9668e+01
Generation 1790:	Best = 2.603989e+04	FES = 179100	Diversity = 3.9667e+01
Generation 1800:	Best = 2.603989e+04	FES = 180100	Diversity = 3.9646e+01
Generation 1810:	Best = 2.603989e+04	FES = 181100	Diversity = 3.9666e+01
Generation 1820:	Best = 2.603989e+04	FES = 182100	Diversity = 3.9649e+01
Generation 1830:	Best = 2.603989e+04	FES = 183100	Diversity = 3.9662e+01
Generation 1840:	Best = 2.603989e+04	FES = 184100	Diversity = 3.9656e+01
Generation 1850:	Best = 2.603989e+04	FES = 185100	Diversity = 3.9647e+01
Generation 1860:	Best = 2.603989e+04	FES = 186100	Diversity = 3.9638e+01
Generation 1870:	Best = 2.560364e+04	FES = 187100	Diversity = 3.9631e+01
Generation 1880:	Best = 2.560364e+04	FES = 188100	Diversity = 3.9625e+01
Generation 1890:	Best = 2.560364e+04	FES = 189100	Diversity = 3.9624e+01
Generation 1900:	Best = 2.560364e+04	FES = 190100	Diversity = 3.9610e+01
Generation 1910:	Best = 2.560364e+04	FES = 191100	Diversity = 3.9603e+01
Generation 1920:	Best = 2.560364e+04	FES = 192100	Diversity = 3.9599e+01
Generation 1930:	Best = 2.560364e+04	FES = 193100	Diversity = 3.9596e+01
Generation 1940:	Best = 2.560364e+04	FES = 194100	Diversity = 3.9596e+01
Generation 1950:	Best = 2.560364e+04	FES = 195100	Diversity = 3.9596e+01
Generation 1960:	Best = 2.560364e+04	FES = 196100	Diversity = 3.9593e+01
Generation 1970:	Best = 2.560364e+04	FES = 197100	Diversity = 3.9578e+01
Generation 1980:	Best = 2.560364e+04	FES = 198100	Diversity = 3.9572e+01
Generation 1990:	Best = 2.560364e+04	FES = 199100	Diversity = 3.9567e+01
Generation 2000:	Best = 2.560364e+04	FES = 200100	Diversity = 3.9562e+01
Generation 2010:	Best = 2.560364e+04	FES = 201100	Diversity = 3.9537e+01
Generation 2020:	Best = 2.560364e+04	FES = 202100	Diversity = 3.9538e+01
Generation 2030:	Best = 2.560364e+04	FES = 203100	Diversity = 3.9538e+01
Generation 2040:	Best = 2.560364e+04	FES = 204100	Diversity = 3.9528e+01
Generation 2050:	Best = 2.560364e+04	FES = 205100	Diversity = 3.9519e+01
Generation 2060:	Best = 2.560364e+04	FES = 206100	Diversity = 3.9522e+01

Generation 2070:	Best = 2.558160e+04	FES = 207100	Diversity = 3.9523e+01
Generation 2080:	Best = 2.558160e+04	FES = 208100	Diversity = 3.9517e+01
Generation 2090:	Best = 2.558160e+04	FES = 209100	Diversity = 3.9503e+01
Generation 2100:	Best = 2.558160e+04	FES = 210100	Diversity = 3.9493e+01
Generation 2110:	Best = 2.558160e+04	FES = 211100	Diversity = 3.9491e+01
Generation 2120:	Best = 2.558160e+04	FES = 212100	Diversity = 3.9489e+01
Generation 2130:	Best = 2.558160e+04	FES = 213100	Diversity = 3.9488e+01
Generation 2140:	Best = 2.558160e+04	FES = 214100	Diversity = 3.9492e+01
Generation 2150:	Best = 2.558160e+04	FES = 215100	Diversity = 3.9492e+01
Generation 2160:	Best = 2.558160e+04	FES = 216100	Diversity = 3.9492e+01
Generation 2170:	Best = 2.558160e+04	FES = 217100	Diversity = 3.9489e+01
Generation 2180:	Best = 2.558160e+04	FES = 218100	Diversity = 3.9482e+01
Generation 2190:	Best = 2.558160e+04	FES = 219100	Diversity = 3.9478e+01
Generation 2200:	Best = 2.558160e+04	FES = 220100	Diversity = 3.9475e+01
Generation 2210:	Best = 2.558160e+04	FES = 221100	Diversity = 3.9477e+01
Generation 2220:	Best = 2.558160e+04	FES = 222100	Diversity = 3.9476e+01
Generation 2230:	Best = 2.558160e+04	FES = 223100	Diversity = 3.9472e+01
Generation 2240:	Best = 2.558160e+04	FES = 224100	Diversity = 3.9473e+01
Generation 2250:	Best = 2.558160e+04	FES = 225100	Diversity = 3.9473e+01
Generation 2260:	Best = 2.558160e+04	FES = 226100	Diversity = 3.9476e+01
Generation 2270:	Best = 2.558160e+04	FES = 227100	Diversity = 3.9493e+01
Generation 2280:	Best = 2.558160e+04	FES = 228100	Diversity = 3.9491e+01
Generation 2290:	Best = 2.558160e+04	FES = 229100	Diversity = 3.9491e+01
Generation 2300:	Best = 2.558160e+04	FES = 230100	Diversity = 3.9482e+01
Generation 2310:	Best = 2.558160e+04	FES = 231100	Diversity = 3.9478e+01
Generation 2320:	Best = 2.510858e+04	FES = 232100	Diversity = 3.9454e+01
Generation 2330:	Best = 2.510858e+04	FES = 233100	Diversity = 3.9448e+01
Generation 2340:	Best = 2.510858e+04	FES = 234100	Diversity = 3.9443e+01
Generation 2350:	Best = 2.510858e+04	FES = 235100	Diversity = 3.9434e+01
Generation 2360:	Best = 2.510858e+04	FES = 236100	Diversity = 3.9422e+01
Generation 2370:	Best = 2.510858e+04	FES = 237100	Diversity = 3.9422e+01
Generation 2380:	Best = 2.510858e+04	FES = 238100	Diversity = 3.9423e+01
Generation 2390:	Best = 2.510858e+04	FES = 239100	Diversity = 3.9419e+01
Generation 2400:	Best = 2.510858e+04	FES = 240100	Diversity = 3.9420e+01
Generation 2410:	Best = 2.510858e+04	FES = 241100	Diversity = 3.9410e+01
Generation 2420:	Best = 2.510858e+04	FES = 242100	Diversity = 3.9270e+01
Generation 2430:	Best = 2.510858e+04	FES = 243100	Diversity = 3.9270e+01
Generation 2440:	Best = 2.510858e+04	FES = 244100	Diversity = 3.9266e+01
Generation 2450:	Best = 2.510858e+04	FES = 245100	Diversity = 3.9258e+01
Generation 2460:	Best = 2.510858e+04	FES = 246100	Diversity = 3.9262e+01
Generation 2470:	Best = 2.510858e+04	FES = 247100	Diversity = 3.9262e+01
Generation 2480:	Best = 2.510858e+04	FES = 248100	Diversity = 3.9262e+01
Generation 2490:	Best = 2.510858e+04	FES = 249100	Diversity = 3.9256e+01
Generation 2500:	Best = 2.510858e+04	FES = 250100	Diversity = 3.9256e+01
Generation 2510:	Best = 2.510858e+04	FES = 251100	Diversity = 3.9258e+01
Generation 2520:	Best = 2.510858e+04	FES = 252100	Diversity = 3.9258e+01
Generation 2530:	Best = 2.510858e+04	FES = 253100	Diversity = 3.9258e+01
Generation 2540:	Best = 2.510858e+04	FES = 254100	Diversity = 3.9258e+01
Generation 2550:	Best = 2.510858e+04	FES = 255100	Diversity = 3.9108e+01
Generation 2560:	Best = 2.510858e+04	FES = 256100	Diversity = 3.9104e+01
Generation 2570:	Best = 2.510858e+04	FES = 257100	Diversity = 3.9102e+01
Generation 2580:	Best = 2.510858e+04	FES = 258100	Diversity = 3.9096e+01
Generation 2590:	Best = 2.510858e+04	FES = 259100	Diversity = 3.9086e+01
Generation 2600:	Best = 2.510858e+04	FES = 260100	Diversity = 3.9083e+01

Generation 2610:	Best = 2.510858e+04	FES = 261100	Diversity = 3.9082e+01
Generation 2620:	Best = 2.510858e+04	FES = 262100	Diversity = 3.9079e+01
Generation 2630:	Best = 2.510858e+04	FES = 263100	Diversity = 3.9073e+01
Generation 2640:	Best = 2.510858e+04	FES = 264100	Diversity = 3.9064e+01
Generation 2650:	Best = 2.510858e+04	FES = 265100	Diversity = 3.9069e+01
Generation 2660:	Best = 2.510858e+04	FES = 266100	Diversity = 3.9070e+01
Generation 2670:	Best = 2.510858e+04	FES = 267100	Diversity = 3.9070e+01
Generation 2680:	Best = 2.510858e+04	FES = 268100	Diversity = 3.9070e+01
Generation 2690:	Best = 2.510858e+04	FES = 269100	Diversity = 3.9064e+01
Generation 2700:	Best = 2.510858e+04	FES = 270100	Diversity = 3.9064e+01
Generation 2710:	Best = 2.510858e+04	FES = 271100	Diversity = 3.9061e+01
Generation 2720:	Best = 2.510858e+04	FES = 272100	Diversity = 3.9055e+01
Generation 2730:	Best = 2.510858e+04	FES = 273100	Diversity = 3.9055e+01
Generation 2740:	Best = 2.510858e+04	FES = 274100	Diversity = 3.9053e+01
Generation 2750:	Best = 2.510858e+04	FES = 275100	Diversity = 3.9050e+01
Generation 2760:	Best = 2.510858e+04	FES = 276100	Diversity = 3.9049e+01
Generation 2770:	Best = 2.510858e+04	FES = 277100	Diversity = 3.9049e+01
Generation 2780:	Best = 2.510858e+04	FES = 278100	Diversity = 3.9048e+01
Generation 2790:	Best = 2.510858e+04	FES = 279100	Diversity = 3.9046e+01
Generation 2800:	Best = 2.510858e+04	FES = 280100	Diversity = 3.9042e+01
Generation 2810:	Best = 2.510858e+04	FES = 281100	Diversity = 3.9039e+01
Generation 2820:	Best = 2.510858e+04	FES = 282100	Diversity = 3.9026e+01
Generation 2830:	Best = 2.510858e+04	FES = 283100	Diversity = 3.9023e+01
Generation 2840:	Best = 2.510858e+04	FES = 284100	Diversity = 3.9013e+01
Generation 2850:	Best = 2.510858e+04	FES = 285100	Diversity = 3.9008e+01
Generation 2860:	Best = 2.510858e+04	FES = 286100	Diversity = 3.8989e+01
Generation 2870:	Best = 2.510858e+04	FES = 287100	Diversity = 3.8987e+01
Generation 2880:	Best = 2.467641e+04	FES = 288100	Diversity = 3.8977e+01
Generation 2890:	Best = 2.467641e+04	FES = 289100	Diversity = 3.8974e+01
Generation 2900:	Best = 2.461643e+04	FES = 290100	Diversity = 3.8966e+01
Generation 2910:	Best = 2.461643e+04	FES = 291100	Diversity = 3.8966e+01
Generation 2920:	Best = 2.461643e+04	FES = 292100	Diversity = 3.8960e+01
Generation 2930:	Best = 2.461643e+04	FES = 293100	Diversity = 3.8959e+01
Generation 2940:	Best = 2.461643e+04	FES = 294100	Diversity = 3.8947e+01
Generation 2950:	Best = 2.435179e+04	FES = 295100	Diversity = 3.8940e+01
Generation 2960:	Best = 2.435179e+04	FES = 296100	Diversity = 3.8925e+01
Generation 2970:	Best = 2.408709e+04	FES = 297100	Diversity = 3.8934e+01
Generation 2980:	Best = 2.408709e+04	FES = 298100	Diversity = 3.8933e+01
Generation 2990:	Best = 2.408709e+04	FES = 299100	Diversity = 3.8930e+01
Generation 3000:	Best = 2.408709e+04	FES = 300100	Diversity = 3.8929e+01
Generation 3010:	Best = 2.408709e+04	FES = 301100	Diversity = 3.8928e+01
Generation 3020:	Best = 2.408709e+04	FES = 302100	Diversity = 3.8913e+01
Generation 3030:	Best = 2.408709e+04	FES = 303100	Diversity = 3.8910e+01
Generation 3040:	Best = 2.408709e+04	FES = 304100	Diversity = 3.8899e+01
Generation 3050:	Best = 2.408709e+04	FES = 305100	Diversity = 3.8898e+01
Generation 3060:	Best = 2.408709e+04	FES = 306100	Diversity = 3.8899e+01
Generation 3070:	Best = 2.408709e+04	FES = 307100	Diversity = 3.8888e+01
Generation 3080:	Best = 2.408709e+04	FES = 308100	Diversity = 3.8884e+01
Generation 3090:	Best = 2.408709e+04	FES = 309100	Diversity = 3.8856e+01
Generation 3100:	Best = 2.408709e+04	FES = 310100	Diversity = 3.8837e+01
Generation 3110:	Best = 2.408709e+04	FES = 311100	Diversity = 3.8843e+01
Generation 3120:	Best = 2.408709e+04	FES = 312100	Diversity = 3.8841e+01
Generation 3130:	Best = 2.408709e+04	FES = 313100	Diversity = 3.8838e+01
Generation 3140:	Best = 2.408709e+04	FES = 314100	Diversity = 3.8838e+01

Generation 3150:	Best = 2.408709e+04	FES = 315100	Diversity = 3.8838e+01
Generation 3160:	Best = 2.408709e+04	FES = 316100	Diversity = 3.8838e+01
Generation 3170:	Best = 2.408709e+04	FES = 317100	Diversity = 3.8825e+01
Generation 3180:	Best = 2.408709e+04	FES = 318100	Diversity = 3.8825e+01
Generation 3190:	Best = 2.408709e+04	FES = 319100	Diversity = 3.8819e+01
Generation 3200:	Best = 2.408709e+04	FES = 320100	Diversity = 3.8817e+01
Generation 3210:	Best = 2.408709e+04	FES = 321100	Diversity = 3.8816e+01
Generation 3220:	Best = 2.408709e+04	FES = 322100	Diversity = 3.8816e+01
Generation 3230:	Best = 2.408709e+04	FES = 323100	Diversity = 3.8818e+01
Generation 3240:	Best = 2.408709e+04	FES = 324100	Diversity = 3.8817e+01
Generation 3250:	Best = 2.408709e+04	FES = 325100	Diversity = 3.8806e+01
Generation 3260:	Best = 2.408709e+04	FES = 326100	Diversity = 3.8806e+01
Generation 3270:	Best = 2.408709e+04	FES = 327100	Diversity = 3.8805e+01
Generation 3280:	Best = 2.408709e+04	FES = 328100	Diversity = 3.8805e+01
Generation 3290:	Best = 2.408709e+04	FES = 329100	Diversity = 3.9016e+01
Generation 3300:	Best = 2.408709e+04	FES = 330100	Diversity = 3.9016e+01
Generation 3310:	Best = 2.408709e+04	FES = 331100	Diversity = 3.9016e+01
Generation 3320:	Best = 2.408709e+04	FES = 332100	Diversity = 3.9011e+01
Generation 3330:	Best = 2.408709e+04	FES = 333100	Diversity = 3.9010e+01
Generation 3340:	Best = 2.408709e+04	FES = 334100	Diversity = 3.9008e+01
Generation 3350:	Best = 2.408709e+04	FES = 335100	Diversity = 3.9004e+01
Generation 3360:	Best = 2.408709e+04	FES = 336100	Diversity = 3.8997e+01
Generation 3370:	Best = 2.408709e+04	FES = 337100	Diversity = 3.8985e+01
Generation 3380:	Best = 2.408709e+04	FES = 338100	Diversity = 3.8983e+01
Generation 3390:	Best = 2.408709e+04	FES = 339100	Diversity = 3.8982e+01
Generation 3400:	Best = 2.408709e+04	FES = 340100	Diversity = 3.8982e+01
Generation 3410:	Best = 2.408709e+04	FES = 341100	Diversity = 3.9026e+01
Generation 3420:	Best = 2.408709e+04	FES = 342100	Diversity = 3.9027e+01
Generation 3430:	Best = 2.408709e+04	FES = 343100	Diversity = 3.9027e+01
Generation 3440:	Best = 2.408709e+04	FES = 344100	Diversity = 3.9027e+01
Generation 3450:	Best = 2.408709e+04	FES = 345100	Diversity = 3.9027e+01
Generation 3460:	Best = 2.408709e+04	FES = 346100	Diversity = 3.9024e+01
Generation 3470:	Best = 2.408709e+04	FES = 347100	Diversity = 3.9027e+01
Generation 3480:	Best = 2.408709e+04	FES = 348100	Diversity = 3.9026e+01
Generation 3490:	Best = 2.408709e+04	FES = 349100	Diversity = 3.9026e+01
Generation 3500:	Best = 2.408709e+04	FES = 350100	Diversity = 3.9024e+01
Generation 3510:	Best = 2.408709e+04	FES = 351100	Diversity = 3.9024e+01
Generation 3520:	Best = 2.408709e+04	FES = 352100	Diversity = 3.9022e+01
Generation 3530:	Best = 2.408709e+04	FES = 353100	Diversity = 3.9020e+01
Generation 3540:	Best = 2.408709e+04	FES = 354100	Diversity = 3.8972e+01
Generation 3550:	Best = 2.408709e+04	FES = 355100	Diversity = 3.8960e+01
Generation 3560:	Best = 2.408709e+04	FES = 356100	Diversity = 3.8960e+01
Generation 3570:	Best = 2.373379e+04	FES = 357100	Diversity = 3.8959e+01
Generation 3580:	Best = 2.373379e+04	FES = 358100	Diversity = 3.8959e+01
Generation 3590:	Best = 2.373379e+04	FES = 359100	Diversity = 3.8960e+01
Generation 3600:	Best = 2.373379e+04	FES = 360100	Diversity = 3.8959e+01
Generation 3610:	Best = 2.373379e+04	FES = 361100	Diversity = 3.8958e+01
Generation 3620:	Best = 2.373379e+04	FES = 362100	Diversity = 3.8950e+01
Generation 3630:	Best = 2.373379e+04	FES = 363100	Diversity = 3.8950e+01
Generation 3640:	Best = 2.373379e+04	FES = 364100	Diversity = 3.8949e+01
Generation 3650:	Best = 2.373379e+04	FES = 365100	Diversity = 3.8947e+01
Generation 3660:	Best = 2.373379e+04	FES = 366100	Diversity = 3.8945e+01
Generation 3670:	Best = 2.373379e+04	FES = 367100	Diversity = 3.8945e+01
Generation 3680:	Best = 2.373379e+04	FES = 368100	Diversity = 3.8943e+01

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Generation 5850:	Best = 2.352105e+04	FES = 585100	Diversity = 3.8400e+01
Generation 5860:	Best = 2.352105e+04	FES = 586100	Diversity = 3.8282e+01
Generation 5870:	Best = 2.347258e+04	FES = 587100	Diversity = 3.8280e+01
Generation 5880:	Best = 2.347122e+04	FES = 588100	Diversity = 3.8093e+01
Generation 5890:	Best = 2.320208e+04	FES = 589100	Diversity = 3.7335e+01
Generation 5900:	Best = 2.320187e+04	FES = 590100	Diversity = 3.7011e+01
Generation 5910:	Best = 2.320187e+04	FES = 591100	Diversity = 3.7011e+01
Generation 5920:	Best = 2.320187e+04	FES = 592100	Diversity = 3.7011e+01
Generation 5930:	Best = 2.320187e+04	FES = 593100	Diversity = 3.7011e+01
Generation 5940:	Best = 2.312063e+04	FES = 594100	Diversity = 3.6684e+01
Generation 5950:	Best = 2.312063e+04	FES = 595100	Diversity = 3.6685e+01
Generation 5960:	Best = 2.312063e+04	FES = 596100	Diversity = 3.6557e+01
Generation 5970:	Best = 2.312005e+04	FES = 597100	Diversity = 3.6557e+01
Generation 5980:	Best = 2.312005e+04	FES = 598100	Diversity = 3.6557e+01
Generation 5990:	Best = 2.312005e+04	FES = 599100	Diversity = 3.6557e+01
Generation 6000:	Best = 2.312005e+04	FES = 600100	Diversity = 3.6557e+01
Generation 6010:	Best = 2.312005e+04	FES = 601100	Diversity = 3.6261e+01
Generation 6020:	Best = 2.312005e+04	FES = 602100	Diversity = 3.6111e+01
Generation 6030:	Best = 2.312005e+04	FES = 603100	Diversity = 3.6111e+01
Generation 6040:	Best = 2.312005e+04	FES = 604100	Diversity = 3.6111e+01
Generation 6050:	Best = 2.312005e+04	FES = 605100	Diversity = 3.6111e+01
Generation 6060:	Best = 2.312005e+04	FES = 606100	Diversity = 3.5982e+01
Generation 6070:	Best = 2.312005e+04	FES = 607100	Diversity = 3.5982e+01
Generation 6080:	Best = 2.312005e+04	FES = 608100	Diversity = 3.5981e+01
Generation 6090:	Best = 2.312005e+04	FES = 609100	Diversity = 3.5907e+01
Generation 6100:	Best = 2.312005e+04	FES = 610100	Diversity = 3.5907e+01
Generation 6110:	Best = 2.312005e+04	FES = 611100	Diversity = 3.5907e+01
Generation 6120:	Best = 2.312005e+04	FES = 612100	Diversity = 3.5907e+01
Generation 6130:	Best = 2.312005e+04	FES = 613100	Diversity = 3.5907e+01
Generation 6140:	Best = 2.312005e+04	FES = 614100	Diversity = 3.5907e+01
Generation 6150:	Best = 2.312005e+04	FES = 615100	Diversity = 3.5907e+01
Generation 6160:	Best = 2.312005e+04	FES = 616100	Diversity = 3.5907e+01
Generation 6170:	Best = 2.312005e+04	FES = 617100	Diversity = 3.5907e+01
Generation 6180:	Best = 2.312005e+04	FES = 618100	Diversity = 3.5907e+01
Generation 6190:	Best = 2.312005e+04	FES = 619100	Diversity = 3.5907e+01
Generation 6200:	Best = 2.310471e+04	FES = 620100	Diversity = 3.5678e+01
Generation 6210:	Best = 2.310264e+04	FES = 621100	Diversity = 3.5158e+01
Generation 6220:	Best = 2.310262e+04	FES = 622100	Diversity = 3.4683e+01
Generation 6230:	Best = 2.310262e+04	FES = 623100	Diversity = 3.4256e+01
Generation 6240:	Best = 2.310262e+04	FES = 624100	Diversity = 3.3689e+01
Generation 6250:	Best = 2.310262e+04	FES = 625100	Diversity = 3.3689e+01
Generation 6260:	Best = 2.310262e+04	FES = 626100	Diversity = 3.3689e+01
Generation 6270:	Best = 2.310262e+04	FES = 627100	Diversity = 3.3489e+01
Generation 6280:	Best = 2.310262e+04	FES = 628100	Diversity = 3.3489e+01
Generation 6290:	Best = 2.310262e+04	FES = 629100	Diversity = 3.3332e+01
Generation 6300:	Best = 2.310262e+04	FES = 630100	Diversity = 3.3332e+01
Generation 6310:	Best = 2.310262e+04	FES = 631100	Diversity = 3.3332e+01
Generation 6320:	Best = 2.310262e+04	FES = 632100	Diversity = 3.3332e+01
Generation 6330:	Best = 2.310262e+04	FES = 633100	Diversity = 3.3166e+01
Generation 6340:	Best = 2.284509e+04	FES = 634100	Diversity = 3.2755e+01
Generation 6350:	Best = 2.283968e+04	FES = 635100	Diversity = 3.2469e+01
Generation 6360:	Best = 2.283960e+04	FES = 636100	Diversity = 3.2469e+01
Generation 6370:	Best = 2.283960e+04	FES = 637100	Diversity = 3.2469e+01
Generation 6380:	Best = 2.283960e+04	FES = 638100	Diversity = 3.2469e+01

Generation 6390:	Best = 2.283960e+04	FES = 639100	Diversity = 3.2198e+01
Generation 6400:	Best = 2.283960e+04	FES = 640100	Diversity = 3.2198e+01
Generation 6410:	Best = 2.283960e+04	FES = 641100	Diversity = 3.1798e+01
Generation 6420:	Best = 2.283960e+04	FES = 642100	Diversity = 3.1380e+01
Generation 6430:	Best = 2.283960e+04	FES = 643100	Diversity = 3.1380e+01
Generation 6440:	Best = 2.283960e+04	FES = 644100	Diversity = 3.1083e+01
Generation 6450:	Best = 2.283960e+04	FES = 645100	Diversity = 3.1083e+01
Generation 6460:	Best = 2.250072e+04	FES = 646100	Diversity = 3.0584e+01
Generation 6470:	Best = 2.248847e+04	FES = 647100	Diversity = 2.9857e+01
Generation 6480:	Best = 2.248733e+04	FES = 648100	Diversity = 2.9857e+01
Generation 6490:	Best = 2.224568e+04	FES = 649100	Diversity = 2.8026e+01
Generation 6500:	Best = 2.180684e+04	FES = 650100	Diversity = 2.6631e+01
Generation 6510:	Best = 2.175412e+04	FES = 651100	Diversity = 2.5802e+01
Generation 6520:	Best = 2.174481e+04	FES = 652100	Diversity = 2.3834e+01
Generation 6530:	Best = 2.174235e+04	FES = 653100	Diversity = 2.3423e+01
Generation 6540:	Best = 2.174185e+04	FES = 654100	Diversity = 2.3078e+01
Generation 6550:	Best = 2.174179e+04	FES = 655100	Diversity = 2.3078e+01
Generation 6560:	Best = 2.137175e+04	FES = 656100	Diversity = 4.4102e+00
Generation 6570:	Best = 2.123872e+04	FES = 657100	Diversity = 2.8925e-02
Generation 6580:	Best = 2.119974e+04	FES = 658100	Diversity = 1.4410e-02
Generation 6590:	Best = 2.116474e+04	FES = 659100	Diversity = 1.4234e-02
Generation 6600:	Best = 2.114003e+04	FES = 660100	Diversity = 8.3632e-03
Generation 6610:	Best = 2.113046e+04	FES = 661100	Diversity = 3.7216e-03
Generation 6620:	Best = 2.112433e+04	FES = 662100	Diversity = 2.8215e-03
Generation 6630:	Best = 2.112040e+04	FES = 663100	Diversity = 1.8006e-03
Generation 6640:	Best = 2.111957e+04	FES = 664100	Diversity = 8.0911e-04
Generation 6650:	Best = 2.111920e+04	FES = 665100	Diversity = 4.2779e-04
Generation 6660:	Best = 2.111896e+04	FES = 666100	Diversity = 3.4799e-04
Generation 6670:	Best = 2.111885e+04	FES = 667100	Diversity = 1.8475e-04
Generation 6680:	Best = 2.111879e+04	FES = 668100	Diversity = 1.6924e-04
Generation 6690:	Best = 2.111874e+04	FES = 669100	Diversity = 1.6438e-04
Generation 6700:	Best = 2.111870e+04	FES = 670100	Diversity = 1.3047e-04
Generation 6710:	Best = 2.111869e+04	FES = 671100	Diversity = 8.0779e-05
Generation 6720:	Best = 2.111869e+04	FES = 672100	Diversity = 3.5102e-05
Generation 6730:	Best = 2.111868e+04	FES = 673100	Diversity = 1.7239e-05
Generation 6740:	Best = 2.111868e+04	FES = 674100	Diversity = 1.2048e-05
Generation 6750:	Best = 2.111868e+04	FES = 675100	Diversity = 6.6869e-06
Generation 6760:	Best = 2.111868e+04	FES = 676100	Diversity = 3.9919e-06
Generation 6770:	Best = 2.111868e+04	FES = 677100	Diversity = 2.1072e-06
Generation 6780:	Best = 2.111868e+04	FES = 678100	Diversity = 1.3634e-06
Generation 6790:	Best = 2.111868e+04	FES = 679100	Diversity = 1.0264e-06
Generation 6800:	Best = 2.111868e+04	FES = 680100	Diversity = 8.0310e-07
Generation 6810:	Best = 2.111868e+04	FES = 681100	Diversity = 3.9411e-07
Generation 6820:	Best = 2.111868e+04	FES = 682100	Diversity = 2.5699e-07
Generation 6830:	Best = 2.111868e+04	FES = 683100	Diversity = 1.2630e-07
Generation 6840:	Best = 2.111868e+04	FES = 684100	Diversity = 8.7502e-08
Generation 6850:	Best = 2.111868e+04	FES = 685100	Diversity = 3.9103e-08
Generation 6860:	Best = 2.111868e+04	FES = 686100	Diversity = 1.7679e-08
Generation 6870:	Best = 2.111868e+04	FES = 687100	Diversity = 1.2813e-08
Generation 6880:	Best = 2.111868e+04	FES = 688100	Diversity = 1.3533e-08
Generation 6890:	Best = 2.111868e+04	FES = 689100	Diversity = 1.1425e-08
Generation 6900:	Best = 2.111868e+04	FES = 690100	Diversity = 8.7565e-09
Generation 6910:	Best = 2.111868e+04	FES = 691100	Diversity = 3.2891e-09
Generation 6920:	Best = 2.111868e+04	FES = 692100	Diversity = 4.9149e-09

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Generation 9630: Best = 2.111868e+04 FES = 963100 Diversity = 3.5658e-14
Generation 9640: Best = 2.111868e+04 FES = 964100 Diversity = 3.5658e-14
Generation 9650: Best = 2.111868e+04 FES = 965100 Diversity = 3.5658e-14
Generation 9660: Best = 2.111868e+04 FES = 966100 Diversity = 3.5658e-14
Generation 9670: Best = 2.111868e+04 FES = 967100 Diversity = 3.5658e-14
Generation 9680: Best = 2.111868e+04 FES = 968100 Diversity = 3.5658e-14
Generation 9690: Best = 2.111868e+04 FES = 969100 Diversity = 3.5658e-14
Generation 9700: Best = 2.111868e+04 FES = 970100 Diversity = 3.5658e-14
Generation 9710: Best = 2.111868e+04 FES = 971100 Diversity = 3.5658e-14
Generation 9720: Best = 2.111868e+04 FES = 972100 Diversity = 3.5658e-14
Generation 9730: Best = 2.111868e+04 FES = 973100 Diversity = 3.5658e-14
Generation 9740: Best = 2.111868e+04 FES = 974100 Diversity = 3.5658e-14
Generation 9750: Best = 2.111868e+04 FES = 975100 Diversity = 3.5658e-14
Generation 9760: Best = 2.111868e+04 FES = 976100 Diversity = 3.5658e-14
Generation 9770: Best = 2.111868e+04 FES = 977100 Diversity = 3.5658e-14
Generation 9780: Best = 2.111868e+04 FES = 978100 Diversity = 3.5658e-14
Generation 9790: Best = 2.111868e+04 FES = 979100 Diversity = 3.5658e-14
Generation 9800: Best = 2.111868e+04 FES = 980100 Diversity = 3.5658e-14
Generation 9810: Best = 2.111868e+04 FES = 981100 Diversity = 3.5658e-14
Generation 9820: Best = 2.111868e+04 FES = 982100 Diversity = 3.5658e-14
Generation 9830: Best = 2.111868e+04 FES = 983100 Diversity = 3.5658e-14
Generation 9840: Best = 2.111868e+04 FES = 984100 Diversity = 3.5658e-14
Generation 9850: Best = 2.111868e+04 FES = 985100 Diversity = 3.5658e-14
Generation 9860: Best = 2.111868e+04 FES = 986100 Diversity = 3.5658e-14
Generation 9870: Best = 2.111868e+04 FES = 987100 Diversity = 3.5658e-14
Generation 9880: Best = 2.111868e+04 FES = 988100 Diversity = 3.5658e-14
Generation 9890: Best = 2.111868e+04 FES = 989100 Diversity = 3.5658e-14
Generation 9900: Best = 2.111868e+04 FES = 990100 Diversity = 3.5658e-14
Generation 9910: Best = 2.111868e+04 FES = 991100 Diversity = 3.5658e-14
Generation 9920: Best = 2.111868e+04 FES = 992100 Diversity = 3.5658e-14
Generation 9930: Best = 2.111868e+04 FES = 993100 Diversity = 3.5658e-14
Generation 9940: Best = 2.111868e+04 FES = 994100 Diversity = 3.5658e-14
Generation 9950: Best = 2.111868e+04 FES = 995100 Diversity = 3.5658e-14
Generation 9960: Best = 2.111868e+04 FES = 996100 Diversity = 3.5658e-14
Generation 9970: Best = 2.111868e+04 FES = 997100 Diversity = 3.5658e-14
Generation 9980: Best = 2.111868e+04 FES = 998100 Diversity = 3.5658e-14
Generation 9990: Best = 2.111868e+04 FES = 999100 Diversity = 3.5658e-14

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Optimization completed!

Final Best Fitness = 2.111868e+04

Total Function Evaluations = 1000000

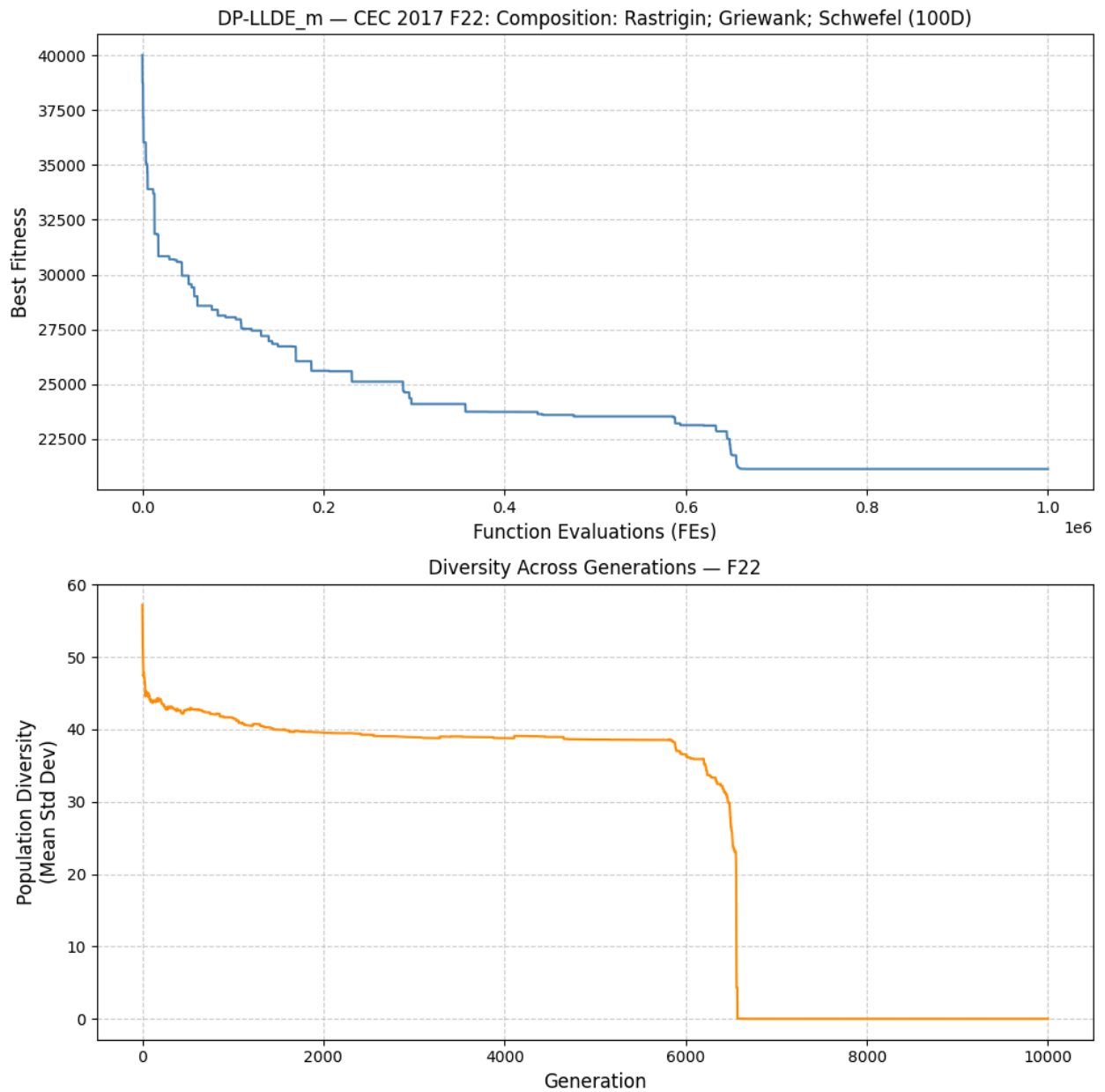
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=
Results:
  Best Fitness          : 2.111868e+04
  Global Optimum        : 2.200000e+03
  Error (|best-optimum|) : 1.891868e+04
  Best Solution (5 dim)  : [ 16.81326888 -30.49518264 -8.35085545 -27.02761882
36.32334533]
  D^(0) (init. diversity): 5.7220e+01
=====
=

```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results_2/all_results_che

ckpoint.pkl



Graph saved as convergence_F22.png

✓ Plot saved to: /content/drive/My Drive/thesis_results_2/plots/convergence_F22.png

```
In [ ]: # =====  
# F23 — Composition Function 3 (Rosenbrock; Ackley; Schwefel; Rastrigin)  
# =====  
run_and_plot('F23', make_wrapper(composition.f23), 2300.0, 'Composition: Rosen
```

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DP-LLDE_m on CEC 2017 F23 – Single Run

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Function : Composition: Rosenbrock; Ackley; Schwefel; Rastrigin (100D)
Bounds : [-100, 100]
Global optimum : 2300.0
Initial $D^*(0) = 5.7220e+01$ | Diversity threshold = 5.7220e+00
Generation 0: Best Fitness = 8.120533e+03 | Diversity = 5.7220e+01

Generation	10:	Best = 5.903640e+03	FES = 1100	Diversity = 4.2603e+01
Generation	20:	Best = 4.926974e+03	FES = 2100	Diversity = 2.6329e+01
Generation	30:	Best = 4.502615e+03	FES = 3100	Diversity = 1.8659e+01
Generation	40:	Best = 4.338565e+03	FES = 4100	Diversity = 1.5199e+01
Generation	50:	Best = 4.213346e+03	FES = 5100	Diversity = 1.2834e+01
Generation	60:	Best = 4.147058e+03	FES = 6100	Diversity = 1.1134e+01
Generation	70:	Best = 4.026827e+03	FES = 7100	Diversity = 9.3142e+00
Generation	80:	Best = 3.942028e+03	FES = 8100	Diversity = 7.5039e+00
Generation	90:	Best = 3.817857e+03	FES = 9100	Diversity = 5.5486e+00
Generation	100:	Best = 3.661481e+03	FES = 10100	Diversity = 2.9570e+00
Generation	110:	Best = 3.584457e+03	FES = 11100	Diversity = 1.7848e+00
Generation	120:	Best = 3.526976e+03	FES = 12100	Diversity = 1.1578e+00
Generation	130:	Best = 3.487499e+03	FES = 13100	Diversity = 6.0600e-01
Generation	140:	Best = 3.472138e+03	FES = 14100	Diversity = 4.0181e-01
Generation	150:	Best = 3.461489e+03	FES = 15100	Diversity = 2.9783e-01
Generation	160:	Best = 3.454313e+03	FES = 16100	Diversity = 2.1972e-01
Generation	170:	Best = 3.450366e+03	FES = 17100	Diversity = 8.6572e-02
Generation	180:	Best = 3.447278e+03	FES = 18100	Diversity = 6.4164e-02
Generation	190:	Best = 3.445498e+03	FES = 19100	Diversity = 4.9885e-02
Generation	200:	Best = 3.443525e+03	FES = 20100	Diversity = 4.4556e-02
Generation	210:	Best = 3.442431e+03	FES = 21100	Diversity = 3.1294e-02
Generation	220:	Best = 3.441423e+03	FES = 22100	Diversity = 2.3346e-02
Generation	230:	Best = 3.440687e+03	FES = 23100	Diversity = 1.8501e-02
Generation	240:	Best = 3.439992e+03	FES = 24100	Diversity = 1.7487e-02
Generation	250:	Best = 3.439429e+03	FES = 25100	Diversity = 1.3947e-02
Generation	260:	Best = 3.439029e+03	FES = 26100	Diversity = 1.1613e-02
Generation	270:	Best = 3.438572e+03	FES = 27100	Diversity = 1.0479e-02
Generation	280:	Best = 3.437885e+03	FES = 28100	Diversity = 1.3766e-02
Generation	290:	Best = 3.437475e+03	FES = 29100	Diversity = 1.2580e-02
Generation	300:	Best = 3.436979e+03	FES = 30100	Diversity = 1.2867e-02
Generation	310:	Best = 3.436541e+03	FES = 31100	Diversity = 1.2232e-02
Generation	320:	Best = 3.436145e+03	FES = 32100	Diversity = 1.2891e-02
Generation	330:	Best = 3.435845e+03	FES = 33100	Diversity = 1.0213e-02
Generation	340:	Best = 3.435670e+03	FES = 34100	Diversity = 7.0646e-03
Generation	350:	Best = 3.435542e+03	FES = 35100	Diversity = 5.2583e-03
Generation	360:	Best = 3.435397e+03	FES = 36100	Diversity = 4.2729e-03
Generation	370:	Best = 3.435306e+03	FES = 37100	Diversity = 4.1922e-03
Generation	380:	Best = 3.435229e+03	FES = 38100	Diversity = 3.8640e-03
Generation	390:	Best = 3.435134e+03	FES = 39100	Diversity = 3.0357e-03
Generation	400:	Best = 3.435089e+03	FES = 40100	Diversity = 2.1605e-03
Generation	410:	Best = 3.435047e+03	FES = 41100	Diversity = 2.1422e-03
Generation	420:	Best = 3.435008e+03	FES = 42100	Diversity = 1.9249e-03
Generation	430:	Best = 3.434970e+03	FES = 43100	Diversity = 1.7224e-03
Generation	440:	Best = 3.434942e+03	FES = 44100	Diversity = 1.5300e-03

Generation	450:	Best = 3.434905e+03	FES = 45100	Diversity = 1.6332e-03
Generation	460:	Best = 3.434864e+03	FES = 46100	Diversity = 1.7702e-03
Generation	470:	Best = 3.434832e+03	FES = 47100	Diversity = 1.7442e-03
Generation	480:	Best = 3.434805e+03	FES = 48100	Diversity = 1.2430e-03
Generation	490:	Best = 3.434778e+03	FES = 49100	Diversity = 1.1487e-03
Generation	500:	Best = 3.434758e+03	FES = 50100	Diversity = 1.1165e-03
Generation	510:	Best = 3.434752e+03	FES = 51100	Diversity = 6.0900e-04
Generation	520:	Best = 3.434747e+03	FES = 52100	Diversity = 4.7035e-04
Generation	530:	Best = 3.434744e+03	FES = 53100	Diversity = 4.2881e-04
Generation	540:	Best = 3.434739e+03	FES = 54100	Diversity = 4.2874e-04
Generation	550:	Best = 3.434736e+03	FES = 55100	Diversity = 3.6252e-04
Generation	560:	Best = 3.434731e+03	FES = 56100	Diversity = 3.9341e-04
Generation	570:	Best = 3.434729e+03	FES = 57100	Diversity = 2.7085e-04
Generation	580:	Best = 3.434727e+03	FES = 58100	Diversity = 2.4402e-04
Generation	590:	Best = 3.434725e+03	FES = 59100	Diversity = 2.4465e-04
Generation	600:	Best = 3.434723e+03	FES = 60100	Diversity = 2.2204e-04
Generation	610:	Best = 3.434722e+03	FES = 61100	Diversity = 1.6867e-04
Generation	620:	Best = 3.434721e+03	FES = 62100	Diversity = 1.5741e-04
Generation	630:	Best = 3.434720e+03	FES = 63100	Diversity = 1.5698e-04
Generation	640:	Best = 3.434720e+03	FES = 64100	Diversity = 1.5203e-04
Generation	650:	Best = 3.434719e+03	FES = 65100	Diversity = 1.1513e-04
Generation	660:	Best = 3.434719e+03	FES = 66100	Diversity = 1.0974e-04
Generation	670:	Best = 3.434718e+03	FES = 67100	Diversity = 7.8973e-05
Generation	680:	Best = 3.434718e+03	FES = 68100	Diversity = 7.3576e-05
Generation	690:	Best = 3.434718e+03	FES = 69100	Diversity = 6.4496e-05
Generation	700:	Best = 3.434718e+03	FES = 70100	Diversity = 5.7760e-05
Generation	710:	Best = 3.434717e+03	FES = 71100	Diversity = 4.0507e-05
Generation	720:	Best = 3.434717e+03	FES = 72100	Diversity = 3.3917e-05
Generation	730:	Best = 3.434717e+03	FES = 73100	Diversity = 2.2159e-05
Generation	740:	Best = 3.434717e+03	FES = 74100	Diversity = 2.2902e-05
Generation	750:	Best = 3.434717e+03	FES = 75100	Diversity = 1.8065e-05
Generation	760:	Best = 3.434717e+03	FES = 76100	Diversity = 1.3224e-05
Generation	770:	Best = 3.434717e+03	FES = 77100	Diversity = 1.3996e-05
Generation	780:	Best = 3.434717e+03	FES = 78100	Diversity = 1.1169e-05
Generation	790:	Best = 3.434717e+03	FES = 79100	Diversity = 7.2465e-06
Generation	800:	Best = 3.434717e+03	FES = 80100	Diversity = 5.7166e-06
Generation	810:	Best = 3.434717e+03	FES = 81100	Diversity = 6.1223e-06
Generation	820:	Best = 3.434717e+03	FES = 82100	Diversity = 4.3685e-06
Generation	830:	Best = 3.434717e+03	FES = 83100	Diversity = 3.4560e-06
Generation	840:	Best = 3.434717e+03	FES = 84100	Diversity = 3.2465e-06
Generation	850:	Best = 3.434717e+03	FES = 85100	Diversity = 2.7982e-06
Generation	860:	Best = 3.434717e+03	FES = 86100	Diversity = 2.1667e-06
Generation	870:	Best = 3.434717e+03	FES = 87100	Diversity = 9.3781e-07
Generation	880:	Best = 3.434717e+03	FES = 88100	Diversity = 6.3017e-07
Generation	890:	Best = 3.434717e+03	FES = 89100	Diversity = 4.8901e-07
Generation	900:	Best = 3.434717e+03	FES = 90100	Diversity = 3.9205e-07
Generation	910:	Best = 3.434717e+03	FES = 91100	Diversity = 2.3697e-07
Generation	920:	Best = 3.434717e+03	FES = 92100	Diversity = 1.7263e-07
Generation	930:	Best = 3.434717e+03	FES = 93100	Diversity = 2.1593e-07
Generation	940:	Best = 3.434717e+03	FES = 94100	Diversity = 1.1234e-07
Generation	950:	Best = 3.434717e+03	FES = 95100	Diversity = 9.3991e-08
Generation	960:	Best = 3.434717e+03	FES = 96100	Diversity = 7.4730e-08
Generation	970:	Best = 3.434717e+03	FES = 97100	Diversity = 6.6843e-08
Generation	980:	Best = 3.434717e+03	FES = 98100	Diversity = 6.1015e-08

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Generation 9630: Best = 3.434717e+03 FES = 963100 Diversity = 5.5556e-14
Generation 9640: Best = 3.434717e+03 FES = 964100 Diversity = 5.5556e-14
Generation 9650: Best = 3.434717e+03 FES = 965100 Diversity = 5.5556e-14
Generation 9660: Best = 3.434717e+03 FES = 966100 Diversity = 5.5556e-14
Generation 9670: Best = 3.434717e+03 FES = 967100 Diversity = 5.5556e-14
Generation 9680: Best = 3.434717e+03 FES = 968100 Diversity = 5.5556e-14
Generation 9690: Best = 3.434717e+03 FES = 969100 Diversity = 5.5556e-14
Generation 9700: Best = 3.434717e+03 FES = 970100 Diversity = 5.5556e-14
Generation 9710: Best = 3.434717e+03 FES = 971100 Diversity = 5.5556e-14
Generation 9720: Best = 3.434717e+03 FES = 972100 Diversity = 5.5556e-14
Generation 9730: Best = 3.434717e+03 FES = 973100 Diversity = 5.5556e-14
Generation 9740: Best = 3.434717e+03 FES = 974100 Diversity = 5.5556e-14
Generation 9750: Best = 3.434717e+03 FES = 975100 Diversity = 5.5556e-14
Generation 9760: Best = 3.434717e+03 FES = 976100 Diversity = 5.5556e-14
Generation 9770: Best = 3.434717e+03 FES = 977100 Diversity = 5.5556e-14
Generation 9780: Best = 3.434717e+03 FES = 978100 Diversity = 5.5556e-14
Generation 9790: Best = 3.434717e+03 FES = 979100 Diversity = 5.5556e-14
Generation 9800: Best = 3.434717e+03 FES = 980100 Diversity = 5.5556e-14
Generation 9810: Best = 3.434717e+03 FES = 981100 Diversity = 5.5556e-14
Generation 9820: Best = 3.434717e+03 FES = 982100 Diversity = 5.5556e-14
Generation 9830: Best = 3.434717e+03 FES = 983100 Diversity = 5.5556e-14
Generation 9840: Best = 3.434717e+03 FES = 984100 Diversity = 5.5556e-14
Generation 9850: Best = 3.434717e+03 FES = 985100 Diversity = 5.5556e-14
Generation 9860: Best = 3.434717e+03 FES = 986100 Diversity = 5.5556e-14
Generation 9870: Best = 3.434717e+03 FES = 987100 Diversity = 5.5556e-14
Generation 9880: Best = 3.434717e+03 FES = 988100 Diversity = 5.5556e-14
Generation 9890: Best = 3.434717e+03 FES = 989100 Diversity = 5.5556e-14
Generation 9900: Best = 3.434717e+03 FES = 990100 Diversity = 5.5556e-14
Generation 9910: Best = 3.434717e+03 FES = 991100 Diversity = 5.5556e-14
Generation 9920: Best = 3.434717e+03 FES = 992100 Diversity = 5.5556e-14
Generation 9930: Best = 3.434717e+03 FES = 993100 Diversity = 5.5556e-14
Generation 9940: Best = 3.434717e+03 FES = 994100 Diversity = 5.5556e-14
Generation 9950: Best = 3.434717e+03 FES = 995100 Diversity = 5.5556e-14
Generation 9960: Best = 3.434717e+03 FES = 996100 Diversity = 5.5556e-14
Generation 9970: Best = 3.434717e+03 FES = 997100 Diversity = 5.5556e-14
Generation 9980: Best = 3.434717e+03 FES = 998100 Diversity = 5.5556e-14
Generation 9990: Best = 3.434717e+03 FES = 999100 Diversity = 5.5556e-14

```

Optimization completed!

Final Best Fitness = 3.434717e+03

Total Function Evaluations = 1000000

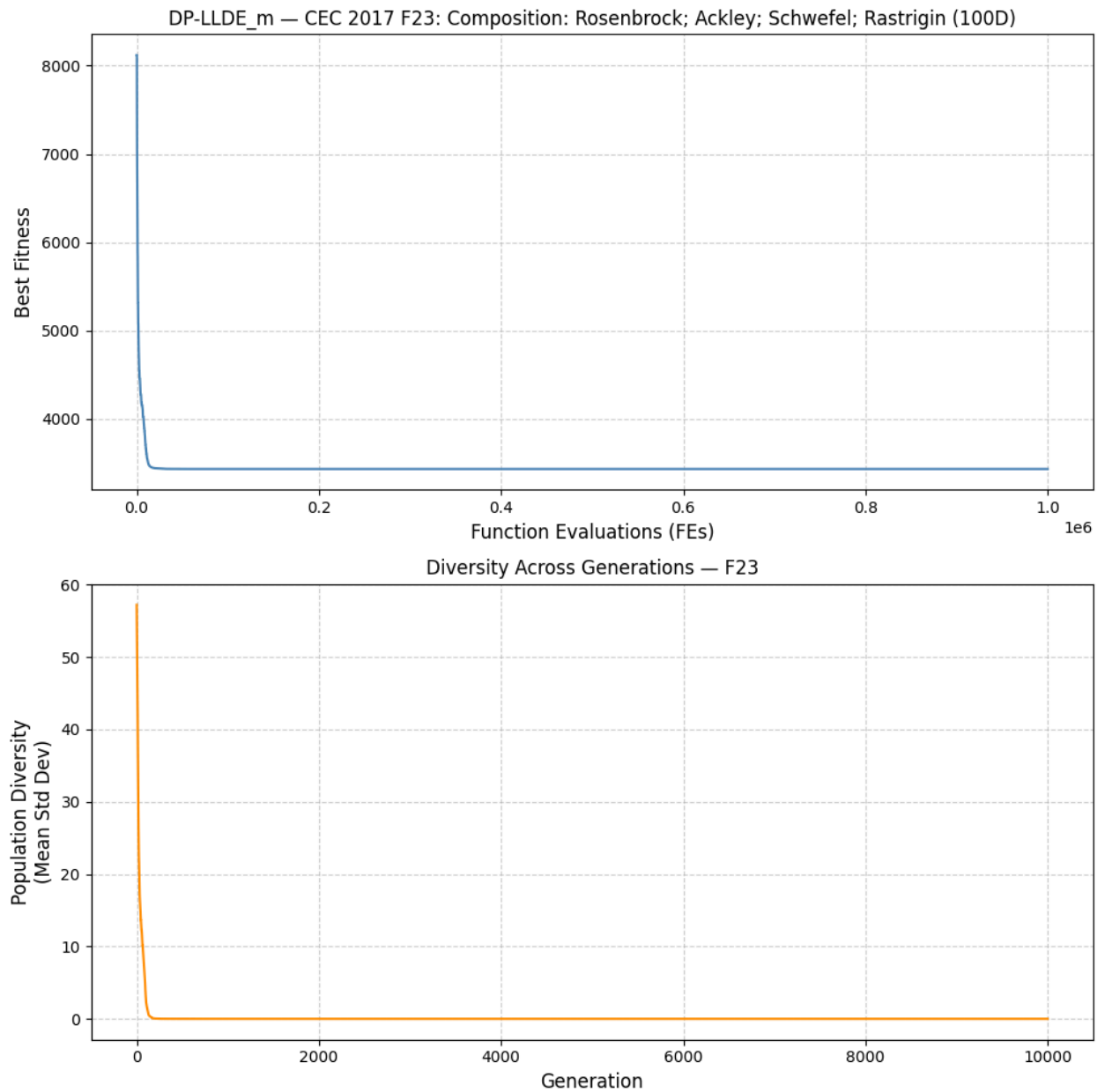
```

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Results:
  Best Fitness          : 3.434717e+03
  Global Optimum        : 2.300000e+03
  Error (|best-optimum|) : 1.134717e+03
  Best Solution (5 dim) : [-62.43438768 -78.30716071 -33.95730215 -22.42948637
-57.68953622]
  D^(0) (init. diversity): 5.7220e+01
=====
=

```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results_2/all_results_che

ckpoint.pkl



Graph saved as convergence_F23.png

✓ Plot saved to: /content/drive/My Drive/thesis_results_2/plots/convergence_F23.png

```
In [ ]: # =====  
# F24 — Composition Function 4 (Ackley; Elliptic; Griewank; Rastrigin)  
# =====  
run_and_plot('F24', make_wrapper(composition.f24), 2400.0, 'Composition: Ackle
```

=====

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DP-LLDE_m on CEC 2017 F24 – Single Run

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=

Function : Composition: Ackley; Elliptic; Griewank; Rastrigin (100D)
Bounds : [-100, 100]
Global optimum : 2400.0

Initial $D^{(0)} = 5.7220e+01$ | Diversity threshold = 5.7220e+00

Generation 0: Best Fitness = 1.419682e+04 | Diversity = 5.7220e+01

Generation 10:	Best = 9.727796e+03	FES = 1100	Diversity = 3.4232e+01
Generation 20:	Best = 7.280022e+03	FES = 2100	Diversity = 2.0698e+01
Generation 30:	Best = 6.246819e+03	FES = 3100	Diversity = 1.2937e+01
Generation 40:	Best = 5.610580e+03	FES = 4100	Diversity = 8.7203e+00
Generation 50:	Best = 5.410488e+03	FES = 5100	Diversity = 7.3219e+00
Generation 60:	Best = 5.101320e+03	FES = 6100	Diversity = 6.3835e+00
Generation 70:	Best = 5.033409e+03	FES = 7100	Diversity = 5.5949e+00
Generation 80:	Best = 4.826447e+03	FES = 8100	Diversity = 4.7324e+00
Generation 90:	Best = 4.709105e+03	FES = 9100	Diversity = 2.1409e+00
Generation 100:	Best = 4.610742e+03	FES = 10100	Diversity = 7.7452e-01
Generation 110:	Best = 4.556541e+03	FES = 11100	Diversity = 4.0032e-01
Generation 120:	Best = 4.534100e+03	FES = 12100	Diversity = 2.5272e-01
Generation 130:	Best = 4.518257e+03	FES = 13100	Diversity = 1.7332e-01
Generation 140:	Best = 4.505574e+03	FES = 14100	Diversity = 1.3738e-01
Generation 150:	Best = 4.499533e+03	FES = 15100	Diversity = 8.7622e-02
Generation 160:	Best = 4.496337e+03	FES = 16100	Diversity = 6.1593e-02
Generation 170:	Best = 4.492843e+03	FES = 17100	Diversity = 5.0450e-02
Generation 180:	Best = 4.490601e+03	FES = 18100	Diversity = 3.9613e-02
Generation 190:	Best = 4.488807e+03	FES = 19100	Diversity = 3.7037e-02
Generation 200:	Best = 4.487178e+03	FES = 20100	Diversity = 3.9851e-02
Generation 210:	Best = 4.484649e+03	FES = 21100	Diversity = 4.1763e-02
Generation 220:	Best = 4.482617e+03	FES = 22100	Diversity = 4.9977e-02
Generation 230:	Best = 4.480979e+03	FES = 23100	Diversity = 3.8398e-02
Generation 240:	Best = 4.479931e+03	FES = 24100	Diversity = 3.8832e-02
Generation 250:	Best = 4.479194e+03	FES = 25100	Diversity = 2.9385e-02
Generation 260:	Best = 4.478292e+03	FES = 26100	Diversity = 2.1702e-02
Generation 270:	Best = 4.477827e+03	FES = 27100	Diversity = 1.4890e-02
Generation 280:	Best = 4.477060e+03	FES = 28100	Diversity = 1.5603e-02
Generation 290:	Best = 4.476649e+03	FES = 29100	Diversity = 1.4006e-02
Generation 300:	Best = 4.476209e+03	FES = 30100	Diversity = 1.1165e-02
Generation 310:	Best = 4.475875e+03	FES = 31100	Diversity = 8.7803e-03
Generation 320:	Best = 4.475730e+03	FES = 32100	Diversity = 6.7230e-03
Generation 330:	Best = 4.475623e+03	FES = 33100	Diversity = 4.1872e-03
Generation 340:	Best = 4.475517e+03	FES = 34100	Diversity = 4.0346e-03
Generation 350:	Best = 4.475437e+03	FES = 35100	Diversity = 4.0405e-03
Generation 360:	Best = 4.475356e+03	FES = 36100	Diversity = 4.1878e-03
Generation 370:	Best = 4.475277e+03	FES = 37100	Diversity = 4.0614e-03
Generation 380:	Best = 4.475206e+03	FES = 38100	Diversity = 3.2363e-03
Generation 390:	Best = 4.475151e+03	FES = 39100	Diversity = 2.5439e-03
Generation 400:	Best = 4.475111e+03	FES = 40100	Diversity = 2.4395e-03
Generation 410:	Best = 4.475086e+03	FES = 41100	Diversity = 1.9397e-03
Generation 420:	Best = 4.475063e+03	FES = 42100	Diversity = 1.3732e-03
Generation 430:	Best = 4.475051e+03	FES = 43100	Diversity = 8.8664e-04
Generation 440:	Best = 4.475041e+03	FES = 44100	Diversity = 6.7954e-04

Generation	450:	Best = 4.475033e+03	FES = 45100	Diversity = 7.5446e-04
Generation	460:	Best = 4.475025e+03	FES = 46100	Diversity = 8.3284e-04
Generation	470:	Best = 4.475016e+03	FES = 47100	Diversity = 8.9303e-04
Generation	480:	Best = 4.475008e+03	FES = 48100	Diversity = 7.2931e-04
Generation	490:	Best = 4.475002e+03	FES = 49100	Diversity = 7.0962e-04
Generation	500:	Best = 4.474996e+03	FES = 50100	Diversity = 8.1937e-04
Generation	510:	Best = 4.474992e+03	FES = 51100	Diversity = 6.0575e-04
Generation	520:	Best = 4.474990e+03	FES = 52100	Diversity = 2.6205e-04
Generation	530:	Best = 4.474988e+03	FES = 53100	Diversity = 2.6741e-04
Generation	540:	Best = 4.474987e+03	FES = 54100	Diversity = 2.0757e-04
Generation	550:	Best = 4.474987e+03	FES = 55100	Diversity = 1.5862e-04
Generation	560:	Best = 4.474986e+03	FES = 56100	Diversity = 1.4357e-04
Generation	570:	Best = 4.474986e+03	FES = 57100	Diversity = 9.5497e-05
Generation	580:	Best = 4.474986e+03	FES = 58100	Diversity = 6.2654e-05
Generation	590:	Best = 4.474986e+03	FES = 59100	Diversity = 4.9261e-05
Generation	600:	Best = 4.474986e+03	FES = 60100	Diversity = 4.6138e-05
Generation	610:	Best = 4.474986e+03	FES = 61100	Diversity = 3.2967e-05
Generation	620:	Best = 4.474986e+03	FES = 62100	Diversity = 3.3724e-05
Generation	630:	Best = 4.474986e+03	FES = 63100	Diversity = 2.6067e-05
Generation	640:	Best = 4.474986e+03	FES = 64100	Diversity = 1.7322e-05
Generation	650:	Best = 4.474986e+03	FES = 65100	Diversity = 1.7776e-05
Generation	660:	Best = 4.474986e+03	FES = 66100	Diversity = 1.4888e-05
Generation	670:	Best = 4.474986e+03	FES = 67100	Diversity = 1.4287e-05
Generation	680:	Best = 4.474986e+03	FES = 68100	Diversity = 8.5482e-06
Generation	690:	Best = 4.474986e+03	FES = 69100	Diversity = 8.0418e-06
Generation	700:	Best = 4.474986e+03	FES = 70100	Diversity = 3.4356e-06
Generation	710:	Best = 4.474986e+03	FES = 71100	Diversity = 4.1822e-06
Generation	720:	Best = 4.474986e+03	FES = 72100	Diversity = 2.9954e-06
Generation	730:	Best = 4.474986e+03	FES = 73100	Diversity = 2.5780e-06
Generation	740:	Best = 4.474986e+03	FES = 74100	Diversity = 1.6414e-06
Generation	750:	Best = 4.474986e+03	FES = 75100	Diversity = 9.4349e-07
Generation	760:	Best = 4.474986e+03	FES = 76100	Diversity = 1.0467e-06
Generation	770:	Best = 4.474986e+03	FES = 77100	Diversity = 6.6386e-07
Generation	780:	Best = 4.474986e+03	FES = 78100	Diversity = 4.0222e-07
Generation	790:	Best = 4.474986e+03	FES = 79100	Diversity = 3.6269e-07
Generation	800:	Best = 4.474986e+03	FES = 80100	Diversity = 2.0235e-07
Generation	810:	Best = 4.474986e+03	FES = 81100	Diversity = 1.6622e-07
Generation	820:	Best = 4.474986e+03	FES = 82100	Diversity = 1.3381e-07
Generation	830:	Best = 4.474986e+03	FES = 83100	Diversity = 7.7217e-08
Generation	840:	Best = 4.474986e+03	FES = 84100	Diversity = 5.8416e-08
Generation	850:	Best = 4.474986e+03	FES = 85100	Diversity = 4.0381e-08
Generation	860:	Best = 4.474986e+03	FES = 86100	Diversity = 2.3271e-08
Generation	870:	Best = 4.474986e+03	FES = 87100	Diversity = 1.7199e-08
Generation	880:	Best = 4.474986e+03	FES = 88100	Diversity = 1.0912e-08
Generation	890:	Best = 4.474986e+03	FES = 89100	Diversity = 6.3131e-09
Generation	900:	Best = 4.474986e+03	FES = 90100	Diversity = 2.1518e-09
Generation	910:	Best = 4.474986e+03	FES = 91100	Diversity = 7.5790e-11
Generation	920:	Best = 4.474986e+03	FES = 92100	Diversity = 2.8713e-11
Generation	930:	Best = 4.474986e+03	FES = 93100	Diversity = 1.6443e-11
Generation	940:	Best = 4.474986e+03	FES = 94100	Diversity = 5.1500e-12
Generation	950:	Best = 4.474986e+03	FES = 95100	Diversity = 5.1084e-14
Generation	960:	Best = 4.474986e+03	FES = 96100	Diversity = 5.1084e-14
Generation	970:	Best = 4.474986e+03	FES = 97100	Diversity = 5.1084e-14
Generation	980:	Best = 4.474986e+03	FES = 98100	Diversity = 5.1084e-14

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Generation 9630: Best = 4.474986e+03 FES = 963100 Diversity = 5.1084e-14
Generation 9640: Best = 4.474986e+03 FES = 964100 Diversity = 5.1084e-14
Generation 9650: Best = 4.474986e+03 FES = 965100 Diversity = 5.1084e-14
Generation 9660: Best = 4.474986e+03 FES = 966100 Diversity = 5.1084e-14
Generation 9670: Best = 4.474986e+03 FES = 967100 Diversity = 5.1084e-14
Generation 9680: Best = 4.474986e+03 FES = 968100 Diversity = 5.1084e-14
Generation 9690: Best = 4.474986e+03 FES = 969100 Diversity = 5.1084e-14
Generation 9700: Best = 4.474986e+03 FES = 970100 Diversity = 5.1084e-14
Generation 9710: Best = 4.474986e+03 FES = 971100 Diversity = 5.1084e-14
Generation 9720: Best = 4.474986e+03 FES = 972100 Diversity = 5.1084e-14
Generation 9730: Best = 4.474986e+03 FES = 973100 Diversity = 5.1084e-14
Generation 9740: Best = 4.474986e+03 FES = 974100 Diversity = 5.1084e-14
Generation 9750: Best = 4.474986e+03 FES = 975100 Diversity = 5.1084e-14
Generation 9760: Best = 4.474986e+03 FES = 976100 Diversity = 5.1084e-14
Generation 9770: Best = 4.474986e+03 FES = 977100 Diversity = 5.1084e-14
Generation 9780: Best = 4.474986e+03 FES = 978100 Diversity = 5.1084e-14
Generation 9790: Best = 4.474986e+03 FES = 979100 Diversity = 5.1084e-14
Generation 9800: Best = 4.474986e+03 FES = 980100 Diversity = 5.1084e-14
Generation 9810: Best = 4.474986e+03 FES = 981100 Diversity = 5.1084e-14
Generation 9820: Best = 4.474986e+03 FES = 982100 Diversity = 5.1084e-14
Generation 9830: Best = 4.474986e+03 FES = 983100 Diversity = 5.1084e-14
Generation 9840: Best = 4.474986e+03 FES = 984100 Diversity = 5.1084e-14
Generation 9850: Best = 4.474986e+03 FES = 985100 Diversity = 5.1084e-14
Generation 9860: Best = 4.474986e+03 FES = 986100 Diversity = 5.1084e-14
Generation 9870: Best = 4.474986e+03 FES = 987100 Diversity = 5.1084e-14
Generation 9880: Best = 4.474986e+03 FES = 988100 Diversity = 5.1084e-14
Generation 9890: Best = 4.474986e+03 FES = 989100 Diversity = 5.1084e-14
Generation 9900: Best = 4.474986e+03 FES = 990100 Diversity = 5.1084e-14
Generation 9910: Best = 4.474986e+03 FES = 991100 Diversity = 5.1084e-14
Generation 9920: Best = 4.474986e+03 FES = 992100 Diversity = 5.1084e-14
Generation 9930: Best = 4.474986e+03 FES = 993100 Diversity = 5.1084e-14
Generation 9940: Best = 4.474986e+03 FES = 994100 Diversity = 5.1084e-14
Generation 9950: Best = 4.474986e+03 FES = 995100 Diversity = 5.1084e-14
Generation 9960: Best = 4.474986e+03 FES = 996100 Diversity = 5.1084e-14
Generation 9970: Best = 4.474986e+03 FES = 997100 Diversity = 5.1084e-14
Generation 9980: Best = 4.474986e+03 FES = 998100 Diversity = 5.1084e-14
Generation 9990: Best = 4.474986e+03 FES = 999100 Diversity = 5.1084e-14

```

Optimization completed!

Final Best Fitness = 4.474986e+03

Total Function Evaluations = 1000000

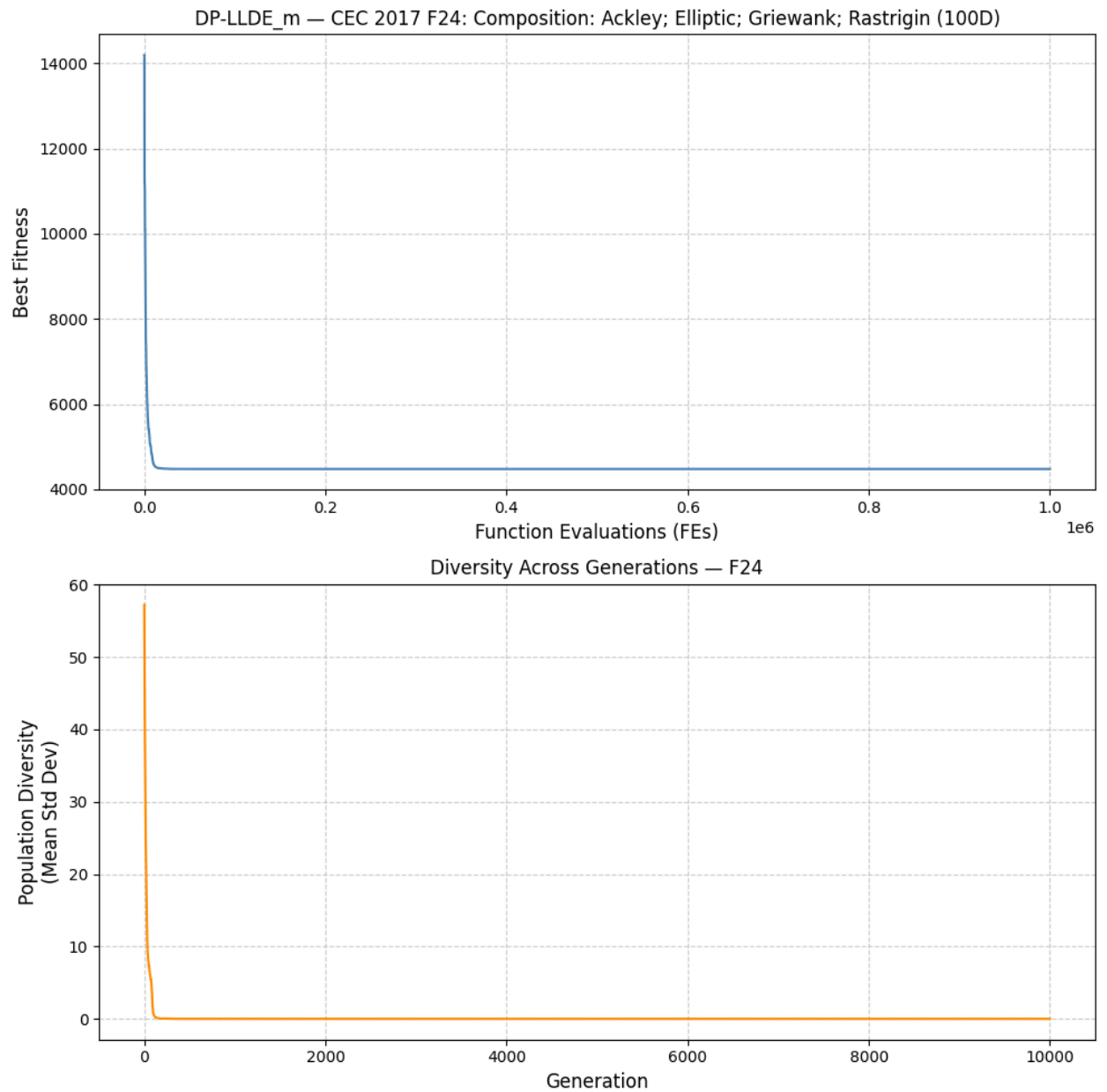
```

=====
=
Results:
  Best Fitness          : 4.474986e+03
  Global Optimum        : 2.400000e+03
  Error (|best-optimum|) : 2.074986e+03
  Best Solution (5 dim)  : [ 45.99409837  88.34493579 -74.44367869 -38.08682215
43.57583899]
  D^(0) (init. diversity): 5.7220e+01
=====
=

```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results_2/all_results_che

ckpoint.pkl



Graph saved as convergence_F24.png

✓ Plot saved to: /content/drive/My Drive/thesis_results_2/plots/convergence_F24.png

```
In [ ]: # =====  
# F25 — Composition Function 5 (Rastrigin; HappyCat; Ackley; Discus; Rosenbrock)  
# =====  
run_and_plot('F25', make_wrapper(composition.f25), 2500.0, 'Composition: Rastr
```

```

=====
=
DP-LLDE_m on CEC 2017 F25 – Single Run
=====
=
Function      : Composition: Rastrigin; HappyCat; Ackley; Discus; Rosenbrock
(100D)
Bounds        : [-100, 100]
Global optimum : 2500.0
Initial  $D^*(0) = 5.7220e+01$  | Diversity threshold = 5.7220e+00
Generation 0: Best Fitness = 1.233610e+05 | Diversity = 5.7220e+01
Generation 10: Best = 1.971690e+04 FES = 1100 Diversity = 2.0446e+01
Generation 20: Best = 1.351338e+04 FES = 2100 Diversity = 7.7528e+00
Generation 30: Best = 1.192663e+04 FES = 3100 Diversity = 3.5885e+00
Generation 40: Best = 1.102952e+04 FES = 4100 Diversity = 2.0299e+00
Generation 50: Best = 1.057325e+04 FES = 5100 Diversity = 1.2701e+00
Generation 60: Best = 1.024994e+04 FES = 6100 Diversity = 9.7354e-01
Generation 70: Best = 1.005576e+04 FES = 7100 Diversity = 7.6500e-01
Generation 80: Best = 9.830648e+03 FES = 8100 Diversity = 7.7234e-01
Generation 90: Best = 9.595509e+03 FES = 9100 Diversity = 6.3877e-01
Generation 100: Best = 9.479383e+03 FES = 10100 Diversity = 5.3038e-01
Generation 110: Best = 9.319921e+03 FES = 11100 Diversity = 5.9865e-01
Generation 120: Best = 9.180953e+03 FES = 12100 Diversity = 5.7997e-01
Generation 130: Best = 9.049579e+03 FES = 13100 Diversity = 5.9517e-01
Generation 140: Best = 8.900765e+03 FES = 14100 Diversity = 6.3867e-01
Generation 150: Best = 8.792582e+03 FES = 15100 Diversity = 5.1586e-01
Generation 160: Best = 8.669198e+03 FES = 16100 Diversity = 5.4906e-01
Generation 170: Best = 8.591125e+03 FES = 17100 Diversity = 4.8698e-01
Generation 180: Best = 8.506796e+03 FES = 18100 Diversity = 4.5903e-01
Generation 190: Best = 8.433177e+03 FES = 19100 Diversity = 3.8563e-01
Generation 200: Best = 8.390704e+03 FES = 20100 Diversity = 4.1696e-01
Generation 210: Best = 8.322816e+03 FES = 21100 Diversity = 3.8432e-01
Generation 220: Best = 8.277717e+03 FES = 22100 Diversity = 3.4919e-01
Generation 230: Best = 8.232191e+03 FES = 23100 Diversity = 3.1874e-01
Generation 240: Best = 8.196448e+03 FES = 24100 Diversity = 3.1339e-01
Generation 250: Best = 8.166721e+03 FES = 25100 Diversity = 2.5007e-01
Generation 260: Best = 8.145294e+03 FES = 26100 Diversity = 2.1158e-01
Generation 270: Best = 8.125730e+03 FES = 27100 Diversity = 1.7861e-01
Generation 280: Best = 8.101309e+03 FES = 28100 Diversity = 1.8676e-01
Generation 290: Best = 8.087387e+03 FES = 29100 Diversity = 1.5179e-01
Generation 300: Best = 8.076911e+03 FES = 30100 Diversity = 1.0947e-01
Generation 310: Best = 8.067649e+03 FES = 31100 Diversity = 8.6349e-02
Generation 320: Best = 8.061205e+03 FES = 32100 Diversity = 9.3220e-02
Generation 330: Best = 8.053877e+03 FES = 33100 Diversity = 8.9636e-02
Generation 340: Best = 8.048094e+03 FES = 34100 Diversity = 7.4261e-02
Generation 350: Best = 8.044792e+03 FES = 35100 Diversity = 5.1749e-02
Generation 360: Best = 8.041248e+03 FES = 36100 Diversity = 4.9512e-02
Generation 370: Best = 8.039294e+03 FES = 37100 Diversity = 3.8514e-02
Generation 380: Best = 8.037531e+03 FES = 38100 Diversity = 2.6808e-02
Generation 390: Best = 8.036269e+03 FES = 39100 Diversity = 3.2197e-02
Generation 400: Best = 8.034434e+03 FES = 40100 Diversity = 3.3084e-02
Generation 410: Best = 8.032972e+03 FES = 41100 Diversity = 2.7041e-02
Generation 420: Best = 8.032273e+03 FES = 42100 Diversity = 1.8082e-02
Generation 430: Best = 8.031781e+03 FES = 43100 Diversity = 1.3108e-02

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Generation	440:	Best = 8.031455e+03	FES = 44100	Diversity = 1.0969e-02
Generation	450:	Best = 8.031223e+03	FES = 45100	Diversity = 4.9770e-03
Generation	460:	Best = 8.031065e+03	FES = 46100	Diversity = 3.7126e-03
Generation	470:	Best = 8.030937e+03	FES = 47100	Diversity = 2.8684e-03
Generation	480:	Best = 8.030856e+03	FES = 48100	Diversity = 2.5108e-03
Generation	490:	Best = 8.030811e+03	FES = 49100	Diversity = 1.6259e-03
Generation	500:	Best = 8.030789e+03	FES = 50100	Diversity = 1.1060e-03
Generation	510:	Best = 8.030773e+03	FES = 51100	Diversity = 9.1263e-04
Generation	520:	Best = 8.030766e+03	FES = 52100	Diversity = 6.4952e-04
Generation	530:	Best = 8.030760e+03	FES = 53100	Diversity = 4.0840e-04
Generation	540:	Best = 8.030755e+03	FES = 54100	Diversity = 4.3844e-04
Generation	550:	Best = 8.030751e+03	FES = 55100	Diversity = 3.3803e-04
Generation	560:	Best = 8.030747e+03	FES = 56100	Diversity = 3.2378e-04
Generation	570:	Best = 8.030745e+03	FES = 57100	Diversity = 2.1330e-04
Generation	580:	Best = 8.030743e+03	FES = 58100	Diversity = 2.2887e-04
Generation	590:	Best = 8.030741e+03	FES = 59100	Diversity = 2.0640e-04
Generation	600:	Best = 8.030740e+03	FES = 60100	Diversity = 1.9810e-04
Generation	610:	Best = 8.030739e+03	FES = 61100	Diversity = 2.4654e-04
Generation	620:	Best = 8.030737e+03	FES = 62100	Diversity = 1.6872e-04
Generation	630:	Best = 8.030737e+03	FES = 63100	Diversity = 1.6389e-04
Generation	640:	Best = 8.030736e+03	FES = 64100	Diversity = 7.3124e-05
Generation	650:	Best = 8.030736e+03	FES = 65100	Diversity = 6.9541e-05
Generation	660:	Best = 8.030736e+03	FES = 66100	Diversity = 5.6159e-05
Generation	670:	Best = 8.030736e+03	FES = 67100	Diversity = 4.7833e-05
Generation	680:	Best = 8.030736e+03	FES = 68100	Diversity = 3.3418e-05
Generation	690:	Best = 8.030736e+03	FES = 69100	Diversity = 2.2876e-05
Generation	700:	Best = 8.030736e+03	FES = 70100	Diversity = 2.2935e-05
Generation	710:	Best = 8.030736e+03	FES = 71100	Diversity = 1.7192e-05
Generation	720:	Best = 8.030736e+03	FES = 72100	Diversity = 1.6915e-05
Generation	730:	Best = 8.030736e+03	FES = 73100	Diversity = 1.2577e-05
Generation	740:	Best = 8.030736e+03	FES = 74100	Diversity = 1.1788e-05
Generation	750:	Best = 8.030736e+03	FES = 75100	Diversity = 1.2070e-05
Generation	760:	Best = 8.030736e+03	FES = 76100	Diversity = 8.7657e-06
Generation	770:	Best = 8.030736e+03	FES = 77100	Diversity = 1.0221e-05
Generation	780:	Best = 8.030736e+03	FES = 78100	Diversity = 4.6625e-06
Generation	790:	Best = 8.030736e+03	FES = 79100	Diversity = 2.5626e-06
Generation	800:	Best = 8.030736e+03	FES = 80100	Diversity = 2.0220e-06
Generation	810:	Best = 8.030736e+03	FES = 81100	Diversity = 9.8540e-07
Generation	820:	Best = 8.030736e+03	FES = 82100	Diversity = 9.3842e-07
Generation	830:	Best = 8.030736e+03	FES = 83100	Diversity = 7.5985e-07
Generation	840:	Best = 8.030736e+03	FES = 84100	Diversity = 9.0761e-07
Generation	850:	Best = 8.030736e+03	FES = 85100	Diversity = 6.9630e-07
Generation	860:	Best = 8.030736e+03	FES = 86100	Diversity = 3.6378e-07
Generation	870:	Best = 8.030736e+03	FES = 87100	Diversity = 3.3902e-07
Generation	880:	Best = 8.030736e+03	FES = 88100	Diversity = 1.6881e-07
Generation	890:	Best = 8.030736e+03	FES = 89100	Diversity = 9.6948e-08
Generation	900:	Best = 8.030736e+03	FES = 90100	Diversity = 4.8234e-08
Generation	910:	Best = 8.030736e+03	FES = 91100	Diversity = 3.7728e-08
Generation	920:	Best = 8.030736e+03	FES = 92100	Diversity = 1.9848e-08
Generation	930:	Best = 8.030736e+03	FES = 93100	Diversity = 1.8912e-08
Generation	940:	Best = 8.030736e+03	FES = 94100	Diversity = 1.0071e-08
Generation	950:	Best = 8.030736e+03	FES = 95100	Diversity = 1.0291e-08
Generation	960:	Best = 8.030736e+03	FES = 96100	Diversity = 1.4570e-08
Generation	970:	Best = 8.030736e+03	FES = 97100	Diversity = 1.6033e-08

Generation 980:	Best = 8.030736e+03	FES = 98100	Diversity = 1.4035e-08
Generation 990:	Best = 8.030736e+03	FES = 99100	Diversity = 1.3005e-08
Generation 1000:	Best = 8.030736e+03	FES = 100100	Diversity = 1.5708e-08
Generation 1010:	Best = 8.030736e+03	FES = 101100	Diversity = 1.5125e-08
Generation 1020:	Best = 8.030736e+03	FES = 102100	Diversity = 1.1535e-08
Generation 1030:	Best = 8.030736e+03	FES = 103100	Diversity = 1.0597e-08
Generation 1040:	Best = 8.030736e+03	FES = 104100	Diversity = 1.0030e-08
Generation 1050:	Best = 8.030736e+03	FES = 105100	Diversity = 1.0065e-08
Generation 1060:	Best = 8.030736e+03	FES = 106100	Diversity = 8.3824e-09
Generation 1070:	Best = 8.030736e+03	FES = 107100	Diversity = 9.2263e-09
Generation 1080:	Best = 8.030736e+03	FES = 108100	Diversity = 9.0019e-09
Generation 1090:	Best = 8.030736e+03	FES = 109100	Diversity = 7.9476e-09
Generation 1100:	Best = 8.030736e+03	FES = 110100	Diversity = 8.1552e-09
Generation 1110:	Best = 8.030736e+03	FES = 111100	Diversity = 7.4652e-09
Generation 1120:	Best = 8.030736e+03	FES = 112100	Diversity = 8.7489e-09
Generation 1130:	Best = 8.030736e+03	FES = 113100	Diversity = 9.2805e-09
Generation 1140:	Best = 8.030736e+03	FES = 114100	Diversity = 1.2941e-08
Generation 1150:	Best = 8.030736e+03	FES = 115100	Diversity = 1.1686e-08
Generation 1160:	Best = 8.030736e+03	FES = 116100	Diversity = 1.2111e-08
Generation 1170:	Best = 8.030736e+03	FES = 117100	Diversity = 1.1540e-08
Generation 1180:	Best = 8.030736e+03	FES = 118100	Diversity = 1.1619e-08
Generation 1190:	Best = 8.030736e+03	FES = 119100	Diversity = 1.0973e-08
Generation 1200:	Best = 8.030736e+03	FES = 120100	Diversity = 8.7125e-09
Generation 1210:	Best = 8.030736e+03	FES = 121100	Diversity = 7.7854e-09
Generation 1220:	Best = 8.030736e+03	FES = 122100	Diversity = 8.0516e-09
Generation 1230:	Best = 8.030736e+03	FES = 123100	Diversity = 4.7791e-09
Generation 1240:	Best = 8.030736e+03	FES = 124100	Diversity = 2.8769e-09
Generation 1250:	Best = 8.030736e+03	FES = 125100	Diversity = 7.3645e-10
Generation 1260:	Best = 8.030736e+03	FES = 126100	Diversity = 2.0210e-14
Generation 1270:	Best = 8.030736e+03	FES = 127100	Diversity = 2.0210e-14
Generation 1280:	Best = 8.030736e+03	FES = 128100	Diversity = 2.0210e-14
Generation 1290:	Best = 8.030736e+03	FES = 129100	Diversity = 2.0210e-14
Generation 1300:	Best = 8.030736e+03	FES = 130100	Diversity = 2.0210e-14
Generation 1310:	Best = 8.030736e+03	FES = 131100	Diversity = 2.0210e-14
Generation 1320:	Best = 8.030736e+03	FES = 132100	Diversity = 2.0210e-14
Generation 1330:	Best = 8.030736e+03	FES = 133100	Diversity = 2.0210e-14
Generation 1340:	Best = 8.030736e+03	FES = 134100	Diversity = 2.0210e-14
Generation 1350:	Best = 8.030736e+03	FES = 135100	Diversity = 2.0210e-14
Generation 1360:	Best = 8.030736e+03	FES = 136100	Diversity = 2.0210e-14
Generation 1370:	Best = 8.030736e+03	FES = 137100	Diversity = 2.0210e-14
Generation 1380:	Best = 8.030736e+03	FES = 138100	Diversity = 2.0210e-14
Generation 1390:	Best = 8.030736e+03	FES = 139100	Diversity = 2.0210e-14
Generation 1400:	Best = 8.030736e+03	FES = 140100	Diversity = 2.0210e-14
Generation 1410:	Best = 8.030736e+03	FES = 141100	Diversity = 2.0210e-14
Generation 1420:	Best = 8.030736e+03	FES = 142100	Diversity = 2.0210e-14
Generation 1430:	Best = 8.030736e+03	FES = 143100	Diversity = 2.0210e-14
Generation 1440:	Best = 8.030736e+03	FES = 144100	Diversity = 2.0210e-14
Generation 1450:	Best = 8.030736e+03	FES = 145100	Diversity = 2.0210e-14
Generation 1460:	Best = 8.030736e+03	FES = 146100	Diversity = 2.0210e-14
Generation 1470:	Best = 8.030736e+03	FES = 147100	Diversity = 2.0210e-14
Generation 1480:	Best = 8.030736e+03	FES = 148100	Diversity = 2.0210e-14
Generation 1490:	Best = 8.030736e+03	FES = 149100	Diversity = 2.0210e-14
Generation 1500:	Best = 8.030736e+03	FES = 150100	Diversity = 2.0210e-14
Generation 1510:	Best = 8.030736e+03	FES = 151100	Diversity = 2.0210e-14

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Generation 9620:	Best = 8.030736e+03	FES = 962100	Diversity = 2.0210e-14
Generation 9630:	Best = 8.030736e+03	FES = 963100	Diversity = 2.0210e-14
Generation 9640:	Best = 8.030736e+03	FES = 964100	Diversity = 2.0210e-14
Generation 9650:	Best = 8.030736e+03	FES = 965100	Diversity = 2.0210e-14
Generation 9660:	Best = 8.030736e+03	FES = 966100	Diversity = 2.0210e-14
Generation 9670:	Best = 8.030736e+03	FES = 967100	Diversity = 2.0210e-14
Generation 9680:	Best = 8.030736e+03	FES = 968100	Diversity = 2.0210e-14
Generation 9690:	Best = 8.030736e+03	FES = 969100	Diversity = 2.0210e-14
Generation 9700:	Best = 8.030736e+03	FES = 970100	Diversity = 2.0210e-14
Generation 9710:	Best = 8.030736e+03	FES = 971100	Diversity = 2.0210e-14
Generation 9720:	Best = 8.030736e+03	FES = 972100	Diversity = 2.0210e-14
Generation 9730:	Best = 8.030736e+03	FES = 973100	Diversity = 2.0210e-14
Generation 9740:	Best = 8.030736e+03	FES = 974100	Diversity = 2.0210e-14
Generation 9750:	Best = 8.030736e+03	FES = 975100	Diversity = 2.0210e-14
Generation 9760:	Best = 8.030736e+03	FES = 976100	Diversity = 2.0210e-14
Generation 9770:	Best = 8.030736e+03	FES = 977100	Diversity = 2.0210e-14
Generation 9780:	Best = 8.030736e+03	FES = 978100	Diversity = 2.0210e-14
Generation 9790:	Best = 8.030736e+03	FES = 979100	Diversity = 2.0210e-14
Generation 9800:	Best = 8.030736e+03	FES = 980100	Diversity = 2.0210e-14
Generation 9810:	Best = 8.030736e+03	FES = 981100	Diversity = 2.0210e-14
Generation 9820:	Best = 8.030736e+03	FES = 982100	Diversity = 2.0210e-14
Generation 9830:	Best = 8.030736e+03	FES = 983100	Diversity = 2.0210e-14
Generation 9840:	Best = 8.030736e+03	FES = 984100	Diversity = 2.0210e-14
Generation 9850:	Best = 8.030736e+03	FES = 985100	Diversity = 2.0210e-14
Generation 9860:	Best = 8.030736e+03	FES = 986100	Diversity = 2.0210e-14
Generation 9870:	Best = 8.030736e+03	FES = 987100	Diversity = 2.0210e-14
Generation 9880:	Best = 8.030736e+03	FES = 988100	Diversity = 2.0210e-14
Generation 9890:	Best = 8.030736e+03	FES = 989100	Diversity = 2.0210e-14
Generation 9900:	Best = 8.030736e+03	FES = 990100	Diversity = 2.0210e-14
Generation 9910:	Best = 8.030736e+03	FES = 991100	Diversity = 2.0210e-14
Generation 9920:	Best = 8.030736e+03	FES = 992100	Diversity = 2.0210e-14
Generation 9930:	Best = 8.030736e+03	FES = 993100	Diversity = 2.0210e-14
Generation 9940:	Best = 8.030736e+03	FES = 994100	Diversity = 2.0210e-14
Generation 9950:	Best = 8.030736e+03	FES = 995100	Diversity = 2.0210e-14
Generation 9960:	Best = 8.030736e+03	FES = 996100	Diversity = 2.0210e-14
Generation 9970:	Best = 8.030736e+03	FES = 997100	Diversity = 2.0210e-14
Generation 9980:	Best = 8.030736e+03	FES = 998100	Diversity = 2.0210e-14
Generation 9990:	Best = 8.030736e+03	FES = 999100	Diversity = 2.0210e-14

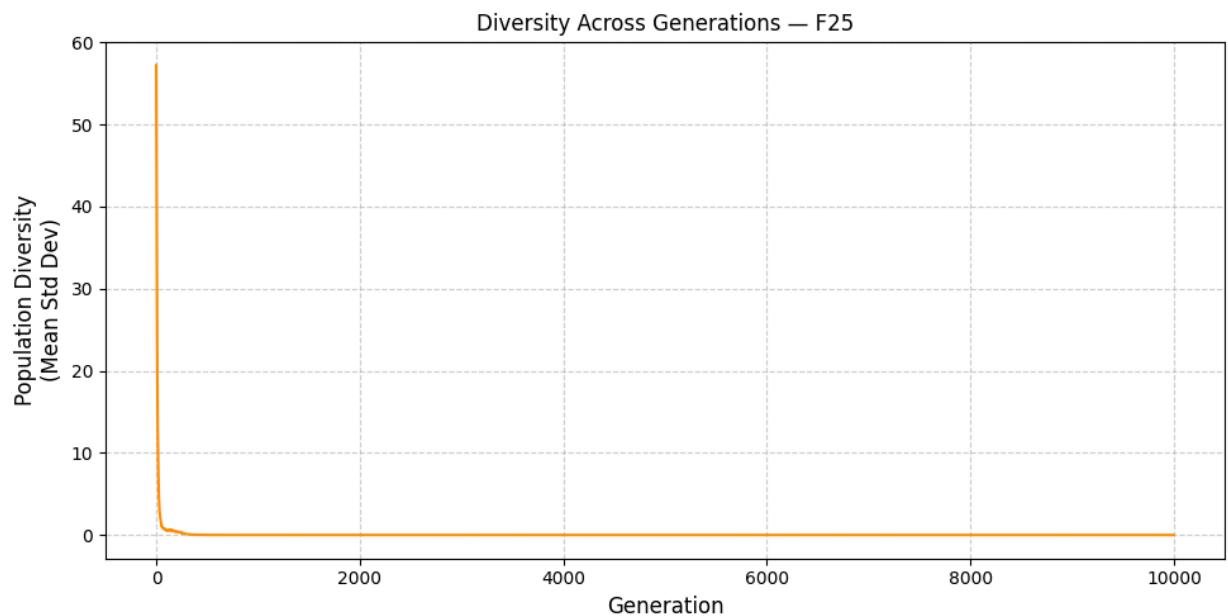
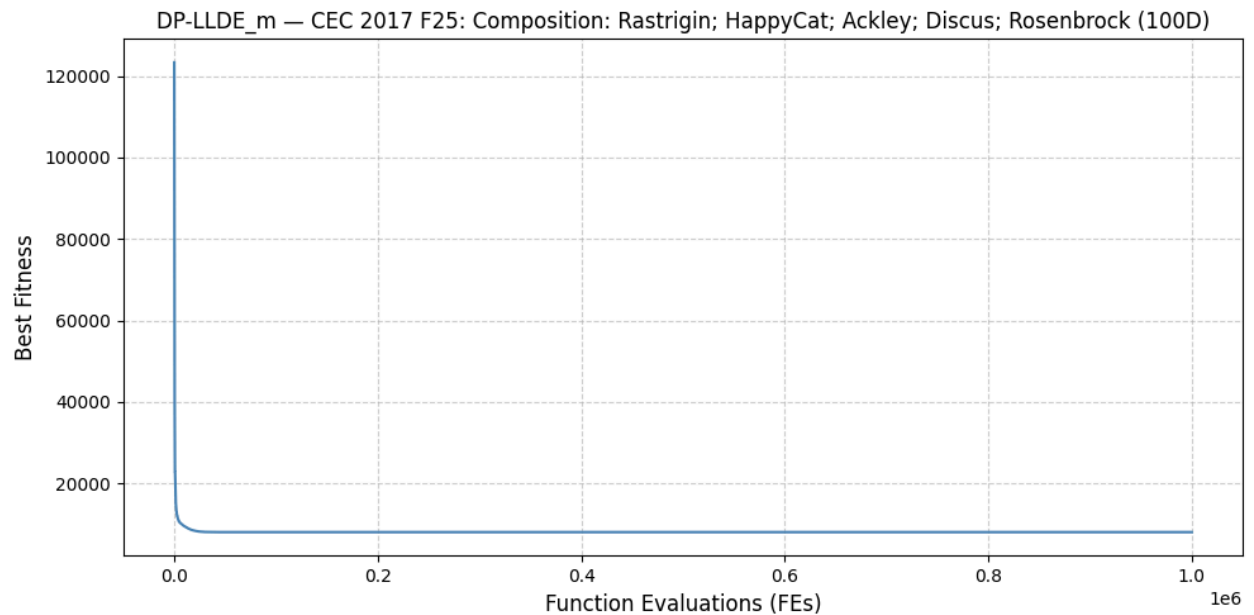
Optimization completed!

Final Best Fitness = 8.030736e+03

Total Function Evaluations = 1000000

```
=====
=
Results:
  Best Fitness           : 8.030736e+03
  Global Optimum         : 2.500000e+03
  Error (|best-optimum|) : 5.530736e+03
  Best Solution (5 dim)  : [-33.9694604  -25.61051485  15.85705943  -14.18435105
-29.4461453 ]
  D^(0) (init. diversity): 5.7220e+01
=====
=
```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results_2/all_results_checkpoint.pkl



Graph saved as convergence_F25.png

✓ Plot saved to: /content/drive/My Drive/thesis_results_2/plots/convergence_F25.png

```
In [ ]: # =====  
# F26 – Composition Function 6 (Schaffer; Schwefel; Griewank; Rosenbrock; Rast  
# =====  
run_and_plot('F26', make_wrapper(composition.f26), 2600.0, 'Composition: Schaf
```

```

=====
=
DP-LLDE_m on CEC 2017 F26 – Single Run
=====
=
Function      : Composition: Schaffer; Schwefel; Griewank; Rosenbrock; Rastrig
in (100D)
Bounds       : [-100, 100]
Global optimum : 2600.0
Initial  $D^*(0) = 5.7220e+01$  | Diversity threshold = 5.7220e+00
Generation 0: Best Fitness = 1.023547e+05 | Diversity = 5.7220e+01
Generation 10: Best = 5.068474e+04 FES = 1100 Diversity = 2.4471e+01
Generation 20: Best = 3.836926e+04 FES = 2100 Diversity = 1.1930e+01
Generation 30: Best = 3.367608e+04 FES = 3100 Diversity = 6.4963e+00
Generation 40: Best = 3.204856e+04 FES = 4100 Diversity = 4.1409e+00
Generation 50: Best = 3.050074e+04 FES = 5100 Diversity = 2.5422e+00
Generation 60: Best = 2.980349e+04 FES = 6100 Diversity = 1.5585e+00
Generation 70: Best = 2.899832e+04 FES = 7100 Diversity = 9.1850e-01
Generation 80: Best = 2.851827e+04 FES = 8100 Diversity = 4.7018e-01
Generation 90: Best = 2.830952e+04 FES = 9100 Diversity = 3.0744e-01
Generation 100: Best = 2.812320e+04 FES = 10100 Diversity = 3.1266e-01
Generation 110: Best = 2.794058e+04 FES = 11100 Diversity = 3.4101e-01
Generation 120: Best = 2.772665e+04 FES = 12100 Diversity = 2.3397e-01
Generation 130: Best = 2.762747e+04 FES = 13100 Diversity = 1.5027e-01
Generation 140: Best = 2.756849e+04 FES = 14100 Diversity = 1.0149e-01
Generation 150: Best = 2.753500e+04 FES = 15100 Diversity = 5.9269e-02
Generation 160: Best = 2.750890e+04 FES = 16100 Diversity = 5.2250e-02
Generation 170: Best = 2.749253e+04 FES = 17100 Diversity = 4.0733e-02
Generation 180: Best = 2.748140e+04 FES = 18100 Diversity = 2.8148e-02
Generation 190: Best = 2.746934e+04 FES = 19100 Diversity = 2.6591e-02
Generation 200: Best = 2.745833e+04 FES = 20100 Diversity = 2.8305e-02
Generation 210: Best = 2.743958e+04 FES = 21100 Diversity = 3.6389e-02
Generation 220: Best = 2.742910e+04 FES = 22100 Diversity = 3.5715e-02
Generation 230: Best = 2.741920e+04 FES = 23100 Diversity = 2.9715e-02
Generation 240: Best = 2.738563e+04 FES = 24100 Diversity = 4.3768e-02
Generation 250: Best = 2.737516e+04 FES = 25100 Diversity = 2.4183e-02
Generation 260: Best = 2.736861e+04 FES = 26100 Diversity = 1.9123e-02
Generation 270: Best = 2.736294e+04 FES = 27100 Diversity = 1.4772e-02
Generation 280: Best = 2.736053e+04 FES = 28100 Diversity = 8.3197e-03
Generation 290: Best = 2.735832e+04 FES = 29100 Diversity = 7.4928e-03
Generation 300: Best = 2.735646e+04 FES = 30100 Diversity = 6.4052e-03
Generation 310: Best = 2.735548e+04 FES = 31100 Diversity = 4.5452e-03
Generation 320: Best = 2.735452e+04 FES = 32100 Diversity = 3.4858e-03
Generation 330: Best = 2.735403e+04 FES = 33100 Diversity = 3.0149e-03
Generation 340: Best = 2.735373e+04 FES = 34100 Diversity = 1.9513e-03
Generation 350: Best = 2.735351e+04 FES = 35100 Diversity = 1.1889e-03
Generation 360: Best = 2.735339e+04 FES = 36100 Diversity = 8.8379e-04
Generation 370: Best = 2.735326e+04 FES = 37100 Diversity = 7.8556e-04
Generation 380: Best = 2.735316e+04 FES = 38100 Diversity = 6.8234e-04
Generation 390: Best = 2.735308e+04 FES = 39100 Diversity = 5.1554e-04
Generation 400: Best = 2.735300e+04 FES = 40100 Diversity = 4.7376e-04
Generation 410: Best = 2.735296e+04 FES = 41100 Diversity = 4.0577e-04
Generation 420: Best = 2.735291e+04 FES = 42100 Diversity = 4.2108e-04
Generation 430: Best = 2.735287e+04 FES = 43100 Diversity = 4.0790e-04

```


Generation	440:	Best = 2.735284e+04	FES = 44100	Diversity = 2.6927e-04
Generation	450:	Best = 2.735283e+04	FES = 45100	Diversity = 3.0576e-04
Generation	460:	Best = 2.735282e+04	FES = 46100	Diversity = 2.3184e-04
Generation	470:	Best = 2.735281e+04	FES = 47100	Diversity = 1.8193e-04
Generation	480:	Best = 2.735280e+04	FES = 48100	Diversity = 1.3448e-04
Generation	490:	Best = 2.735280e+04	FES = 49100	Diversity = 7.7331e-05
Generation	500:	Best = 2.735280e+04	FES = 50100	Diversity = 6.3036e-05
Generation	510:	Best = 2.735280e+04	FES = 51100	Diversity = 5.1654e-05
Generation	520:	Best = 2.735279e+04	FES = 52100	Diversity = 3.8000e-05
Generation	530:	Best = 2.735279e+04	FES = 53100	Diversity = 2.8780e-05
Generation	540:	Best = 2.735279e+04	FES = 54100	Diversity = 1.9453e-05
Generation	550:	Best = 2.735279e+04	FES = 55100	Diversity = 1.5325e-05
Generation	560:	Best = 2.735279e+04	FES = 56100	Diversity = 2.3420e-05
Generation	570:	Best = 2.735279e+04	FES = 57100	Diversity = 1.5014e-05
Generation	580:	Best = 2.735279e+04	FES = 58100	Diversity = 1.3328e-05
Generation	590:	Best = 2.735279e+04	FES = 59100	Diversity = 1.2549e-05
Generation	600:	Best = 2.735279e+04	FES = 60100	Diversity = 1.0646e-05
Generation	610:	Best = 2.735279e+04	FES = 61100	Diversity = 5.9591e-06
Generation	620:	Best = 2.735279e+04	FES = 62100	Diversity = 4.0262e-06
Generation	630:	Best = 2.735279e+04	FES = 63100	Diversity = 2.6211e-06
Generation	640:	Best = 2.735279e+04	FES = 64100	Diversity = 1.6109e-06
Generation	650:	Best = 2.735279e+04	FES = 65100	Diversity = 1.7872e-06
Generation	660:	Best = 2.735279e+04	FES = 66100	Diversity = 8.3051e-07
Generation	670:	Best = 2.735279e+04	FES = 67100	Diversity = 8.4569e-07
Generation	680:	Best = 2.735279e+04	FES = 68100	Diversity = 7.8814e-07
Generation	690:	Best = 2.735279e+04	FES = 69100	Diversity = 4.0196e-07
Generation	700:	Best = 2.735279e+04	FES = 70100	Diversity = 3.0580e-07
Generation	710:	Best = 2.735279e+04	FES = 71100	Diversity = 1.7127e-07
Generation	720:	Best = 2.735279e+04	FES = 72100	Diversity = 1.8953e-07
Generation	730:	Best = 2.735279e+04	FES = 73100	Diversity = 1.1476e-07
Generation	740:	Best = 2.735279e+04	FES = 74100	Diversity = 8.8402e-08
Generation	750:	Best = 2.735279e+04	FES = 75100	Diversity = 6.2833e-08
Generation	760:	Best = 2.735279e+04	FES = 76100	Diversity = 6.3967e-08
Generation	770:	Best = 2.735279e+04	FES = 77100	Diversity = 1.5628e-08
Generation	780:	Best = 2.735279e+04	FES = 78100	Diversity = 1.2563e-08
Generation	790:	Best = 2.735279e+04	FES = 79100	Diversity = 7.7391e-09
Generation	800:	Best = 2.735279e+04	FES = 80100	Diversity = 6.3991e-09
Generation	810:	Best = 2.735279e+04	FES = 81100	Diversity = 4.0012e-09
Generation	820:	Best = 2.735279e+04	FES = 82100	Diversity = 4.6210e-09
Generation	830:	Best = 2.735279e+04	FES = 83100	Diversity = 4.1202e-09
Generation	840:	Best = 2.735279e+04	FES = 84100	Diversity = 4.9109e-09
Generation	850:	Best = 2.735279e+04	FES = 85100	Diversity = 6.4251e-09
Generation	860:	Best = 2.735279e+04	FES = 86100	Diversity = 4.8492e-09
Generation	870:	Best = 2.735279e+04	FES = 87100	Diversity = 4.8130e-09
Generation	880:	Best = 2.735279e+04	FES = 88100	Diversity = 4.2087e-09
Generation	890:	Best = 2.735279e+04	FES = 89100	Diversity = 3.4111e-09
Generation	900:	Best = 2.735279e+04	FES = 90100	Diversity = 3.3138e-09
Generation	910:	Best = 2.735279e+04	FES = 91100	Diversity = 2.4944e-09
Generation	920:	Best = 2.735279e+04	FES = 92100	Diversity = 1.5791e-09
Generation	930:	Best = 2.735279e+04	FES = 93100	Diversity = 4.0898e-12
Generation	940:	Best = 2.735279e+04	FES = 94100	Diversity = 2.3693e-14
Generation	950:	Best = 2.735279e+04	FES = 95100	Diversity = 2.3693e-14
Generation	960:	Best = 2.735279e+04	FES = 96100	Diversity = 2.3693e-14
Generation	970:	Best = 2.735279e+04	FES = 97100	Diversity = 2.3693e-14

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Generation 9620:	Best = 2.735279e+04	FES = 962100	Diversity = 2.3693e-14
Generation 9630:	Best = 2.735279e+04	FES = 963100	Diversity = 2.3693e-14
Generation 9640:	Best = 2.735279e+04	FES = 964100	Diversity = 2.3693e-14
Generation 9650:	Best = 2.735279e+04	FES = 965100	Diversity = 2.3693e-14
Generation 9660:	Best = 2.735279e+04	FES = 966100	Diversity = 2.3693e-14
Generation 9670:	Best = 2.735279e+04	FES = 967100	Diversity = 2.3693e-14
Generation 9680:	Best = 2.735279e+04	FES = 968100	Diversity = 2.3693e-14
Generation 9690:	Best = 2.735279e+04	FES = 969100	Diversity = 2.3693e-14
Generation 9700:	Best = 2.735279e+04	FES = 970100	Diversity = 2.3693e-14
Generation 9710:	Best = 2.735279e+04	FES = 971100	Diversity = 2.3693e-14
Generation 9720:	Best = 2.735279e+04	FES = 972100	Diversity = 2.3693e-14
Generation 9730:	Best = 2.735279e+04	FES = 973100	Diversity = 2.3693e-14
Generation 9740:	Best = 2.735279e+04	FES = 974100	Diversity = 2.3693e-14
Generation 9750:	Best = 2.735279e+04	FES = 975100	Diversity = 2.3693e-14
Generation 9760:	Best = 2.735279e+04	FES = 976100	Diversity = 2.3693e-14
Generation 9770:	Best = 2.735279e+04	FES = 977100	Diversity = 2.3693e-14
Generation 9780:	Best = 2.735279e+04	FES = 978100	Diversity = 2.3693e-14
Generation 9790:	Best = 2.735279e+04	FES = 979100	Diversity = 2.3693e-14
Generation 9800:	Best = 2.735279e+04	FES = 980100	Diversity = 2.3693e-14
Generation 9810:	Best = 2.735279e+04	FES = 981100	Diversity = 2.3693e-14
Generation 9820:	Best = 2.735279e+04	FES = 982100	Diversity = 2.3693e-14
Generation 9830:	Best = 2.735279e+04	FES = 983100	Diversity = 2.3693e-14
Generation 9840:	Best = 2.735279e+04	FES = 984100	Diversity = 2.3693e-14
Generation 9850:	Best = 2.735279e+04	FES = 985100	Diversity = 2.3693e-14
Generation 9860:	Best = 2.735279e+04	FES = 986100	Diversity = 2.3693e-14
Generation 9870:	Best = 2.735279e+04	FES = 987100	Diversity = 2.3693e-14
Generation 9880:	Best = 2.735279e+04	FES = 988100	Diversity = 2.3693e-14
Generation 9890:	Best = 2.735279e+04	FES = 989100	Diversity = 2.3693e-14
Generation 9900:	Best = 2.735279e+04	FES = 990100	Diversity = 2.3693e-14
Generation 9910:	Best = 2.735279e+04	FES = 991100	Diversity = 2.3693e-14
Generation 9920:	Best = 2.735279e+04	FES = 992100	Diversity = 2.3693e-14
Generation 9930:	Best = 2.735279e+04	FES = 993100	Diversity = 2.3693e-14
Generation 9940:	Best = 2.735279e+04	FES = 994100	Diversity = 2.3693e-14
Generation 9950:	Best = 2.735279e+04	FES = 995100	Diversity = 2.3693e-14
Generation 9960:	Best = 2.735279e+04	FES = 996100	Diversity = 2.3693e-14
Generation 9970:	Best = 2.735279e+04	FES = 997100	Diversity = 2.3693e-14
Generation 9980:	Best = 2.735279e+04	FES = 998100	Diversity = 2.3693e-14
Generation 9990:	Best = 2.735279e+04	FES = 999100	Diversity = 2.3693e-14

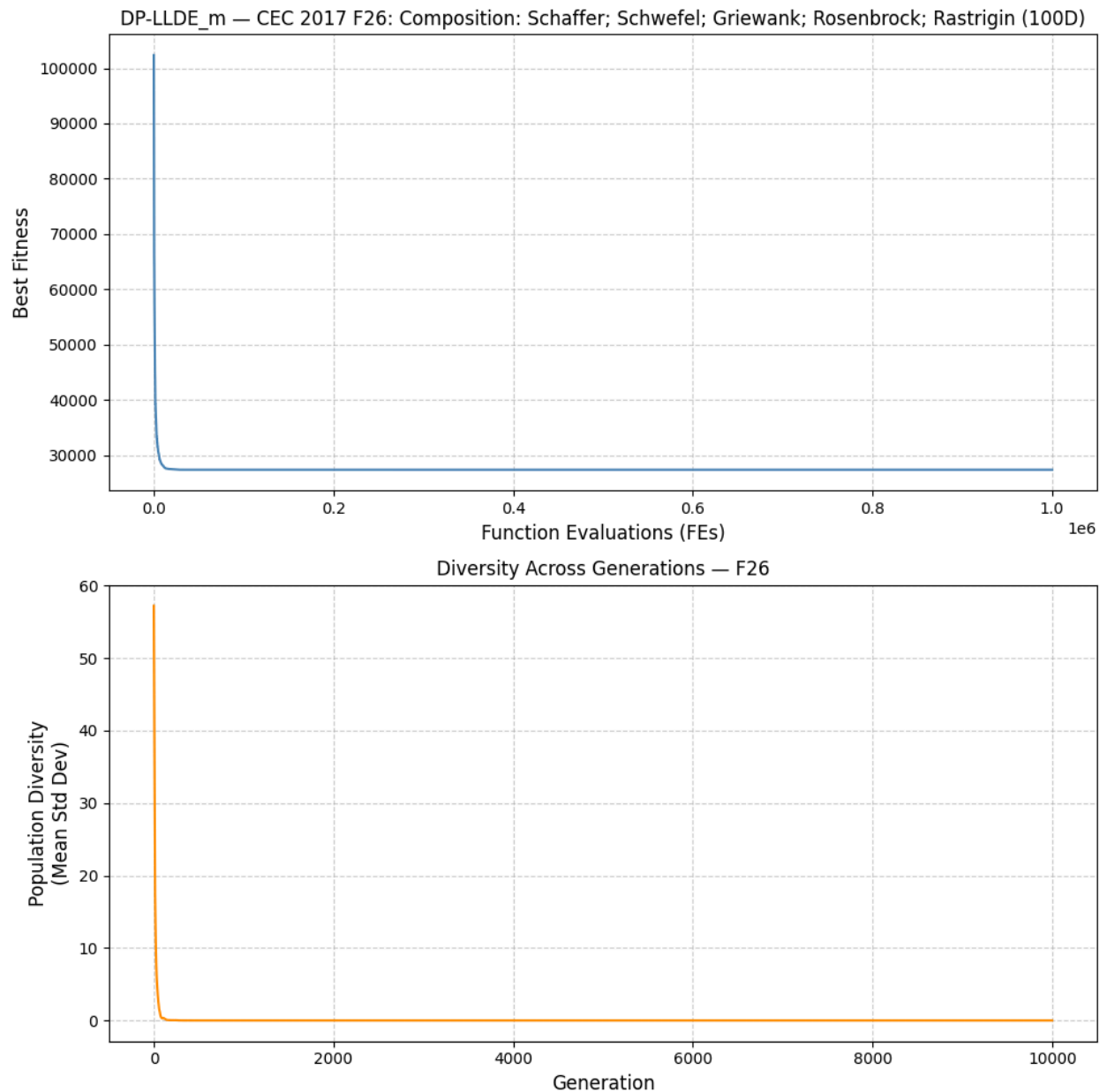
Optimization completed!

Final Best Fitness = 2.735279e+04

Total Function Evaluations = 1000000

```
=====
=
Results:
  Best Fitness           : 2.735279e+04
  Global Optimum         : 2.600000e+03
  Error (|best-optimum|) : 2.475279e+04
  Best Solution (5 dim)  : [ 14.7327709  -24.95778384 -12.56866591 -45.50297912
-0.37676163]
  D^(0) (init. diversity): 5.7220e+01
=====
=
```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results_2/all_results_checkpoint.pkl



Graph saved as convergence_F26.png

✓ Plot saved to: /content/drive/My Drive/thesis_results_2/plots/convergence_F26.png

```
In [ ]: # =====  
# F27 — Composition Function 7 (HGBat; Rastrigin; Schwefel; Bent Cigar; Ellipt  
# =====  
run_and_plot('F27', make_wrapper(composition.f27), 2700.0, 'Composition: HGBat
```

=====

=

DP-LLDE_m on CEC 2017 F27 – Single Run

=====

=

Function : Composition: HGBat; Rastrigin; Schwefel; Bent Cigar; Elliptic;
Schaffer (100D)

Bounds : [-100, 100]

Global optimum : 2700.0

Initial $D^*(0) = 5.7220e+01$ | Diversity threshold = 5.7220e+00

Generation 0: Best Fitness = 1.447050e+04 | Diversity = 5.7220e+01

Generation 10: Best = 1.152316e+04 FES = 1100 Diversity = 3.2172e+01

Generation 20: Best = 8.159647e+03 FES = 2100 Diversity = 2.2045e+01

Generation 30: Best = 6.639856e+03 FES = 3100 Diversity = 1.4679e+01

Generation 40: Best = 5.978808e+03 FES = 4100 Diversity = 1.0726e+01

Generation 50: Best = 5.396536e+03 FES = 5100 Diversity = 8.7366e+00

Generation 60: Best = 5.057883e+03 FES = 6100 Diversity = 7.4949e+00

Generation 70: Best = 4.733243e+03 FES = 7100 Diversity = 6.4468e+00

Generation 80: Best = 4.611703e+03 FES = 8100 Diversity = 4.7240e+00

Generation 90: Best = 4.396723e+03 FES = 9100 Diversity = 2.4860e+00

Generation 100: Best = 4.314994e+03 FES = 10100 Diversity = 1.2909e+00

Generation 110: Best = 4.266749e+03 FES = 11100 Diversity = 6.1369e-01

Generation 120: Best = 4.234994e+03 FES = 12100 Diversity = 4.0574e-01

Generation 130: Best = 4.219254e+03 FES = 13100 Diversity = 2.8542e-01

Generation 140: Best = 4.210352e+03 FES = 14100 Diversity = 1.9560e-01

Generation 150: Best = 4.204672e+03 FES = 15100 Diversity = 1.1085e-01

Generation 160: Best = 4.202339e+03 FES = 16100 Diversity = 6.7558e-02

Generation 170: Best = 4.200729e+03 FES = 17100 Diversity = 4.2723e-02

Generation 180: Best = 4.199590e+03 FES = 18100 Diversity = 2.7152e-02

Generation 190: Best = 4.198832e+03 FES = 19100 Diversity = 1.9145e-02

Generation 200: Best = 4.198350e+03 FES = 20100 Diversity = 1.5898e-02

Generation 210: Best = 4.197833e+03 FES = 21100 Diversity = 1.4105e-02

Generation 220: Best = 4.197464e+03 FES = 22100 Diversity = 1.0392e-02

Generation 230: Best = 4.197026e+03 FES = 23100 Diversity = 9.1293e-03

Generation 240: Best = 4.196799e+03 FES = 24100 Diversity = 6.6939e-03

Generation 250: Best = 4.196554e+03 FES = 25100 Diversity = 6.1114e-03

Generation 260: Best = 4.196367e+03 FES = 26100 Diversity = 5.5427e-03

Generation 270: Best = 4.196206e+03 FES = 27100 Diversity = 5.1514e-03

Generation 280: Best = 4.196000e+03 FES = 28100 Diversity = 5.7498e-03

Generation 290: Best = 4.195867e+03 FES = 29100 Diversity = 4.4735e-03

Generation 300: Best = 4.195710e+03 FES = 30100 Diversity = 3.9125e-03

Generation 310: Best = 4.195581e+03 FES = 31100 Diversity = 3.6921e-03

Generation 320: Best = 4.195518e+03 FES = 32100 Diversity = 2.7073e-03

Generation 330: Best = 4.195455e+03 FES = 33100 Diversity = 2.1165e-03

Generation 340: Best = 4.195419e+03 FES = 34100 Diversity = 1.8550e-03

Generation 350: Best = 4.195403e+03 FES = 35100 Diversity = 1.3576e-03

Generation 360: Best = 4.195382e+03 FES = 36100 Diversity = 1.3400e-03

Generation 370: Best = 4.195363e+03 FES = 37100 Diversity = 1.1197e-03

Generation 380: Best = 4.195346e+03 FES = 38100 Diversity = 8.5881e-04

Generation 390: Best = 4.195337e+03 FES = 39100 Diversity = 6.7386e-04

Generation 400: Best = 4.195330e+03 FES = 40100 Diversity = 6.2588e-04

Generation 410: Best = 4.195323e+03 FES = 41100 Diversity = 5.3833e-04

Generation 420: Best = 4.195319e+03 FES = 42100 Diversity = 4.5662e-04

Generation 430: Best = 4.195317e+03 FES = 43100 Diversity = 4.1942e-04

Generation	440:	Best = 4.195315e+03	FES = 44100	Diversity = 3.3512e-04
Generation	450:	Best = 4.195314e+03	FES = 45100	Diversity = 2.9902e-04
Generation	460:	Best = 4.195312e+03	FES = 46100	Diversity = 3.1293e-04
Generation	470:	Best = 4.195311e+03	FES = 47100	Diversity = 2.7816e-04
Generation	480:	Best = 4.195311e+03	FES = 48100	Diversity = 2.3106e-04
Generation	490:	Best = 4.195310e+03	FES = 49100	Diversity = 1.8817e-04
Generation	500:	Best = 4.195309e+03	FES = 50100	Diversity = 2.1061e-04
Generation	510:	Best = 4.195308e+03	FES = 51100	Diversity = 1.4719e-04
Generation	520:	Best = 4.195308e+03	FES = 52100	Diversity = 9.3488e-05
Generation	530:	Best = 4.195308e+03	FES = 53100	Diversity = 1.3164e-04
Generation	540:	Best = 4.195308e+03	FES = 54100	Diversity = 8.1926e-05
Generation	550:	Best = 4.195307e+03	FES = 55100	Diversity = 6.9830e-05
Generation	560:	Best = 4.195307e+03	FES = 56100	Diversity = 5.1528e-05
Generation	570:	Best = 4.195307e+03	FES = 57100	Diversity = 4.2272e-05
Generation	580:	Best = 4.195307e+03	FES = 58100	Diversity = 2.7959e-05
Generation	590:	Best = 4.195307e+03	FES = 59100	Diversity = 3.8903e-05
Generation	600:	Best = 4.195307e+03	FES = 60100	Diversity = 2.4758e-05
Generation	610:	Best = 4.195307e+03	FES = 61100	Diversity = 1.4956e-05
Generation	620:	Best = 4.195307e+03	FES = 62100	Diversity = 8.8103e-06
Generation	630:	Best = 4.195307e+03	FES = 63100	Diversity = 8.9305e-06
Generation	640:	Best = 4.195307e+03	FES = 64100	Diversity = 5.7206e-06
Generation	650:	Best = 4.195307e+03	FES = 65100	Diversity = 1.2569e-06
Generation	660:	Best = 4.195307e+03	FES = 66100	Diversity = 8.6100e-07
Generation	670:	Best = 4.195307e+03	FES = 67100	Diversity = 4.6306e-07
Generation	680:	Best = 4.195307e+03	FES = 68100	Diversity = 4.5368e-07
Generation	690:	Best = 4.195307e+03	FES = 69100	Diversity = 1.8244e-07
Generation	700:	Best = 4.195307e+03	FES = 70100	Diversity = 1.6555e-07
Generation	710:	Best = 4.195307e+03	FES = 71100	Diversity = 6.7122e-08
Generation	720:	Best = 4.195307e+03	FES = 72100	Diversity = 4.3196e-08
Generation	730:	Best = 4.195307e+03	FES = 73100	Diversity = 6.5464e-08
Generation	740:	Best = 4.195307e+03	FES = 74100	Diversity = 2.3563e-08
Generation	750:	Best = 4.195307e+03	FES = 75100	Diversity = 9.1457e-09
Generation	760:	Best = 4.195307e+03	FES = 76100	Diversity = 1.9335e-08
Generation	770:	Best = 4.195307e+03	FES = 77100	Diversity = 1.2545e-08
Generation	780:	Best = 4.195307e+03	FES = 78100	Diversity = 6.1570e-09
Generation	790:	Best = 4.195307e+03	FES = 79100	Diversity = 1.0083e-09
Generation	800:	Best = 4.195307e+03	FES = 80100	Diversity = 7.9805e-10
Generation	810:	Best = 4.195307e+03	FES = 81100	Diversity = 9.6882e-10
Generation	820:	Best = 4.195307e+03	FES = 82100	Diversity = 8.2116e-10
Generation	830:	Best = 4.195307e+03	FES = 83100	Diversity = 8.5386e-10
Generation	840:	Best = 4.195307e+03	FES = 84100	Diversity = 8.3401e-10
Generation	850:	Best = 4.195307e+03	FES = 85100	Diversity = 5.7819e-10
Generation	860:	Best = 4.195307e+03	FES = 86100	Diversity = 4.3802e-10
Generation	870:	Best = 4.195307e+03	FES = 87100	Diversity = 3.7292e-10
Generation	880:	Best = 4.195307e+03	FES = 88100	Diversity = 9.8689e-11
Generation	890:	Best = 4.195307e+03	FES = 89100	Diversity = 5.4270e-14
Generation	900:	Best = 4.195307e+03	FES = 90100	Diversity = 5.4270e-14
Generation	910:	Best = 4.195307e+03	FES = 91100	Diversity = 5.4270e-14
Generation	920:	Best = 4.195307e+03	FES = 92100	Diversity = 5.4270e-14
Generation	930:	Best = 4.195307e+03	FES = 93100	Diversity = 5.4270e-14
Generation	940:	Best = 4.195307e+03	FES = 94100	Diversity = 5.4270e-14
Generation	950:	Best = 4.195307e+03	FES = 95100	Diversity = 5.4270e-14
Generation	960:	Best = 4.195307e+03	FES = 96100	Diversity = 5.4270e-14
Generation	970:	Best = 4.195307e+03	FES = 97100	Diversity = 5.4270e-14

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Generation 9620: Best = 4.195307e+03 FES = 962100 Diversity = 5.4270e-14
Generation 9630: Best = 4.195307e+03 FES = 963100 Diversity = 5.4270e-14
Generation 9640: Best = 4.195307e+03 FES = 964100 Diversity = 5.4270e-14
Generation 9650: Best = 4.195307e+03 FES = 965100 Diversity = 5.4270e-14
Generation 9660: Best = 4.195307e+03 FES = 966100 Diversity = 5.4270e-14
Generation 9670: Best = 4.195307e+03 FES = 967100 Diversity = 5.4270e-14
Generation 9680: Best = 4.195307e+03 FES = 968100 Diversity = 5.4270e-14
Generation 9690: Best = 4.195307e+03 FES = 969100 Diversity = 5.4270e-14
Generation 9700: Best = 4.195307e+03 FES = 970100 Diversity = 5.4270e-14
Generation 9710: Best = 4.195307e+03 FES = 971100 Diversity = 5.4270e-14
Generation 9720: Best = 4.195307e+03 FES = 972100 Diversity = 5.4270e-14
Generation 9730: Best = 4.195307e+03 FES = 973100 Diversity = 5.4270e-14
Generation 9740: Best = 4.195307e+03 FES = 974100 Diversity = 5.4270e-14
Generation 9750: Best = 4.195307e+03 FES = 975100 Diversity = 5.4270e-14
Generation 9760: Best = 4.195307e+03 FES = 976100 Diversity = 5.4270e-14
Generation 9770: Best = 4.195307e+03 FES = 977100 Diversity = 5.4270e-14
Generation 9780: Best = 4.195307e+03 FES = 978100 Diversity = 5.4270e-14
Generation 9790: Best = 4.195307e+03 FES = 979100 Diversity = 5.4270e-14
Generation 9800: Best = 4.195307e+03 FES = 980100 Diversity = 5.4270e-14
Generation 9810: Best = 4.195307e+03 FES = 981100 Diversity = 5.4270e-14
Generation 9820: Best = 4.195307e+03 FES = 982100 Diversity = 5.4270e-14
Generation 9830: Best = 4.195307e+03 FES = 983100 Diversity = 5.4270e-14
Generation 9840: Best = 4.195307e+03 FES = 984100 Diversity = 5.4270e-14
Generation 9850: Best = 4.195307e+03 FES = 985100 Diversity = 5.4270e-14
Generation 9860: Best = 4.195307e+03 FES = 986100 Diversity = 5.4270e-14
Generation 9870: Best = 4.195307e+03 FES = 987100 Diversity = 5.4270e-14
Generation 9880: Best = 4.195307e+03 FES = 988100 Diversity = 5.4270e-14
Generation 9890: Best = 4.195307e+03 FES = 989100 Diversity = 5.4270e-14
Generation 9900: Best = 4.195307e+03 FES = 990100 Diversity = 5.4270e-14
Generation 9910: Best = 4.195307e+03 FES = 991100 Diversity = 5.4270e-14
Generation 9920: Best = 4.195307e+03 FES = 992100 Diversity = 5.4270e-14
Generation 9930: Best = 4.195307e+03 FES = 993100 Diversity = 5.4270e-14
Generation 9940: Best = 4.195307e+03 FES = 994100 Diversity = 5.4270e-14
Generation 9950: Best = 4.195307e+03 FES = 995100 Diversity = 5.4270e-14
Generation 9960: Best = 4.195307e+03 FES = 996100 Diversity = 5.4270e-14
Generation 9970: Best = 4.195307e+03 FES = 997100 Diversity = 5.4270e-14
Generation 9980: Best = 4.195307e+03 FES = 998100 Diversity = 5.4270e-14
Generation 9990: Best = 4.195307e+03 FES = 999100 Diversity = 5.4270e-14

```

Optimization completed!

Final Best Fitness = 4.195307e+03

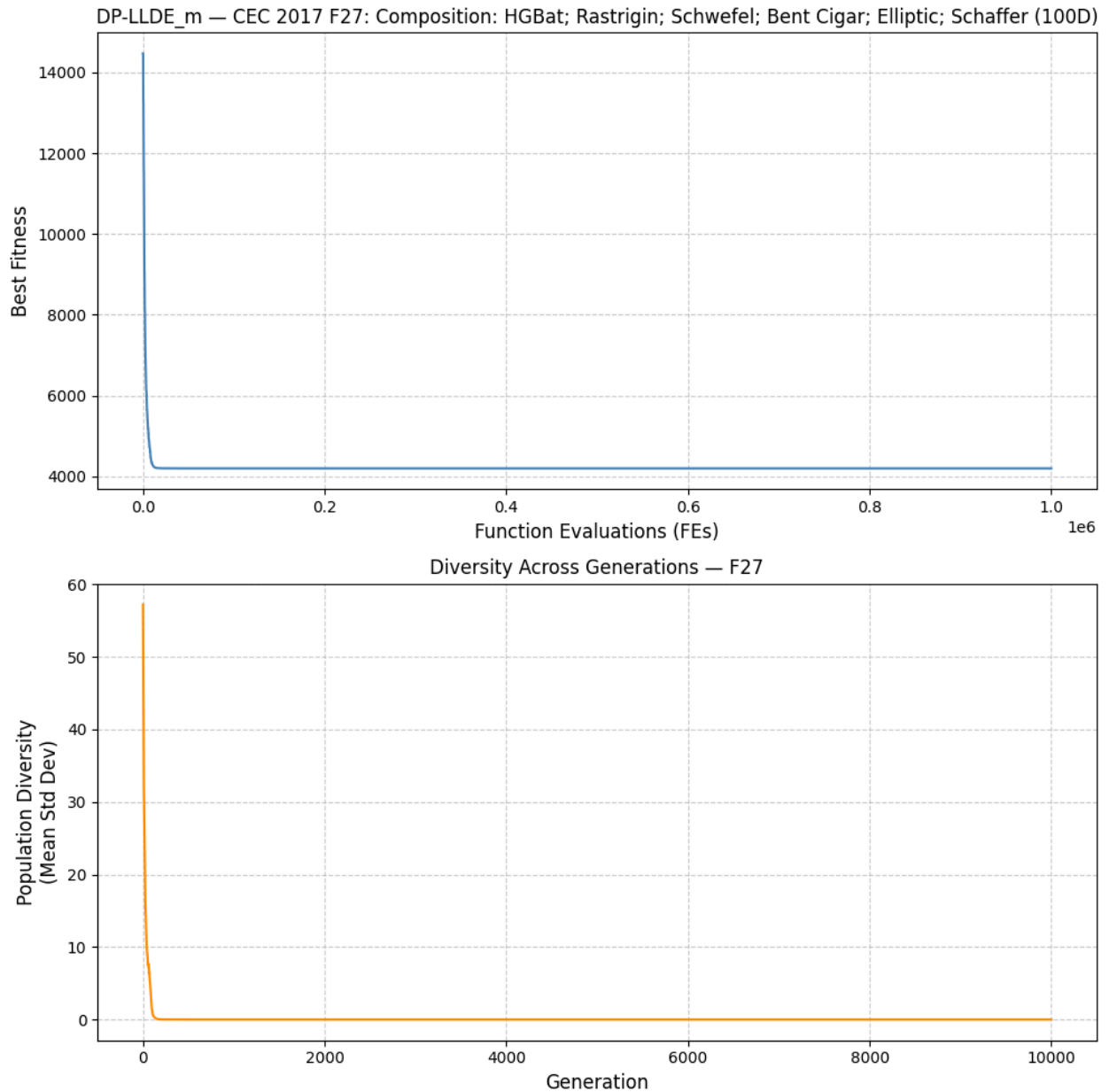
Total Function Evaluations = 1000000

```

=====
=
Results:
  Best Fitness           : 4.195307e+03
  Global Optimum         : 2.700000e+03
  Error (|best-optimum|) : 1.495307e+03
  Best Solution (5 dim)  : [ -1.32260517  60.87225913 -52.85922867 -12.89719825
8.16359162]
  D^(0) (init. diversity): 5.7220e+01
=====
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```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results_2/all_results_checkpoint.pkl



Graph saved as convergence_F27.png

✓ Plot saved to: /content/drive/My Drive/thesis_results_2/plots/convergence_F27.png

```
In [ ]: # =====  
# F28 — Composition Function 8 (Ackley; Griewank; Discus; Rosenbrock; HappyCat  
# =====  
run_and_plot('F28', make_wrapper(composition.f28), 2800.0, 'Composition: Ackle
```

=====

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DP-LLDE_m on CEC 2017 F28 – Single Run

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Function : Composition: Ackley; Griewank; Discus; Rosenbrock; HappyCat; S
chaffer (100D)

Bounds : [-100, 100]

Global optimum : 2800.0

Initial $D^*(0) = 5.7220e+01$ | Diversity threshold = 5.7220e+00

Generation 0: Best Fitness = 6.506878e+04 | Diversity = 5.7220e+01

Generation 10: Best = 2.632001e+04 FES = 1100 Diversity = 2.2461e+01

Generation 20: Best = 1.928448e+04 FES = 2100 Diversity = 9.0577e+00

Generation 30: Best = 1.738424e+04 FES = 3100 Diversity = 4.6568e+00

Generation 40: Best = 1.618535e+04 FES = 4100 Diversity = 2.6923e+00

Generation 50: Best = 1.548052e+04 FES = 5100 Diversity = 1.8340e+00

Generation 60: Best = 1.504108e+04 FES = 6100 Diversity = 1.3550e+00

Generation 70: Best = 1.462758e+04 FES = 7100 Diversity = 9.9263e-01

Generation 80: Best = 1.425568e+04 FES = 8100 Diversity = 9.1814e-01

Generation 90: Best = 1.398441e+04 FES = 9100 Diversity = 8.5421e-01

Generation 100: Best = 1.372457e+04 FES = 10100 Diversity = 8.1885e-01

Generation 110: Best = 1.344252e+04 FES = 11100 Diversity = 8.0972e-01

Generation 120: Best = 1.320981e+04 FES = 12100 Diversity = 6.7423e-01

Generation 130: Best = 1.306412e+04 FES = 13100 Diversity = 6.2781e-01

Generation 140: Best = 1.289067e+04 FES = 14100 Diversity = 6.9743e-01

Generation 150: Best = 1.271640e+04 FES = 15100 Diversity = 6.7300e-01

Generation 160: Best = 1.255809e+04 FES = 16100 Diversity = 6.1342e-01

Generation 170: Best = 1.243876e+04 FES = 17100 Diversity = 5.0451e-01

Generation 180: Best = 1.231991e+04 FES = 18100 Diversity = 4.8953e-01

Generation 190: Best = 1.222135e+04 FES = 19100 Diversity = 4.2454e-01

Generation 200: Best = 1.212144e+04 FES = 20100 Diversity = 3.7697e-01

Generation 210: Best = 1.203504e+04 FES = 21100 Diversity = 4.2414e-01

Generation 220: Best = 1.193548e+04 FES = 22100 Diversity = 4.2528e-01

Generation 230: Best = 1.182109e+04 FES = 23100 Diversity = 4.6715e-01

Generation 240: Best = 1.172925e+04 FES = 24100 Diversity = 5.1784e-01

Generation 250: Best = 1.163895e+04 FES = 25100 Diversity = 3.7935e-01

Generation 260: Best = 1.155788e+04 FES = 26100 Diversity = 3.7920e-01

Generation 270: Best = 1.149991e+04 FES = 27100 Diversity = 3.5237e-01

Generation 280: Best = 1.142646e+04 FES = 28100 Diversity = 3.9510e-01

Generation 290: Best = 1.134942e+04 FES = 29100 Diversity = 4.1401e-01

Generation 300: Best = 1.125783e+04 FES = 30100 Diversity = 4.0568e-01

Generation 310: Best = 1.115172e+04 FES = 31100 Diversity = 4.2269e-01

Generation 320: Best = 1.105676e+04 FES = 32100 Diversity = 4.0003e-01

Generation 330: Best = 1.095628e+04 FES = 33100 Diversity = 4.1355e-01

Generation 340: Best = 1.086748e+04 FES = 34100 Diversity = 4.0949e-01

Generation 350: Best = 1.076450e+04 FES = 35100 Diversity = 4.0529e-01

Generation 360: Best = 1.068805e+04 FES = 36100 Diversity = 3.7272e-01

Generation 370: Best = 1.060565e+04 FES = 37100 Diversity = 3.8719e-01

Generation 380: Best = 1.052550e+04 FES = 38100 Diversity = 3.8469e-01

Generation 390: Best = 1.043051e+04 FES = 39100 Diversity = 3.7260e-01

Generation 400: Best = 1.034974e+04 FES = 40100 Diversity = 3.7794e-01

Generation 410: Best = 1.027867e+04 FES = 41100 Diversity = 3.3812e-01

Generation 420: Best = 1.022826e+04 FES = 42100 Diversity = 2.8849e-01

Generation 430: Best = 1.018783e+04 FES = 43100 Diversity = 2.4372e-01

Generation	440:	Best = 1.015277e+04	FES = 44100	Diversity = 2.1167e-01
Generation	450:	Best = 1.012410e+04	FES = 45100	Diversity = 1.7811e-01
Generation	460:	Best = 1.009450e+04	FES = 46100	Diversity = 1.6619e-01
Generation	470:	Best = 1.007421e+04	FES = 47100	Diversity = 1.4760e-01
Generation	480:	Best = 1.005807e+04	FES = 48100	Diversity = 1.1833e-01
Generation	490:	Best = 1.004914e+04	FES = 49100	Diversity = 7.5674e-02
Generation	500:	Best = 1.004083e+04	FES = 50100	Diversity = 6.2453e-02
Generation	510:	Best = 1.003535e+04	FES = 51100	Diversity = 4.1820e-02
Generation	520:	Best = 1.003240e+04	FES = 52100	Diversity = 3.1697e-02
Generation	530:	Best = 1.002957e+04	FES = 53100	Diversity = 2.3370e-02
Generation	540:	Best = 1.002854e+04	FES = 54100	Diversity = 1.5413e-02
Generation	550:	Best = 1.002763e+04	FES = 55100	Diversity = 1.1423e-02
Generation	560:	Best = 1.002671e+04	FES = 56100	Diversity = 1.2570e-02
Generation	570:	Best = 1.002637e+04	FES = 57100	Diversity = 6.7231e-03
Generation	580:	Best = 1.002617e+04	FES = 58100	Diversity = 5.5470e-03
Generation	590:	Best = 1.002603e+04	FES = 59100	Diversity = 4.5271e-03
Generation	600:	Best = 1.002593e+04	FES = 60100	Diversity = 4.4708e-03
Generation	610:	Best = 1.002586e+04	FES = 61100	Diversity = 2.4654e-03
Generation	620:	Best = 1.002583e+04	FES = 62100	Diversity = 1.9652e-03
Generation	630:	Best = 1.002581e+04	FES = 63100	Diversity = 1.4414e-03
Generation	640:	Best = 1.002580e+04	FES = 64100	Diversity = 1.1085e-03
Generation	650:	Best = 1.002579e+04	FES = 65100	Diversity = 8.7346e-04
Generation	660:	Best = 1.002579e+04	FES = 66100	Diversity = 8.4930e-04
Generation	670:	Best = 1.002578e+04	FES = 67100	Diversity = 5.4553e-04
Generation	680:	Best = 1.002578e+04	FES = 68100	Diversity = 5.3941e-04
Generation	690:	Best = 1.002578e+04	FES = 69100	Diversity = 4.8260e-04
Generation	700:	Best = 1.002578e+04	FES = 70100	Diversity = 2.8062e-04
Generation	710:	Best = 1.002578e+04	FES = 71100	Diversity = 1.8570e-04
Generation	720:	Best = 1.002578e+04	FES = 72100	Diversity = 2.0891e-04
Generation	730:	Best = 1.002577e+04	FES = 73100	Diversity = 1.7306e-04
Generation	740:	Best = 1.002577e+04	FES = 74100	Diversity = 1.5712e-04
Generation	750:	Best = 1.002577e+04	FES = 75100	Diversity = 1.0921e-04
Generation	760:	Best = 1.002577e+04	FES = 76100	Diversity = 6.6265e-05
Generation	770:	Best = 1.002577e+04	FES = 77100	Diversity = 3.8561e-05
Generation	780:	Best = 1.002577e+04	FES = 78100	Diversity = 3.1255e-05
Generation	790:	Best = 1.002577e+04	FES = 79100	Diversity = 2.9136e-05
Generation	800:	Best = 1.002577e+04	FES = 80100	Diversity = 1.7105e-05
Generation	810:	Best = 1.002577e+04	FES = 81100	Diversity = 1.1743e-05
Generation	820:	Best = 1.002577e+04	FES = 82100	Diversity = 1.0833e-05
Generation	830:	Best = 1.002577e+04	FES = 83100	Diversity = 5.7819e-06
Generation	840:	Best = 1.002577e+04	FES = 84100	Diversity = 4.7464e-06
Generation	850:	Best = 1.002577e+04	FES = 85100	Diversity = 4.0191e-06
Generation	860:	Best = 1.002577e+04	FES = 86100	Diversity = 3.6942e-06
Generation	870:	Best = 1.002577e+04	FES = 87100	Diversity = 1.7611e-06
Generation	880:	Best = 1.002577e+04	FES = 88100	Diversity = 1.0692e-06
Generation	890:	Best = 1.002577e+04	FES = 89100	Diversity = 8.6941e-07
Generation	900:	Best = 1.002577e+04	FES = 90100	Diversity = 5.9314e-07
Generation	910:	Best = 1.002577e+04	FES = 91100	Diversity = 5.7519e-07
Generation	920:	Best = 1.002577e+04	FES = 92100	Diversity = 2.7290e-07
Generation	930:	Best = 1.002577e+04	FES = 93100	Diversity = 2.3089e-07
Generation	940:	Best = 1.002577e+04	FES = 94100	Diversity = 1.9372e-07
Generation	950:	Best = 1.002577e+04	FES = 95100	Diversity = 1.2060e-07
Generation	960:	Best = 1.002577e+04	FES = 96100	Diversity = 1.1753e-07
Generation	970:	Best = 1.002577e+04	FES = 97100	Diversity = 6.6608e-08

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Generation 9620: Best = 1.002577e+04 FES = 962100 Diversity = 3.1942e-14
Generation 9630: Best = 1.002577e+04 FES = 963100 Diversity = 3.1942e-14
Generation 9640: Best = 1.002577e+04 FES = 964100 Diversity = 3.1942e-14
Generation 9650: Best = 1.002577e+04 FES = 965100 Diversity = 3.1942e-14
Generation 9660: Best = 1.002577e+04 FES = 966100 Diversity = 3.1942e-14
Generation 9670: Best = 1.002577e+04 FES = 967100 Diversity = 3.1942e-14
Generation 9680: Best = 1.002577e+04 FES = 968100 Diversity = 3.1942e-14
Generation 9690: Best = 1.002577e+04 FES = 969100 Diversity = 3.1942e-14
Generation 9700: Best = 1.002577e+04 FES = 970100 Diversity = 3.1942e-14
Generation 9710: Best = 1.002577e+04 FES = 971100 Diversity = 3.1942e-14
Generation 9720: Best = 1.002577e+04 FES = 972100 Diversity = 3.1942e-14
Generation 9730: Best = 1.002577e+04 FES = 973100 Diversity = 3.1942e-14
Generation 9740: Best = 1.002577e+04 FES = 974100 Diversity = 3.1942e-14
Generation 9750: Best = 1.002577e+04 FES = 975100 Diversity = 3.1942e-14
Generation 9760: Best = 1.002577e+04 FES = 976100 Diversity = 3.1942e-14
Generation 9770: Best = 1.002577e+04 FES = 977100 Diversity = 3.1942e-14
Generation 9780: Best = 1.002577e+04 FES = 978100 Diversity = 3.1942e-14
Generation 9790: Best = 1.002577e+04 FES = 979100 Diversity = 3.1942e-14
Generation 9800: Best = 1.002577e+04 FES = 980100 Diversity = 3.1942e-14
Generation 9810: Best = 1.002577e+04 FES = 981100 Diversity = 3.1942e-14
Generation 9820: Best = 1.002577e+04 FES = 982100 Diversity = 3.1942e-14
Generation 9830: Best = 1.002577e+04 FES = 983100 Diversity = 3.1942e-14
Generation 9840: Best = 1.002577e+04 FES = 984100 Diversity = 3.1942e-14
Generation 9850: Best = 1.002577e+04 FES = 985100 Diversity = 3.1942e-14
Generation 9860: Best = 1.002577e+04 FES = 986100 Diversity = 3.1942e-14
Generation 9870: Best = 1.002577e+04 FES = 987100 Diversity = 3.1942e-14
Generation 9880: Best = 1.002577e+04 FES = 988100 Diversity = 3.1942e-14
Generation 9890: Best = 1.002577e+04 FES = 989100 Diversity = 3.1942e-14
Generation 9900: Best = 1.002577e+04 FES = 990100 Diversity = 3.1942e-14
Generation 9910: Best = 1.002577e+04 FES = 991100 Diversity = 3.1942e-14
Generation 9920: Best = 1.002577e+04 FES = 992100 Diversity = 3.1942e-14
Generation 9930: Best = 1.002577e+04 FES = 993100 Diversity = 3.1942e-14
Generation 9940: Best = 1.002577e+04 FES = 994100 Diversity = 3.1942e-14
Generation 9950: Best = 1.002577e+04 FES = 995100 Diversity = 3.1942e-14
Generation 9960: Best = 1.002577e+04 FES = 996100 Diversity = 3.1942e-14
Generation 9970: Best = 1.002577e+04 FES = 997100 Diversity = 3.1942e-14
Generation 9980: Best = 1.002577e+04 FES = 998100 Diversity = 3.1942e-14
Generation 9990: Best = 1.002577e+04 FES = 999100 Diversity = 3.1942e-14

```

Optimization completed!

Final Best Fitness = 1.002577e+04

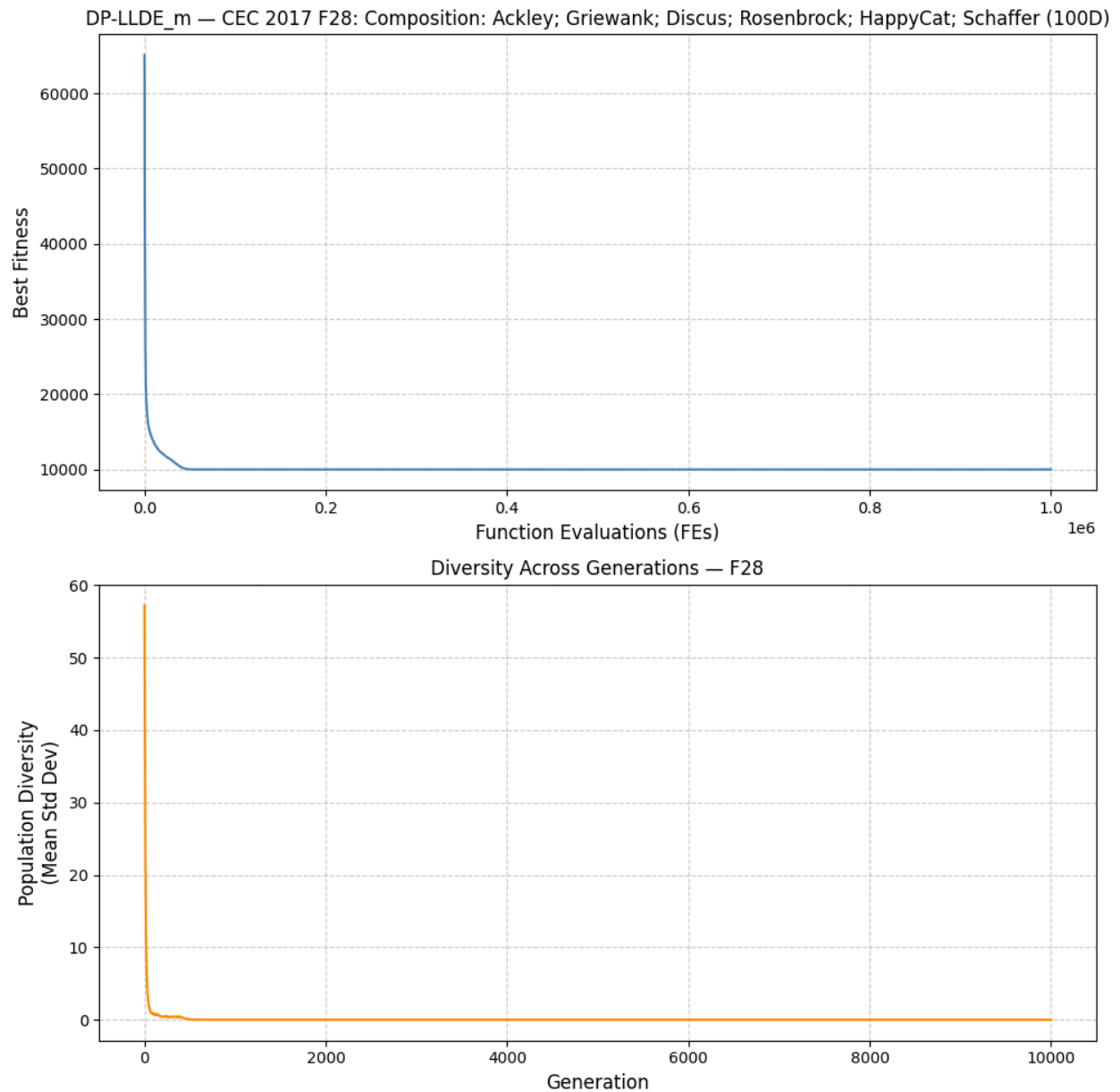
Total Function Evaluations = 1000000

```

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Results:
  Best Fitness           : 1.002577e+04
  Global Optimum         : 2.800000e+03
  Error (|best-optimum|) : 7.225774e+03
  Best Solution (5 dim)  : [-10.42739404  11.54764852  14.16252091  21.46981837
-0.28794739]
  D^(0) (init. diversity): 5.7220e+01
=====
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```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results_2/all_results_checkpoint.pkl



Graph saved as convergence_F28.png

✓ Plot saved to: /content/drive/My Drive/thesis_results_2/plots/convergence_F28.png

```
In [30]: # =====  
# F29 — Composition Function 9 (F15; F16; F17)  
# =====  
run_and_plot('F29', make_wrapper(composition.f29), 2900.0, 'Composition: F15;
```

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DP-LLDE_m on CEC 2017 F29 – Single Run

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Function : Composition: F15; F16; F17 (100D)

Bounds : [-100, 100]

Global optimum : 2900.0

Initial $D^{(0)} = 5.7220e+01$ | Diversity threshold = 5.7220e+00

Generation 0: Best Fitness = 1.114421e+07 | Diversity = 5.7220e+01

Generation	10:	Best = 6.176195e+04	FES = 1100	Diversity = 2.7139e+01
Generation	20:	Best = 2.048237e+04	FES = 2100	Diversity = 1.5543e+01
Generation	30:	Best = 1.635298e+04	FES = 3100	Diversity = 1.1593e+01
Generation	40:	Best = 1.389532e+04	FES = 4100	Diversity = 9.7264e+00
Generation	50:	Best = 1.310795e+04	FES = 5100	Diversity = 8.5138e+00
Generation	60:	Best = 1.289172e+04	FES = 6100	Diversity = 7.9637e+00
Generation	70:	Best = 1.267570e+04	FES = 7100	Diversity = 7.4462e+00
Generation	80:	Best = 1.194032e+04	FES = 8100	Diversity = 6.9869e+00
Generation	90:	Best = 1.126624e+04	FES = 9100	Diversity = 6.3125e+00
Generation	100:	Best = 1.070553e+04	FES = 10100	Diversity = 5.9270e+00
Generation	110:	Best = 1.004407e+04	FES = 11100	Diversity = 5.5202e+00
Generation	120:	Best = 1.004407e+04	FES = 12100	Diversity = 5.2126e+00
Generation	130:	Best = 9.980948e+03	FES = 13100	Diversity = 4.9231e+00
Generation	140:	Best = 9.894509e+03	FES = 14100	Diversity = 4.6654e+00
Generation	150:	Best = 9.794693e+03	FES = 15100	Diversity = 4.5350e+00
Generation	160:	Best = 9.751456e+03	FES = 16100	Diversity = 4.3172e+00
Generation	170:	Best = 9.658002e+03	FES = 17100	Diversity = 4.2223e+00
Generation	180:	Best = 9.608339e+03	FES = 18100	Diversity = 4.0489e+00
Generation	190:	Best = 9.538472e+03	FES = 19100	Diversity = 4.0406e+00
Generation	200:	Best = 9.491416e+03	FES = 20100	Diversity = 3.9274e+00
Generation	210:	Best = 9.388052e+03	FES = 21100	Diversity = 3.7878e+00
Generation	220:	Best = 9.326605e+03	FES = 22100	Diversity = 3.7498e+00
Generation	230:	Best = 9.303620e+03	FES = 23100	Diversity = 3.7470e+00
Generation	240:	Best = 9.303620e+03	FES = 24100	Diversity = 3.7153e+00
Generation	250:	Best = 9.303169e+03	FES = 25100	Diversity = 3.6918e+00
Generation	260:	Best = 9.273913e+03	FES = 26100	Diversity = 3.6222e+00
Generation	270:	Best = 9.191917e+03	FES = 27100	Diversity = 3.5873e+00
Generation	280:	Best = 9.104561e+03	FES = 28100	Diversity = 3.5275e+00
Generation	290:	Best = 9.034070e+03	FES = 29100	Diversity = 3.5064e+00
Generation	300:	Best = 8.949373e+03	FES = 30100	Diversity = 3.5149e+00
Generation	310:	Best = 8.947965e+03	FES = 31100	Diversity = 3.4715e+00
Generation	320:	Best = 8.945493e+03	FES = 32100	Diversity = 3.5532e+00
Generation	330:	Best = 8.870172e+03	FES = 33100	Diversity = 3.5947e+00
Generation	340:	Best = 8.857939e+03	FES = 34100	Diversity = 3.6123e+00
Generation	350:	Best = 8.623012e+03	FES = 35100	Diversity = 3.6326e+00
Generation	360:	Best = 8.556992e+03	FES = 36100	Diversity = 3.6436e+00
Generation	370:	Best = 8.457483e+03	FES = 37100	Diversity = 3.6611e+00
Generation	380:	Best = 8.278528e+03	FES = 38100	Diversity = 3.5622e+00
Generation	390:	Best = 8.040148e+03	FES = 39100	Diversity = 3.5179e+00
Generation	400:	Best = 8.018216e+03	FES = 40100	Diversity = 3.5873e+00
Generation	410:	Best = 7.988161e+03	FES = 41100	Diversity = 3.6238e+00
Generation	420:	Best = 7.958973e+03	FES = 42100	Diversity = 3.5665e+00
Generation	430:	Best = 7.897695e+03	FES = 43100	Diversity = 3.5414e+00
Generation	440:	Best = 7.890678e+03	FES = 44100	Diversity = 3.4605e+00

Generation	450:	Best = 7.860951e+03	FES = 45100	Diversity = 3.3219e+00
Generation	460:	Best = 7.853510e+03	FES = 46100	Diversity = 3.1976e+00
Generation	470:	Best = 7.833612e+03	FES = 47100	Diversity = 3.0425e+00
Generation	480:	Best = 7.815382e+03	FES = 48100	Diversity = 2.6766e+00
Generation	490:	Best = 7.784885e+03	FES = 49100	Diversity = 2.1100e+00
Generation	500:	Best = 7.736523e+03	FES = 50100	Diversity = 1.4623e+00
Generation	510:	Best = 7.680186e+03	FES = 51100	Diversity = 8.1128e-01
Generation	520:	Best = 7.573581e+03	FES = 52100	Diversity = 6.0170e-01
Generation	530:	Best = 7.516820e+03	FES = 53100	Diversity = 5.6226e-01
Generation	540:	Best = 7.450256e+03	FES = 54100	Diversity = 5.2697e-01
Generation	550:	Best = 7.420874e+03	FES = 55100	Diversity = 4.5693e-01
Generation	560:	Best = 7.366101e+03	FES = 56100	Diversity = 4.3037e-01
Generation	570:	Best = 7.310646e+03	FES = 57100	Diversity = 4.4719e-01
Generation	580:	Best = 7.254152e+03	FES = 58100	Diversity = 4.6629e-01
Generation	590:	Best = 7.173402e+03	FES = 59100	Diversity = 5.5924e-01
Generation	600:	Best = 7.110945e+03	FES = 60100	Diversity = 4.6334e-01
Generation	610:	Best = 7.061977e+03	FES = 61100	Diversity = 4.5080e-01
Generation	620:	Best = 7.031313e+03	FES = 62100	Diversity = 4.4968e-01
Generation	630:	Best = 6.982152e+03	FES = 63100	Diversity = 4.4314e-01
Generation	640:	Best = 6.956860e+03	FES = 64100	Diversity = 4.3310e-01
Generation	650:	Best = 6.932918e+03	FES = 65100	Diversity = 3.9206e-01
Generation	660:	Best = 6.905609e+03	FES = 66100	Diversity = 3.8865e-01
Generation	670:	Best = 6.893915e+03	FES = 67100	Diversity = 3.6701e-01
Generation	680:	Best = 6.875006e+03	FES = 68100	Diversity = 3.4184e-01
Generation	690:	Best = 6.861819e+03	FES = 69100	Diversity = 3.3546e-01
Generation	700:	Best = 6.846555e+03	FES = 70100	Diversity = 3.2278e-01
Generation	710:	Best = 6.830748e+03	FES = 71100	Diversity = 3.4127e-01
Generation	720:	Best = 6.821610e+03	FES = 72100	Diversity = 3.2745e-01
Generation	730:	Best = 6.812267e+03	FES = 73100	Diversity = 3.1400e-01
Generation	740:	Best = 6.800354e+03	FES = 74100	Diversity = 3.0849e-01
Generation	750:	Best = 6.795190e+03	FES = 75100	Diversity = 2.9912e-01
Generation	760:	Best = 6.787445e+03	FES = 76100	Diversity = 3.0383e-01
Generation	770:	Best = 6.778530e+03	FES = 77100	Diversity = 2.9153e-01
Generation	780:	Best = 6.776050e+03	FES = 78100	Diversity = 2.7544e-01
Generation	790:	Best = 6.762776e+03	FES = 79100	Diversity = 2.6859e-01
Generation	800:	Best = 6.759738e+03	FES = 80100	Diversity = 2.6002e-01
Generation	810:	Best = 6.752839e+03	FES = 81100	Diversity = 2.4223e-01
Generation	820:	Best = 6.748942e+03	FES = 82100	Diversity = 2.2273e-01
Generation	830:	Best = 6.748022e+03	FES = 83100	Diversity = 2.1680e-01
Generation	840:	Best = 6.744477e+03	FES = 84100	Diversity = 2.1139e-01
Generation	850:	Best = 6.740761e+03	FES = 85100	Diversity = 2.0648e-01
Generation	860:	Best = 6.737143e+03	FES = 86100	Diversity = 2.0190e-01
Generation	870:	Best = 6.736141e+03	FES = 87100	Diversity = 1.9439e-01
Generation	880:	Best = 6.736141e+03	FES = 88100	Diversity = 1.8398e-01
Generation	890:	Best = 6.732672e+03	FES = 89100	Diversity = 1.7664e-01
Generation	900:	Best = 6.731339e+03	FES = 90100	Diversity = 1.7233e-01
Generation	910:	Best = 6.731339e+03	FES = 91100	Diversity = 1.7430e-01
Generation	920:	Best = 6.728830e+03	FES = 92100	Diversity = 1.6458e-01
Generation	930:	Best = 6.727741e+03	FES = 93100	Diversity = 1.6256e-01
Generation	940:	Best = 6.727741e+03	FES = 94100	Diversity = 1.5934e-01
Generation	950:	Best = 6.725363e+03	FES = 95100	Diversity = 1.5879e-01
Generation	960:	Best = 6.724899e+03	FES = 96100	Diversity = 1.5919e-01
Generation	970:	Best = 6.723271e+03	FES = 97100	Diversity = 1.6128e-01
Generation	980:	Best = 6.722392e+03	FES = 98100	Diversity = 1.6029e-01

Generation 990:	Best = 6.721797e+03	FES = 99100	Diversity = 1.5985e-01
Generation 1000:	Best = 6.721341e+03	FES = 100100	Diversity = 1.5962e-01
Generation 1010:	Best = 6.720267e+03	FES = 101100	Diversity = 1.5744e-01
Generation 1020:	Best = 6.719820e+03	FES = 102100	Diversity = 1.5303e-01
Generation 1030:	Best = 6.719382e+03	FES = 103100	Diversity = 1.4854e-01
Generation 1040:	Best = 6.718358e+03	FES = 104100	Diversity = 1.5207e-01
Generation 1050:	Best = 6.717685e+03	FES = 105100	Diversity = 1.5035e-01
Generation 1060:	Best = 6.717231e+03	FES = 106100	Diversity = 1.4655e-01
Generation 1070:	Best = 6.715772e+03	FES = 107100	Diversity = 1.4608e-01
Generation 1080:	Best = 6.715438e+03	FES = 108100	Diversity = 1.4547e-01
Generation 1090:	Best = 6.713744e+03	FES = 109100	Diversity = 1.4404e-01
Generation 1100:	Best = 6.713744e+03	FES = 110100	Diversity = 1.4049e-01
Generation 1110:	Best = 6.713501e+03	FES = 111100	Diversity = 1.4121e-01
Generation 1120:	Best = 6.712809e+03	FES = 112100	Diversity = 1.3668e-01
Generation 1130:	Best = 6.711703e+03	FES = 113100	Diversity = 1.3474e-01
Generation 1140:	Best = 6.710818e+03	FES = 114100	Diversity = 1.3179e-01
Generation 1150:	Best = 6.710571e+03	FES = 115100	Diversity = 1.2971e-01
Generation 1160:	Best = 6.710223e+03	FES = 116100	Diversity = 1.2660e-01
Generation 1170:	Best = 6.710014e+03	FES = 117100	Diversity = 1.2595e-01
Generation 1180:	Best = 6.709793e+03	FES = 118100	Diversity = 1.2323e-01
Generation 1190:	Best = 6.708896e+03	FES = 119100	Diversity = 1.2072e-01
Generation 1200:	Best = 6.708111e+03	FES = 120100	Diversity = 1.1959e-01
Generation 1210:	Best = 6.708070e+03	FES = 121100	Diversity = 1.1917e-01
Generation 1220:	Best = 6.707991e+03	FES = 122100	Diversity = 1.1708e-01
Generation 1230:	Best = 6.707622e+03	FES = 123100	Diversity = 1.1452e-01
Generation 1240:	Best = 6.707325e+03	FES = 124100	Diversity = 1.1474e-01
Generation 1250:	Best = 6.706922e+03	FES = 125100	Diversity = 1.1176e-01
Generation 1260:	Best = 6.706638e+03	FES = 126100	Diversity = 1.1291e-01
Generation 1270:	Best = 6.706638e+03	FES = 127100	Diversity = 1.1459e-01
Generation 1280:	Best = 6.705895e+03	FES = 128100	Diversity = 1.1603e-01
Generation 1290:	Best = 6.705570e+03	FES = 129100	Diversity = 1.3494e-01
Generation 1300:	Best = 6.704836e+03	FES = 130100	Diversity = 1.4151e-01
Generation 1310:	Best = 6.702560e+03	FES = 131100	Diversity = 1.4908e-01
Generation 1320:	Best = 6.700453e+03	FES = 132100	Diversity = 1.6267e-01
Generation 1330:	Best = 6.695300e+03	FES = 133100	Diversity = 1.9997e-01
Generation 1340:	Best = 6.690369e+03	FES = 134100	Diversity = 2.4719e-01
Generation 1350:	Best = 6.661157e+03	FES = 135100	Diversity = 3.0141e-01
Generation 1360:	Best = 6.655664e+03	FES = 136100	Diversity = 3.1669e-01
Generation 1370:	Best = 6.654315e+03	FES = 137100	Diversity = 2.9510e-01
Generation 1380:	Best = 6.653826e+03	FES = 138100	Diversity = 2.8096e-01
Generation 1390:	Best = 6.650482e+03	FES = 139100	Diversity = 2.7794e-01
Generation 1400:	Best = 6.647755e+03	FES = 140100	Diversity = 2.6143e-01
Generation 1410:	Best = 6.647280e+03	FES = 141100	Diversity = 2.5636e-01
Generation 1420:	Best = 6.644675e+03	FES = 142100	Diversity = 2.5053e-01
Generation 1430:	Best = 6.641381e+03	FES = 143100	Diversity = 2.4532e-01
Generation 1440:	Best = 6.638595e+03	FES = 144100	Diversity = 2.4220e-01
Generation 1450:	Best = 6.638274e+03	FES = 145100	Diversity = 2.4613e-01
Generation 1460:	Best = 6.636445e+03	FES = 146100	Diversity = 2.4937e-01
Generation 1470:	Best = 6.633283e+03	FES = 147100	Diversity = 2.4120e-01
Generation 1480:	Best = 6.632598e+03	FES = 148100	Diversity = 2.4163e-01
Generation 1490:	Best = 6.632598e+03	FES = 149100	Diversity = 2.3881e-01
Generation 1500:	Best = 6.632495e+03	FES = 150100	Diversity = 2.3575e-01
Generation 1510:	Best = 6.630347e+03	FES = 151100	Diversity = 2.2629e-01
Generation 1520:	Best = 6.630299e+03	FES = 152100	Diversity = 2.2156e-01

Generation 1530:	Best = 6.628676e+03	FES = 153100	Diversity = 2.2986e-01
Generation 1540:	Best = 6.628676e+03	FES = 154100	Diversity = 2.2716e-01
Generation 1550:	Best = 6.625946e+03	FES = 155100	Diversity = 2.2193e-01
Generation 1560:	Best = 6.625830e+03	FES = 156100	Diversity = 2.2200e-01
Generation 1570:	Best = 6.623770e+03	FES = 157100	Diversity = 2.1632e-01
Generation 1580:	Best = 6.622346e+03	FES = 158100	Diversity = 2.1534e-01
Generation 1590:	Best = 6.622346e+03	FES = 159100	Diversity = 2.1068e-01
Generation 1600:	Best = 6.620751e+03	FES = 160100	Diversity = 2.0660e-01
Generation 1610:	Best = 6.620155e+03	FES = 161100	Diversity = 2.0208e-01
Generation 1620:	Best = 6.620102e+03	FES = 162100	Diversity = 2.0108e-01
Generation 1630:	Best = 6.619827e+03	FES = 163100	Diversity = 2.0341e-01
Generation 1640:	Best = 6.618583e+03	FES = 164100	Diversity = 1.9899e-01
Generation 1650:	Best = 6.617823e+03	FES = 165100	Diversity = 2.0020e-01
Generation 1660:	Best = 6.614192e+03	FES = 166100	Diversity = 2.0116e-01
Generation 1670:	Best = 6.614046e+03	FES = 167100	Diversity = 1.8814e-01
Generation 1680:	Best = 6.613033e+03	FES = 168100	Diversity = 1.9236e-01
Generation 1690:	Best = 6.612770e+03	FES = 169100	Diversity = 1.9943e-01
Generation 1700:	Best = 6.612207e+03	FES = 170100	Diversity = 1.9785e-01
Generation 1710:	Best = 6.611395e+03	FES = 171100	Diversity = 1.9578e-01
Generation 1720:	Best = 6.609263e+03	FES = 172100	Diversity = 2.0689e-01
Generation 1730:	Best = 6.607466e+03	FES = 173100	Diversity = 2.0539e-01
Generation 1740:	Best = 6.605325e+03	FES = 174100	Diversity = 2.0394e-01
Generation 1750:	Best = 6.601261e+03	FES = 175100	Diversity = 2.0663e-01
Generation 1760:	Best = 6.601130e+03	FES = 176100	Diversity = 1.9620e-01
Generation 1770:	Best = 6.600404e+03	FES = 177100	Diversity = 1.9474e-01
Generation 1780:	Best = 6.600040e+03	FES = 178100	Diversity = 1.9368e-01
Generation 1790:	Best = 6.599639e+03	FES = 179100	Diversity = 1.8287e-01
Generation 1800:	Best = 6.598617e+03	FES = 180100	Diversity = 1.8760e-01
Generation 1810:	Best = 6.598072e+03	FES = 181100	Diversity = 1.8508e-01
Generation 1820:	Best = 6.597746e+03	FES = 182100	Diversity = 1.8384e-01
Generation 1830:	Best = 6.596979e+03	FES = 183100	Diversity = 1.8362e-01
Generation 1840:	Best = 6.596535e+03	FES = 184100	Diversity = 1.8039e-01
Generation 1850:	Best = 6.594184e+03	FES = 185100	Diversity = 1.9199e-01
Generation 1860:	Best = 6.593378e+03	FES = 186100	Diversity = 1.9573e-01
Generation 1870:	Best = 6.592789e+03	FES = 187100	Diversity = 1.9701e-01
Generation 1880:	Best = 6.588845e+03	FES = 188100	Diversity = 1.9118e-01
Generation 1890:	Best = 6.588530e+03	FES = 189100	Diversity = 1.7661e-01
Generation 1900:	Best = 6.588129e+03	FES = 190100	Diversity = 1.8189e-01
Generation 1910:	Best = 6.586747e+03	FES = 191100	Diversity = 1.8098e-01
Generation 1920:	Best = 6.585984e+03	FES = 192100	Diversity = 1.7100e-01
Generation 1930:	Best = 6.585309e+03	FES = 193100	Diversity = 1.6948e-01
Generation 1940:	Best = 6.583546e+03	FES = 194100	Diversity = 1.7188e-01
Generation 1950:	Best = 6.582956e+03	FES = 195100	Diversity = 1.7624e-01
Generation 1960:	Best = 6.581196e+03	FES = 196100	Diversity = 1.8103e-01
Generation 1970:	Best = 6.581196e+03	FES = 197100	Diversity = 1.8681e-01
Generation 1980:	Best = 6.580504e+03	FES = 198100	Diversity = 1.9006e-01
Generation 1990:	Best = 6.579699e+03	FES = 199100	Diversity = 1.8812e-01
Generation 2000:	Best = 6.576023e+03	FES = 200100	Diversity = 1.9553e-01
Generation 2010:	Best = 6.570988e+03	FES = 201100	Diversity = 2.0956e-01
Generation 2020:	Best = 6.565299e+03	FES = 202100	Diversity = 2.3657e-01
Generation 2030:	Best = 6.561167e+03	FES = 203100	Diversity = 2.6215e-01
Generation 2040:	Best = 6.558238e+03	FES = 204100	Diversity = 2.5868e-01
Generation 2050:	Best = 6.553459e+03	FES = 205100	Diversity = 2.4916e-01
Generation 2060:	Best = 6.550526e+03	FES = 206100	Diversity = 2.4672e-01

Generation 2070:	Best = 6.549001e+03	FES = 207100	Diversity = 2.3579e-01
Generation 2080:	Best = 6.545741e+03	FES = 208100	Diversity = 2.3945e-01
Generation 2090:	Best = 6.544396e+03	FES = 209100	Diversity = 2.4040e-01
Generation 2100:	Best = 6.541983e+03	FES = 210100	Diversity = 2.4146e-01
Generation 2110:	Best = 6.535871e+03	FES = 211100	Diversity = 2.4844e-01
Generation 2120:	Best = 6.532815e+03	FES = 212100	Diversity = 2.4326e-01
Generation 2130:	Best = 6.529047e+03	FES = 213100	Diversity = 2.4322e-01
Generation 2140:	Best = 6.527650e+03	FES = 214100	Diversity = 2.3037e-01
Generation 2150:	Best = 6.522159e+03	FES = 215100	Diversity = 2.5097e-01
Generation 2160:	Best = 6.510519e+03	FES = 216100	Diversity = 2.9848e-01
Generation 2170:	Best = 6.510519e+03	FES = 217100	Diversity = 3.0912e-01
Generation 2180:	Best = 6.501735e+03	FES = 218100	Diversity = 2.9656e-01
Generation 2190:	Best = 6.496121e+03	FES = 219100	Diversity = 2.9674e-01
Generation 2200:	Best = 6.493863e+03	FES = 220100	Diversity = 2.7884e-01
Generation 2210:	Best = 6.491474e+03	FES = 221100	Diversity = 2.8067e-01
Generation 2220:	Best = 6.487623e+03	FES = 222100	Diversity = 2.6446e-01
Generation 2230:	Best = 6.483347e+03	FES = 223100	Diversity = 2.3599e-01
Generation 2240:	Best = 6.482699e+03	FES = 224100	Diversity = 2.1011e-01
Generation 2250:	Best = 6.479862e+03	FES = 225100	Diversity = 2.1248e-01
Generation 2260:	Best = 6.477830e+03	FES = 226100	Diversity = 1.9949e-01
Generation 2270:	Best = 6.477211e+03	FES = 227100	Diversity = 2.0423e-01
Generation 2280:	Best = 6.476962e+03	FES = 228100	Diversity = 1.9290e-01
Generation 2290:	Best = 6.476337e+03	FES = 229100	Diversity = 1.8230e-01
Generation 2300:	Best = 6.475930e+03	FES = 230100	Diversity = 1.8268e-01
Generation 2310:	Best = 6.475270e+03	FES = 231100	Diversity = 1.8244e-01
Generation 2320:	Best = 6.473698e+03	FES = 232100	Diversity = 1.8509e-01
Generation 2330:	Best = 6.472056e+03	FES = 233100	Diversity = 1.7559e-01
Generation 2340:	Best = 6.471999e+03	FES = 234100	Diversity = 1.7205e-01
Generation 2350:	Best = 6.471996e+03	FES = 235100	Diversity = 1.6769e-01
Generation 2360:	Best = 6.471744e+03	FES = 236100	Diversity = 1.6325e-01
Generation 2370:	Best = 6.471528e+03	FES = 237100	Diversity = 1.6621e-01
Generation 2380:	Best = 6.471516e+03	FES = 238100	Diversity = 1.6804e-01
Generation 2390:	Best = 6.470870e+03	FES = 239100	Diversity = 1.7606e-01
Generation 2400:	Best = 6.470581e+03	FES = 240100	Diversity = 1.7719e-01
Generation 2410:	Best = 6.470116e+03	FES = 241100	Diversity = 1.7671e-01
Generation 2420:	Best = 6.469875e+03	FES = 242100	Diversity = 1.7452e-01
Generation 2430:	Best = 6.468870e+03	FES = 243100	Diversity = 1.7737e-01
Generation 2440:	Best = 6.466986e+03	FES = 244100	Diversity = 1.9991e-01
Generation 2450:	Best = 6.463584e+03	FES = 245100	Diversity = 2.5902e-01
Generation 2460:	Best = 6.452653e+03	FES = 246100	Diversity = 3.1276e-01
Generation 2470:	Best = 6.448424e+03	FES = 247100	Diversity = 2.2678e-01
Generation 2480:	Best = 6.445890e+03	FES = 248100	Diversity = 1.9721e-01
Generation 2490:	Best = 6.444910e+03	FES = 249100	Diversity = 1.9178e-01
Generation 2500:	Best = 6.444910e+03	FES = 250100	Diversity = 1.8373e-01
Generation 2510:	Best = 6.441966e+03	FES = 251100	Diversity = 1.8168e-01
Generation 2520:	Best = 6.441749e+03	FES = 252100	Diversity = 1.8105e-01
Generation 2530:	Best = 6.441749e+03	FES = 253100	Diversity = 1.7899e-01
Generation 2540:	Best = 6.440382e+03	FES = 254100	Diversity = 1.8541e-01
Generation 2550:	Best = 6.440382e+03	FES = 255100	Diversity = 1.8682e-01
Generation 2560:	Best = 6.438144e+03	FES = 256100	Diversity = 1.8712e-01
Generation 2570:	Best = 6.438144e+03	FES = 257100	Diversity = 1.8670e-01
Generation 2580:	Best = 6.437864e+03	FES = 258100	Diversity = 1.9406e-01
Generation 2590:	Best = 6.436994e+03	FES = 259100	Diversity = 1.9376e-01
Generation 2600:	Best = 6.436951e+03	FES = 260100	Diversity = 1.9787e-01

Generation 2610:	Best = 6.436093e+03	FES = 261100	Diversity = 2.0604e-01
Generation 2620:	Best = 6.432851e+03	FES = 262100	Diversity = 2.0960e-01
Generation 2630:	Best = 6.432585e+03	FES = 263100	Diversity = 2.2846e-01
Generation 2640:	Best = 6.432553e+03	FES = 264100	Diversity = 2.4877e-01
Generation 2650:	Best = 6.432065e+03	FES = 265100	Diversity = 2.5274e-01
Generation 2660:	Best = 6.431899e+03	FES = 266100	Diversity = 2.5460e-01
Generation 2670:	Best = 6.430433e+03	FES = 267100	Diversity = 2.5932e-01
Generation 2680:	Best = 6.430127e+03	FES = 268100	Diversity = 2.6419e-01
Generation 2690:	Best = 6.426642e+03	FES = 269100	Diversity = 2.4482e-01
Generation 2700:	Best = 6.425678e+03	FES = 270100	Diversity = 2.3918e-01
Generation 2710:	Best = 6.425144e+03	FES = 271100	Diversity = 2.2470e-01
Generation 2720:	Best = 6.423384e+03	FES = 272100	Diversity = 2.0361e-01
Generation 2730:	Best = 6.421872e+03	FES = 273100	Diversity = 1.9642e-01
Generation 2740:	Best = 6.421856e+03	FES = 274100	Diversity = 1.9198e-01
Generation 2750:	Best = 6.418355e+03	FES = 275100	Diversity = 1.9830e-01
Generation 2760:	Best = 6.416768e+03	FES = 276100	Diversity = 2.0074e-01
Generation 2770:	Best = 6.416690e+03	FES = 277100	Diversity = 1.9641e-01
Generation 2780:	Best = 6.416690e+03	FES = 278100	Diversity = 1.9589e-01
Generation 2790:	Best = 6.416612e+03	FES = 279100	Diversity = 1.9291e-01
Generation 2800:	Best = 6.415958e+03	FES = 280100	Diversity = 1.9328e-01
Generation 2810:	Best = 6.414584e+03	FES = 281100	Diversity = 1.8810e-01
Generation 2820:	Best = 6.414550e+03	FES = 282100	Diversity = 1.8467e-01
Generation 2830:	Best = 6.414466e+03	FES = 283100	Diversity = 1.7958e-01
Generation 2840:	Best = 6.413568e+03	FES = 284100	Diversity = 1.7645e-01
Generation 2850:	Best = 6.413568e+03	FES = 285100	Diversity = 1.7770e-01
Generation 2860:	Best = 6.413565e+03	FES = 286100	Diversity = 1.7930e-01
Generation 2870:	Best = 6.412114e+03	FES = 287100	Diversity = 1.7665e-01
Generation 2880:	Best = 6.412096e+03	FES = 288100	Diversity = 1.7250e-01
Generation 2890:	Best = 6.411036e+03	FES = 289100	Diversity = 1.7845e-01
Generation 2900:	Best = 6.410654e+03	FES = 290100	Diversity = 1.7931e-01
Generation 2910:	Best = 6.409433e+03	FES = 291100	Diversity = 1.7987e-01
Generation 2920:	Best = 6.409433e+03	FES = 292100	Diversity = 1.8439e-01
Generation 2930:	Best = 6.408703e+03	FES = 293100	Diversity = 1.8114e-01
Generation 2940:	Best = 6.408676e+03	FES = 294100	Diversity = 1.7535e-01
Generation 2950:	Best = 6.408613e+03	FES = 295100	Diversity = 1.7884e-01
Generation 2960:	Best = 6.407903e+03	FES = 296100	Diversity = 1.8726e-01
Generation 2970:	Best = 6.406226e+03	FES = 297100	Diversity = 1.8580e-01
Generation 2980:	Best = 6.406122e+03	FES = 298100	Diversity = 1.8495e-01
Generation 2990:	Best = 6.405537e+03	FES = 299100	Diversity = 1.9184e-01
Generation 3000:	Best = 6.405362e+03	FES = 300100	Diversity = 1.8851e-01
Generation 3010:	Best = 6.405362e+03	FES = 301100	Diversity = 1.9453e-01
Generation 3020:	Best = 6.403760e+03	FES = 302100	Diversity = 1.9212e-01
Generation 3030:	Best = 6.403370e+03	FES = 303100	Diversity = 2.1275e-01
Generation 3040:	Best = 6.401558e+03	FES = 304100	Diversity = 2.2774e-01
Generation 3050:	Best = 6.398857e+03	FES = 305100	Diversity = 2.4751e-01
Generation 3060:	Best = 6.393941e+03	FES = 306100	Diversity = 2.6619e-01
Generation 3070:	Best = 6.392771e+03	FES = 307100	Diversity = 2.4913e-01
Generation 3080:	Best = 6.392272e+03	FES = 308100	Diversity = 2.4767e-01
Generation 3090:	Best = 6.391009e+03	FES = 309100	Diversity = 2.4253e-01
Generation 3100:	Best = 6.388991e+03	FES = 310100	Diversity = 2.4120e-01
Generation 3110:	Best = 6.388991e+03	FES = 311100	Diversity = 2.5043e-01
Generation 3120:	Best = 6.387323e+03	FES = 312100	Diversity = 2.4612e-01
Generation 3130:	Best = 6.383265e+03	FES = 313100	Diversity = 2.4972e-01
Generation 3140:	Best = 6.380498e+03	FES = 314100	Diversity = 2.6275e-01

Generation 3150:	Best = 6.376686e+03	FES = 315100	Diversity = 2.8215e-01
Generation 3160:	Best = 6.375635e+03	FES = 316100	Diversity = 2.9357e-01
Generation 3170:	Best = 6.371075e+03	FES = 317100	Diversity = 3.1359e-01
Generation 3180:	Best = 6.362511e+03	FES = 318100	Diversity = 3.5351e-01
Generation 3190:	Best = 6.361034e+03	FES = 319100	Diversity = 3.5082e-01
Generation 3200:	Best = 6.357772e+03	FES = 320100	Diversity = 3.3936e-01
Generation 3210:	Best = 6.353009e+03	FES = 321100	Diversity = 3.1697e-01
Generation 3220:	Best = 6.351745e+03	FES = 322100	Diversity = 2.7413e-01
Generation 3230:	Best = 6.347805e+03	FES = 323100	Diversity = 2.7390e-01
Generation 3240:	Best = 6.343470e+03	FES = 324100	Diversity = 2.5924e-01
Generation 3250:	Best = 6.343087e+03	FES = 325100	Diversity = 2.4105e-01
Generation 3260:	Best = 6.341460e+03	FES = 326100	Diversity = 2.2231e-01
Generation 3270:	Best = 6.340666e+03	FES = 327100	Diversity = 2.1725e-01
Generation 3280:	Best = 6.339261e+03	FES = 328100	Diversity = 2.1348e-01
Generation 3290:	Best = 6.337505e+03	FES = 329100	Diversity = 2.0475e-01
Generation 3300:	Best = 6.337016e+03	FES = 330100	Diversity = 2.0173e-01
Generation 3310:	Best = 6.334684e+03	FES = 331100	Diversity = 2.0752e-01
Generation 3320:	Best = 6.334115e+03	FES = 332100	Diversity = 2.0383e-01
Generation 3330:	Best = 6.333664e+03	FES = 333100	Diversity = 2.0276e-01
Generation 3340:	Best = 6.332863e+03	FES = 334100	Diversity = 2.0591e-01
Generation 3350:	Best = 6.332165e+03	FES = 335100	Diversity = 2.0179e-01
Generation 3360:	Best = 6.330328e+03	FES = 336100	Diversity = 2.1002e-01
Generation 3370:	Best = 6.329429e+03	FES = 337100	Diversity = 2.1540e-01
Generation 3380:	Best = 6.329266e+03	FES = 338100	Diversity = 2.1905e-01
Generation 3390:	Best = 6.329262e+03	FES = 339100	Diversity = 2.2843e-01
Generation 3400:	Best = 6.328281e+03	FES = 340100	Diversity = 2.2771e-01
Generation 3410:	Best = 6.325495e+03	FES = 341100	Diversity = 2.2527e-01
Generation 3420:	Best = 6.323418e+03	FES = 342100	Diversity = 2.2568e-01
Generation 3430:	Best = 6.323172e+03	FES = 343100	Diversity = 2.1672e-01
Generation 3440:	Best = 6.321598e+03	FES = 344100	Diversity = 2.1159e-01
Generation 3450:	Best = 6.321466e+03	FES = 345100	Diversity = 2.1210e-01
Generation 3460:	Best = 6.321365e+03	FES = 346100	Diversity = 2.1357e-01
Generation 3470:	Best = 6.321239e+03	FES = 347100	Diversity = 2.1877e-01
Generation 3480:	Best = 6.320776e+03	FES = 348100	Diversity = 2.1810e-01
Generation 3490:	Best = 6.319573e+03	FES = 349100	Diversity = 2.1582e-01
Generation 3500:	Best = 6.319573e+03	FES = 350100	Diversity = 2.1218e-01
Generation 3510:	Best = 6.318468e+03	FES = 351100	Diversity = 2.2444e-01
Generation 3520:	Best = 6.316013e+03	FES = 352100	Diversity = 2.2430e-01
Generation 3530:	Best = 6.315642e+03	FES = 353100	Diversity = 2.2180e-01
Generation 3540:	Best = 6.313223e+03	FES = 354100	Diversity = 2.3264e-01
Generation 3550:	Best = 6.312187e+03	FES = 355100	Diversity = 2.5636e-01
Generation 3560:	Best = 6.308565e+03	FES = 356100	Diversity = 2.7824e-01
Generation 3570:	Best = 6.301625e+03	FES = 357100	Diversity = 3.0990e-01
Generation 3580:	Best = 6.300881e+03	FES = 358100	Diversity = 3.5012e-01
Generation 3590:	Best = 6.300881e+03	FES = 359100	Diversity = 3.5522e-01
Generation 3600:	Best = 6.299369e+03	FES = 360100	Diversity = 2.7641e-01
Generation 3610:	Best = 6.298079e+03	FES = 361100	Diversity = 2.5587e-01
Generation 3620:	Best = 6.296390e+03	FES = 362100	Diversity = 2.4343e-01
Generation 3630:	Best = 6.296390e+03	FES = 363100	Diversity = 2.5196e-01
Generation 3640:	Best = 6.295021e+03	FES = 364100	Diversity = 2.5496e-01
Generation 3650:	Best = 6.294678e+03	FES = 365100	Diversity = 2.5574e-01
Generation 3660:	Best = 6.292710e+03	FES = 366100	Diversity = 2.6641e-01
Generation 3670:	Best = 6.292228e+03	FES = 367100	Diversity = 2.7343e-01
Generation 3680:	Best = 6.292228e+03	FES = 368100	Diversity = 2.7869e-01

Generation 3690:	Best = 6.291962e+03	FES = 369100	Diversity = 2.9025e-01
Generation 3700:	Best = 6.290631e+03	FES = 370100	Diversity = 2.9178e-01
Generation 3710:	Best = 6.288307e+03	FES = 371100	Diversity = 2.8270e-01
Generation 3720:	Best = 6.288169e+03	FES = 372100	Diversity = 2.8258e-01
Generation 3730:	Best = 6.288148e+03	FES = 373100	Diversity = 2.6466e-01
Generation 3740:	Best = 6.287831e+03	FES = 374100	Diversity = 2.6195e-01
Generation 3750:	Best = 6.286916e+03	FES = 375100	Diversity = 2.6255e-01
Generation 3760:	Best = 6.286870e+03	FES = 376100	Diversity = 2.6192e-01
Generation 3770:	Best = 6.286175e+03	FES = 377100	Diversity = 2.6237e-01
Generation 3780:	Best = 6.286175e+03	FES = 378100	Diversity = 2.6085e-01
Generation 3790:	Best = 6.286175e+03	FES = 379100	Diversity = 2.6094e-01
Generation 3800:	Best = 6.286112e+03	FES = 380100	Diversity = 2.6459e-01
Generation 3810:	Best = 6.285642e+03	FES = 381100	Diversity = 2.6346e-01
Generation 3820:	Best = 6.285357e+03	FES = 382100	Diversity = 2.6478e-01
Generation 3830:	Best = 6.285291e+03	FES = 383100	Diversity = 2.5906e-01
Generation 3840:	Best = 6.285039e+03	FES = 384100	Diversity = 2.5506e-01
Generation 3850:	Best = 6.284845e+03	FES = 385100	Diversity = 2.5146e-01
Generation 3860:	Best = 6.283636e+03	FES = 386100	Diversity = 2.5034e-01
Generation 3870:	Best = 6.283415e+03	FES = 387100	Diversity = 2.4268e-01
Generation 3880:	Best = 6.283381e+03	FES = 388100	Diversity = 2.3368e-01
Generation 3890:	Best = 6.283231e+03	FES = 389100	Diversity = 2.4385e-01
Generation 3900:	Best = 6.283193e+03	FES = 390100	Diversity = 2.5349e-01
Generation 3910:	Best = 6.283100e+03	FES = 391100	Diversity = 2.5872e-01
Generation 3920:	Best = 6.283100e+03	FES = 392100	Diversity = 2.5554e-01
Generation 3930:	Best = 6.283032e+03	FES = 393100	Diversity = 2.5041e-01
Generation 3940:	Best = 6.281676e+03	FES = 394100	Diversity = 2.4021e-01
Generation 3950:	Best = 6.280936e+03	FES = 395100	Diversity = 2.3532e-01
Generation 3960:	Best = 6.280876e+03	FES = 396100	Diversity = 2.3547e-01
Generation 3970:	Best = 6.280593e+03	FES = 397100	Diversity = 2.4274e-01
Generation 3980:	Best = 6.280436e+03	FES = 398100	Diversity = 2.3959e-01
Generation 3990:	Best = 6.280099e+03	FES = 399100	Diversity = 2.3628e-01
Generation 4000:	Best = 6.280093e+03	FES = 400100	Diversity = 2.3102e-01
Generation 4010:	Best = 6.280093e+03	FES = 401100	Diversity = 2.2987e-01
Generation 4020:	Best = 6.279033e+03	FES = 402100	Diversity = 2.2368e-01
Generation 4030:	Best = 6.279033e+03	FES = 403100	Diversity = 2.2374e-01
Generation 4040:	Best = 6.278938e+03	FES = 404100	Diversity = 2.2515e-01
Generation 4050:	Best = 6.278907e+03	FES = 405100	Diversity = 2.2864e-01
Generation 4060:	Best = 6.278516e+03	FES = 406100	Diversity = 2.3131e-01
Generation 4070:	Best = 6.278293e+03	FES = 407100	Diversity = 2.3173e-01
Generation 4080:	Best = 6.278136e+03	FES = 408100	Diversity = 2.3285e-01
Generation 4090:	Best = 6.278136e+03	FES = 409100	Diversity = 2.3330e-01
Generation 4100:	Best = 6.277758e+03	FES = 410100	Diversity = 2.2457e-01
Generation 4110:	Best = 6.277574e+03	FES = 411100	Diversity = 2.2232e-01
Generation 4120:	Best = 6.277504e+03	FES = 412100	Diversity = 2.2154e-01
Generation 4130:	Best = 6.277504e+03	FES = 413100	Diversity = 2.2558e-01
Generation 4140:	Best = 6.277142e+03	FES = 414100	Diversity = 2.2377e-01
Generation 4150:	Best = 6.277142e+03	FES = 415100	Diversity = 2.2635e-01
Generation 4160:	Best = 6.277045e+03	FES = 416100	Diversity = 2.2258e-01
Generation 4170:	Best = 6.277020e+03	FES = 417100	Diversity = 2.2376e-01
Generation 4180:	Best = 6.276957e+03	FES = 418100	Diversity = 2.2313e-01
Generation 4190:	Best = 6.276957e+03	FES = 419100	Diversity = 2.1837e-01
Generation 4200:	Best = 6.276913e+03	FES = 420100	Diversity = 2.1929e-01
Generation 4210:	Best = 6.276913e+03	FES = 421100	Diversity = 2.1888e-01
Generation 4220:	Best = 6.276806e+03	FES = 422100	Diversity = 2.1833e-01

Generation 4230:	Best = 6.276760e+03	FES = 423100	Diversity = 2.1618e-01
Generation 4240:	Best = 6.276760e+03	FES = 424100	Diversity = 2.1405e-01
Generation 4250:	Best = 6.276682e+03	FES = 425100	Diversity = 2.1449e-01
Generation 4260:	Best = 6.276661e+03	FES = 426100	Diversity = 2.2145e-01
Generation 4270:	Best = 6.276552e+03	FES = 427100	Diversity = 2.2190e-01
Generation 4280:	Best = 6.276541e+03	FES = 428100	Diversity = 2.2195e-01
Generation 4290:	Best = 6.276530e+03	FES = 429100	Diversity = 2.2058e-01
Generation 4300:	Best = 6.276518e+03	FES = 430100	Diversity = 2.2067e-01
Generation 4310:	Best = 6.276518e+03	FES = 431100	Diversity = 2.1640e-01
Generation 4320:	Best = 6.276518e+03	FES = 432100	Diversity = 2.1502e-01
Generation 4330:	Best = 6.276518e+03	FES = 433100	Diversity = 2.1686e-01
Generation 4340:	Best = 6.276471e+03	FES = 434100	Diversity = 2.1521e-01
Generation 4350:	Best = 6.276166e+03	FES = 435100	Diversity = 2.1556e-01
Generation 4360:	Best = 6.276139e+03	FES = 436100	Diversity = 2.1033e-01
Generation 4370:	Best = 6.276082e+03	FES = 437100	Diversity = 2.1097e-01
Generation 4380:	Best = 6.275863e+03	FES = 438100	Diversity = 2.1274e-01
Generation 4390:	Best = 6.275790e+03	FES = 439100	Diversity = 2.1322e-01
Generation 4400:	Best = 6.275546e+03	FES = 440100	Diversity = 2.1492e-01
Generation 4410:	Best = 6.275331e+03	FES = 441100	Diversity = 2.1857e-01
Generation 4420:	Best = 6.275225e+03	FES = 442100	Diversity = 2.1554e-01
Generation 4430:	Best = 6.275225e+03	FES = 443100	Diversity = 2.1313e-01
Generation 4440:	Best = 6.275225e+03	FES = 444100	Diversity = 2.1241e-01
Generation 4450:	Best = 6.275151e+03	FES = 445100	Diversity = 2.1161e-01
Generation 4460:	Best = 6.275121e+03	FES = 446100	Diversity = 2.0985e-01
Generation 4470:	Best = 6.275117e+03	FES = 447100	Diversity = 2.1203e-01
Generation 4480:	Best = 6.275059e+03	FES = 448100	Diversity = 2.0963e-01
Generation 4490:	Best = 6.275056e+03	FES = 449100	Diversity = 2.1169e-01
Generation 4500:	Best = 6.274873e+03	FES = 450100	Diversity = 2.1028e-01
Generation 4510:	Best = 6.274873e+03	FES = 451100	Diversity = 2.1207e-01
Generation 4520:	Best = 6.274768e+03	FES = 452100	Diversity = 2.1302e-01
Generation 4530:	Best = 6.274751e+03	FES = 453100	Diversity = 2.1244e-01
Generation 4540:	Best = 6.274751e+03	FES = 454100	Diversity = 2.0813e-01
Generation 4550:	Best = 6.274737e+03	FES = 455100	Diversity = 2.0950e-01
Generation 4560:	Best = 6.274592e+03	FES = 456100	Diversity = 2.1004e-01
Generation 4570:	Best = 6.274161e+03	FES = 457100	Diversity = 2.0804e-01
Generation 4580:	Best = 6.274161e+03	FES = 458100	Diversity = 2.0940e-01
Generation 4590:	Best = 6.274141e+03	FES = 459100	Diversity = 2.0693e-01
Generation 4600:	Best = 6.274027e+03	FES = 460100	Diversity = 2.0598e-01
Generation 4610:	Best = 6.273283e+03	FES = 461100	Diversity = 2.1154e-01
Generation 4620:	Best = 6.273277e+03	FES = 462100	Diversity = 2.1036e-01
Generation 4630:	Best = 6.273277e+03	FES = 463100	Diversity = 2.1007e-01
Generation 4640:	Best = 6.273277e+03	FES = 464100	Diversity = 2.0923e-01
Generation 4650:	Best = 6.273277e+03	FES = 465100	Diversity = 2.0642e-01
Generation 4660:	Best = 6.273277e+03	FES = 466100	Diversity = 2.0517e-01
Generation 4670:	Best = 6.273277e+03	FES = 467100	Diversity = 2.0455e-01
Generation 4680:	Best = 6.273211e+03	FES = 468100	Diversity = 2.0531e-01
Generation 4690:	Best = 6.273211e+03	FES = 469100	Diversity = 2.0647e-01
Generation 4700:	Best = 6.273193e+03	FES = 470100	Diversity = 2.0192e-01
Generation 4710:	Best = 6.273193e+03	FES = 471100	Diversity = 2.0224e-01
Generation 4720:	Best = 6.273193e+03	FES = 472100	Diversity = 2.0120e-01
Generation 4730:	Best = 6.272912e+03	FES = 473100	Diversity = 2.0084e-01
Generation 4740:	Best = 6.272889e+03	FES = 474100	Diversity = 2.0284e-01
Generation 4750:	Best = 6.272848e+03	FES = 475100	Diversity = 1.9800e-01
Generation 4760:	Best = 6.272707e+03	FES = 476100	Diversity = 1.9808e-01

Generation 4770:	Best = 6.272707e+03	FES = 477100	Diversity = 1.9442e-01
Generation 4780:	Best = 6.272676e+03	FES = 478100	Diversity = 1.9405e-01
Generation 4790:	Best = 6.272591e+03	FES = 479100	Diversity = 1.9035e-01
Generation 4800:	Best = 6.272537e+03	FES = 480100	Diversity = 1.9194e-01
Generation 4810:	Best = 6.271793e+03	FES = 481100	Diversity = 1.9606e-01
Generation 4820:	Best = 6.271710e+03	FES = 482100	Diversity = 1.9737e-01
Generation 4830:	Best = 6.271710e+03	FES = 483100	Diversity = 2.0018e-01
Generation 4840:	Best = 6.271699e+03	FES = 484100	Diversity = 2.0469e-01
Generation 4850:	Best = 6.271596e+03	FES = 485100	Diversity = 2.0431e-01
Generation 4860:	Best = 6.271517e+03	FES = 486100	Diversity = 2.0486e-01
Generation 4870:	Best = 6.271513e+03	FES = 487100	Diversity = 2.0523e-01
Generation 4880:	Best = 6.271356e+03	FES = 488100	Diversity = 2.0885e-01
Generation 4890:	Best = 6.271349e+03	FES = 489100	Diversity = 2.0807e-01
Generation 4900:	Best = 6.271346e+03	FES = 490100	Diversity = 2.0842e-01
Generation 4910:	Best = 6.271203e+03	FES = 491100	Diversity = 2.0795e-01
Generation 4920:	Best = 6.271203e+03	FES = 492100	Diversity = 2.0737e-01
Generation 4930:	Best = 6.271195e+03	FES = 493100	Diversity = 2.0296e-01
Generation 4940:	Best = 6.271195e+03	FES = 494100	Diversity = 2.0348e-01
Generation 4950:	Best = 6.271122e+03	FES = 495100	Diversity = 2.0130e-01
Generation 4960:	Best = 6.271105e+03	FES = 496100	Diversity = 2.0117e-01
Generation 4970:	Best = 6.271105e+03	FES = 497100	Diversity = 2.0160e-01
Generation 4980:	Best = 6.271068e+03	FES = 498100	Diversity = 1.9844e-01
Generation 4990:	Best = 6.271013e+03	FES = 499100	Diversity = 1.9739e-01
Generation 5000:	Best = 6.270919e+03	FES = 500100	Diversity = 1.9413e-01
Generation 5010:	Best = 6.270857e+03	FES = 501100	Diversity = 1.9636e-01
Generation 5020:	Best = 6.270852e+03	FES = 502100	Diversity = 1.9344e-01
Generation 5030:	Best = 6.270806e+03	FES = 503100	Diversity = 1.8736e-01
Generation 5040:	Best = 6.270806e+03	FES = 504100	Diversity = 1.8669e-01
Generation 5050:	Best = 6.270438e+03	FES = 505100	Diversity = 1.8550e-01
Generation 5060:	Best = 6.270438e+03	FES = 506100	Diversity = 1.8642e-01
Generation 5070:	Best = 6.270416e+03	FES = 507100	Diversity = 1.8459e-01
Generation 5080:	Best = 6.270412e+03	FES = 508100	Diversity = 1.8320e-01
Generation 5090:	Best = 6.270412e+03	FES = 509100	Diversity = 1.8075e-01
Generation 5100:	Best = 6.270112e+03	FES = 510100	Diversity = 1.8250e-01
Generation 5110:	Best = 6.270112e+03	FES = 511100	Diversity = 1.7633e-01
Generation 5120:	Best = 6.269978e+03	FES = 512100	Diversity = 1.7568e-01
Generation 5130:	Best = 6.269922e+03	FES = 513100	Diversity = 1.7569e-01
Generation 5140:	Best = 6.269874e+03	FES = 514100	Diversity = 1.7683e-01
Generation 5150:	Best = 6.269874e+03	FES = 515100	Diversity = 1.7685e-01
Generation 5160:	Best = 6.269874e+03	FES = 516100	Diversity = 1.7591e-01
Generation 5170:	Best = 6.269447e+03	FES = 517100	Diversity = 1.7742e-01
Generation 5180:	Best = 6.269445e+03	FES = 518100	Diversity = 1.7495e-01
Generation 5190:	Best = 6.269445e+03	FES = 519100	Diversity = 1.7375e-01
Generation 5200:	Best = 6.269443e+03	FES = 520100	Diversity = 1.7078e-01
Generation 5210:	Best = 6.269435e+03	FES = 521100	Diversity = 1.6587e-01
Generation 5220:	Best = 6.269422e+03	FES = 522100	Diversity = 1.6525e-01
Generation 5230:	Best = 6.269379e+03	FES = 523100	Diversity = 1.6418e-01
Generation 5240:	Best = 6.269354e+03	FES = 524100	Diversity = 1.6449e-01
Generation 5250:	Best = 6.269354e+03	FES = 525100	Diversity = 1.6438e-01
Generation 5260:	Best = 6.269354e+03	FES = 526100	Diversity = 1.6278e-01
Generation 5270:	Best = 6.269350e+03	FES = 527100	Diversity = 1.6071e-01
Generation 5280:	Best = 6.269350e+03	FES = 528100	Diversity = 1.6108e-01
Generation 5290:	Best = 6.269350e+03	FES = 529100	Diversity = 1.6086e-01
Generation 5300:	Best = 6.269244e+03	FES = 530100	Diversity = 1.6183e-01

Generation 5310:	Best = 6.269214e+03	FES = 531100	Diversity = 1.6122e-01
Generation 5320:	Best = 6.269021e+03	FES = 532100	Diversity = 1.5882e-01
Generation 5330:	Best = 6.269021e+03	FES = 533100	Diversity = 1.5848e-01
Generation 5340:	Best = 6.268960e+03	FES = 534100	Diversity = 1.5805e-01
Generation 5350:	Best = 6.268849e+03	FES = 535100	Diversity = 1.5710e-01
Generation 5360:	Best = 6.268849e+03	FES = 536100	Diversity = 1.5404e-01
Generation 5370:	Best = 6.268778e+03	FES = 537100	Diversity = 1.5199e-01
Generation 5380:	Best = 6.268776e+03	FES = 538100	Diversity = 1.5149e-01
Generation 5390:	Best = 6.268588e+03	FES = 539100	Diversity = 1.5267e-01
Generation 5400:	Best = 6.268588e+03	FES = 540100	Diversity = 1.5142e-01
Generation 5410:	Best = 6.268586e+03	FES = 541100	Diversity = 1.5352e-01
Generation 5420:	Best = 6.268546e+03	FES = 542100	Diversity = 1.5360e-01
Generation 5430:	Best = 6.268546e+03	FES = 543100	Diversity = 1.5167e-01
Generation 5440:	Best = 6.268408e+03	FES = 544100	Diversity = 1.5214e-01
Generation 5450:	Best = 6.268408e+03	FES = 545100	Diversity = 1.5227e-01
Generation 5460:	Best = 6.268382e+03	FES = 546100	Diversity = 1.5413e-01
Generation 5470:	Best = 6.268348e+03	FES = 547100	Diversity = 1.5339e-01
Generation 5480:	Best = 6.268087e+03	FES = 548100	Diversity = 1.5491e-01
Generation 5490:	Best = 6.267827e+03	FES = 549100	Diversity = 1.5673e-01
Generation 5500:	Best = 6.267808e+03	FES = 550100	Diversity = 1.5653e-01
Generation 5510:	Best = 6.267632e+03	FES = 551100	Diversity = 1.5847e-01
Generation 5520:	Best = 6.267632e+03	FES = 552100	Diversity = 1.5764e-01
Generation 5530:	Best = 6.267543e+03	FES = 553100	Diversity = 1.5813e-01
Generation 5540:	Best = 6.267400e+03	FES = 554100	Diversity = 1.5895e-01
Generation 5550:	Best = 6.267360e+03	FES = 555100	Diversity = 1.5637e-01
Generation 5560:	Best = 6.267356e+03	FES = 556100	Diversity = 1.5793e-01
Generation 5570:	Best = 6.267301e+03	FES = 557100	Diversity = 1.5493e-01
Generation 5580:	Best = 6.267289e+03	FES = 558100	Diversity = 1.5584e-01
Generation 5590:	Best = 6.267152e+03	FES = 559100	Diversity = 1.5652e-01
Generation 5600:	Best = 6.267080e+03	FES = 560100	Diversity = 1.5480e-01
Generation 5610:	Best = 6.267080e+03	FES = 561100	Diversity = 1.5440e-01
Generation 5620:	Best = 6.267080e+03	FES = 562100	Diversity = 1.5436e-01
Generation 5630:	Best = 6.267028e+03	FES = 563100	Diversity = 1.5533e-01
Generation 5640:	Best = 6.266904e+03	FES = 564100	Diversity = 1.5721e-01
Generation 5650:	Best = 6.266863e+03	FES = 565100	Diversity = 1.5820e-01
Generation 5660:	Best = 6.266863e+03	FES = 566100	Diversity = 1.5538e-01
Generation 5670:	Best = 6.266863e+03	FES = 567100	Diversity = 1.5657e-01
Generation 5680:	Best = 6.266822e+03	FES = 568100	Diversity = 1.5751e-01
Generation 5690:	Best = 6.266780e+03	FES = 569100	Diversity = 1.5888e-01
Generation 5700:	Best = 6.266767e+03	FES = 570100	Diversity = 1.5701e-01
Generation 5710:	Best = 6.266756e+03	FES = 571100	Diversity = 1.5542e-01
Generation 5720:	Best = 6.266751e+03	FES = 572100	Diversity = 1.5591e-01
Generation 5730:	Best = 6.266721e+03	FES = 573100	Diversity = 1.5765e-01
Generation 5740:	Best = 6.266668e+03	FES = 574100	Diversity = 1.5623e-01
Generation 5750:	Best = 6.266668e+03	FES = 575100	Diversity = 1.5322e-01
Generation 5760:	Best = 6.266562e+03	FES = 576100	Diversity = 1.5299e-01
Generation 5770:	Best = 6.266542e+03	FES = 577100	Diversity = 1.5425e-01
Generation 5780:	Best = 6.266526e+03	FES = 578100	Diversity = 1.5334e-01
Generation 5790:	Best = 6.266526e+03	FES = 579100	Diversity = 1.5026e-01
Generation 5800:	Best = 6.266526e+03	FES = 580100	Diversity = 1.4634e-01
Generation 5810:	Best = 6.266356e+03	FES = 581100	Diversity = 1.4774e-01
Generation 5820:	Best = 6.266356e+03	FES = 582100	Diversity = 1.4657e-01
Generation 5830:	Best = 6.266354e+03	FES = 583100	Diversity = 1.4433e-01
Generation 5840:	Best = 6.266354e+03	FES = 584100	Diversity = 1.4916e-01

Generation 5850:	Best = 6.266349e+03	FES = 585100	Diversity = 1.4871e-01
Generation 5860:	Best = 6.266319e+03	FES = 586100	Diversity = 1.4700e-01
Generation 5870:	Best = 6.265509e+03	FES = 587100	Diversity = 1.4498e-01
Generation 5880:	Best = 6.265374e+03	FES = 588100	Diversity = 1.4749e-01
Generation 5890:	Best = 6.265314e+03	FES = 589100	Diversity = 1.4413e-01
Generation 5900:	Best = 6.264940e+03	FES = 590100	Diversity = 1.4586e-01
Generation 5910:	Best = 6.264929e+03	FES = 591100	Diversity = 1.4570e-01
Generation 5920:	Best = 6.264929e+03	FES = 592100	Diversity = 1.4611e-01
Generation 5930:	Best = 6.264784e+03	FES = 593100	Diversity = 1.4071e-01
Generation 5940:	Best = 6.264770e+03	FES = 594100	Diversity = 1.3998e-01
Generation 5950:	Best = 6.264770e+03	FES = 595100	Diversity = 1.3837e-01
Generation 5960:	Best = 6.264731e+03	FES = 596100	Diversity = 1.4151e-01
Generation 5970:	Best = 6.264731e+03	FES = 597100	Diversity = 1.4017e-01
Generation 5980:	Best = 6.264644e+03	FES = 598100	Diversity = 1.3893e-01
Generation 5990:	Best = 6.264607e+03	FES = 599100	Diversity = 1.3699e-01
Generation 6000:	Best = 6.264607e+03	FES = 600100	Diversity = 1.3180e-01
Generation 6010:	Best = 6.264607e+03	FES = 601100	Diversity = 1.3364e-01
Generation 6020:	Best = 6.264079e+03	FES = 602100	Diversity = 1.3499e-01
Generation 6030:	Best = 6.264076e+03	FES = 603100	Diversity = 1.3371e-01
Generation 6040:	Best = 6.264076e+03	FES = 604100	Diversity = 1.3233e-01
Generation 6050:	Best = 6.264076e+03	FES = 605100	Diversity = 1.3122e-01
Generation 6060:	Best = 6.264076e+03	FES = 606100	Diversity = 1.3294e-01
Generation 6070:	Best = 6.263936e+03	FES = 607100	Diversity = 1.3070e-01
Generation 6080:	Best = 6.263469e+03	FES = 608100	Diversity = 1.2798e-01
Generation 6090:	Best = 6.263420e+03	FES = 609100	Diversity = 1.3074e-01
Generation 6100:	Best = 6.263419e+03	FES = 610100	Diversity = 1.3115e-01
Generation 6110:	Best = 6.263339e+03	FES = 611100	Diversity = 1.3003e-01
Generation 6120:	Best = 6.263324e+03	FES = 612100	Diversity = 1.2924e-01
Generation 6130:	Best = 6.262972e+03	FES = 613100	Diversity = 1.3115e-01
Generation 6140:	Best = 6.262794e+03	FES = 614100	Diversity = 1.2816e-01
Generation 6150:	Best = 6.262730e+03	FES = 615100	Diversity = 1.3044e-01
Generation 6160:	Best = 6.262656e+03	FES = 616100	Diversity = 1.3114e-01
Generation 6170:	Best = 6.262605e+03	FES = 617100	Diversity = 1.2875e-01
Generation 6180:	Best = 6.262605e+03	FES = 618100	Diversity = 1.2934e-01
Generation 6190:	Best = 6.262589e+03	FES = 619100	Diversity = 1.2648e-01
Generation 6200:	Best = 6.262564e+03	FES = 620100	Diversity = 1.1916e-01
Generation 6210:	Best = 6.262562e+03	FES = 621100	Diversity = 1.1819e-01
Generation 6220:	Best = 6.261618e+03	FES = 622100	Diversity = 1.1866e-01
Generation 6230:	Best = 6.261618e+03	FES = 623100	Diversity = 1.1796e-01
Generation 6240:	Best = 6.261597e+03	FES = 624100	Diversity = 1.1524e-01
Generation 6250:	Best = 6.261597e+03	FES = 625100	Diversity = 1.1845e-01
Generation 6260:	Best = 6.261572e+03	FES = 626100	Diversity = 1.1654e-01
Generation 6270:	Best = 6.261494e+03	FES = 627100	Diversity = 1.1565e-01
Generation 6280:	Best = 6.261494e+03	FES = 628100	Diversity = 1.1596e-01
Generation 6290:	Best = 6.261492e+03	FES = 629100	Diversity = 1.1908e-01
Generation 6300:	Best = 6.261492e+03	FES = 630100	Diversity = 1.2012e-01
Generation 6310:	Best = 6.261492e+03	FES = 631100	Diversity = 1.2220e-01
Generation 6320:	Best = 6.261492e+03	FES = 632100	Diversity = 1.2113e-01
Generation 6330:	Best = 6.261477e+03	FES = 633100	Diversity = 1.2030e-01
Generation 6340:	Best = 6.261444e+03	FES = 634100	Diversity = 1.1940e-01
Generation 6350:	Best = 6.261440e+03	FES = 635100	Diversity = 1.1719e-01
Generation 6360:	Best = 6.261422e+03	FES = 636100	Diversity = 1.1538e-01
Generation 6370:	Best = 6.261422e+03	FES = 637100	Diversity = 1.1592e-01
Generation 6380:	Best = 6.261270e+03	FES = 638100	Diversity = 1.1322e-01

Generation 6390:	Best = 6.261270e+03	FES = 639100	Diversity = 1.1231e-01
Generation 6400:	Best = 6.261270e+03	FES = 640100	Diversity = 1.1163e-01
Generation 6410:	Best = 6.261249e+03	FES = 641100	Diversity = 1.1243e-01
Generation 6420:	Best = 6.261221e+03	FES = 642100	Diversity = 1.1108e-01
Generation 6430:	Best = 6.261200e+03	FES = 643100	Diversity = 1.1018e-01
Generation 6440:	Best = 6.261190e+03	FES = 644100	Diversity = 1.0893e-01
Generation 6450:	Best = 6.261128e+03	FES = 645100	Diversity = 1.1043e-01
Generation 6460:	Best = 6.261001e+03	FES = 646100	Diversity = 1.0864e-01
Generation 6470:	Best = 6.260999e+03	FES = 647100	Diversity = 1.0998e-01
Generation 6480:	Best = 6.260980e+03	FES = 648100	Diversity = 1.1486e-01
Generation 6490:	Best = 6.260956e+03	FES = 649100	Diversity = 1.1565e-01
Generation 6500:	Best = 6.260919e+03	FES = 650100	Diversity = 1.1516e-01
Generation 6510:	Best = 6.260462e+03	FES = 651100	Diversity = 1.1519e-01
Generation 6520:	Best = 6.260443e+03	FES = 652100	Diversity = 1.1439e-01
Generation 6530:	Best = 6.260443e+03	FES = 653100	Diversity = 1.1289e-01
Generation 6540:	Best = 6.260353e+03	FES = 654100	Diversity = 1.1354e-01
Generation 6550:	Best = 6.260353e+03	FES = 655100	Diversity = 1.1298e-01
Generation 6560:	Best = 6.260353e+03	FES = 656100	Diversity = 1.1251e-01
Generation 6570:	Best = 6.260353e+03	FES = 657100	Diversity = 1.1148e-01
Generation 6580:	Best = 6.260347e+03	FES = 658100	Diversity = 1.1138e-01
Generation 6590:	Best = 6.260347e+03	FES = 659100	Diversity = 1.1116e-01
Generation 6600:	Best = 6.260346e+03	FES = 660100	Diversity = 1.1062e-01
Generation 6610:	Best = 6.260280e+03	FES = 661100	Diversity = 1.0936e-01
Generation 6620:	Best = 6.260280e+03	FES = 662100	Diversity = 1.0931e-01
Generation 6630:	Best = 6.260251e+03	FES = 663100	Diversity = 1.0928e-01
Generation 6640:	Best = 6.260248e+03	FES = 664100	Diversity = 1.0935e-01
Generation 6650:	Best = 6.260225e+03	FES = 665100	Diversity = 1.0975e-01
Generation 6660:	Best = 6.260210e+03	FES = 666100	Diversity = 1.1120e-01
Generation 6670:	Best = 6.260210e+03	FES = 667100	Diversity = 1.1121e-01
Generation 6680:	Best = 6.260204e+03	FES = 668100	Diversity = 1.1116e-01
Generation 6690:	Best = 6.260188e+03	FES = 669100	Diversity = 1.1098e-01
Generation 6700:	Best = 6.260171e+03	FES = 670100	Diversity = 1.0913e-01
Generation 6710:	Best = 6.260171e+03	FES = 671100	Diversity = 1.0884e-01
Generation 6720:	Best = 6.260171e+03	FES = 672100	Diversity = 1.0926e-01
Generation 6730:	Best = 6.260171e+03	FES = 673100	Diversity = 1.0923e-01
Generation 6740:	Best = 6.260101e+03	FES = 674100	Diversity = 1.0904e-01
Generation 6750:	Best = 6.260075e+03	FES = 675100	Diversity = 1.0904e-01
Generation 6760:	Best = 6.260064e+03	FES = 676100	Diversity = 1.0813e-01
Generation 6770:	Best = 6.260064e+03	FES = 677100	Diversity = 1.0598e-01
Generation 6780:	Best = 6.260060e+03	FES = 678100	Diversity = 1.0399e-01
Generation 6790:	Best = 6.260060e+03	FES = 679100	Diversity = 1.0397e-01
Generation 6800:	Best = 6.260059e+03	FES = 680100	Diversity = 1.0523e-01
Generation 6810:	Best = 6.260056e+03	FES = 681100	Diversity = 1.0512e-01
Generation 6820:	Best = 6.259983e+03	FES = 682100	Diversity = 1.0509e-01
Generation 6830:	Best = 6.259983e+03	FES = 683100	Diversity = 1.0404e-01
Generation 6840:	Best = 6.259983e+03	FES = 684100	Diversity = 1.0541e-01
Generation 6850:	Best = 6.259978e+03	FES = 685100	Diversity = 1.0495e-01
Generation 6860:	Best = 6.259978e+03	FES = 686100	Diversity = 1.0249e-01
Generation 6870:	Best = 6.259977e+03	FES = 687100	Diversity = 1.0244e-01
Generation 6880:	Best = 6.259975e+03	FES = 688100	Diversity = 1.0242e-01
Generation 6890:	Best = 6.259973e+03	FES = 689100	Diversity = 1.0236e-01
Generation 6900:	Best = 6.259967e+03	FES = 690100	Diversity = 1.0135e-01
Generation 6910:	Best = 6.259966e+03	FES = 691100	Diversity = 1.0132e-01
Generation 6920:	Best = 6.259963e+03	FES = 692100	Diversity = 1.0123e-01

Generation 6930:	Best = 6.259961e+03	FES = 693100	Diversity = 1.0208e-01
Generation 6940:	Best = 6.259951e+03	FES = 694100	Diversity = 1.0207e-01
Generation 6950:	Best = 6.259544e+03	FES = 695100	Diversity = 1.0124e-01
Generation 6960:	Best = 6.259508e+03	FES = 696100	Diversity = 1.0130e-01
Generation 6970:	Best = 6.259484e+03	FES = 697100	Diversity = 1.0028e-01
Generation 6980:	Best = 6.259484e+03	FES = 698100	Diversity = 9.9251e-02
Generation 6990:	Best = 6.259484e+03	FES = 699100	Diversity = 9.8830e-02
Generation 7000:	Best = 6.259482e+03	FES = 700100	Diversity = 9.8109e-02
Generation 7010:	Best = 6.259482e+03	FES = 701100	Diversity = 9.7697e-02
Generation 7020:	Best = 6.259482e+03	FES = 702100	Diversity = 9.7703e-02
Generation 7030:	Best = 6.259482e+03	FES = 703100	Diversity = 9.7670e-02
Generation 7040:	Best = 6.259361e+03	FES = 704100	Diversity = 9.6332e-02
Generation 7050:	Best = 6.259361e+03	FES = 705100	Diversity = 9.5943e-02
Generation 7060:	Best = 6.259361e+03	FES = 706100	Diversity = 9.6112e-02
Generation 7070:	Best = 6.259361e+03	FES = 707100	Diversity = 9.5051e-02
Generation 7080:	Best = 6.259361e+03	FES = 708100	Diversity = 9.4071e-02
Generation 7090:	Best = 6.259360e+03	FES = 709100	Diversity = 9.3785e-02
Generation 7100:	Best = 6.259348e+03	FES = 710100	Diversity = 9.3527e-02
Generation 7110:	Best = 6.259344e+03	FES = 711100	Diversity = 9.3425e-02
Generation 7120:	Best = 6.259339e+03	FES = 712100	Diversity = 9.3626e-02
Generation 7130:	Best = 6.259338e+03	FES = 713100	Diversity = 9.2778e-02
Generation 7140:	Best = 6.259338e+03	FES = 714100	Diversity = 9.2896e-02
Generation 7150:	Best = 6.259328e+03	FES = 715100	Diversity = 9.2601e-02
Generation 7160:	Best = 6.259328e+03	FES = 716100	Diversity = 9.2581e-02
Generation 7170:	Best = 6.259328e+03	FES = 717100	Diversity = 9.2564e-02
Generation 7180:	Best = 6.259327e+03	FES = 718100	Diversity = 9.2933e-02
Generation 7190:	Best = 6.259324e+03	FES = 719100	Diversity = 9.2733e-02
Generation 7200:	Best = 6.259324e+03	FES = 720100	Diversity = 9.2700e-02
Generation 7210:	Best = 6.259315e+03	FES = 721100	Diversity = 9.2440e-02
Generation 7220:	Best = 6.259315e+03	FES = 722100	Diversity = 9.2419e-02
Generation 7230:	Best = 6.259315e+03	FES = 723100	Diversity = 9.2385e-02
Generation 7240:	Best = 6.259313e+03	FES = 724100	Diversity = 9.1727e-02
Generation 7250:	Best = 6.259304e+03	FES = 725100	Diversity = 9.1697e-02
Generation 7260:	Best = 6.259304e+03	FES = 726100	Diversity = 9.1253e-02
Generation 7270:	Best = 6.259304e+03	FES = 727100	Diversity = 9.1314e-02
Generation 7280:	Best = 6.259304e+03	FES = 728100	Diversity = 9.1284e-02
Generation 7290:	Best = 6.259267e+03	FES = 729100	Diversity = 8.8914e-02
Generation 7300:	Best = 6.259265e+03	FES = 730100	Diversity = 8.8624e-02
Generation 7310:	Best = 6.259264e+03	FES = 731100	Diversity = 8.8442e-02
Generation 7320:	Best = 6.259260e+03	FES = 732100	Diversity = 8.8795e-02
Generation 7330:	Best = 6.259256e+03	FES = 733100	Diversity = 8.9432e-02
Generation 7340:	Best = 6.259254e+03	FES = 734100	Diversity = 8.9251e-02
Generation 7350:	Best = 6.259242e+03	FES = 735100	Diversity = 8.9879e-02
Generation 7360:	Best = 6.259242e+03	FES = 736100	Diversity = 8.9633e-02
Generation 7370:	Best = 6.259242e+03	FES = 737100	Diversity = 8.7945e-02
Generation 7380:	Best = 6.259242e+03	FES = 738100	Diversity = 8.7180e-02
Generation 7390:	Best = 6.259242e+03	FES = 739100	Diversity = 8.7031e-02
Generation 7400:	Best = 6.259234e+03	FES = 740100	Diversity = 8.6837e-02
Generation 7410:	Best = 6.259234e+03	FES = 741100	Diversity = 8.7630e-02
Generation 7420:	Best = 6.259225e+03	FES = 742100	Diversity = 8.6972e-02
Generation 7430:	Best = 6.259225e+03	FES = 743100	Diversity = 8.6435e-02
Generation 7440:	Best = 6.259225e+03	FES = 744100	Diversity = 8.5839e-02
Generation 7450:	Best = 6.259223e+03	FES = 745100	Diversity = 8.5830e-02
Generation 7460:	Best = 6.259222e+03	FES = 746100	Diversity = 7.9167e-02

Generation 7470:	Best = 6.259222e+03	FES = 747100	Diversity = 7.8252e-02
Generation 7480:	Best = 6.259219e+03	FES = 748100	Diversity = 7.7143e-02
Generation 7490:	Best = 6.259211e+03	FES = 749100	Diversity = 7.6074e-02
Generation 7500:	Best = 6.259211e+03	FES = 750100	Diversity = 7.5921e-02
Generation 7510:	Best = 6.259211e+03	FES = 751100	Diversity = 7.6009e-02
Generation 7520:	Best = 6.259210e+03	FES = 752100	Diversity = 7.5530e-02
Generation 7530:	Best = 6.259209e+03	FES = 753100	Diversity = 7.3557e-02
Generation 7540:	Best = 6.259209e+03	FES = 754100	Diversity = 7.0705e-02
Generation 7550:	Best = 6.259209e+03	FES = 755100	Diversity = 7.0449e-02
Generation 7560:	Best = 6.259209e+03	FES = 756100	Diversity = 7.0443e-02
Generation 7570:	Best = 6.259206e+03	FES = 757100	Diversity = 6.9481e-02
Generation 7580:	Best = 6.259206e+03	FES = 758100	Diversity = 6.9340e-02
Generation 7590:	Best = 6.259206e+03	FES = 759100	Diversity = 7.0074e-02
Generation 7600:	Best = 6.259203e+03	FES = 760100	Diversity = 7.0090e-02
Generation 7610:	Best = 6.259203e+03	FES = 761100	Diversity = 7.0073e-02
Generation 7620:	Best = 6.259202e+03	FES = 762100	Diversity = 6.9647e-02
Generation 7630:	Best = 6.259202e+03	FES = 763100	Diversity = 6.9643e-02
Generation 7640:	Best = 6.259202e+03	FES = 764100	Diversity = 6.9351e-02
Generation 7650:	Best = 6.259202e+03	FES = 765100	Diversity = 6.9341e-02
Generation 7660:	Best = 6.259201e+03	FES = 766100	Diversity = 6.9327e-02
Generation 7670:	Best = 6.259201e+03	FES = 767100	Diversity = 6.9809e-02
Generation 7680:	Best = 6.259201e+03	FES = 768100	Diversity = 6.9431e-02
Generation 7690:	Best = 6.259201e+03	FES = 769100	Diversity = 6.9321e-02
Generation 7700:	Best = 6.259201e+03	FES = 770100	Diversity = 6.9345e-02
Generation 7710:	Best = 6.259201e+03	FES = 771100	Diversity = 6.9297e-02
Generation 7720:	Best = 6.259201e+03	FES = 772100	Diversity = 6.9183e-02
Generation 7730:	Best = 6.259201e+03	FES = 773100	Diversity = 6.8864e-02
Generation 7740:	Best = 6.259201e+03	FES = 774100	Diversity = 6.8019e-02
Generation 7750:	Best = 6.259201e+03	FES = 775100	Diversity = 6.7885e-02
Generation 7760:	Best = 6.259201e+03	FES = 776100	Diversity = 6.7688e-02
Generation 7770:	Best = 6.259201e+03	FES = 777100	Diversity = 6.7631e-02
Generation 7780:	Best = 6.259201e+03	FES = 778100	Diversity = 6.6953e-02
Generation 7790:	Best = 6.259200e+03	FES = 779100	Diversity = 6.7382e-02
Generation 7800:	Best = 6.259194e+03	FES = 780100	Diversity = 6.7247e-02
Generation 7810:	Best = 6.259194e+03	FES = 781100	Diversity = 6.7247e-02
Generation 7820:	Best = 6.259194e+03	FES = 782100	Diversity = 6.7239e-02
Generation 7830:	Best = 6.259194e+03	FES = 783100	Diversity = 6.8115e-02
Generation 7840:	Best = 6.259191e+03	FES = 784100	Diversity = 6.8374e-02
Generation 7850:	Best = 6.259191e+03	FES = 785100	Diversity = 6.8372e-02
Generation 7860:	Best = 6.259191e+03	FES = 786100	Diversity = 6.8991e-02
Generation 7870:	Best = 6.259191e+03	FES = 787100	Diversity = 6.8971e-02
Generation 7880:	Best = 6.259191e+03	FES = 788100	Diversity = 6.8976e-02
Generation 7890:	Best = 6.259191e+03	FES = 789100	Diversity = 6.8829e-02
Generation 7900:	Best = 6.259191e+03	FES = 790100	Diversity = 6.8194e-02
Generation 7910:	Best = 6.259191e+03	FES = 791100	Diversity = 6.8331e-02
Generation 7920:	Best = 6.259191e+03	FES = 792100	Diversity = 6.8332e-02
Generation 7930:	Best = 6.259189e+03	FES = 793100	Diversity = 6.8331e-02
Generation 7940:	Best = 6.259176e+03	FES = 794100	Diversity = 6.8333e-02
Generation 7950:	Best = 6.259176e+03	FES = 795100	Diversity = 6.7908e-02
Generation 7960:	Best = 6.259176e+03	FES = 796100	Diversity = 6.7909e-02
Generation 7970:	Best = 6.259175e+03	FES = 797100	Diversity = 6.7914e-02
Generation 7980:	Best = 6.259175e+03	FES = 798100	Diversity = 6.6749e-02
Generation 7990:	Best = 6.259175e+03	FES = 799100	Diversity = 6.5946e-02
Generation 8000:	Best = 6.259175e+03	FES = 800100	Diversity = 6.6138e-02

Generation 8010:	Best = 6.259175e+03	FES = 801100	Diversity = 6.6143e-02
Generation 8020:	Best = 6.259175e+03	FES = 802100	Diversity = 6.5725e-02
Generation 8030:	Best = 6.259174e+03	FES = 803100	Diversity = 6.5811e-02
Generation 8040:	Best = 6.259173e+03	FES = 804100	Diversity = 6.5733e-02
Generation 8050:	Best = 6.259173e+03	FES = 805100	Diversity = 6.5731e-02
Generation 8060:	Best = 6.259173e+03	FES = 806100	Diversity = 6.4878e-02
Generation 8070:	Best = 6.259173e+03	FES = 807100	Diversity = 6.4870e-02
Generation 8080:	Best = 6.259169e+03	FES = 808100	Diversity = 6.4859e-02
Generation 8090:	Best = 6.259167e+03	FES = 809100	Diversity = 6.3935e-02
Generation 8100:	Best = 6.259167e+03	FES = 810100	Diversity = 6.3509e-02
Generation 8110:	Best = 6.259157e+03	FES = 811100	Diversity = 6.3564e-02
Generation 8120:	Best = 6.259157e+03	FES = 812100	Diversity = 6.2309e-02
Generation 8130:	Best = 6.259156e+03	FES = 813100	Diversity = 6.1682e-02
Generation 8140:	Best = 6.259156e+03	FES = 814100	Diversity = 6.1881e-02
Generation 8150:	Best = 6.259156e+03	FES = 815100	Diversity = 6.2026e-02
Generation 8160:	Best = 6.259154e+03	FES = 816100	Diversity = 6.1912e-02
Generation 8170:	Best = 6.259153e+03	FES = 817100	Diversity = 6.1909e-02
Generation 8180:	Best = 6.259153e+03	FES = 818100	Diversity = 6.1905e-02
Generation 8190:	Best = 6.259153e+03	FES = 819100	Diversity = 6.1941e-02
Generation 8200:	Best = 6.259148e+03	FES = 820100	Diversity = 6.1738e-02
Generation 8210:	Best = 6.259138e+03	FES = 821100	Diversity = 6.1738e-02
Generation 8220:	Best = 6.259134e+03	FES = 822100	Diversity = 6.1308e-02
Generation 8230:	Best = 6.259134e+03	FES = 823100	Diversity = 6.1389e-02
Generation 8240:	Best = 6.259132e+03	FES = 824100	Diversity = 6.1396e-02
Generation 8250:	Best = 6.259131e+03	FES = 825100	Diversity = 6.1162e-02
Generation 8260:	Best = 6.259131e+03	FES = 826100	Diversity = 6.1163e-02
Generation 8270:	Best = 6.259131e+03	FES = 827100	Diversity = 6.1160e-02
Generation 8280:	Best = 6.259128e+03	FES = 828100	Diversity = 6.1154e-02
Generation 8290:	Best = 6.259128e+03	FES = 829100	Diversity = 6.1152e-02
Generation 8300:	Best = 6.259127e+03	FES = 830100	Diversity = 6.1148e-02
Generation 8310:	Best = 6.259126e+03	FES = 831100	Diversity = 6.1121e-02
Generation 8320:	Best = 6.259126e+03	FES = 832100	Diversity = 6.0909e-02
Generation 8330:	Best = 6.259126e+03	FES = 833100	Diversity = 6.0916e-02
Generation 8340:	Best = 6.259126e+03	FES = 834100	Diversity = 6.0918e-02
Generation 8350:	Best = 6.259125e+03	FES = 835100	Diversity = 6.0924e-02
Generation 8360:	Best = 6.259122e+03	FES = 836100	Diversity = 6.0930e-02
Generation 8370:	Best = 6.259122e+03	FES = 837100	Diversity = 6.0926e-02
Generation 8380:	Best = 6.259122e+03	FES = 838100	Diversity = 6.0804e-02
Generation 8390:	Best = 6.259122e+03	FES = 839100	Diversity = 6.0755e-02
Generation 8400:	Best = 6.259120e+03	FES = 840100	Diversity = 6.0617e-02
Generation 8410:	Best = 6.259116e+03	FES = 841100	Diversity = 6.0143e-02
Generation 8420:	Best = 6.259108e+03	FES = 842100	Diversity = 6.0139e-02
Generation 8430:	Best = 6.259108e+03	FES = 843100	Diversity = 6.0158e-02
Generation 8440:	Best = 6.259108e+03	FES = 844100	Diversity = 5.9678e-02
Generation 8450:	Best = 6.259069e+03	FES = 845100	Diversity = 5.9683e-02
Generation 8460:	Best = 6.259069e+03	FES = 846100	Diversity = 5.9679e-02
Generation 8470:	Best = 6.259069e+03	FES = 847100	Diversity = 5.9872e-02
Generation 8480:	Best = 6.259069e+03	FES = 848100	Diversity = 5.9866e-02
Generation 8490:	Best = 6.259069e+03	FES = 849100	Diversity = 5.9854e-02
Generation 8500:	Best = 6.259069e+03	FES = 850100	Diversity = 5.9849e-02
Generation 8510:	Best = 6.259065e+03	FES = 851100	Diversity = 5.9320e-02
Generation 8520:	Best = 6.259009e+03	FES = 852100	Diversity = 5.9623e-02
Generation 8530:	Best = 6.259009e+03	FES = 853100	Diversity = 5.9617e-02
Generation 8540:	Best = 6.259009e+03	FES = 854100	Diversity = 5.9480e-02

Generation 8550:	Best = 6.259009e+03	FES = 855100	Diversity = 5.9278e-02
Generation 8560:	Best = 6.259009e+03	FES = 856100	Diversity = 5.9189e-02
Generation 8570:	Best = 6.259009e+03	FES = 857100	Diversity = 5.9695e-02
Generation 8580:	Best = 6.259008e+03	FES = 858100	Diversity = 5.9261e-02
Generation 8590:	Best = 6.259007e+03	FES = 859100	Diversity = 5.8618e-02
Generation 8600:	Best = 6.259007e+03	FES = 860100	Diversity = 5.8099e-02
Generation 8610:	Best = 6.259007e+03	FES = 861100	Diversity = 5.5032e-02
Generation 8620:	Best = 6.259007e+03	FES = 862100	Diversity = 5.2646e-02
Generation 8630:	Best = 6.259006e+03	FES = 863100	Diversity = 5.2559e-02
Generation 8640:	Best = 6.259006e+03	FES = 864100	Diversity = 5.0929e-02
Generation 8650:	Best = 6.259006e+03	FES = 865100	Diversity = 5.0768e-02
Generation 8660:	Best = 6.259006e+03	FES = 866100	Diversity = 4.8015e-02
Generation 8670:	Best = 6.259006e+03	FES = 867100	Diversity = 4.5892e-02
Generation 8680:	Best = 6.258880e+03	FES = 868100	Diversity = 4.4448e-02
Generation 8690:	Best = 6.258879e+03	FES = 869100	Diversity = 4.4252e-02
Generation 8700:	Best = 6.258879e+03	FES = 870100	Diversity = 4.4354e-02
Generation 8710:	Best = 6.258873e+03	FES = 871100	Diversity = 4.3009e-02
Generation 8720:	Best = 6.258873e+03	FES = 872100	Diversity = 4.3049e-02
Generation 8730:	Best = 6.258873e+03	FES = 873100	Diversity = 4.3051e-02
Generation 8740:	Best = 6.258871e+03	FES = 874100	Diversity = 4.2909e-02
Generation 8750:	Best = 6.258871e+03	FES = 875100	Diversity = 4.2061e-02
Generation 8760:	Best = 6.258868e+03	FES = 876100	Diversity = 4.1443e-02
Generation 8770:	Best = 6.258868e+03	FES = 877100	Diversity = 4.0532e-02
Generation 8780:	Best = 6.258868e+03	FES = 878100	Diversity = 3.9057e-02
Generation 8790:	Best = 6.258868e+03	FES = 879100	Diversity = 3.7431e-02
Generation 8800:	Best = 6.258867e+03	FES = 880100	Diversity = 3.5844e-02
Generation 8810:	Best = 6.258862e+03	FES = 881100	Diversity = 3.6381e-02
Generation 8820:	Best = 6.258856e+03	FES = 882100	Diversity = 3.5431e-02
Generation 8830:	Best = 6.258794e+03	FES = 883100	Diversity = 3.3574e-02
Generation 8840:	Best = 6.258794e+03	FES = 884100	Diversity = 3.3898e-02
Generation 8850:	Best = 6.258794e+03	FES = 885100	Diversity = 3.2932e-02
Generation 8860:	Best = 6.258793e+03	FES = 886100	Diversity = 3.1728e-02
Generation 8870:	Best = 6.258793e+03	FES = 887100	Diversity = 3.0934e-02
Generation 8880:	Best = 6.258792e+03	FES = 888100	Diversity = 3.0311e-02
Generation 8890:	Best = 6.258792e+03	FES = 889100	Diversity = 2.9522e-02
Generation 8900:	Best = 6.258729e+03	FES = 890100	Diversity = 2.8615e-02
Generation 8910:	Best = 6.258729e+03	FES = 891100	Diversity = 2.3707e-02
Generation 8920:	Best = 6.258724e+03	FES = 892100	Diversity = 2.3996e-02
Generation 8930:	Best = 6.258724e+03	FES = 893100	Diversity = 2.3479e-02
Generation 8940:	Best = 6.258714e+03	FES = 894100	Diversity = 2.0551e-02
Generation 8950:	Best = 6.258714e+03	FES = 895100	Diversity = 1.7607e-02
Generation 8960:	Best = 6.258712e+03	FES = 896100	Diversity = 1.7036e-02
Generation 8970:	Best = 6.258712e+03	FES = 897100	Diversity = 1.6515e-02
Generation 8980:	Best = 6.258706e+03	FES = 898100	Diversity = 1.6019e-02
Generation 8990:	Best = 6.258699e+03	FES = 899100	Diversity = 1.4398e-02
Generation 9000:	Best = 6.258697e+03	FES = 900100	Diversity = 1.2520e-02
Generation 9010:	Best = 6.258697e+03	FES = 901100	Diversity = 1.1464e-02
Generation 9020:	Best = 6.258694e+03	FES = 902100	Diversity = 1.1195e-02
Generation 9030:	Best = 6.258694e+03	FES = 903100	Diversity = 1.0619e-02
Generation 9040:	Best = 6.258680e+03	FES = 904100	Diversity = 1.0264e-02
Generation 9050:	Best = 6.258667e+03	FES = 905100	Diversity = 1.0085e-02
Generation 9060:	Best = 6.258667e+03	FES = 906100	Diversity = 1.0004e-02
Generation 9070:	Best = 6.258652e+03	FES = 907100	Diversity = 9.0289e-03
Generation 9080:	Best = 6.258652e+03	FES = 908100	Diversity = 8.3603e-03

Generation 9090:	Best = 6.258652e+03	FES = 909100	Diversity = 8.3786e-03
Generation 9100:	Best = 6.258651e+03	FES = 910100	Diversity = 8.3633e-03
Generation 9110:	Best = 6.258647e+03	FES = 911100	Diversity = 8.3130e-03
Generation 9120:	Best = 6.258643e+03	FES = 912100	Diversity = 8.2737e-03
Generation 9130:	Best = 6.258643e+03	FES = 913100	Diversity = 6.2361e-03
Generation 9140:	Best = 6.258633e+03	FES = 914100	Diversity = 4.9274e-03
Generation 9150:	Best = 6.258633e+03	FES = 915100	Diversity = 4.3612e-03
Generation 9160:	Best = 6.258633e+03	FES = 916100	Diversity = 4.0409e-03
Generation 9170:	Best = 6.258630e+03	FES = 917100	Diversity = 3.9440e-03
Generation 9180:	Best = 6.258630e+03	FES = 918100	Diversity = 3.8987e-03
Generation 9190:	Best = 6.258628e+03	FES = 919100	Diversity = 3.6581e-03
Generation 9200:	Best = 6.258621e+03	FES = 920100	Diversity = 3.6297e-03
Generation 9210:	Best = 6.258615e+03	FES = 921100	Diversity = 3.4825e-03
Generation 9220:	Best = 6.258615e+03	FES = 922100	Diversity = 3.3067e-03
Generation 9230:	Best = 6.258614e+03	FES = 923100	Diversity = 3.0226e-03
Generation 9240:	Best = 6.258607e+03	FES = 924100	Diversity = 2.8624e-03
Generation 9250:	Best = 6.258602e+03	FES = 925100	Diversity = 2.8782e-03
Generation 9260:	Best = 6.258600e+03	FES = 926100	Diversity = 2.9007e-03
Generation 9270:	Best = 6.258596e+03	FES = 927100	Diversity = 2.7551e-03
Generation 9280:	Best = 6.258591e+03	FES = 928100	Diversity = 2.9296e-03
Generation 9290:	Best = 6.258587e+03	FES = 929100	Diversity = 3.0067e-03
Generation 9300:	Best = 6.258582e+03	FES = 930100	Diversity = 2.8714e-03
Generation 9310:	Best = 6.258581e+03	FES = 931100	Diversity = 2.7821e-03
Generation 9320:	Best = 6.258580e+03	FES = 932100	Diversity = 2.9076e-03
Generation 9330:	Best = 6.258572e+03	FES = 933100	Diversity = 2.8868e-03
Generation 9340:	Best = 6.258572e+03	FES = 934100	Diversity = 2.9329e-03
Generation 9350:	Best = 6.258570e+03	FES = 935100	Diversity = 2.8802e-03
Generation 9360:	Best = 6.258568e+03	FES = 936100	Diversity = 2.9067e-03
Generation 9370:	Best = 6.258567e+03	FES = 937100	Diversity = 2.7940e-03
Generation 9380:	Best = 6.258565e+03	FES = 938100	Diversity = 2.8121e-03
Generation 9390:	Best = 6.258562e+03	FES = 939100	Diversity = 2.6261e-03
Generation 9400:	Best = 6.258555e+03	FES = 940100	Diversity = 2.5493e-03
Generation 9410:	Best = 6.258550e+03	FES = 941100	Diversity = 2.6634e-03
Generation 9420:	Best = 6.258550e+03	FES = 942100	Diversity = 2.4097e-03
Generation 9430:	Best = 6.258547e+03	FES = 943100	Diversity = 2.3950e-03
Generation 9440:	Best = 6.258547e+03	FES = 944100	Diversity = 2.2572e-03
Generation 9450:	Best = 6.258545e+03	FES = 945100	Diversity = 1.7772e-03
Generation 9460:	Best = 6.258540e+03	FES = 946100	Diversity = 2.0579e-03
Generation 9470:	Best = 6.258540e+03	FES = 947100	Diversity = 2.1047e-03
Generation 9480:	Best = 6.258538e+03	FES = 948100	Diversity = 2.0401e-03
Generation 9490:	Best = 6.258538e+03	FES = 949100	Diversity = 1.8983e-03
Generation 9500:	Best = 6.258536e+03	FES = 950100	Diversity = 1.9468e-03
Generation 9510:	Best = 6.258533e+03	FES = 951100	Diversity = 1.7201e-03
Generation 9520:	Best = 6.258532e+03	FES = 952100	Diversity = 1.6222e-03
Generation 9530:	Best = 6.258531e+03	FES = 953100	Diversity = 1.4660e-03
Generation 9540:	Best = 6.258531e+03	FES = 954100	Diversity = 1.4682e-03
Generation 9550:	Best = 6.258530e+03	FES = 955100	Diversity = 1.4252e-03
Generation 9560:	Best = 6.258530e+03	FES = 956100	Diversity = 1.6283e-03
Generation 9570:	Best = 6.258528e+03	FES = 957100	Diversity = 1.6783e-03
Generation 9580:	Best = 6.258528e+03	FES = 958100	Diversity = 1.4431e-03
Generation 9590:	Best = 6.258525e+03	FES = 959100	Diversity = 1.3811e-03
Generation 9600:	Best = 6.258525e+03	FES = 960100	Diversity = 1.5275e-03
Generation 9610:	Best = 6.258515e+03	FES = 961100	Diversity = 1.7174e-03
Generation 9620:	Best = 6.258515e+03	FES = 962100	Diversity = 1.7621e-03

Generation 9630:	Best = 6.258515e+03	FES = 963100	Diversity = 1.6809e-03
Generation 9640:	Best = 6.258515e+03	FES = 964100	Diversity = 1.7617e-03
Generation 9650:	Best = 6.258514e+03	FES = 965100	Diversity = 1.6674e-03
Generation 9660:	Best = 6.258514e+03	FES = 966100	Diversity = 1.6359e-03
Generation 9670:	Best = 6.258514e+03	FES = 967100	Diversity = 1.5316e-03
Generation 9680:	Best = 6.258514e+03	FES = 968100	Diversity = 1.4891e-03
Generation 9690:	Best = 6.258512e+03	FES = 969100	Diversity = 1.5019e-03
Generation 9700:	Best = 6.258512e+03	FES = 970100	Diversity = 1.3343e-03
Generation 9710:	Best = 6.258508e+03	FES = 971100	Diversity = 1.3952e-03
Generation 9720:	Best = 6.258507e+03	FES = 972100	Diversity = 1.4812e-03
Generation 9730:	Best = 6.258504e+03	FES = 973100	Diversity = 1.5632e-03
Generation 9740:	Best = 6.258504e+03	FES = 974100	Diversity = 1.6844e-03
Generation 9750:	Best = 6.258504e+03	FES = 975100	Diversity = 1.4617e-03
Generation 9760:	Best = 6.258504e+03	FES = 976100	Diversity = 1.3810e-03
Generation 9770:	Best = 6.258501e+03	FES = 977100	Diversity = 1.1721e-03
Generation 9780:	Best = 6.258501e+03	FES = 978100	Diversity = 1.2168e-03
Generation 9790:	Best = 6.258500e+03	FES = 979100	Diversity = 1.2295e-03
Generation 9800:	Best = 6.258499e+03	FES = 980100	Diversity = 1.1105e-03
Generation 9810:	Best = 6.258497e+03	FES = 981100	Diversity = 9.5738e-04
Generation 9820:	Best = 6.258493e+03	FES = 982100	Diversity = 1.1231e-03
Generation 9830:	Best = 6.258493e+03	FES = 983100	Diversity = 1.1694e-03
Generation 9840:	Best = 6.258493e+03	FES = 984100	Diversity = 1.0736e-03
Generation 9850:	Best = 6.258492e+03	FES = 985100	Diversity = 1.0477e-03
Generation 9860:	Best = 6.258492e+03	FES = 986100	Diversity = 9.4701e-04
Generation 9870:	Best = 6.258491e+03	FES = 987100	Diversity = 8.8132e-04
Generation 9880:	Best = 6.258490e+03	FES = 988100	Diversity = 8.3668e-04
Generation 9890:	Best = 6.258488e+03	FES = 989100	Diversity = 7.7075e-04
Generation 9900:	Best = 6.258488e+03	FES = 990100	Diversity = 8.1088e-04
Generation 9910:	Best = 6.258486e+03	FES = 991100	Diversity = 8.4630e-04
Generation 9920:	Best = 6.258485e+03	FES = 992100	Diversity = 1.0453e-03
Generation 9930:	Best = 6.258485e+03	FES = 993100	Diversity = 9.8934e-04
Generation 9940:	Best = 6.258481e+03	FES = 994100	Diversity = 1.0402e-03
Generation 9950:	Best = 6.258480e+03	FES = 995100	Diversity = 9.7585e-04
Generation 9960:	Best = 6.258480e+03	FES = 996100	Diversity = 8.8944e-04
Generation 9970:	Best = 6.258480e+03	FES = 997100	Diversity = 8.9574e-04
Generation 9980:	Best = 6.258480e+03	FES = 998100	Diversity = 8.7746e-04
Generation 9990:	Best = 6.258479e+03	FES = 999100	Diversity = 7.9547e-04

Optimization completed!

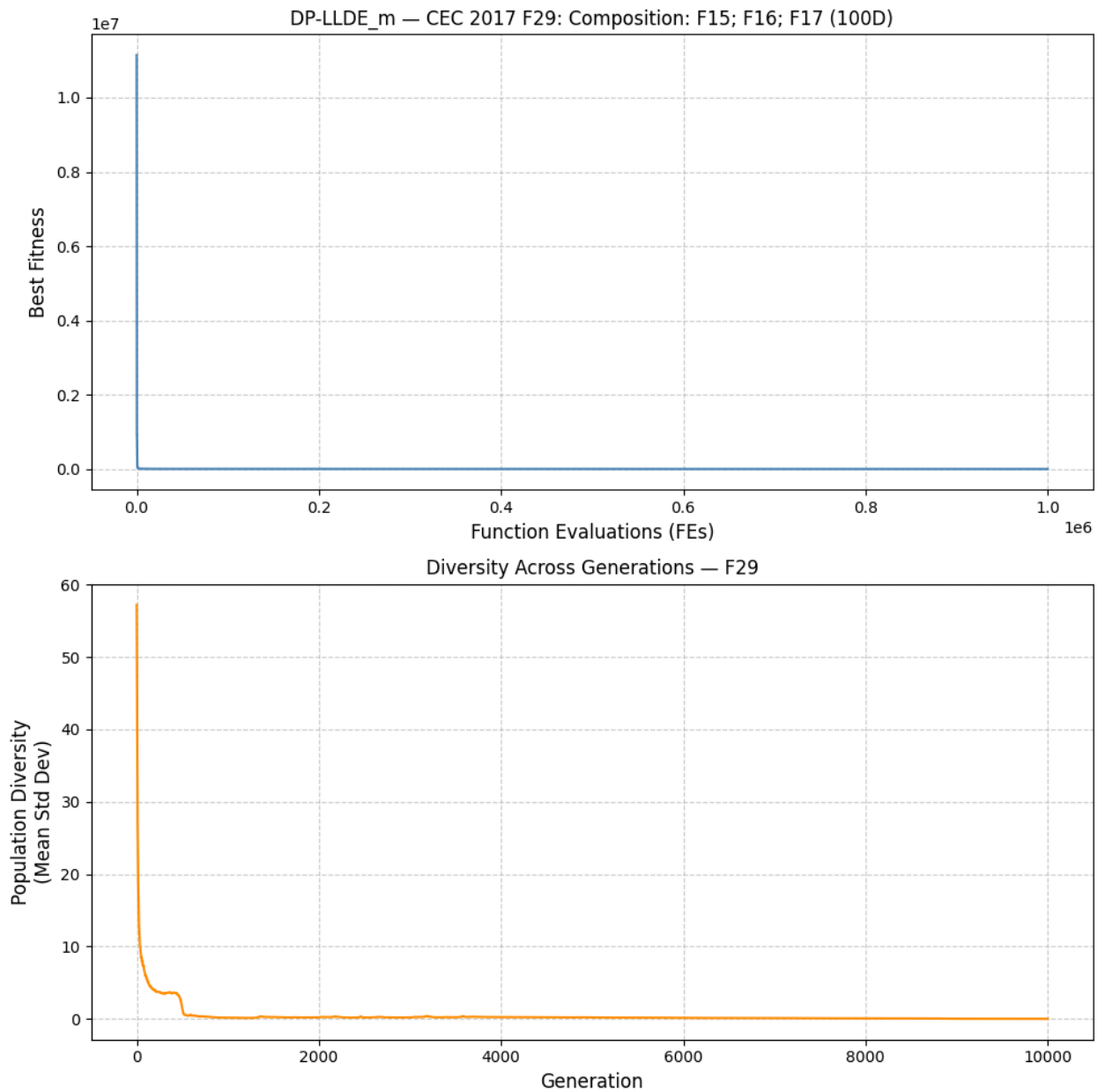
Final Best Fitness = 6.258478e+03

Total Function Evaluations = 1000000

```
=====
=
Results:
  Best Fitness          : 6.258478e+03
  Global Optimum        : 2.900000e+03
  Error (|best-optimum|) : 3.358478e+03
  Best Solution (5 dim)  : [ -5.80074748 -32.92033455 -28.14921772  80.57232518
12.75508179]
  D^(0) (init. diversity): 5.7220e+01
=====
```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results_2/all_results_che

ckpoint.pkl



Graph saved as convergence_F29.png

✓ Plot saved to: /content/drive/My Drive/thesis_results_2/plots/convergence_F29.png

```
In [31]: # =====  
# F30 — Composition Function 10 (F15; F18; F19)  
# =====  
run_and_plot('F30', make_wrapper(composition.f30), 3000.0, 'Composition: F15;
```

=====

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DP-LLDE_m on CEC 2017 F30 – Single Run

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Function : Composition: F15; F18; F19 (100D)

Bounds : [-100, 100]

Global optimum : 3000.0

Initial $D^{(0)} = 5.7220e+01$ | Diversity threshold = 5.7220e+00

Generation 0: Best Fitness = 7.713945e+11 | Diversity = 5.7220e+01

Generation	10:	Best = 2.061333e+11	FES = 1100	Diversity = 2.4771e+01
Generation	20:	Best = 8.699910e+10	FES = 2100	Diversity = 1.2348e+01
Generation	30:	Best = 6.822560e+10	FES = 3100	Diversity = 7.6902e+00
Generation	40:	Best = 5.207712e+10	FES = 4100	Diversity = 5.3684e+00
Generation	50:	Best = 4.284238e+10	FES = 5100	Diversity = 4.5071e+00
Generation	60:	Best = 3.858634e+10	FES = 6100	Diversity = 3.7327e+00
Generation	70:	Best = 3.427946e+10	FES = 7100	Diversity = 3.2702e+00
Generation	80:	Best = 3.046772e+10	FES = 8100	Diversity = 3.4109e+00
Generation	90:	Best = 2.730062e+10	FES = 9100	Diversity = 3.1028e+00
Generation	100:	Best = 2.520947e+10	FES = 10100	Diversity = 2.6928e+00
Generation	110:	Best = 2.322205e+10	FES = 11100	Diversity = 2.9393e+00
Generation	120:	Best = 2.192804e+10	FES = 12100	Diversity = 2.3658e+00
Generation	130:	Best = 2.080060e+10	FES = 13100	Diversity = 2.0410e+00
Generation	140:	Best = 1.955336e+10	FES = 14100	Diversity = 2.1205e+00
Generation	150:	Best = 1.866925e+10	FES = 15100	Diversity = 2.1275e+00
Generation	160:	Best = 1.781813e+10	FES = 16100	Diversity = 2.3591e+00
Generation	170:	Best = 1.699894e+10	FES = 17100	Diversity = 1.9815e+00
Generation	180:	Best = 1.628773e+10	FES = 18100	Diversity = 2.0478e+00
Generation	190:	Best = 1.568888e+10	FES = 19100	Diversity = 1.8459e+00
Generation	200:	Best = 1.519795e+10	FES = 20100	Diversity = 1.5709e+00
Generation	210:	Best = 1.468337e+10	FES = 21100	Diversity = 1.6116e+00
Generation	220:	Best = 1.422483e+10	FES = 22100	Diversity = 1.5011e+00
Generation	230:	Best = 1.377851e+10	FES = 23100	Diversity = 1.5526e+00
Generation	240:	Best = 1.325224e+10	FES = 24100	Diversity = 1.5076e+00
Generation	250:	Best = 1.284576e+10	FES = 25100	Diversity = 1.2819e+00
Generation	260:	Best = 1.244687e+10	FES = 26100	Diversity = 1.1947e+00
Generation	270:	Best = 1.213044e+10	FES = 27100	Diversity = 9.7458e-01
Generation	280:	Best = 1.186302e+10	FES = 28100	Diversity = 9.9899e-01
Generation	290:	Best = 1.166548e+10	FES = 29100	Diversity = 9.0944e-01
Generation	300:	Best = 1.150201e+10	FES = 30100	Diversity = 7.8273e-01
Generation	310:	Best = 1.134407e+10	FES = 31100	Diversity = 7.2048e-01
Generation	320:	Best = 1.122584e+10	FES = 32100	Diversity = 6.5640e-01
Generation	330:	Best = 1.108540e+10	FES = 33100	Diversity = 6.7090e-01
Generation	340:	Best = 1.096186e+10	FES = 34100	Diversity = 5.9014e-01
Generation	350:	Best = 1.086381e+10	FES = 35100	Diversity = 5.7848e-01
Generation	360:	Best = 1.075002e+10	FES = 36100	Diversity = 5.4379e-01
Generation	370:	Best = 1.064383e+10	FES = 37100	Diversity = 5.8084e-01
Generation	380:	Best = 1.055829e+10	FES = 38100	Diversity = 5.1358e-01
Generation	390:	Best = 1.048430e+10	FES = 39100	Diversity = 5.2008e-01
Generation	400:	Best = 1.042040e+10	FES = 40100	Diversity = 4.8417e-01
Generation	410:	Best = 1.034620e+10	FES = 41100	Diversity = 4.9660e-01
Generation	420:	Best = 1.030808e+10	FES = 42100	Diversity = 4.4993e-01
Generation	430:	Best = 1.026625e+10	FES = 43100	Diversity = 3.4431e-01
Generation	440:	Best = 1.022404e+10	FES = 44100	Diversity = 3.8835e-01

Generation	450:	Best = 1.018334e+10	FES = 45100	Diversity = 3.6003e-01
Generation	460:	Best = 1.013652e+10	FES = 46100	Diversity = 3.6118e-01
Generation	470:	Best = 1.009312e+10	FES = 47100	Diversity = 3.3444e-01
Generation	480:	Best = 1.004976e+10	FES = 48100	Diversity = 3.3027e-01
Generation	490:	Best = 1.001701e+10	FES = 49100	Diversity = 2.7121e-01
Generation	500:	Best = 9.993021e+09	FES = 50100	Diversity = 2.5577e-01
Generation	510:	Best = 9.966085e+09	FES = 51100	Diversity = 2.5081e-01
Generation	520:	Best = 9.948788e+09	FES = 52100	Diversity = 2.0582e-01
Generation	530:	Best = 9.934235e+09	FES = 53100	Diversity = 1.8237e-01
Generation	540:	Best = 9.925752e+09	FES = 54100	Diversity = 1.4835e-01
Generation	550:	Best = 9.920531e+09	FES = 55100	Diversity = 1.1536e-01
Generation	560:	Best = 9.916394e+09	FES = 56100	Diversity = 8.2758e-02
Generation	570:	Best = 9.913370e+09	FES = 57100	Diversity = 6.8487e-02
Generation	580:	Best = 9.911022e+09	FES = 58100	Diversity = 5.3703e-02
Generation	590:	Best = 9.909595e+09	FES = 59100	Diversity = 3.4333e-02
Generation	600:	Best = 9.908964e+09	FES = 60100	Diversity = 2.5064e-02
Generation	610:	Best = 9.908378e+09	FES = 61100	Diversity = 1.7872e-02
Generation	620:	Best = 9.908158e+09	FES = 62100	Diversity = 1.2499e-02
Generation	630:	Best = 9.908034e+09	FES = 63100	Diversity = 9.2338e-03
Generation	640:	Best = 9.907942e+09	FES = 64100	Diversity = 5.0815e-03
Generation	650:	Best = 9.907886e+09	FES = 65100	Diversity = 4.3921e-03
Generation	660:	Best = 9.907851e+09	FES = 66100	Diversity = 2.8765e-03
Generation	670:	Best = 9.907822e+09	FES = 67100	Diversity = 2.6184e-03
Generation	680:	Best = 9.907799e+09	FES = 68100	Diversity = 2.1908e-03
Generation	690:	Best = 9.907783e+09	FES = 69100	Diversity = 2.2739e-03
Generation	700:	Best = 9.907778e+09	FES = 70100	Diversity = 1.0131e-03
Generation	710:	Best = 9.907775e+09	FES = 71100	Diversity = 5.9096e-04
Generation	720:	Best = 9.907773e+09	FES = 72100	Diversity = 6.7361e-04
Generation	730:	Best = 9.907772e+09	FES = 73100	Diversity = 3.4742e-04
Generation	740:	Best = 9.907772e+09	FES = 74100	Diversity = 2.4365e-04
Generation	750:	Best = 9.907772e+09	FES = 75100	Diversity = 1.6176e-04
Generation	760:	Best = 9.907772e+09	FES = 76100	Diversity = 1.3686e-04
Generation	770:	Best = 9.907772e+09	FES = 77100	Diversity = 1.0948e-04
Generation	780:	Best = 9.907772e+09	FES = 78100	Diversity = 9.2741e-05
Generation	790:	Best = 9.907772e+09	FES = 79100	Diversity = 6.7476e-05
Generation	800:	Best = 9.907772e+09	FES = 80100	Diversity = 4.5970e-05
Generation	810:	Best = 9.907772e+09	FES = 81100	Diversity = 2.7666e-05
Generation	820:	Best = 9.907772e+09	FES = 82100	Diversity = 1.7691e-05
Generation	830:	Best = 9.907772e+09	FES = 83100	Diversity = 1.0812e-05
Generation	840:	Best = 9.907772e+09	FES = 84100	Diversity = 7.1065e-06
Generation	850:	Best = 9.907772e+09	FES = 85100	Diversity = 6.0793e-06
Generation	860:	Best = 9.907772e+09	FES = 86100	Diversity = 8.4474e-06
Generation	870:	Best = 9.907772e+09	FES = 87100	Diversity = 4.2114e-06
Generation	880:	Best = 9.907772e+09	FES = 88100	Diversity = 3.5779e-06
Generation	890:	Best = 9.907772e+09	FES = 89100	Diversity = 2.6586e-06
Generation	900:	Best = 9.907772e+09	FES = 90100	Diversity = 1.0822e-06
Generation	910:	Best = 9.907772e+09	FES = 91100	Diversity = 1.1404e-06
Generation	920:	Best = 9.907772e+09	FES = 92100	Diversity = 6.6063e-07
Generation	930:	Best = 9.907772e+09	FES = 93100	Diversity = 7.4390e-07
Generation	940:	Best = 9.907772e+09	FES = 94100	Diversity = 3.8688e-07
Generation	950:	Best = 9.907772e+09	FES = 95100	Diversity = 2.5220e-07
Generation	960:	Best = 9.907772e+09	FES = 96100	Diversity = 2.1274e-07
Generation	970:	Best = 9.907772e+09	FES = 97100	Diversity = 7.5319e-08
Generation	980:	Best = 9.907772e+09	FES = 98100	Diversity = 4.2844e-08

[illegible]

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[illegible]

[illegible]

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```

Generation 9630: Best = 9.907772e+09 FES = 963100 Diversity = 5.0618e-14
Generation 9640: Best = 9.907772e+09 FES = 964100 Diversity = 5.0618e-14
Generation 9650: Best = 9.907772e+09 FES = 965100 Diversity = 5.0618e-14
Generation 9660: Best = 9.907772e+09 FES = 966100 Diversity = 5.0618e-14
Generation 9670: Best = 9.907772e+09 FES = 967100 Diversity = 5.0618e-14
Generation 9680: Best = 9.907772e+09 FES = 968100 Diversity = 5.0618e-14
Generation 9690: Best = 9.907772e+09 FES = 969100 Diversity = 5.0618e-14
Generation 9700: Best = 9.907772e+09 FES = 970100 Diversity = 5.0618e-14
Generation 9710: Best = 9.907772e+09 FES = 971100 Diversity = 5.0618e-14
Generation 9720: Best = 9.907772e+09 FES = 972100 Diversity = 5.0618e-14
Generation 9730: Best = 9.907772e+09 FES = 973100 Diversity = 5.0618e-14
Generation 9740: Best = 9.907772e+09 FES = 974100 Diversity = 5.0618e-14
Generation 9750: Best = 9.907772e+09 FES = 975100 Diversity = 5.0618e-14
Generation 9760: Best = 9.907772e+09 FES = 976100 Diversity = 5.0618e-14
Generation 9770: Best = 9.907772e+09 FES = 977100 Diversity = 5.0618e-14
Generation 9780: Best = 9.907772e+09 FES = 978100 Diversity = 5.0618e-14
Generation 9790: Best = 9.907772e+09 FES = 979100 Diversity = 5.0618e-14
Generation 9800: Best = 9.907772e+09 FES = 980100 Diversity = 5.0618e-14
Generation 9810: Best = 9.907772e+09 FES = 981100 Diversity = 5.0618e-14
Generation 9820: Best = 9.907772e+09 FES = 982100 Diversity = 5.0618e-14
Generation 9830: Best = 9.907772e+09 FES = 983100 Diversity = 5.0618e-14
Generation 9840: Best = 9.907772e+09 FES = 984100 Diversity = 5.0618e-14
Generation 9850: Best = 9.907772e+09 FES = 985100 Diversity = 5.0618e-14
Generation 9860: Best = 9.907772e+09 FES = 986100 Diversity = 5.0618e-14
Generation 9870: Best = 9.907772e+09 FES = 987100 Diversity = 5.0618e-14
Generation 9880: Best = 9.907772e+09 FES = 988100 Diversity = 5.0618e-14
Generation 9890: Best = 9.907772e+09 FES = 989100 Diversity = 5.0618e-14
Generation 9900: Best = 9.907772e+09 FES = 990100 Diversity = 5.0618e-14
Generation 9910: Best = 9.907772e+09 FES = 991100 Diversity = 5.0618e-14
Generation 9920: Best = 9.907772e+09 FES = 992100 Diversity = 5.0618e-14
Generation 9930: Best = 9.907772e+09 FES = 993100 Diversity = 5.0618e-14
Generation 9940: Best = 9.907772e+09 FES = 994100 Diversity = 5.0618e-14
Generation 9950: Best = 9.907772e+09 FES = 995100 Diversity = 5.0618e-14
Generation 9960: Best = 9.907772e+09 FES = 996100 Diversity = 5.0618e-14
Generation 9970: Best = 9.907772e+09 FES = 997100 Diversity = 5.0618e-14
Generation 9980: Best = 9.907772e+09 FES = 998100 Diversity = 5.0618e-14
Generation 9990: Best = 9.907772e+09 FES = 999100 Diversity = 5.0618e-14

```

Optimization completed!

Final Best Fitness = 9.907772e+09

Total Function Evaluations = 1000000

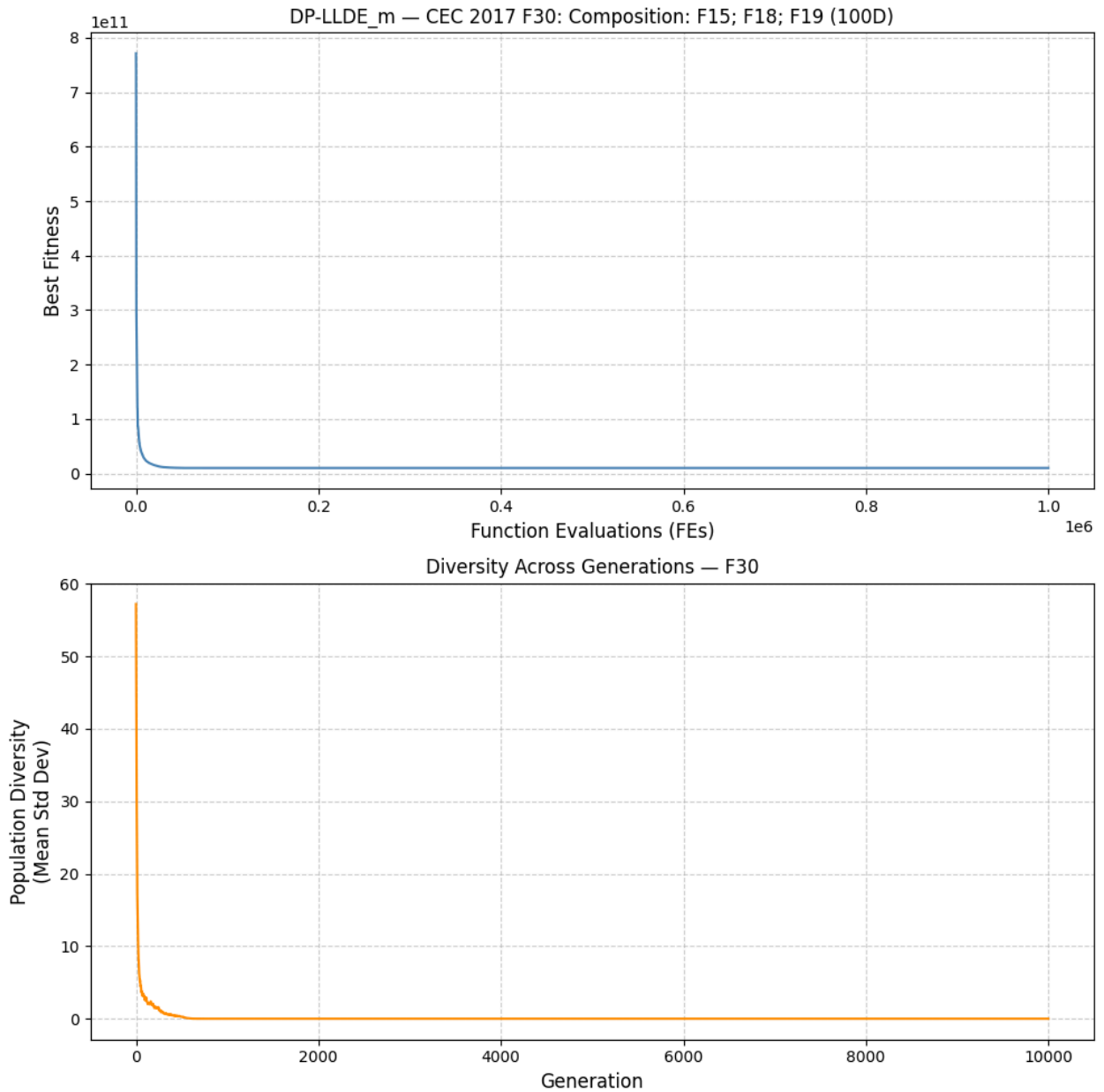
```

=====
=
Results:
  Best Fitness          : 9.907772e+09
  Global Optimum        : 3.000000e+03
  Error (|best-optimum|) : 9.907769e+09
  Best Solution (5 dim) : [ 99.95694126 -26.09338057 -34.1854924 -21.64822772
-25.36387828]
  D^(0) (init. diversity): 5.7220e+01
=====
=

```

✓ Checkpoint saved to: /content/drive/My Drive/thesis_results_2/all_results_che

ckpoint.pkl



Graph saved as convergence_F30.png

✓ Plot saved to: /content/drive/My Drive/thesis_results_2/plots/convergence_F30.png

Section 7. Compiled Results

Print out a summary of all results in a tabular format

```
In [32]: print('=' * 80)
print('SUMMARY — DP-LLDE_m on CEC 2017 (100D), Single Run, Seed=42')
print('=' * 80)
```

```

print(f'{"Function":<10} {"Best Fitness":>18} {"Global Optimum":>18} {"Error |
print('-' * 70)
for fn in [f'F{i}' for i in range(1, 31)]:
    if fn not in all_results:
        continue
    r = all_results[fn]
    if r['best_fit'] is None:
        print(f'{fn:<10} {"N/A (deprecated)":>18} {r["f_global"]:>18.2f} {"N/A
    else:
        print(f'{fn:<10} {r["best_fit"]:>18.6e} {r["f_global"]:>18.2f} {r["err
print('=' * 80)

```

=====

=

SUMMARY – DP-LLDE_m on CEC 2017 (100D), Single Run, Seed=42

=====

=

Function	Best Fitness	Global Optimum	Error best-opt
F1	5.780968e+11	100.00	5.780968e+11
F2	6.067083e+127	200.00	6.067083e+127
F3	1.724335e+04	300.00	1.694335e+04
F4	7.175389e+03	400.00	6.775389e+03
F5	1.027984e+03	500.00	5.279842e+02
F6	6.381420e+02	600.00	3.814201e+01
F7	1.463655e+03	700.00	7.636549e+02
F8	1.277922e+03	800.00	4.779219e+02
F9	6.894113e+03	900.00	5.994113e+03
F10	2.024832e+04	1000.00	1.924832e+04
F11	2.118840e+03	1100.00	1.018840e+03
F12	9.291957e+10	1200.00	9.291957e+10
F13	2.156417e+10	1300.00	2.156417e+10
F14	4.807895e+04	1400.00	4.667895e+04
F15	1.355182e+10	1500.00	1.355182e+10
F16	4.136130e+03	1600.00	2.536130e+03
F17	3.837005e+03	1700.00	2.137005e+03
F18	4.010990e+05	1800.00	3.992990e+05
F19	3.456128e+09	1900.00	3.456126e+09
F20	4.330084e+03	2000.00	2.330084e+03
F21	2.706285e+03	2100.00	6.062849e+02
F22	2.111868e+04	2200.00	1.891868e+04
F23	3.434717e+03	2300.00	1.134717e+03
F24	4.474986e+03	2400.00	2.074986e+03
F25	8.030736e+03	2500.00	5.530736e+03
F26	2.735279e+04	2600.00	2.475279e+04
F27	4.195307e+03	2700.00	1.495307e+03
F28	1.002577e+04	2800.00	7.225774e+03
F29	6.258478e+03	2900.00	3.358478e+03
F30	9.907772e+09	3000.00	9.907769e+09

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